SATA 1D: Along-Track Ramps

Targets 200 m and 2.6 km Apart, Normal and Large Heights

Lucas Ayub Rodrigues Tuler
DLR–DAAD Fellowship No. 734 — Microwaves and Radar Institute (DLR–HR)

September 2026

Setup. SATA 1D, $N_{rx} = 4$, PRF = 2000 Hz, $\mathrm{PRF}_{\mathrm{ch}} = 500$ Hz, DPCA along-track baselines ($\Delta b_{at} = 7.69$ m), $b_{xt} = 63.7, 27.0, 4.1, 1.7$ m ($U(0, 100)$, seed 0), $r_0 = 766.2$ km, $\theta = 20°$, $\lambda = 0.25$ m. All targets are iso-range (same slant range r_0, displaced in ground range with their height). Each scene is simulated twice with the *same heights*: targets 200 m apart in azimuth, and targets 2.6 km apart. The 2.6 km spacing is larger than the width of one block of the ΔC_0 map ($2h_w = 2.52$ km), so every target has its own correction, and keeps every pair of targets at least 1 km away from a multiple of 6.23 km, where two targets would share Doppler bins (the Doppler-alias collision); this limits the separated scenes to $N \leq 5$. Each case is reconstructed without SATA, with SATA whole band and with SATA per sub-band.
Scenes. Along-track ramp: $N \in \{2, 3, 5, 7, 9, 13\}$ iso-range targets, $\alpha \in \{0.1, 1, 1.5, 2, 3, 4, 5, 10, 15, 20, 30, 40, 50$ in four configurations:

configuration	spacing	heights	N	cases	IDs
A: close, normal heights	200 m	$h_j = j \cdot 200\,\mathrm{m} \cdot \tan\alpha$ (up to 2.9 km)	2–13	78	RA
B: separated, normal heights	2.6 km	$h_j = j \cdot 200\,\mathrm{m} \cdot \tan\alpha$ (up to 0.95 km)	2–5	39	RAS
C: close, large heights	200 m	$h_j = j \cdot 2600\,\mathrm{m} \cdot \tan\alpha$ (up to 37 km)	2–13	78	RAH
D: separated, large heights	2.6 km	$h_j = j \cdot 2600\,\mathrm{m} \cdot \tan\alpha$ (up to 12.4 km)	2–5	39	RAHS

Table 1: The four ramp configurations (234 cases). Normal heights: the ramp of slope α over the 200-m spacing. Large heights: the heights of a ramp of slope α over 2.6-km steps, i.e. 13 times the normal heights. In C the height difference between neighbours 200 m apart is $2600 \tan\alpha$ (45 m at 1°, 950 m at 20°); D is the ramp of slope α itself, sampled every 2.6 km. Heights above ~ 9 km (C: $\alpha \geq 10°$ for $N \geq 5$; D: $\alpha \geq 40°$ for $N = 5$) are beyond real terrain and are kept only to complete the grid.

1 The ambiguity metric

The earlier heatmaps used the *worst ambiguity*: the largest value of the focused response outside ±40 samples (±154 m) of every target. That maximum is not always an ambiguity. With SATA it is often the pedestal around the main lobe (the skirt at $|t| \lesssim 0.3$ s, about −36 dB), which does not depend on the correction; without SATA it is the ambiguity at ±0.81 s. In case RA-007 ($N = 2$, $\alpha = 5°$, 200 m apart) both happen to be at −36.5 dB, so the improvement read 0 dB although the ambiguities differ by 12 dB (Fig. 1).

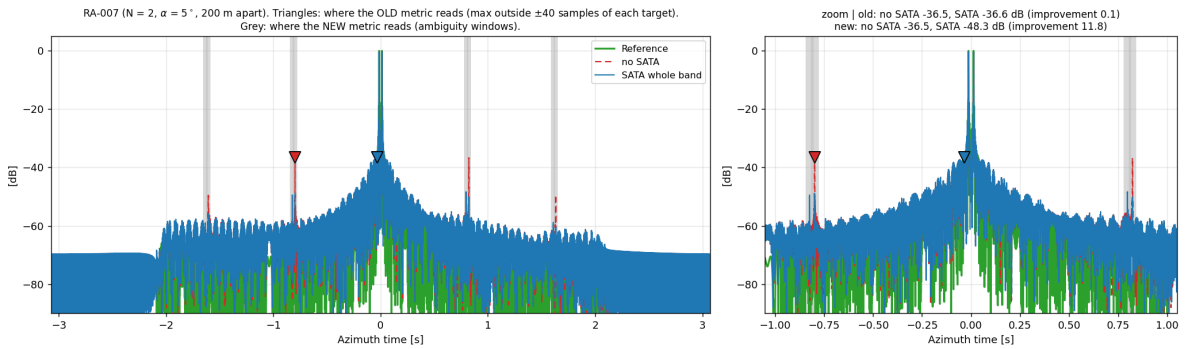


Figure 1: Case RA-007. Triangles: where the old metric reads, for no SATA (the ambiguity at +0.81 s) and for SATA (the main-lobe pedestal). Grey bands: the windows of the metric used here.

Ambiguity peak. A Doppler shift of $m\,\mathrm{PRF}_{\mathrm{ch}}$ moves the focused copy of a target by $m\,\Delta t_{amb}$, with

$$\Delta t_{amb} = \frac{\mathrm{PRF}_{\mathrm{ch}}}{K_a} = \frac{\mathrm{PRF}_{\mathrm{ch}}\,\lambda\,r_0}{2\,v^2} = 0.810\ \mathrm{s} \quad (6.23\ \mathrm{km}). \tag{1}$$

The ambiguity peak is the largest |IRF| within ±20 ms of $x_j/v + m\,\Delta t_{amb}$, $m = \pm1, \pm2$, over all targets j, leaving out ±20 ms around every target's own main lobe, in dB relative to the reference peak. The grey bands in every IRF page show where it is read. Improvement = ambiguity peak (no SATA) − ambiguity peak (SATA). For the reference, which has no ambiguities, it reads the level of the IRF floor in the windows (−52 to −63 dB); with targets 2.6 km apart and $N \geq 3$ the windows of one target also see the skirt of the main lobe of a neighbour, which sets a floor of −46 dB ($N = 3$) and −43 dB ($N = 5$) common to all methods.

2 Results

configuration	cases	no SATA median	SATA whole band median (range)	improvement median (max)	impr., $1° \leq \alpha \leq 5°$ median (min)	|sub − whole| max
A: 200 m, normal heights	78	−22.5 dB	−44.8 (−56.7 . . . − 24.2) dB	18.8 (30.7) dB	18.8 (7.0) dB	0.09 dB
B: 2.6 km, normal heights	39	−28.6 dB	−45.9 (−58.2 . . . − 43.0) dB	20.2 (44.2) dB	12.8 (1.1) dB	0.39 dB
C: 200 m, large heights	78	−7.3 dB	−25.1 (−56.6 . . . − 13.3) dB	12.6 (30.8) dB	20.7 (12.4) dB	0.20 dB
D: 2.6 km, large heights	39	−8.9 dB	−46.0 (−58.2 . . . − 42.9) dB	36.7 (49.2) dB	35.0 (23.8) dB	0.39 dB

Table 2: Ambiguity peak, medians over the cases. With the old metric the improvement at $1° \leq \alpha \leq 5°$ in A had a median of 2.1 dB and was exactly 0.0 in 15 of the 36 cells.

- **Small slopes (A).** SATA reduces the ambiguities already at $\alpha = 1°$ (heights up to 42 m for $N = 13$): by 7.0 dB for $N = 2$ and by 20 dB for $N = 13$. Only $\alpha = 0.1°$ (heights below 5 m) stays at 0–3 dB, where there is almost nothing to correct.
- **Targets 200 m apart (A, C).** The ambiguity peak after SATA rises with the height difference between neighbours: in A from −57 dB at small α to −24 dB at 50°; in C, where that difference is 13 times larger, from −57 to −48 dB at 0.1° to −40 dB at 1°, −25 dB at 5° and −13 to −18 dB from 20° on. Neighbours 200 m apart fall into one sub-aperture cell ($\delta x = 309$ m) and receive one common correction. SATA still improves every case of C, by 5.1 to 30.8 dB, because no SATA saturates much earlier: at −5 to −9 dB already from $\alpha \approx 3°$ ($N \geq 5$), the level where the channels are fully decorrelated.
- **Targets 2.6 km apart (B, D).** The ambiguity peak after SATA does not depend on the heights: −56 to −58 dB for $N = 2$ and the −46/−43 dB floor of the metric for $N = 3/5$,

both with normal heights (up to 0.95 km) and with large heights (up to 12.4 km). Every target has its own block in the ΔC_0 map and its own correction, which is exact whatever its height. Without SATA the large heights push the ambiguities to −5 to −9 dB, so the improvement in D reaches 49 dB (median 36.7 dB).

- **Spacing against height** (same scenes, $N \leq 5$). Moving the targets from 200 m to 2.6 km changes the median ambiguity after SATA from −47.1 to −45.9 dB with normal heights (A is already near the floor, and B sits on the −46/−43 dB floor of the metric) and from −25.0 to −46.0 dB with large heights. Making the heights 13 times larger costs about 22 dB at 200 m and nothing at 2.6 km.
- **Whole band and per sub-band** agree within 0.4 dB in every case.
- **Spacing sweep** (Fig. 2). With the heights fixed, the ambiguity after SATA drops as the targets move apart, except at 1.6 and 2 km, where one pair of targets lies within 0.63 km of 6.23 km (4×1.6 and 3×2 km) and shares Doppler bins.

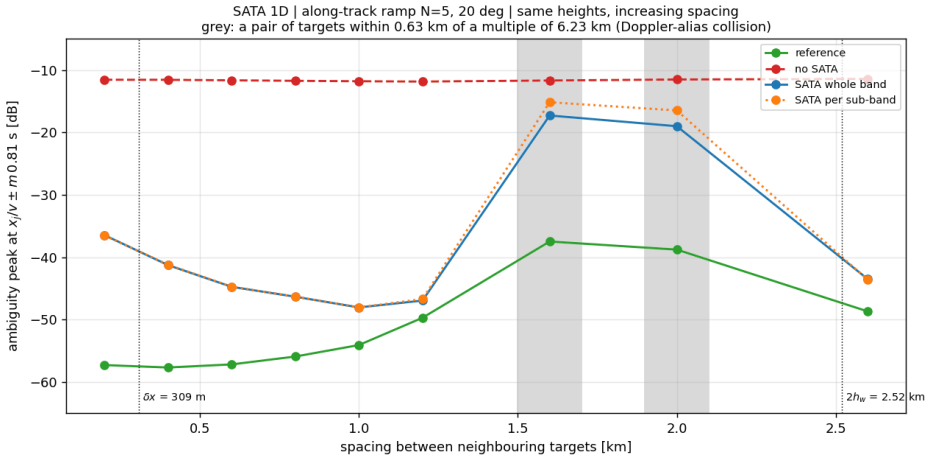


Figure 2: Along-track ramp $N = 5$, $\alpha = 20°$ (heights 0–291 m): ambiguity peak against the spacing between neighbouring targets. Grey: collision spacings; there the ambiguity of one target falls 20–30 ms from the main lobe of another target, so even the reference reads −38 dB in the windows.

3 Heatmaps

For each configuration, two pages: the improvement over no SATA (SATA whole band and SATA per sub-band), and the ambiguity peak itself (no SATA, SATA whole band, SATA per sub-band).

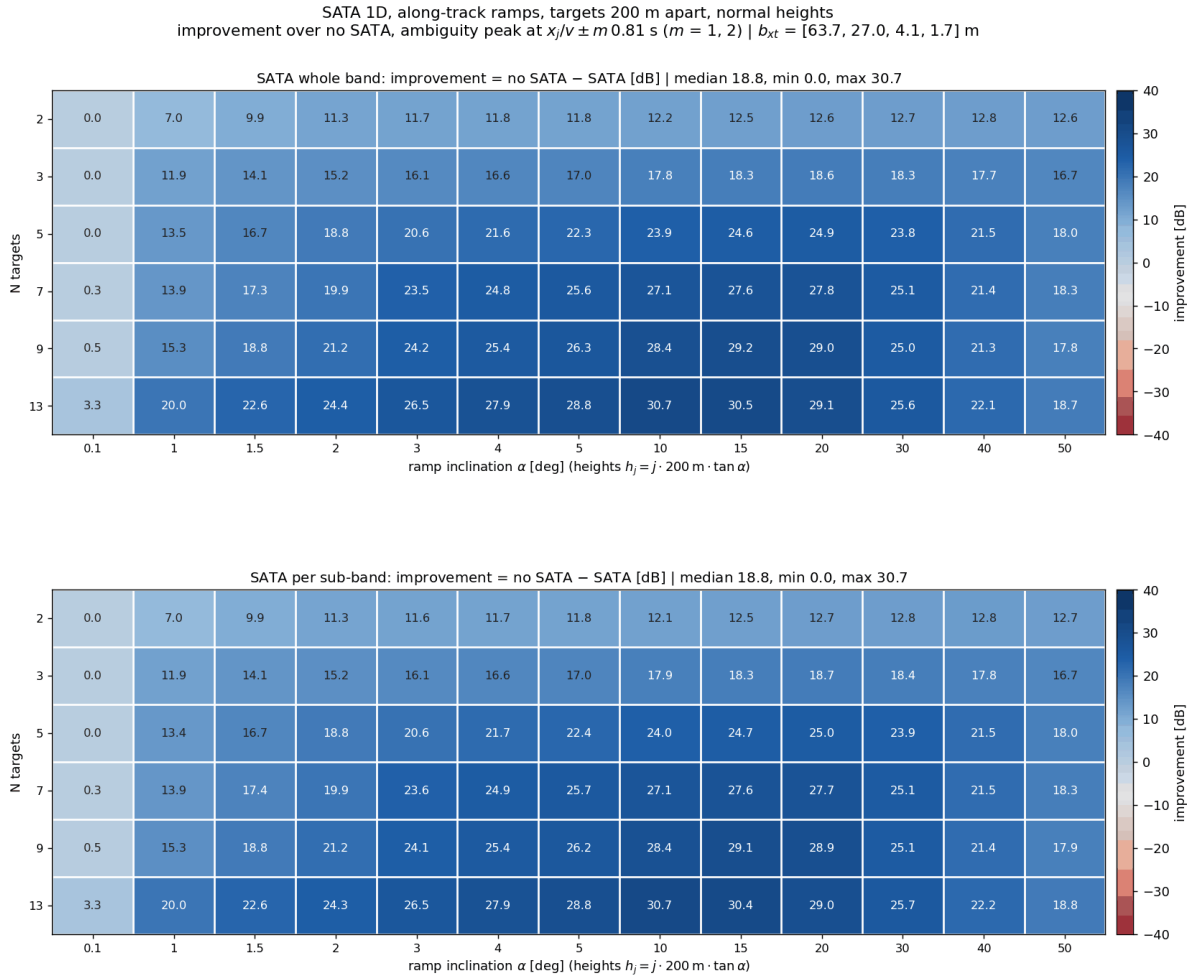


Figure 3: Along-track ramps, A: targets 200 m apart, normal heights ($h_j = j \cdot 200 \tan\alpha$): improvement over no SATA, whole band (top) and per sub-band (bottom).

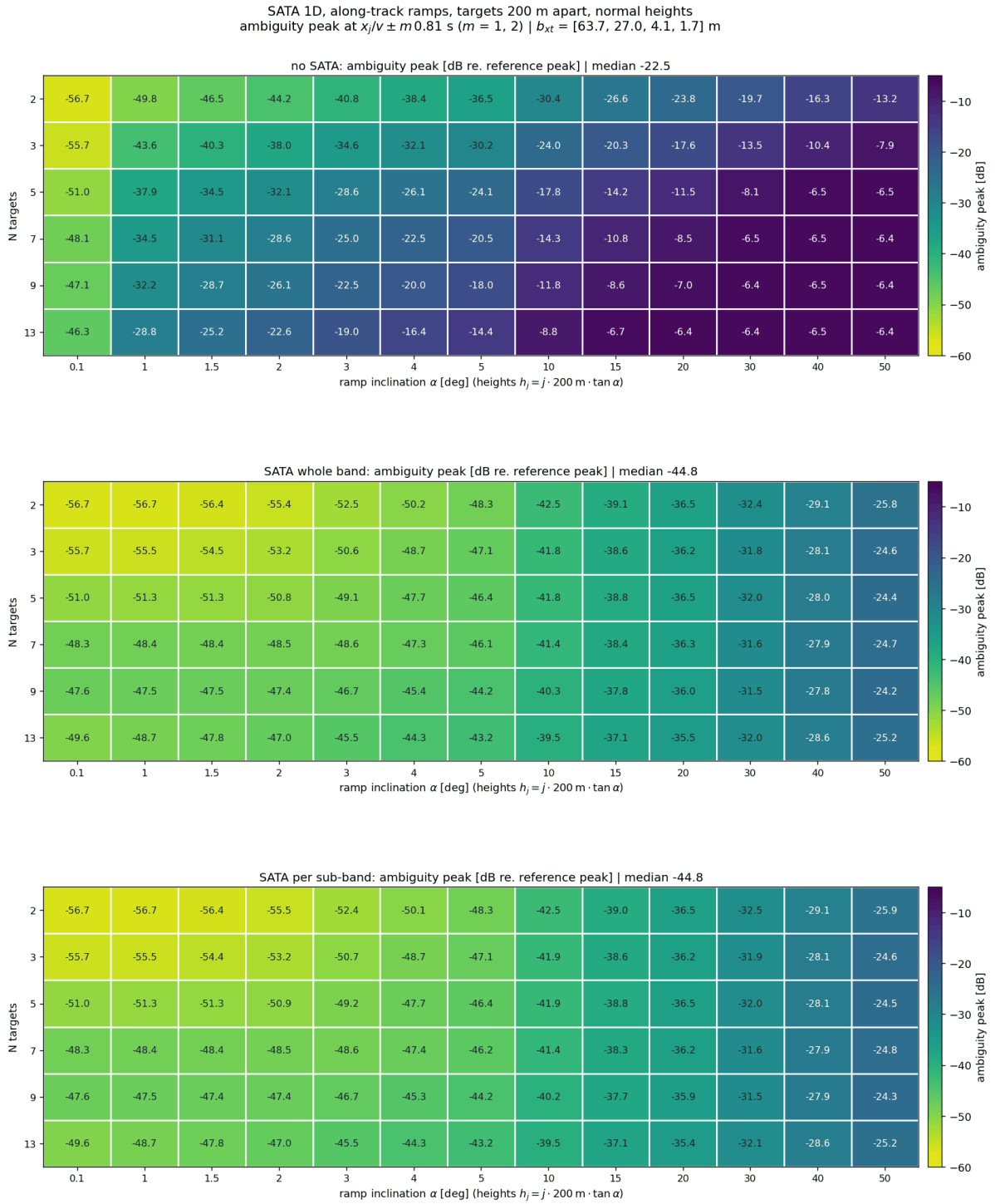


Figure 4: Along-track ramps, A: targets 200 m apart, normal heights ($h_j = j \cdot 200 \tan\alpha$): ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

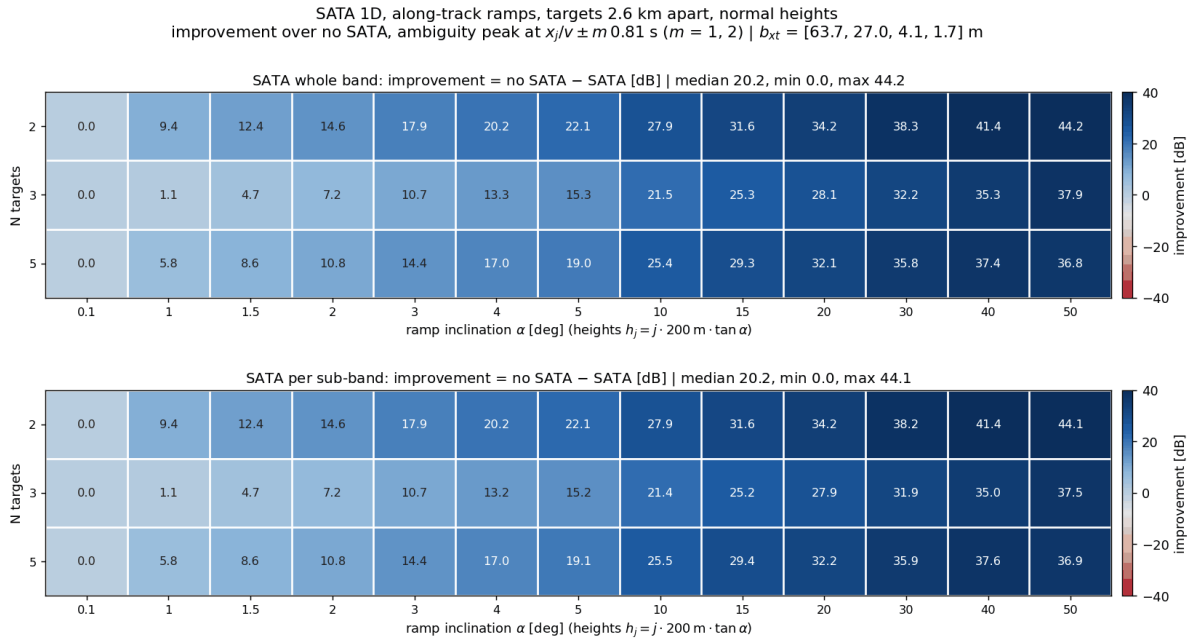


Figure 5: Along-track ramps, B: targets 2.6 km apart, normal heights ($h_j = j \cdot 200 \tan\alpha$): improvement over no SATA, whole band (top) and per sub-band (bottom).

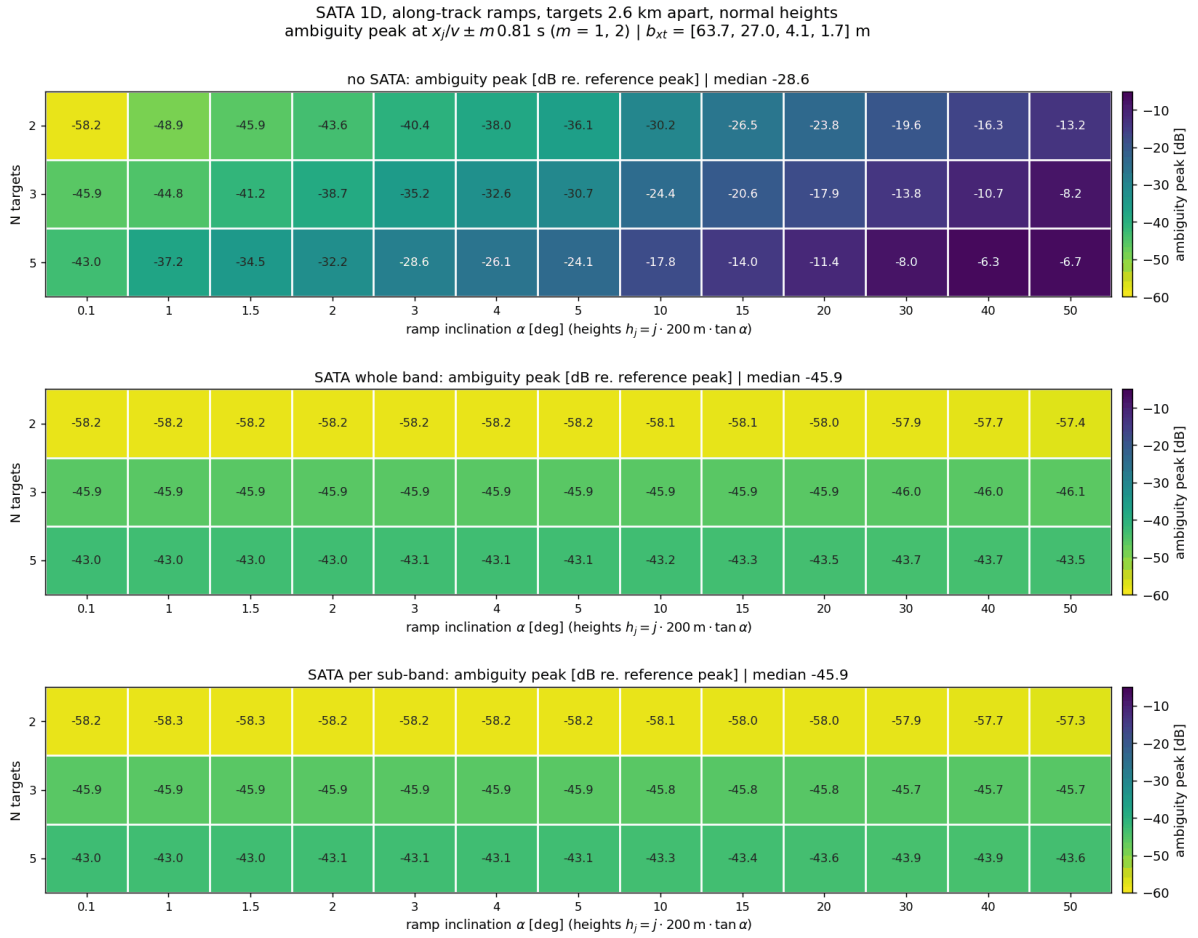


Figure 6: Along-track ramps, B: targets 2.6 km apart, normal heights ($h_j = j \cdot 200 \tan\alpha$): ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

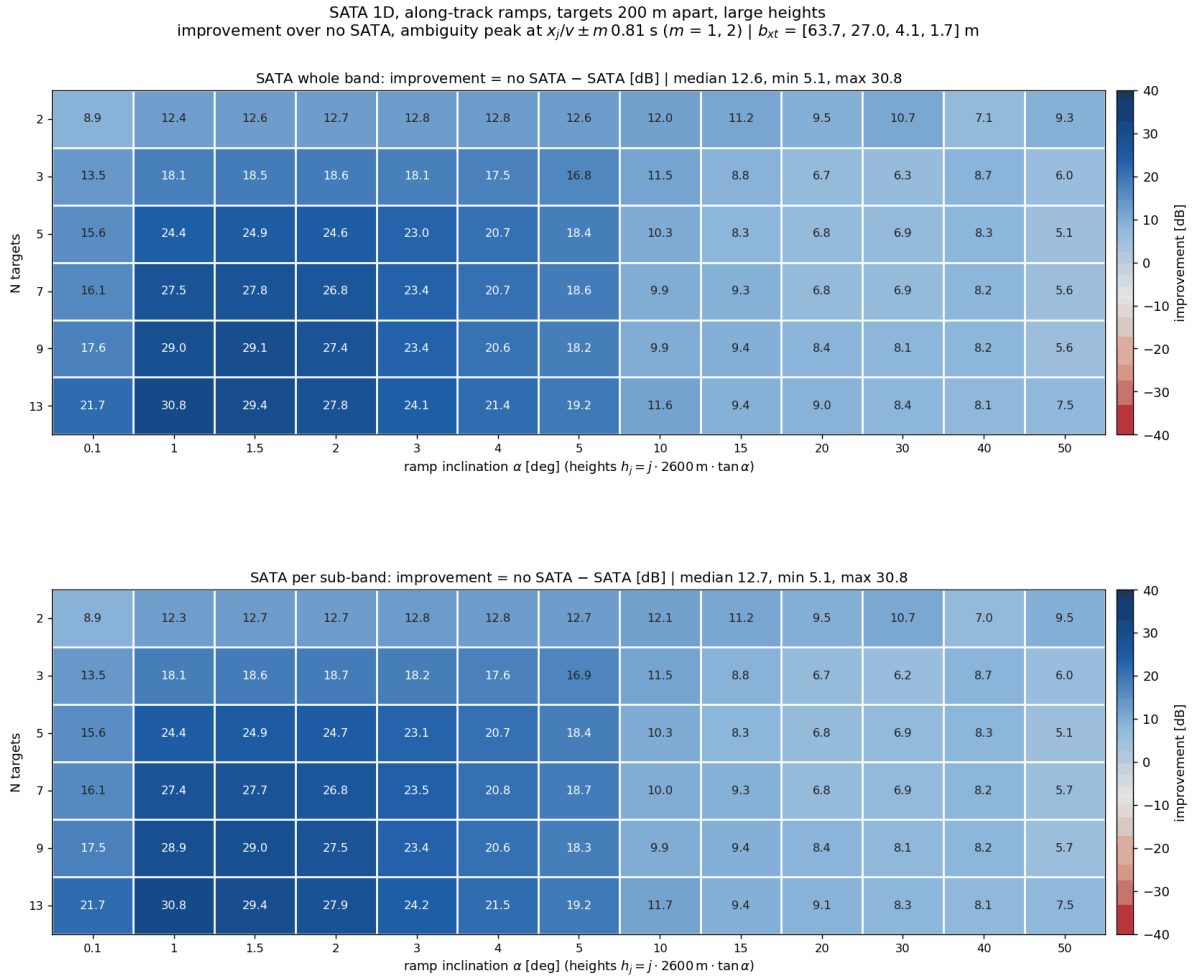


Figure 7: Along-track ramps, C: targets 200 m apart, large heights ($h_j = j \cdot 2600 \tan\alpha$): improvement over no SATA, whole band (top) and per sub-band (bottom).

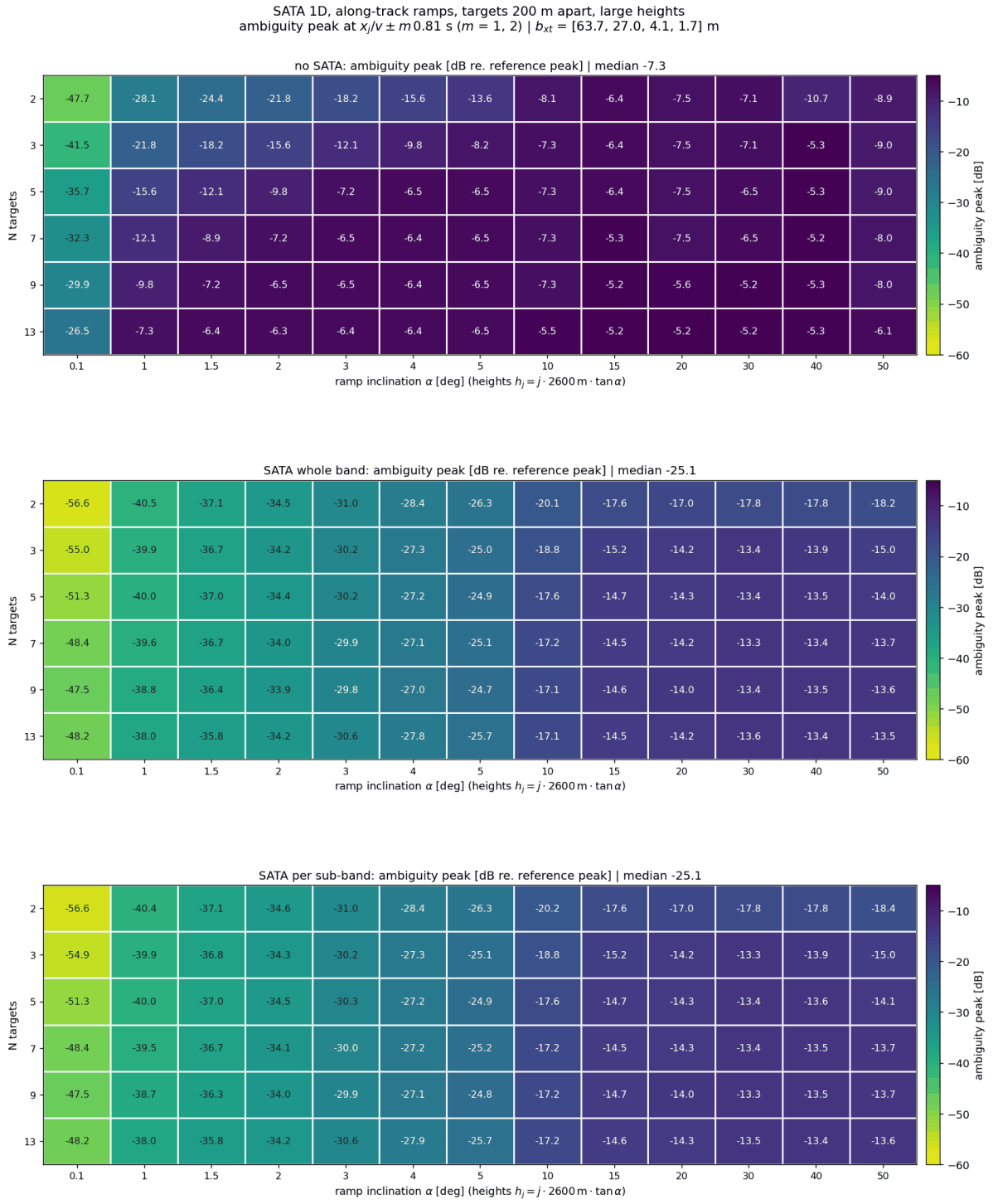


Figure 8: Along-track ramps, C: targets 200 m apart, large heights ($h_j = j \cdot 2600 \tan\alpha$): ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

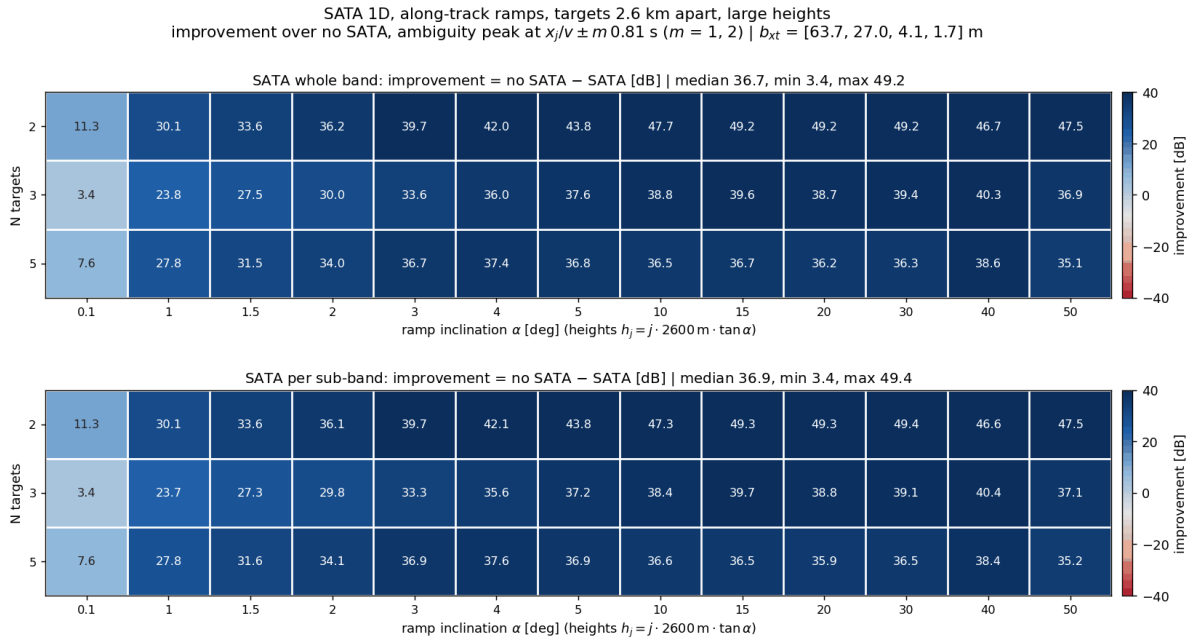


Figure 9: Along-track ramps, D: targets 2.6 km apart, large heights ($h_j = j \cdot 2600 \tan\alpha$): improvement over no SATA, whole band (top) and per sub-band (bottom).



Figure 10: Along-track ramps, D: targets 2.6 km apart, large heights ($h_j = j \cdot 2600 \tan\alpha$): ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

4 IRFs, A: targets 200 m apart, normal heights

Top: reference, SATA whole band and SATA per sub-band. Bottom: reference and no SATA. Grey bands: the ambiguity windows. Right: target geometry in 2D and in 3D (azimuth, ground-range offset, height).

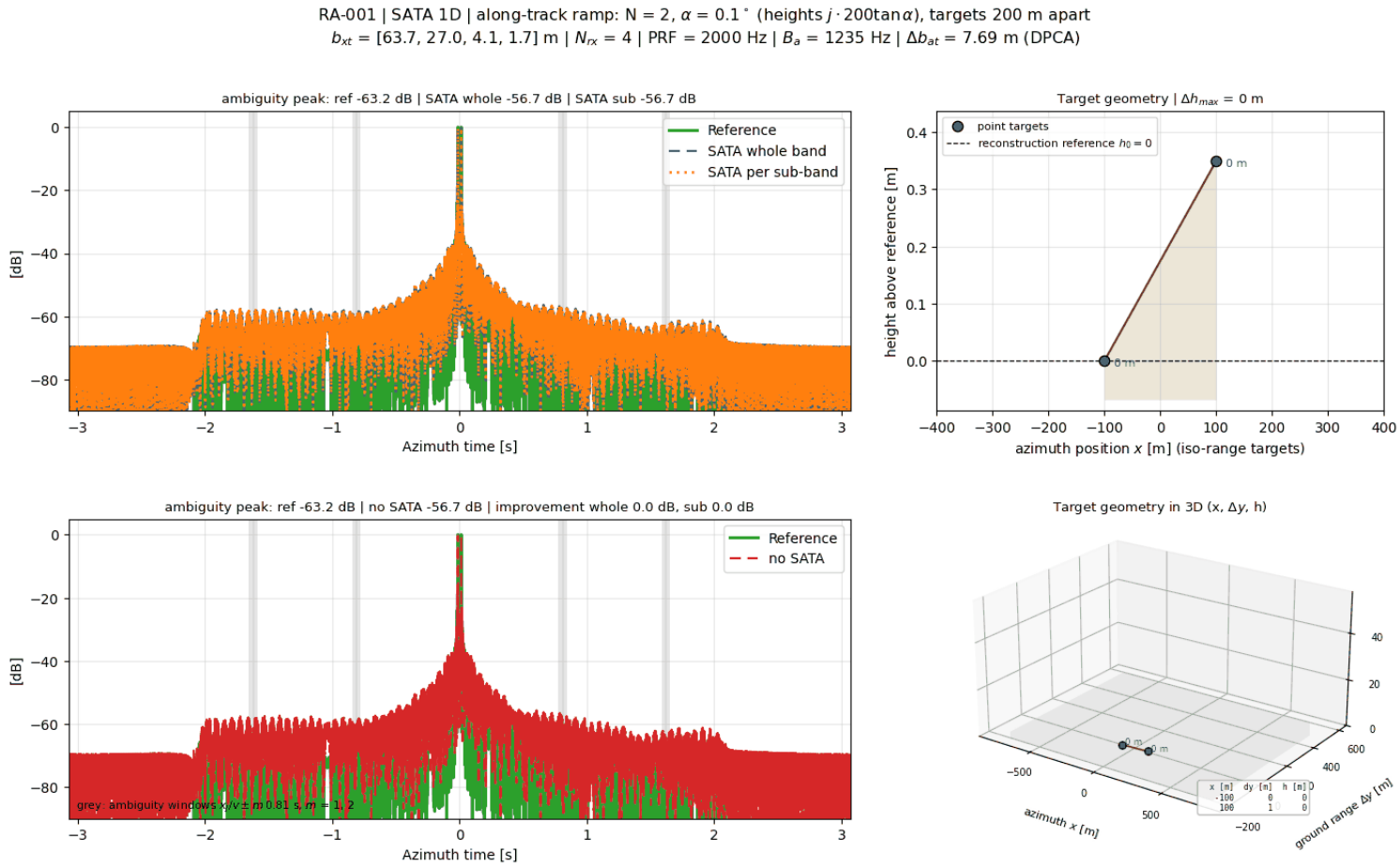


Figure 11: Along-track ramp, 200 m apart, normal heights: RA-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −56.7, SATA whole band −56.7, per sub-band −56.7 dB (improvement 0.0 dB).

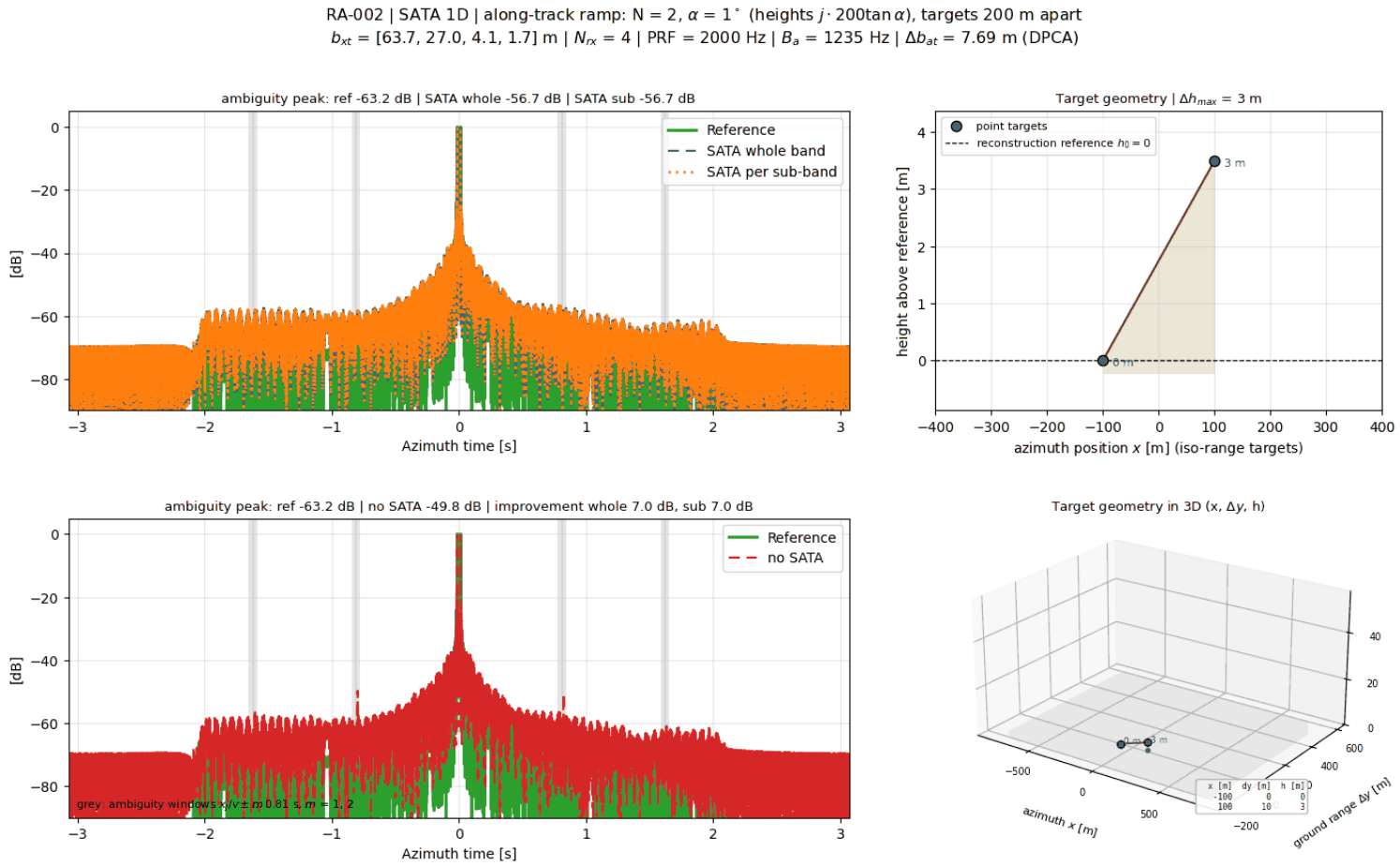


Figure 12: Along-track ramp, 200 m apart, normal heights: RA-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −49.8, SATA whole band −56.7, per sub-band −56.7 dB (improvement 7.0 dB).

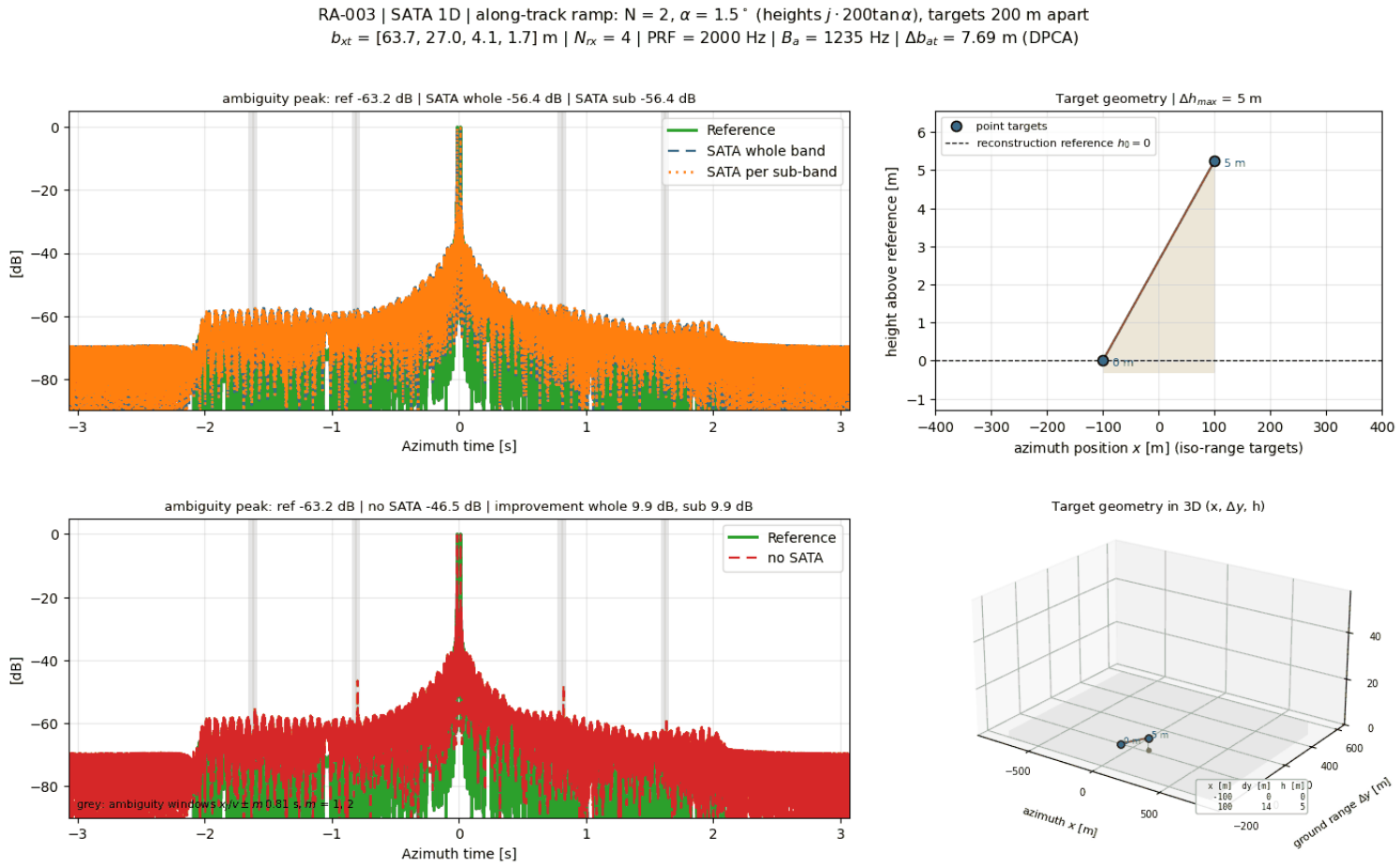


Figure 13: Along-track ramp, 200 m apart, normal heights: RA-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −46.5, SATA whole band −56.4, per sub-band −56.4 dB (improvement 9.9 dB).

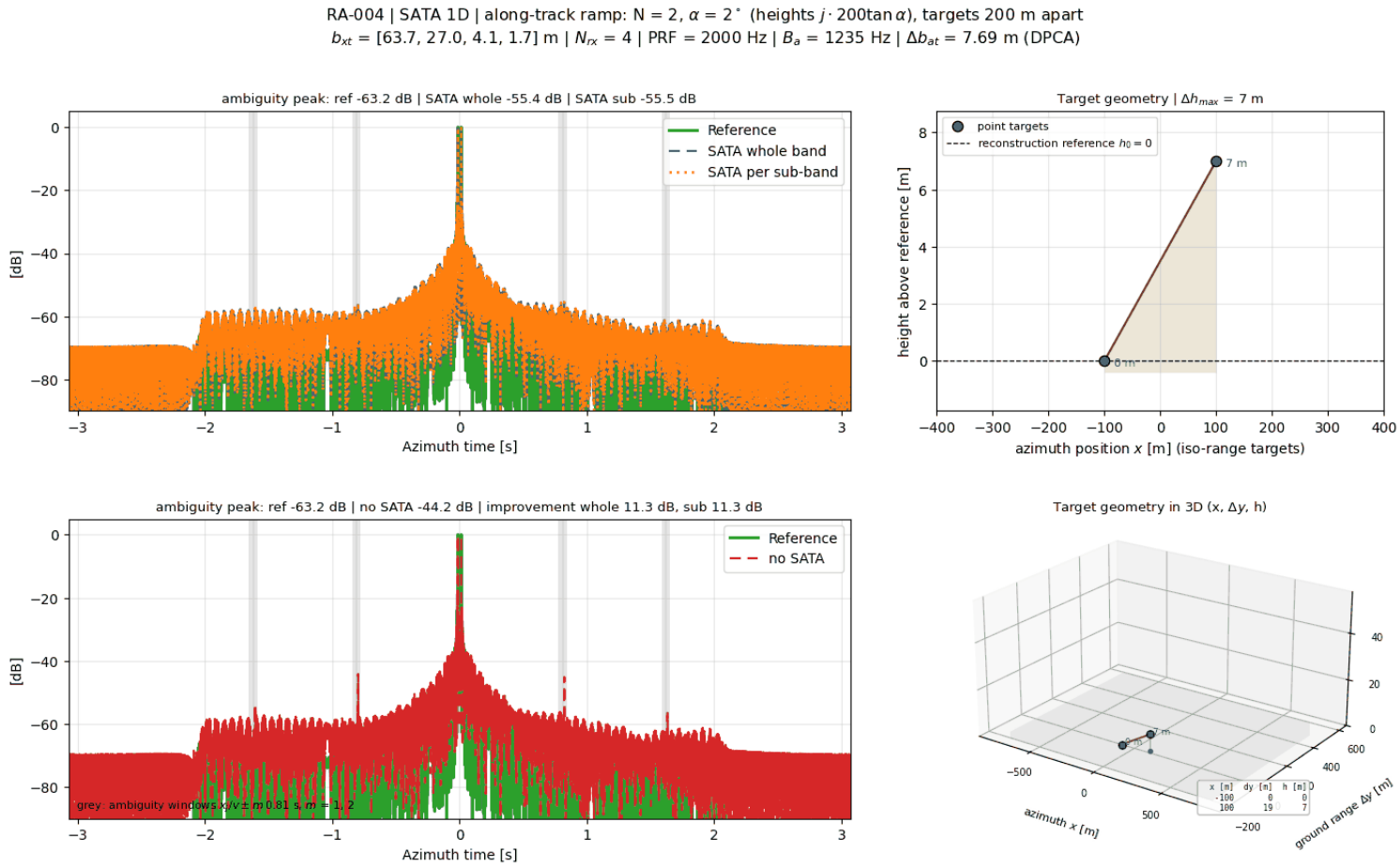


Figure 14: Along-track ramp, 200 m apart, normal heights: RA-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -44.2, SATA whole band -55.4, per sub-band -55.5 dB (improvement 11.3 dB).

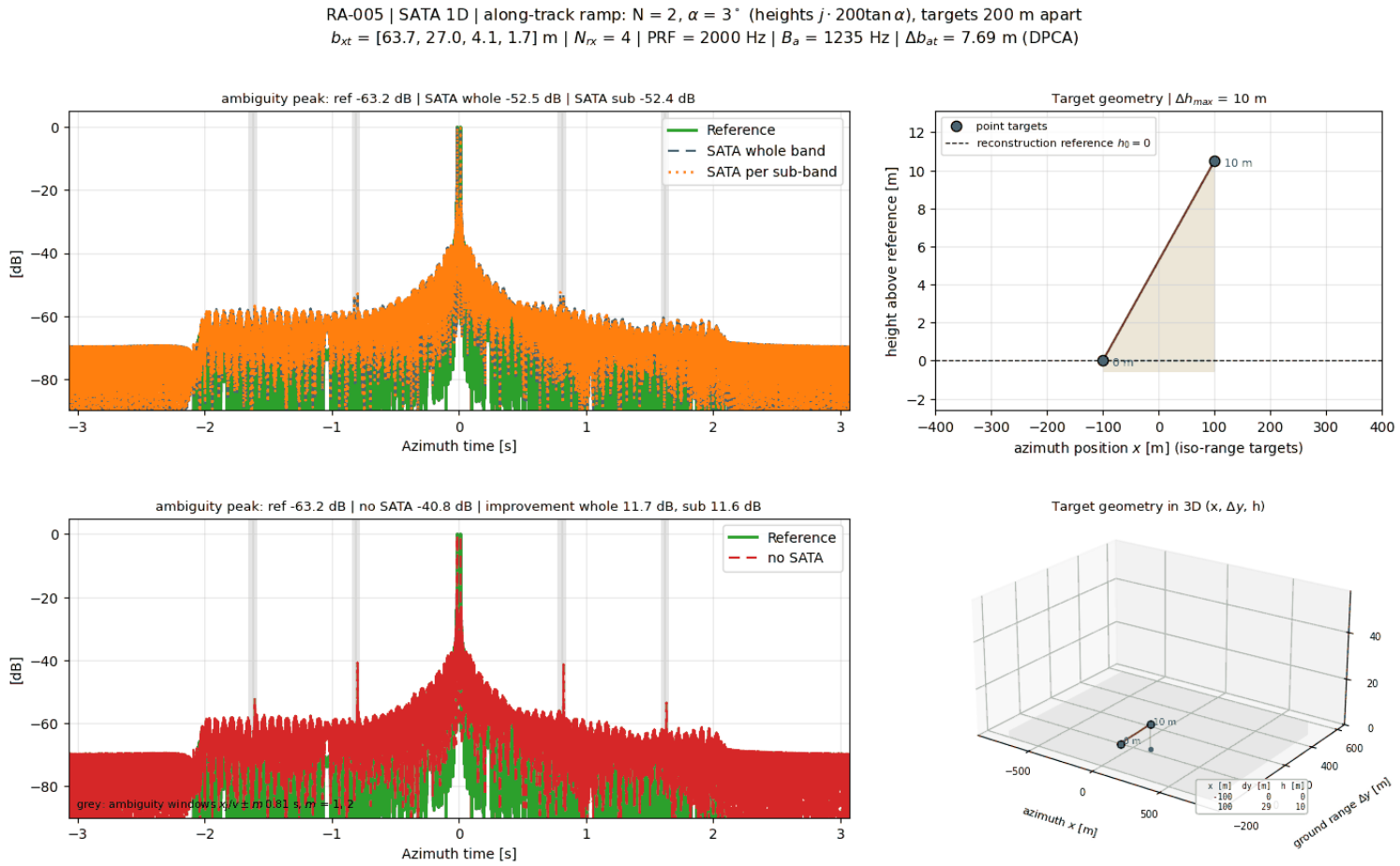


Figure 15: Along-track ramp, 200 m apart, normal heights: RA-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −40.8, SATA whole band −52.5, per sub-band −52.4 dB (improvement 11.7 dB).

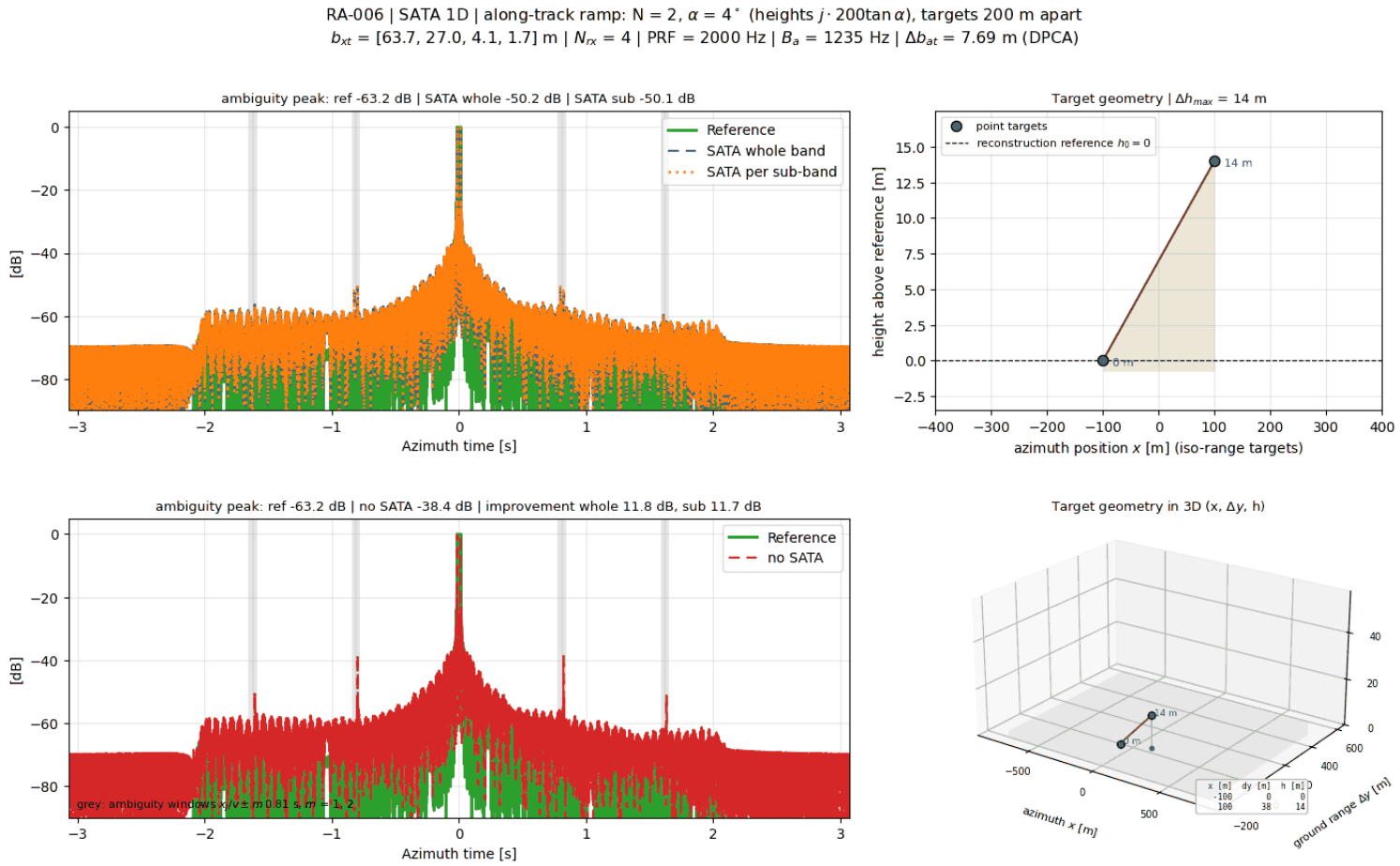


Figure 16: Along-track ramp, 200 m apart, normal heights: RA-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −38.4, SATA whole band −50.2, per sub-band −50.1 dB (improvement 11.8 dB).

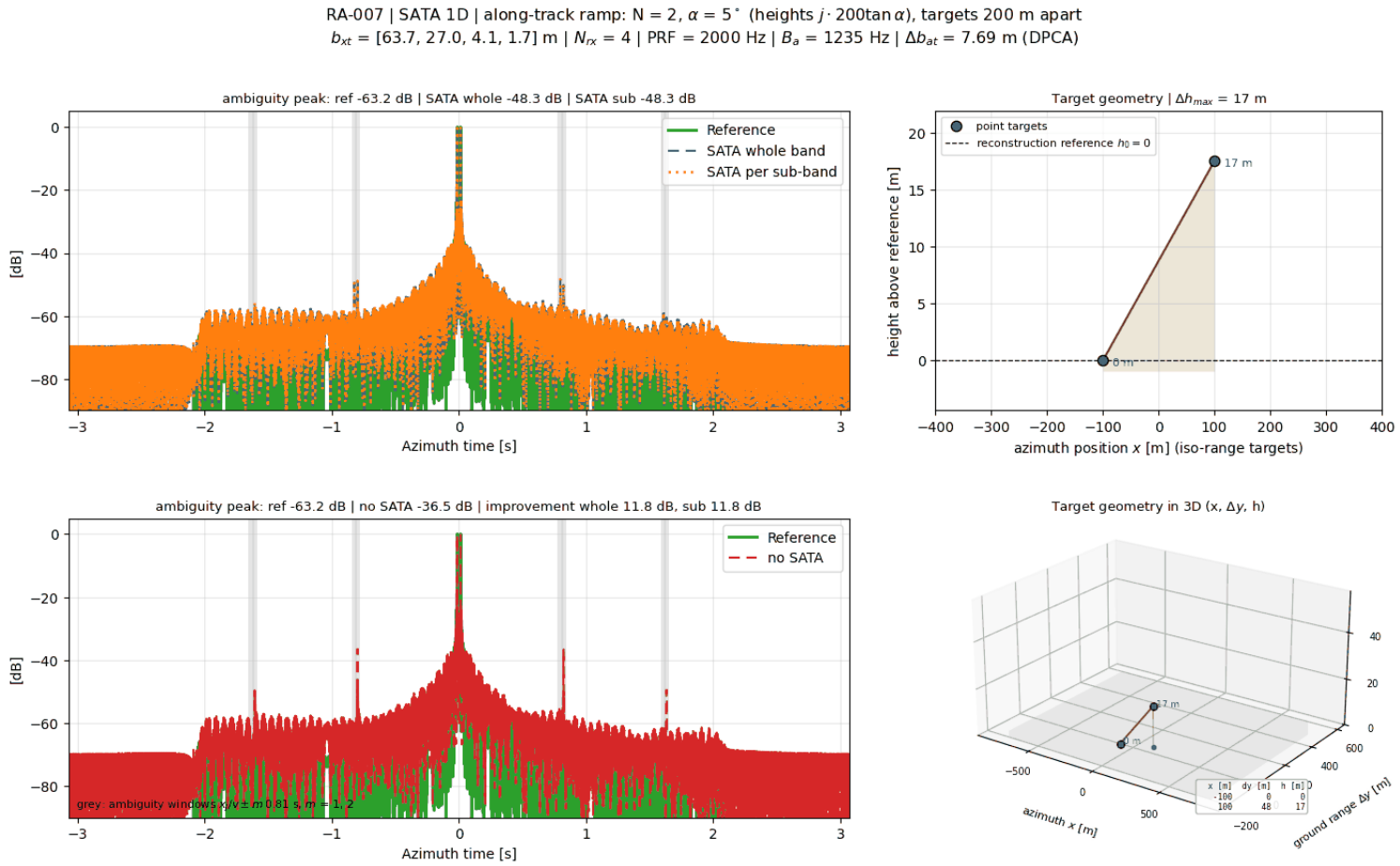


Figure 17: Along-track ramp, 200 m apart, normal heights: RA-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −36.5, SATA whole band −48.3, per sub-band −48.3 dB (improvement 11.8 dB).

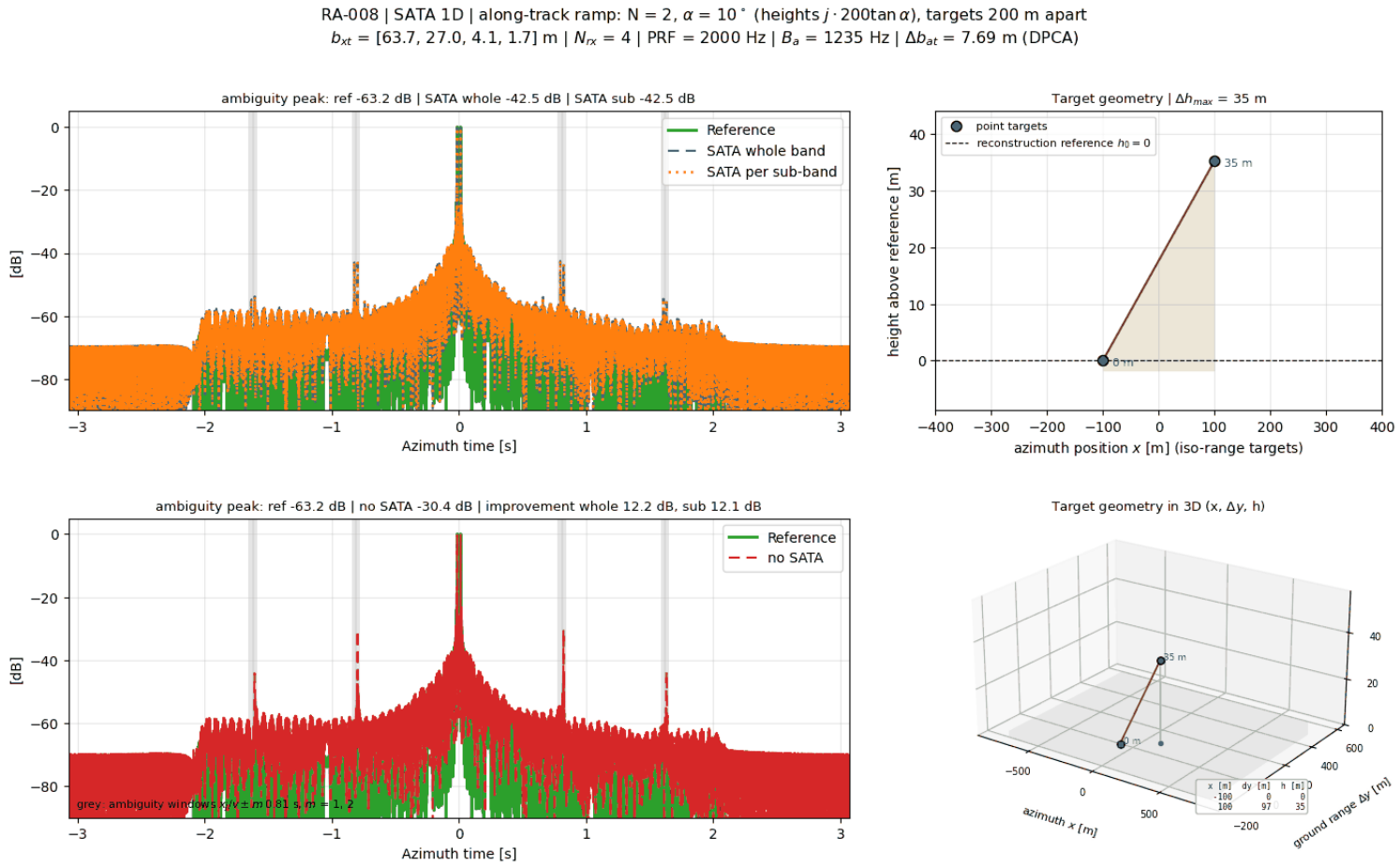


Figure 18: Along-track ramp, 200 m apart, normal heights: RA-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −30.4, SATA whole band −42.5, per sub-band −42.5 dB (improvement 12.2 dB).

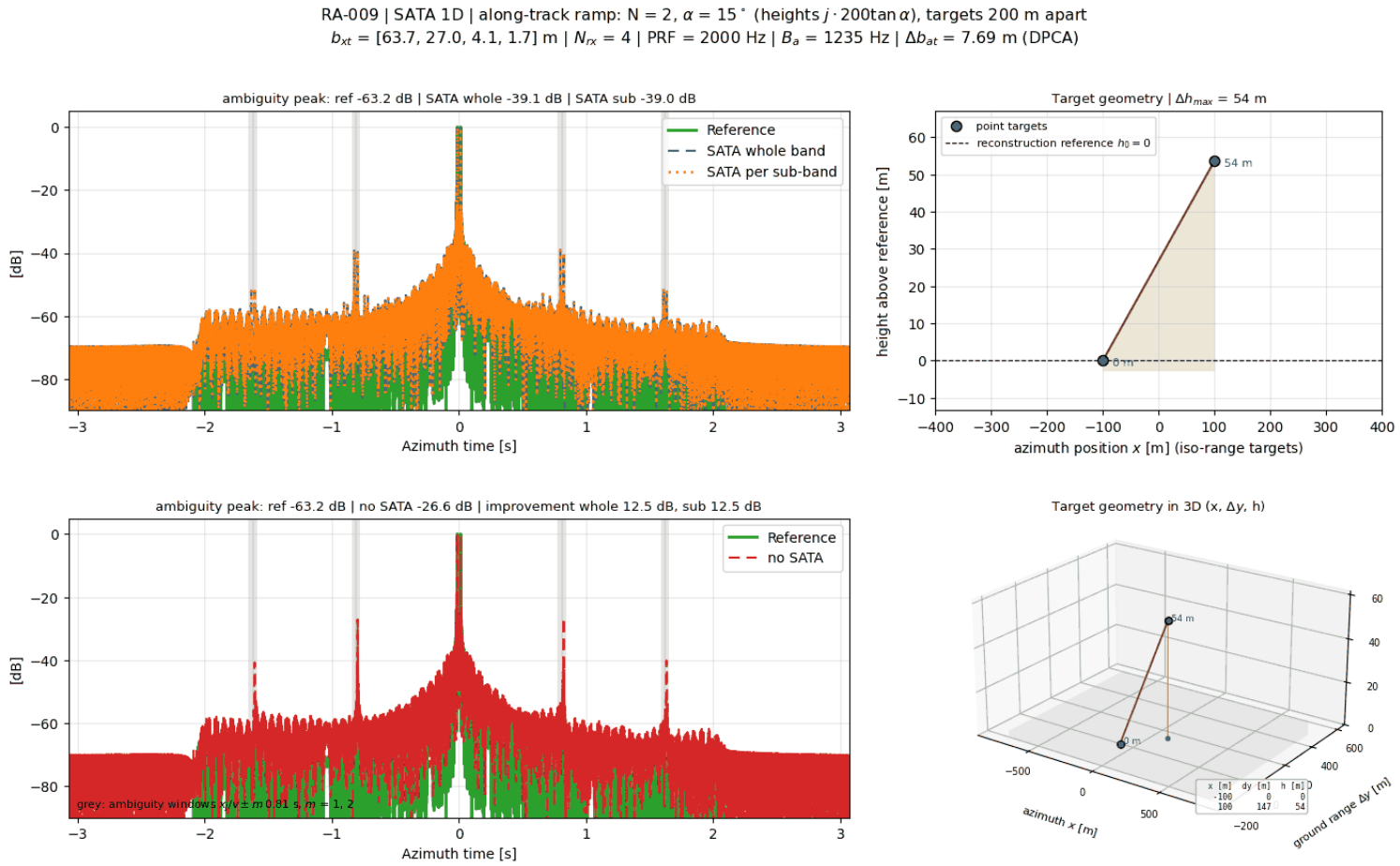


Figure 19: Along-track ramp, 200 m apart, normal heights: RA-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −26.6, SATA whole band −39.1, per sub-band −39.0 dB (improvement 12.5 dB).

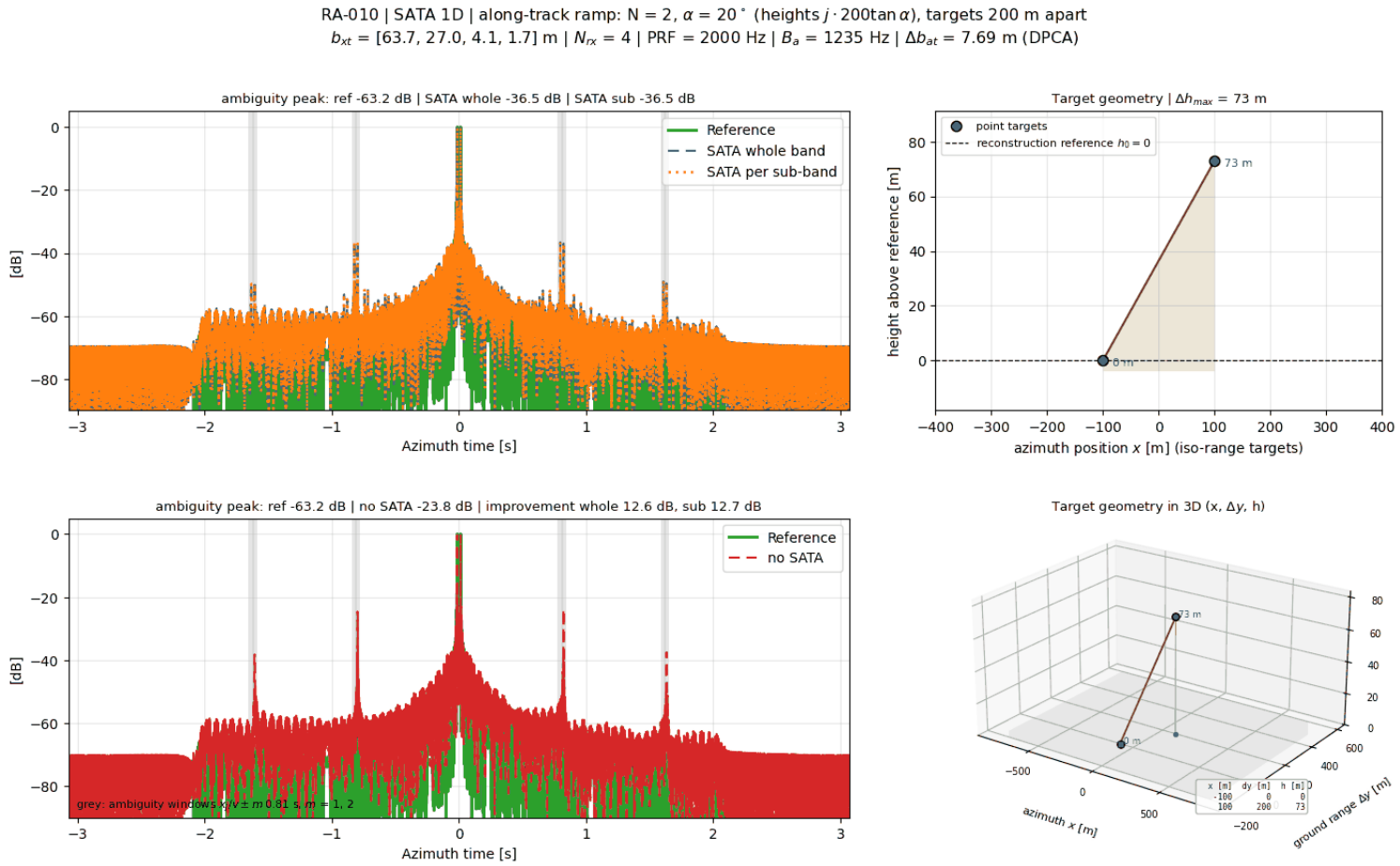


Figure 20: Along-track ramp, 200 m apart, normal heights: RA-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -23.8, SATA whole band -36.5, per sub-band -36.5 dB (improvement 12.6 dB).

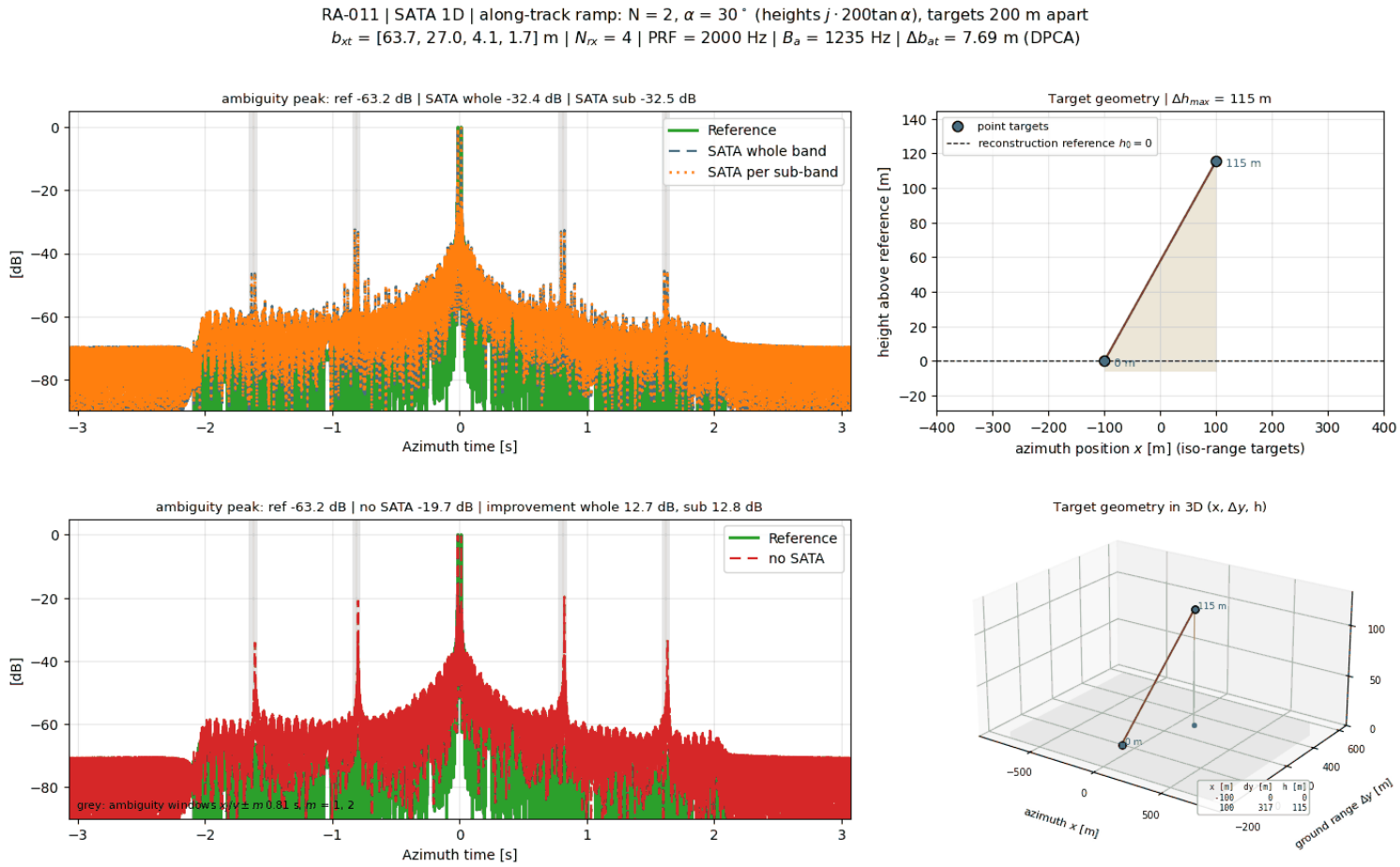


Figure 21: Along-track ramp, 200 m apart, normal heights: RA-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −19.7, SATA whole band −32.4, per sub-band −32.5 dB (improvement 12.7 dB).

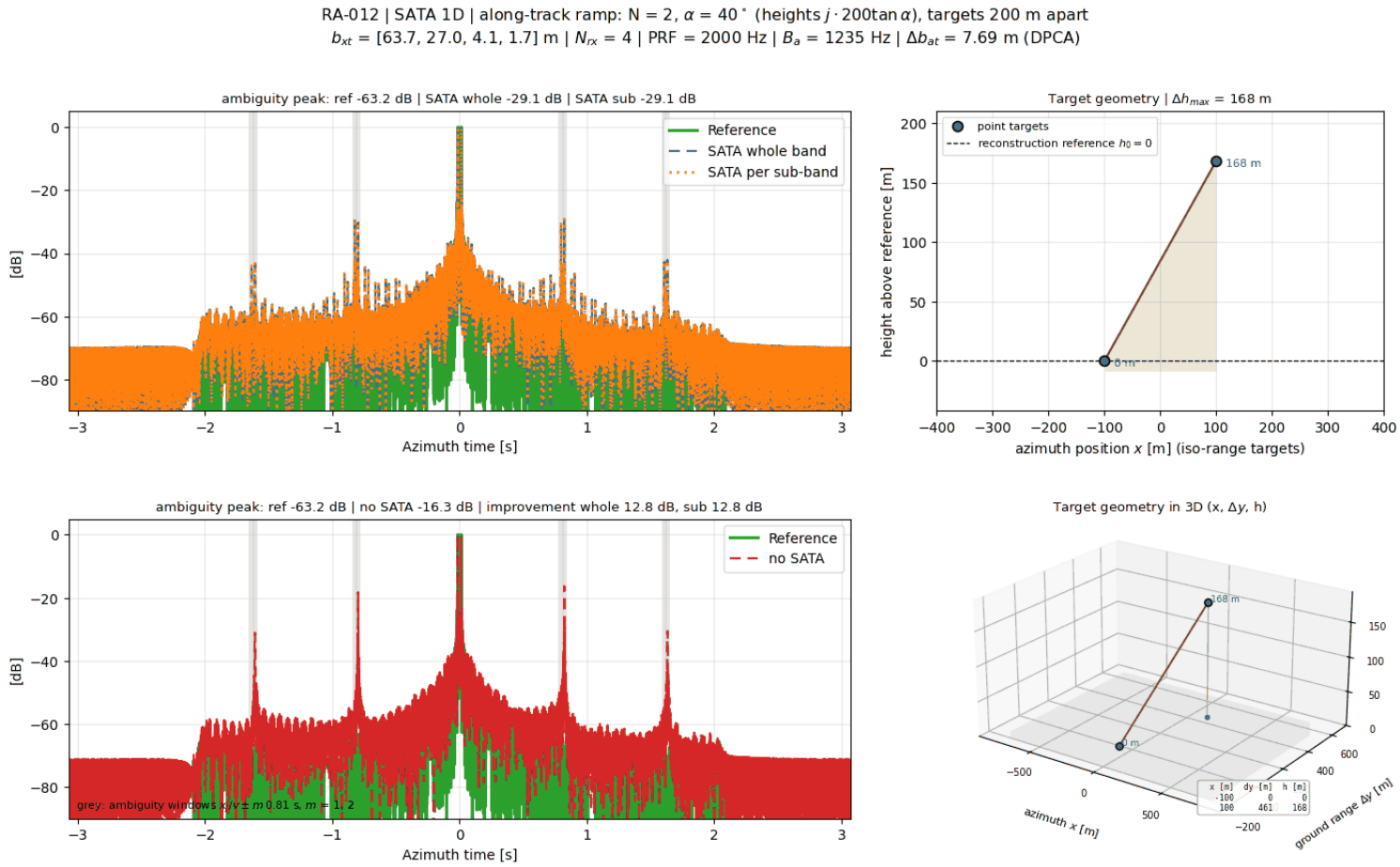


Figure 22: Along-track ramp, 200 m apart, normal heights: RA-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -16.3, SATA whole band -29.1, per sub-band -29.1 dB (improvement 12.8 dB).

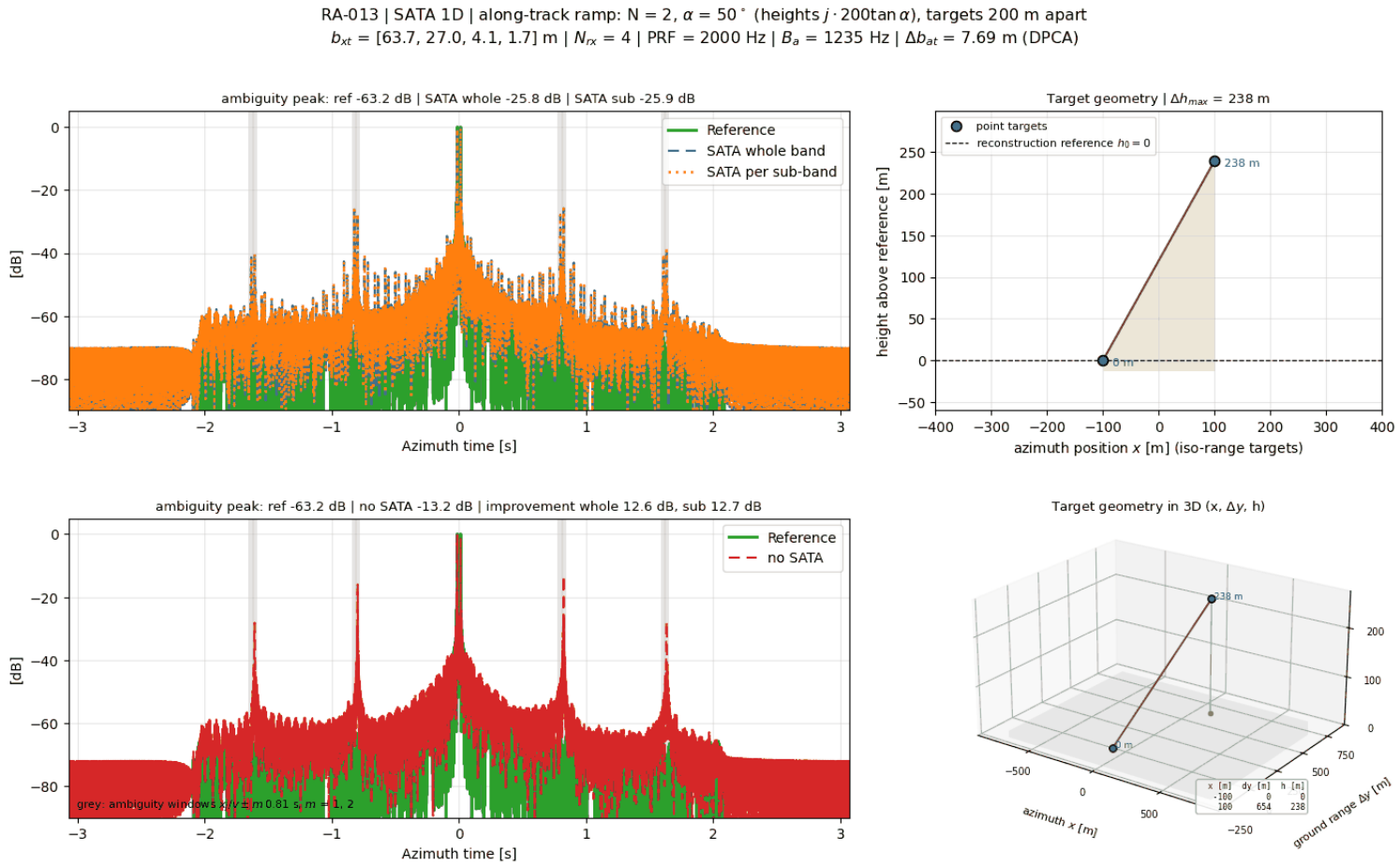


Figure 23: Along-track ramp, 200 m apart, normal heights: RA-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −13.2, SATA whole band −25.8, per sub-band −25.9 dB (improvement 12.6 dB).

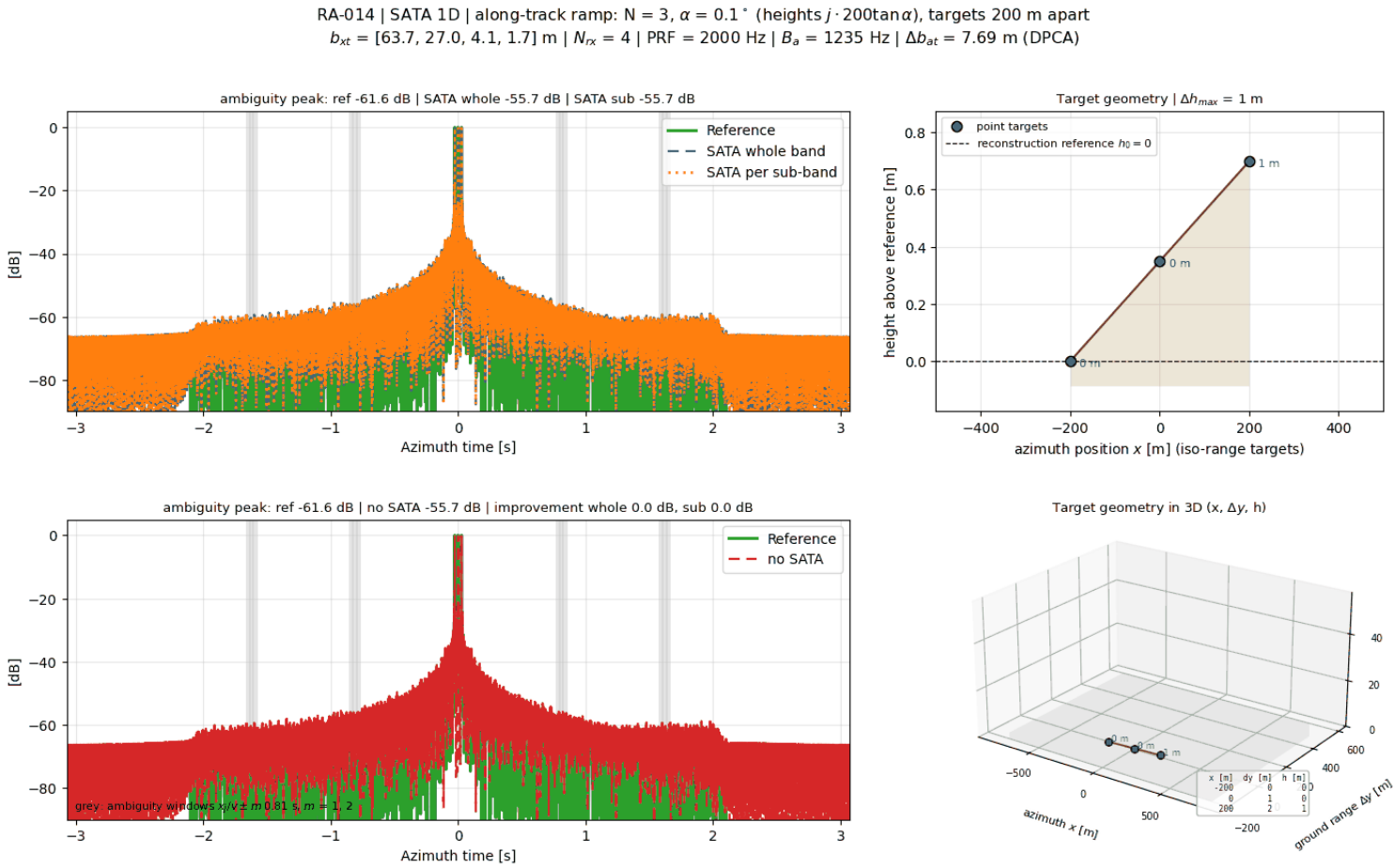


Figure 24: Along-track ramp, 200 m apart, normal heights: RA-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −55.7, SATA whole band −55.7, per sub-band −55.7 dB (improvement 0.0 dB).

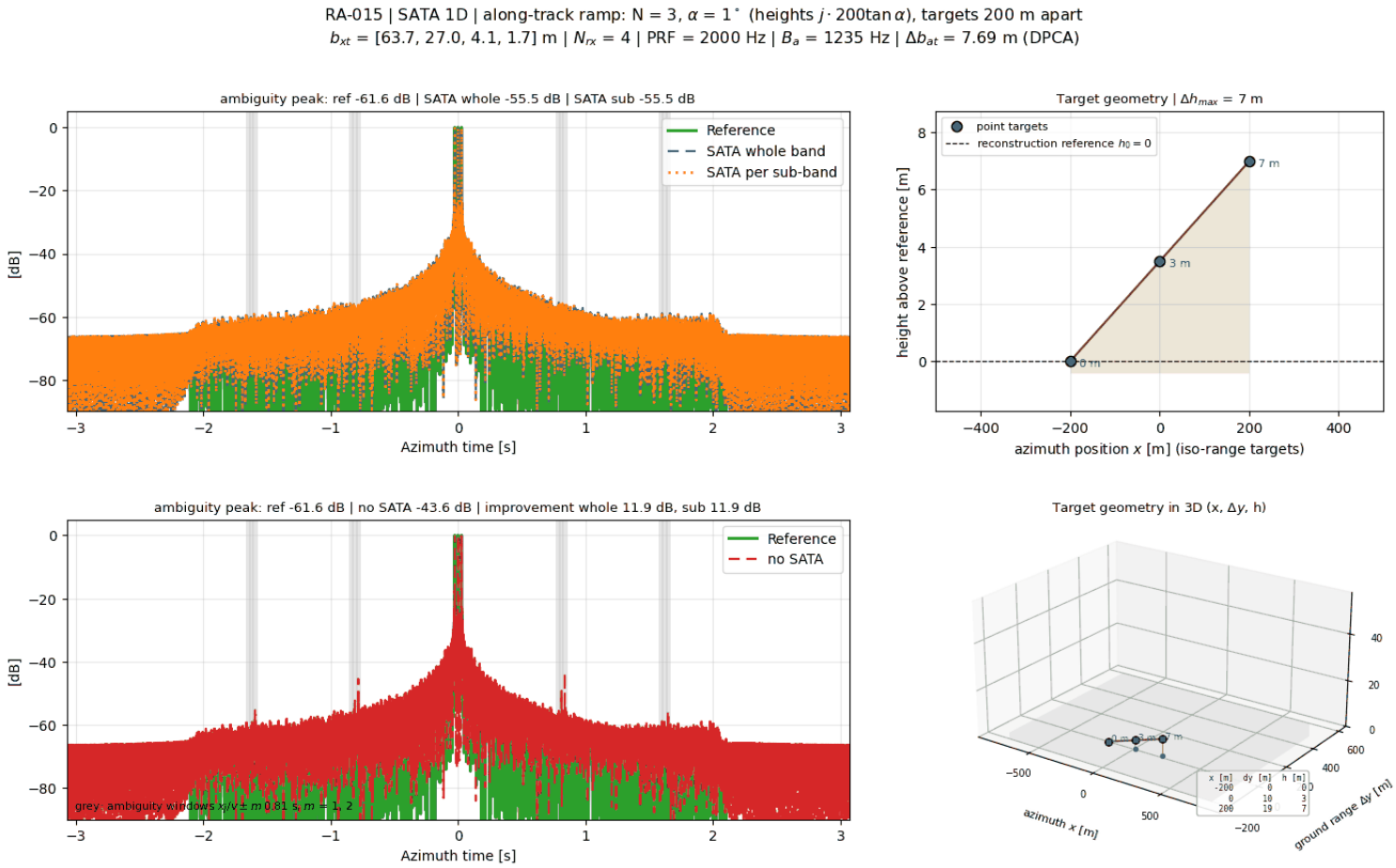


Figure 25: Along-track ramp, 200 m apart, normal heights: RA-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −43.6, SATA whole band −55.5, per sub-band −55.5 dB (improvement 11.9 dB).

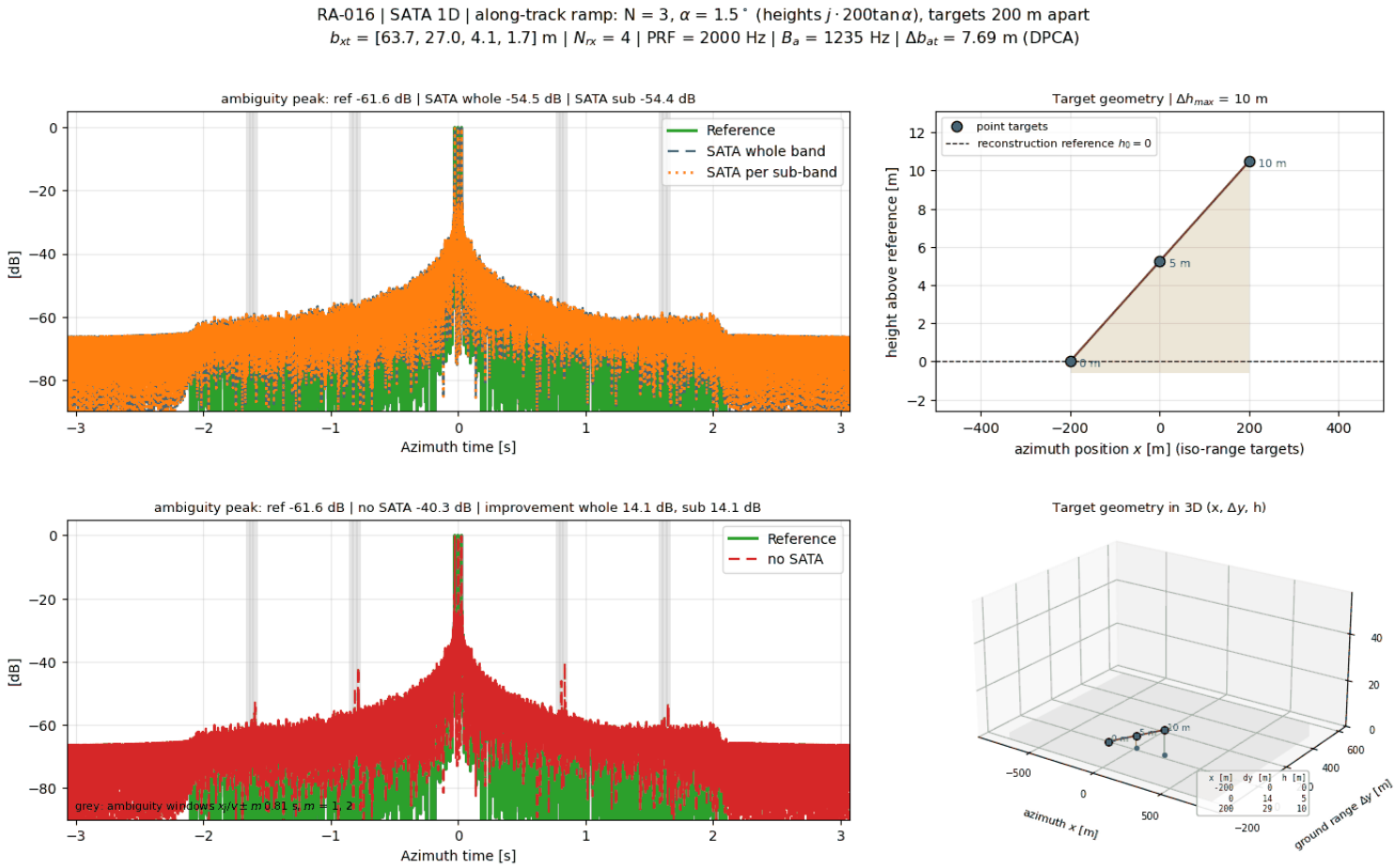


Figure 26: Along-track ramp, 200 m apart, normal heights: RA-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -40.3, SATA whole band -54.5, per sub-band -54.4 dB (improvement 14.1 dB).

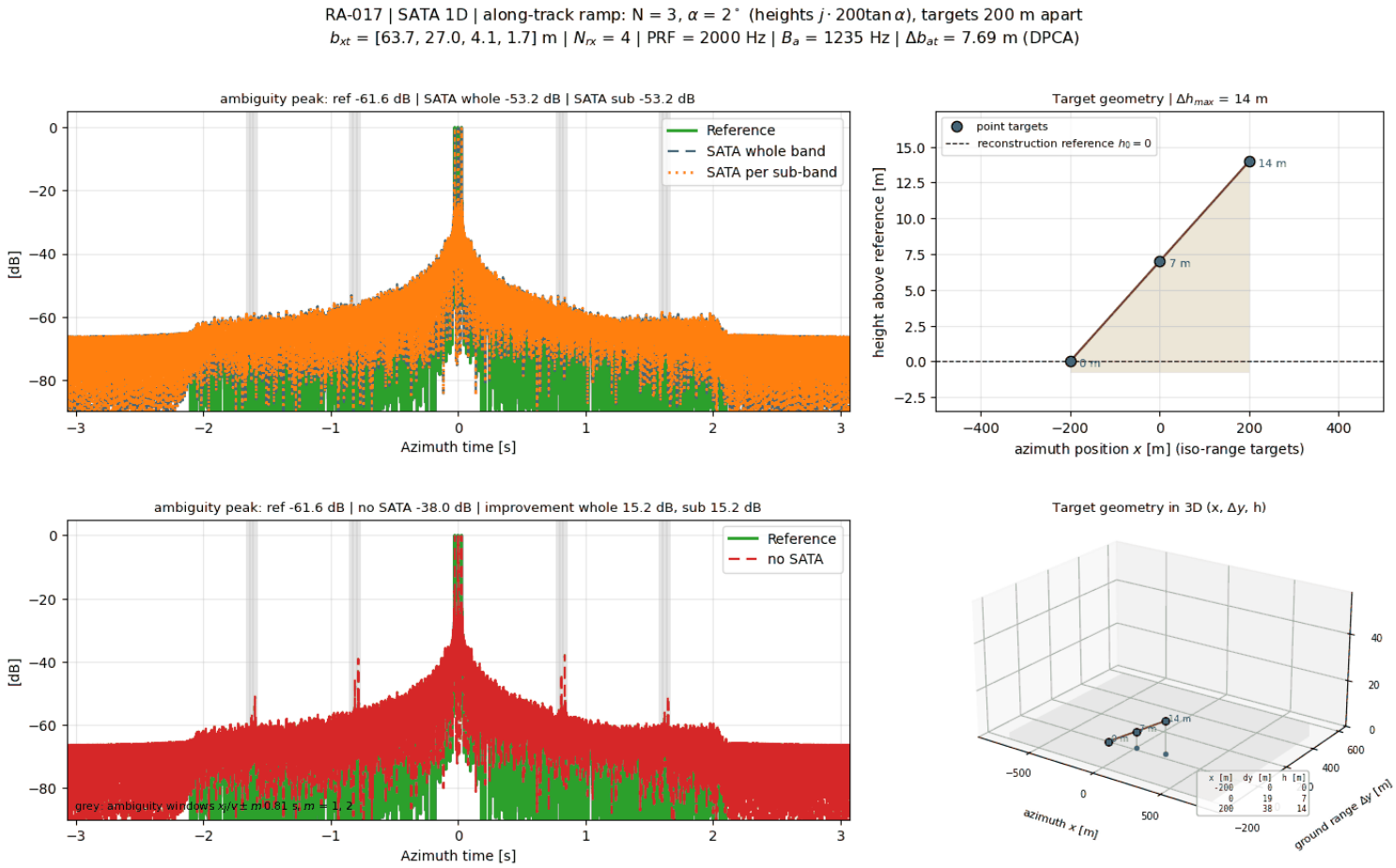


Figure 27: Along-track ramp, 200 m apart, normal heights: RA-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -38.0, SATA whole band -53.2, per sub-band -53.2 dB (improvement 15.2 dB).

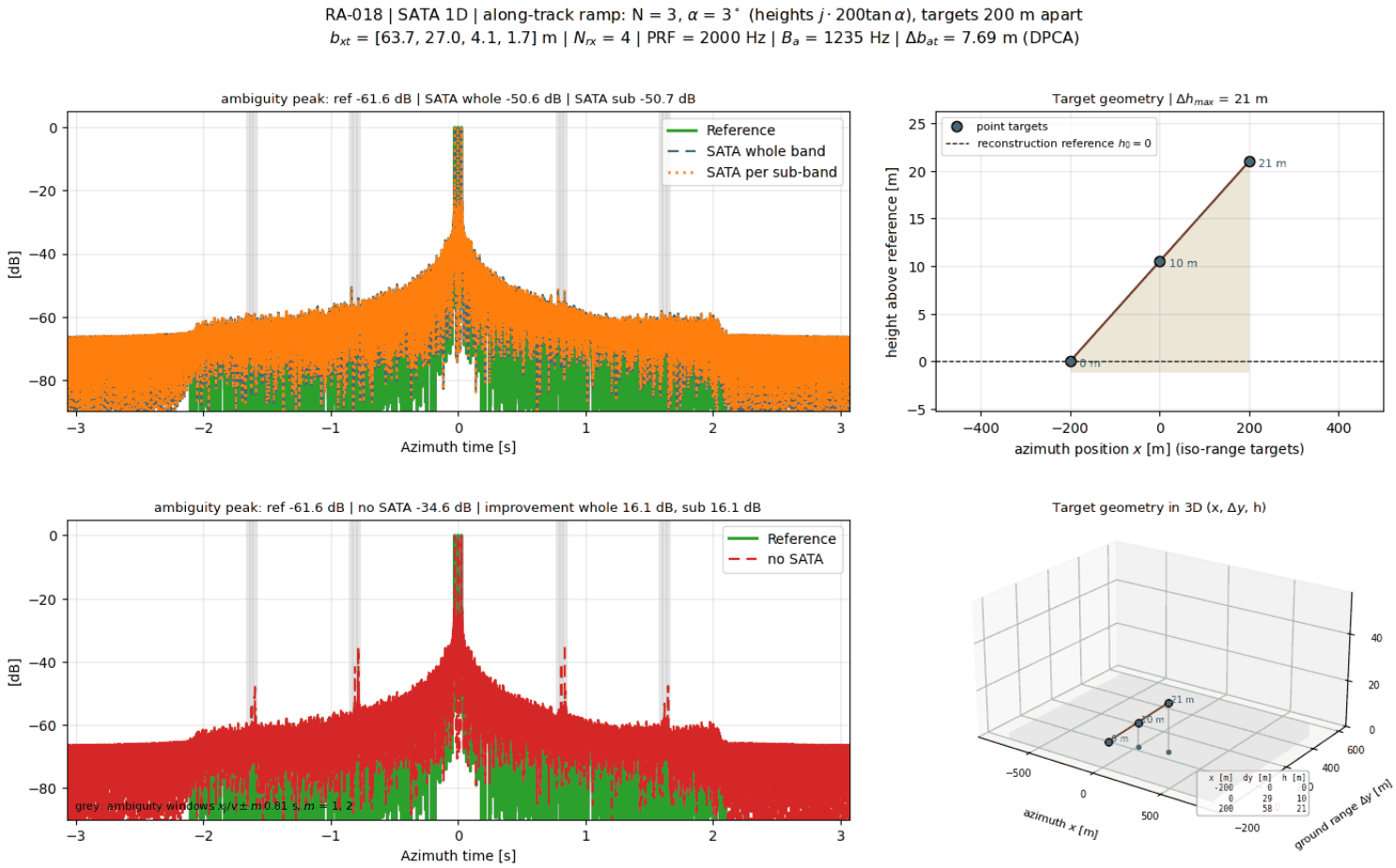


Figure 28: Along-track ramp, 200 m apart, normal heights: RA-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -34.6, SATA whole band -50.6, per sub-band -50.7 dB (improvement 16.1 dB).

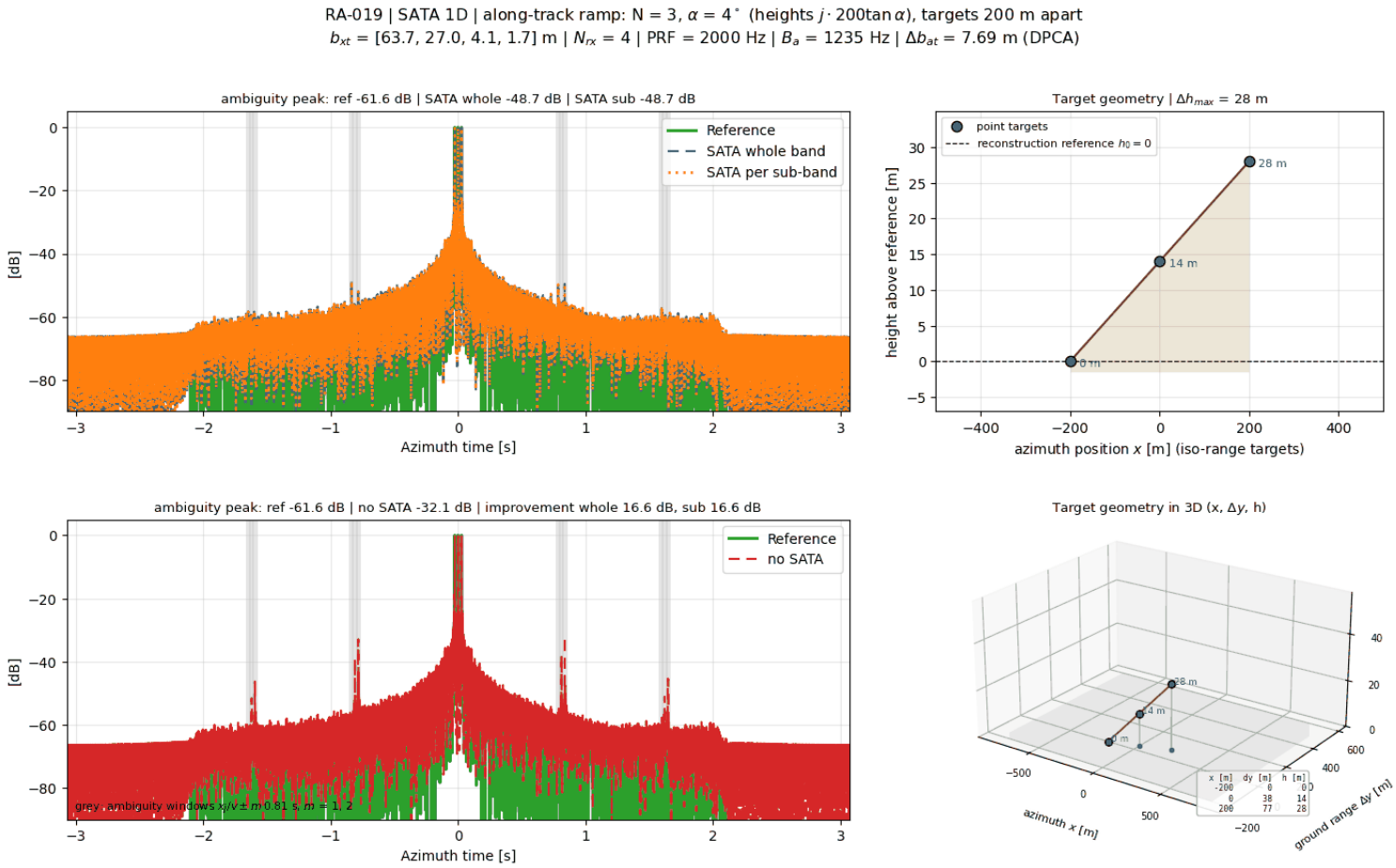


Figure 29: Along-track ramp, 200 m apart, normal heights: RA-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −32.1, SATA whole band −48.7, per sub-band −48.7 dB (improvement 16.6 dB).

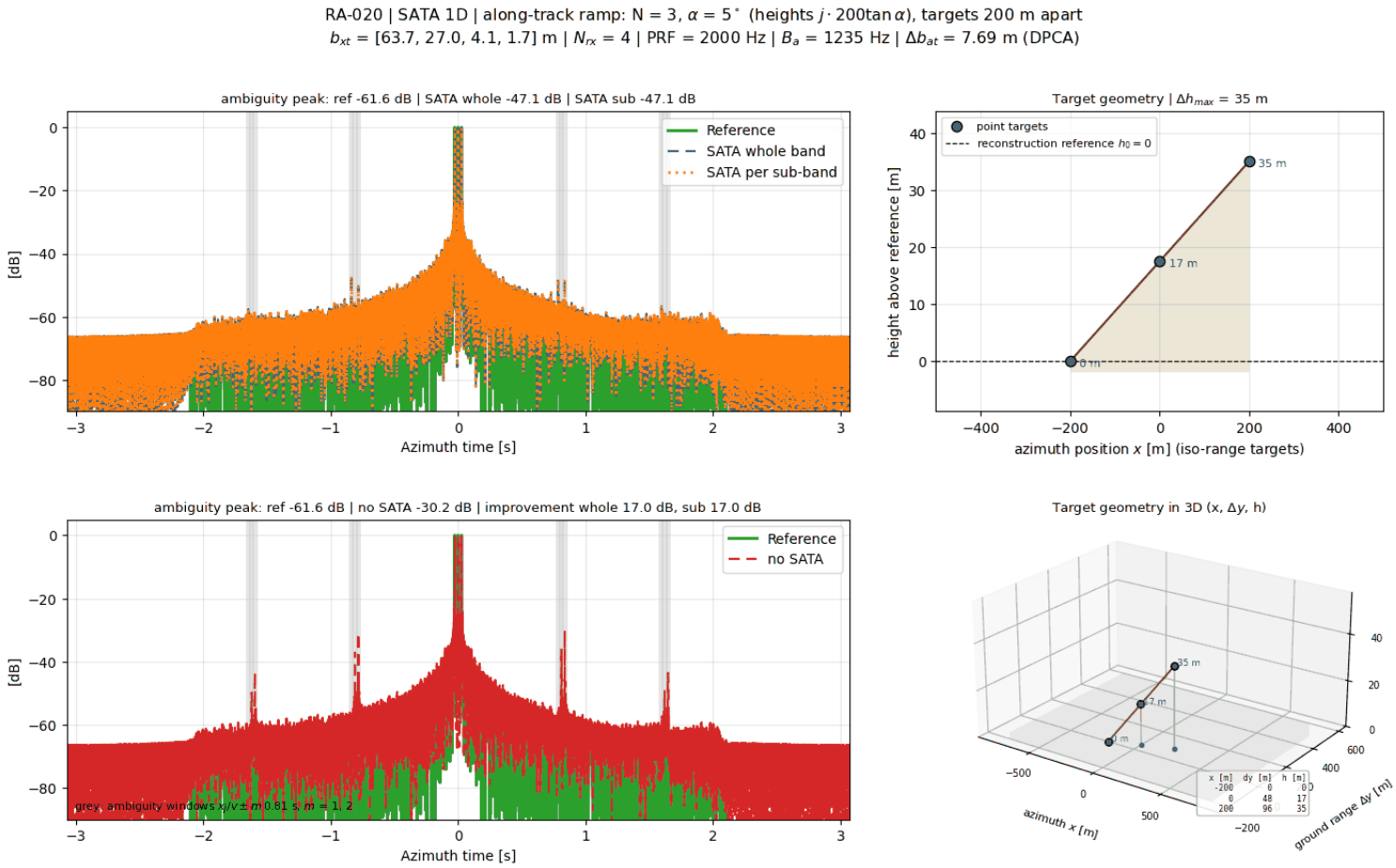


Figure 30: Along-track ramp, 200 m apart, normal heights: RA-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −30.2, SATA whole band −47.1, per sub-band −47.1 dB (improvement 17.0 dB).

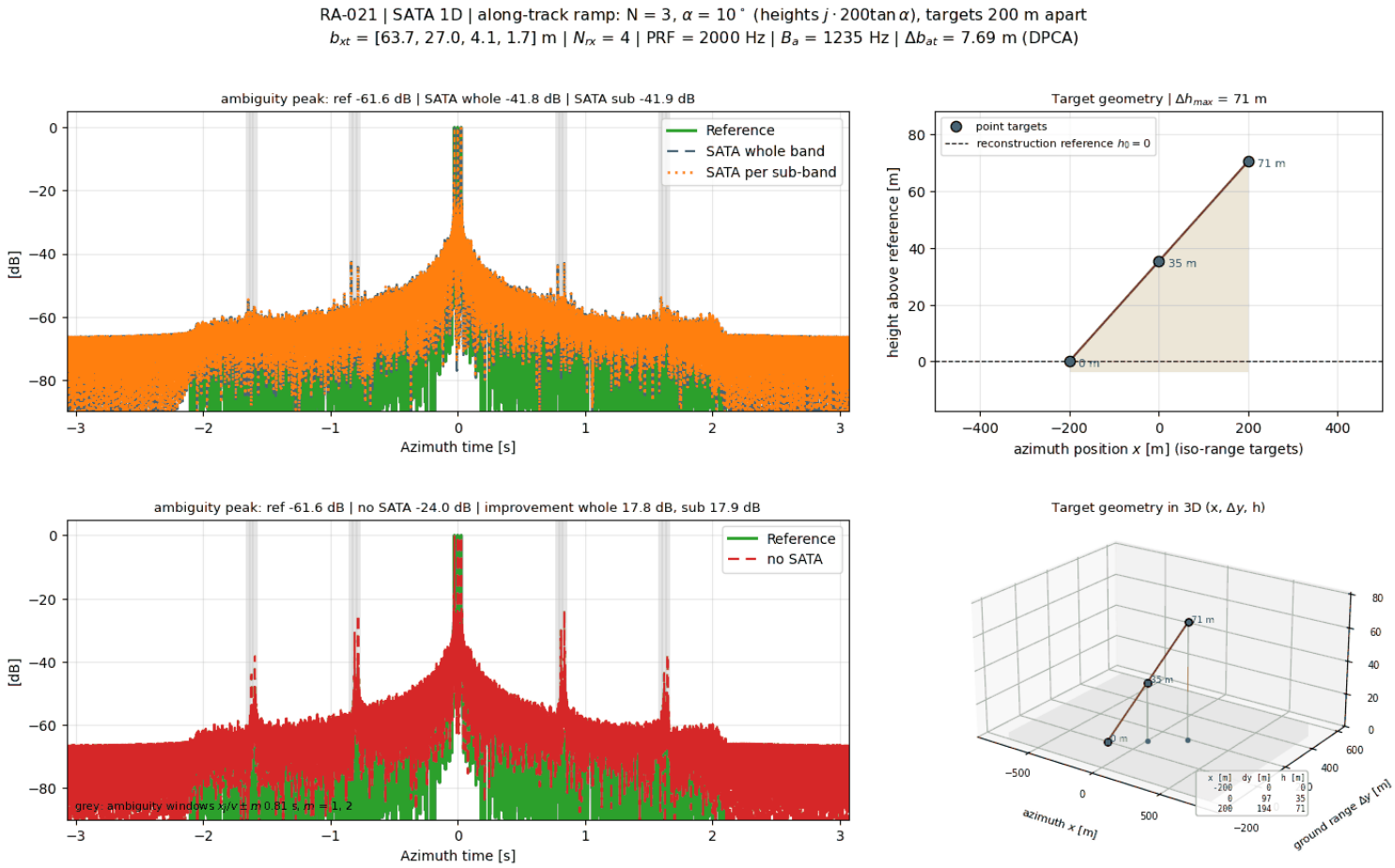


Figure 31: Along-track ramp, 200 m apart, normal heights: RA-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -24.0, SATA whole band -41.8, per sub-band -41.9 dB (improvement 17.8 dB).

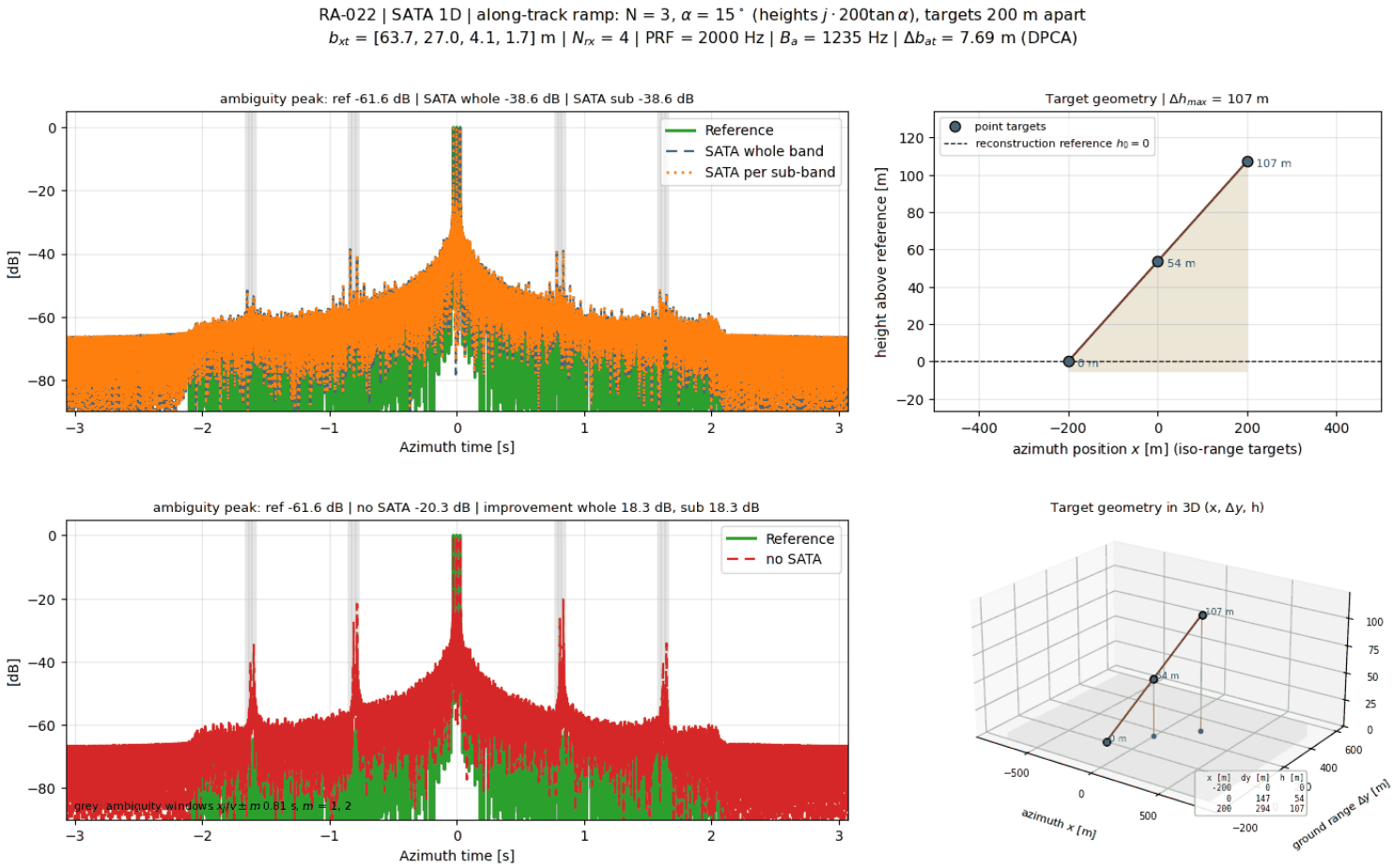


Figure 32: Along-track ramp, 200 m apart, normal heights: RA-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -20.3, SATA whole band -38.6, per sub-band -38.6 dB (improvement 18.3 dB).

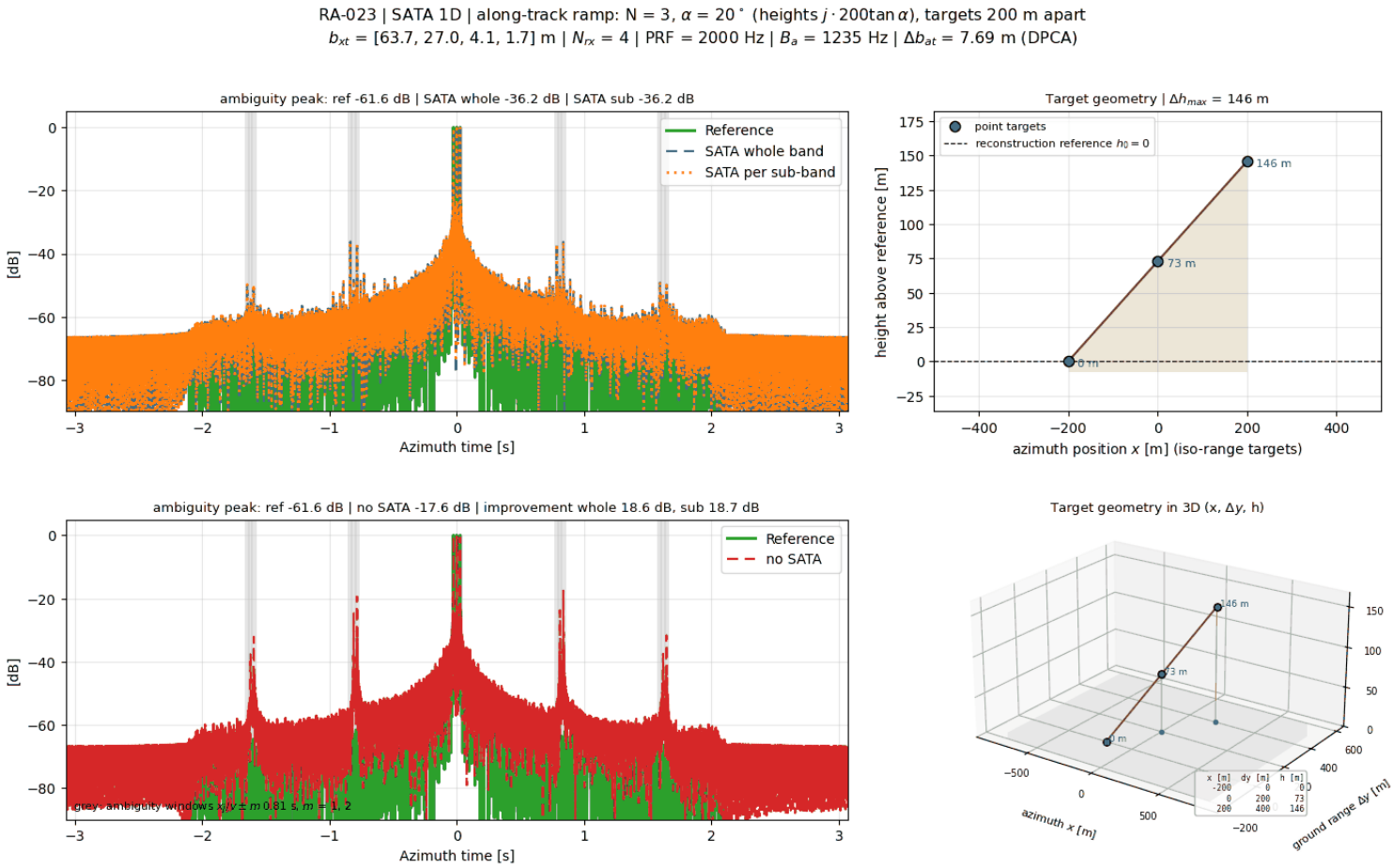


Figure 33: Along-track ramp, 200 m apart, normal heights: RA-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −17.6, SATA whole band −36.2, per sub-band −36.2 dB (improvement 18.6 dB).

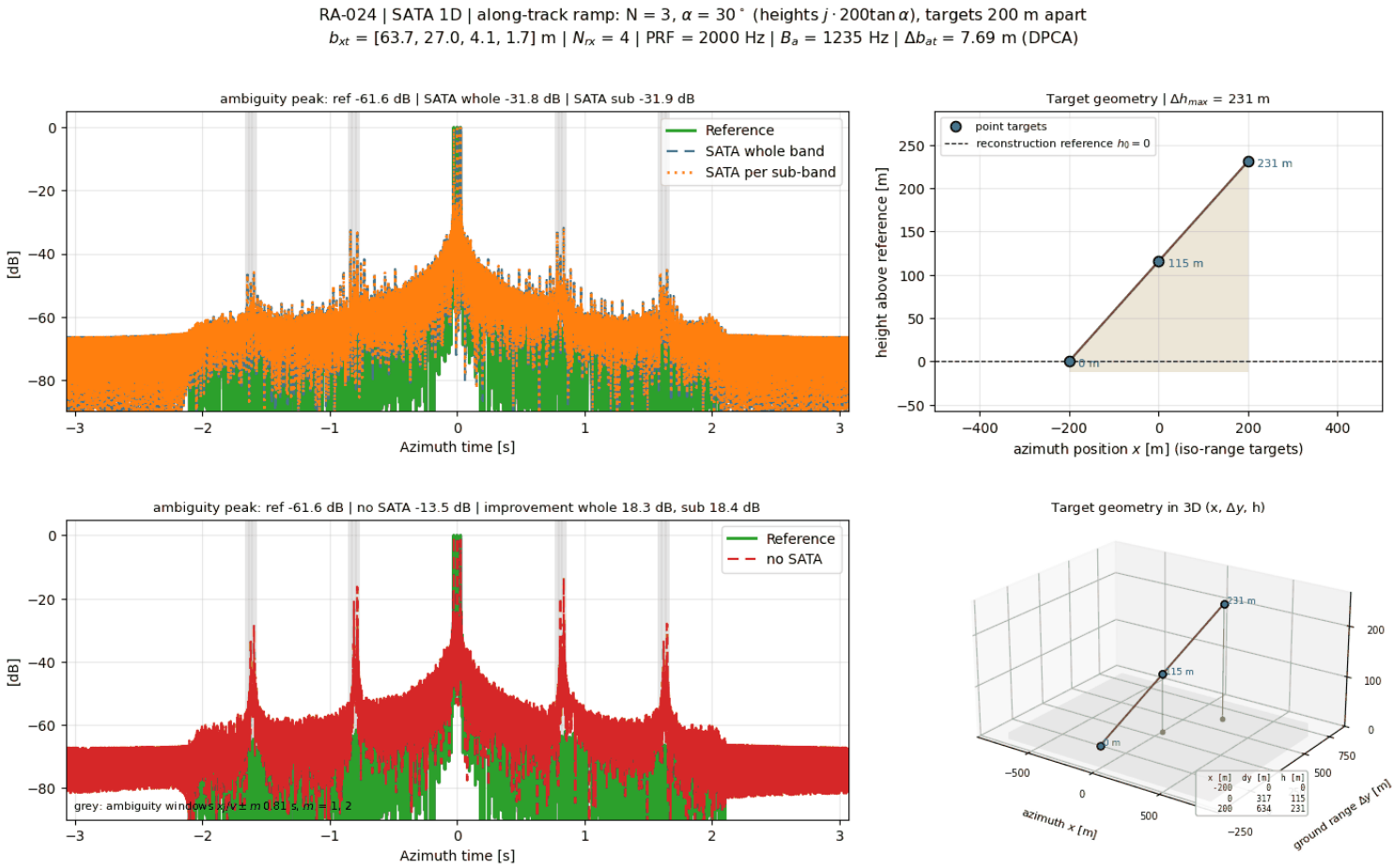


Figure 34: Along-track ramp, 200 m apart, normal heights: RA-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -13.5, SATA whole band -31.8, per sub-band -31.9 dB (improvement 18.3 dB).

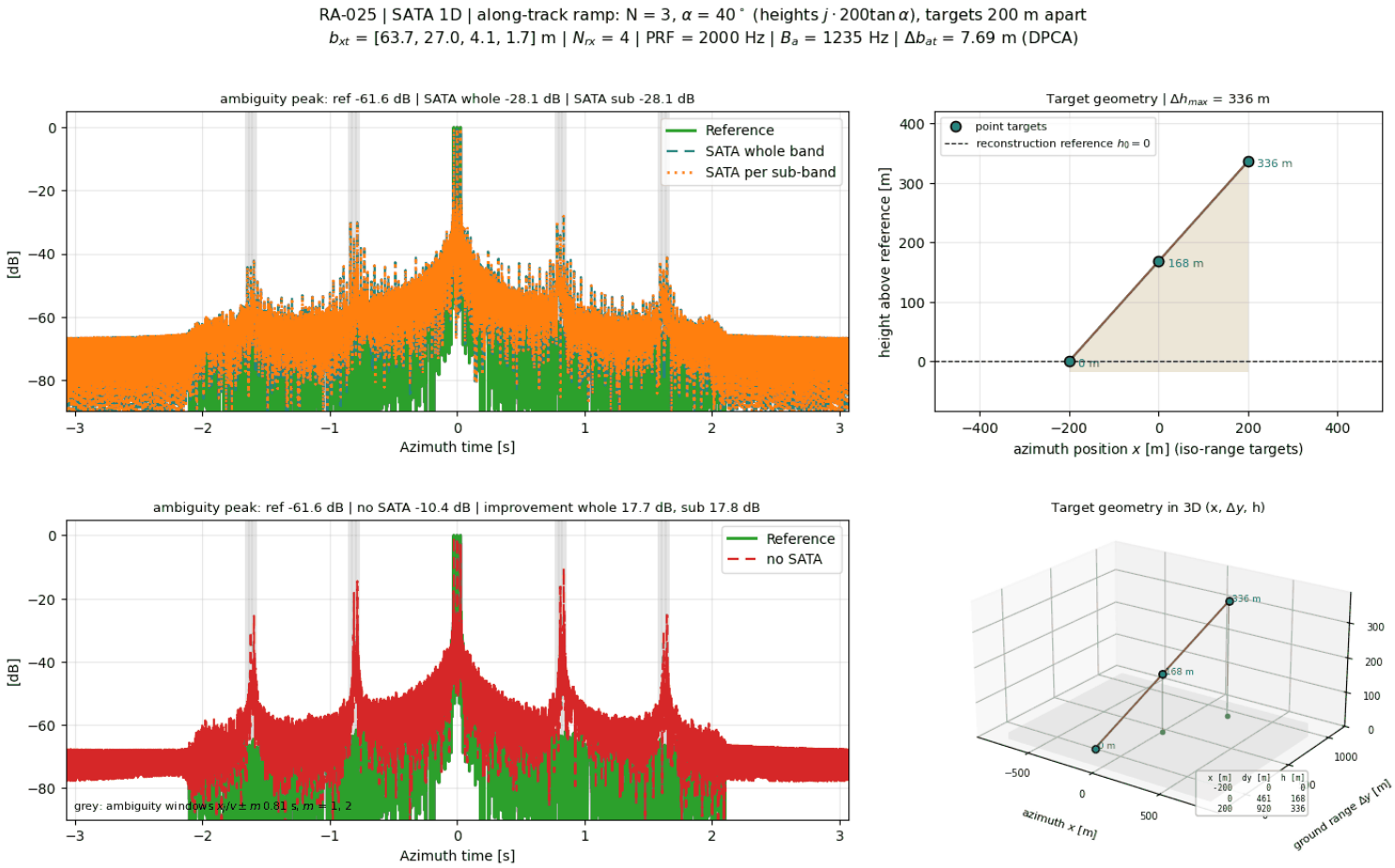


Figure 35: Along-track ramp, 200 m apart, normal heights: RA-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -10.4, SATA whole band -28.1, per sub-band -28.1 dB (improvement 17.7 dB).

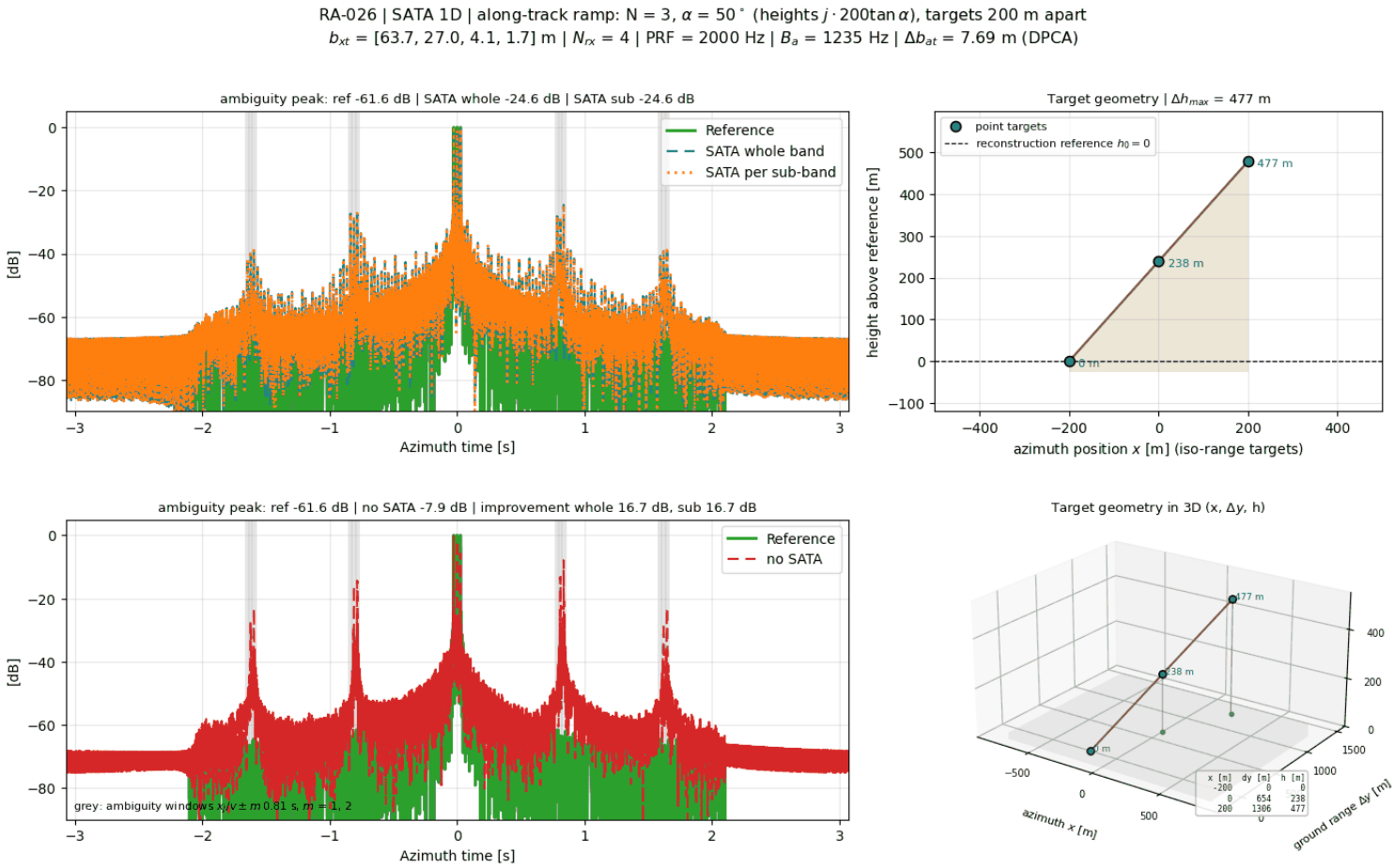


Figure 36: Along-track ramp, 200 m apart, normal heights: RA-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −7.9, SATA whole band −24.6, per sub-band −24.6 dB (improvement 16.7 dB).

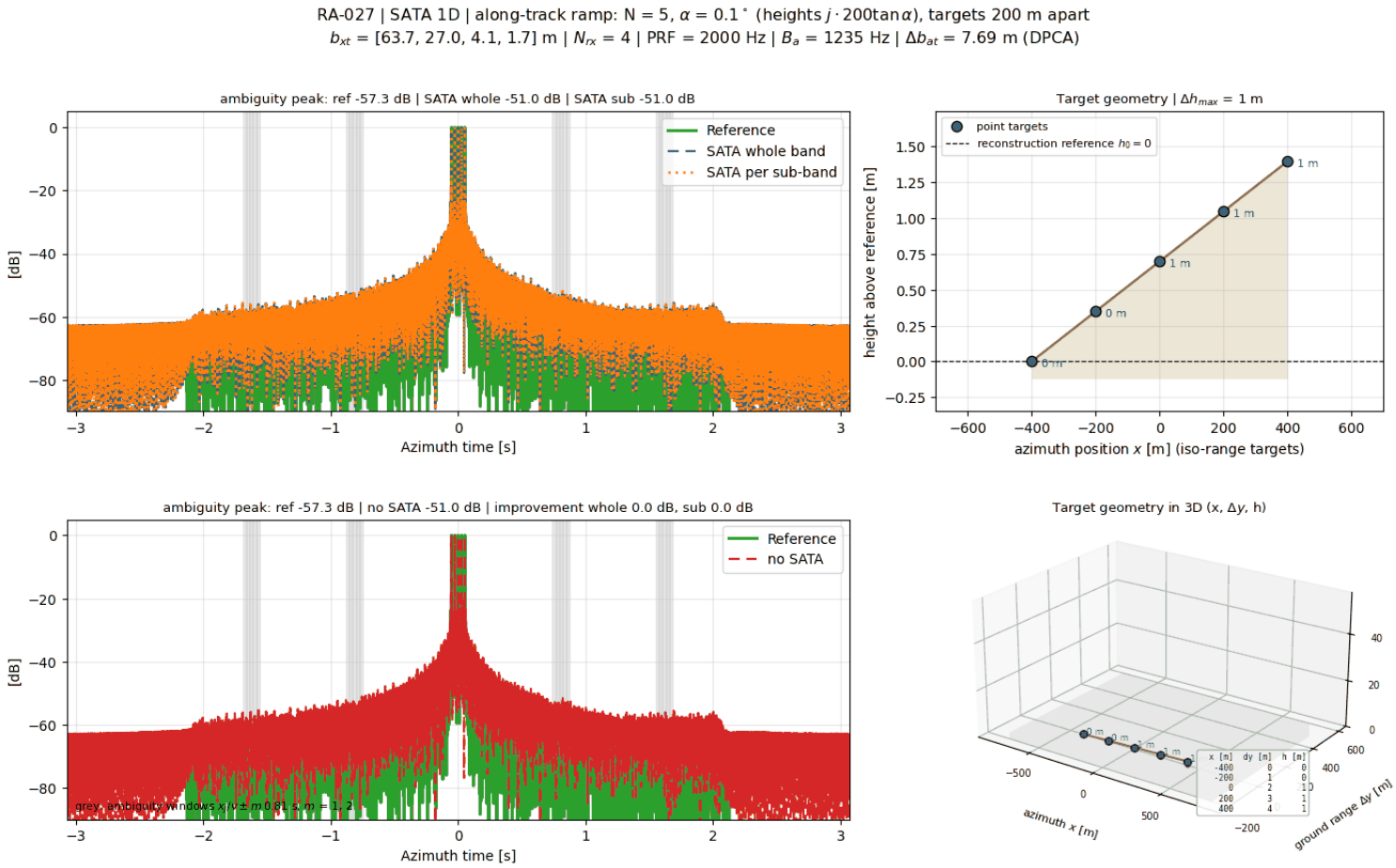


Figure 37: Along-track ramp, 200 m apart, normal heights: RA-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −51.0, SATA whole band −51.0, per sub-band −51.0 dB (improvement 0.0 dB).

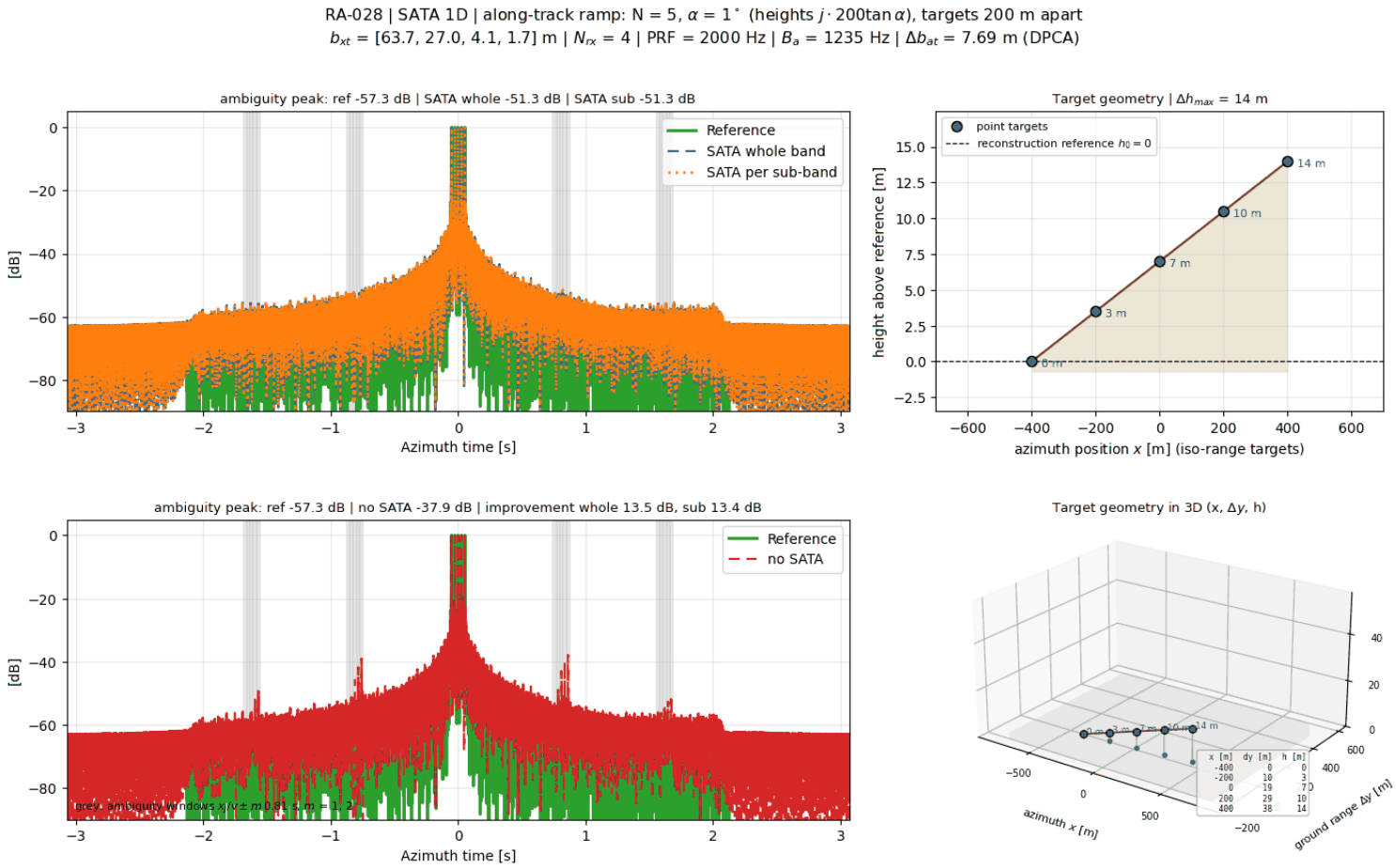


Figure 38: Along-track ramp, 200 m apart, normal heights: RA-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -37.9, SATA whole band -51.3, per sub-band -51.3 dB (improvement 13.5 dB).

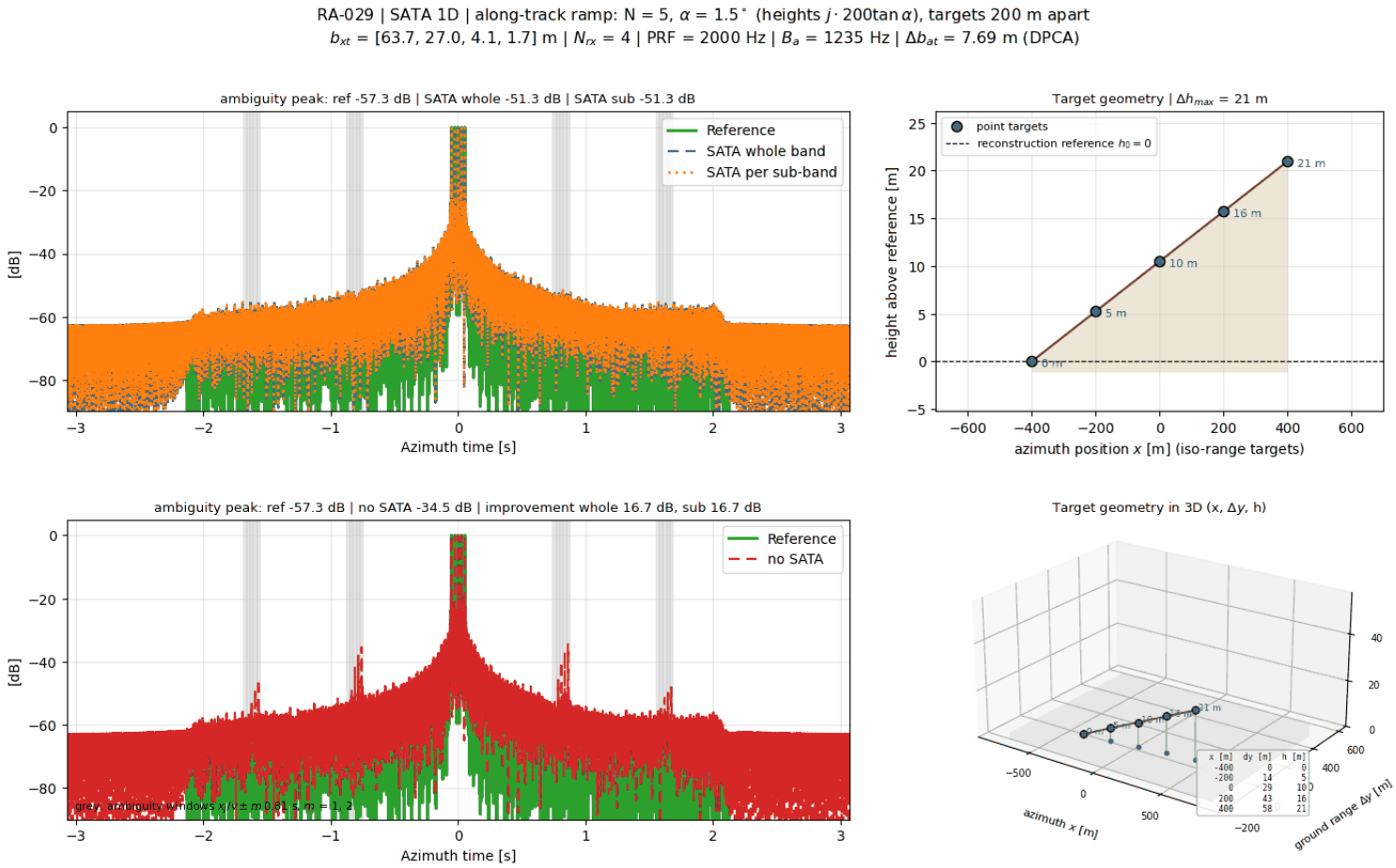


Figure 39: Along-track ramp, 200 m apart, normal heights: RA-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −34.5, SATA whole band −51.3, per sub-band −51.3 dB (improvement 16.7 dB).

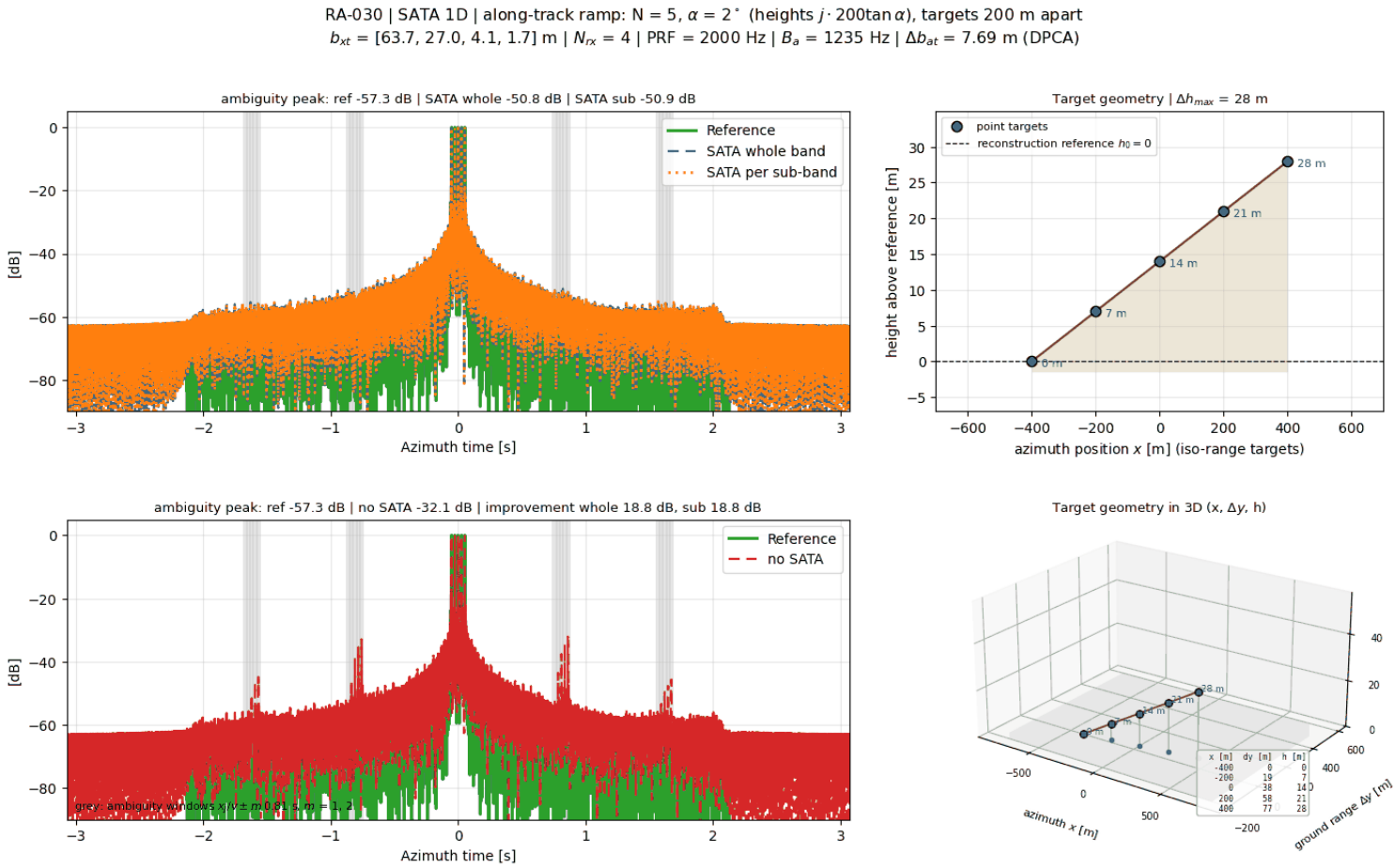


Figure 40: Along-track ramp, 200 m apart, normal heights: RA-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -32.1, SATA whole band -50.8, per sub-band -50.9 dB (improvement 18.8 dB).

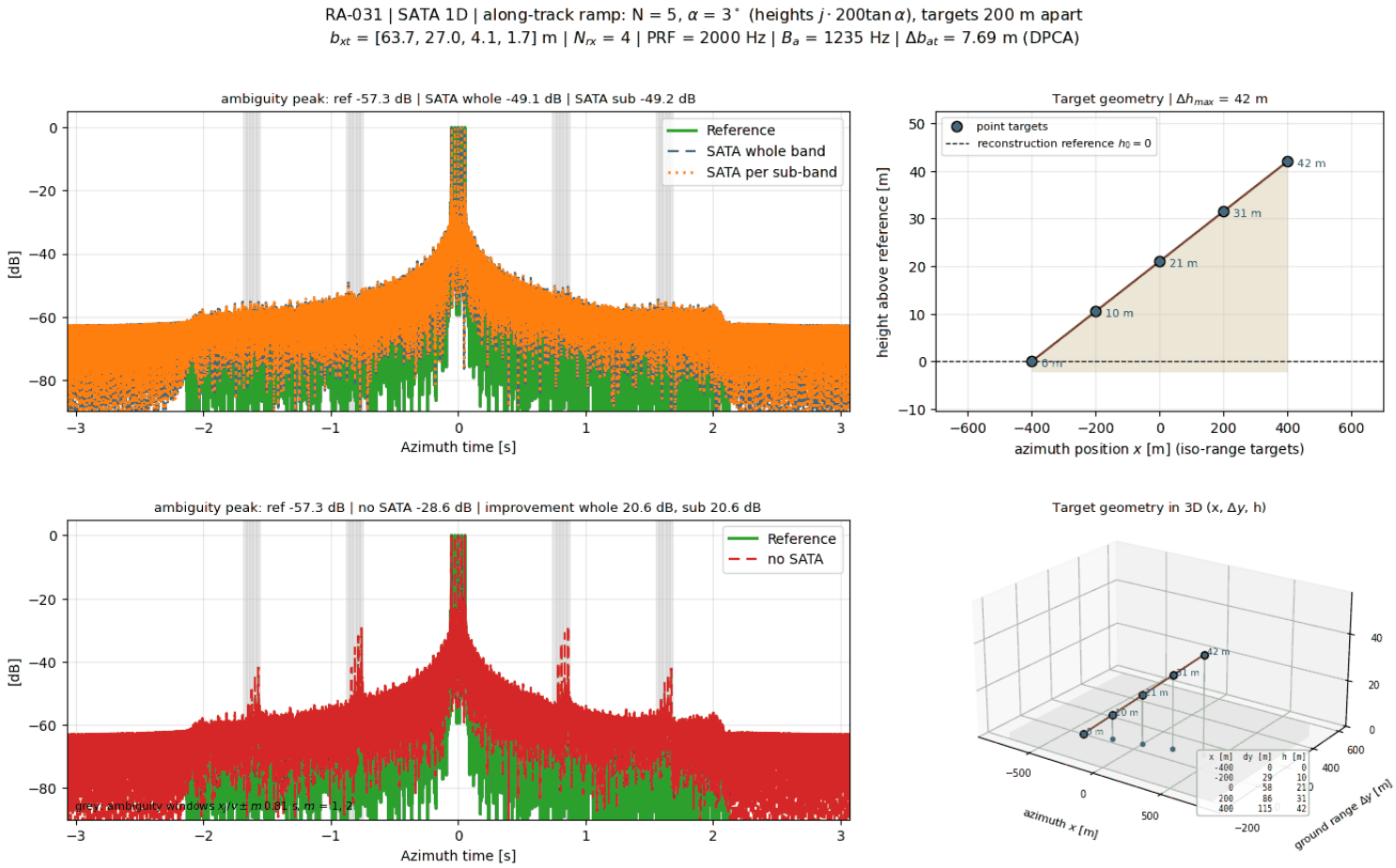


Figure 41: Along-track ramp, 200 m apart, normal heights: RA-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −28.6, SATA whole band −49.1, per sub-band −49.2 dB (improvement 20.6 dB).

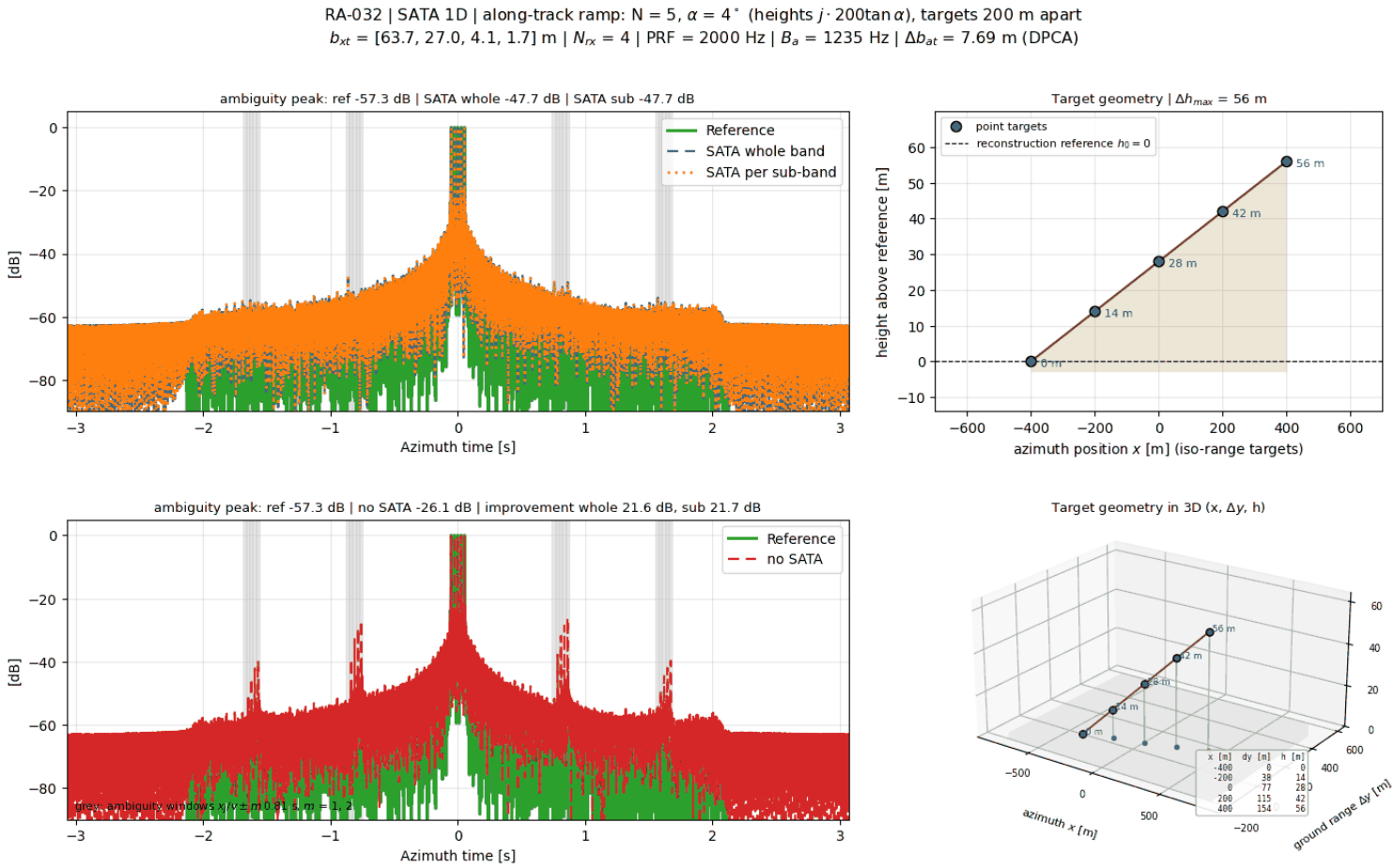


Figure 42: Along-track ramp, 200 m apart, normal heights: RA-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −26.1, SATA whole band −47.7, per sub-band −47.7 dB (improvement 21.6 dB).

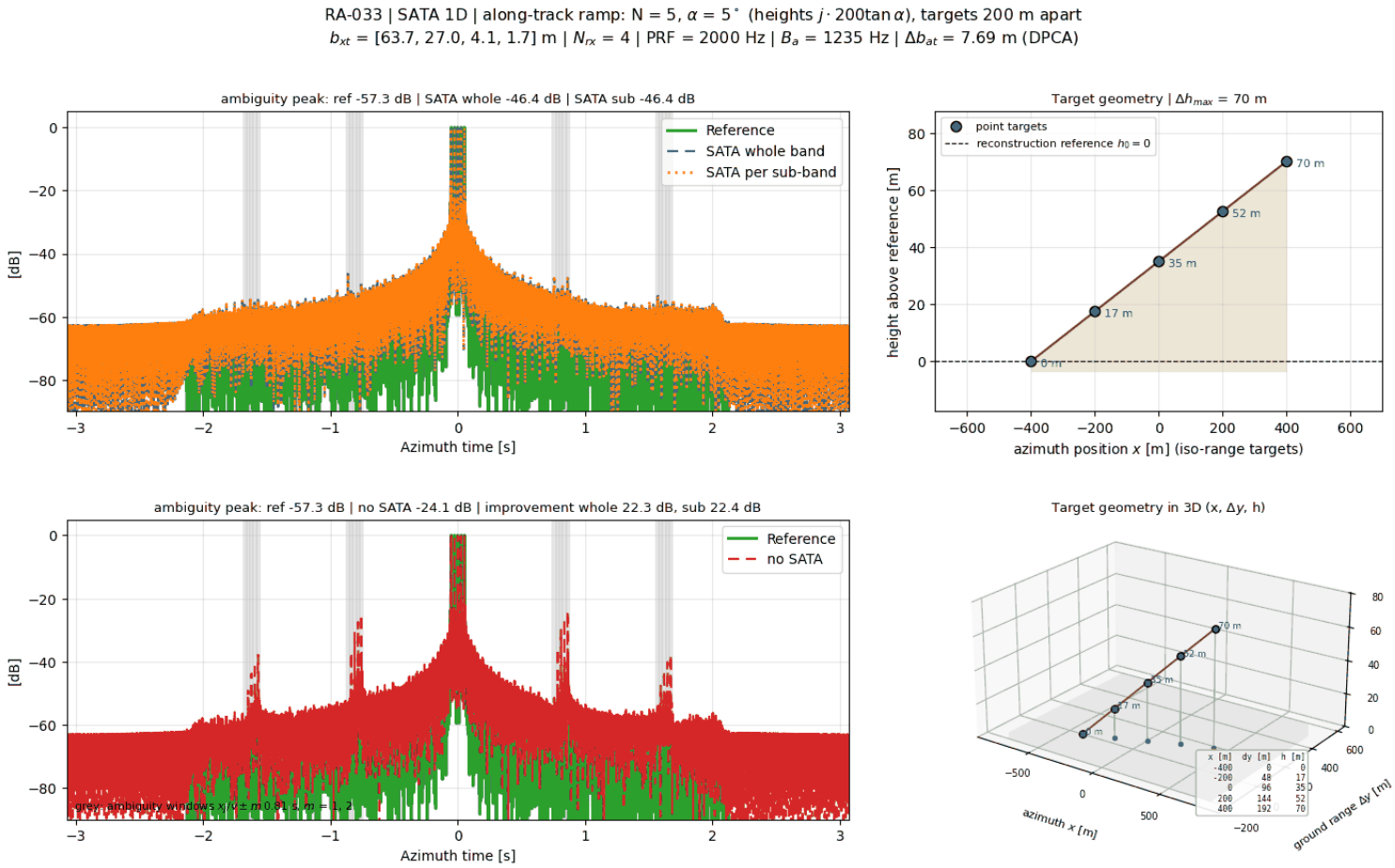


Figure 43: Along-track ramp, 200 m apart, normal heights: RA-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −24.1, SATA whole band −46.4, per sub-band −46.4 dB (improvement 22.3 dB).

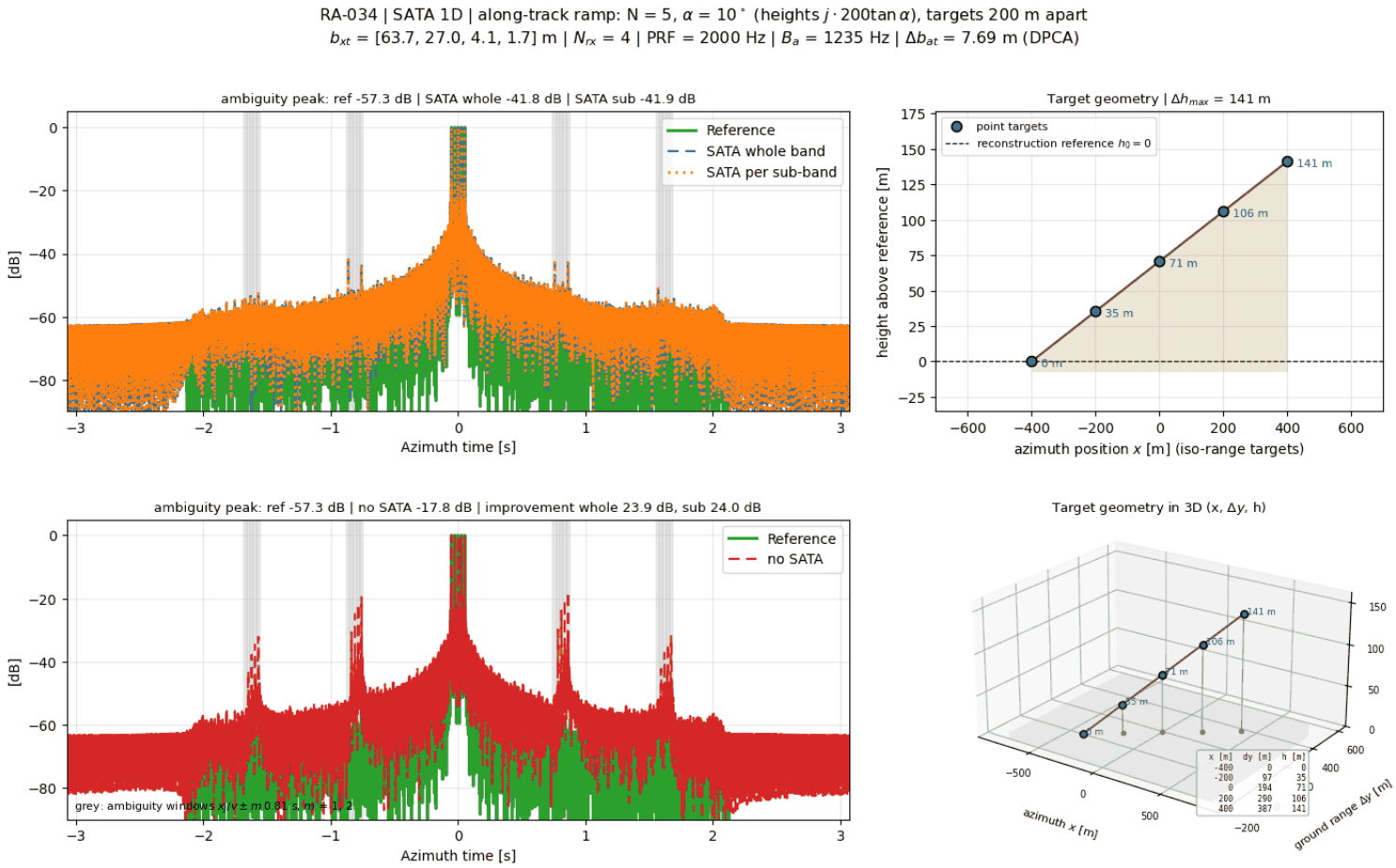


Figure 44: Along-track ramp, 200 m apart, normal heights: RA-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -17.8, SATA whole band -41.8, per sub-band -41.9 dB (improvement 23.9 dB).

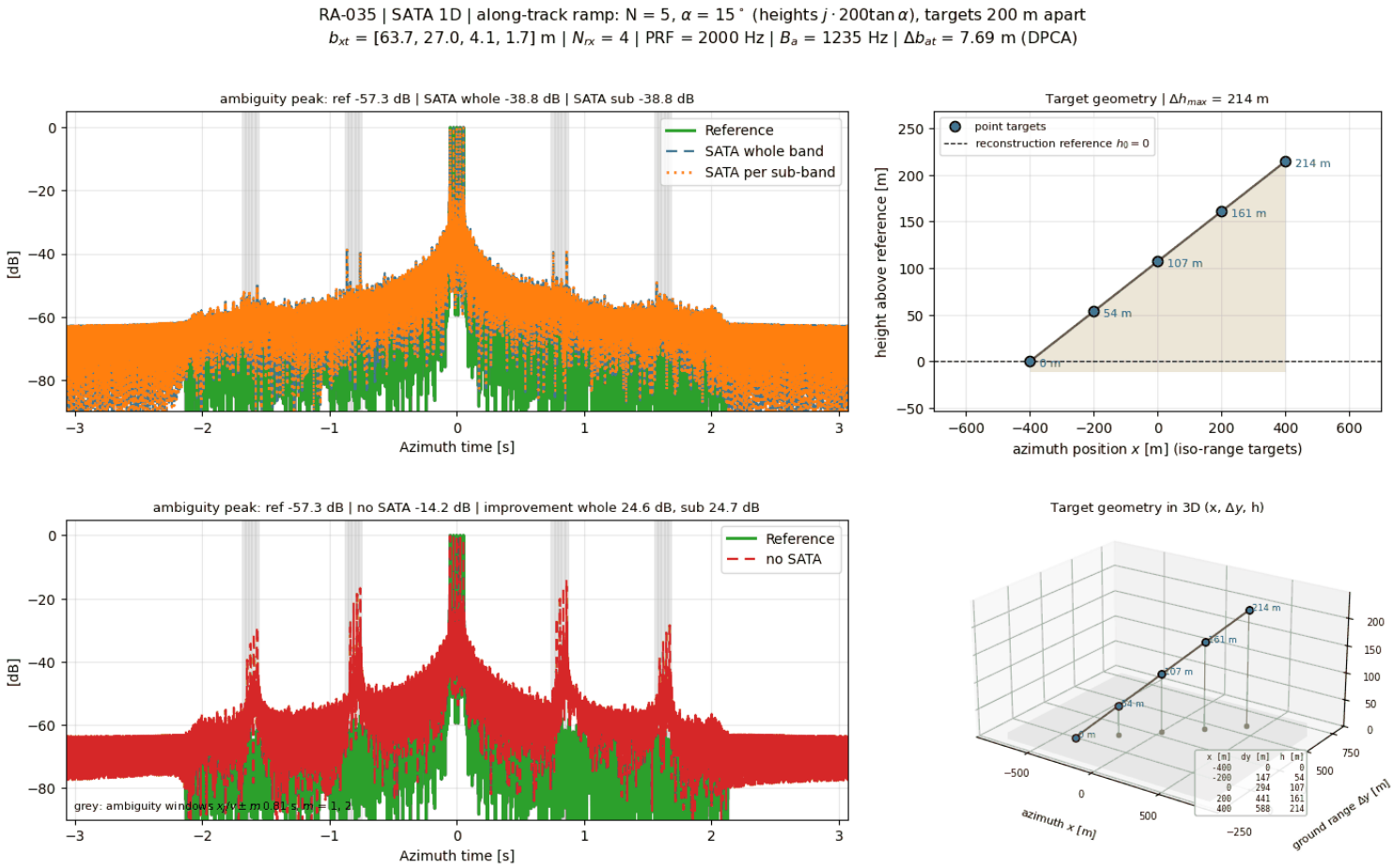


Figure 45: Along-track ramp, 200 m apart, normal heights: RA-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -14.2, SATA whole band -38.8, per sub-band -38.8 dB (improvement 24.6 dB).

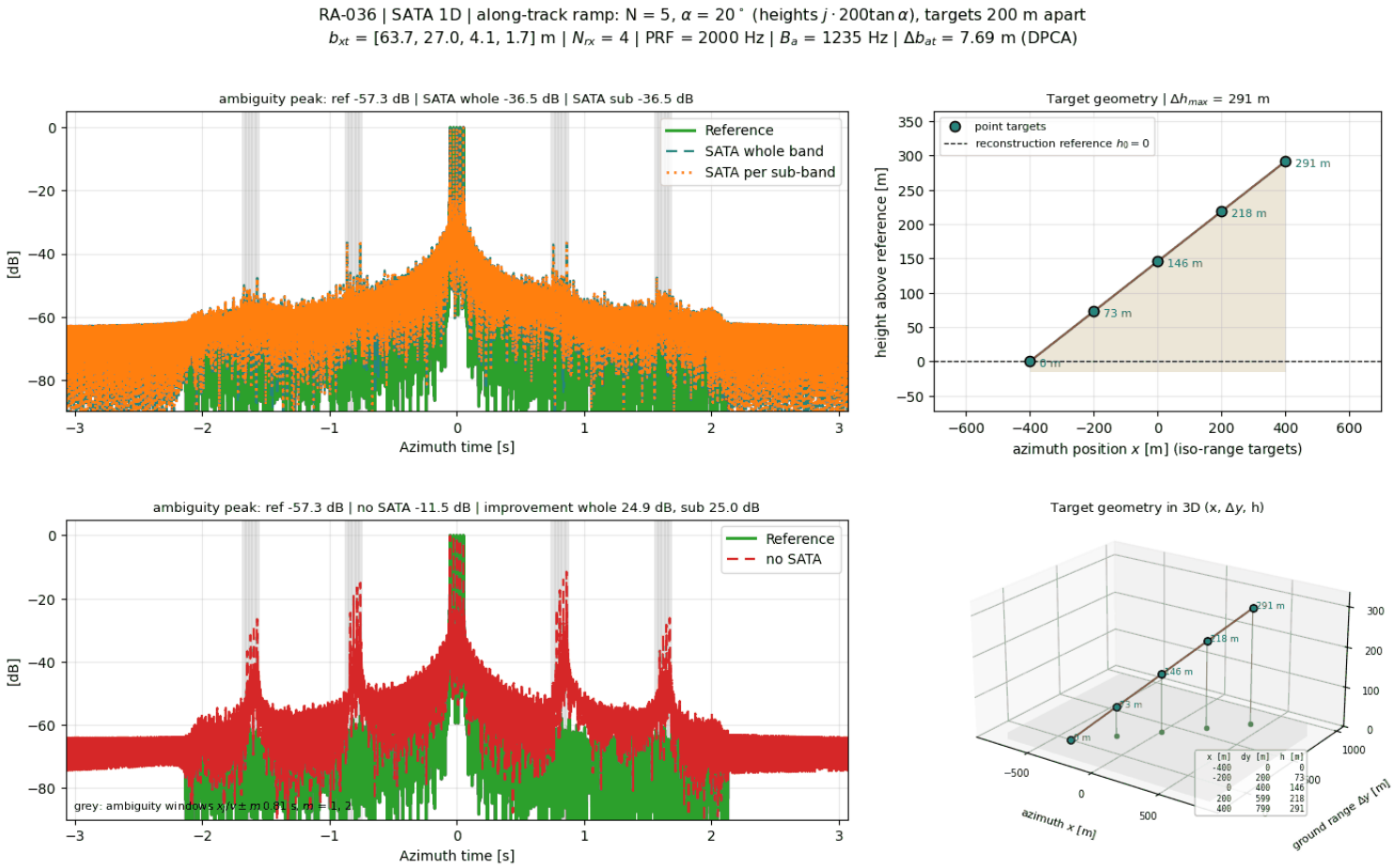


Figure 46: Along-track ramp, 200 m apart, normal heights: RA-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -11.5, SATA whole band -36.5, per sub-band -36.5 dB (improvement 24.9 dB).

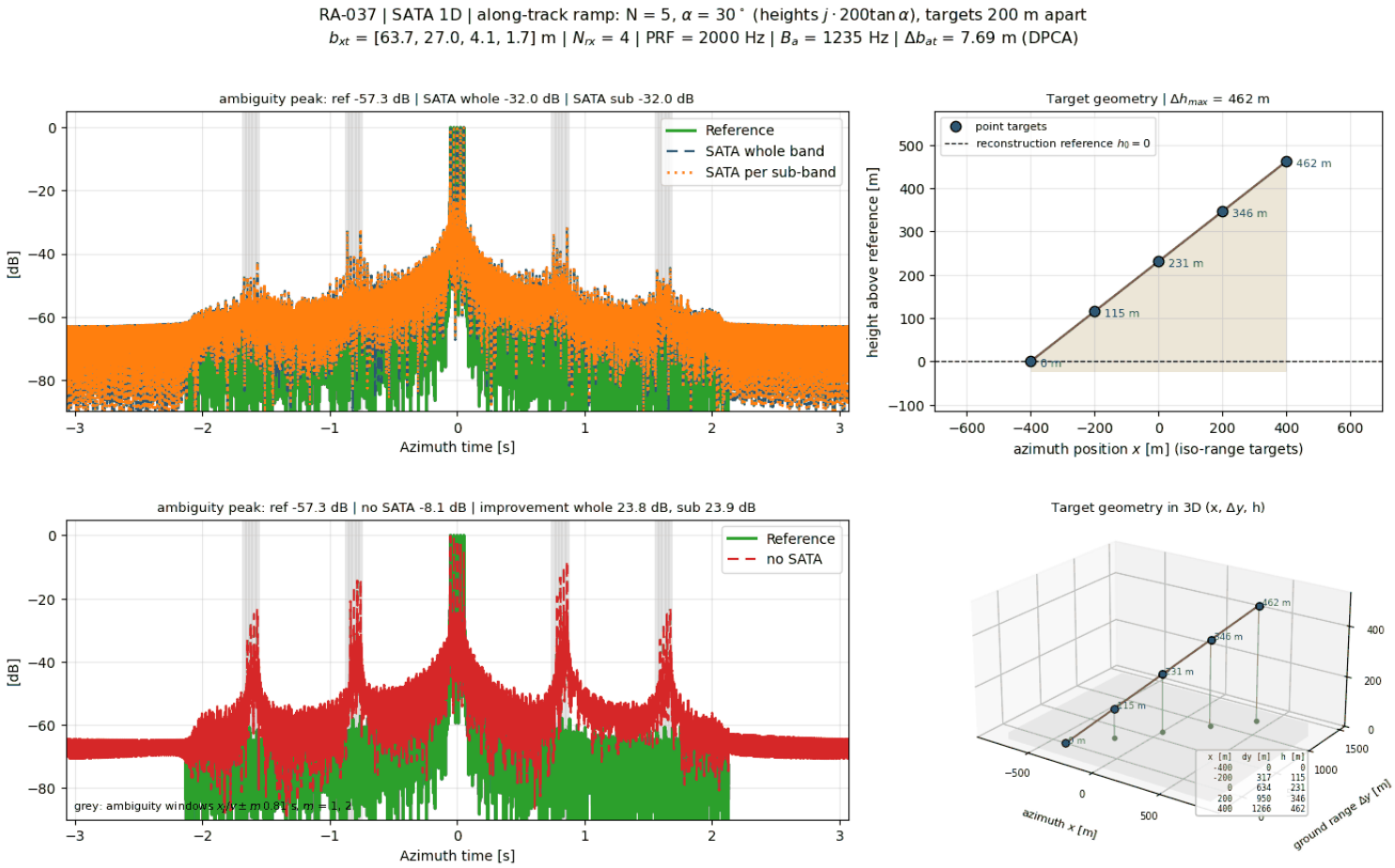


Figure 47: Along-track ramp, 200 m apart, normal heights: RA-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -8.1, SATA whole band -32.0, per sub-band -32.0 dB (improvement 23.8 dB).

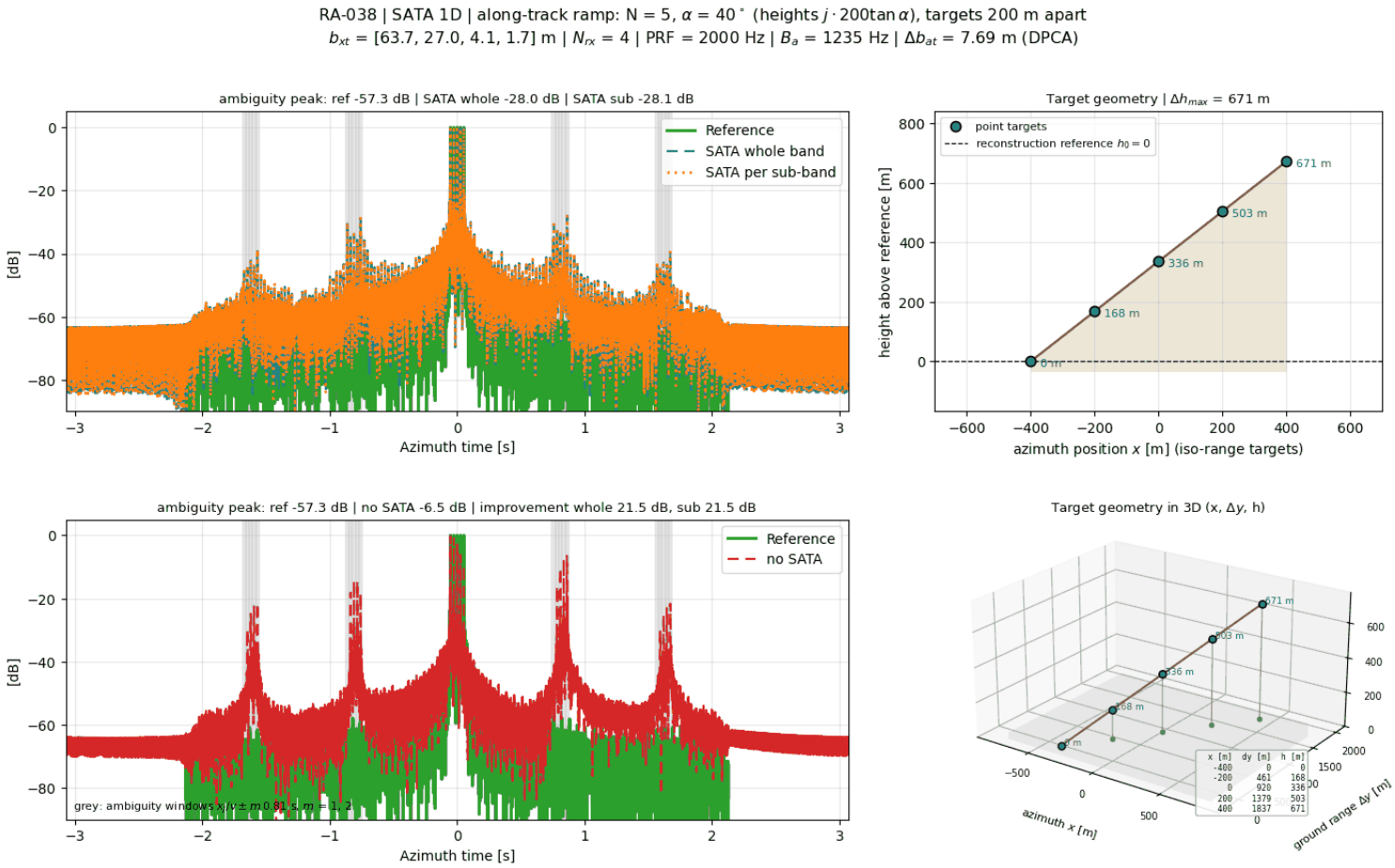


Figure 48: Along-track ramp, 200 m apart, normal heights: RA-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −28.0, per sub-band −28.1 dB (improvement 21.5 dB).

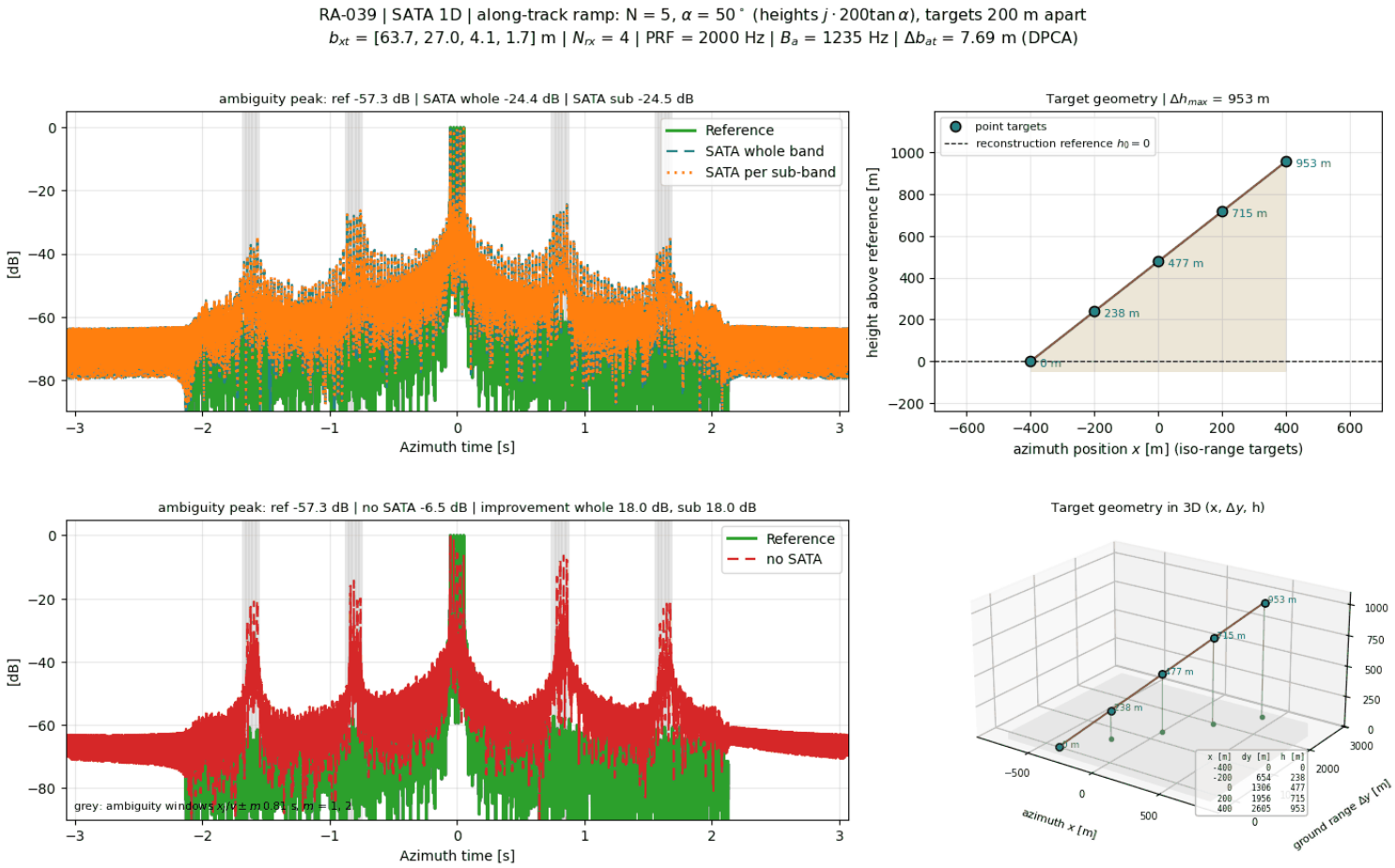


Figure 49: Along-track ramp, 200 m apart, normal heights: RA-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −24.4, per sub-band −24.5 dB (improvement 18.0 dB).

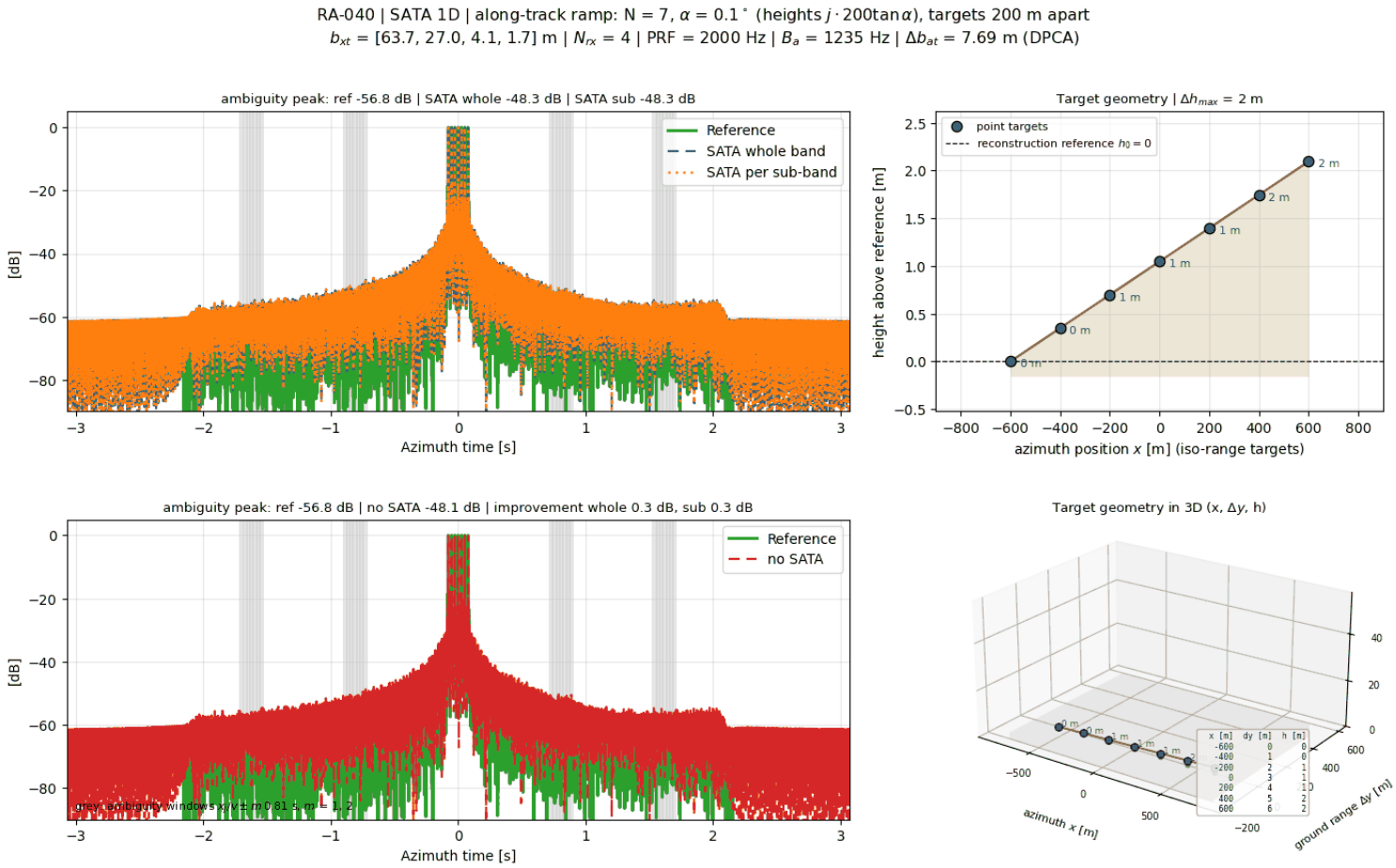


Figure 50: Along-track ramp, 200 m apart, normal heights: RA-040, $N = 7$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −48.1, SATA whole band −48.3, per sub-band −48.3 dB (improvement 0.3 dB).

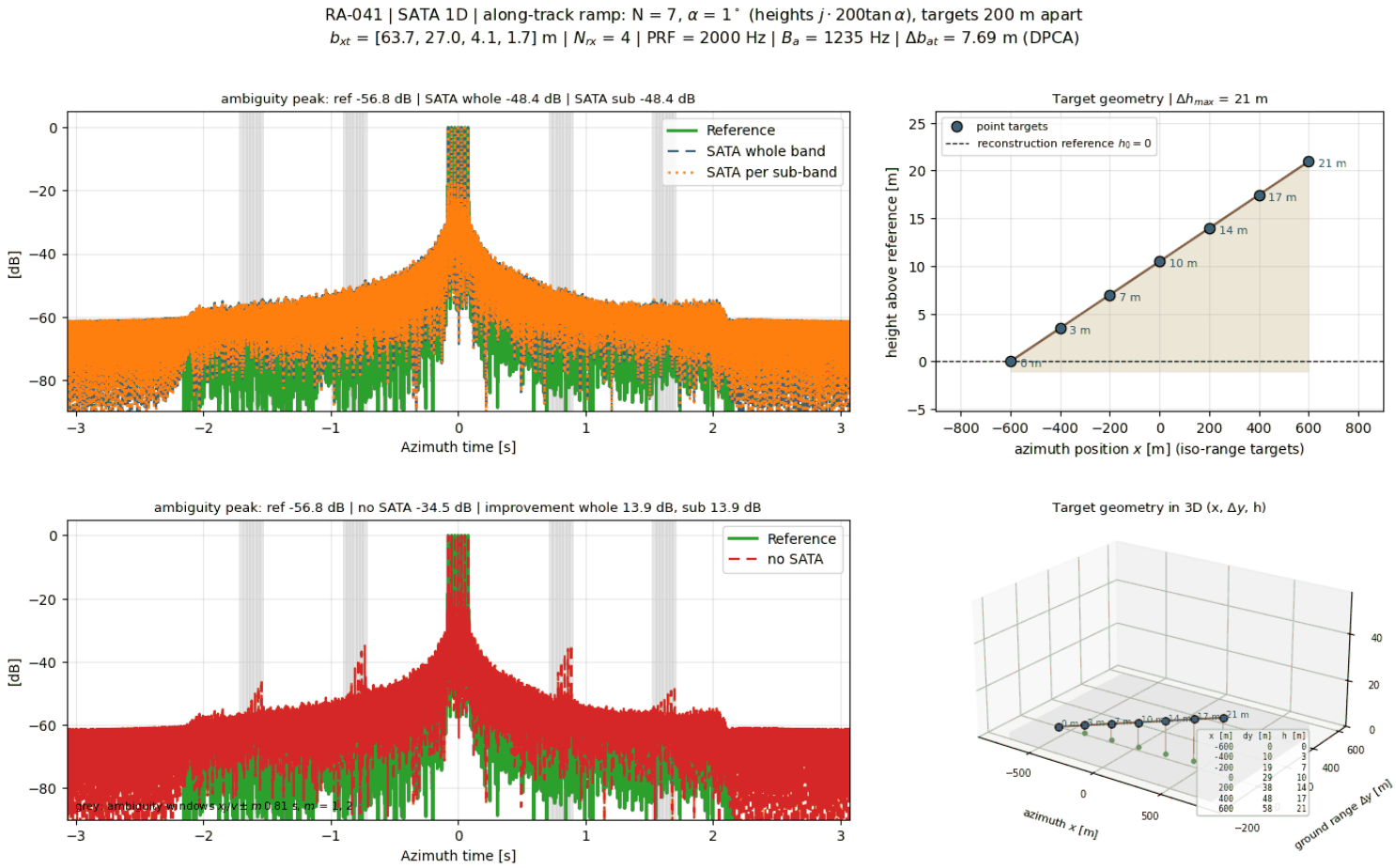


Figure 51: Along-track ramp, 200 m apart, normal heights: RA-041, $N = 7$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −34.5, SATA whole band −48.4, per sub-band −48.4 dB (improvement 13.9 dB).

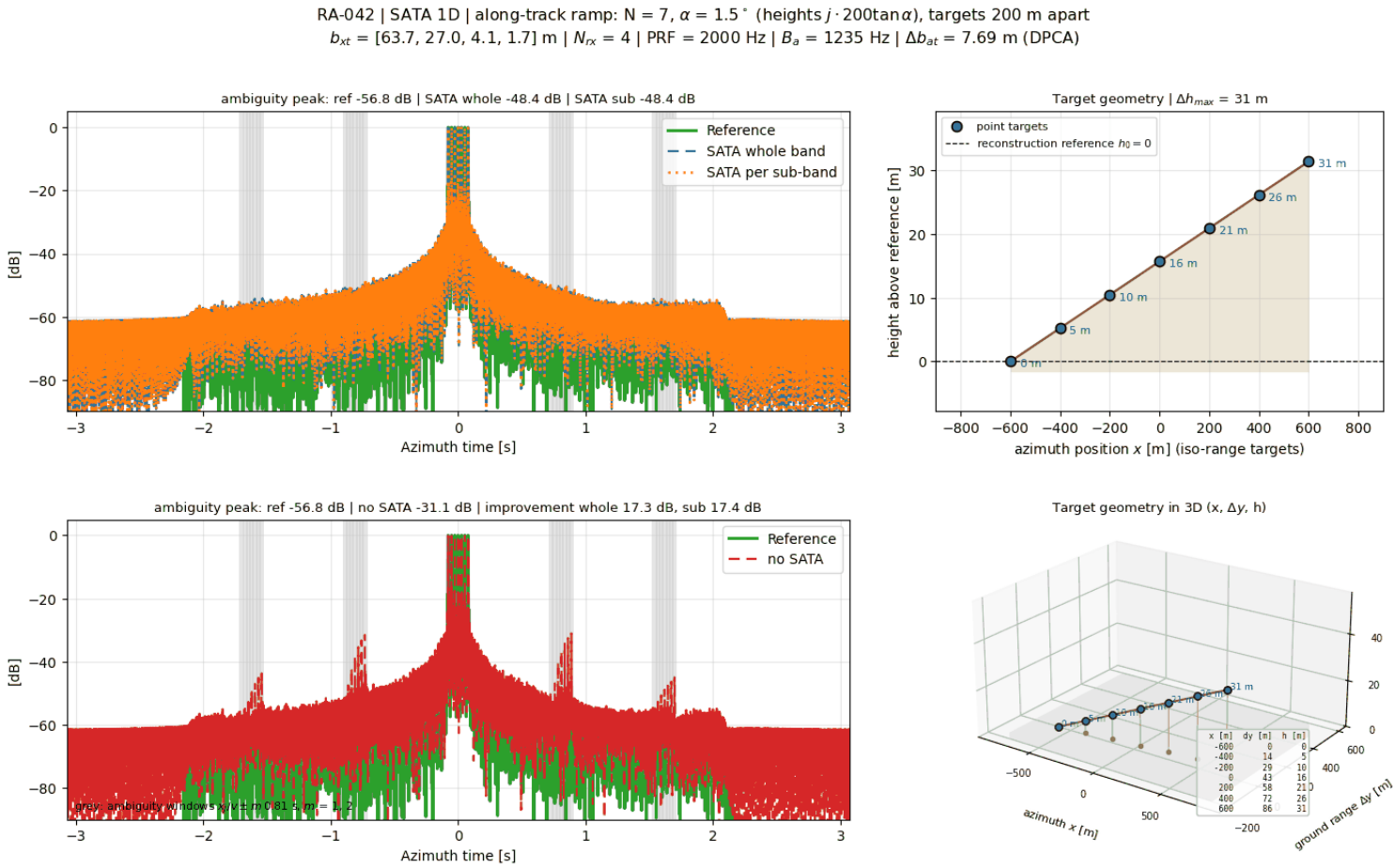


Figure 52: Along-track ramp, 200 m apart, normal heights: RA-042, $N = 7$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −31.1, SATA whole band −48.4, per sub-band −48.4 dB (improvement 17.3 dB).

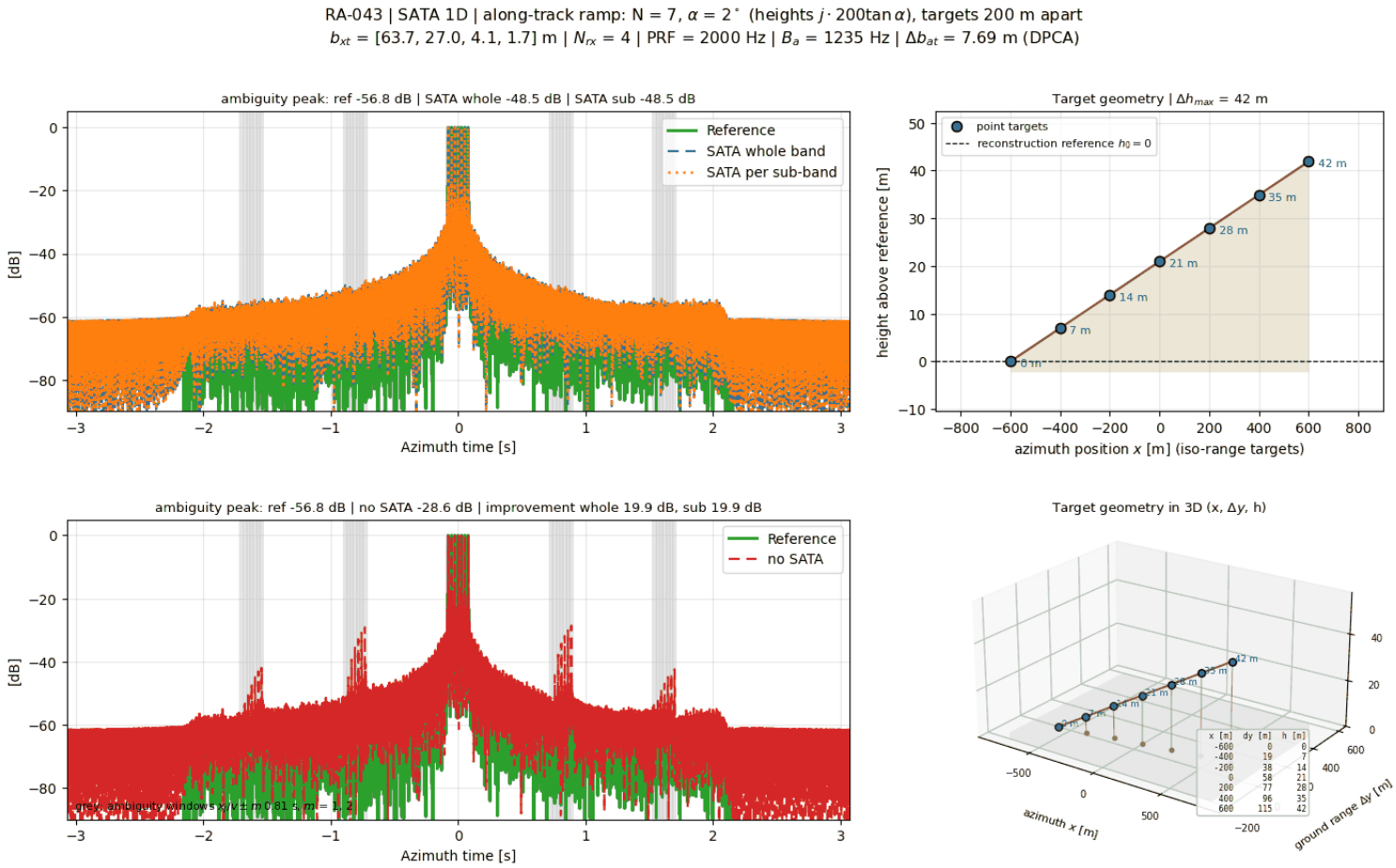


Figure 53: Along-track ramp, 200 m apart, normal heights: RA-043, $N = 7$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −28.6, SATA whole band −48.5, per sub-band −48.5 dB (improvement 19.9 dB).

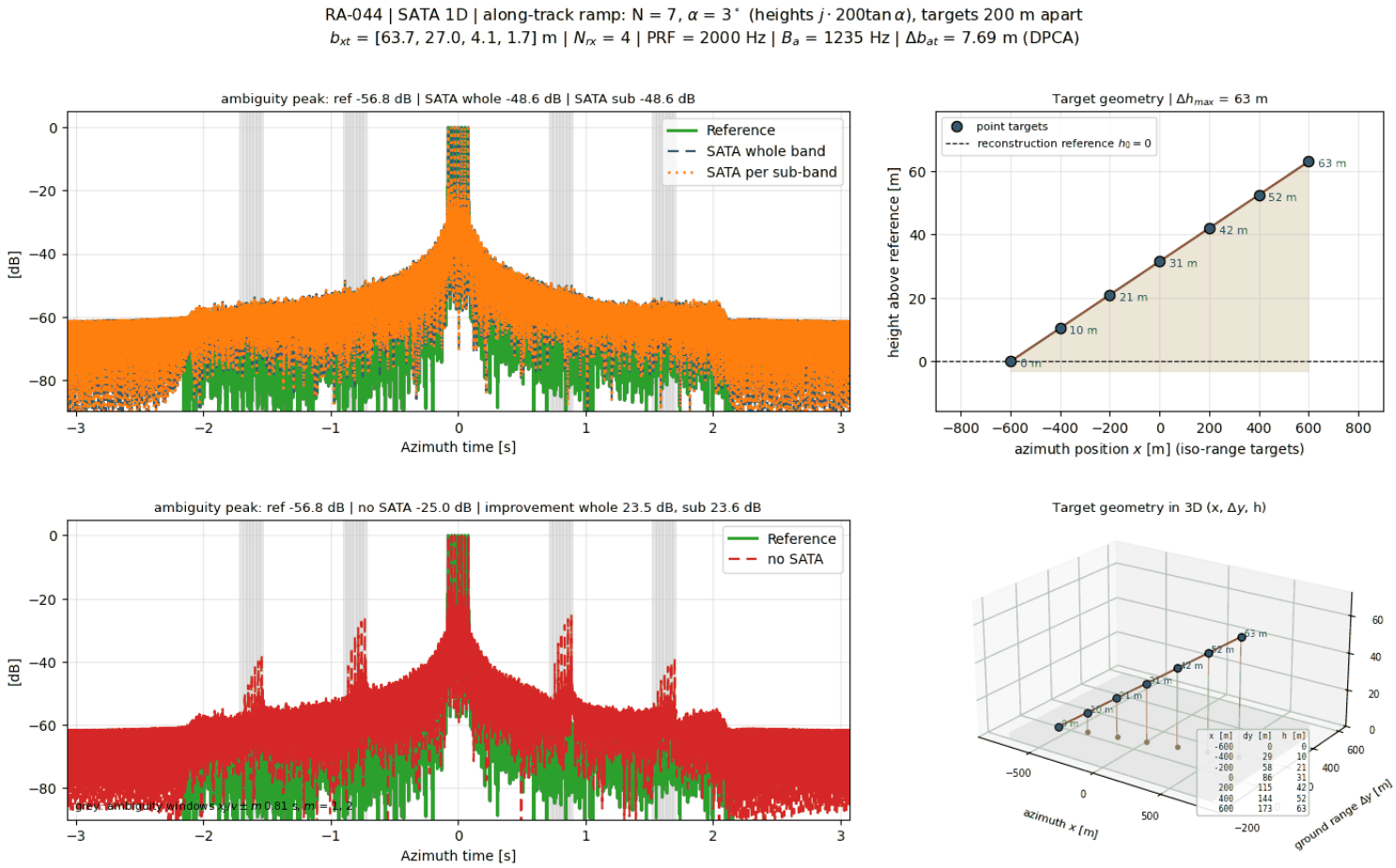


Figure 54: Along-track ramp, 200 m apart, normal heights: RA-044, $N = 7$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −25.0, SATA whole band −48.6, per sub-band −48.6 dB (improvement 23.5 dB).

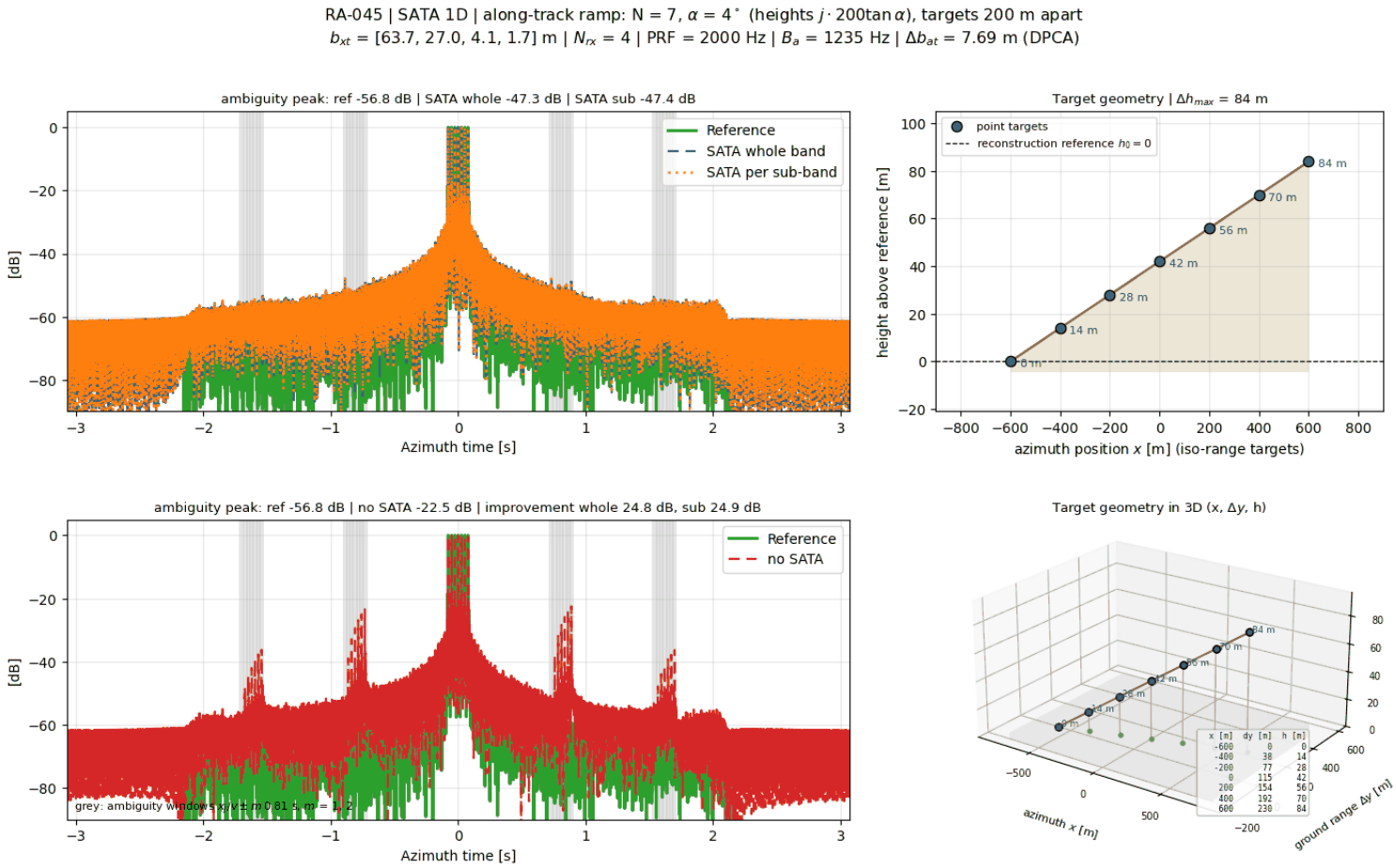


Figure 55: Along-track ramp, 200 m apart, normal heights: RA-045, $N = 7$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -22.5, SATA whole band -47.3, per sub-band -47.4 dB (improvement 24.8 dB).

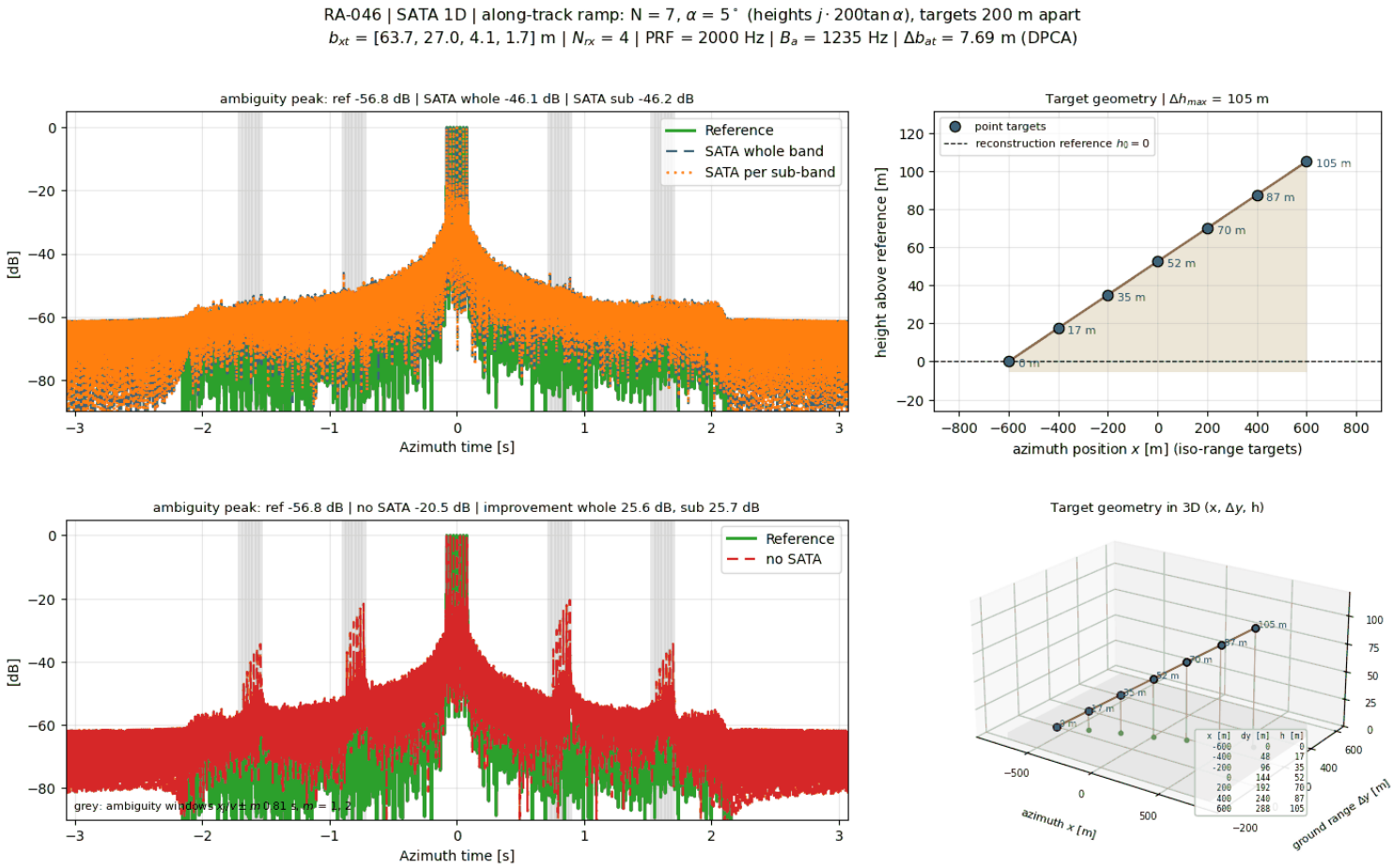


Figure 56: Along-track ramp, 200 m apart, normal heights: RA-046, $N = 7$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -20.5, SATA whole band -46.1, per sub-band -46.2 dB (improvement 25.6 dB).

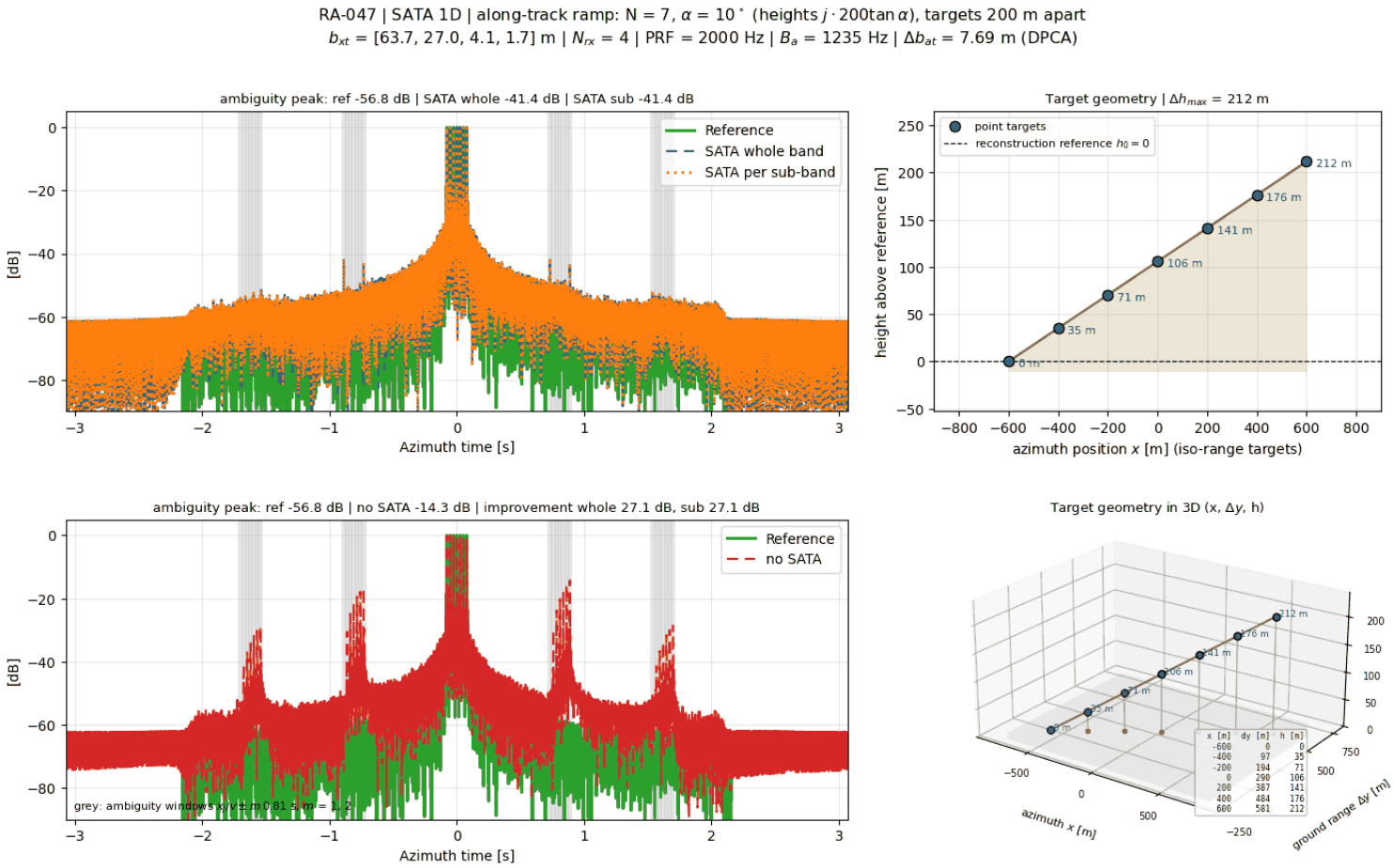


Figure 57: Along-track ramp, 200 m apart, normal heights: RA-047, $N = 7$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -14.3, SATA whole band -41.4, per sub-band -41.4 dB (improvement 27.1 dB).

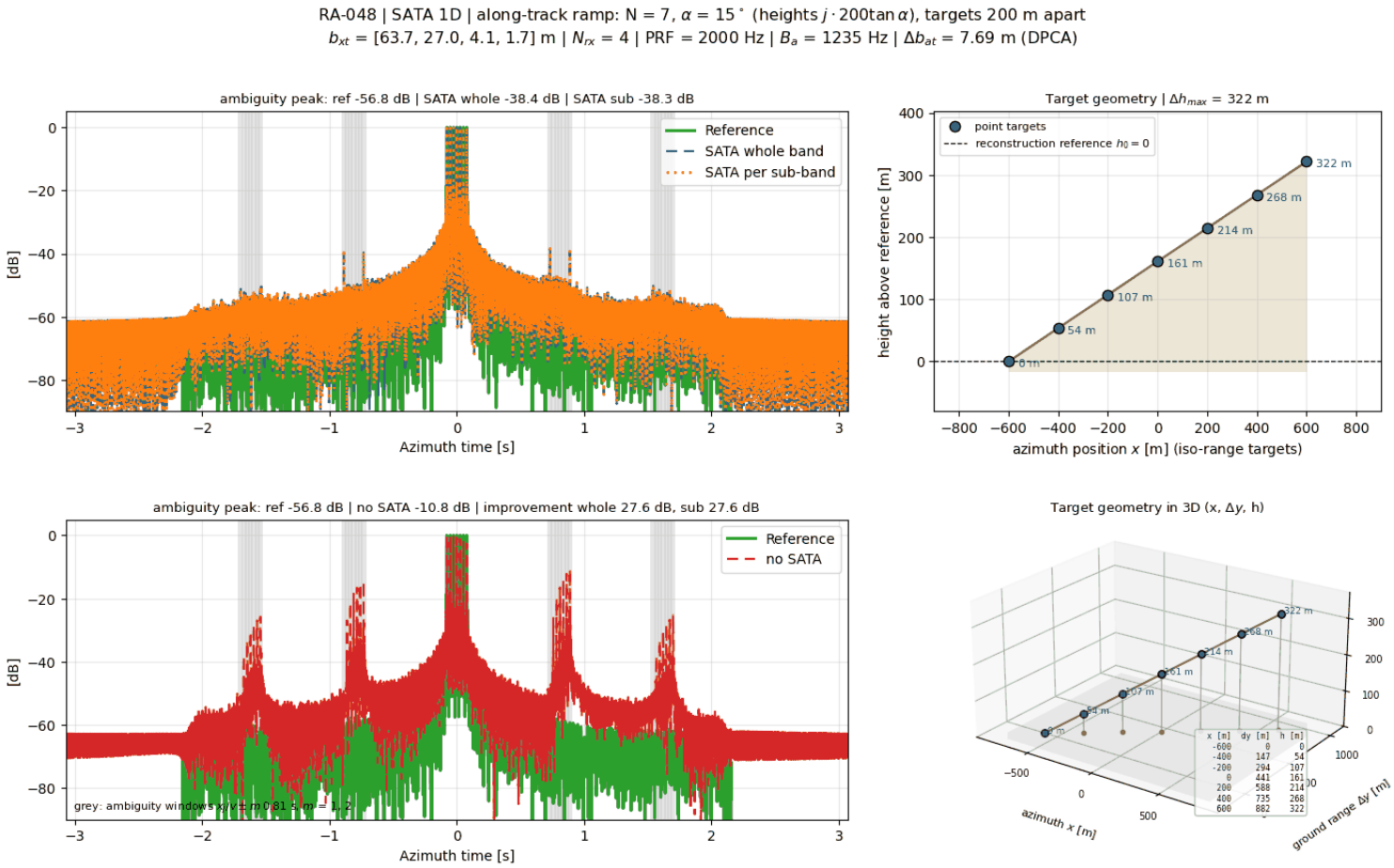


Figure 58: Along-track ramp, 200 m apart, normal heights: RA-048, $N = 7$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -10.8, SATA whole band -38.4, per sub-band -38.3 dB (improvement 27.6 dB).

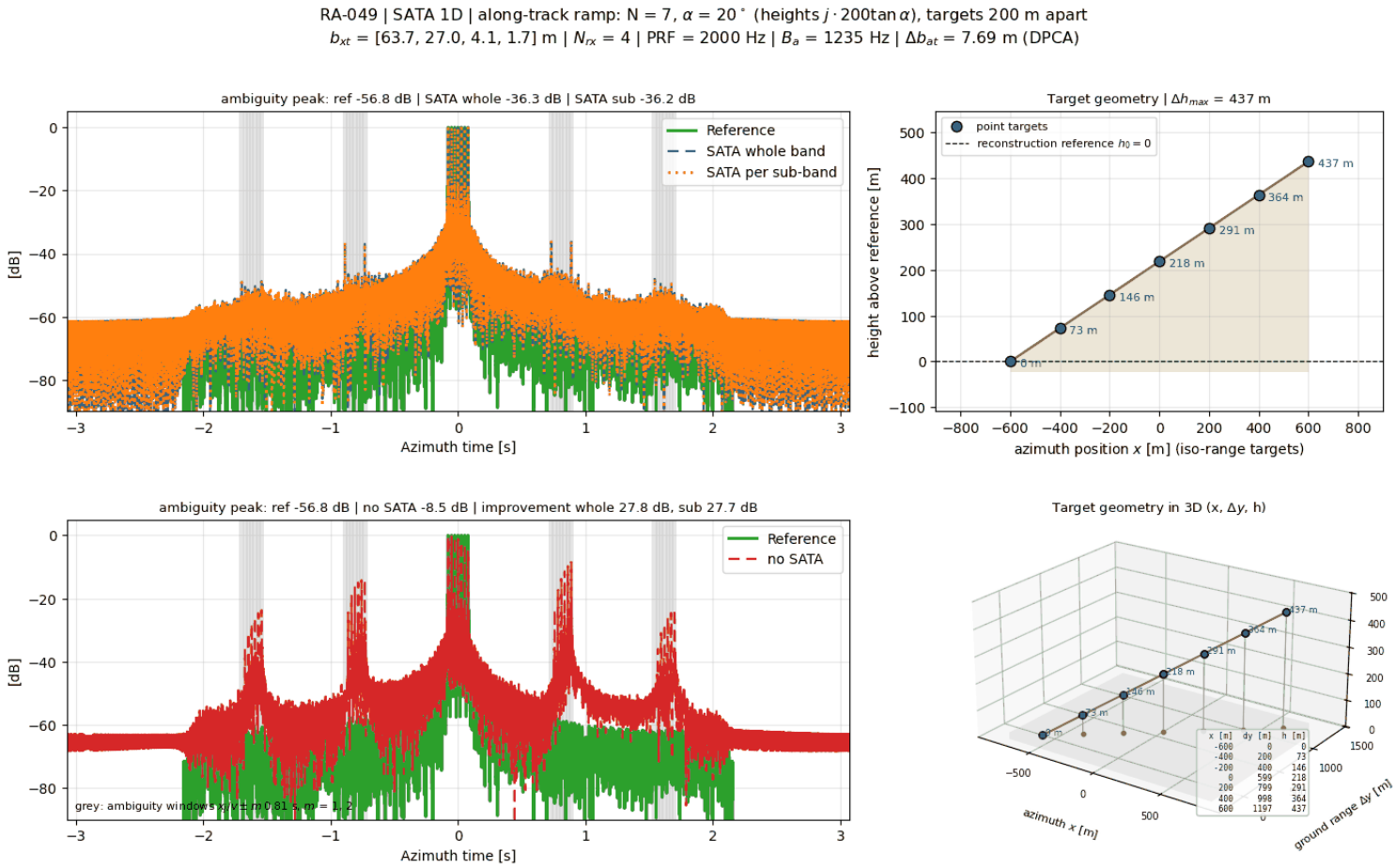


Figure 59: Along-track ramp, 200 m apart, normal heights: RA-049, $N = 7$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -8.5, SATA whole band -36.3, per sub-band -36.2 dB (improvement 27.8 dB).

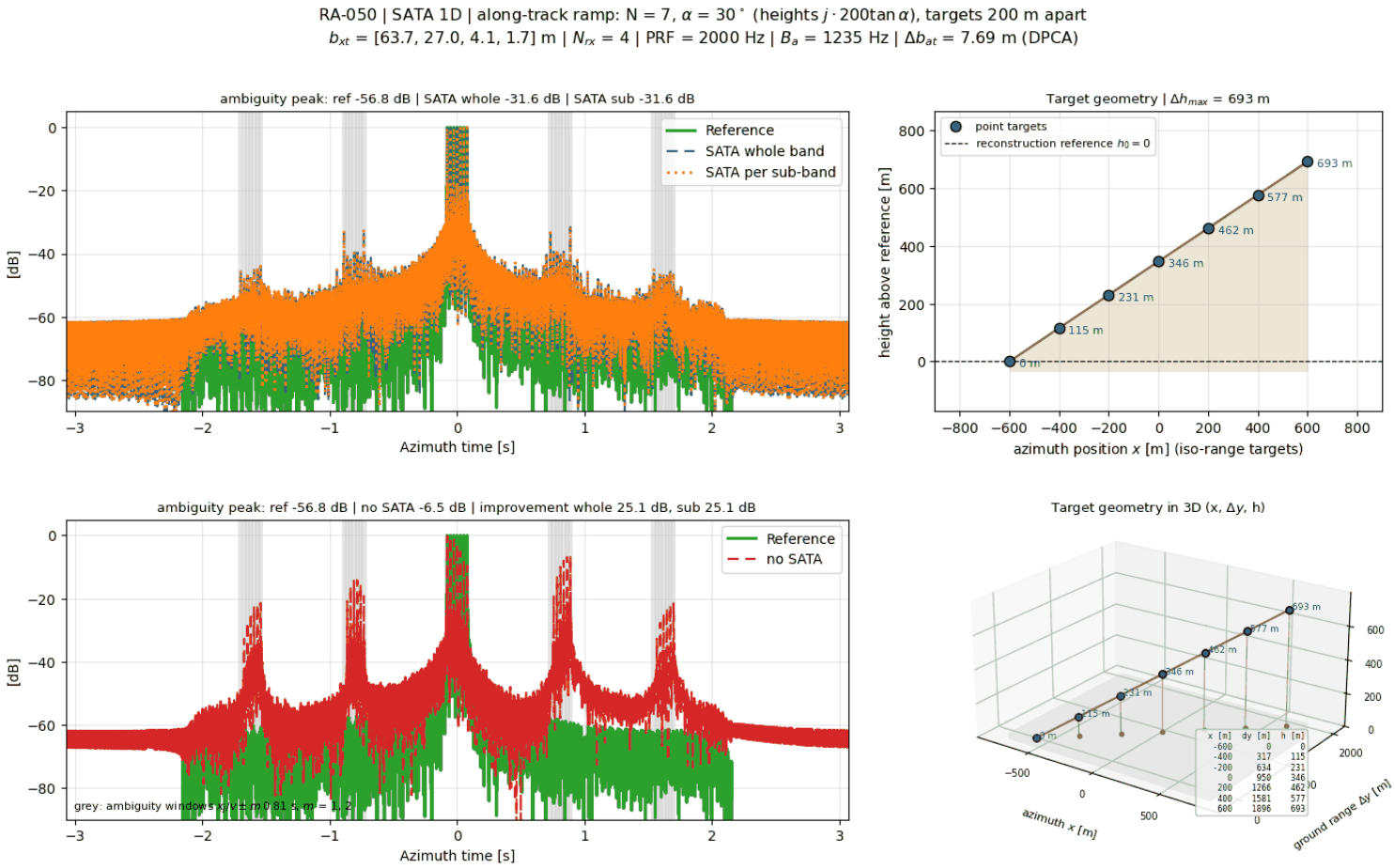


Figure 60: Along-track ramp, 200 m apart, normal heights: RA-050, $N = 7$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −31.6, per sub-band −31.6 dB (improvement 25.1 dB).

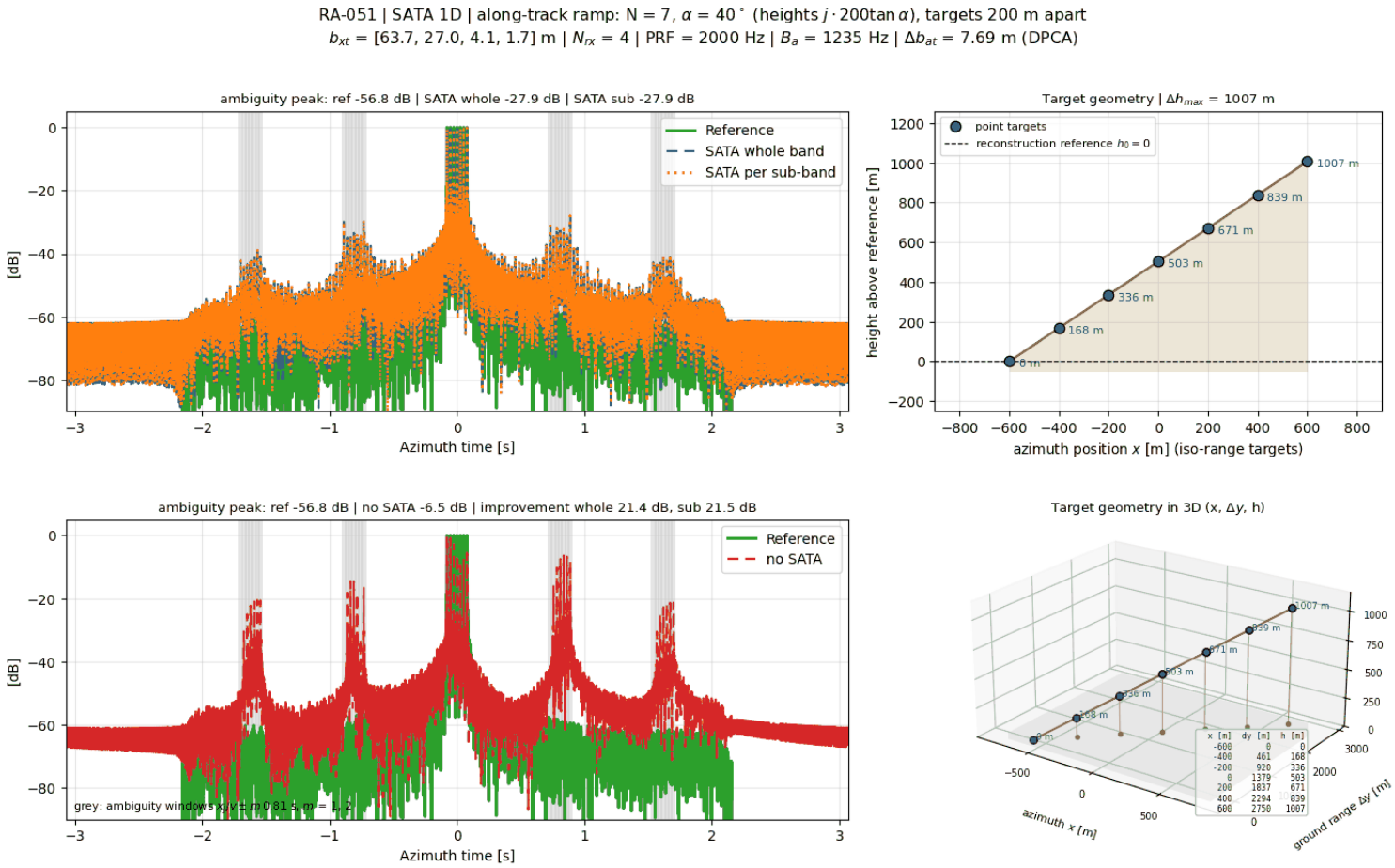


Figure 61: Along-track ramp, 200 m apart, normal heights: RA-051, $N = 7$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -6.5, SATA whole band -27.9, per sub-band -27.9 dB (improvement 21.4 dB).

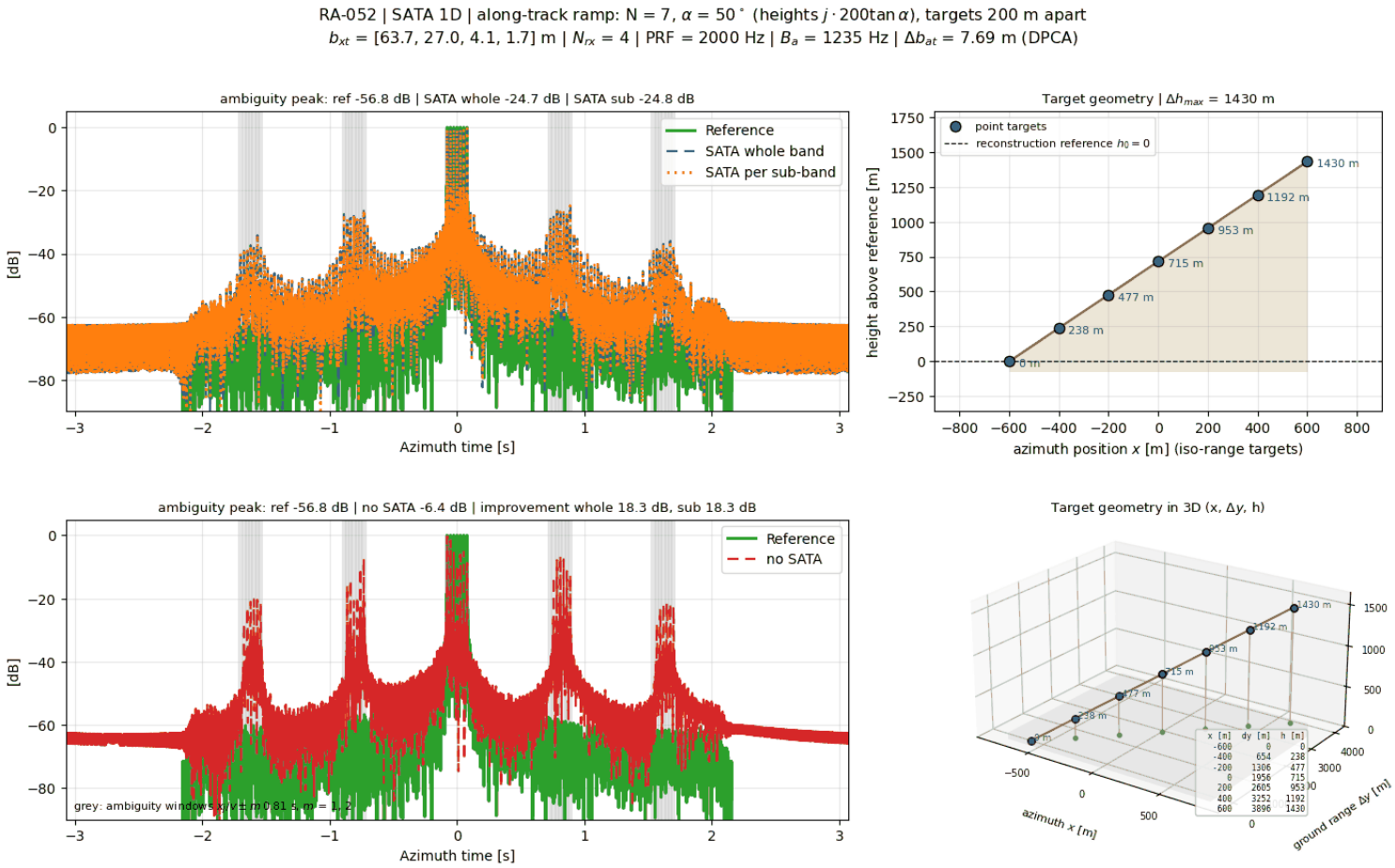


Figure 62: Along-track ramp, 200 m apart, normal heights: RA-052, $N = 7$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -6.4, SATA whole band -24.7, per sub-band -24.8 dB (improvement 18.3 dB).

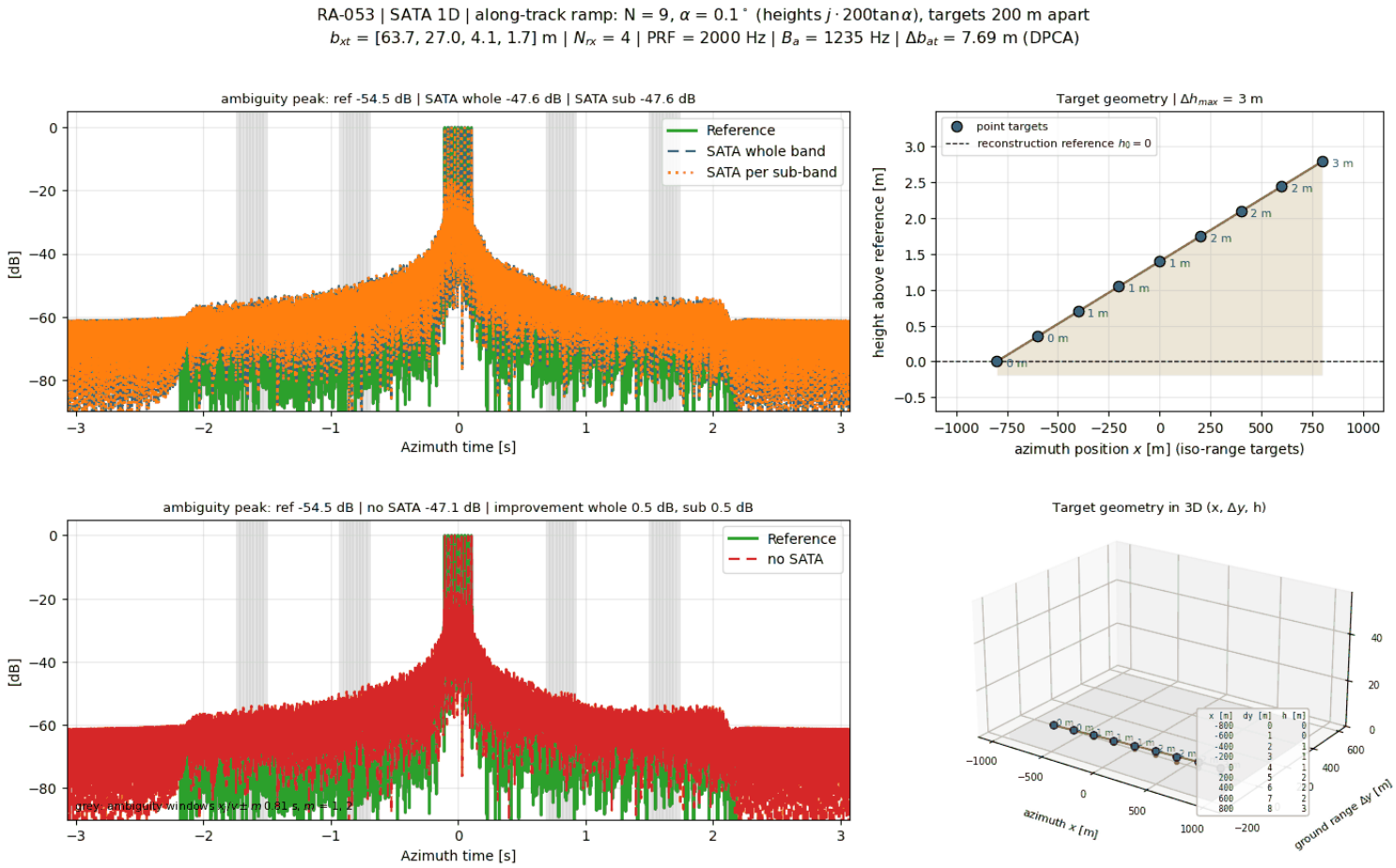


Figure 63: Along-track ramp, 200 m apart, normal heights: RA-053, $N = 9$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −47.1, SATA whole band −47.6, per sub-band −47.6 dB (improvement 0.5 dB).

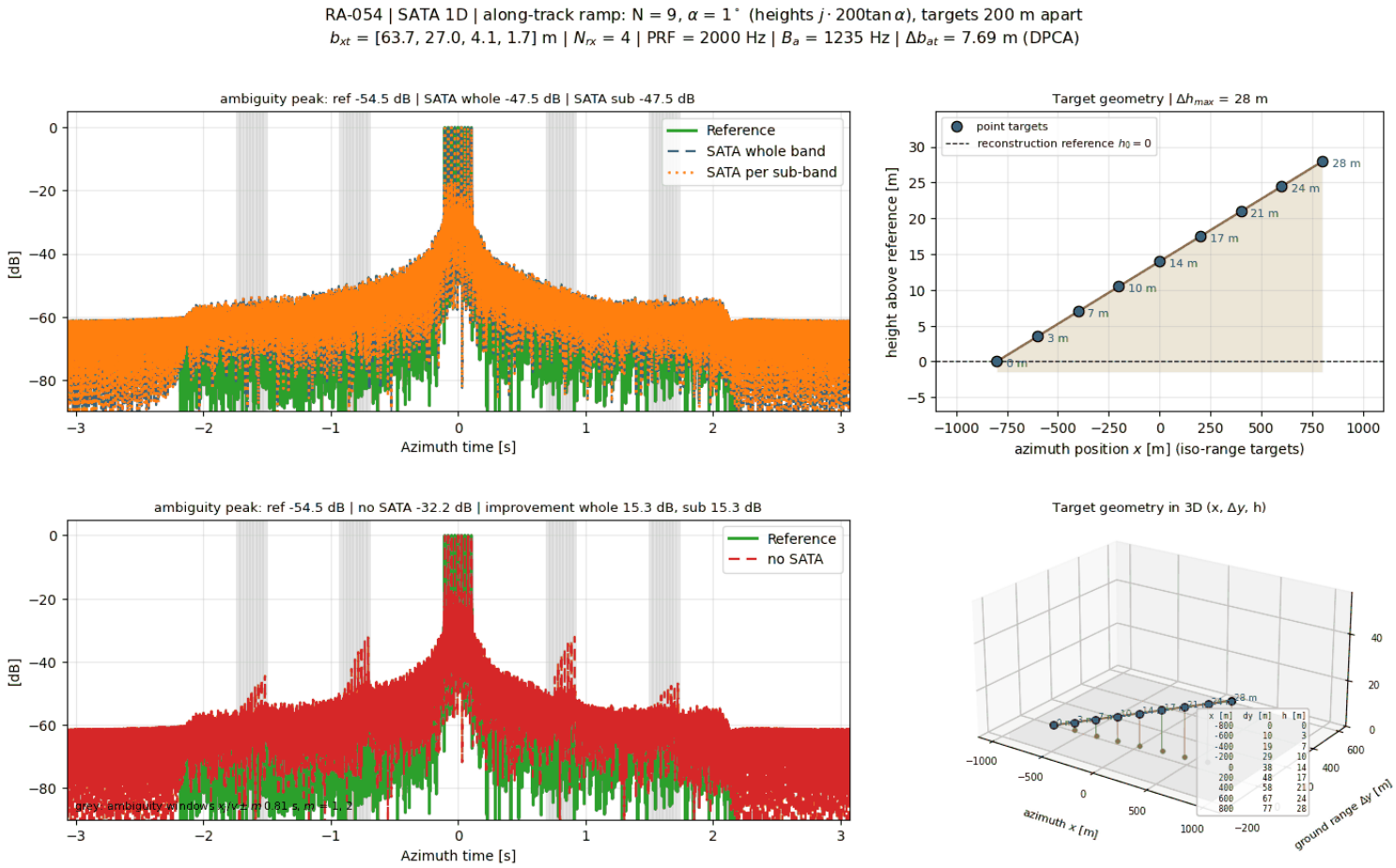


Figure 64: Along-track ramp, 200 m apart, normal heights: RA-054, $N = 9$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -32.2, SATA whole band -47.5, per sub-band -47.5 dB (improvement 15.3 dB).

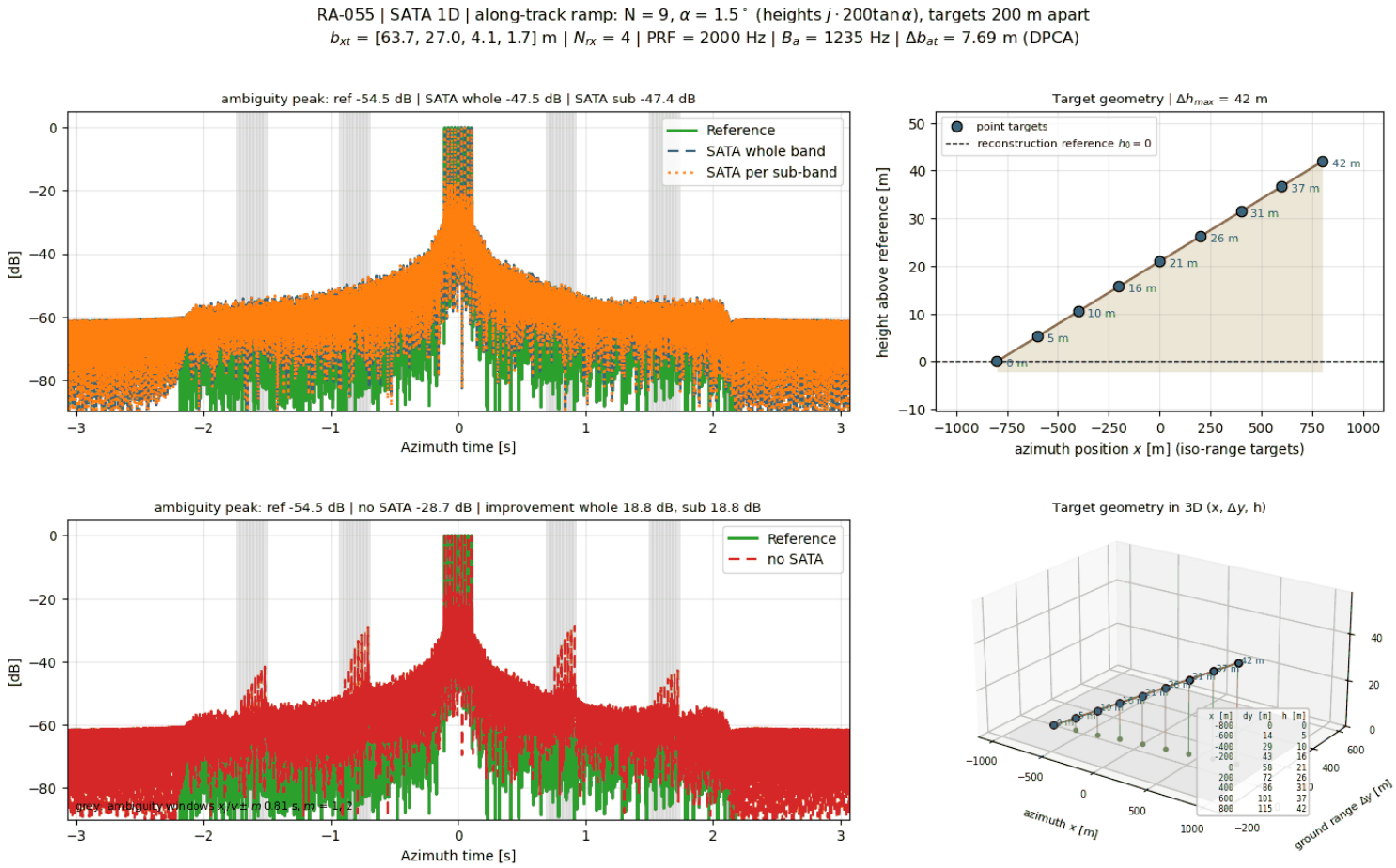


Figure 65: Along-track ramp, 200 m apart, normal heights: RA-055, $N = 9$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -28.7, SATA whole band -47.5, per sub-band -47.4 dB (improvement 18.8 dB).

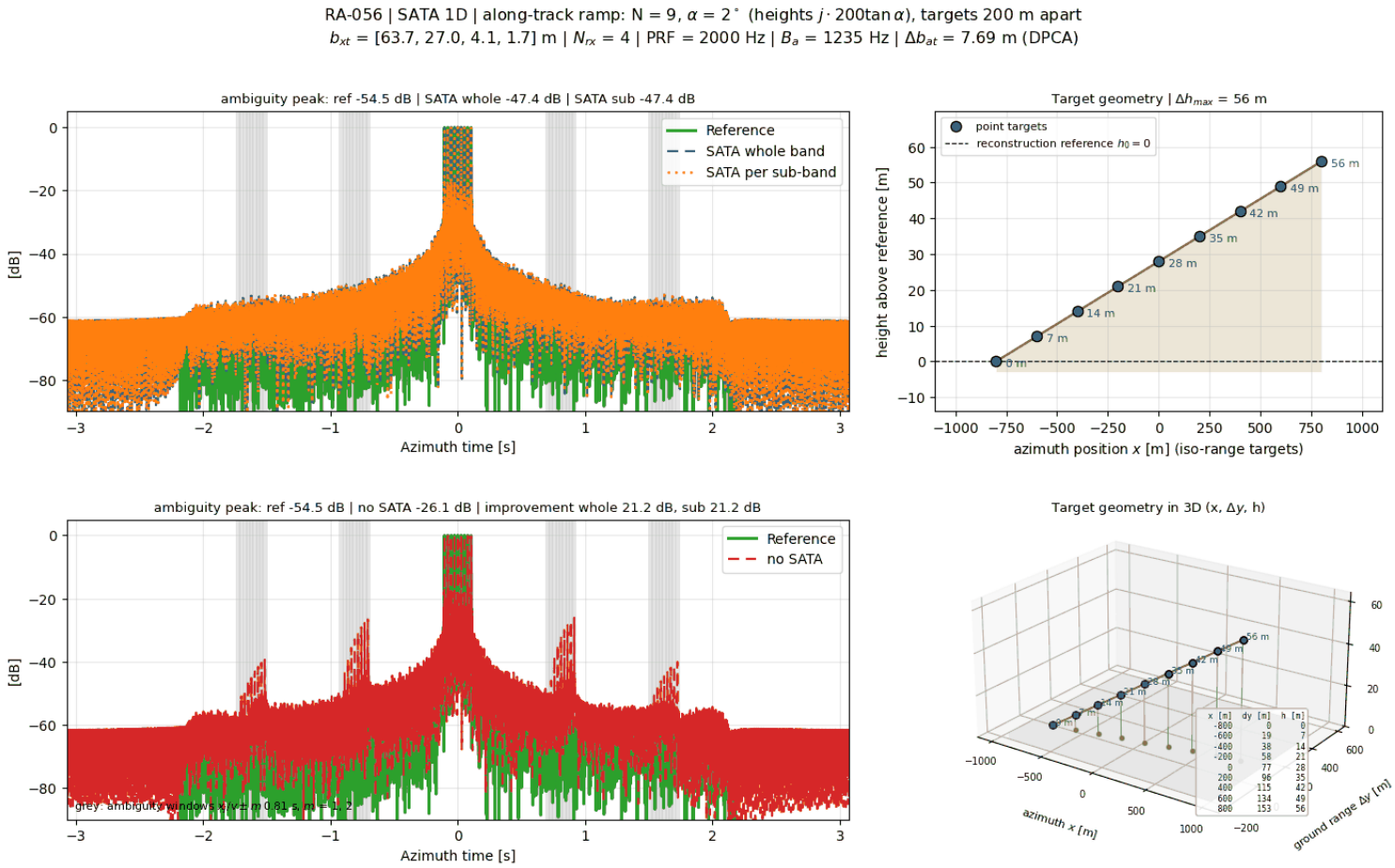


Figure 66: Along-track ramp, 200 m apart, normal heights: RA-056, $N = 9$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -26.1, SATA whole band -47.4, per sub-band -47.4 dB (improvement 21.2 dB).

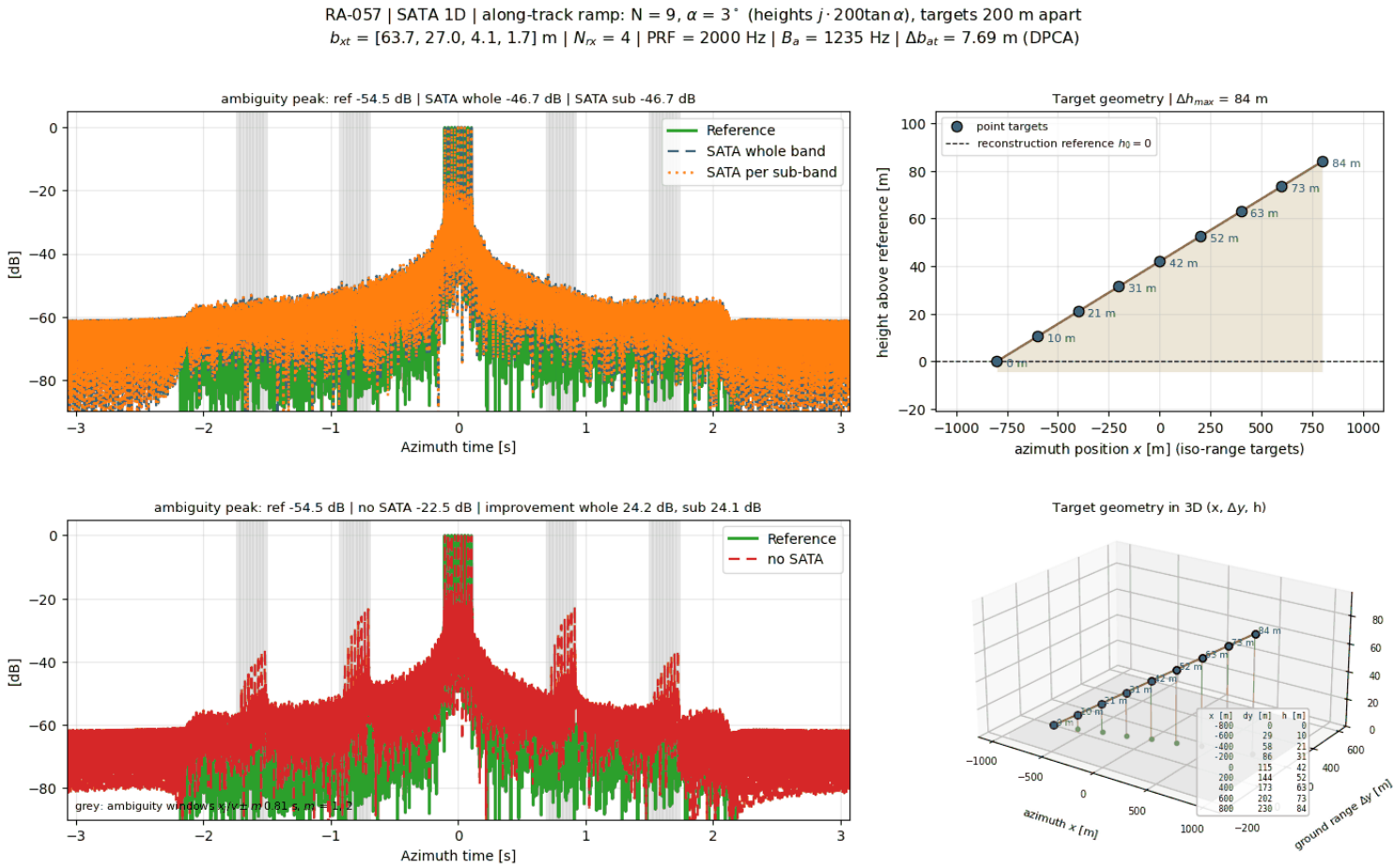


Figure 67: Along-track ramp, 200 m apart, normal heights: RA-057, $N = 9$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -22.5, SATA whole band -46.7, per sub-band -46.7 dB (improvement 24.2 dB).

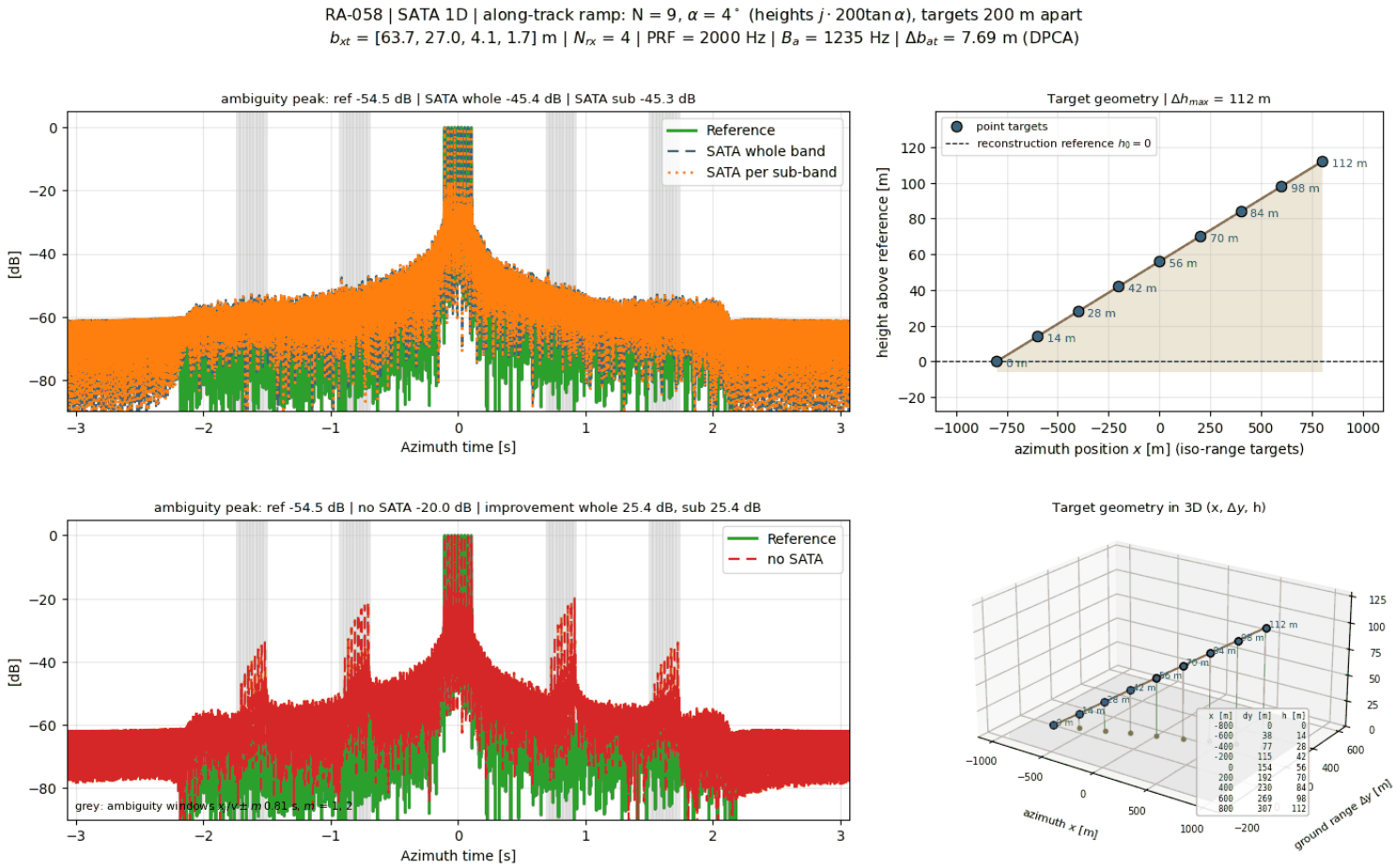


Figure 68: Along-track ramp, 200 m apart, normal heights: RA-058, $N = 9$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -20.0, SATA whole band -45.4, per sub-band -45.3 dB (improvement 25.4 dB).

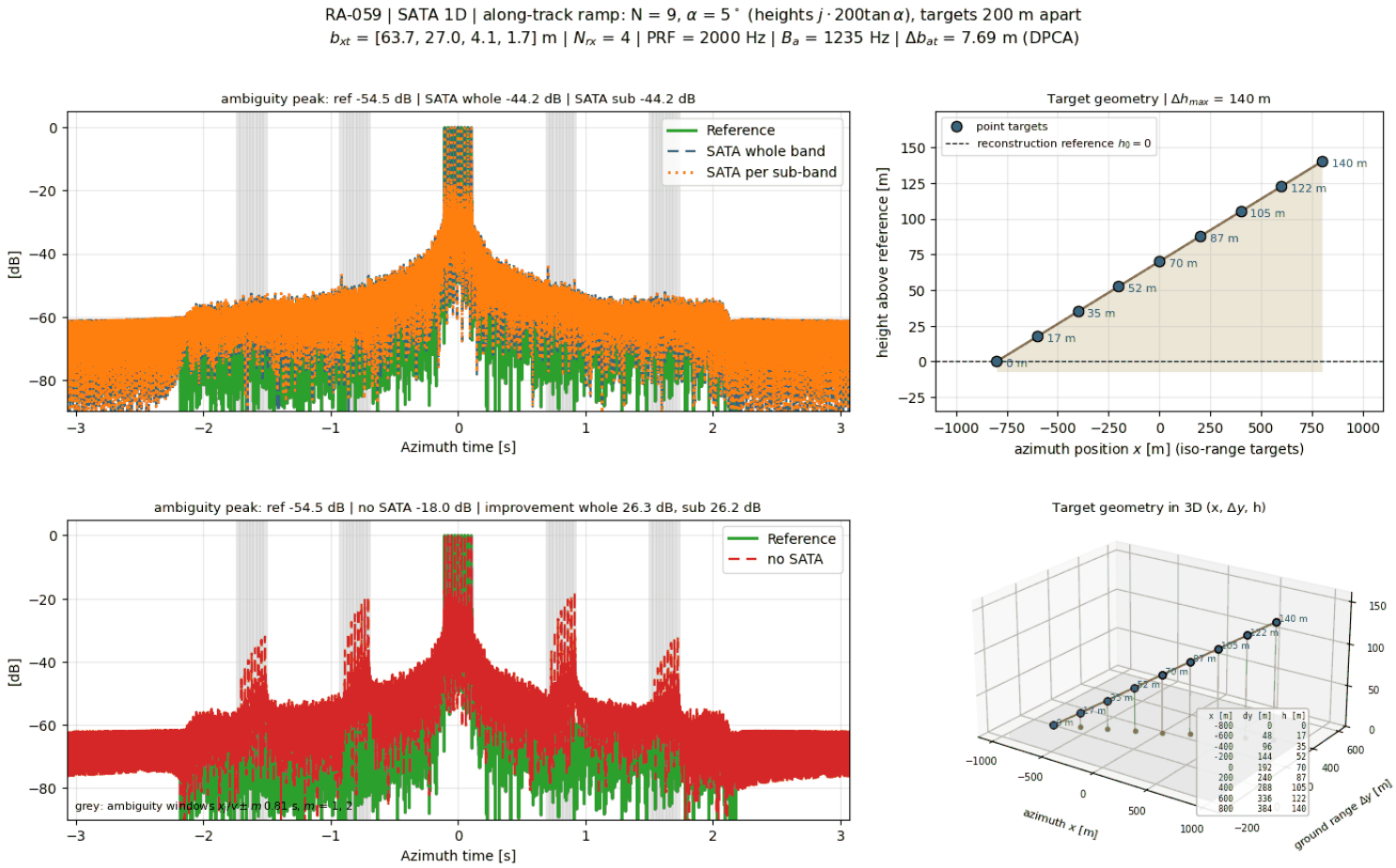


Figure 69: Along-track ramp, 200 m apart, normal heights: RA-059, $N = 9$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -18.0, SATA whole band -44.2, per sub-band -44.2 dB (improvement 26.3 dB).

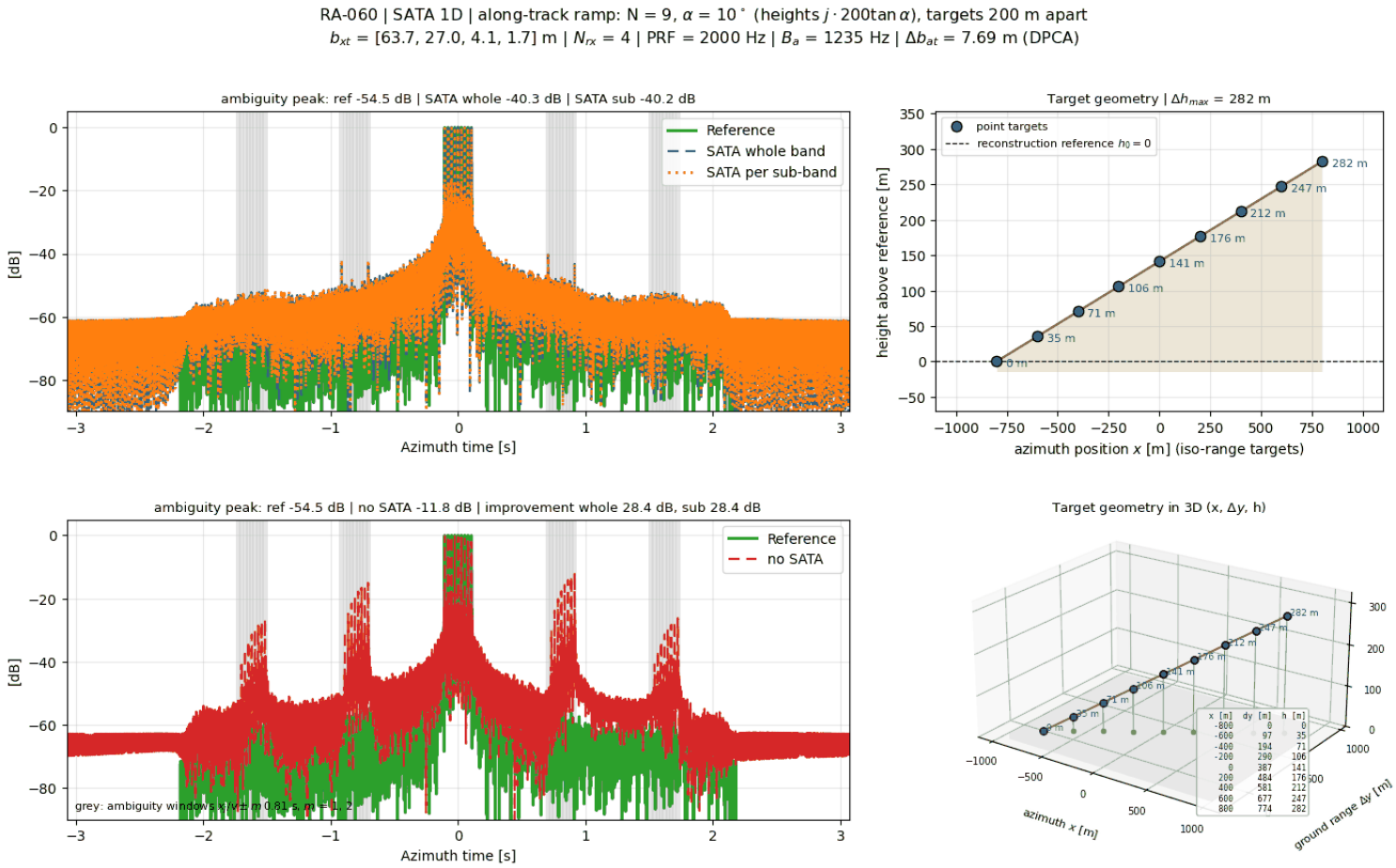


Figure 70: Along-track ramp, 200 m apart, normal heights: RA-060, $N = 9$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -11.8, SATA whole band -40.3, per sub-band -40.2 dB (improvement 28.4 dB).

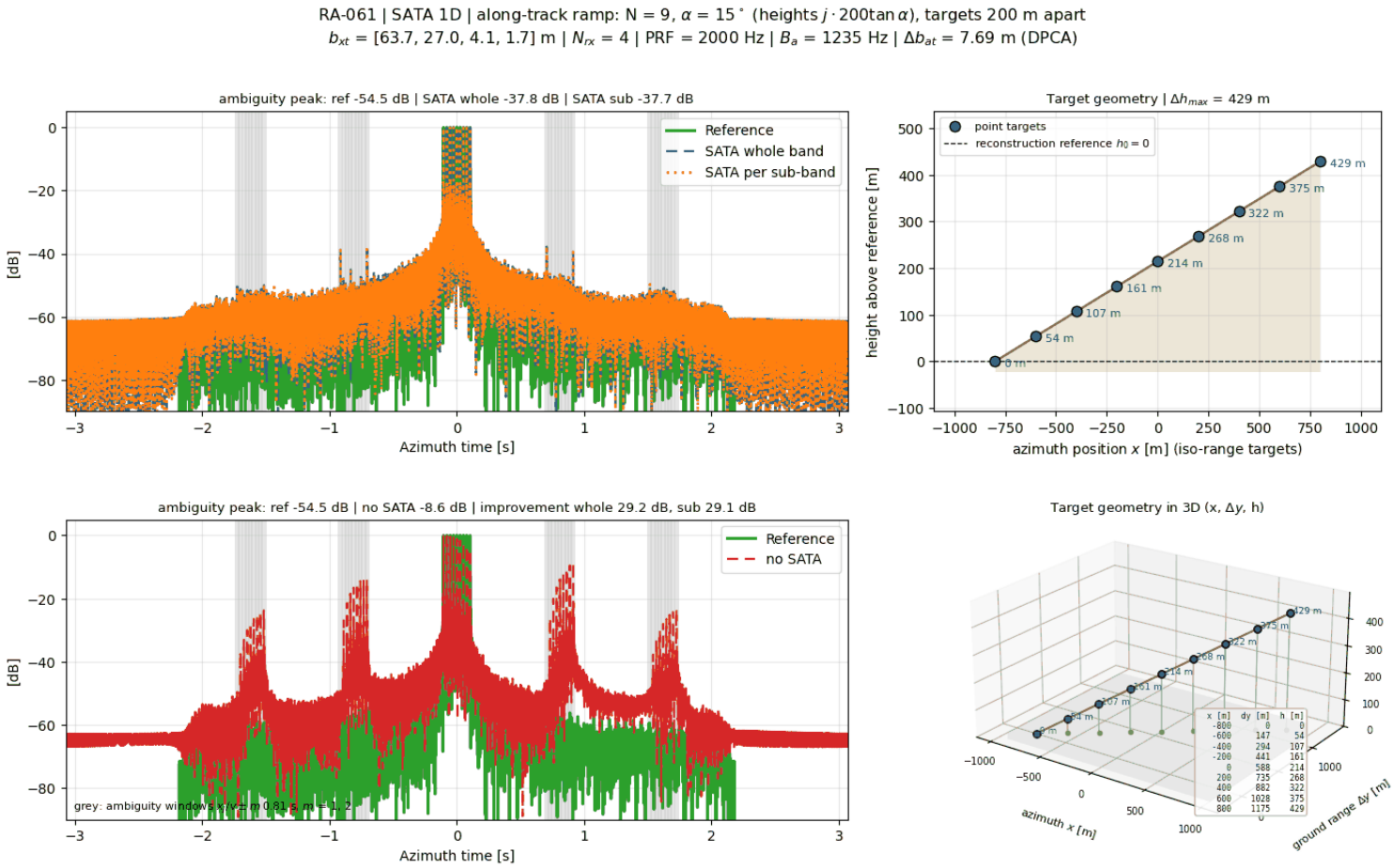


Figure 71: Along-track ramp, 200 m apart, normal heights: RA-061, $N = 9$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -8.6, SATA whole band -37.8, per sub-band -37.7 dB (improvement 29.2 dB).

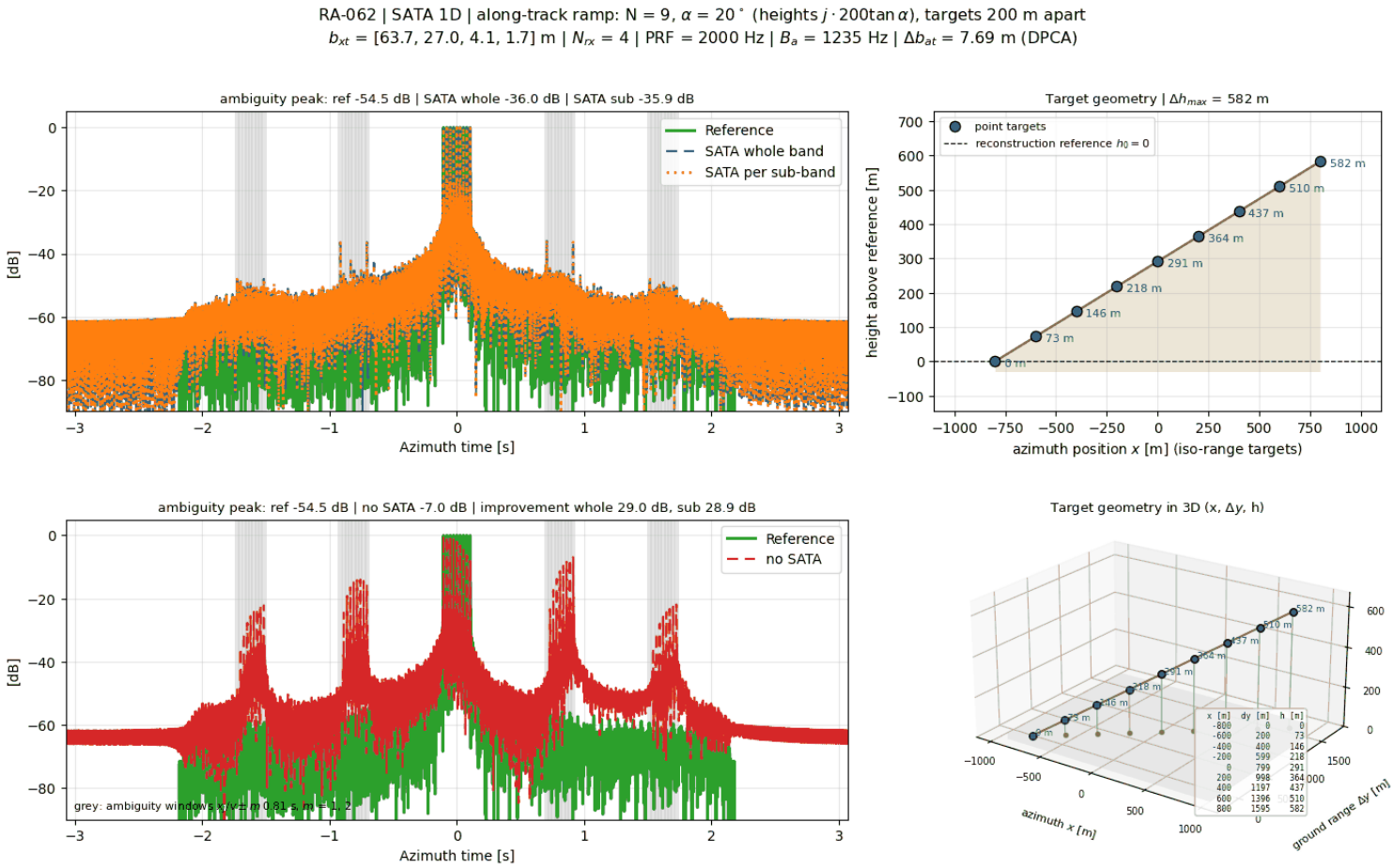


Figure 72: Along-track ramp, 200 m apart, normal heights: RA-062, $N = 9$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -7.0, SATA whole band -36.0, per sub-band -35.9 dB (improvement 29.0 dB).

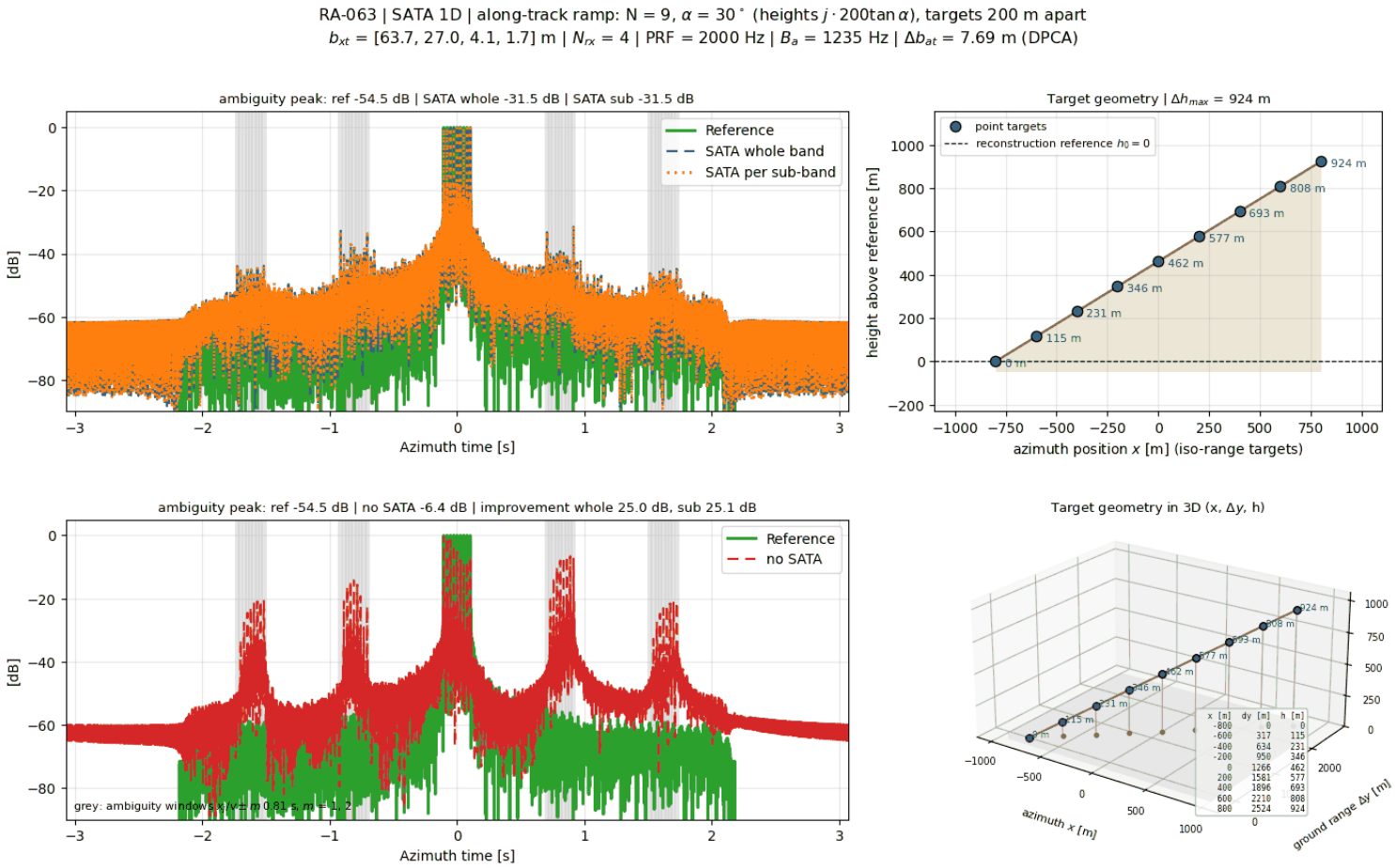


Figure 73: Along-track ramp, 200 m apart, normal heights: RA-063, $N = 9$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -6.4, SATA whole band -31.5, per sub-band -31.5 dB (improvement 25.0 dB).

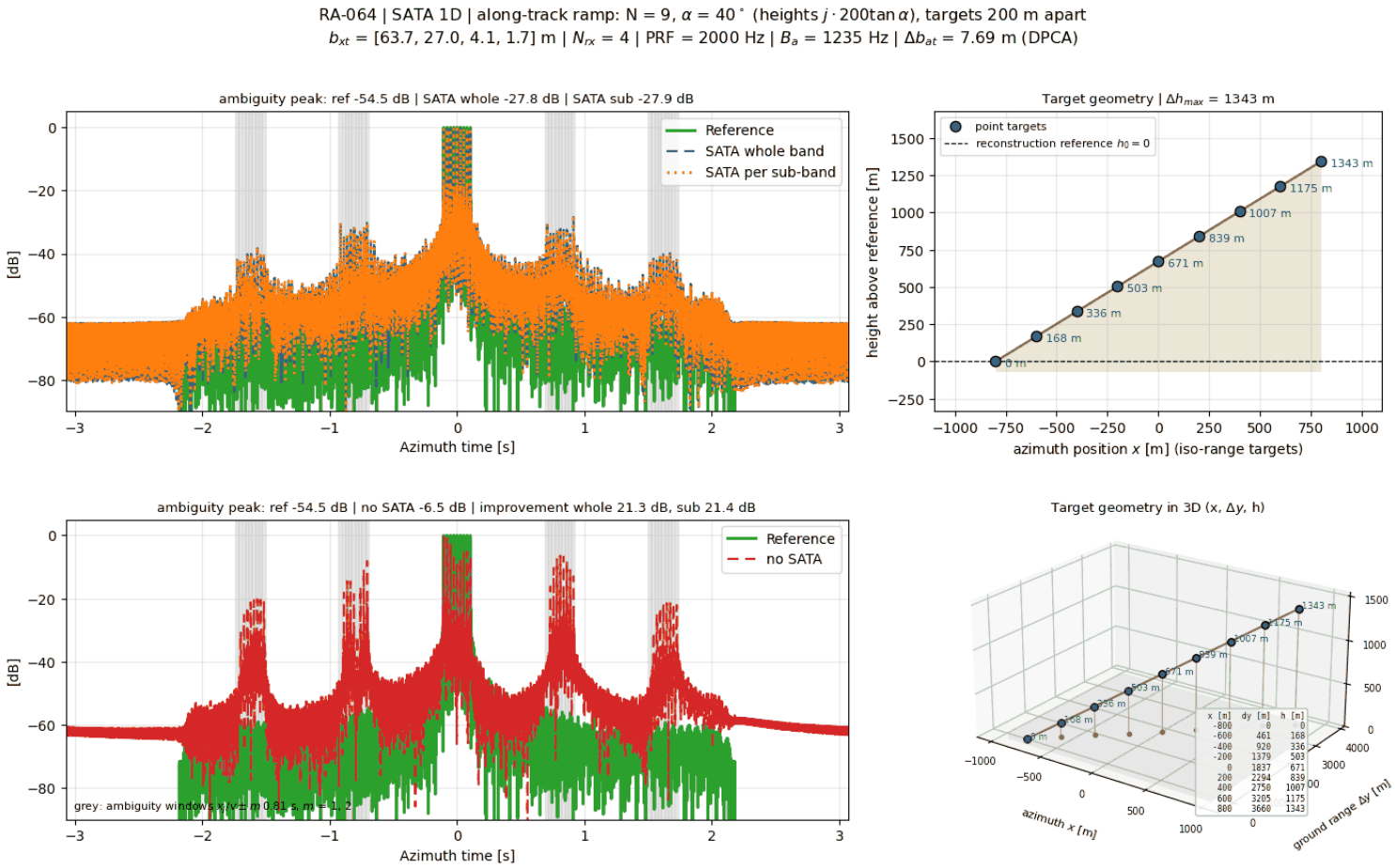


Figure 74: Along-track ramp, 200 m apart, normal heights: RA-064, $N = 9$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -6.5, SATA whole band -27.8, per sub-band -27.9 dB (improvement 21.3 dB).

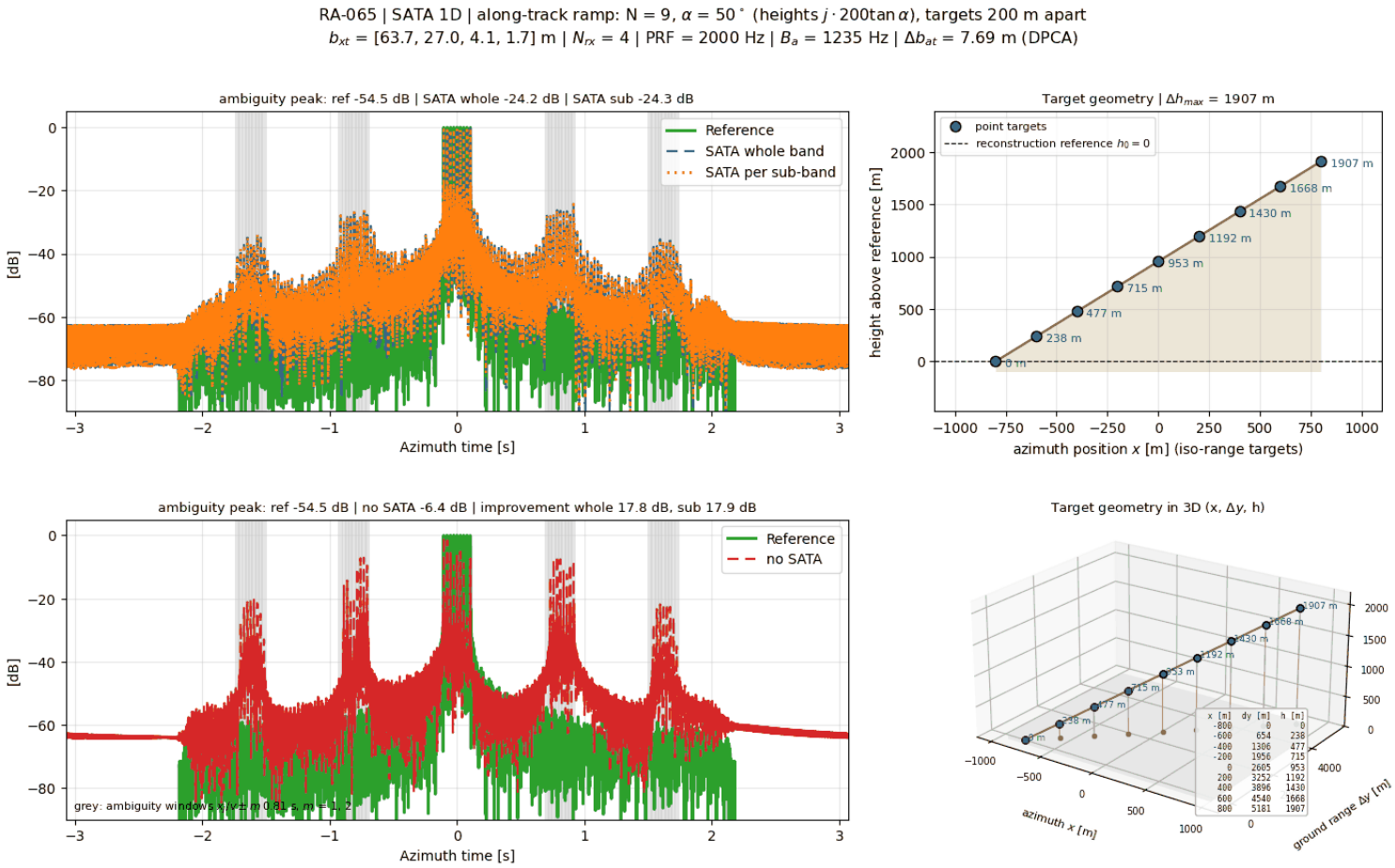


Figure 75: Along-track ramp, 200 m apart, normal heights: RA-065, $N = 9$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −24.2, per sub-band −24.3 dB (improvement 17.8 dB).

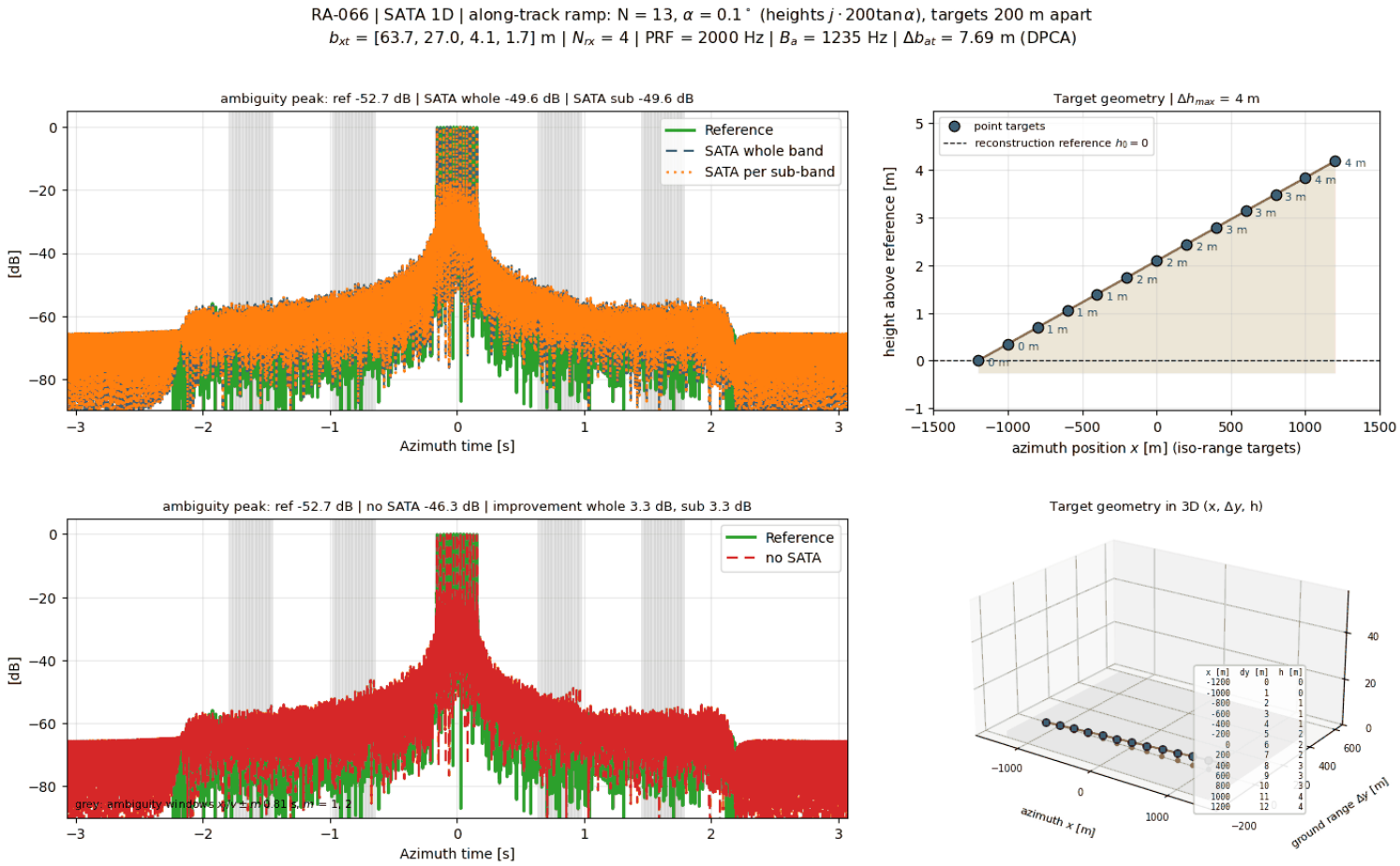


Figure 76: Along-track ramp, 200 m apart, normal heights: RA-066, $N = 13$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −46.3, SATA whole band −49.6, per sub-band −49.6 dB (improvement 3.3 dB).

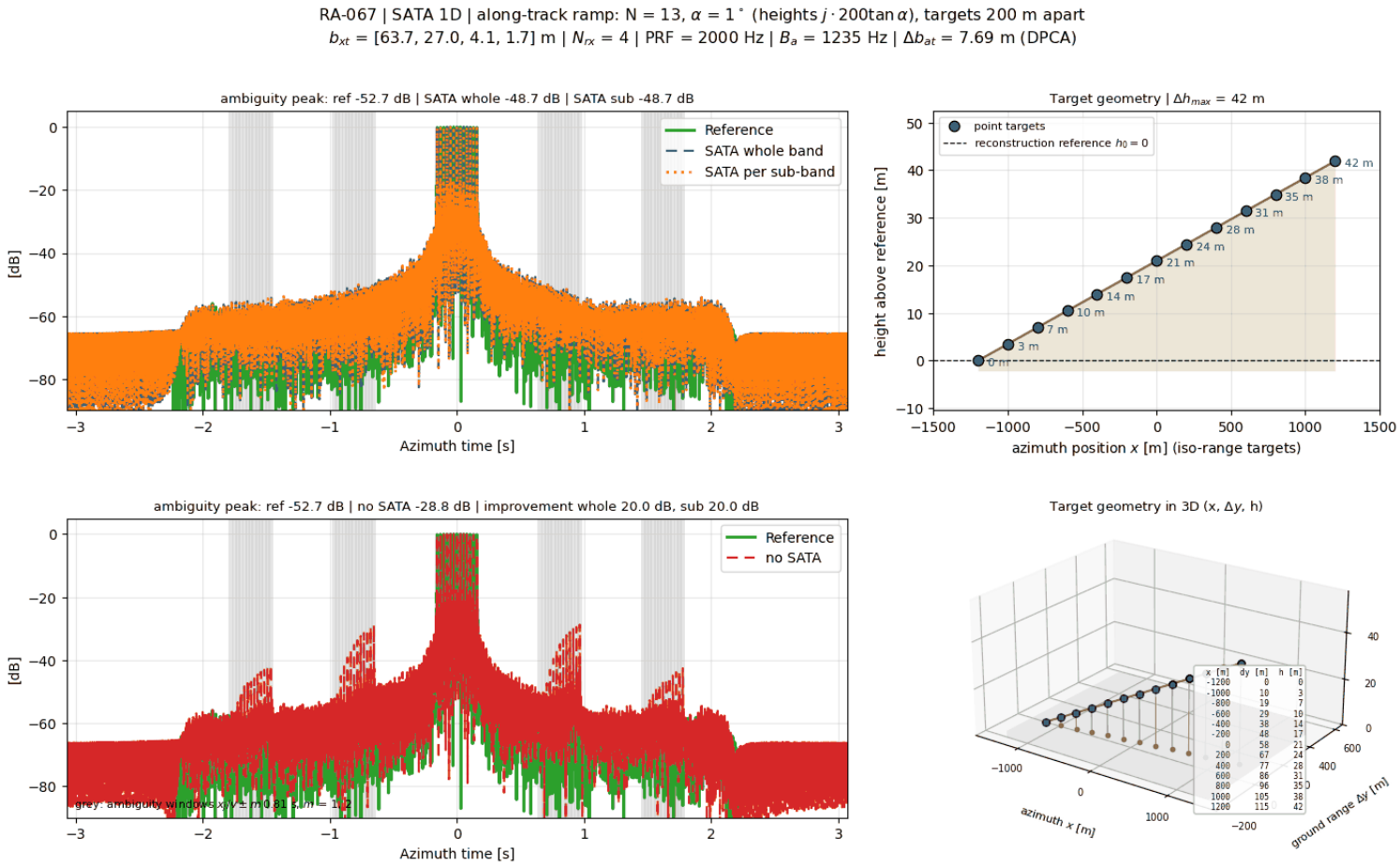


Figure 77: Along-track ramp, 200 m apart, normal heights: RA-067, $N = 13$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −28.8, SATA whole band −48.7, per sub-band −48.7 dB (improvement 20.0 dB).

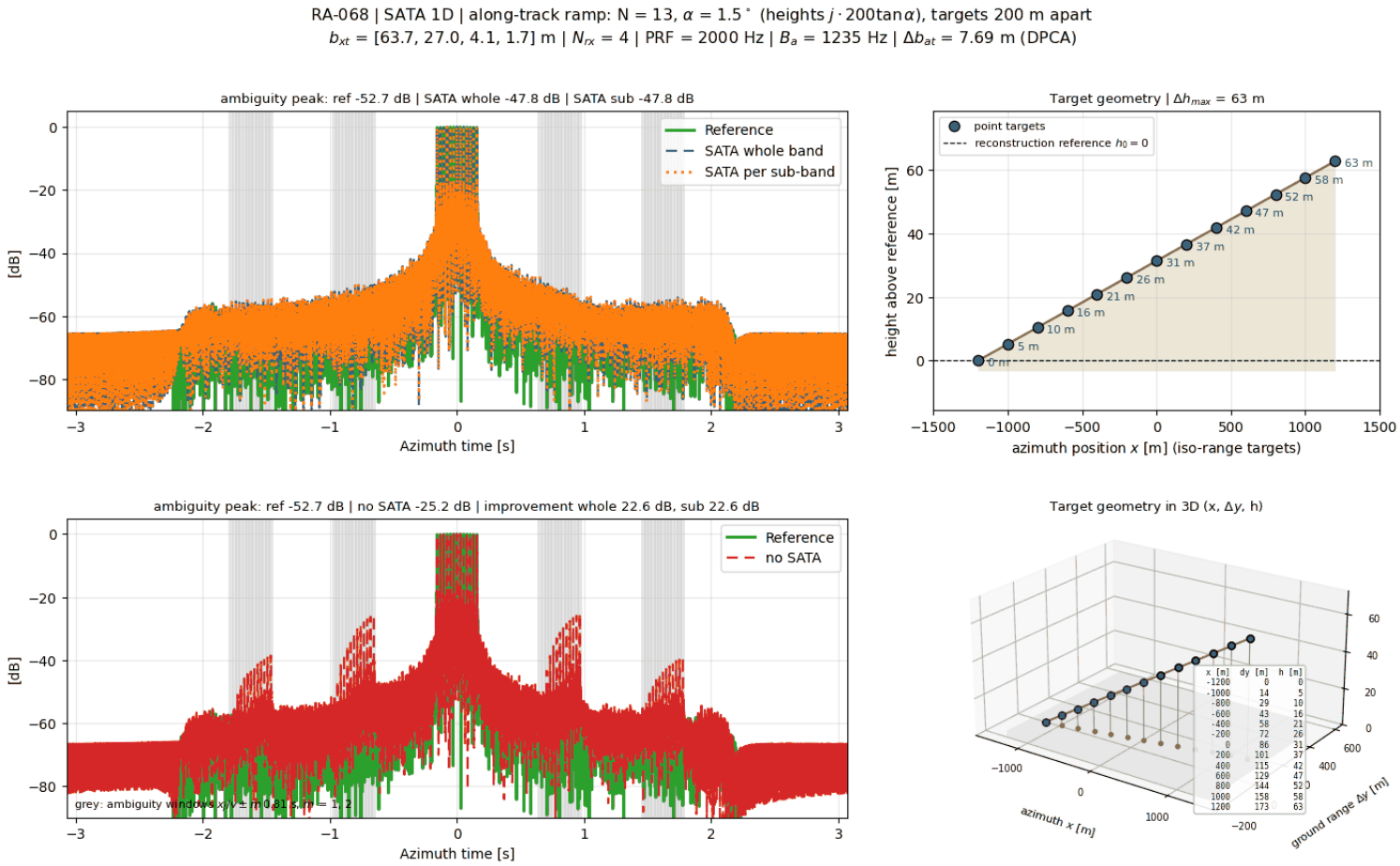


Figure 78: Along-track ramp, 200 m apart, normal heights: RA-068, $N = 13$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −25.2, SATA whole band −47.8, per sub-band −47.8 dB (improvement 22.6 dB).

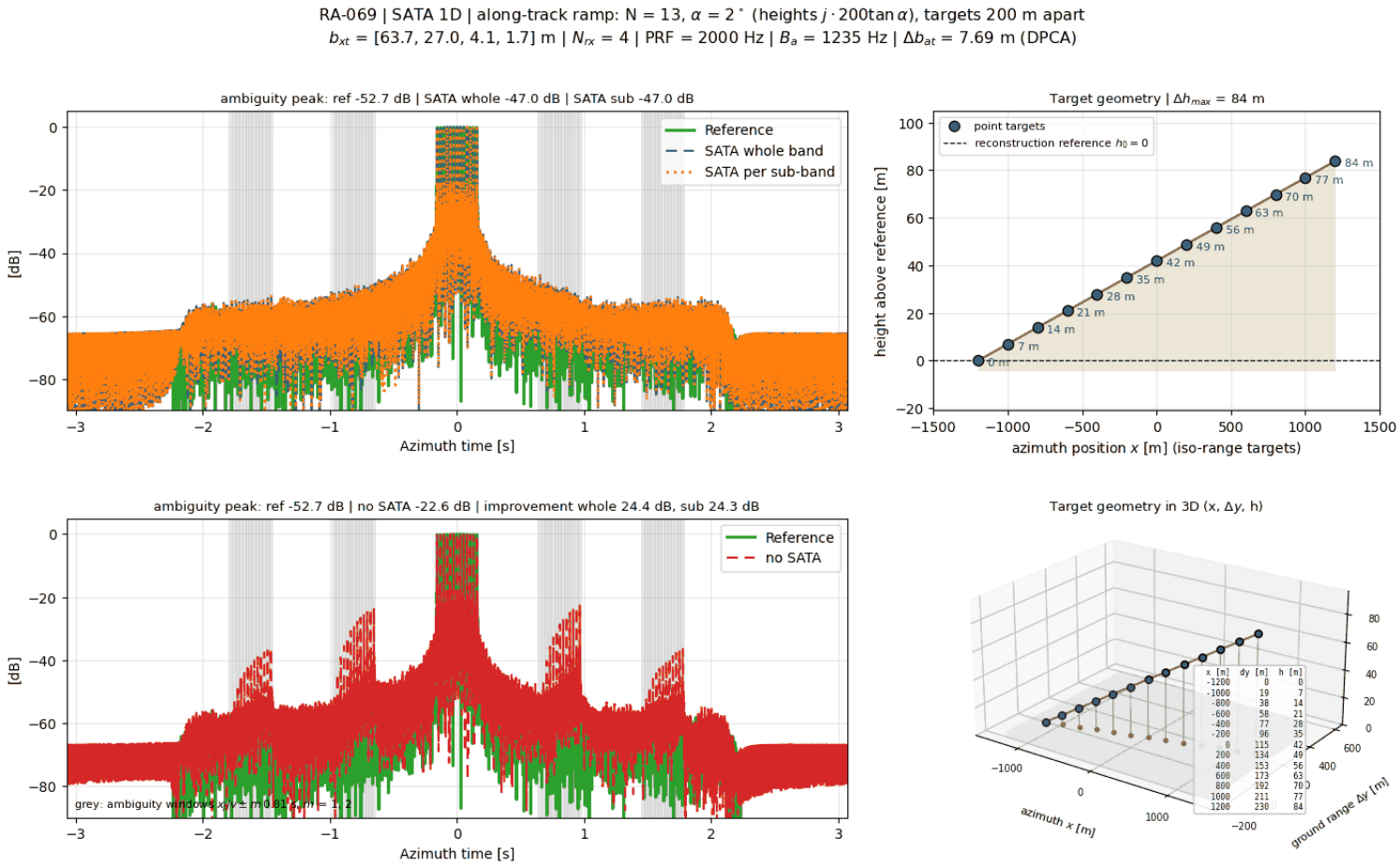


Figure 79: Along-track ramp, 200 m apart, normal heights: RA-069, $N = 13$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -22.6, SATA whole band -47.0, per sub-band -47.0 dB (improvement 24.4 dB).

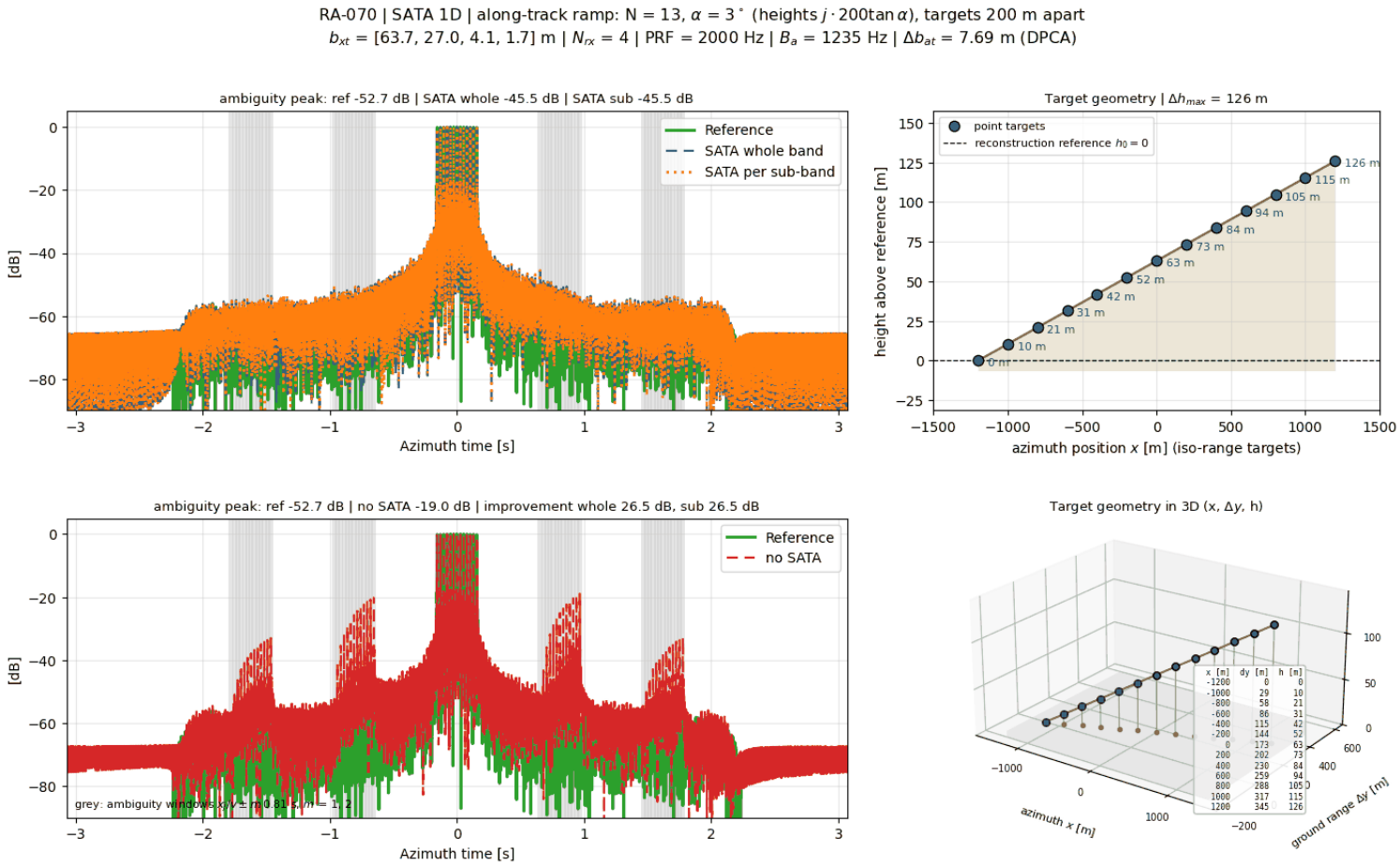


Figure 80: Along-track ramp, 200 m apart, normal heights: RA-070, $N = 13$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −19.0, SATA whole band −45.5, per sub-band −45.5 dB (improvement 26.5 dB).

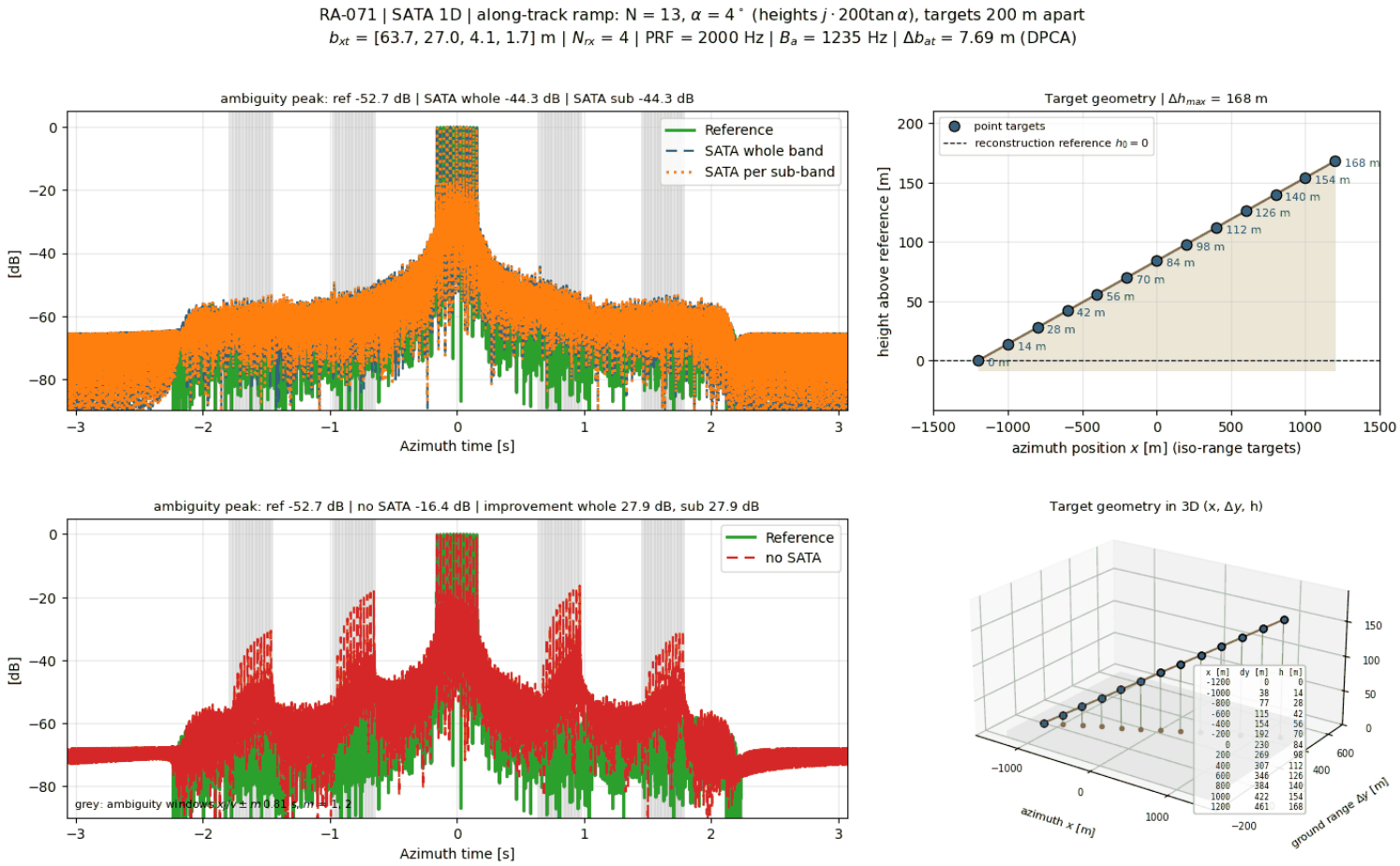


Figure 81: Along-track ramp, 200 m apart, normal heights: RA-071, $N = 13$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −16.4, SATA whole band −44.3, per sub-band −44.3 dB (improvement 27.9 dB).

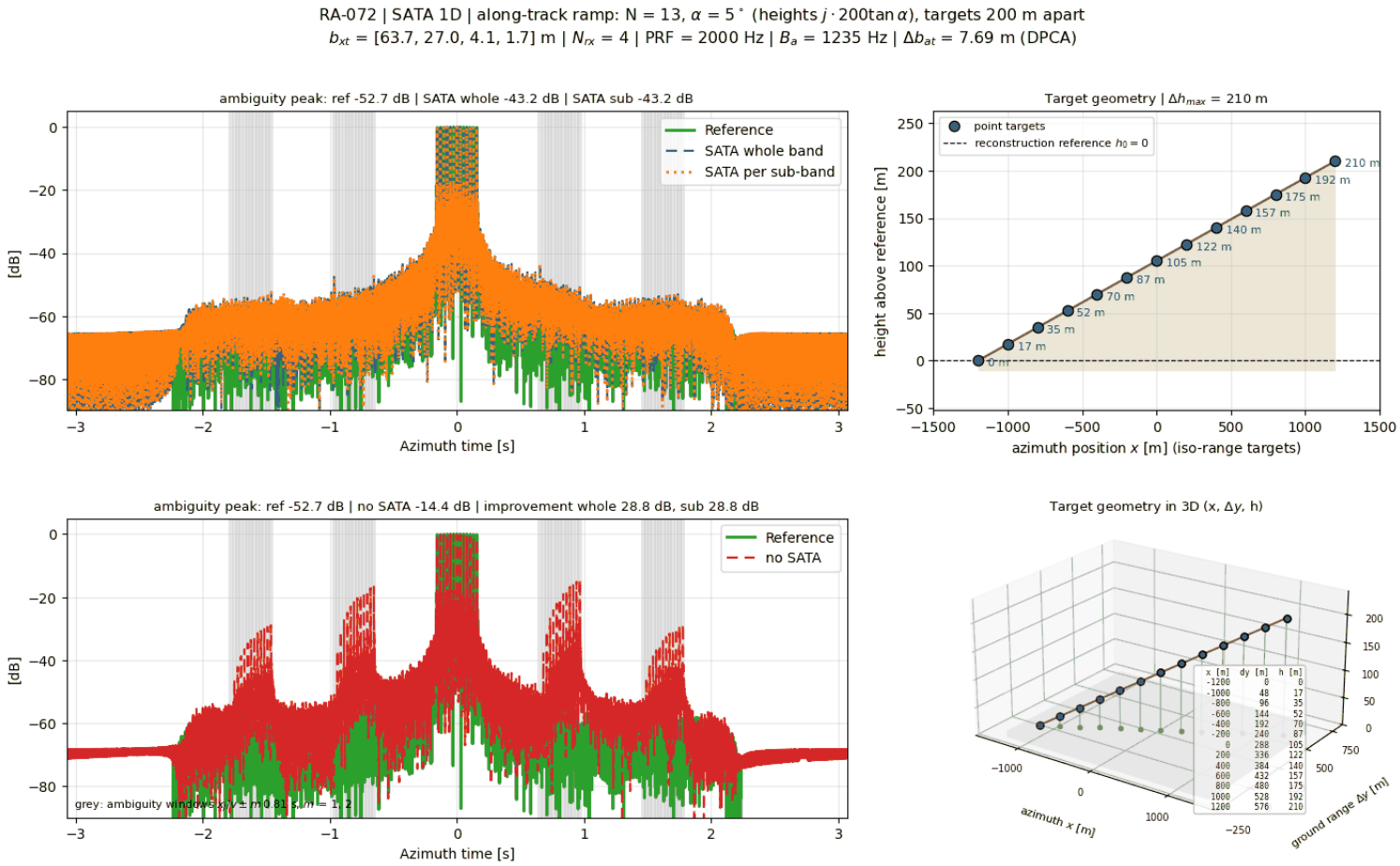


Figure 82: Along-track ramp, 200 m apart, normal heights: RA-072, $N = 13$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -14.4, SATA whole band -43.2, per sub-band -43.2 dB (improvement 28.8 dB).

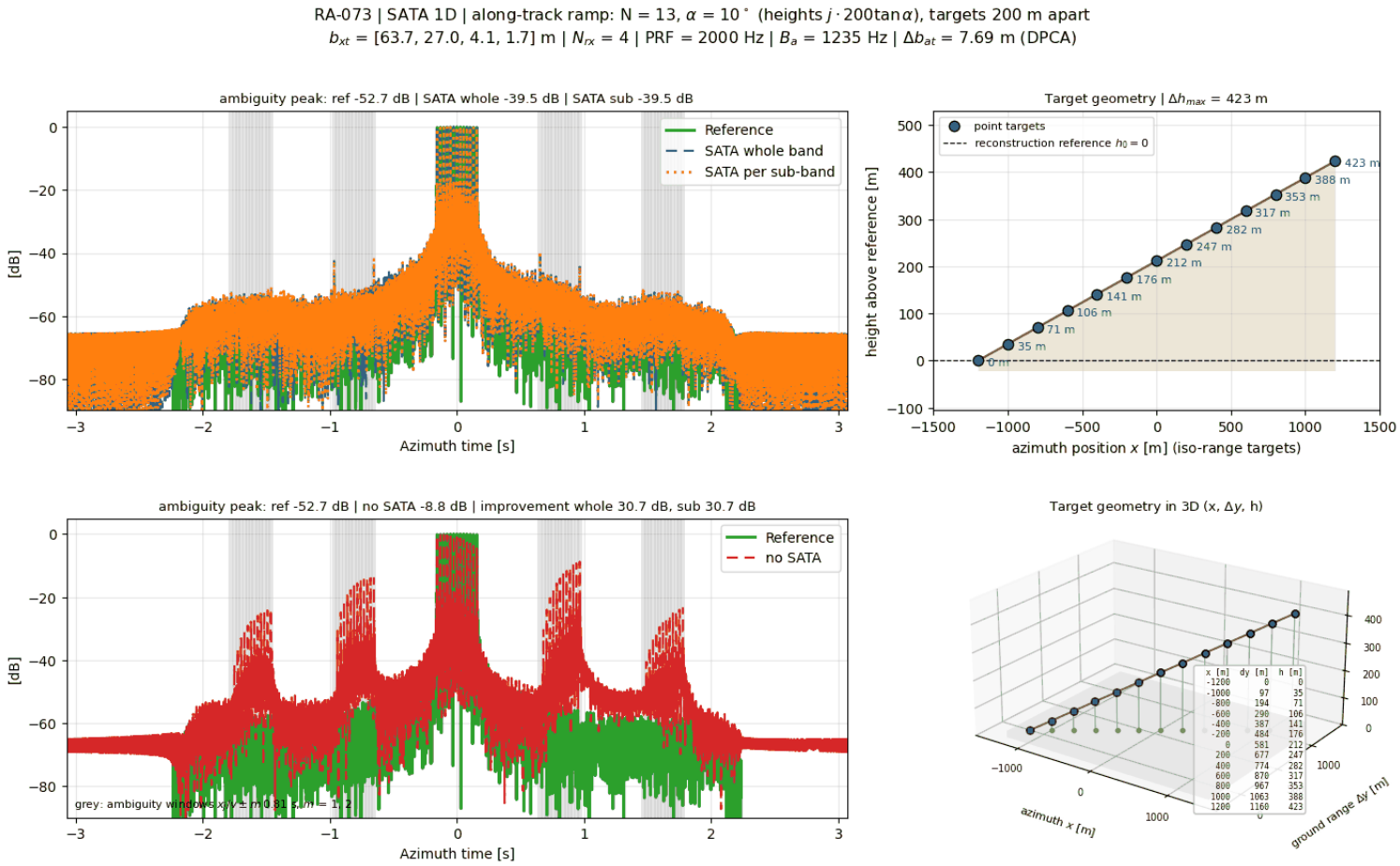


Figure 83: Along-track ramp, 200 m apart, normal heights: RA-073, $N = 13$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −8.8, SATA whole band −39.5, per sub-band −39.5 dB (improvement 30.7 dB).

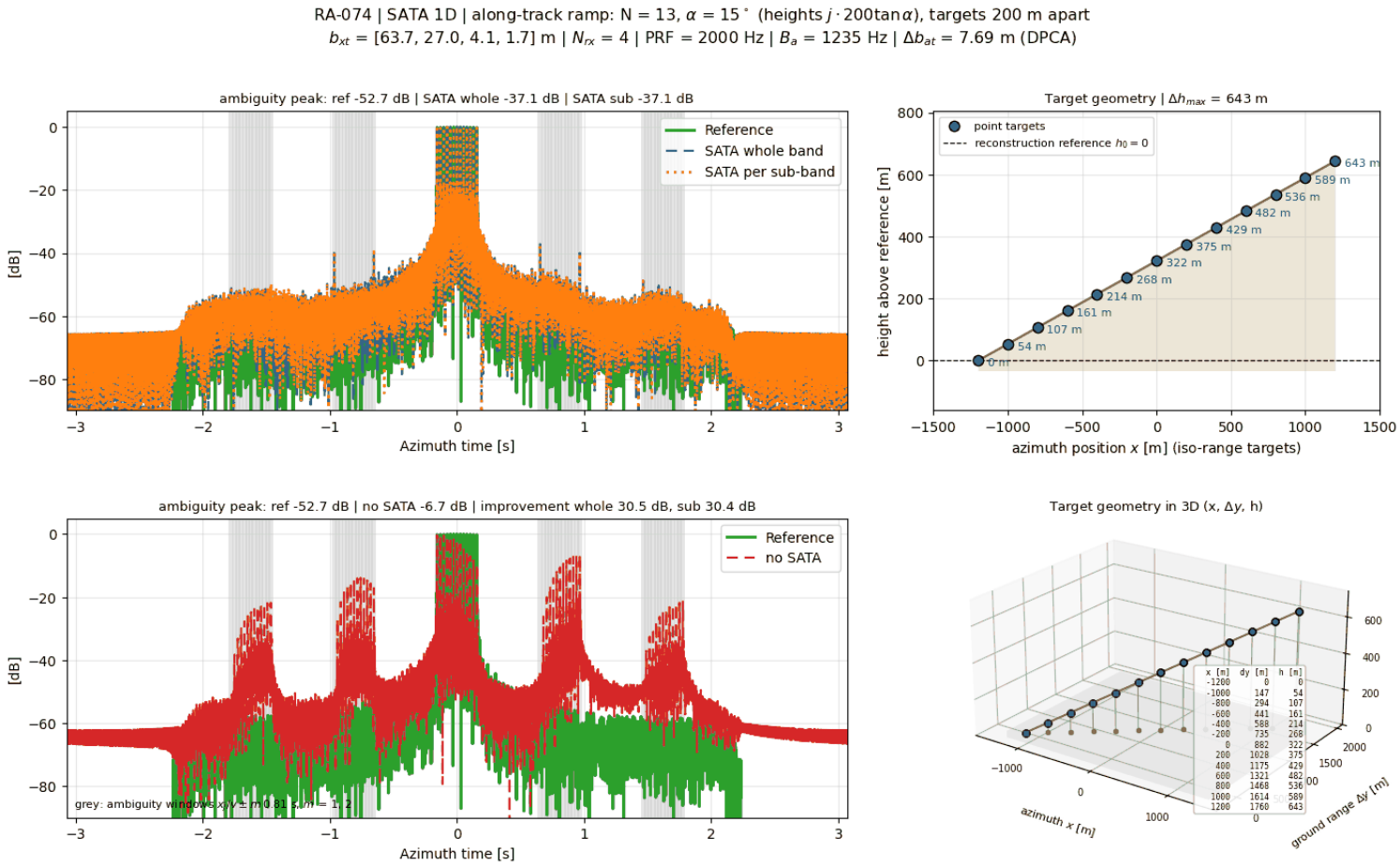


Figure 84: Along-track ramp, 200 m apart, normal heights: RA-074, $N = 13$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.7, SATA whole band −37.1, per sub-band −37.1 dB (improvement 30.5 dB).

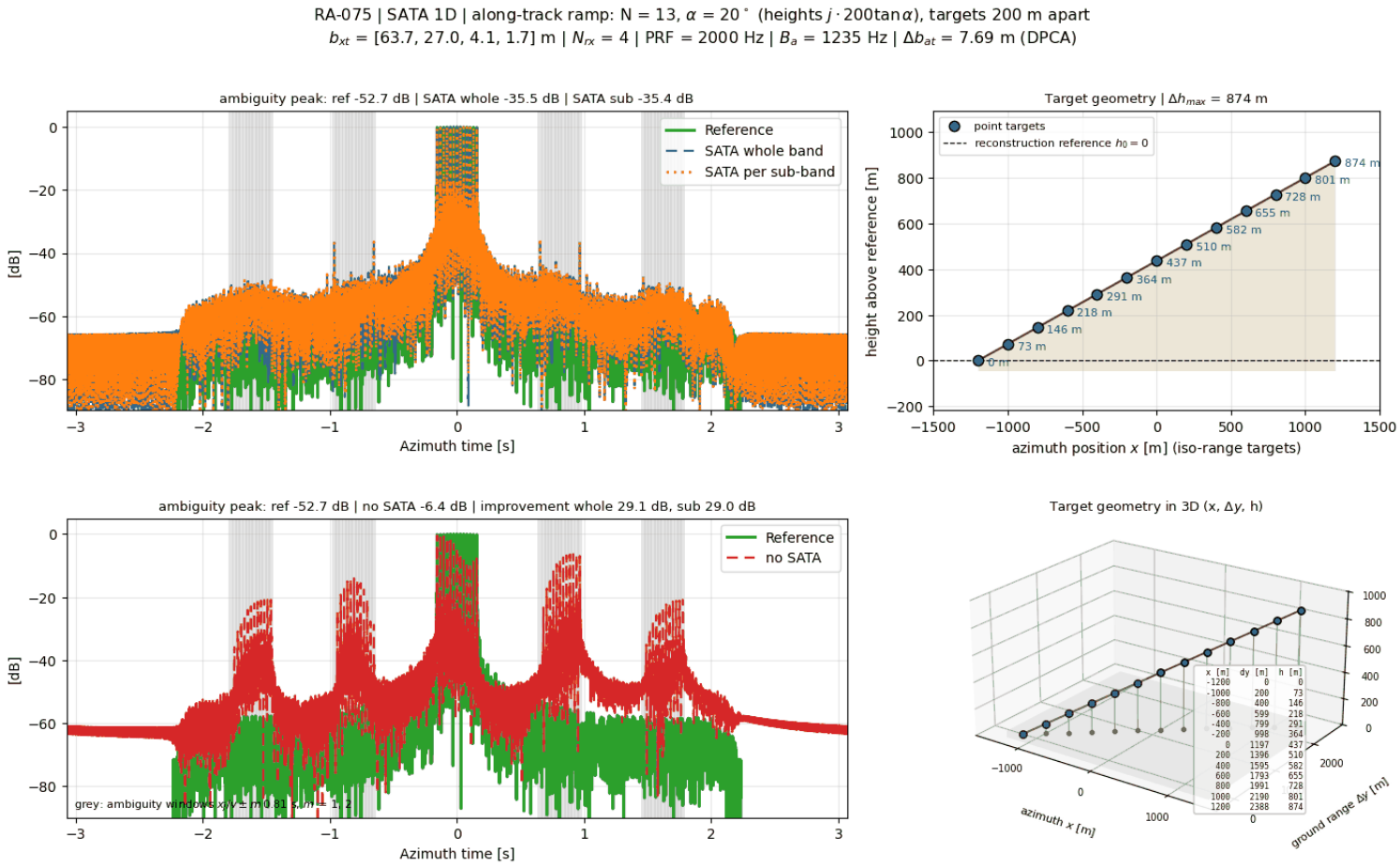


Figure 85: Along-track ramp, 200 m apart, normal heights: RA-075, $N = 13$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -6.4, SATA whole band -35.5, per sub-band -35.4 dB (improvement 29.1 dB).

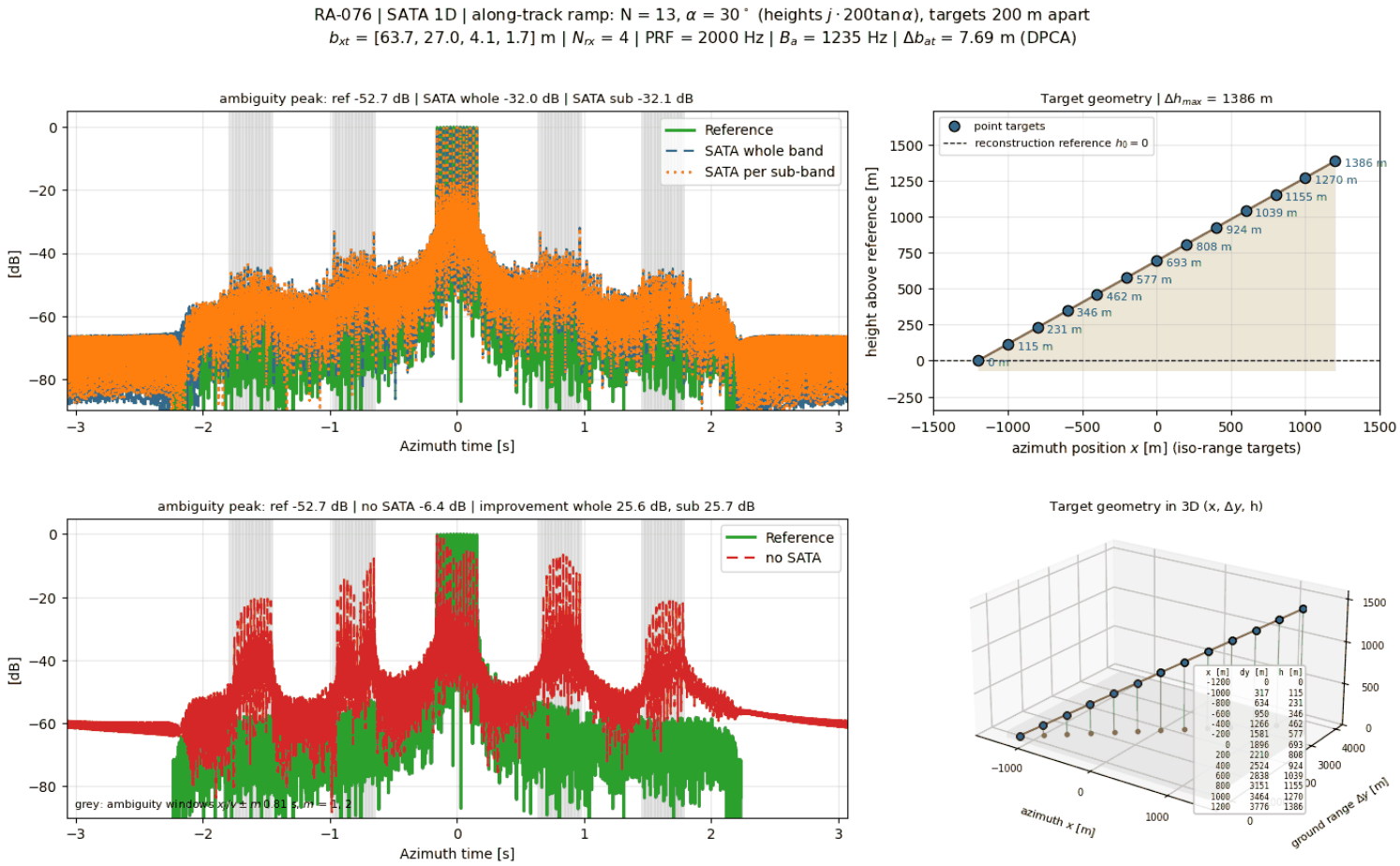


Figure 86: Along-track ramp, 200 m apart, normal heights: RA-076, $N = 13$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −32.0, per sub-band −32.1 dB (improvement 25.6 dB).

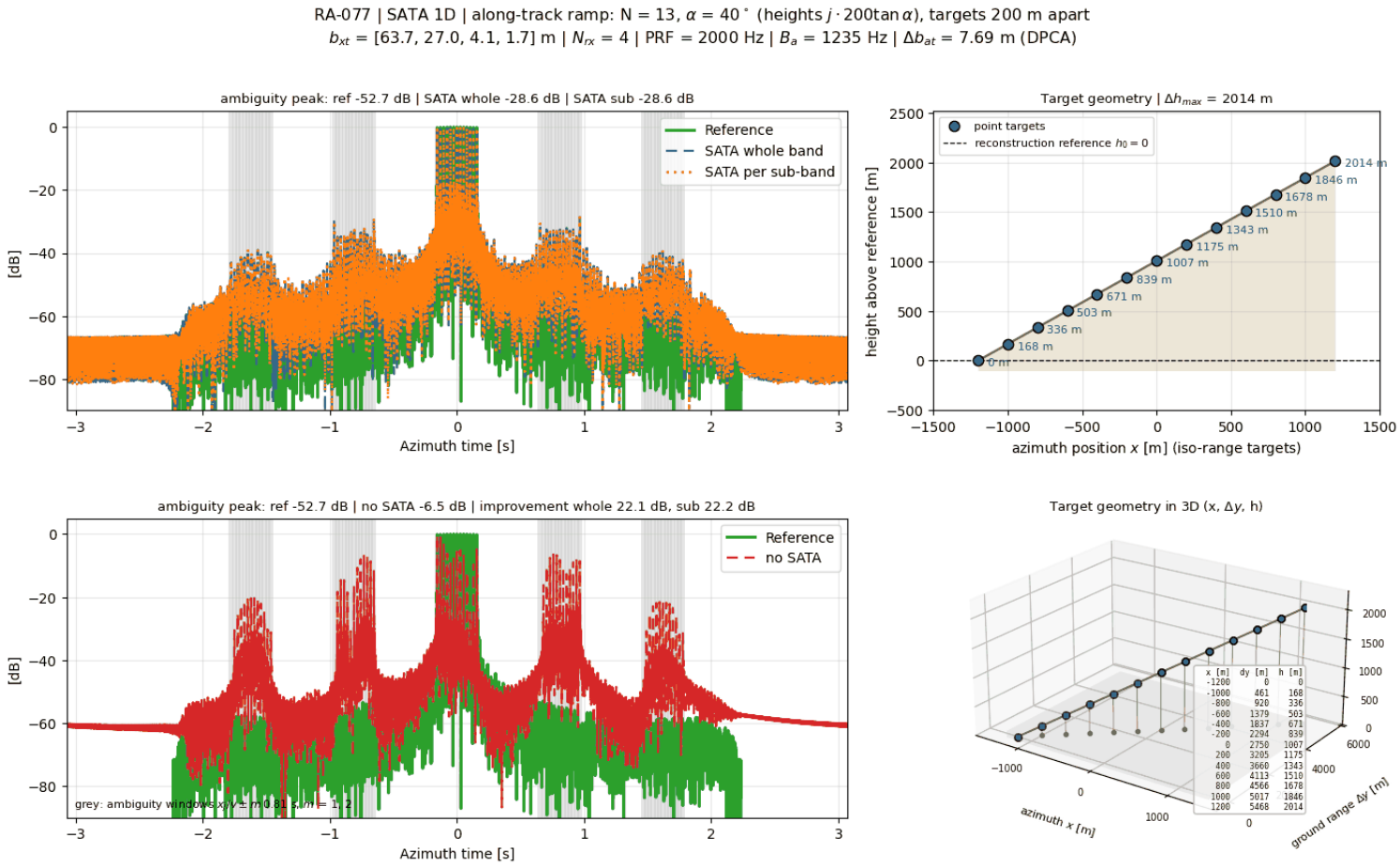


Figure 87: Along-track ramp, 200 m apart, normal heights: RA-077, $N = 13$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −28.6, per sub-band −28.6 dB (improvement 22.1 dB).

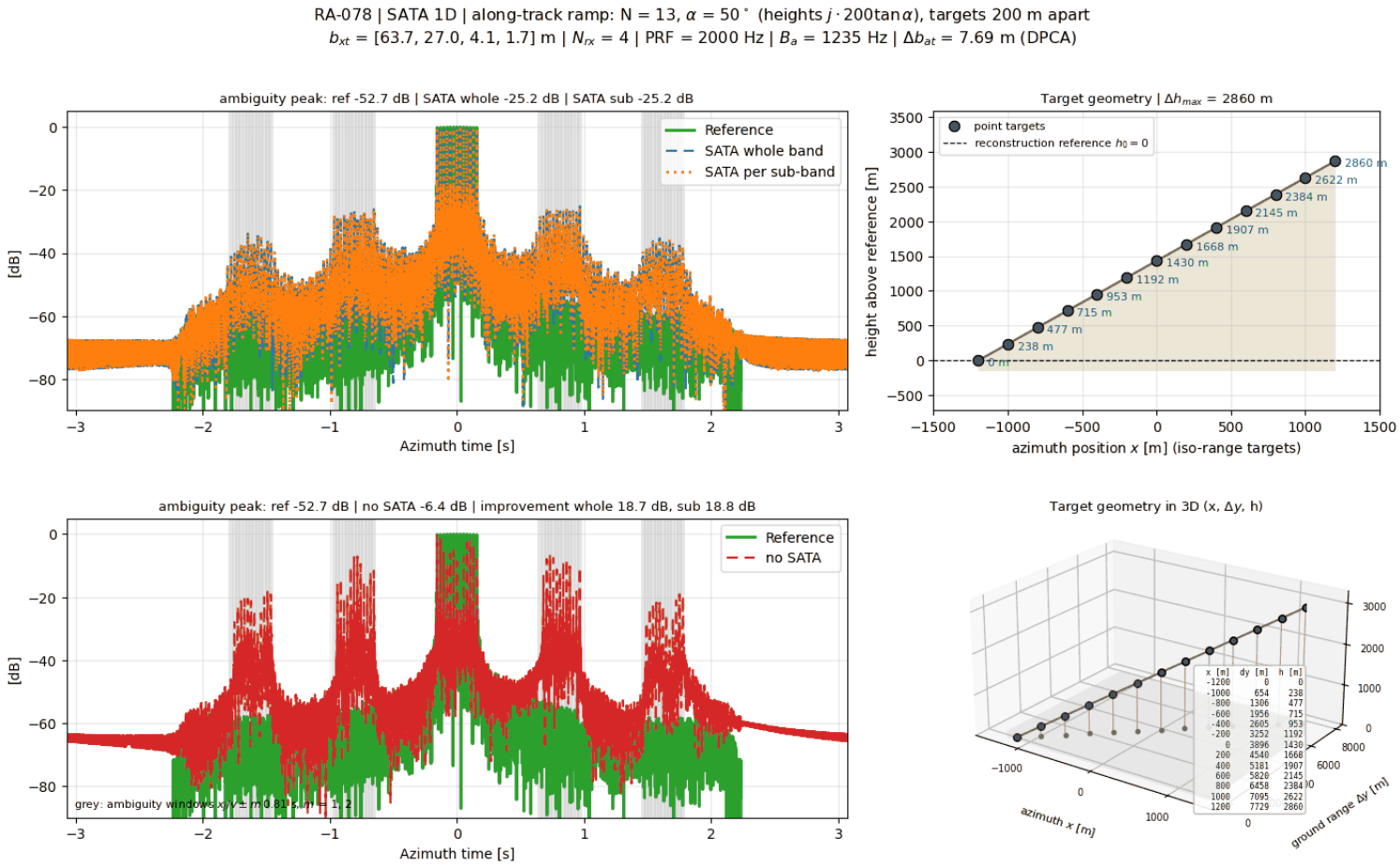


Figure 88: Along-track ramp, 200 m apart, normal heights: RA-078, $N = 13$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -6.4, SATA whole band -25.2, per sub-band -25.2 dB (improvement 18.7 dB).

5 IRFs, B: targets 2.6 km apart, normal heights

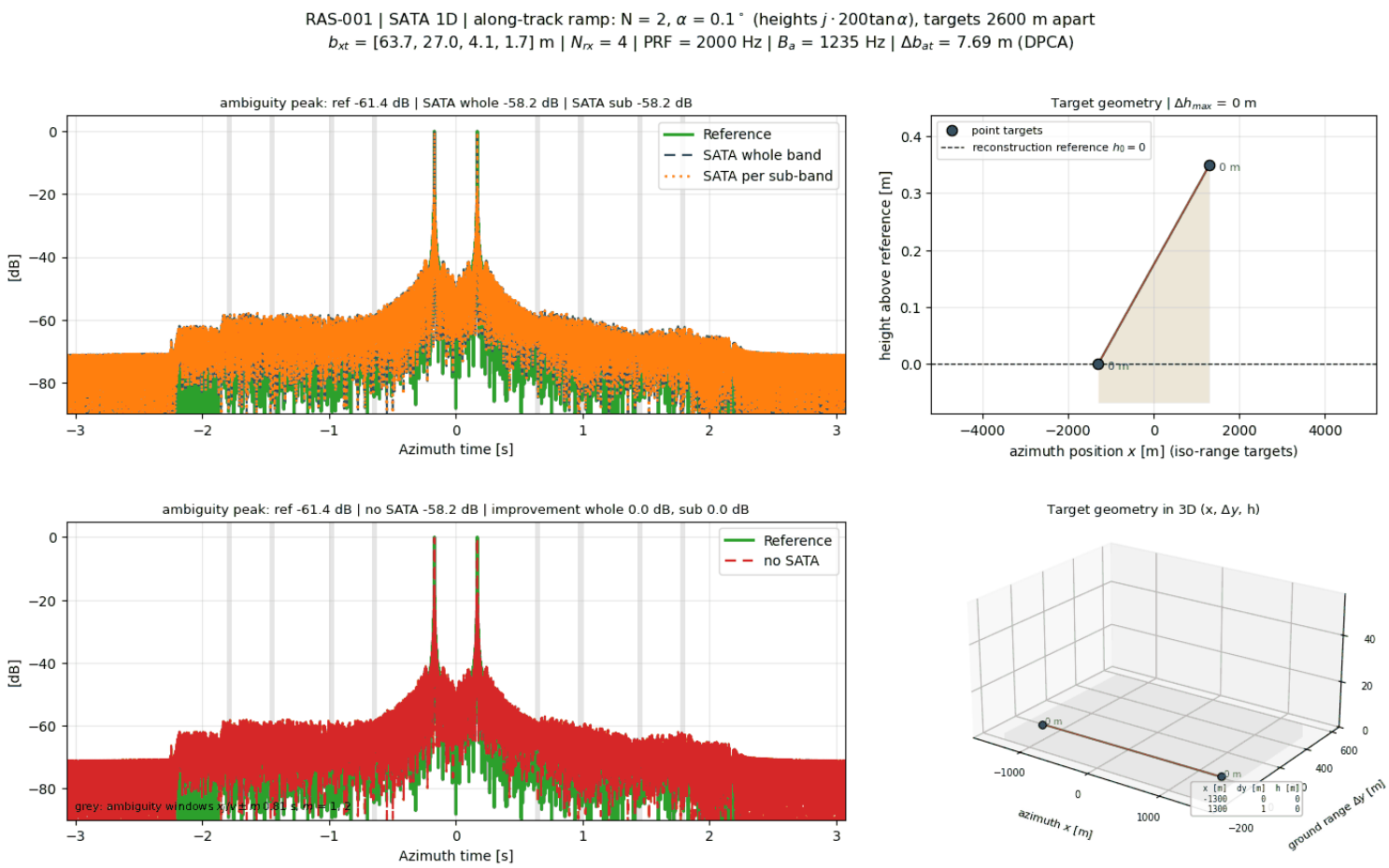


Figure 89: Along-track ramp, 2.6 km apart, normal heights: RAS-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −58.2, SATA whole band −58.2, per sub-band −58.2 dB (improvement 0.0 dB).

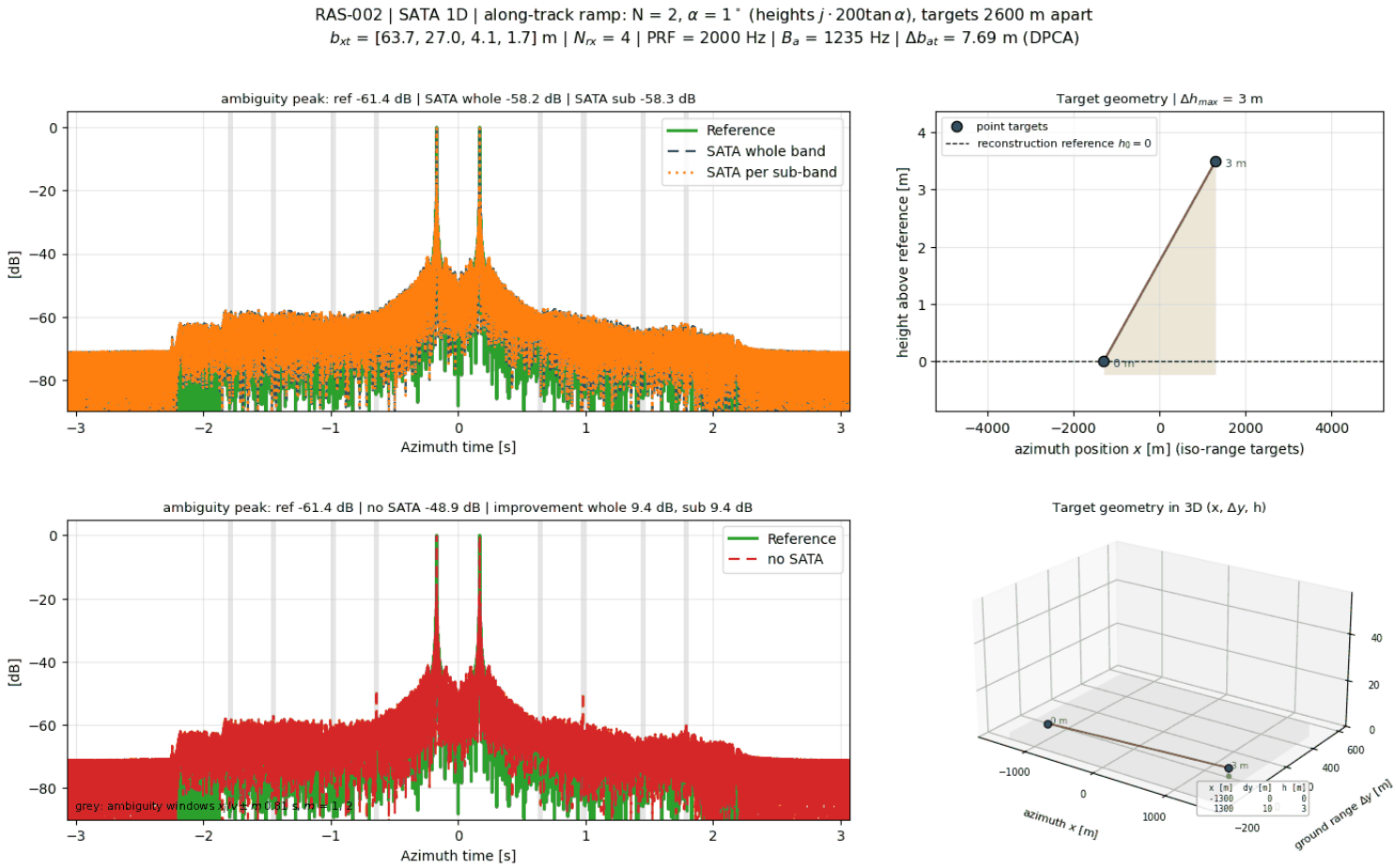


Figure 90: Along-track ramp, 2.6 km apart, normal heights: RAS-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −48.9, SATA whole band −58.2, per sub-band −58.3 dB (improvement 9.4 dB).

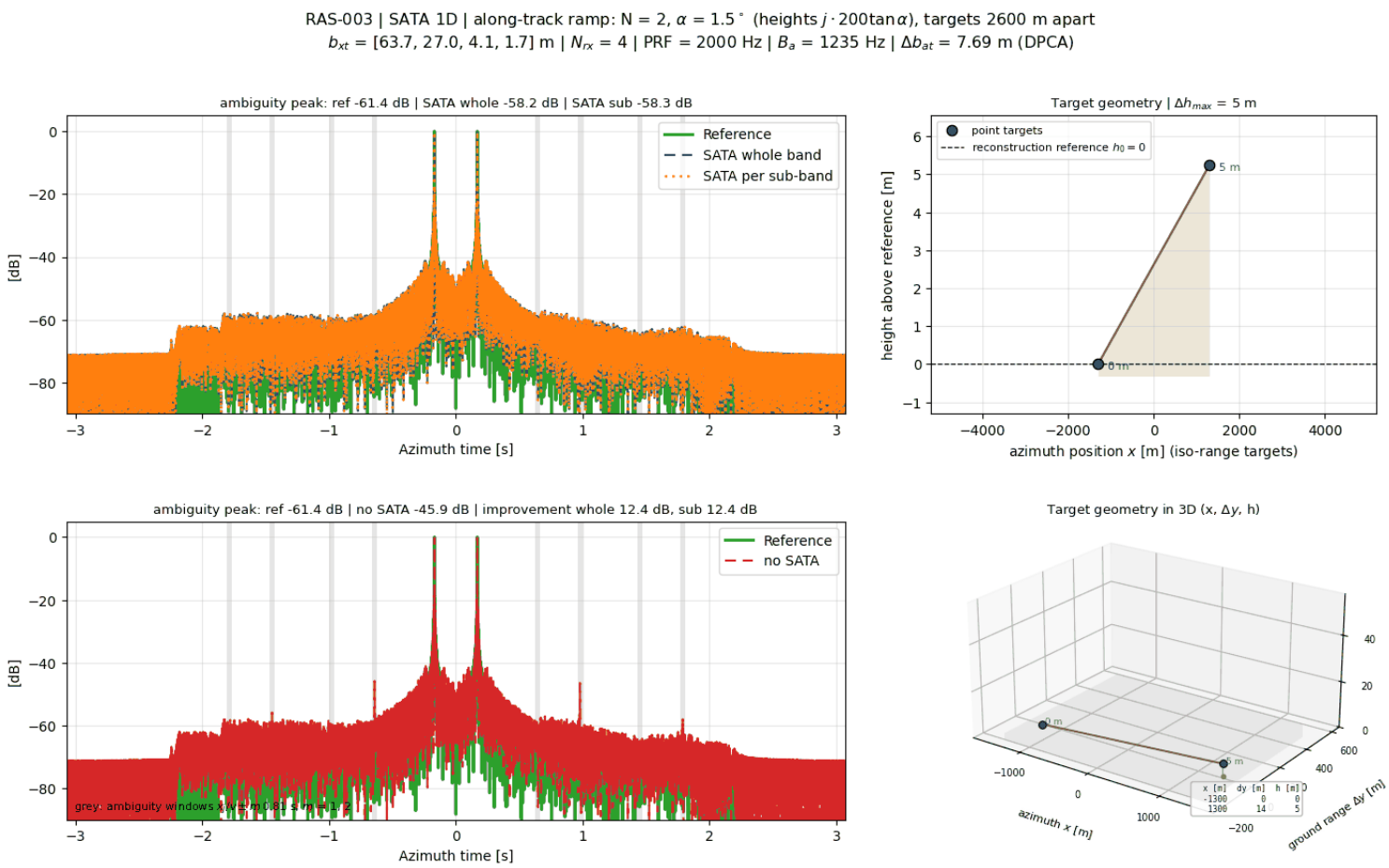


Figure 91: Along-track ramp, 2.6 km apart, normal heights: RAS-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −45.9, SATA whole band −58.2, per sub-band −58.3 dB (improvement 12.4 dB).

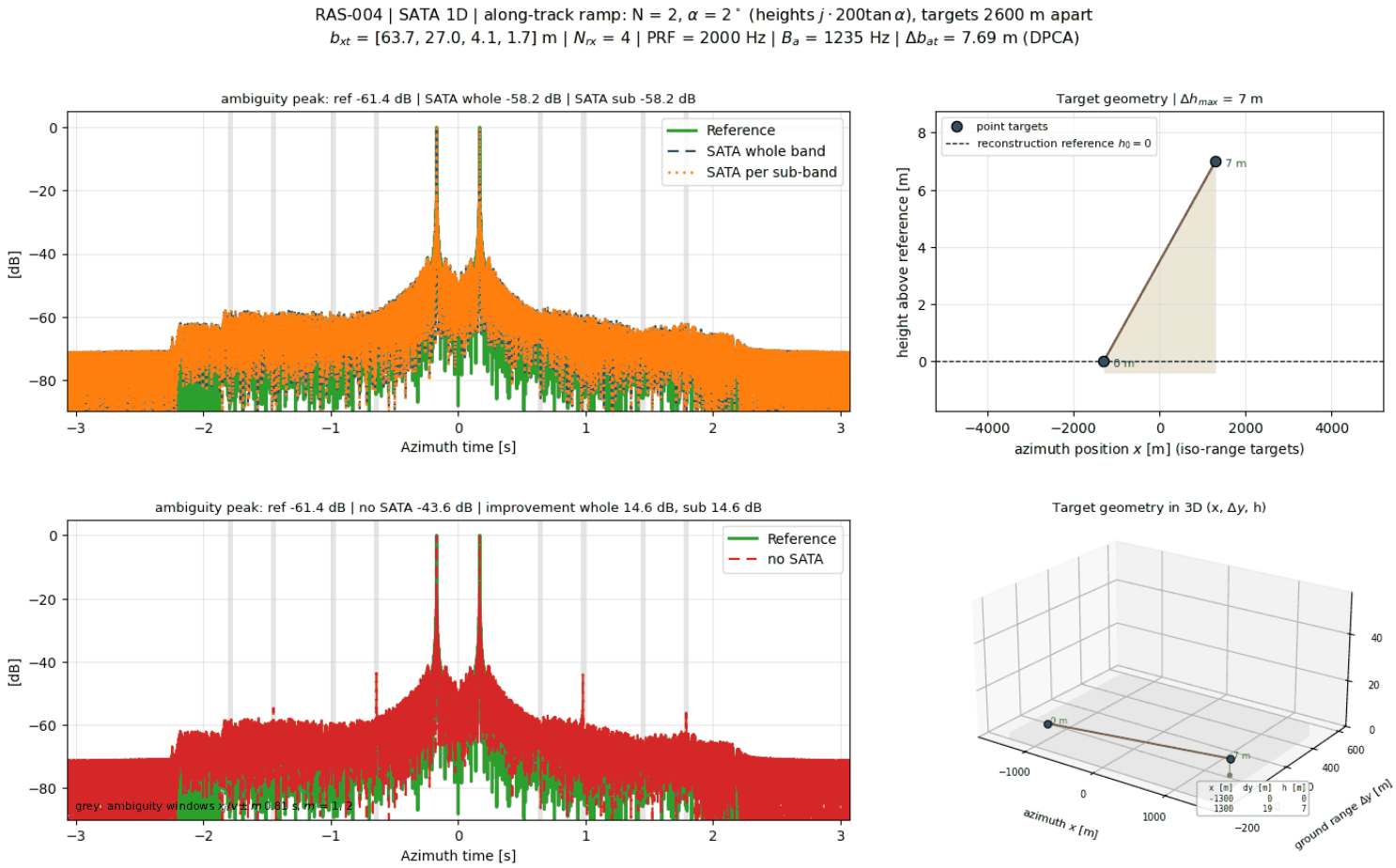


Figure 92: Along-track ramp, 2.6 km apart, normal heights: RAS-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −43.6, SATA whole band −58.2, per sub-band −58.2 dB (improvement 14.6 dB).

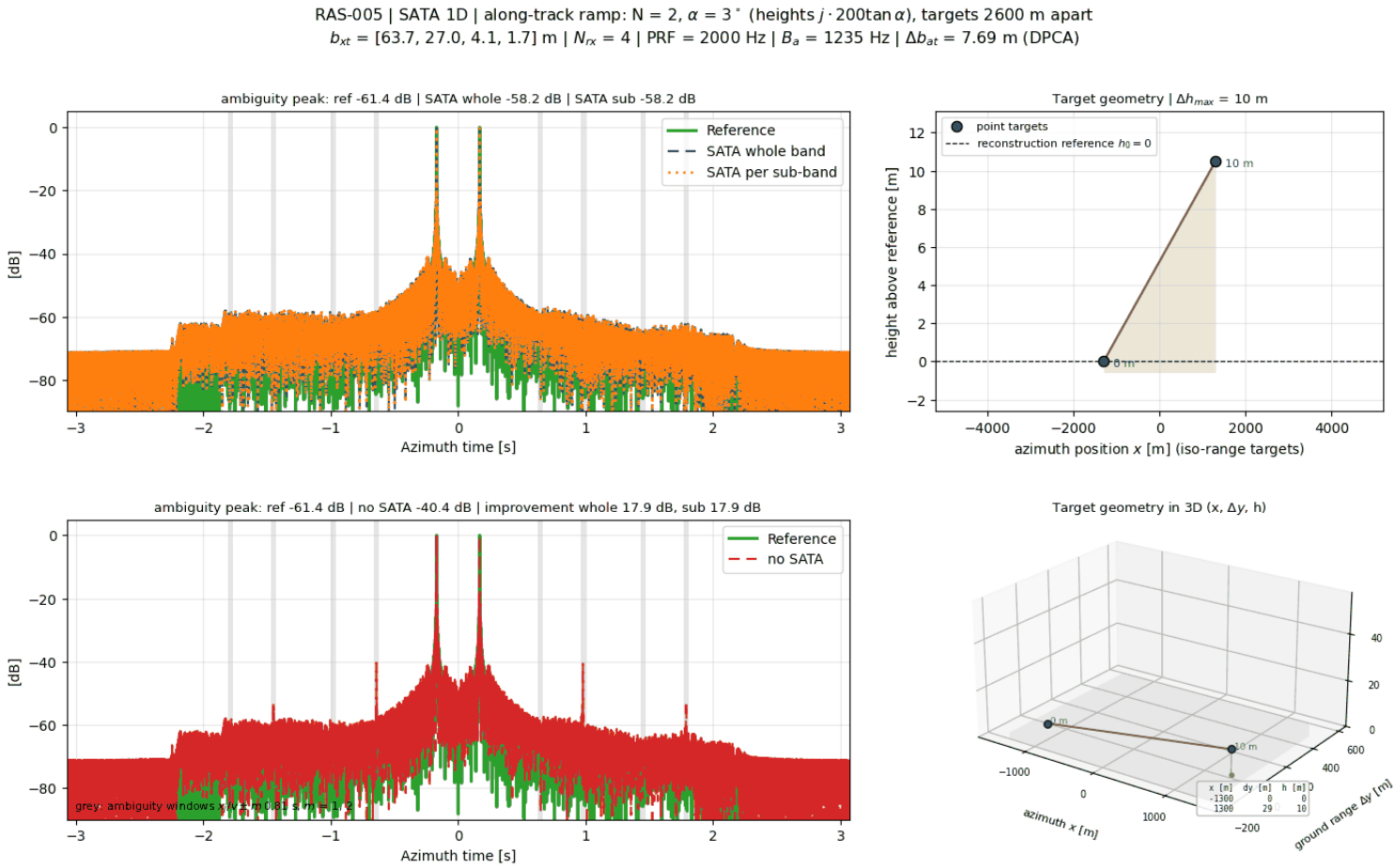


Figure 93: Along-track ramp, 2.6 km apart, normal heights: RAS-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −40.4, SATA whole band −58.2, per sub-band −58.2 dB (improvement 17.9 dB).

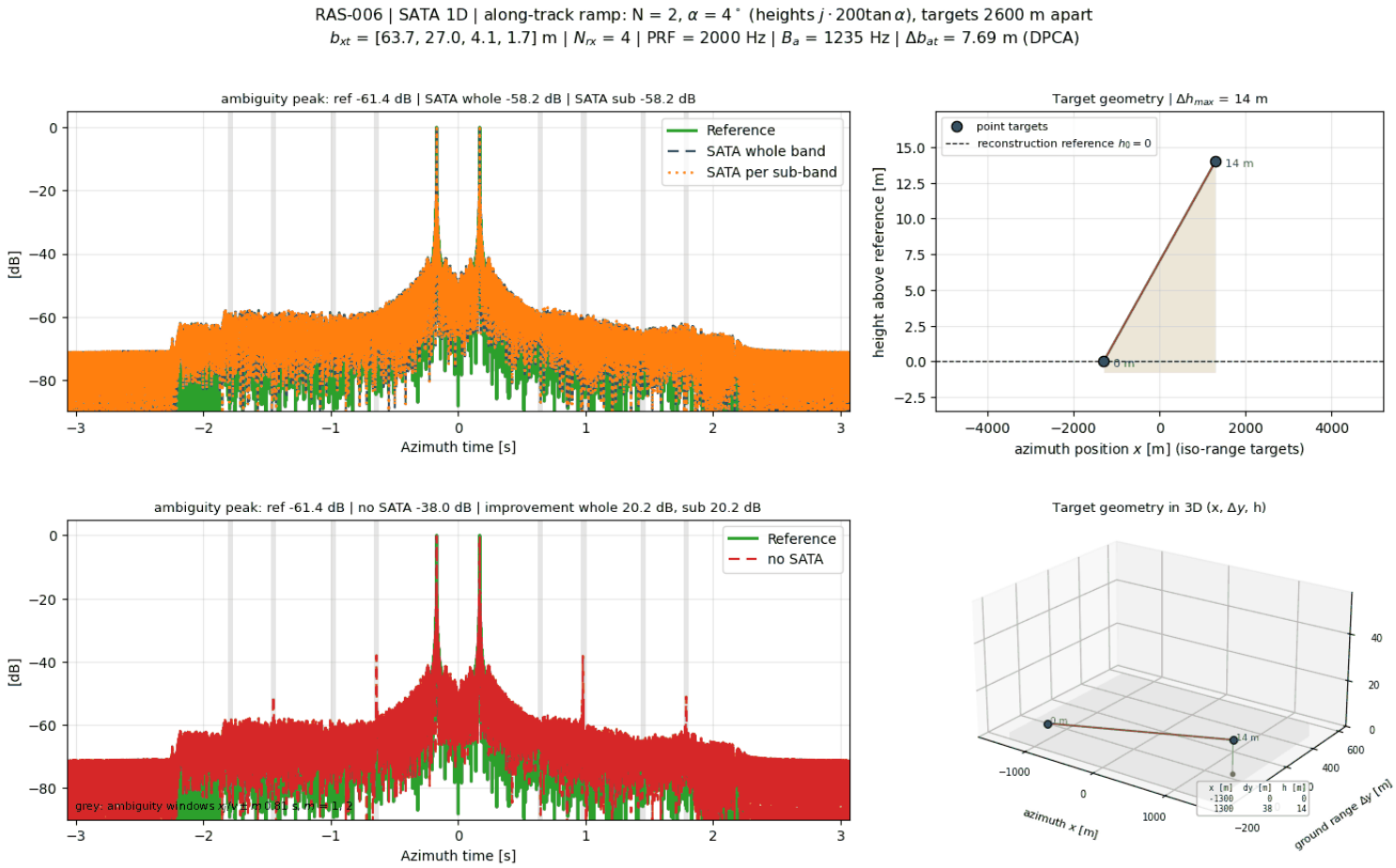


Figure 94: Along-track ramp, 2.6 km apart, normal heights: RAS-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −38.0, SATA whole band −58.2, per sub-band −58.2 dB (improvement 20.2 dB).

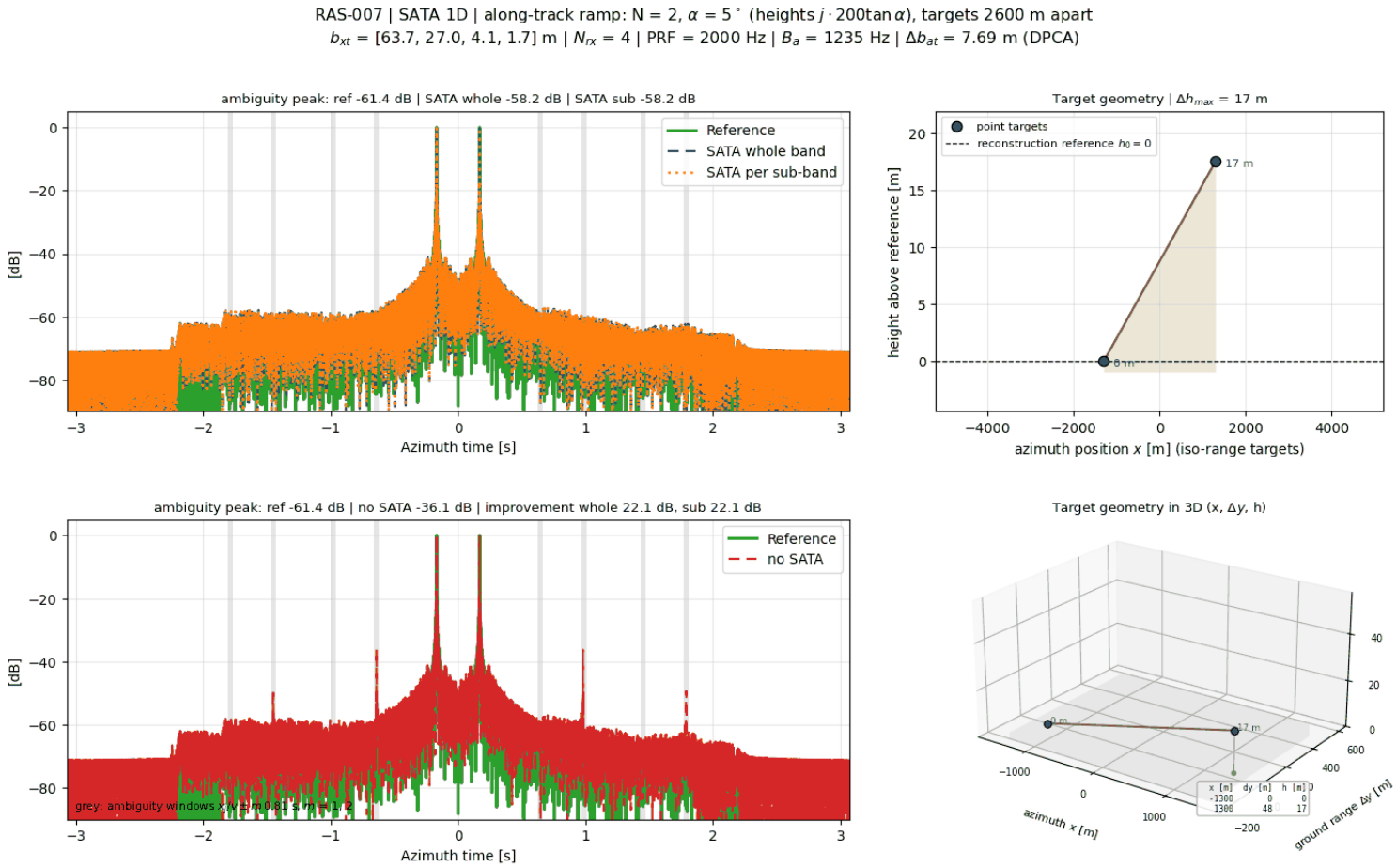


Figure 95: Along-track ramp, 2.6 km apart, normal heights: RAS-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −36.1, SATA whole band −58.2, per sub-band −58.2 dB (improvement 22.1 dB).

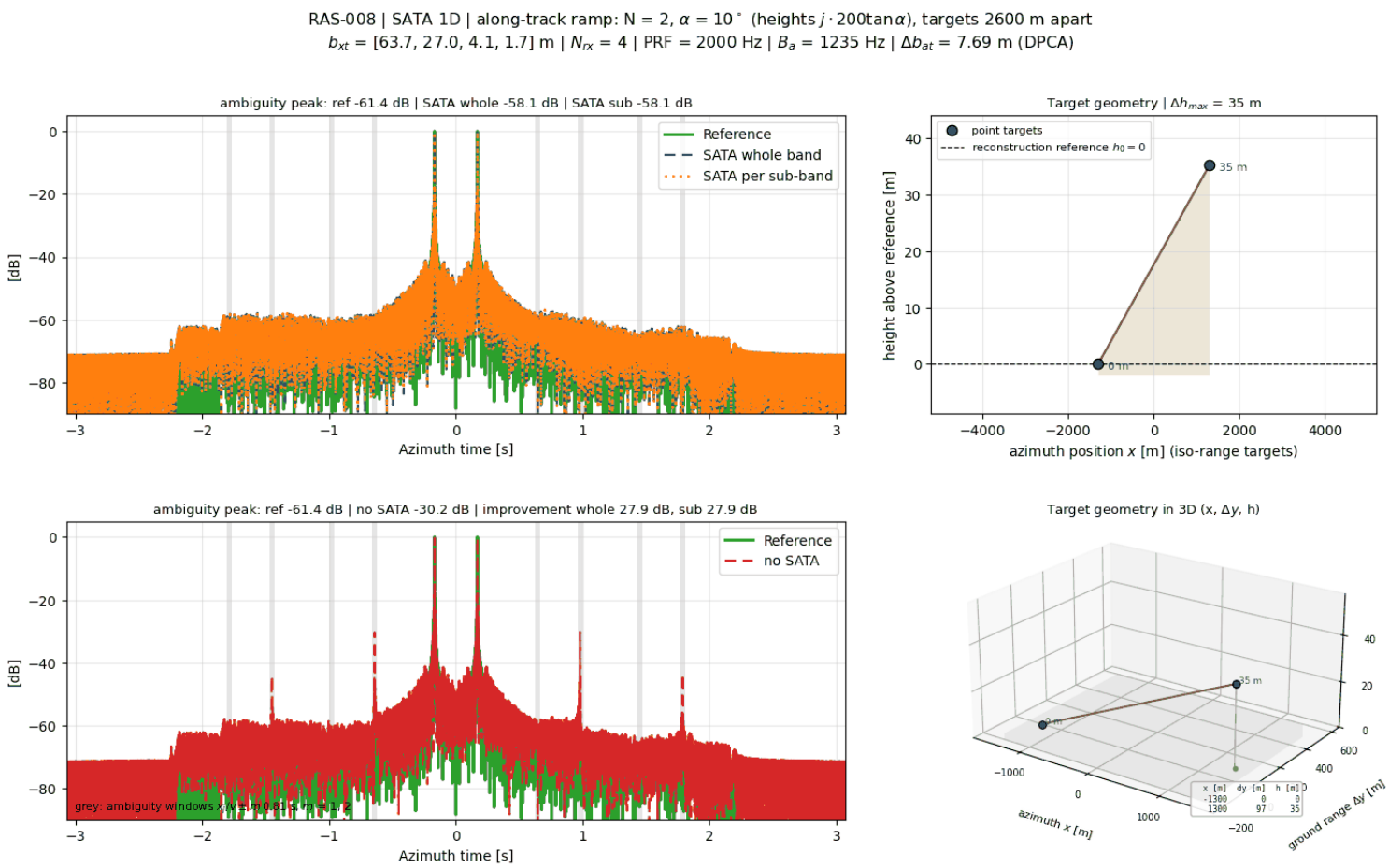


Figure 96: Along-track ramp, 2.6 km apart, normal heights: RAS-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −30.2, SATA whole band −58.1, per sub-band −58.1 dB (improvement 27.9 dB).

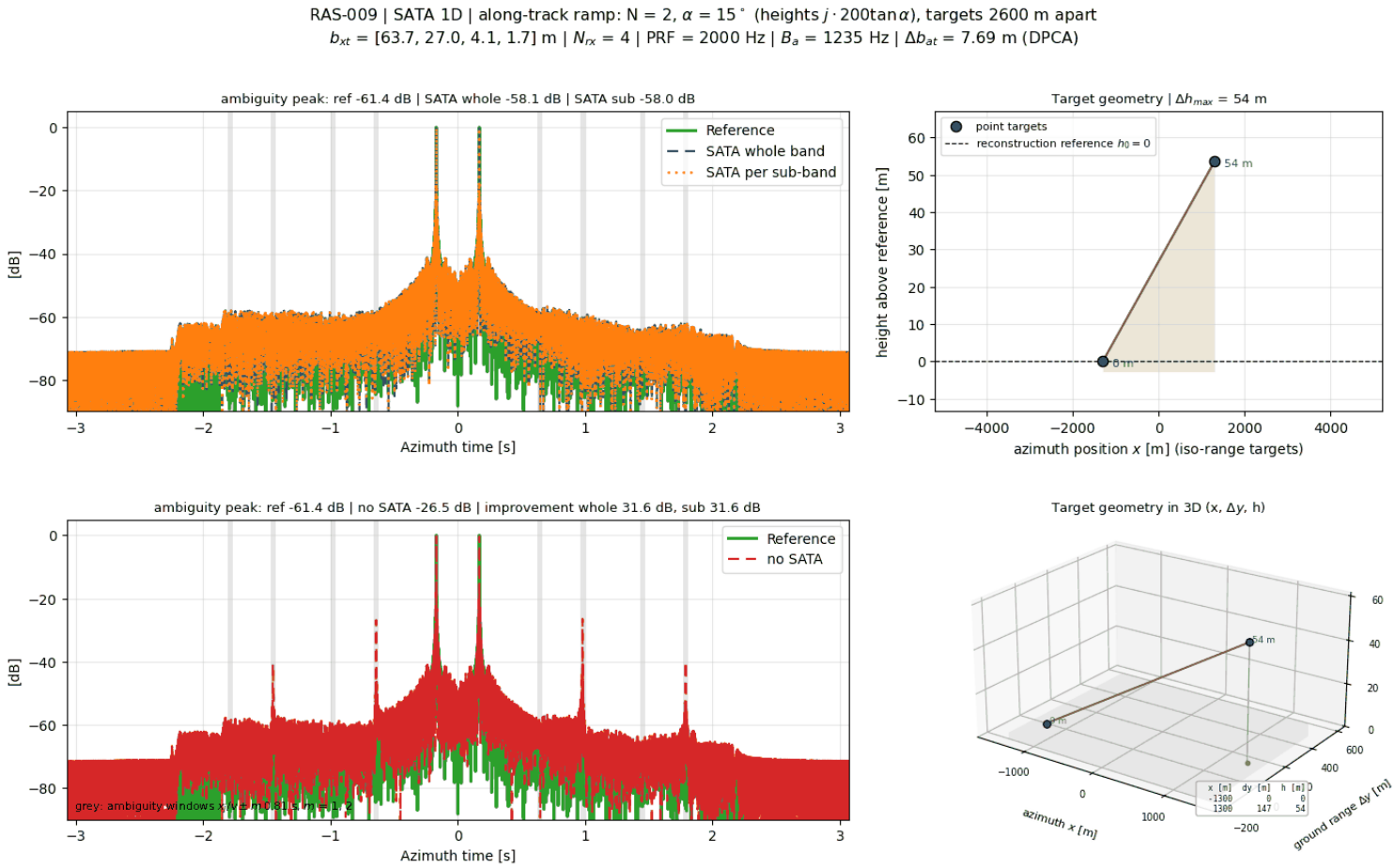


Figure 97: Along-track ramp, 2.6 km apart, normal heights: RAS-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −26.5, SATA whole band −58.1, per sub-band −58.0 dB (improvement 31.6 dB).

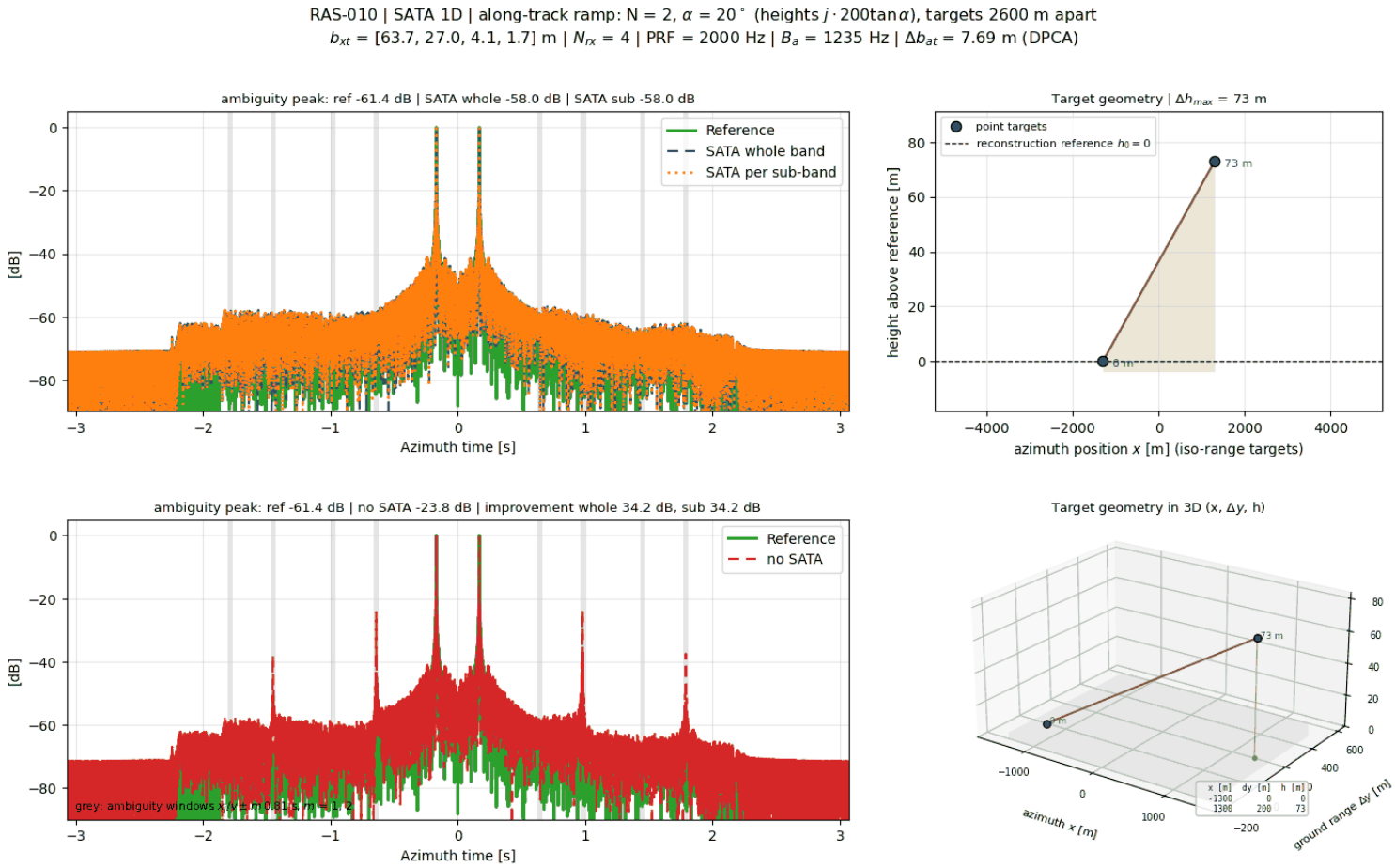


Figure 98: Along-track ramp, 2.6 km apart, normal heights: RAS-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −23.8, SATA whole band −58.0, per sub-band −58.0 dB (improvement 34.2 dB).

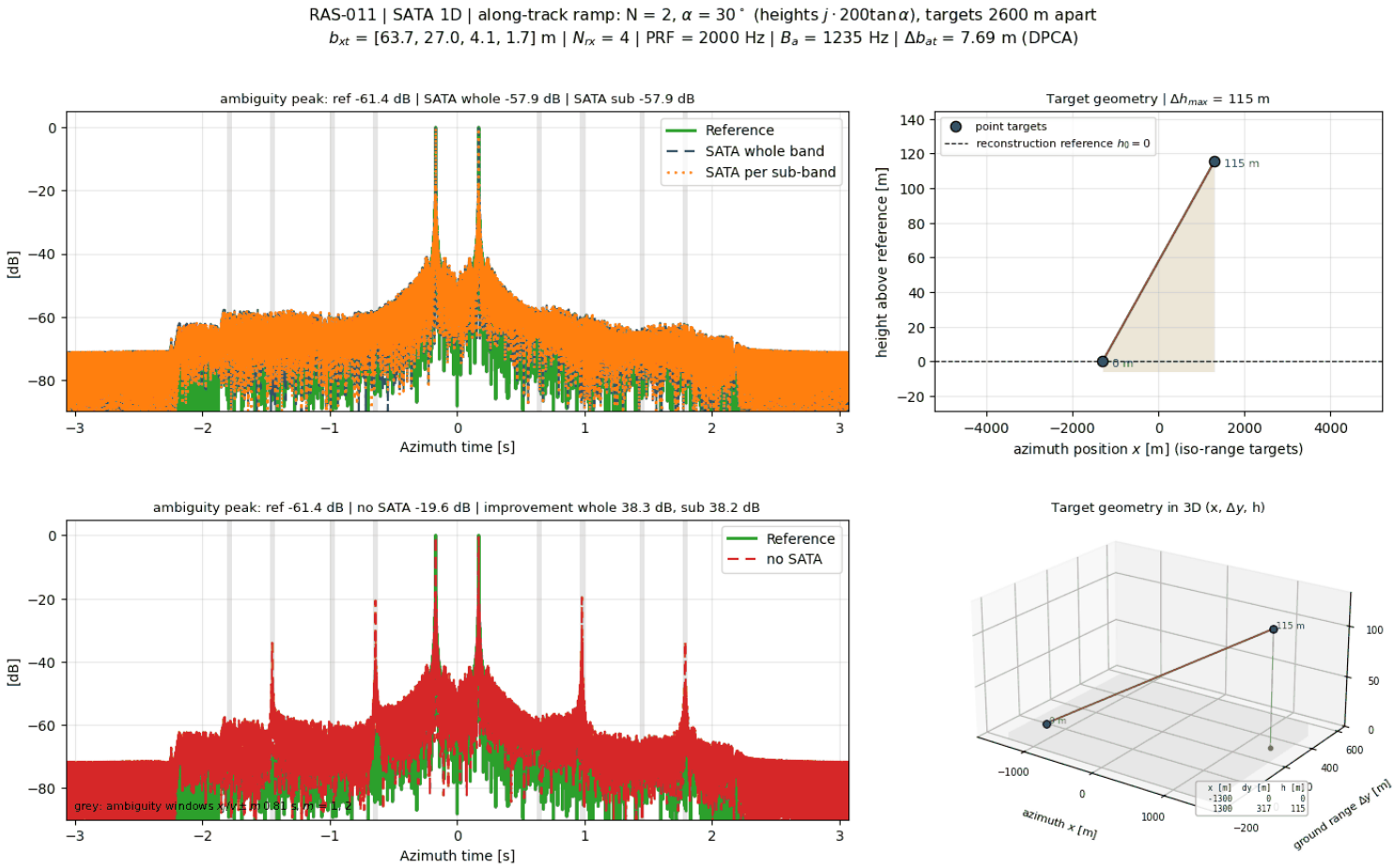


Figure 99: Along-track ramp, 2.6 km apart, normal heights: RAS-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −19.6, SATA whole band −57.9, per sub-band −57.9 dB (improvement 38.3 dB).

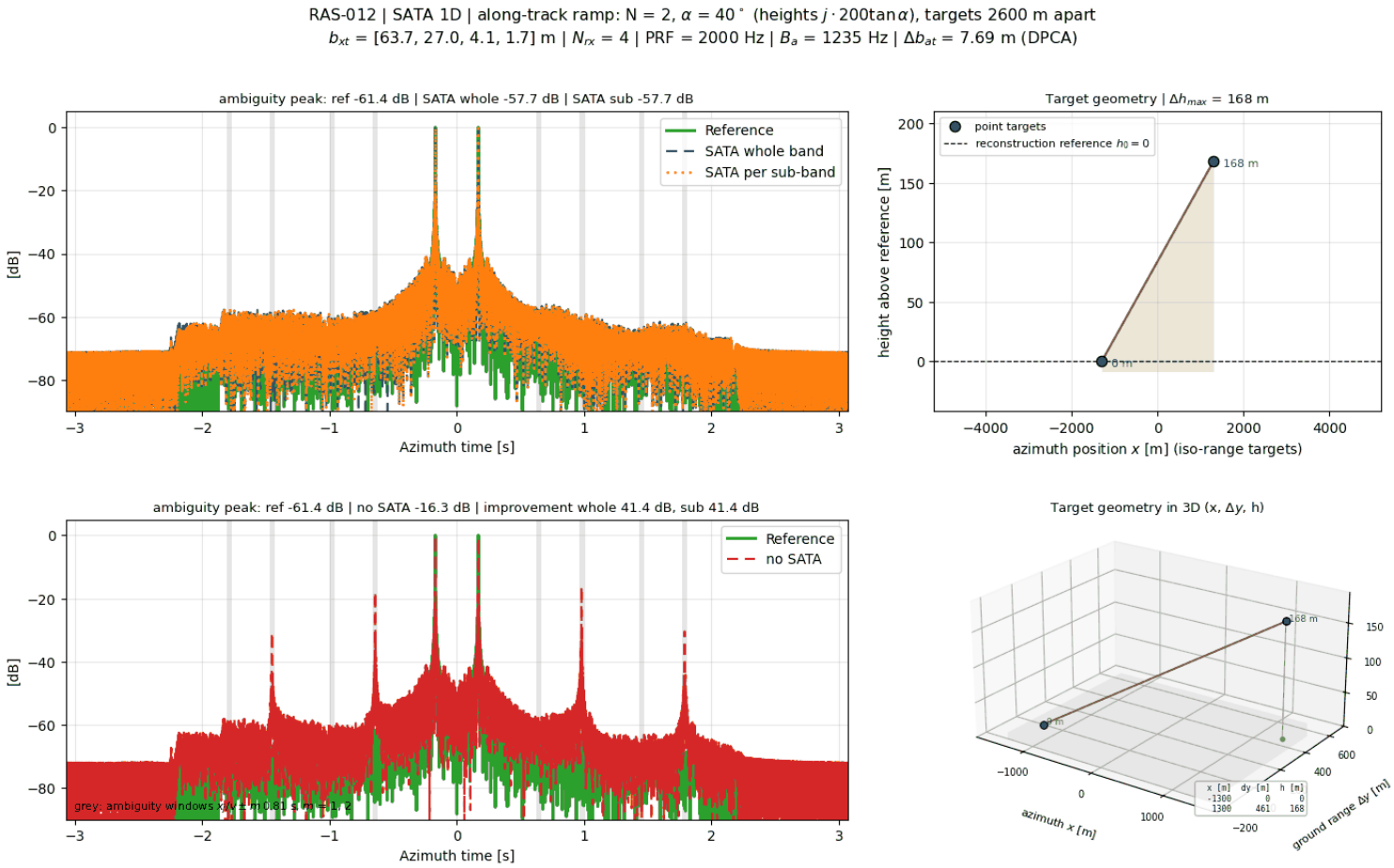


Figure 100: Along-track ramp, 2.6 km apart, normal heights: RAS-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −16.3, SATA whole band −57.7, per sub-band −57.7 dB (improvement 41.4 dB).

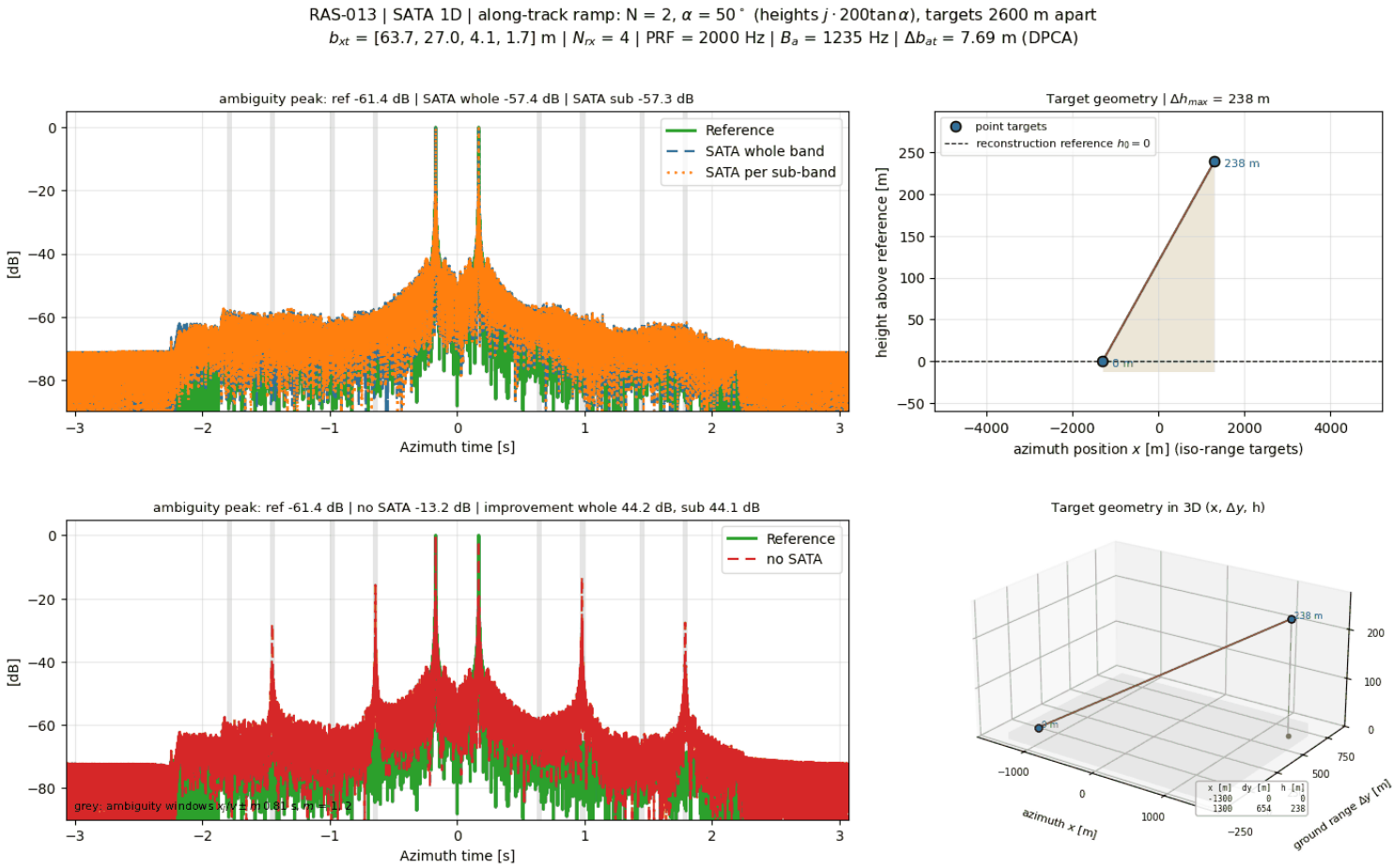


Figure 101: Along-track ramp, 2.6 km apart, normal heights: RAS-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −13.2, SATA whole band −57.4, per sub-band −57.3 dB (improvement 44.2 dB).

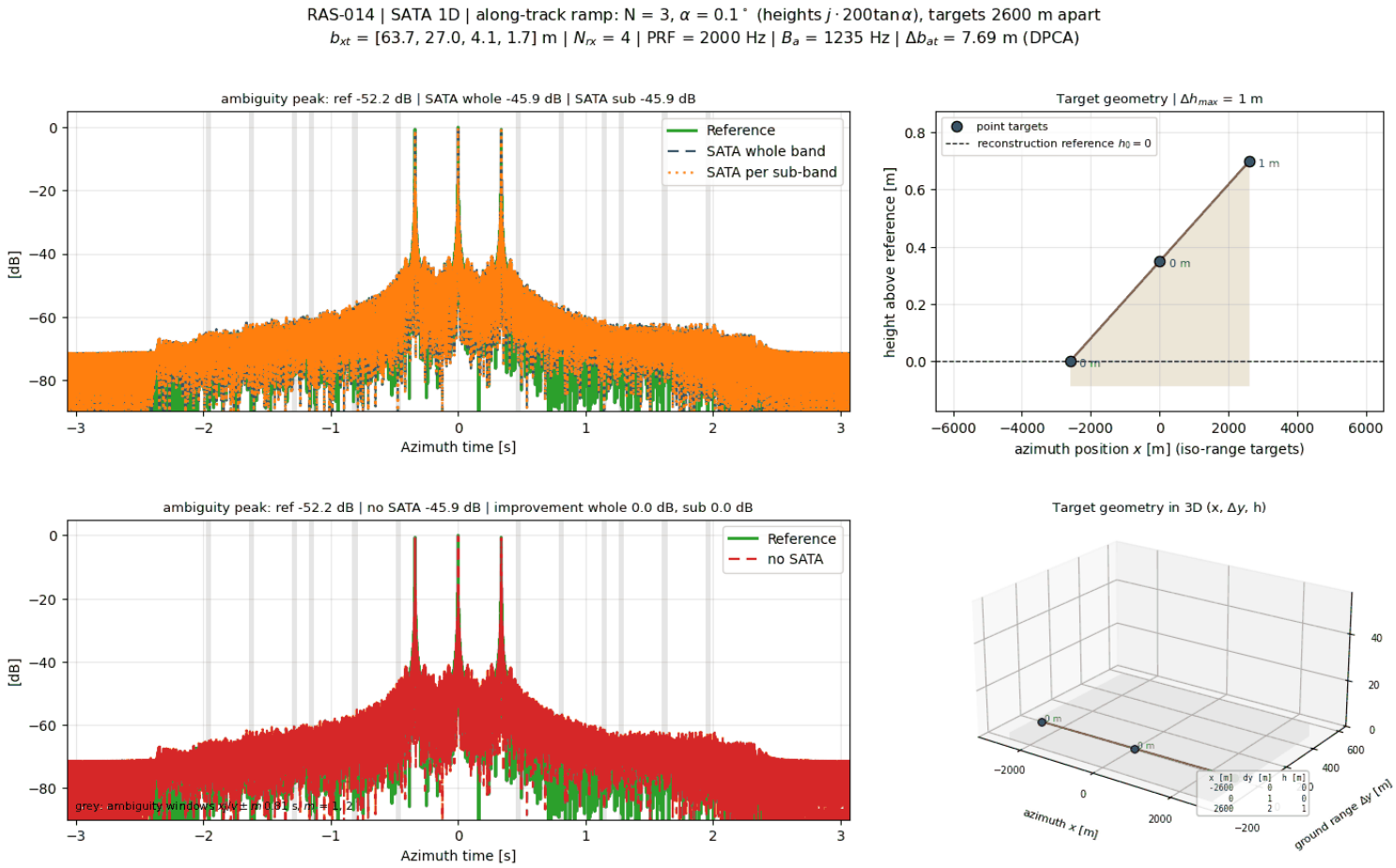


Figure 102: Along-track ramp, 2.6 km apart, normal heights: RAS-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −45.9, SATA whole band −45.9, per sub-band −45.9 dB (improvement 0.0 dB).

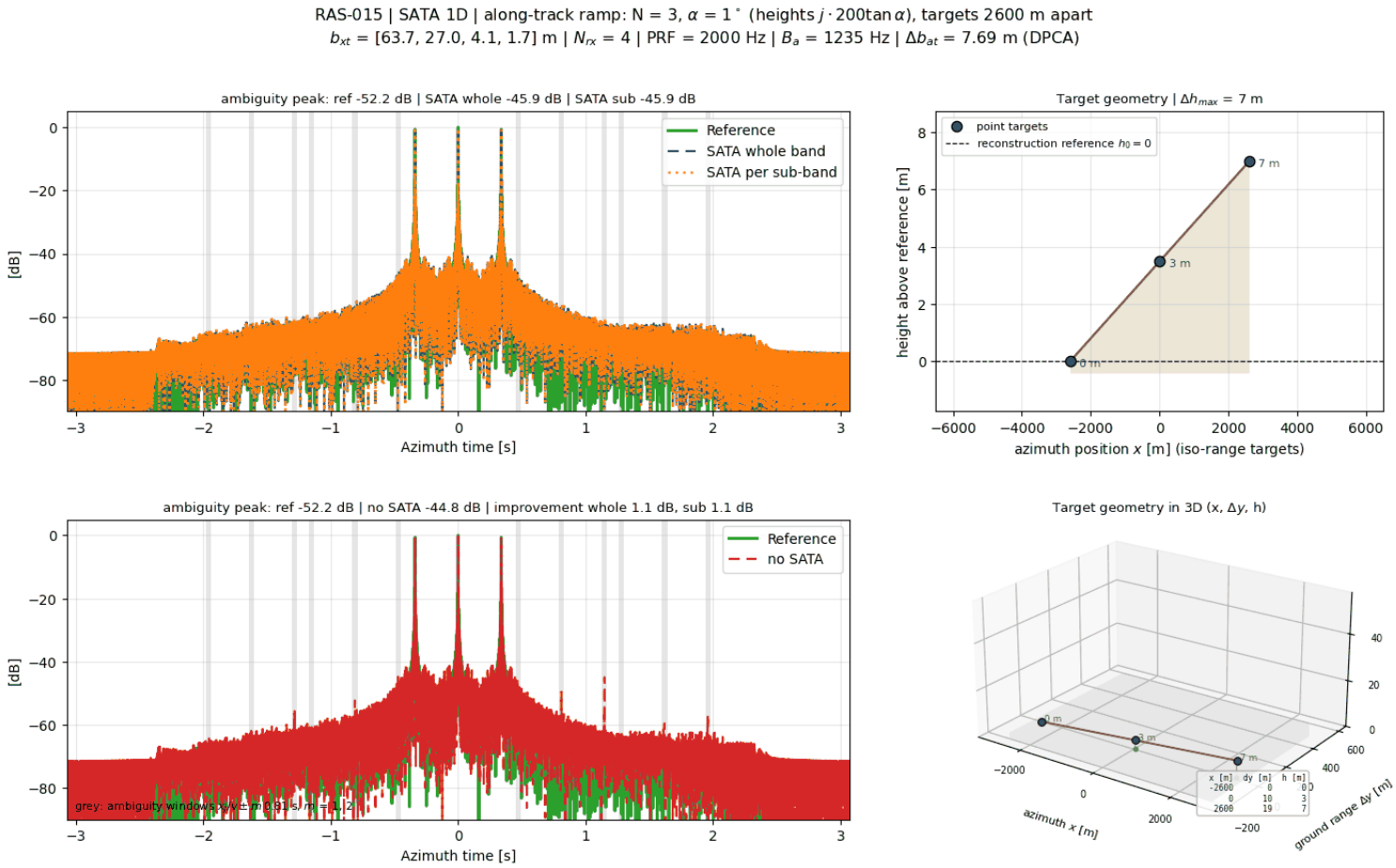


Figure 103: Along-track ramp, 2.6 km apart, normal heights: RAS-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −44.8, SATA whole band −45.9, per sub-band −45.9 dB (improvement 1.1 dB).

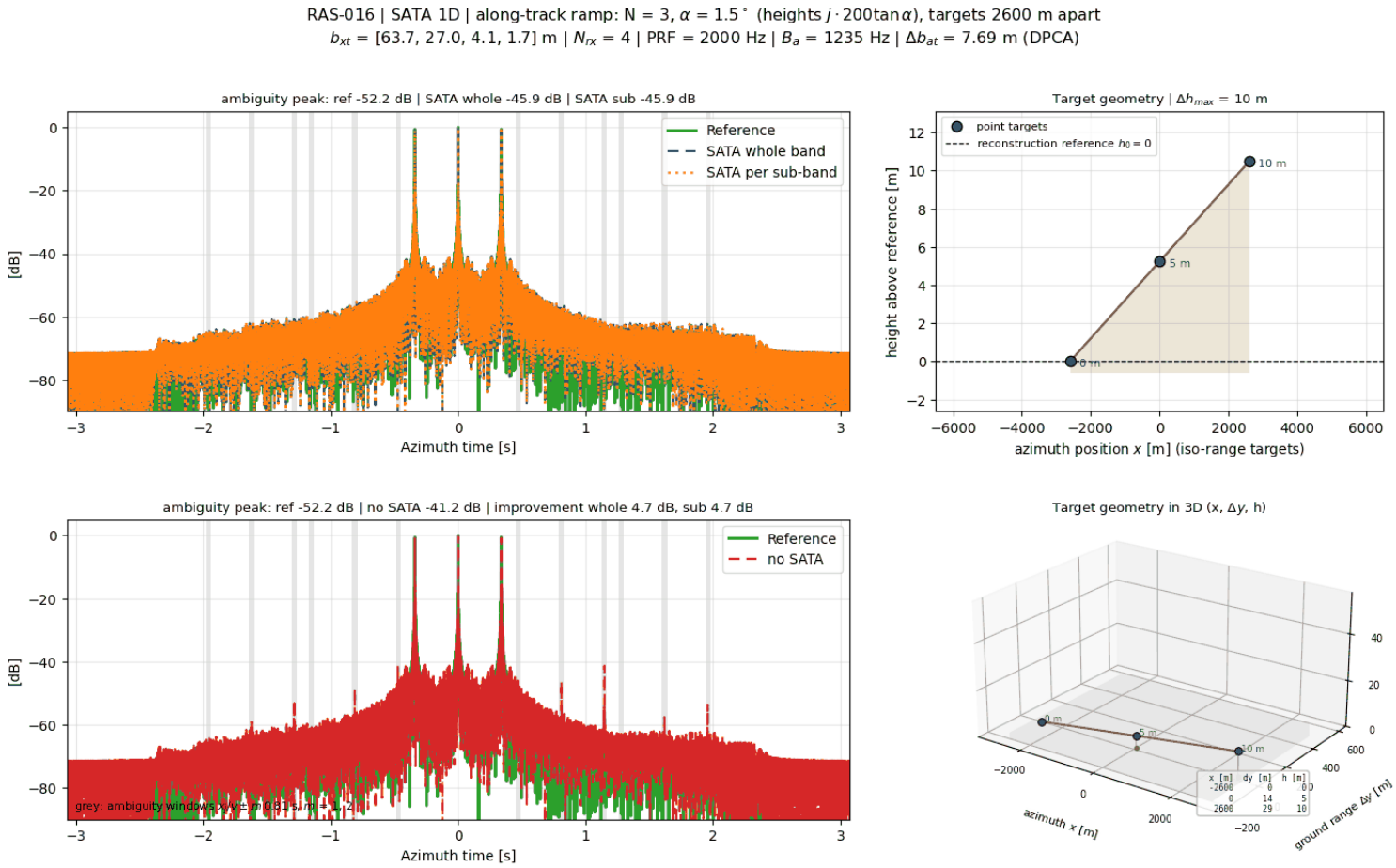


Figure 104: Along-track ramp, 2.6 km apart, normal heights: RAS-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −41.2, SATA whole band −45.9, per sub-band −45.9 dB (improvement 4.7 dB).

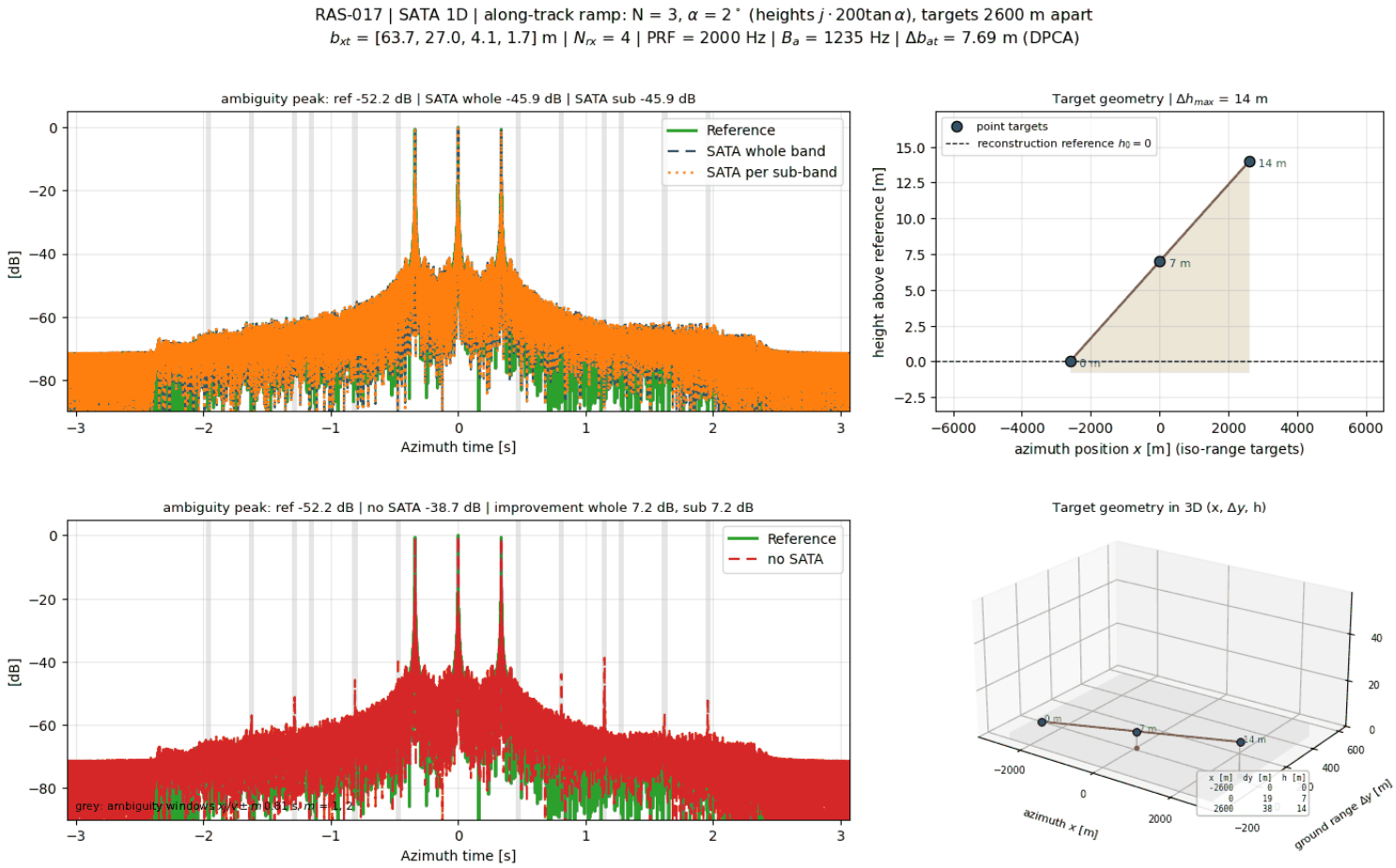


Figure 105: Along-track ramp, 2.6 km apart, normal heights: RAS-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −38.7, SATA whole band −45.9, per sub-band −45.9 dB (improvement 7.2 dB).

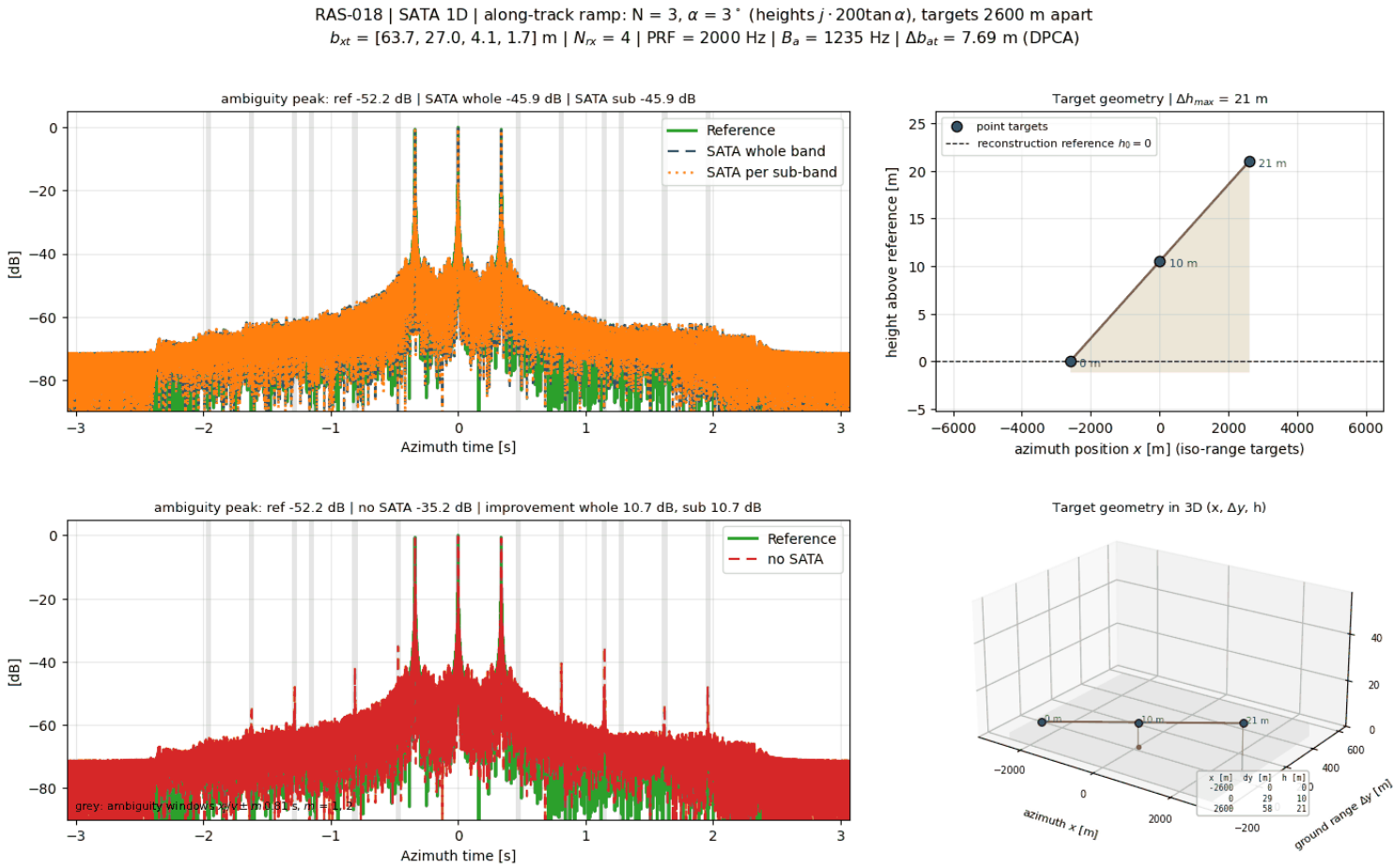


Figure 106: Along-track ramp, 2.6 km apart, normal heights: RAS-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −35.2, SATA whole band −45.9, per sub-band −45.9 dB (improvement 10.7 dB).

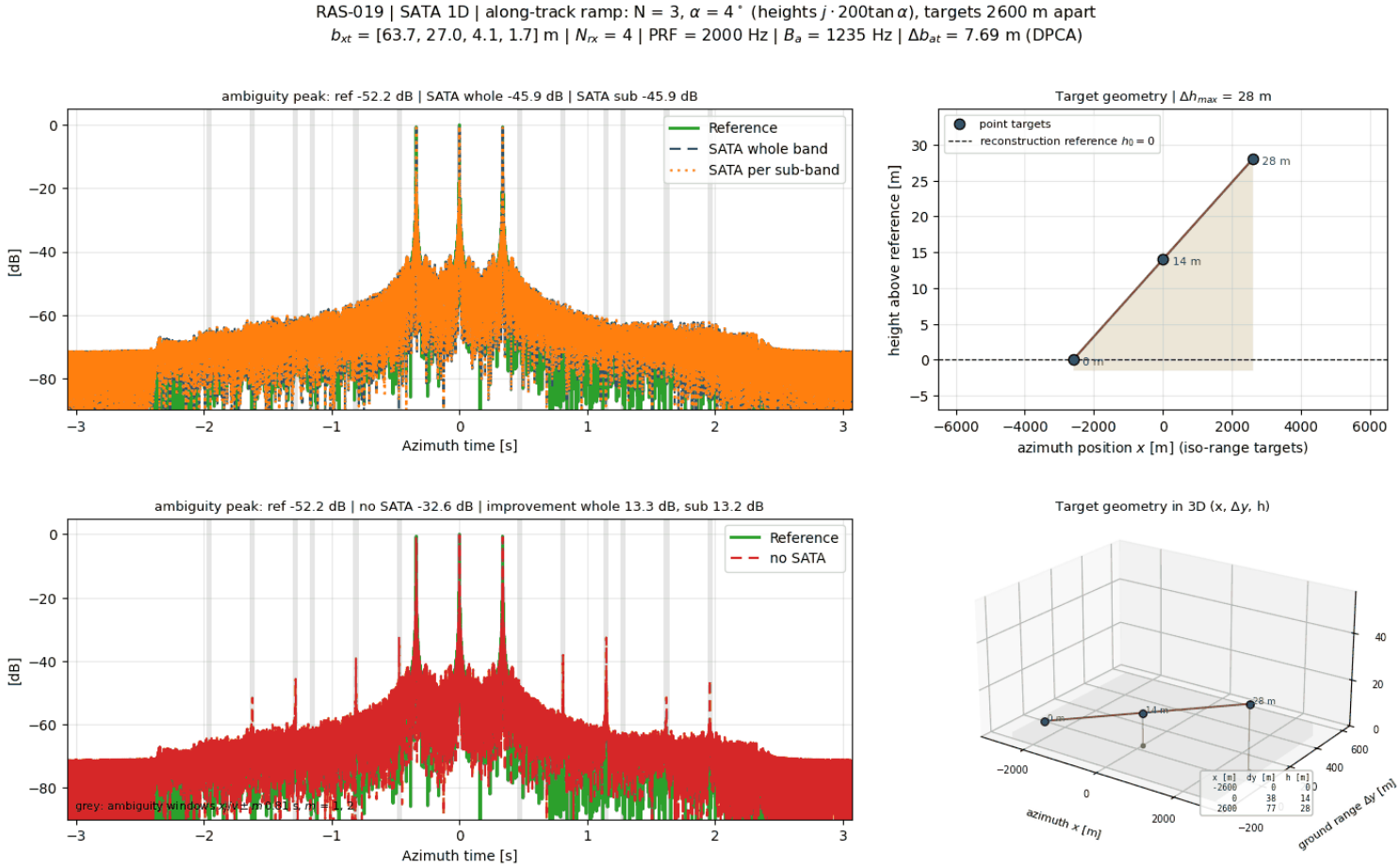


Figure 107: Along-track ramp, 2.6 km apart, normal heights: RAS-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −32.6, SATA whole band −45.9, per sub-band −45.9 dB (improvement 13.3 dB).

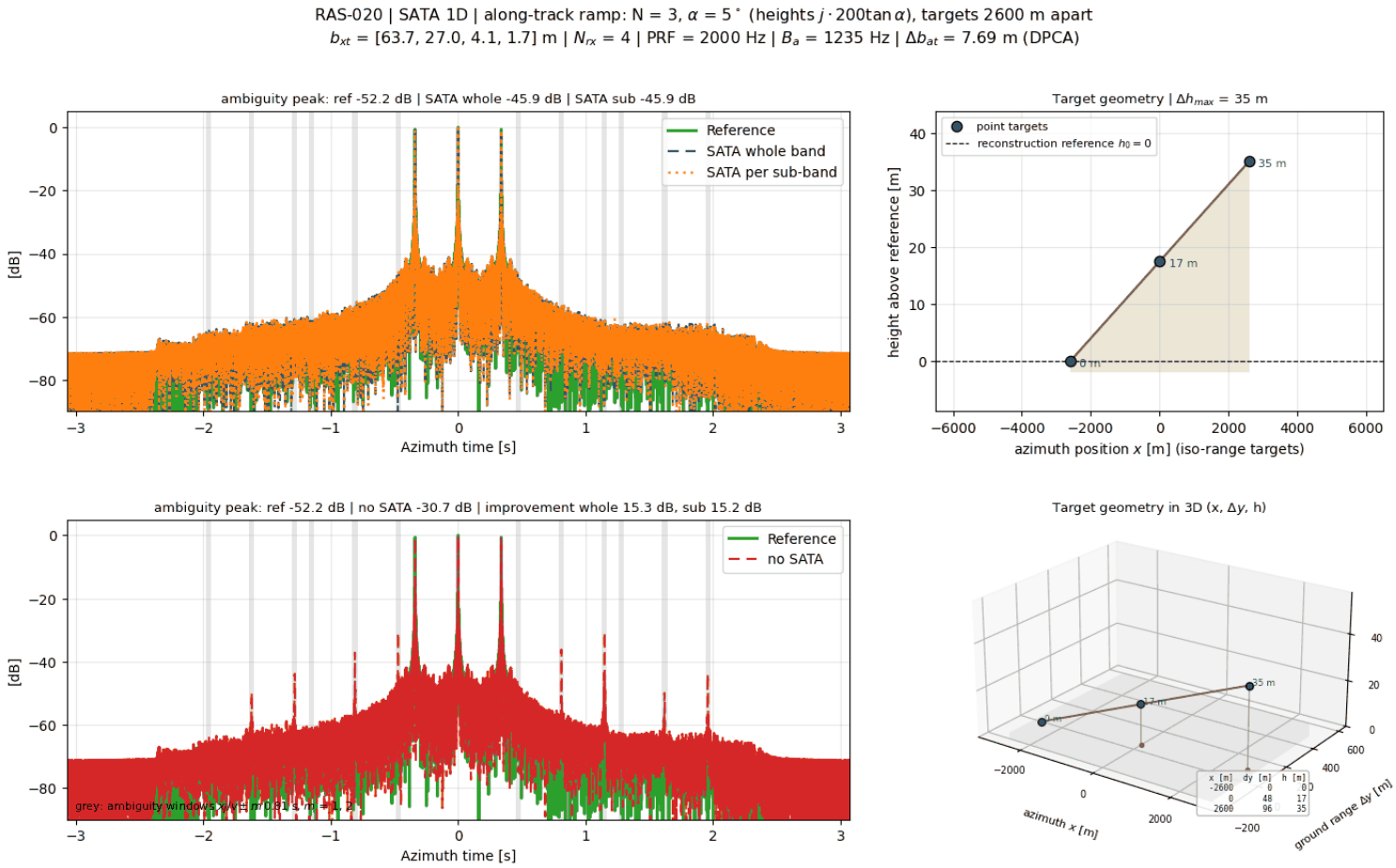


Figure 108: Along-track ramp, 2.6 km apart, normal heights: RAS-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −30.7, SATA whole band −45.9, per sub-band −45.9 dB (improvement 15.3 dB).

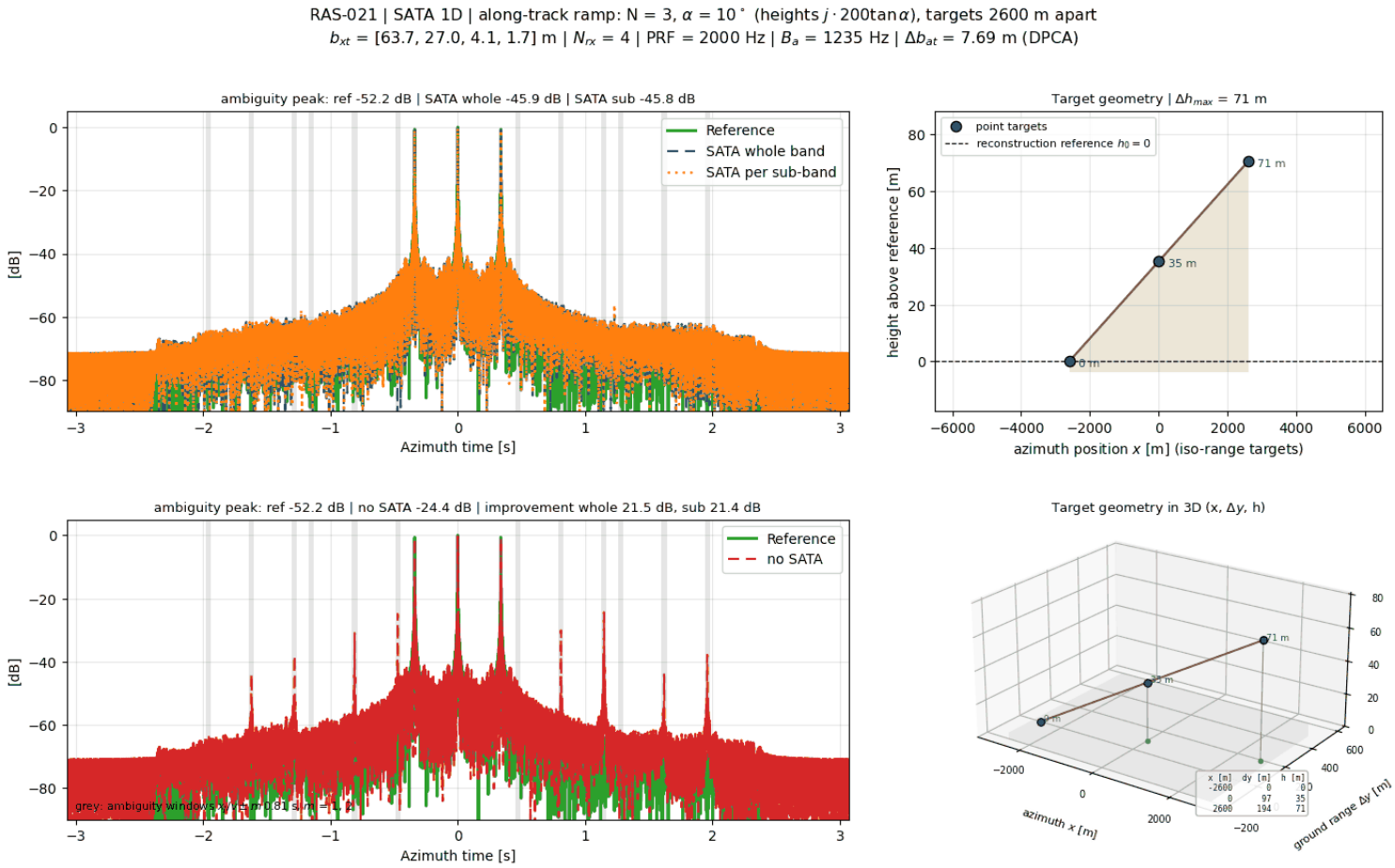


Figure 109: Along-track ramp, 2.6 km apart, normal heights: RAS-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −24.4, SATA whole band −45.9, per sub-band −45.8 dB (improvement 21.5 dB).

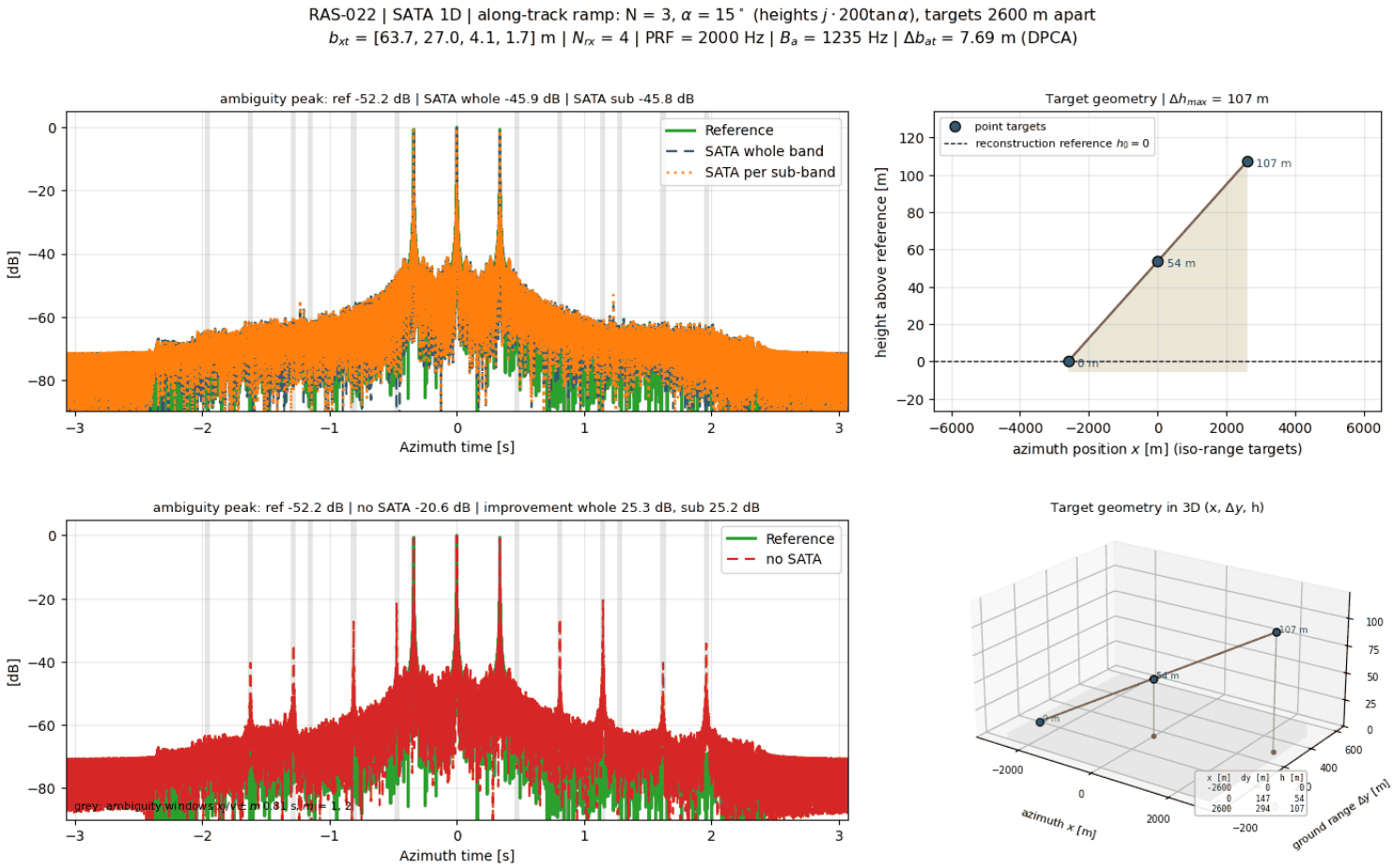


Figure 110: Along-track ramp, 2.6 km apart, normal heights: RAS-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −20.6, SATA whole band −45.9, per sub-band −45.8 dB (improvement 25.3 dB).

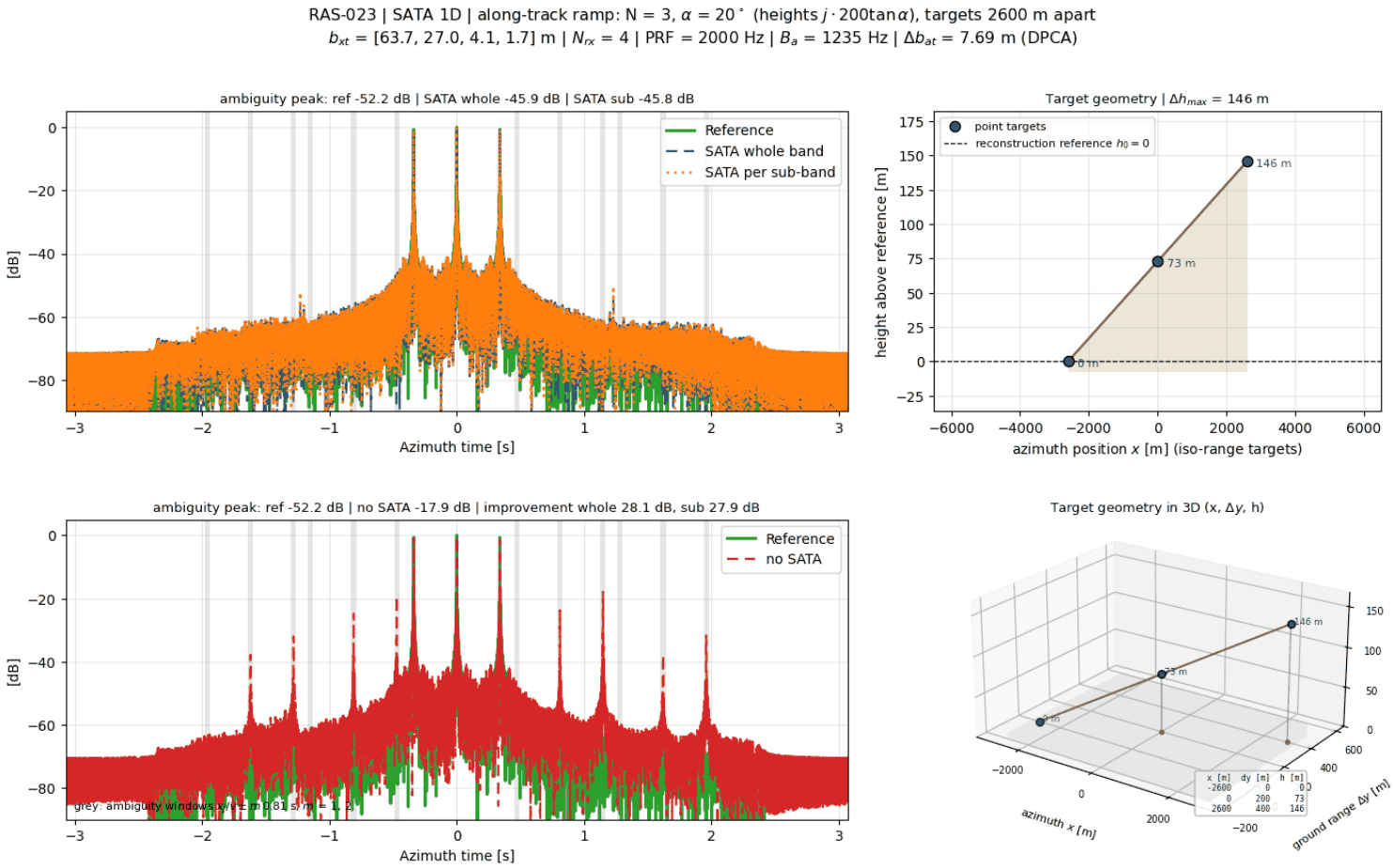


Figure 111: Along-track ramp, 2.6 km apart, normal heights: RAS-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA -17.9, SATA whole band -45.9, per sub-band -45.8 dB (improvement 28.1 dB).

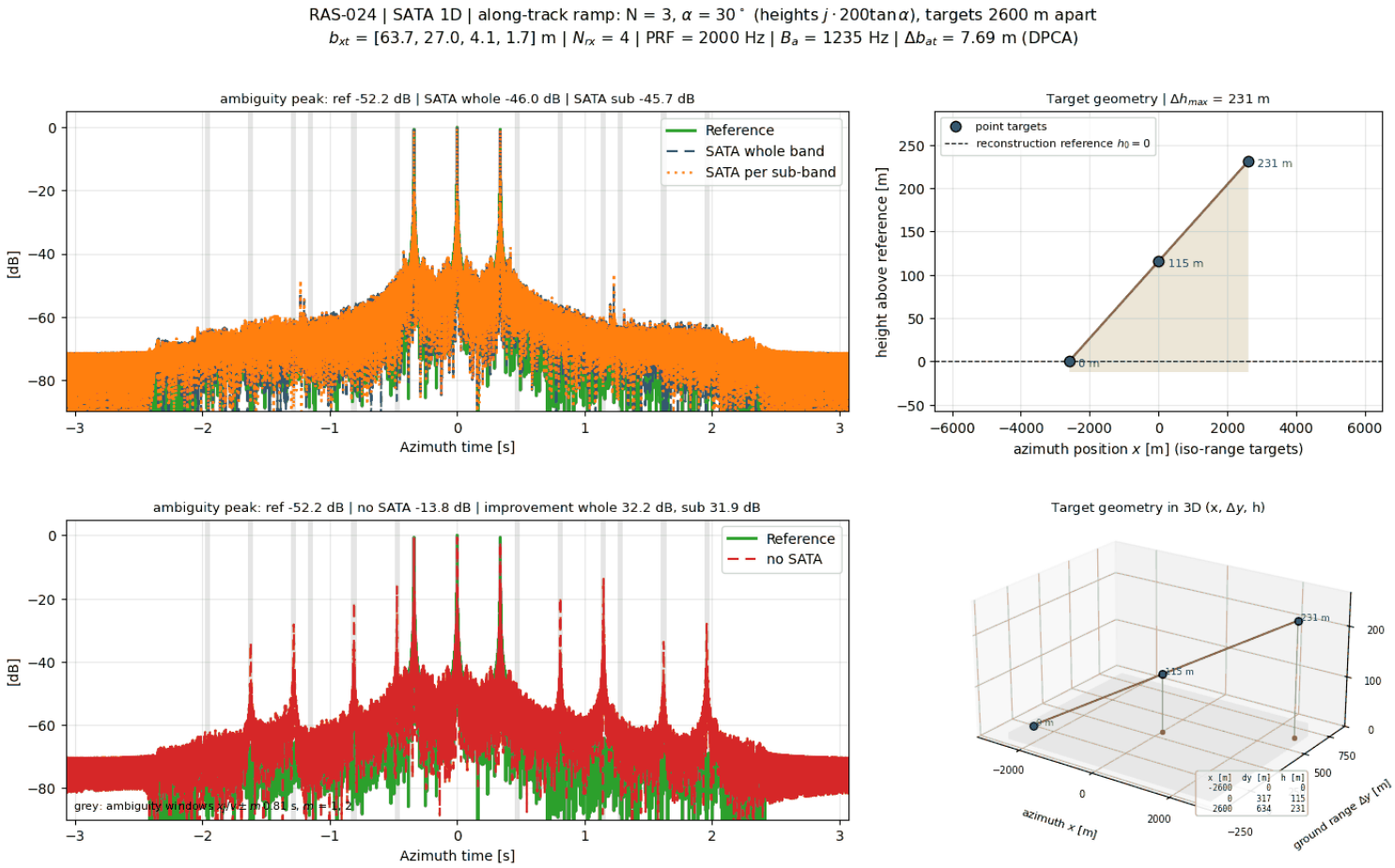


Figure 112: Along-track ramp, 2.6 km apart, normal heights: RAS-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −13.8, SATA whole band −46.0, per sub-band −45.7 dB (improvement 32.2 dB).

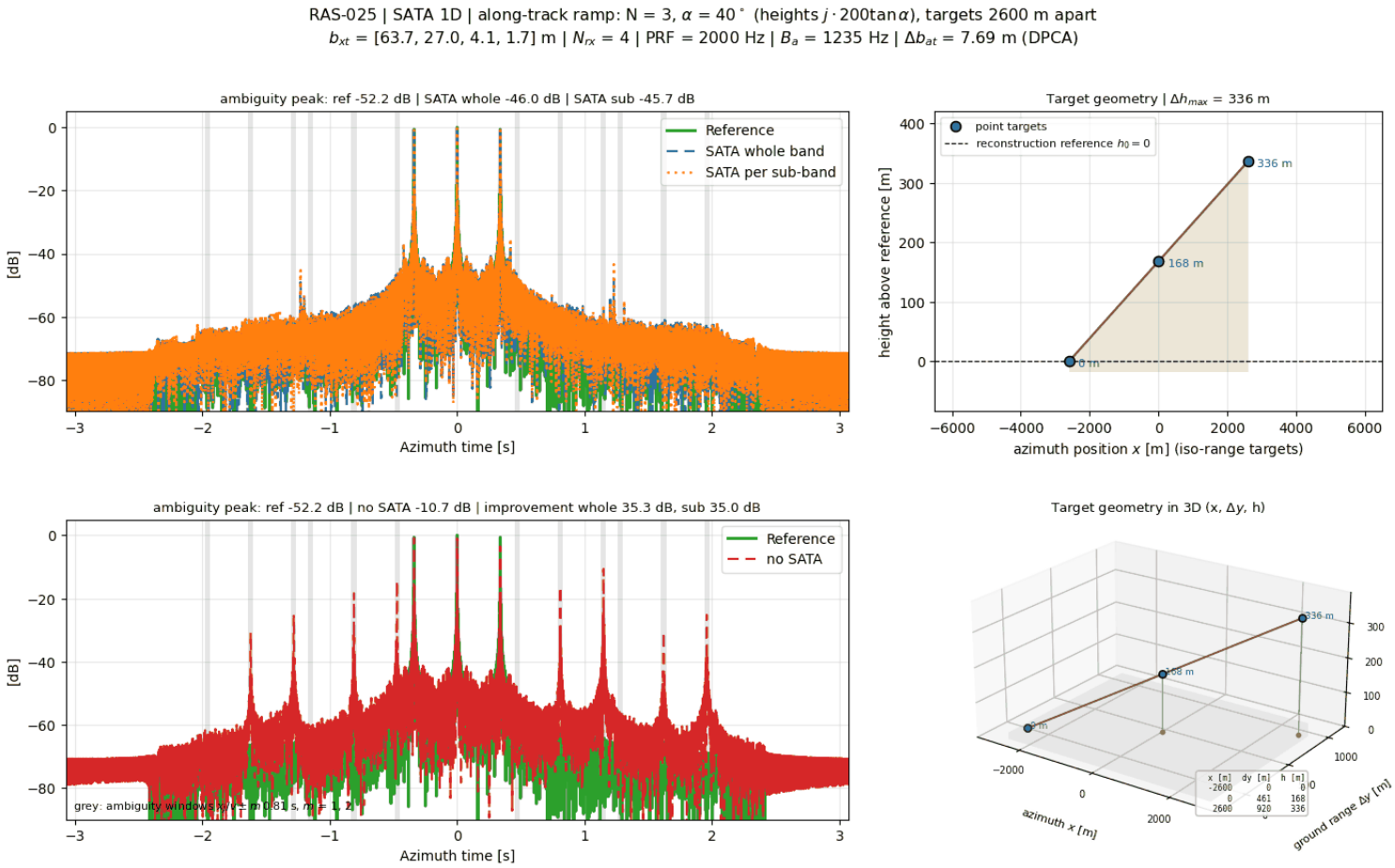


Figure 113: Along-track ramp, 2.6 km apart, normal heights: RAS-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −10.7, SATA whole band −46.0, per sub-band −45.7 dB (improvement 35.3 dB).

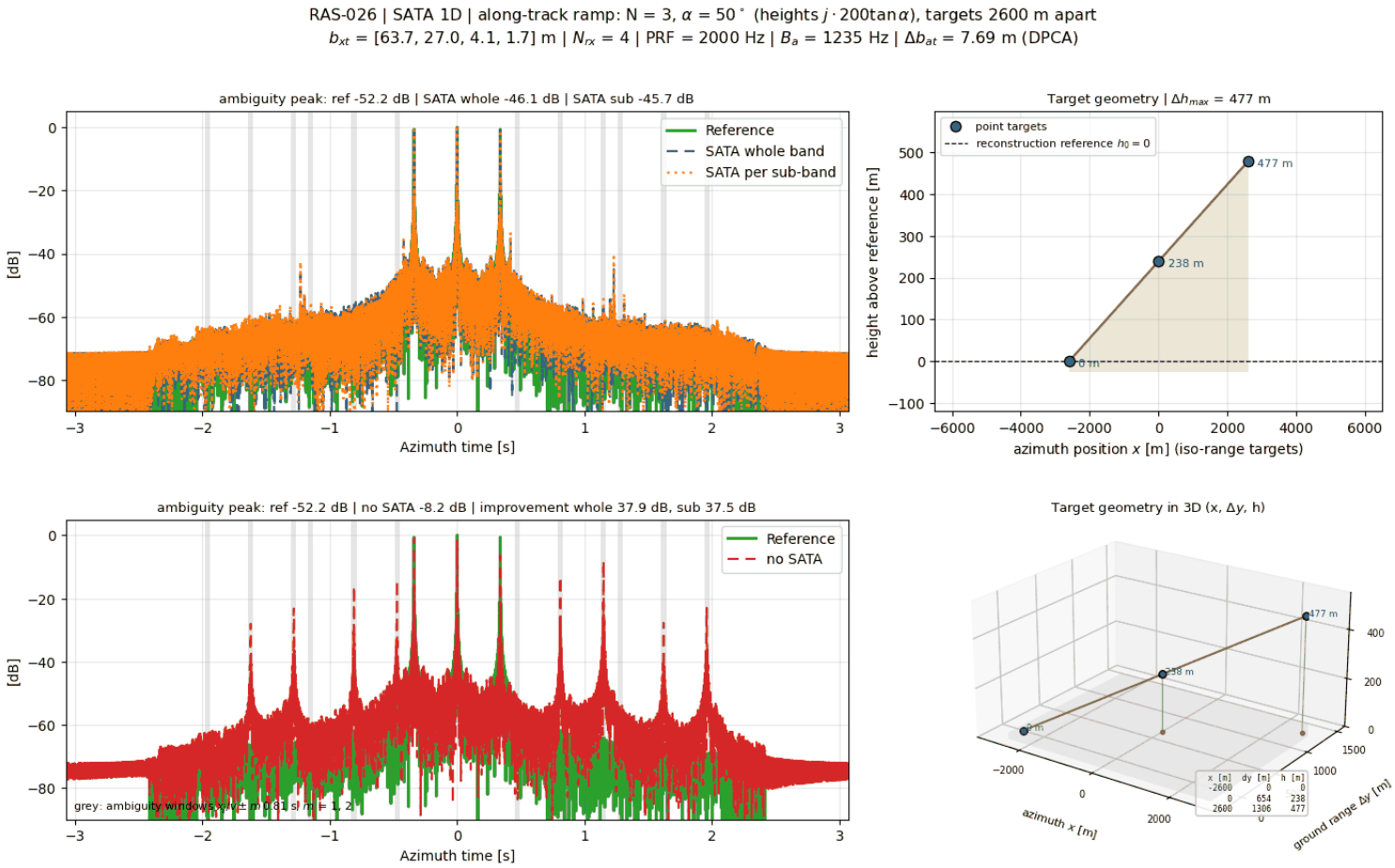


Figure 114: Along-track ramp, 2.6 km apart, normal heights: RAS-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −8.2, SATA whole band −46.1, per sub-band −45.7 dB (improvement 37.9 dB).

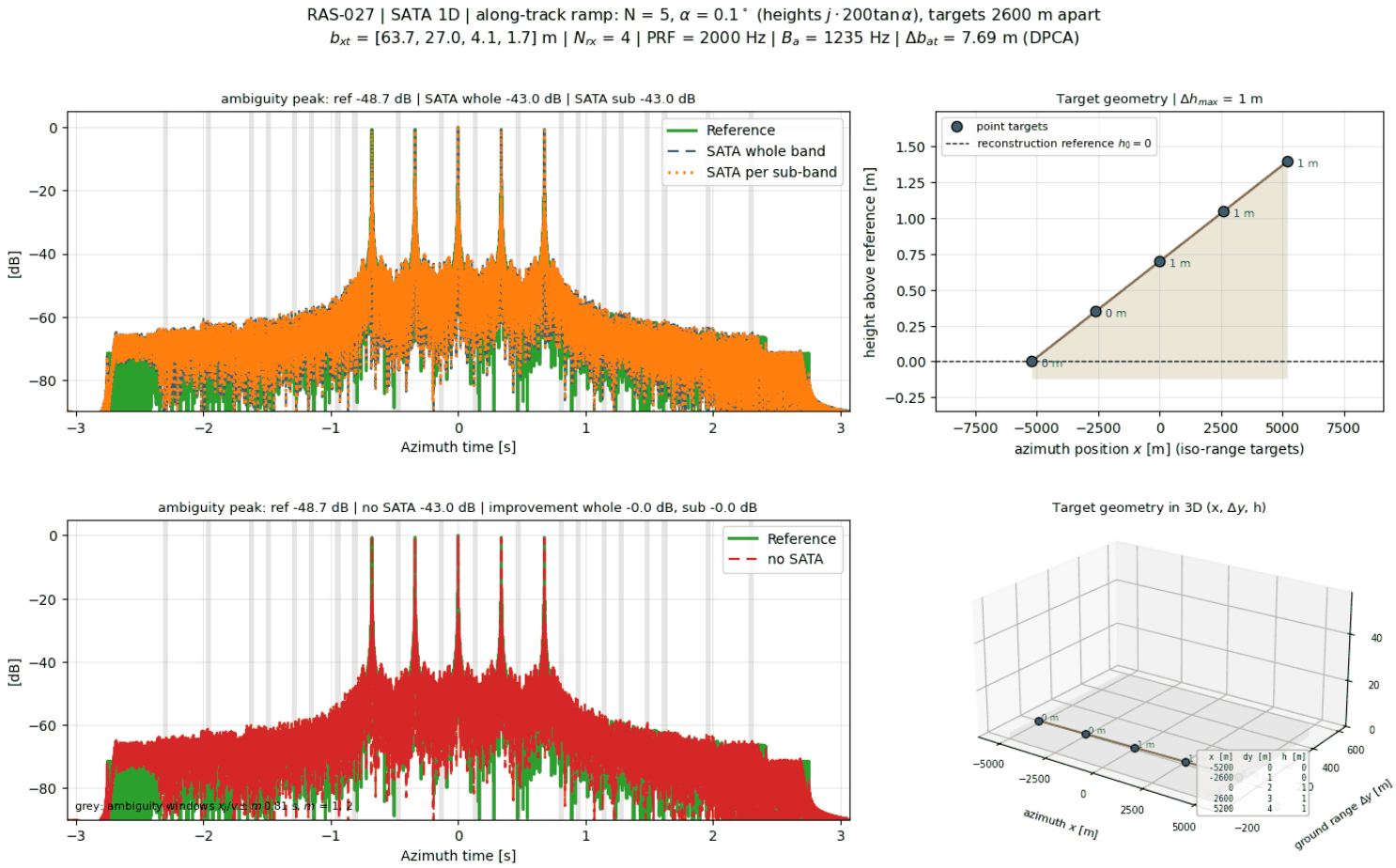


Figure 115: Along-track ramp, 2.6 km apart, normal heights: RAS-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −43.0, SATA whole band −43.0, per sub-band −43.0 dB (improvement 0.0 dB).

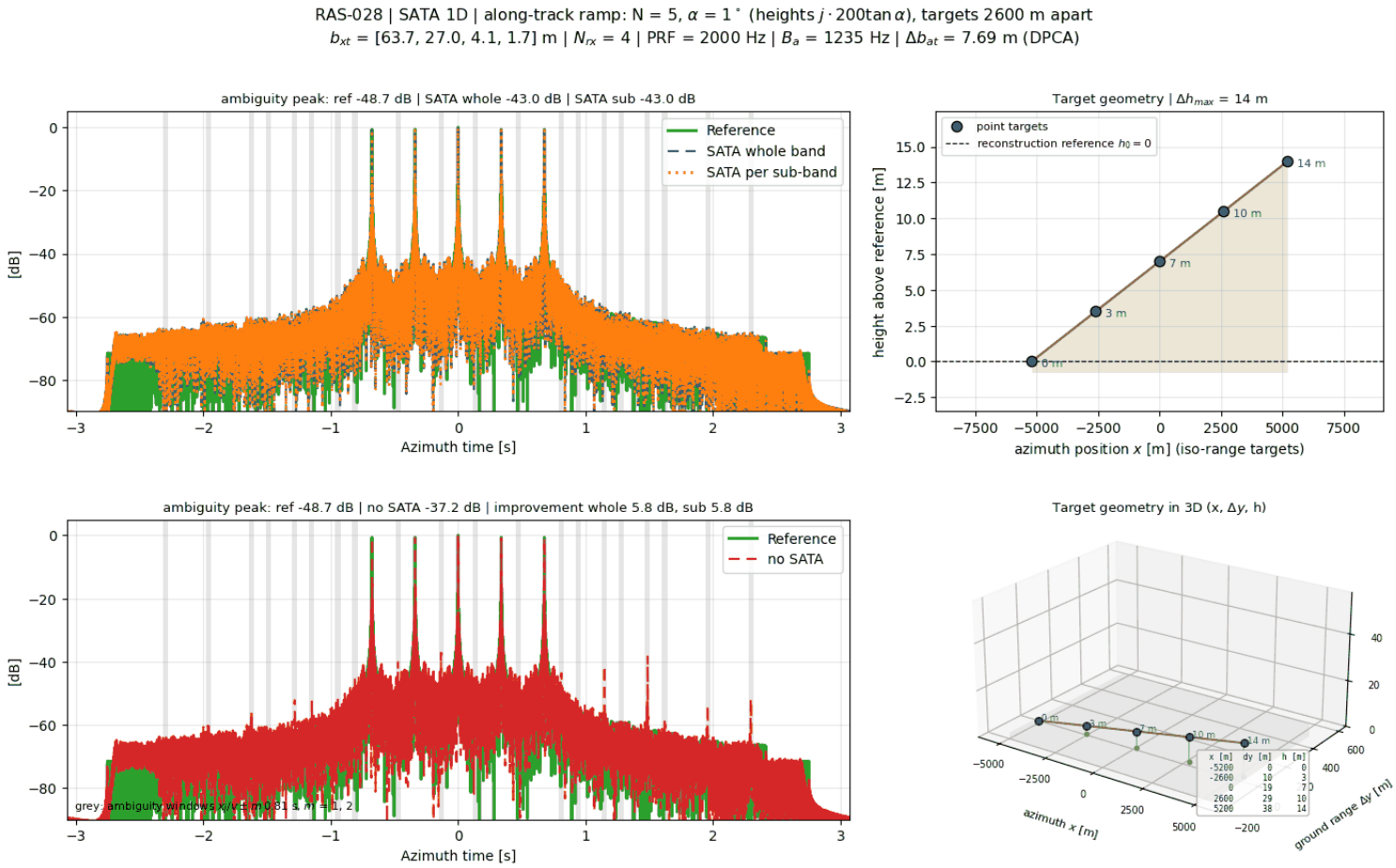


Figure 116: Along-track ramp, 2.6 km apart, normal heights: RAS-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −37.2, SATA whole band −43.0, per sub-band −43.0 dB (improvement 5.8 dB).

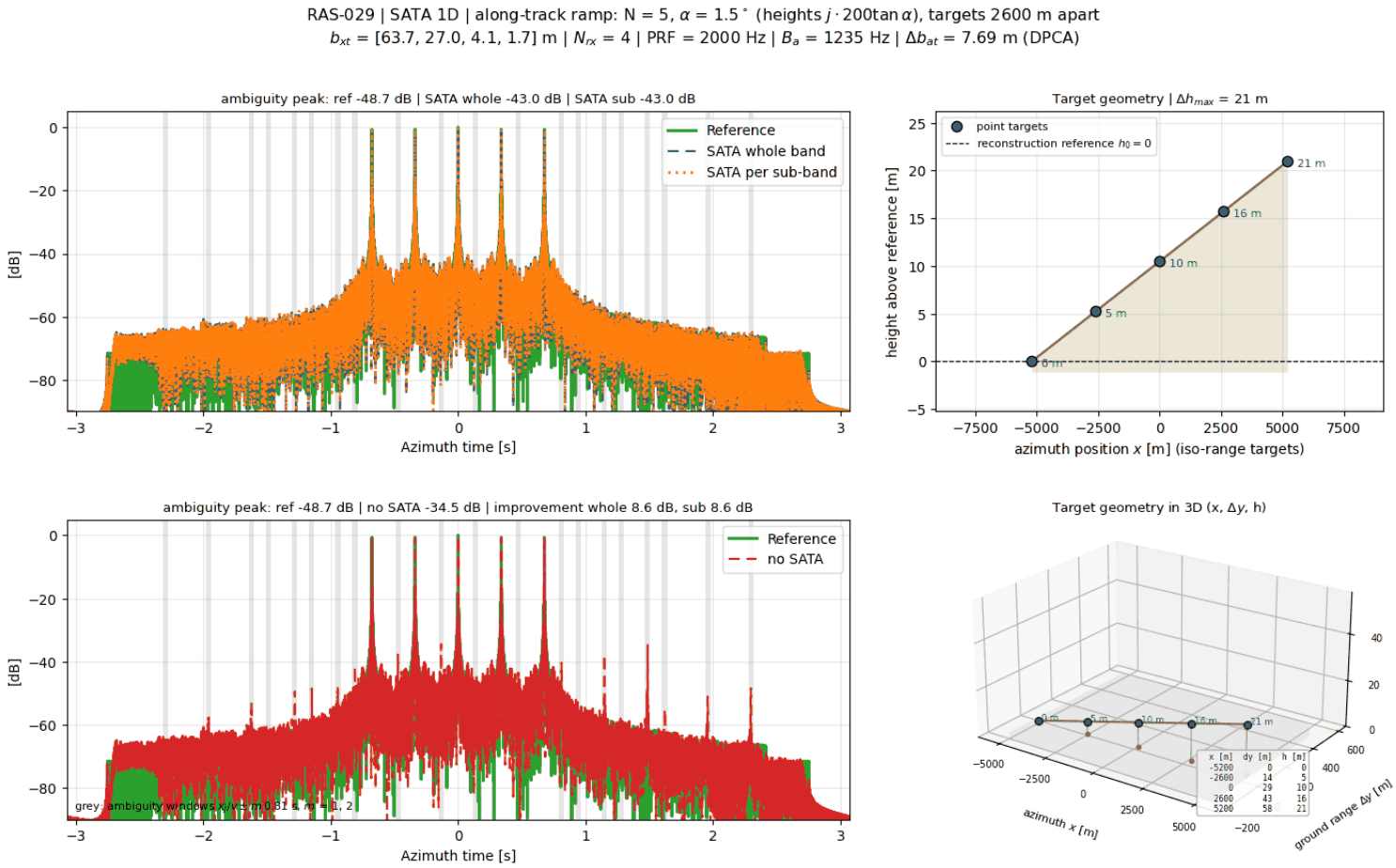


Figure 117: Along-track ramp, 2.6 km apart, normal heights: RAS-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA -34.5, SATA whole band -43.0, per sub-band -43.0 dB (improvement 8.6 dB).

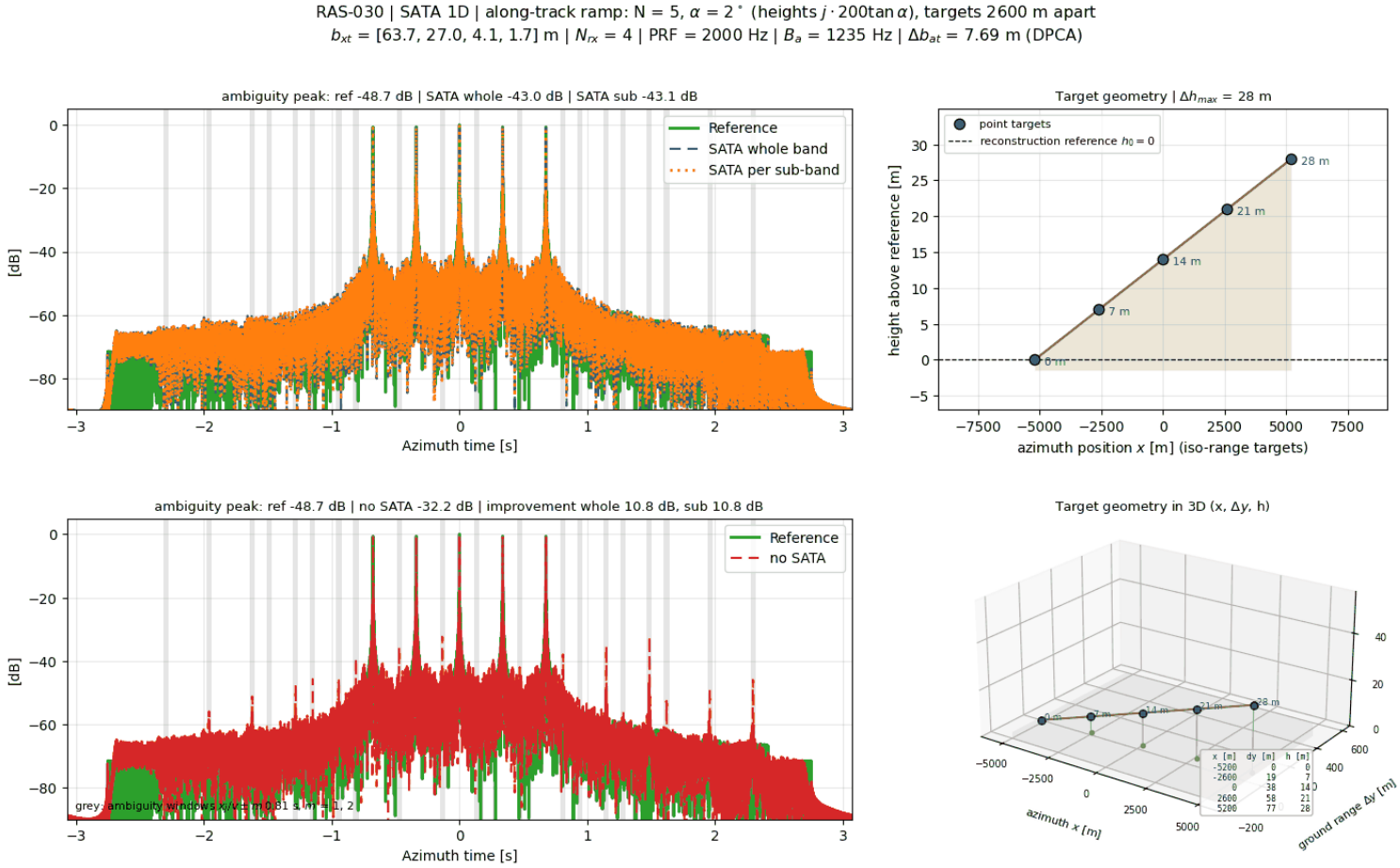


Figure 118: Along-track ramp, 2.6 km apart, normal heights: RAS-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −32.2, SATA whole band −43.0, per sub-band −43.1 dB (improvement 10.8 dB).

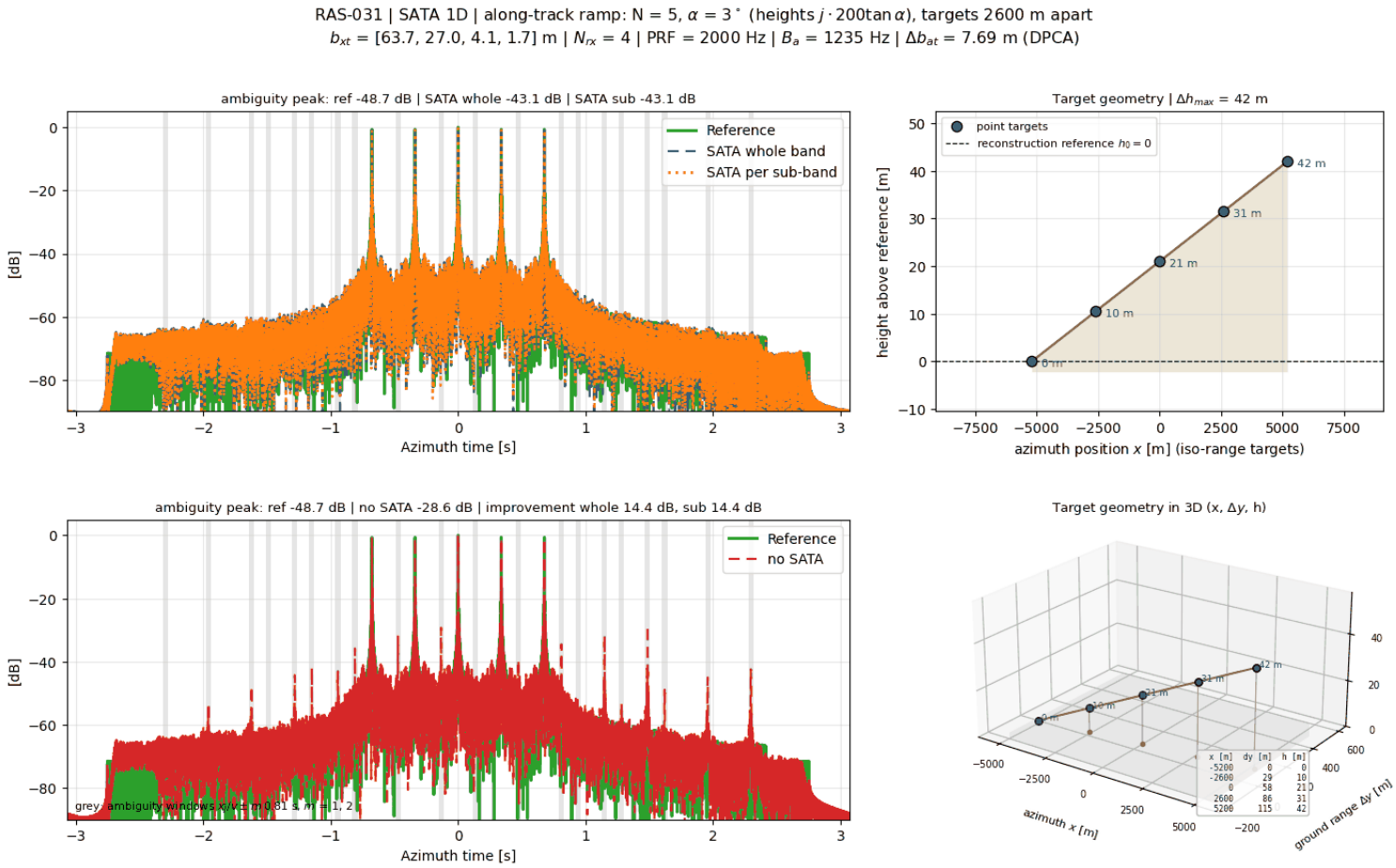


Figure 119: Along-track ramp, 2.6 km apart, normal heights: RAS-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −28.6, SATA whole band −43.1, per sub-band −43.1 dB (improvement 14.4 dB).

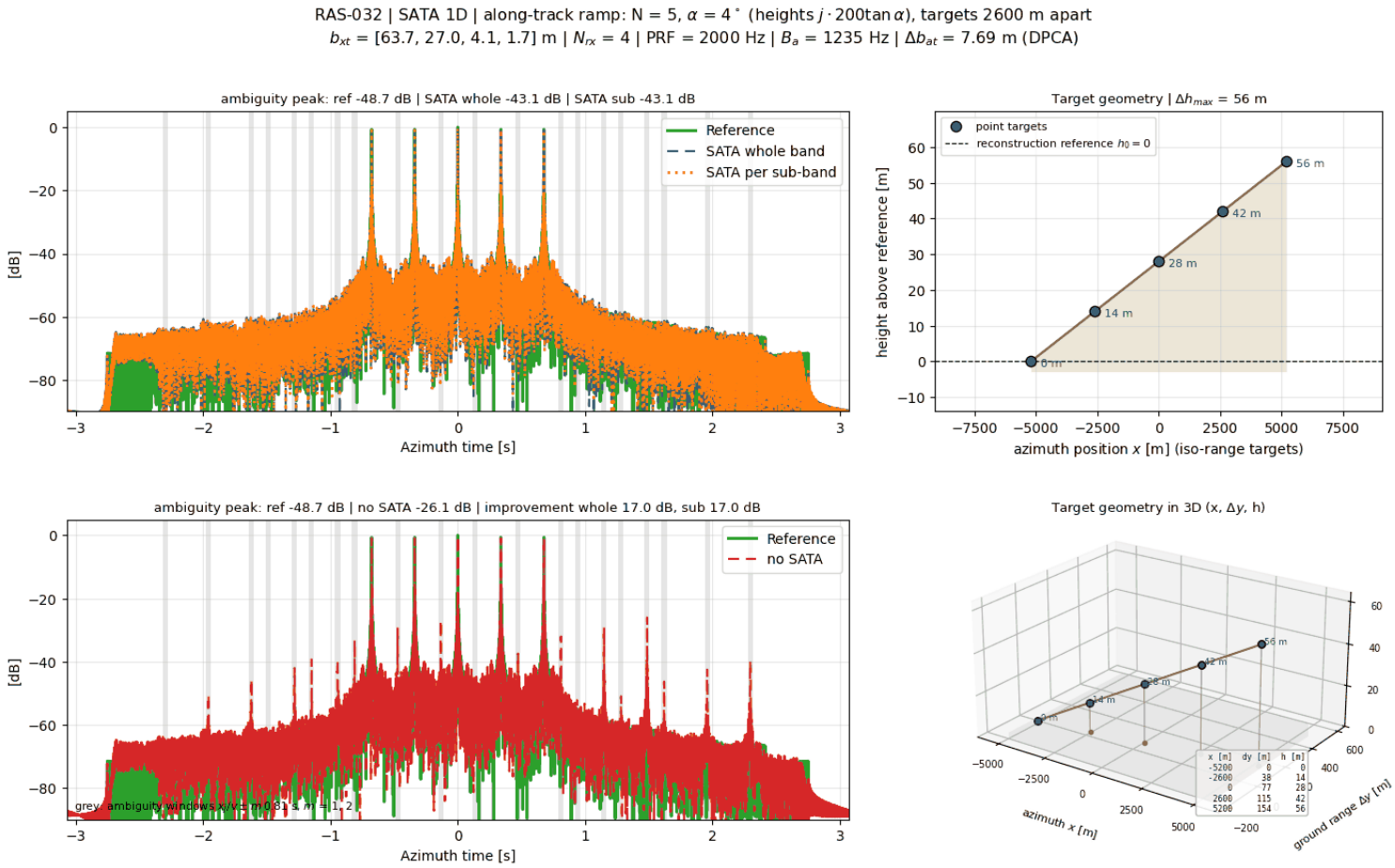


Figure 120: Along-track ramp, 2.6 km apart, normal heights: RAS-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −26.1, SATA whole band −43.1, per sub-band −43.1 dB (improvement 17.0 dB).

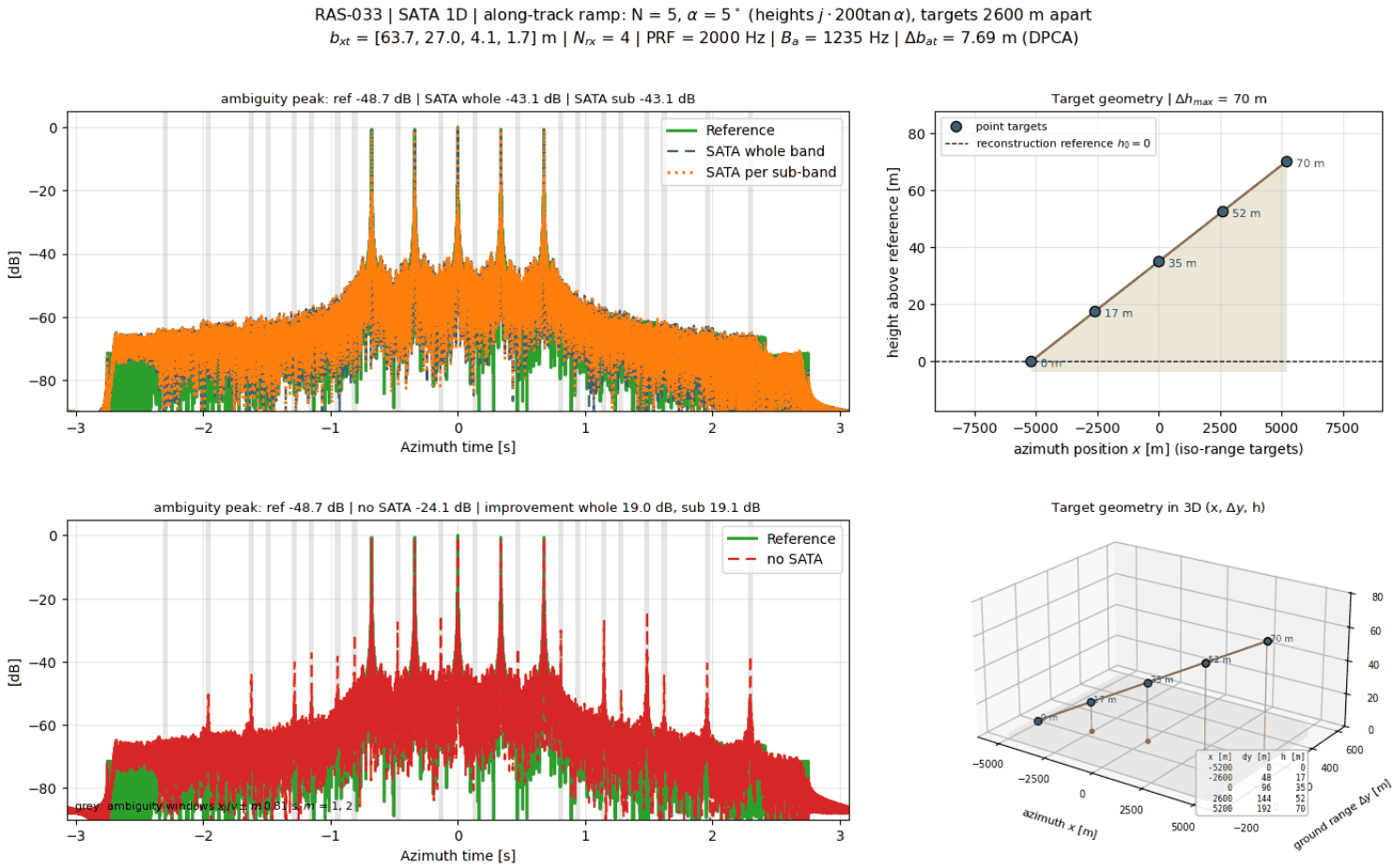


Figure 121: Along-track ramp, 2.6 km apart, normal heights: RAS-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −24.1, SATA whole band −43.1, per sub-band −43.1 dB (improvement 19.0 dB).

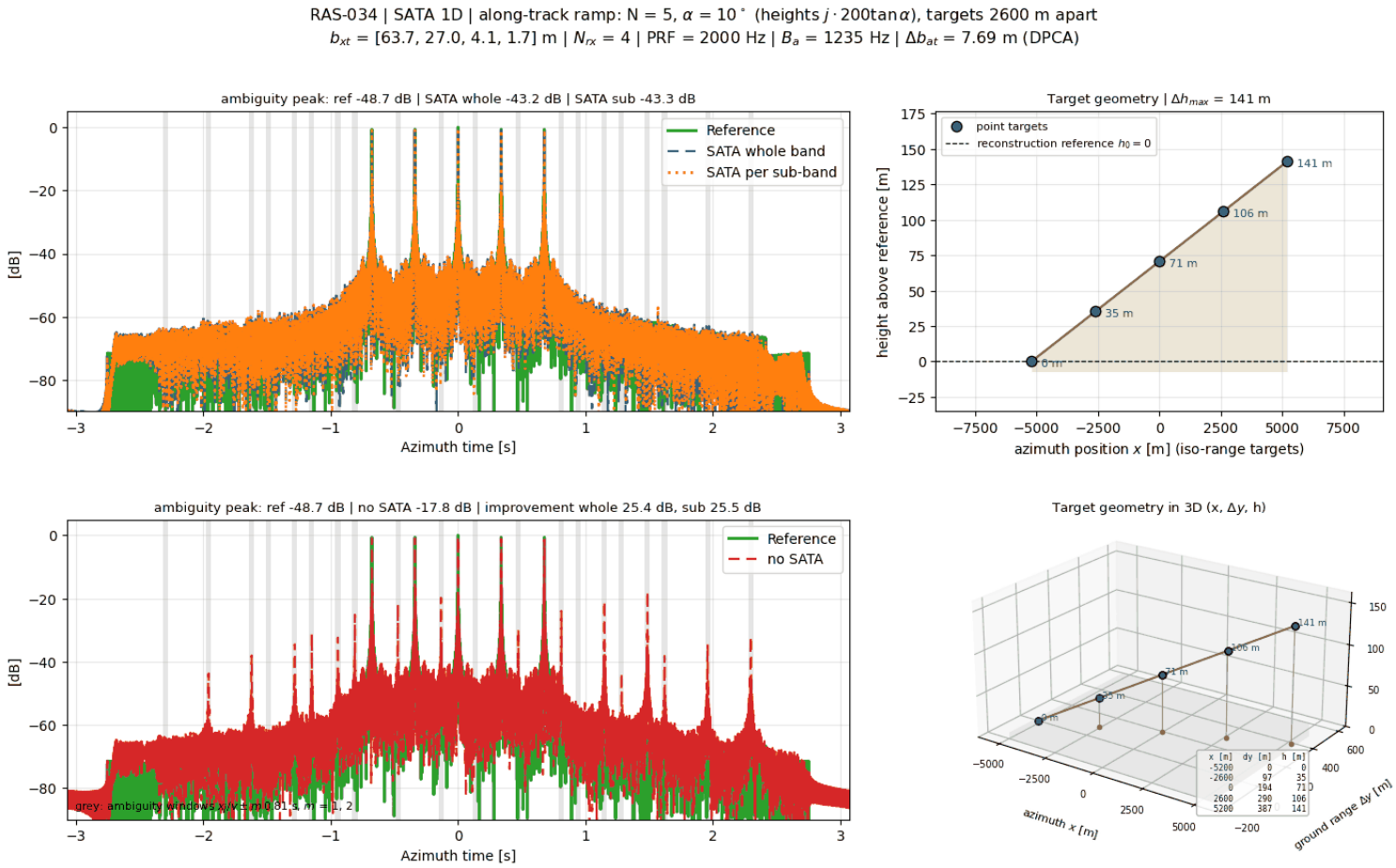


Figure 122: Along-track ramp, 2.6 km apart, normal heights: RAS-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −17.8, SATA whole band −43.2, per sub-band −43.3 dB (improvement 25.4 dB).

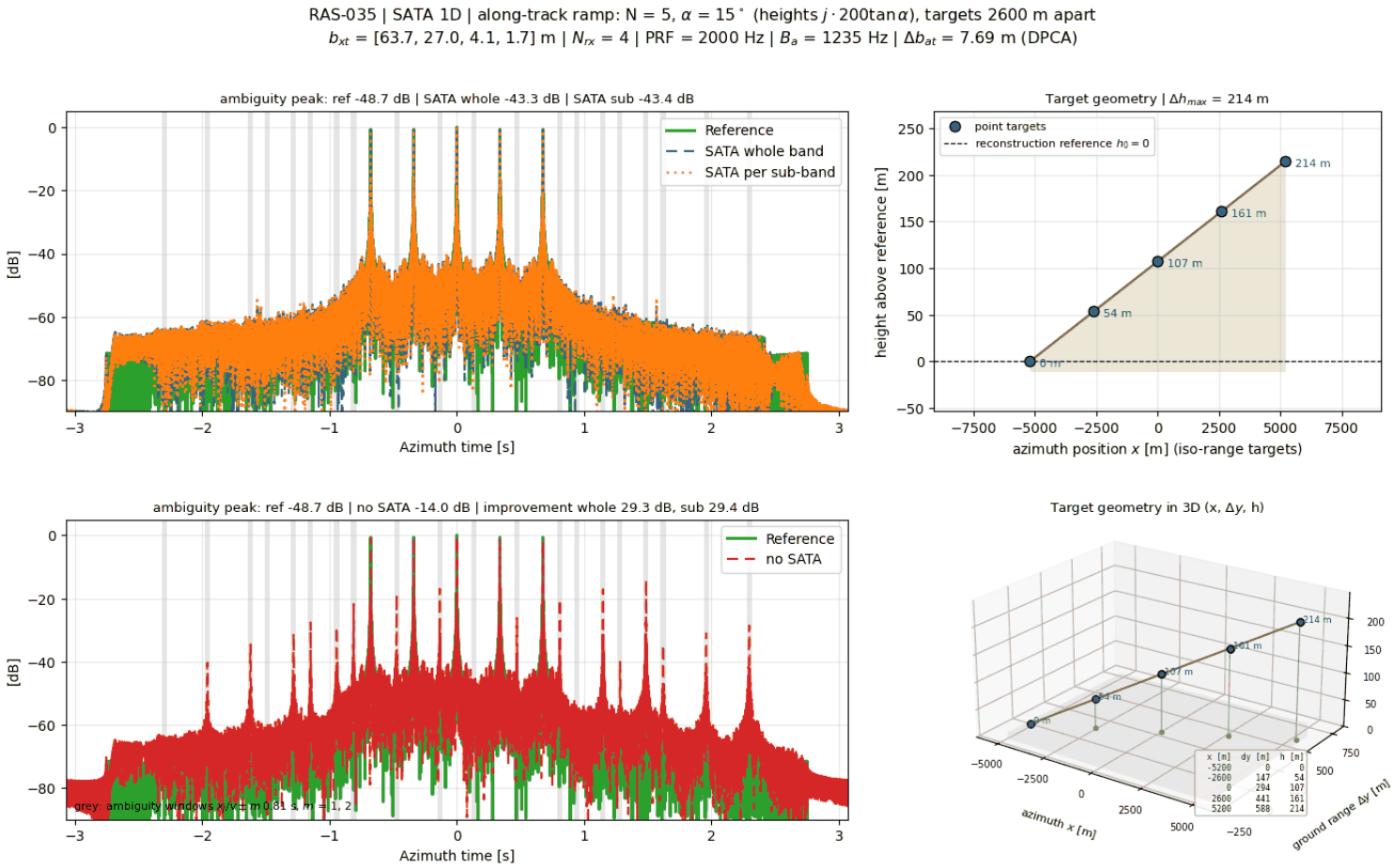


Figure 123: Along-track ramp, 2.6 km apart, normal heights: RAS-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −14.0, SATA whole band −43.3, per sub-band −43.4 dB (improvement 29.3 dB).

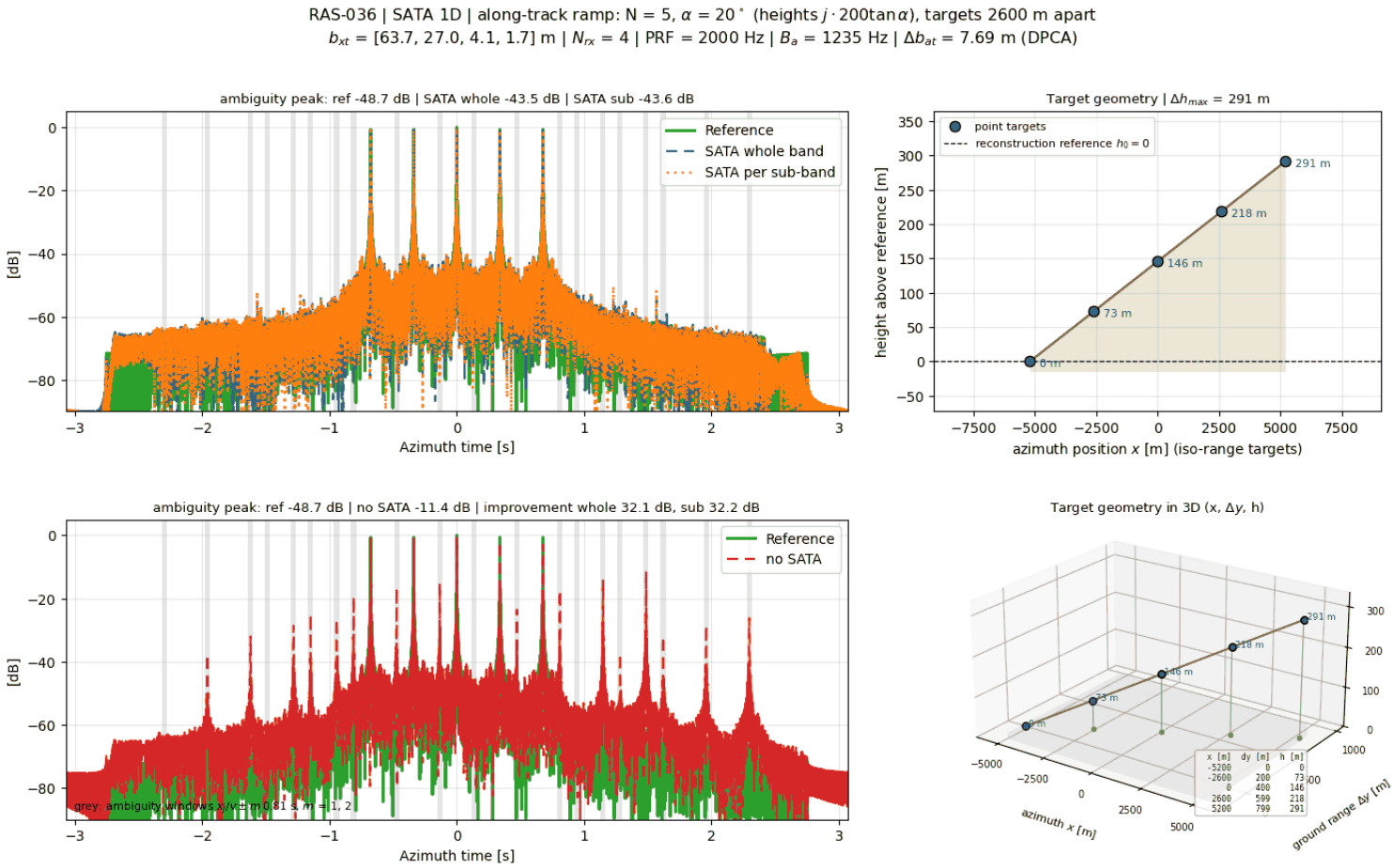


Figure 124: Along-track ramp, 2.6 km apart, normal heights: RAS-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −11.4, SATA whole band −43.5, per sub-band −43.6 dB (improvement 32.1 dB).

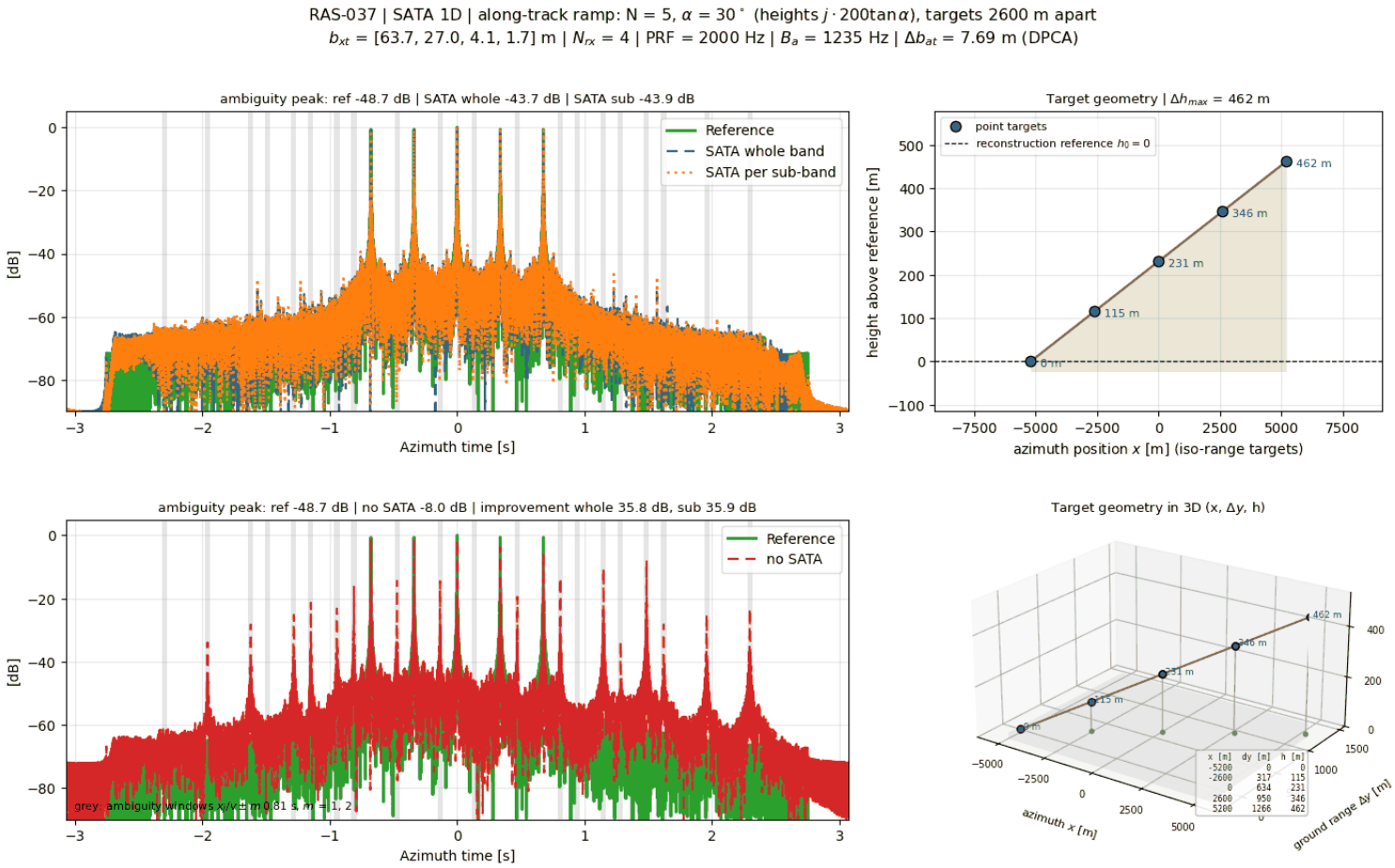


Figure 125: Along-track ramp, 2.6 km apart, normal heights: RAS-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 200 \tan \alpha$). Ambiguity peak: no SATA −8.0, SATA whole band −43.7, per sub-band −43.9 dB (improvement 35.8 dB).

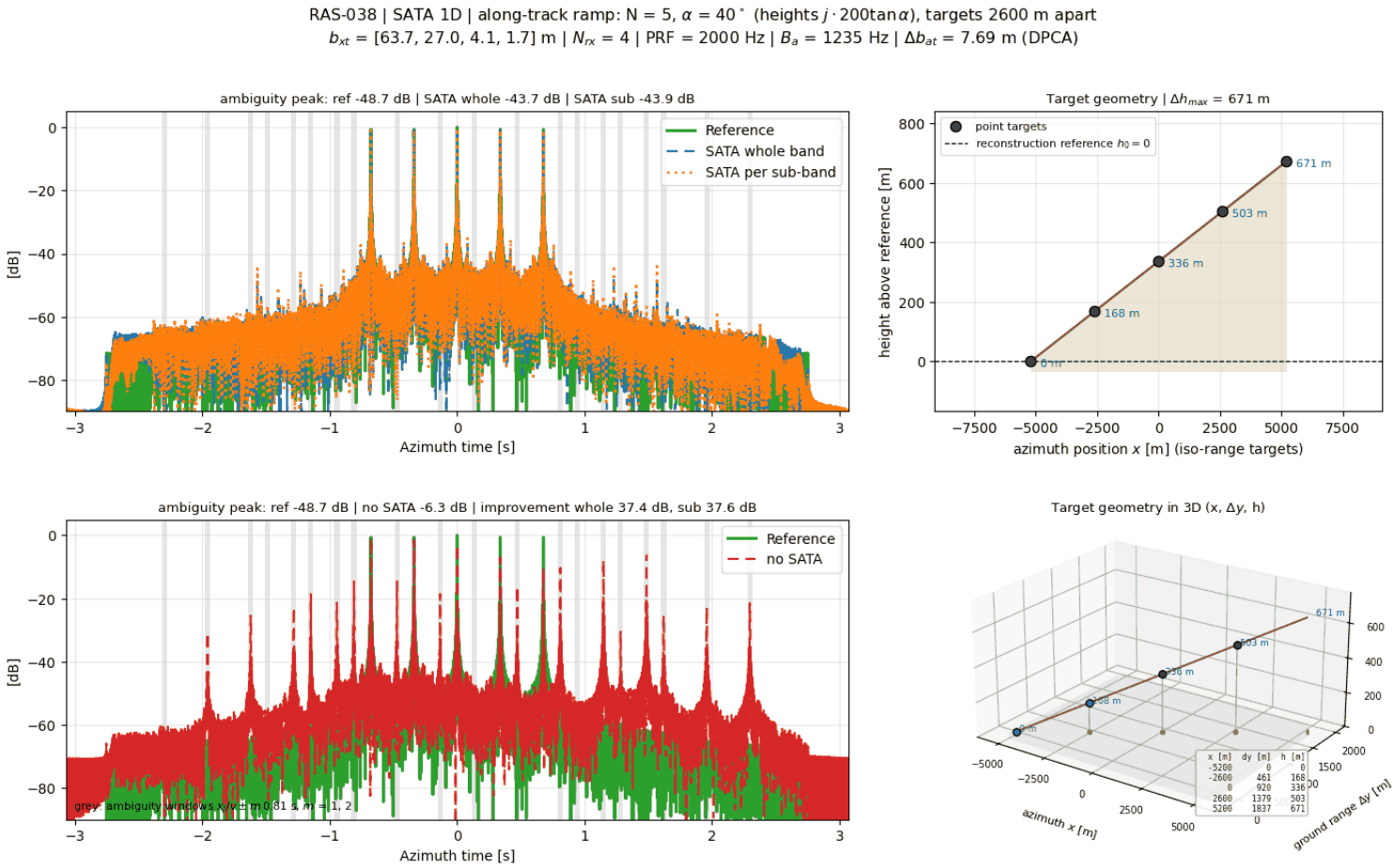


Figure 126: Along-track ramp, 2.6 km apart, normal heights: RAS-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.3, SATA whole band −43.7, per sub-band −43.9 dB (improvement 37.4 dB).

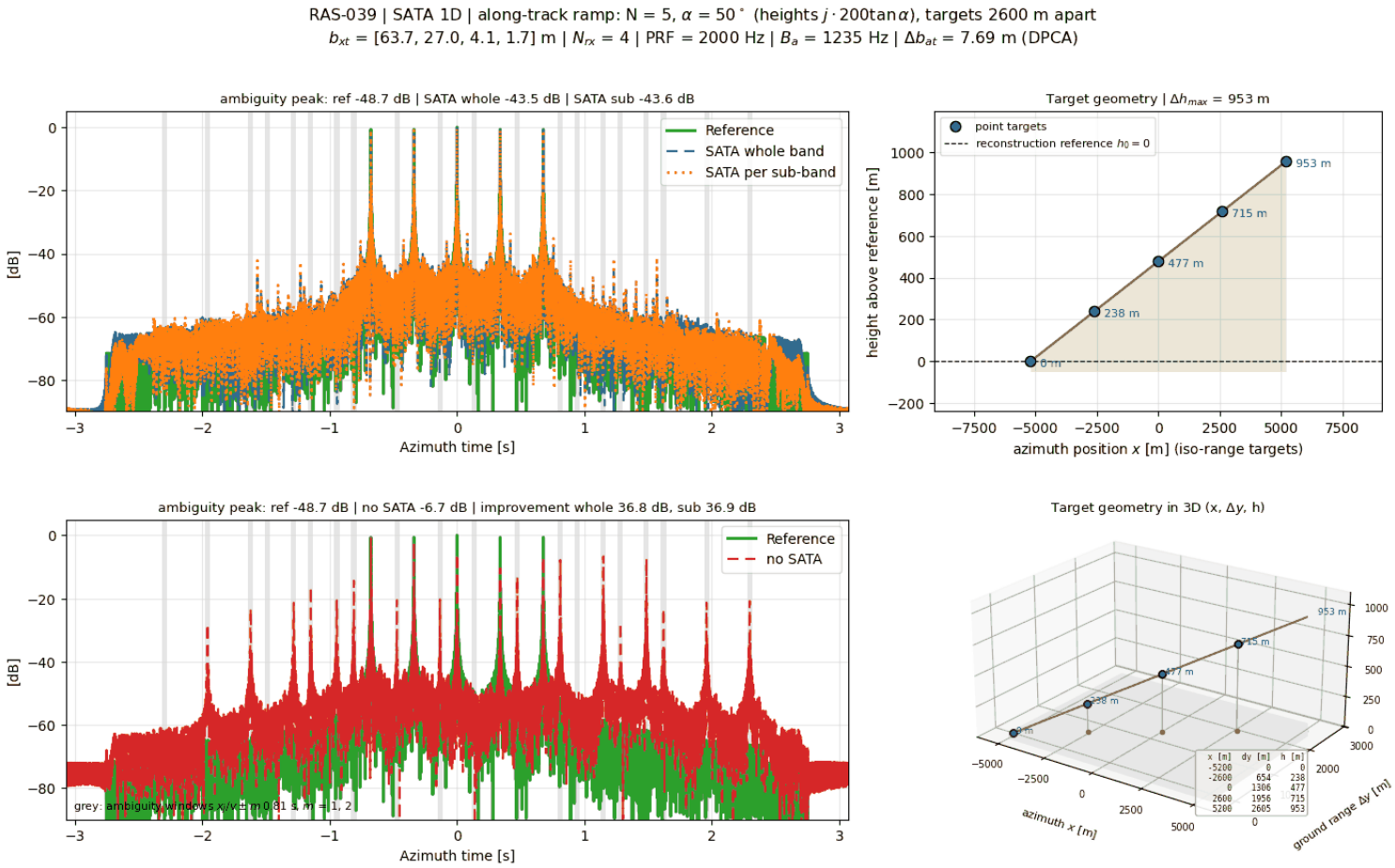


Figure 127: Along-track ramp, 2.6 km apart, normal heights: RAS-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 200 \tan\alpha$). Ambiguity peak: no SATA −6.7, SATA whole band −43.5, per sub-band −43.6 dB (improvement 36.8 dB).

6 IRFs, C: targets 200 m apart, large heights

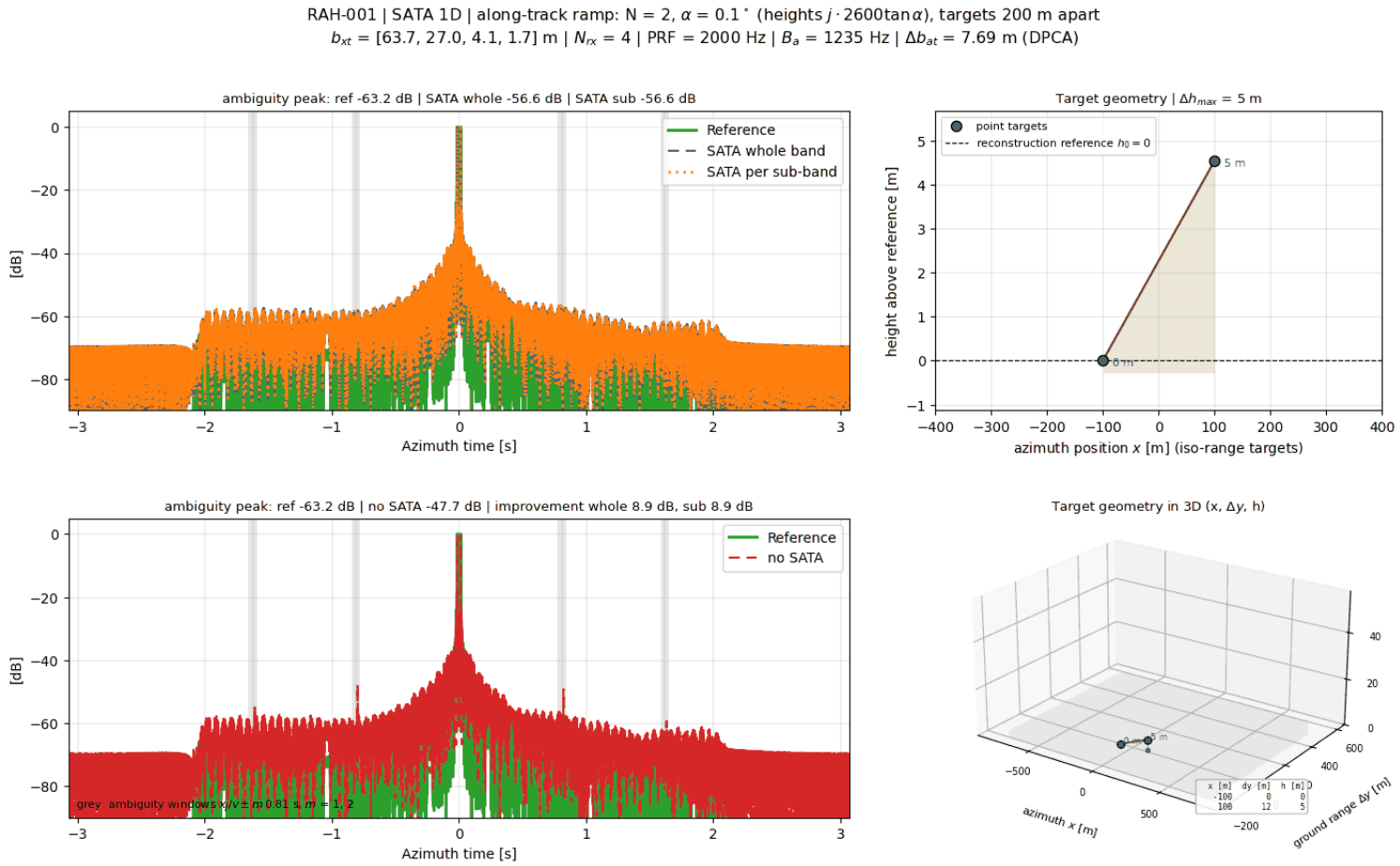


Figure 128: Along-track ramp, 200 m apart, large heights: RAH-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −47.7, SATA whole band −56.6, per sub-band −56.6 dB (improvement 8.9 dB).

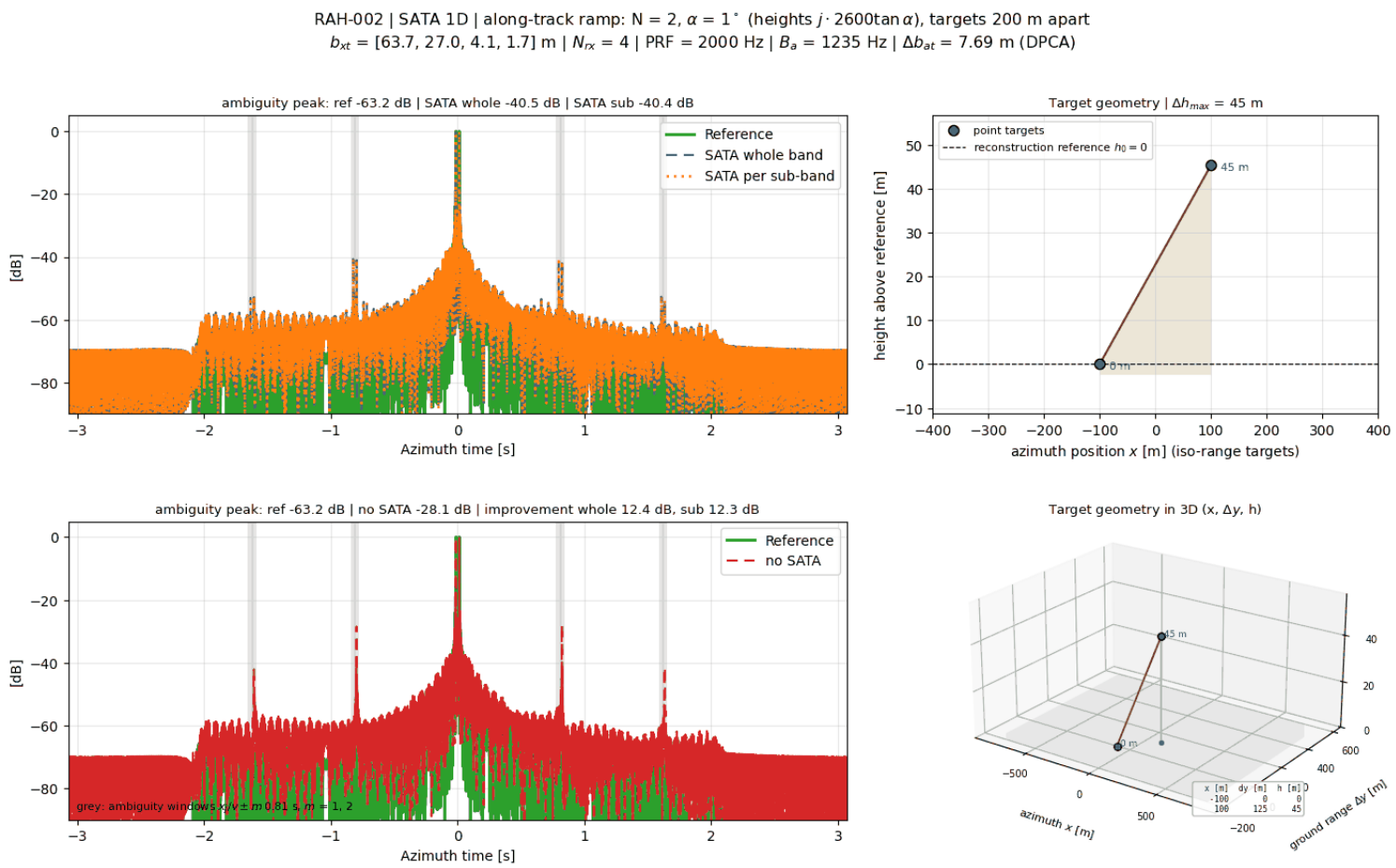


Figure 129: Along-track ramp, 200 m apart, large heights: RAH-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −28.1, SATA whole band −40.5, per sub-band −40.4 dB (improvement 12.4 dB).

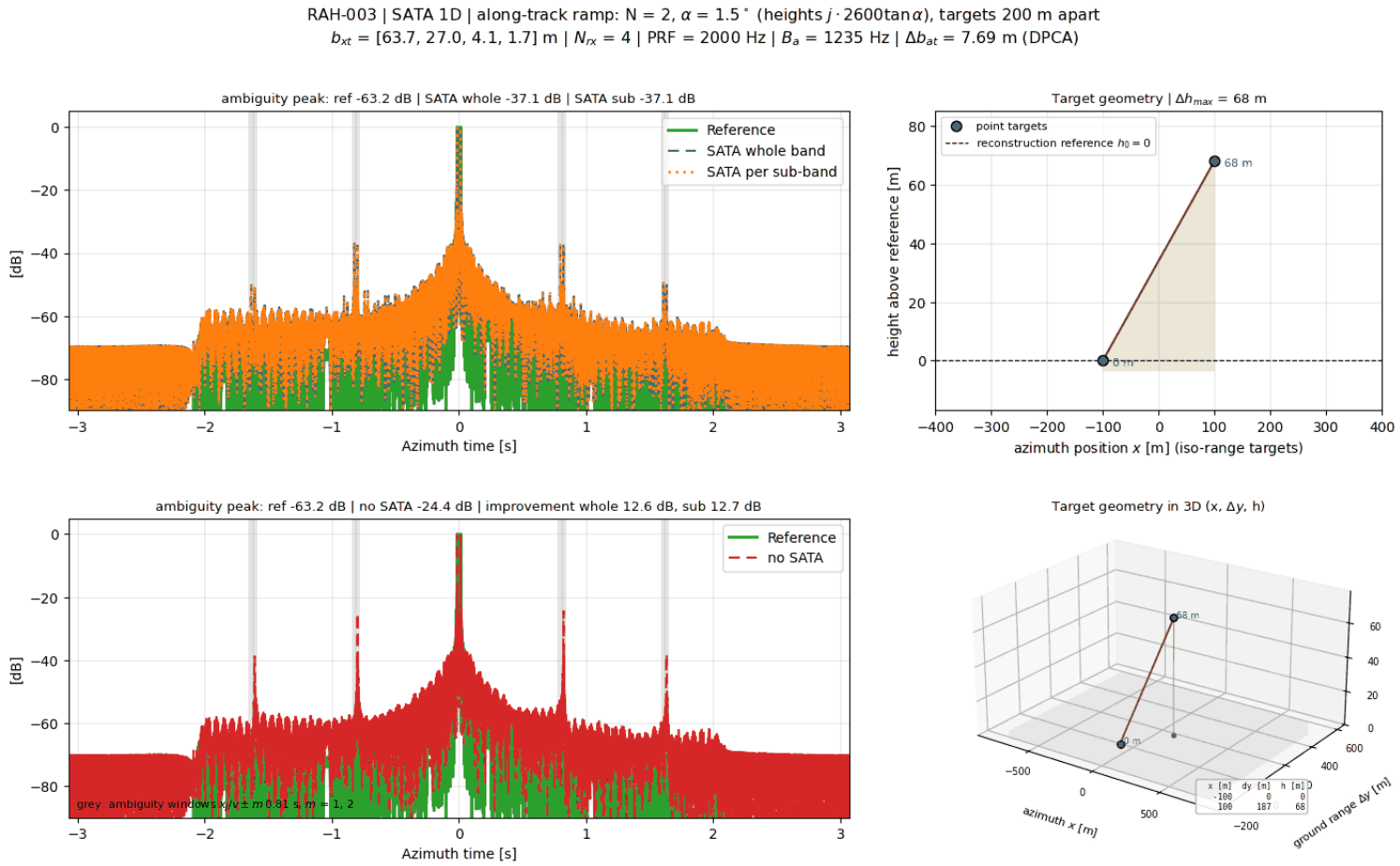


Figure 130: Along-track ramp, 200 m apart, large heights: RAH-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −24.4, SATA whole band −37.1, per sub-band −37.1 dB (improvement 12.6 dB).

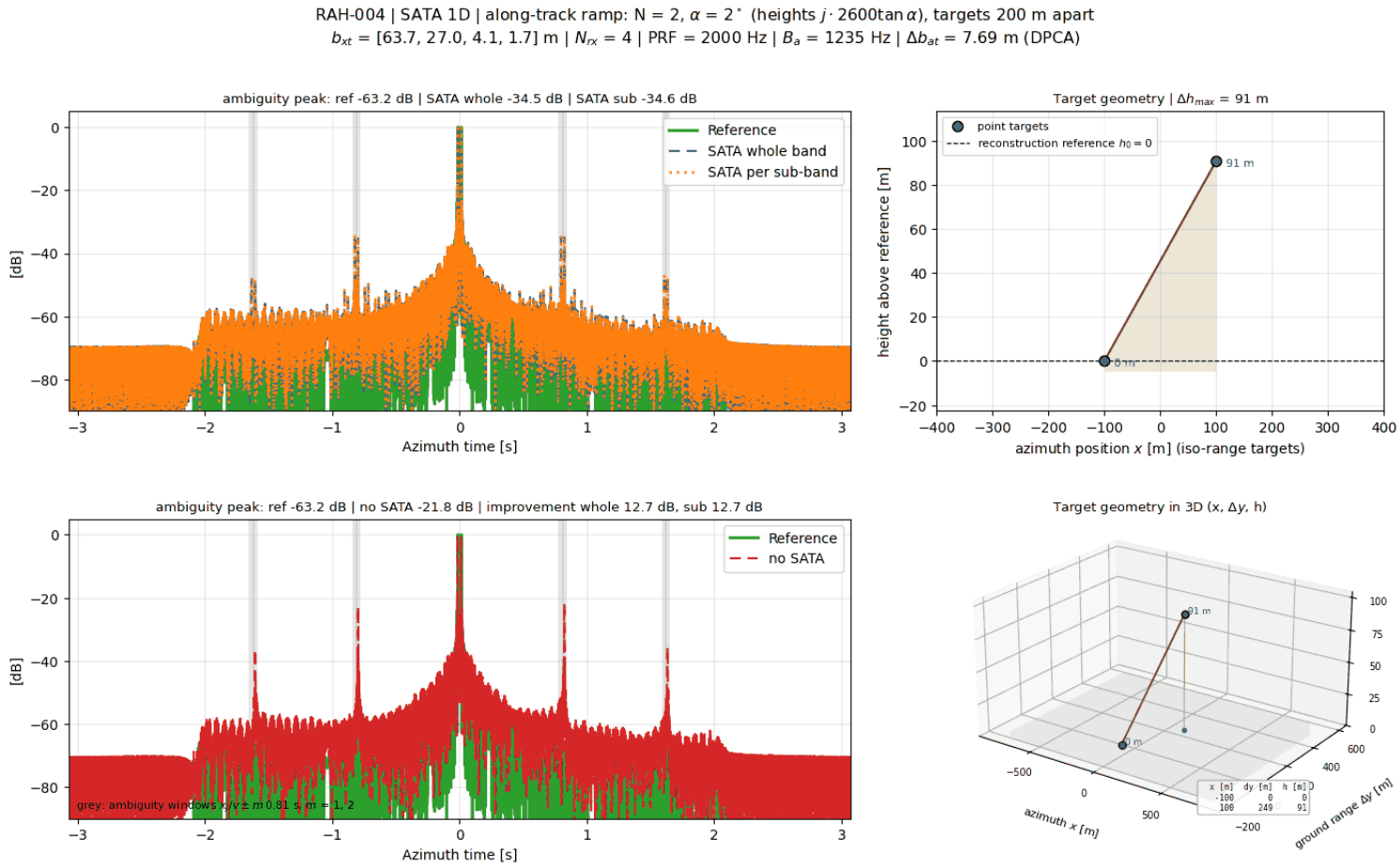


Figure 131: Along-track ramp, 200 m apart, large heights: RAH-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −21.8, SATA whole band −34.5, per sub-band −34.6 dB (improvement 12.7 dB).

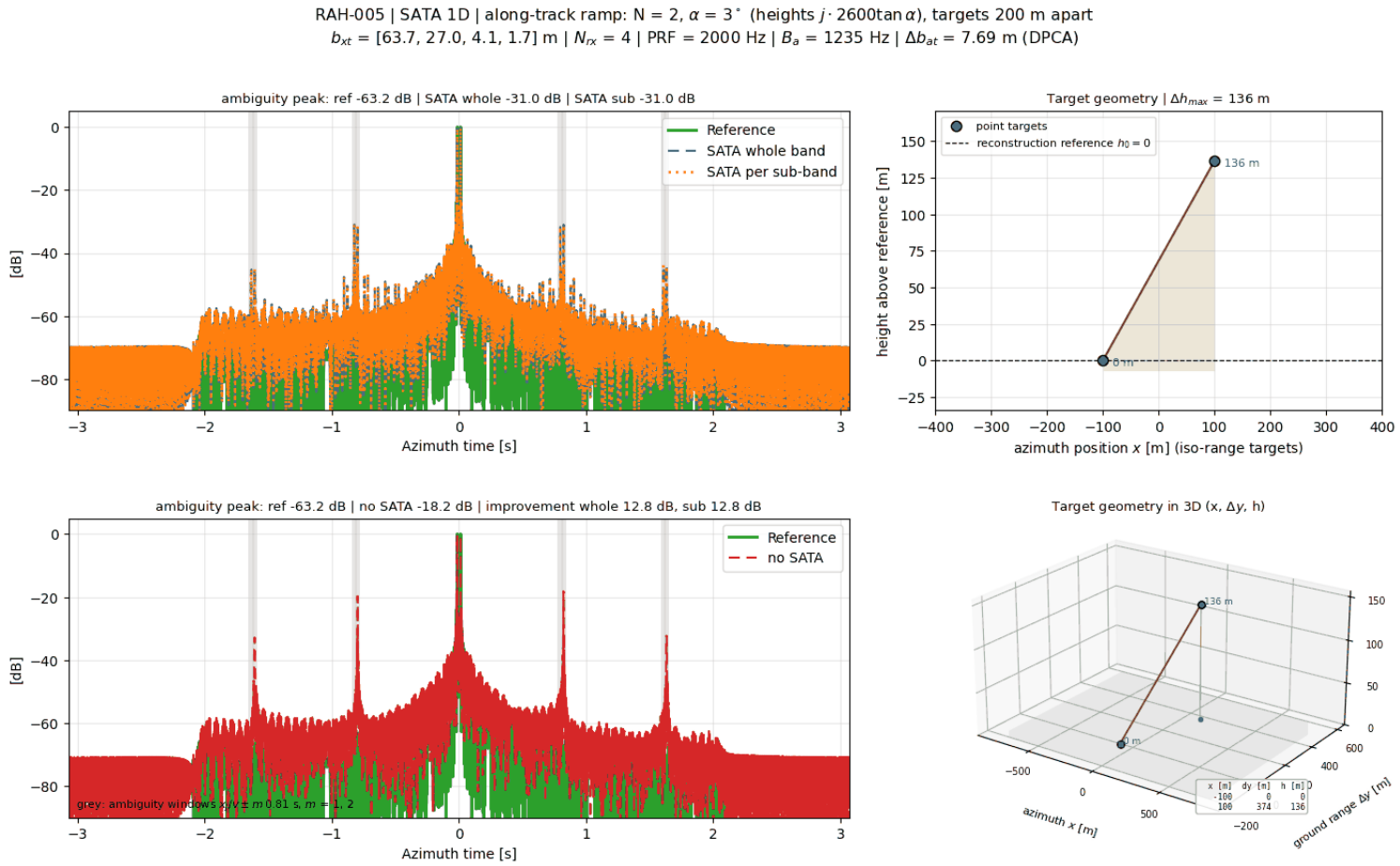


Figure 132: Along-track ramp, 200 m apart, large heights: RAH-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −18.2, SATA whole band −31.0, per sub-band −31.0 dB (improvement 12.8 dB).

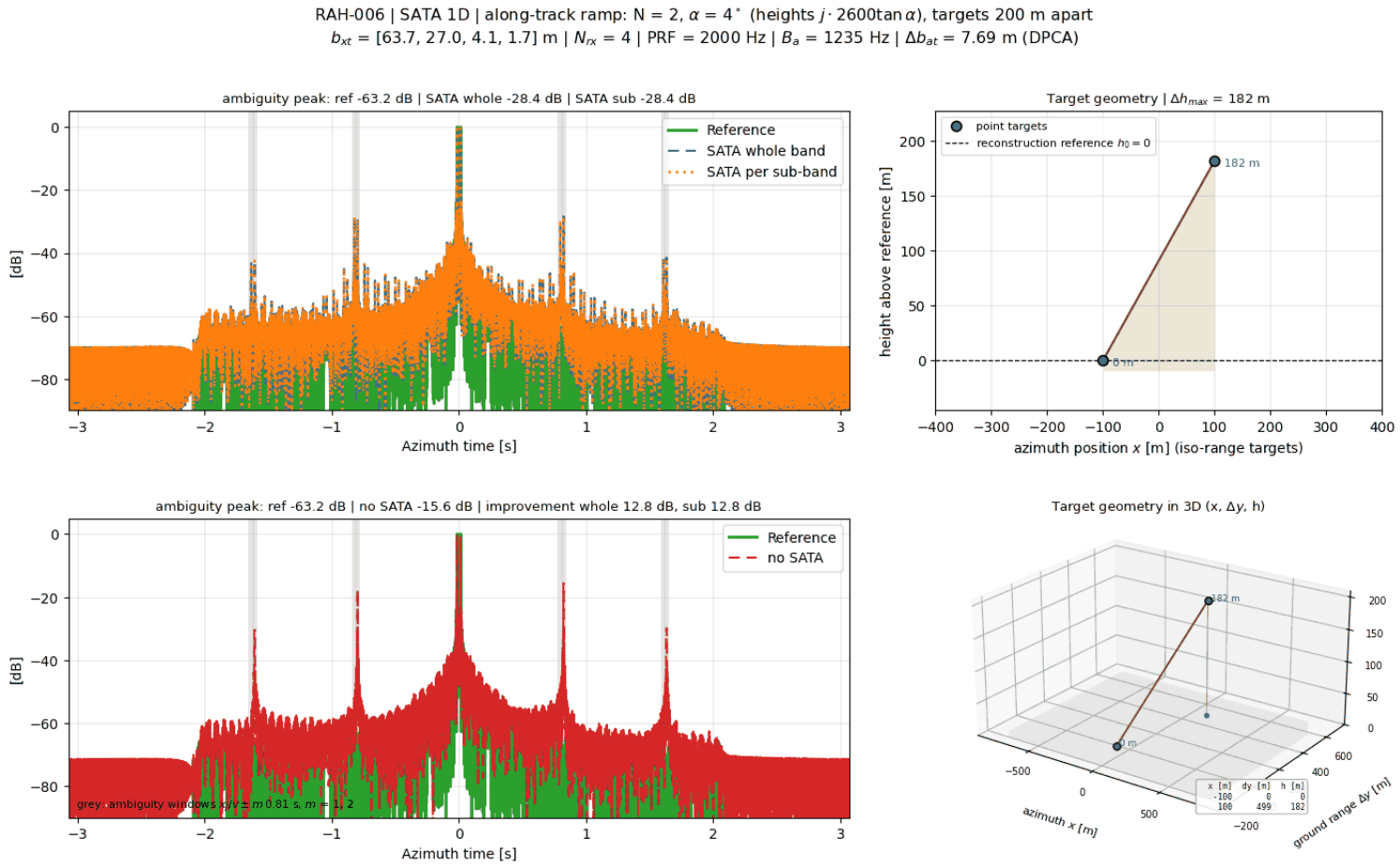


Figure 133: Along-track ramp, 200 m apart, large heights: RAH-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −15.6, SATA whole band −28.4, per sub-band −28.4 dB (improvement 12.8 dB).

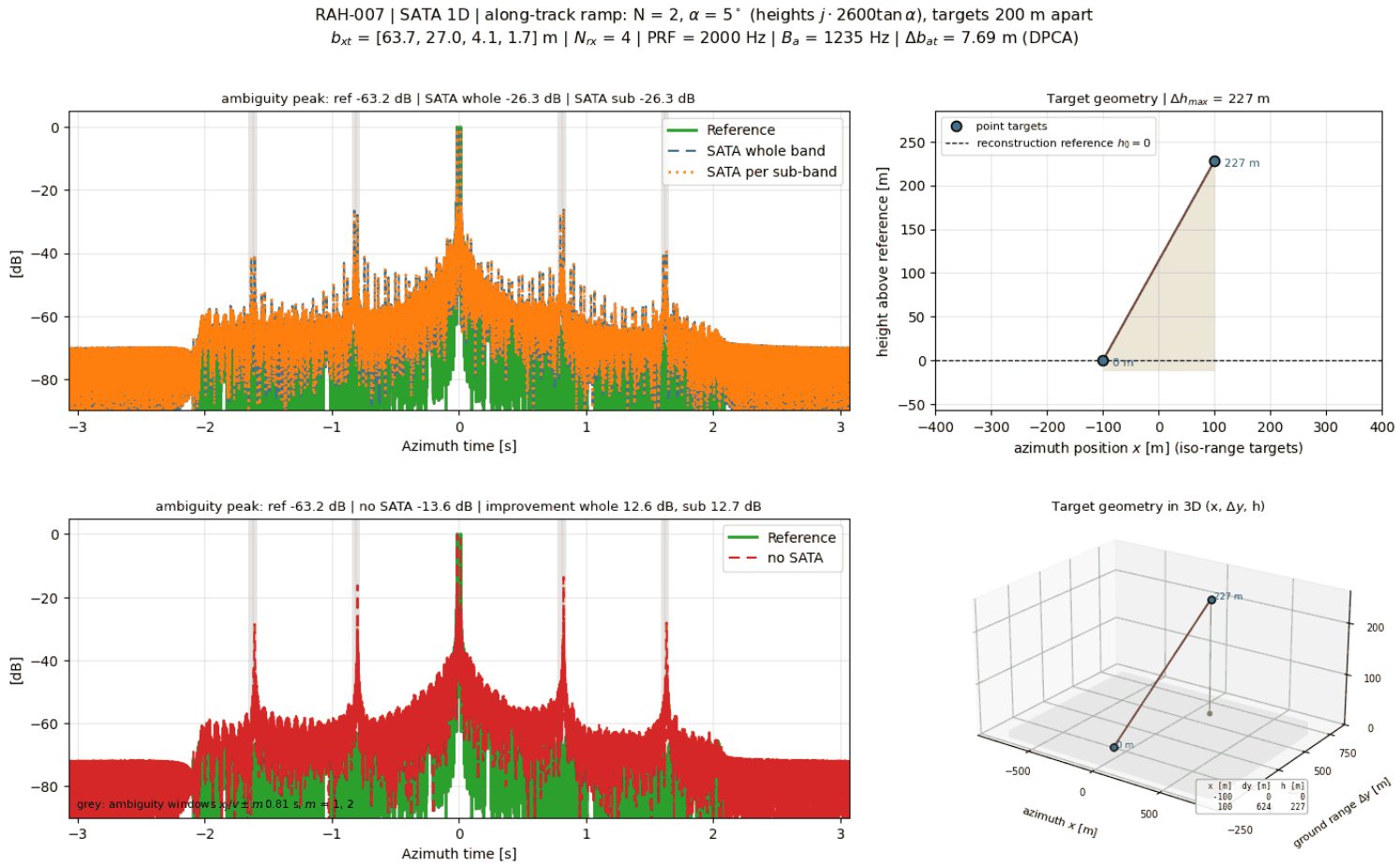


Figure 134: Along-track ramp, 200 m apart, large heights: RAH-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −13.6, SATA whole band −26.3, per sub-band −26.3 dB (improvement 12.6 dB).

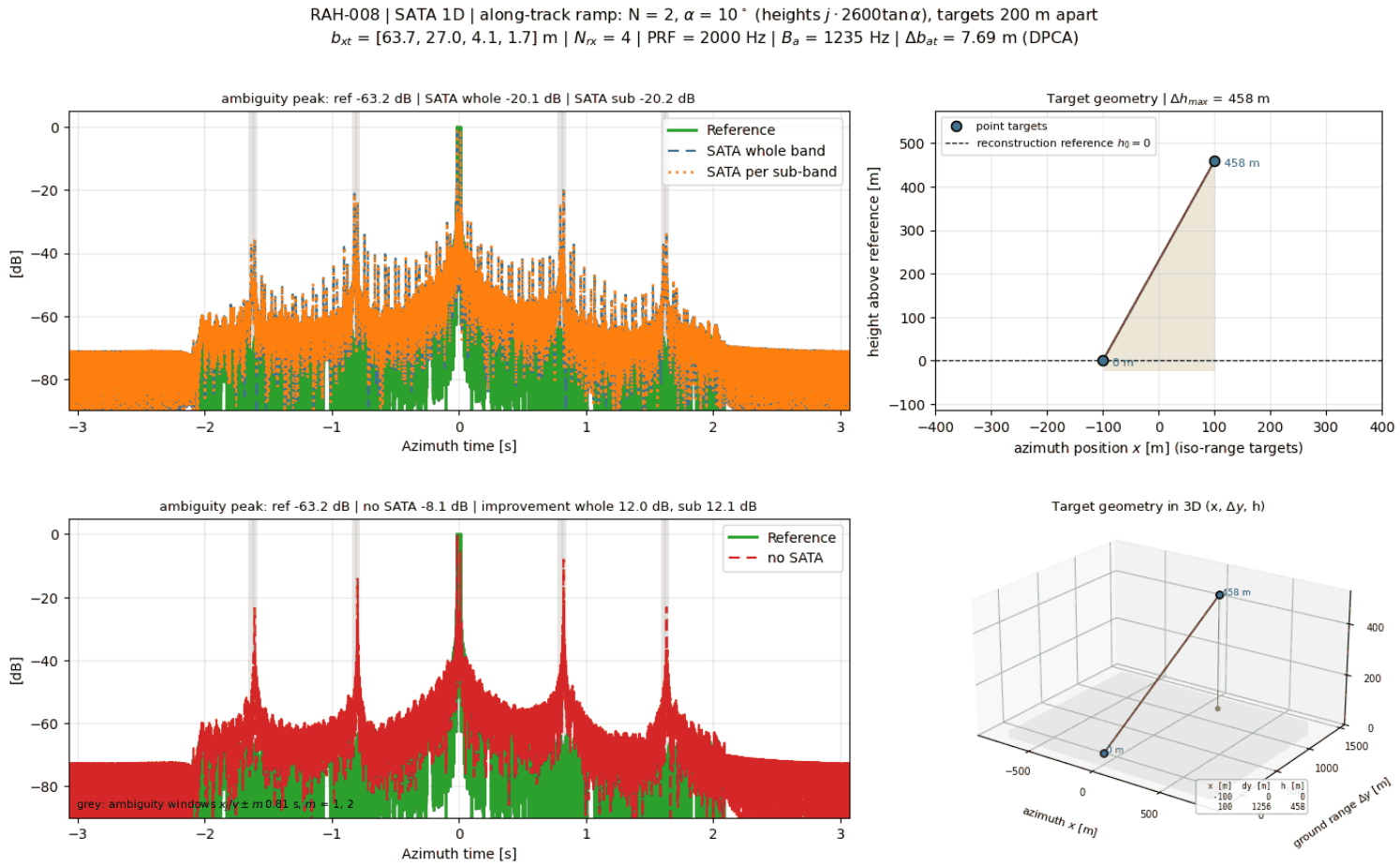


Figure 135: Along-track ramp, 200 m apart, large heights: RAH-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.1, SATA whole band −20.1, per sub-band −20.2 dB (improvement 12.0 dB).

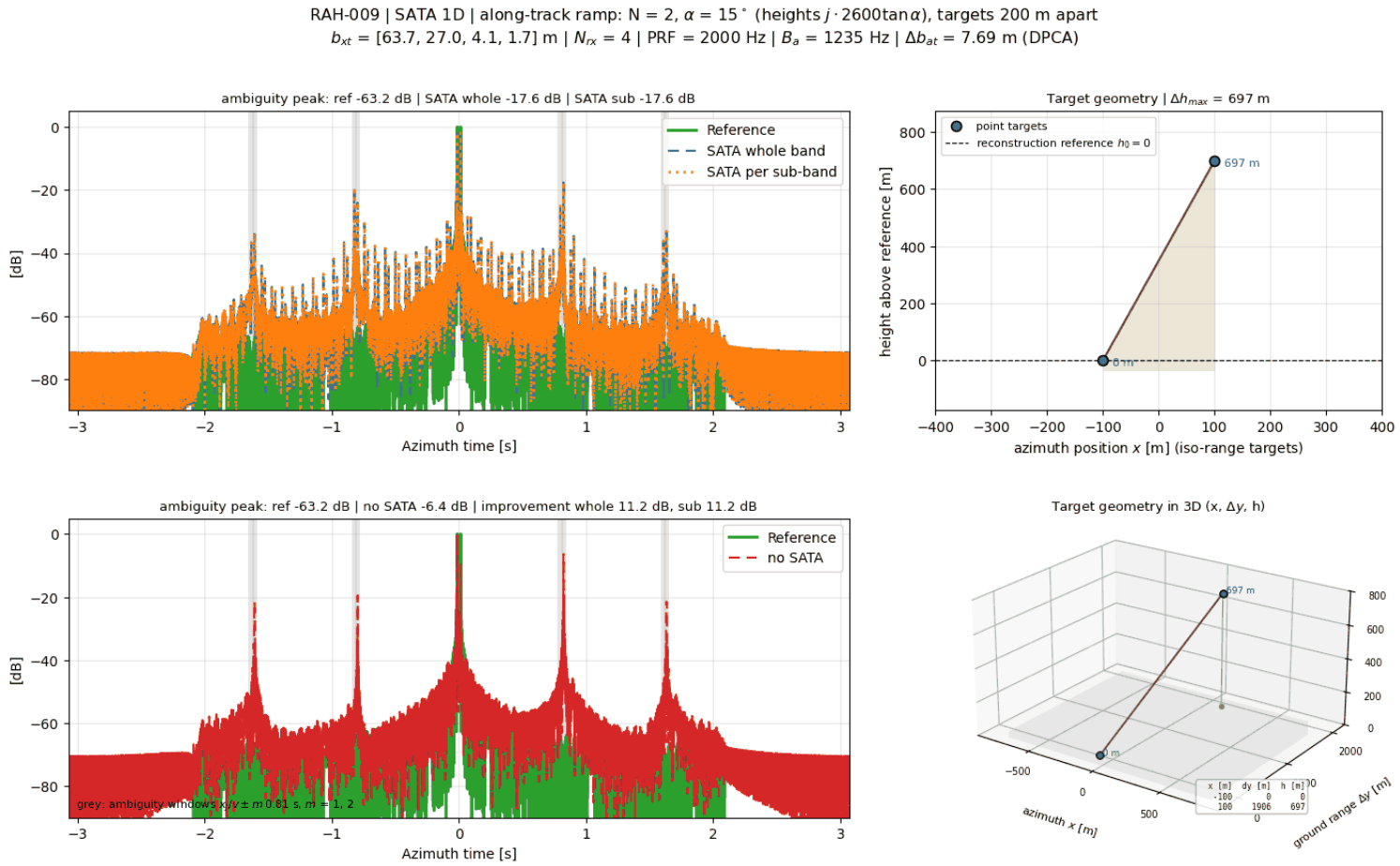


Figure 136: Along-track ramp, 200 m apart, large heights: RAH-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −17.6, per sub-band −17.6 dB (improvement 11.2 dB).

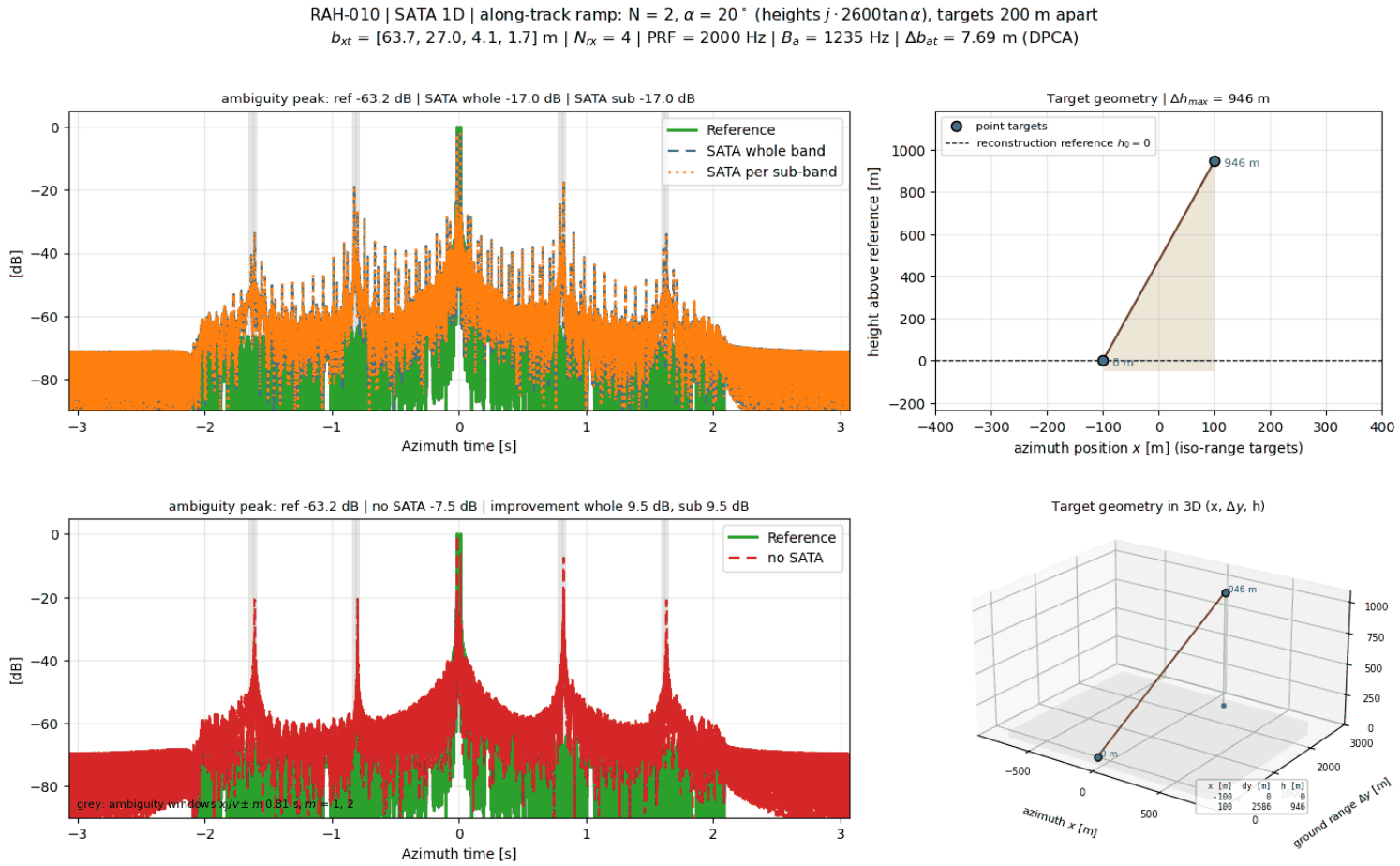


Figure 137: Along-track ramp, 200 m apart, large heights: RAH-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.5, SATA whole band −17.0, per sub-band −17.0 dB (improvement 9.5 dB).

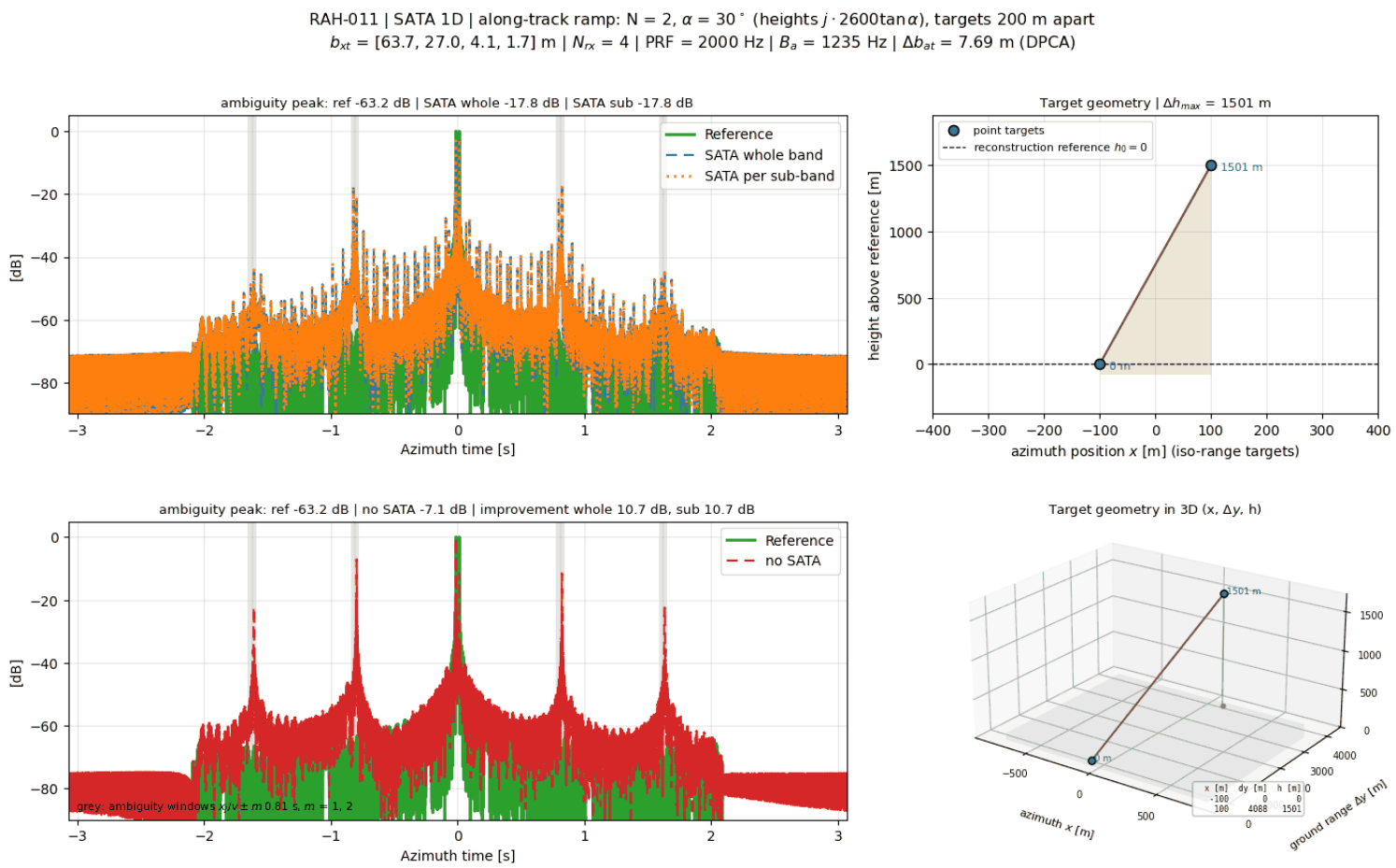


Figure 138: Along-track ramp, 200 m apart, large heights: RAH-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.1, SATA whole band −17.8, per sub-band −17.8 dB (improvement 10.7 dB).

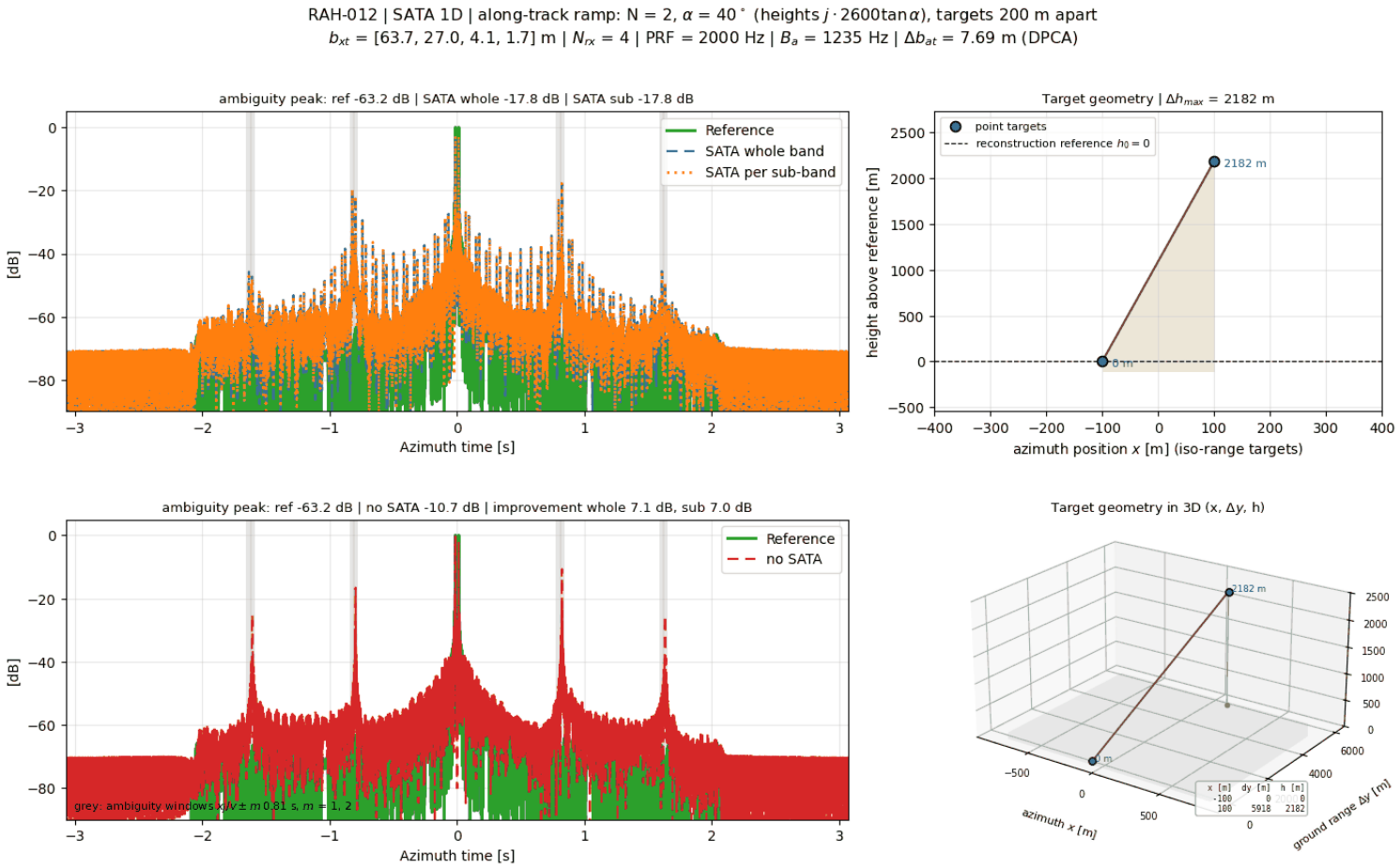


Figure 139: Along-track ramp, 200 m apart, large heights: RAH-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −10.7, SATA whole band −17.8, per sub-band −17.8 dB (improvement 7.1 dB).

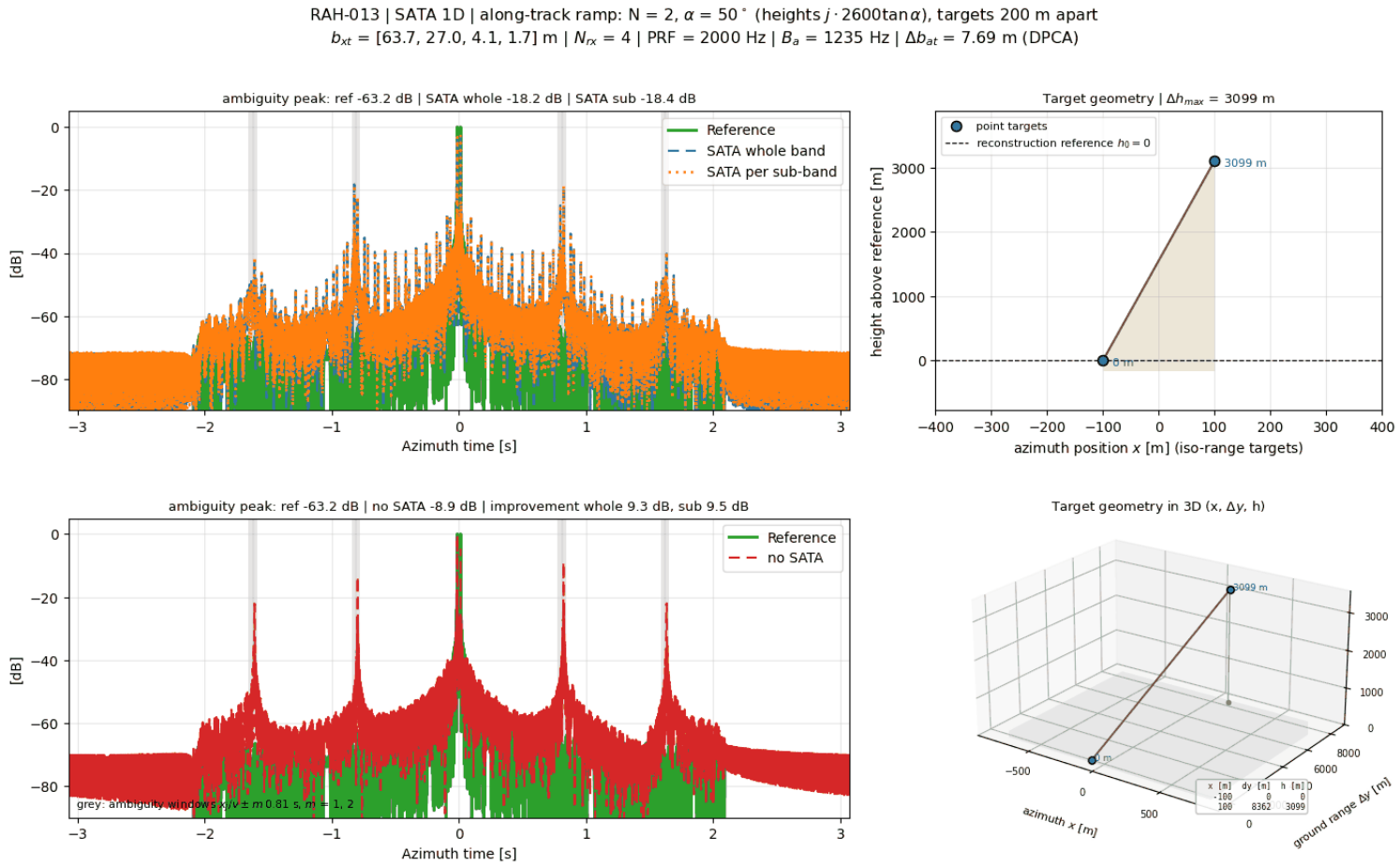


Figure 140: Along-track ramp, 200 m apart, large heights: RAH-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.9, SATA whole band −18.2, per sub-band −18.4 dB (improvement 9.3 dB).

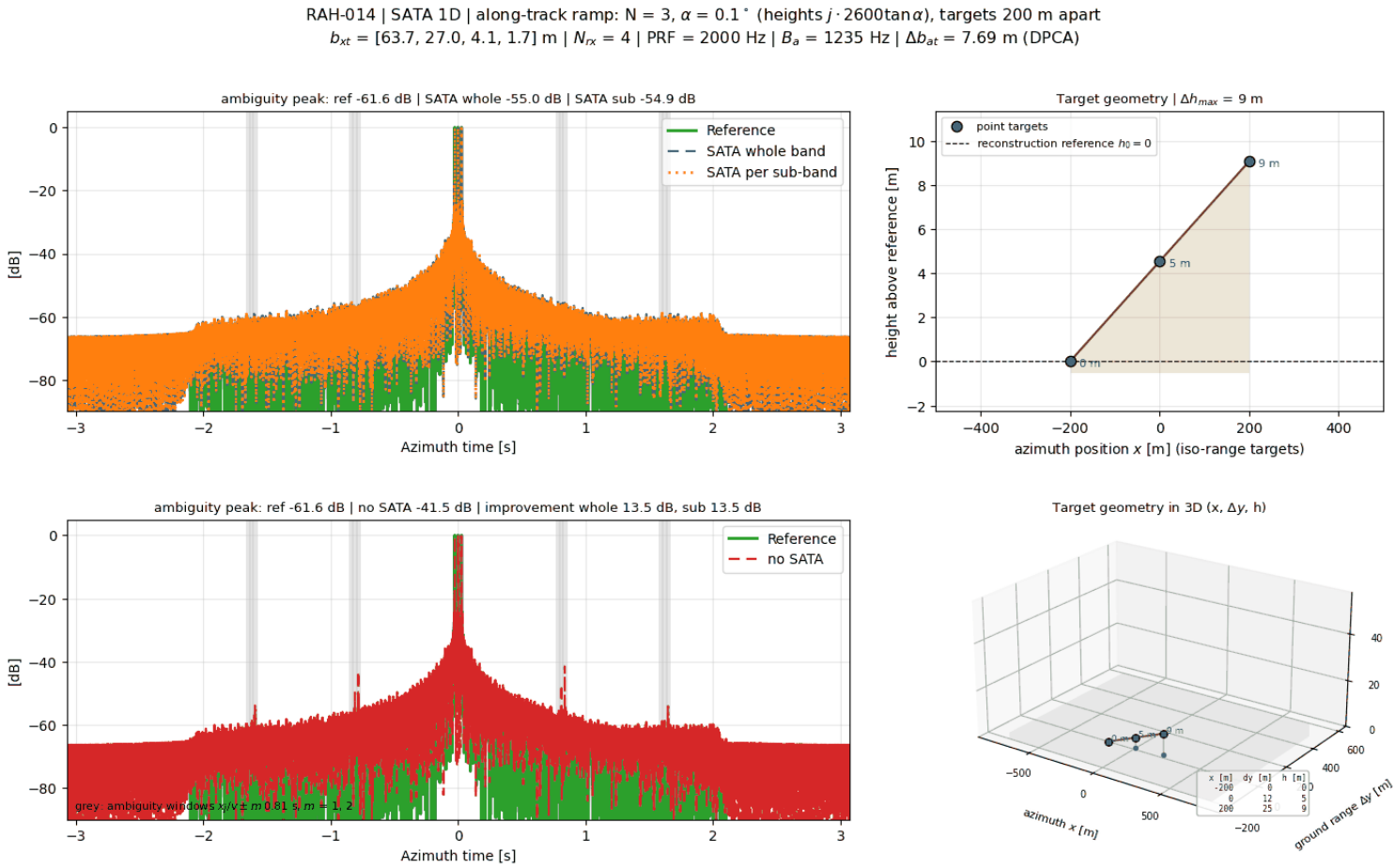


Figure 141: Along-track ramp, 200 m apart, large heights: RAH-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −41.5, SATA whole band −55.0, per sub-band −54.9 dB (improvement 13.5 dB).

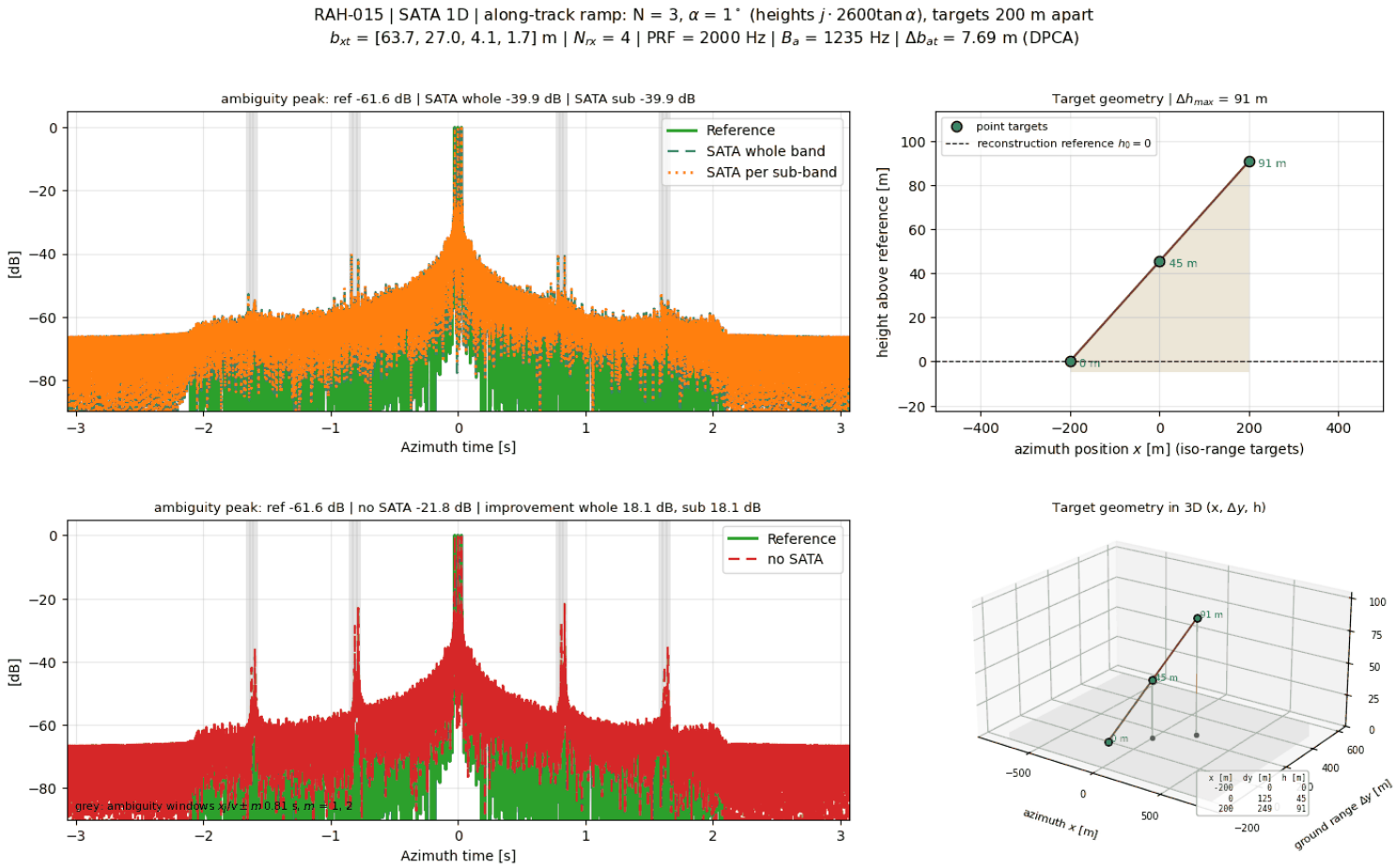


Figure 142: Along-track ramp, 200 m apart, large heights: RAH-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −21.8, SATA whole band −39.9, per sub-band −39.9 dB (improvement 18.1 dB).

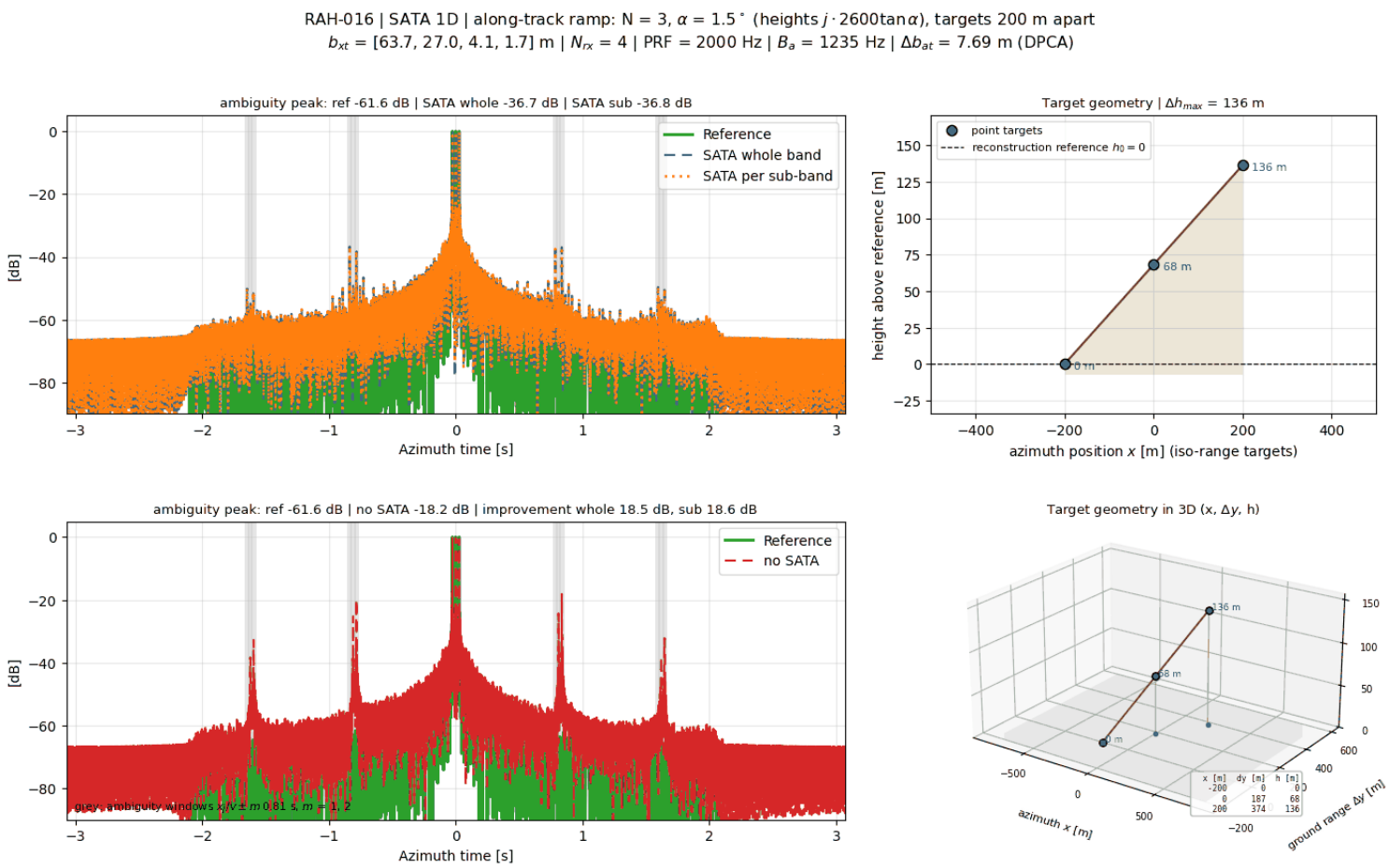


Figure 143: Along-track ramp, 200 m apart, large heights: RAH-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −18.2, SATA whole band −36.7, per sub-band −36.8 dB (improvement 18.5 dB).

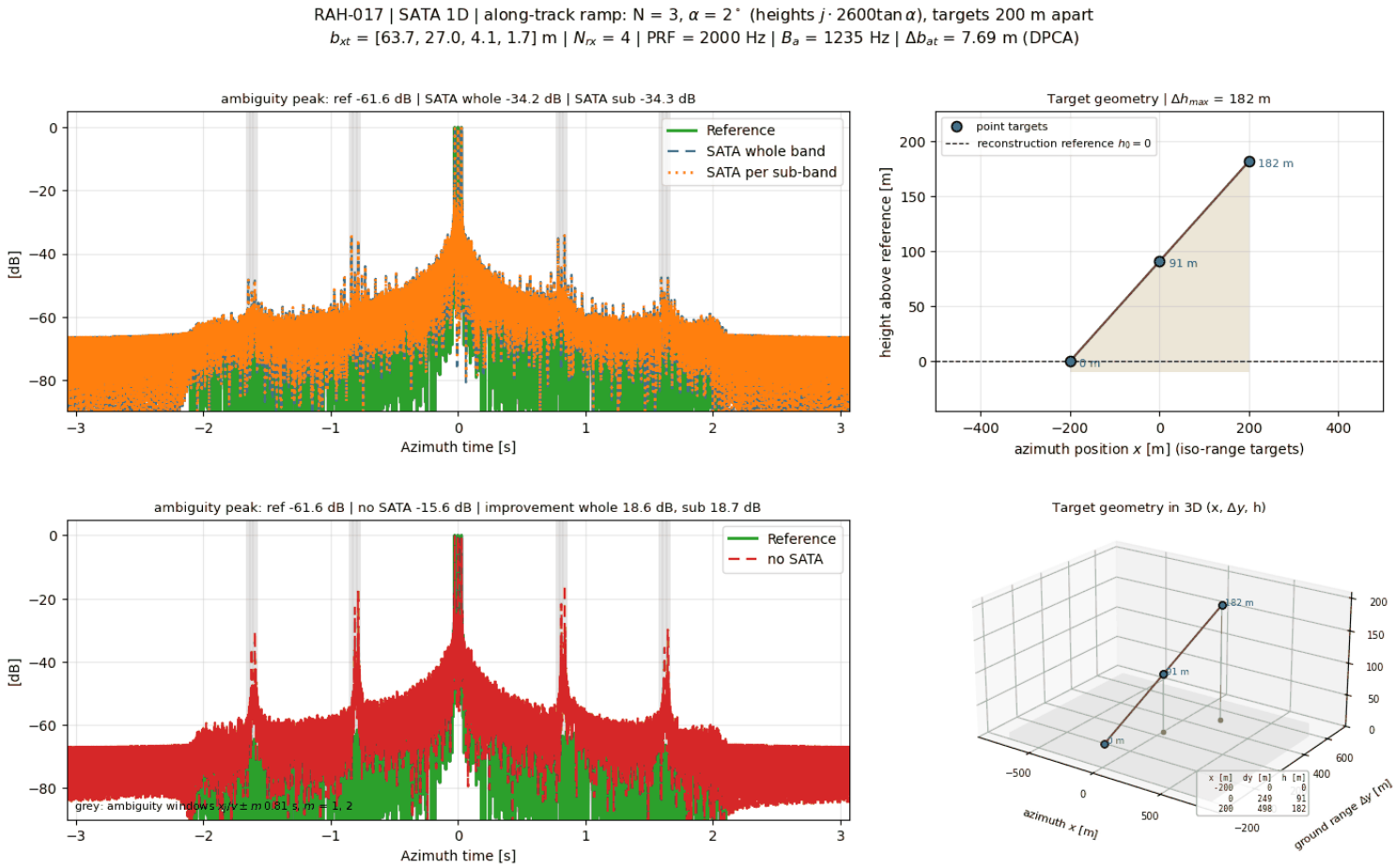


Figure 144: Along-track ramp, 200 m apart, large heights: RAH-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −15.6, SATA whole band −34.2, per sub-band −34.3 dB (improvement 18.6 dB).

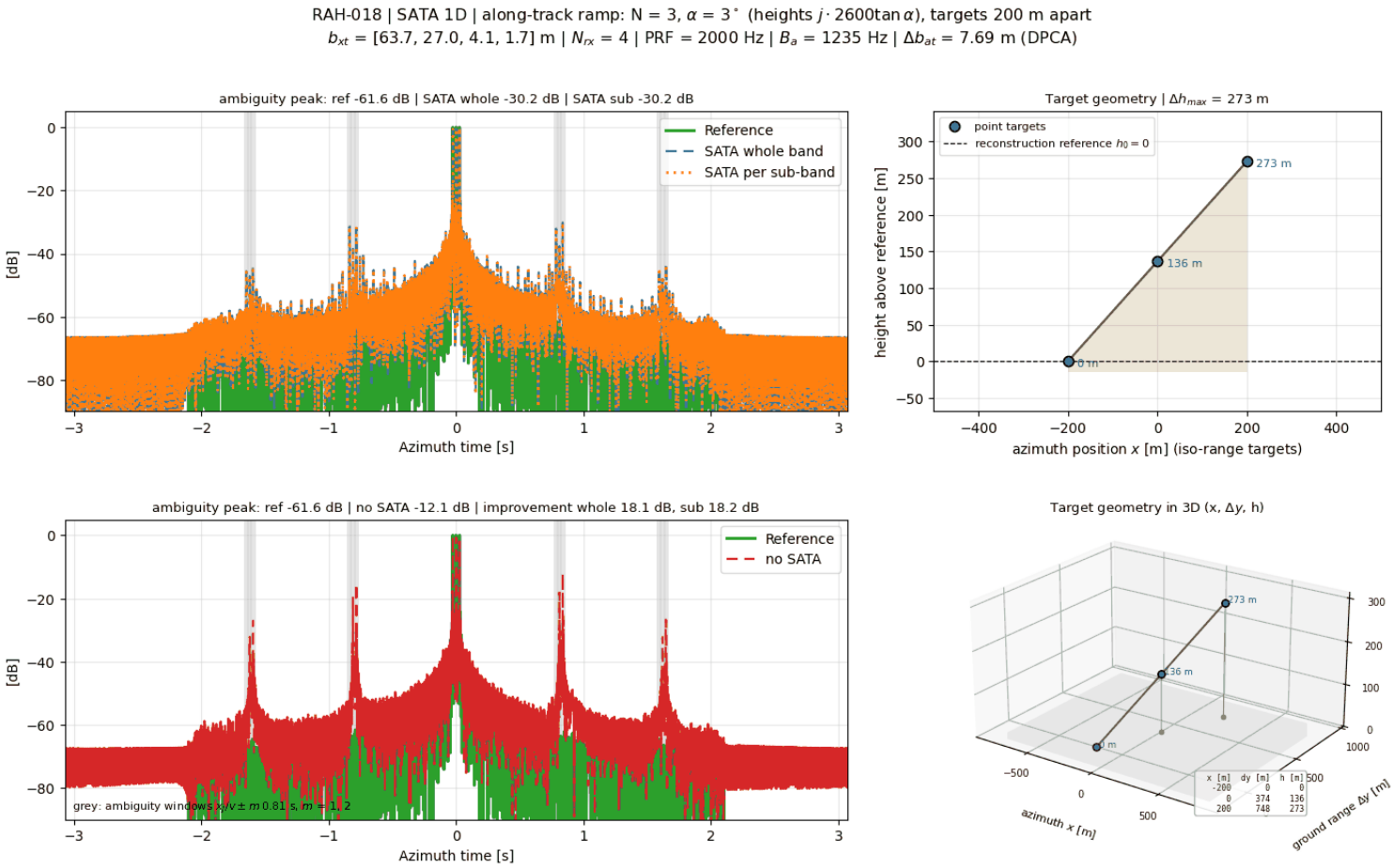


Figure 145: Along-track ramp, 200 m apart, large heights: RAH-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −12.1, SATA whole band −30.2, per sub-band −30.2 dB (improvement 18.1 dB).

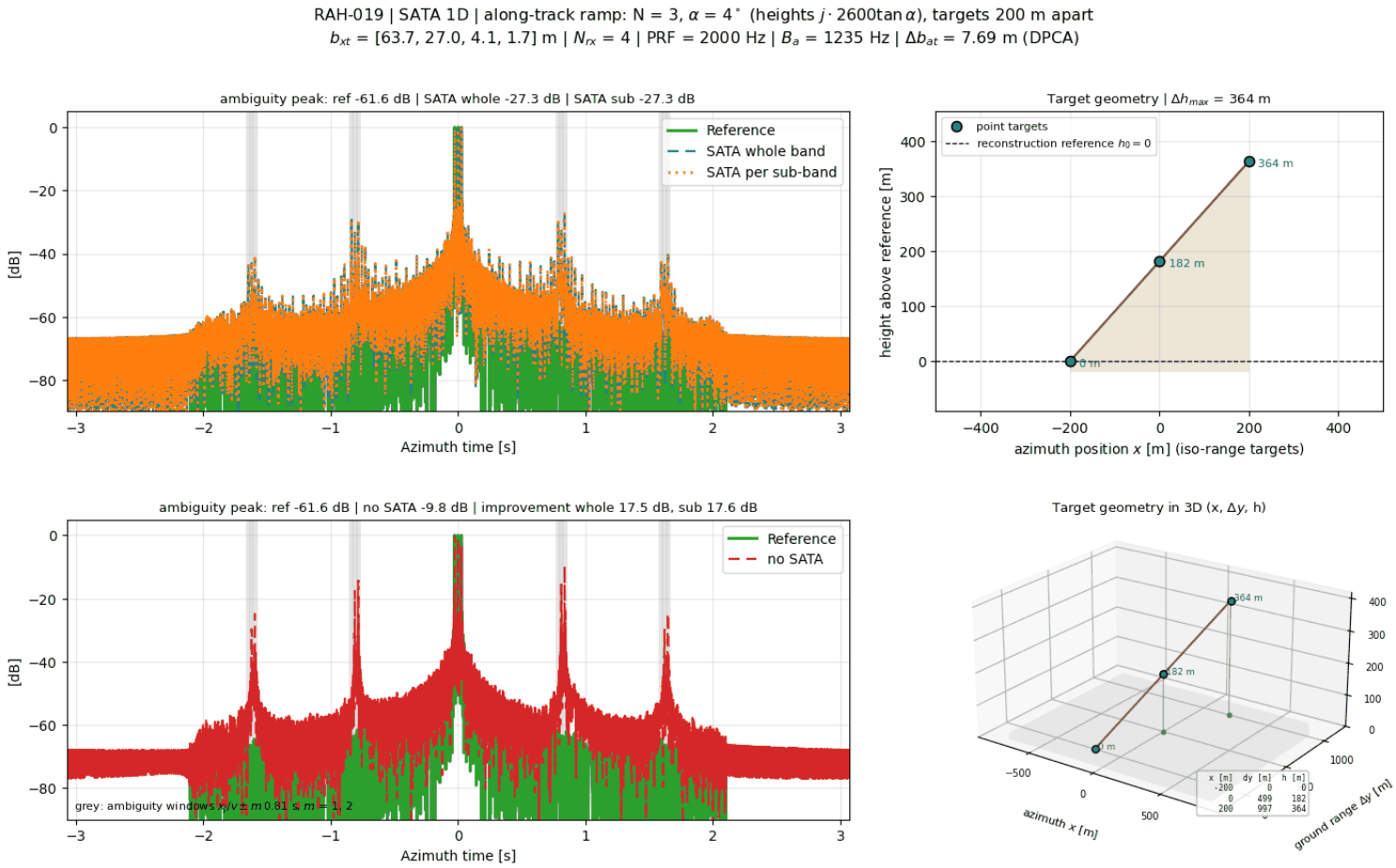


Figure 146: Along-track ramp, 200 m apart, large heights: RAH-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.8, SATA whole band −27.3, per sub-band −27.3 dB (improvement 17.5 dB).

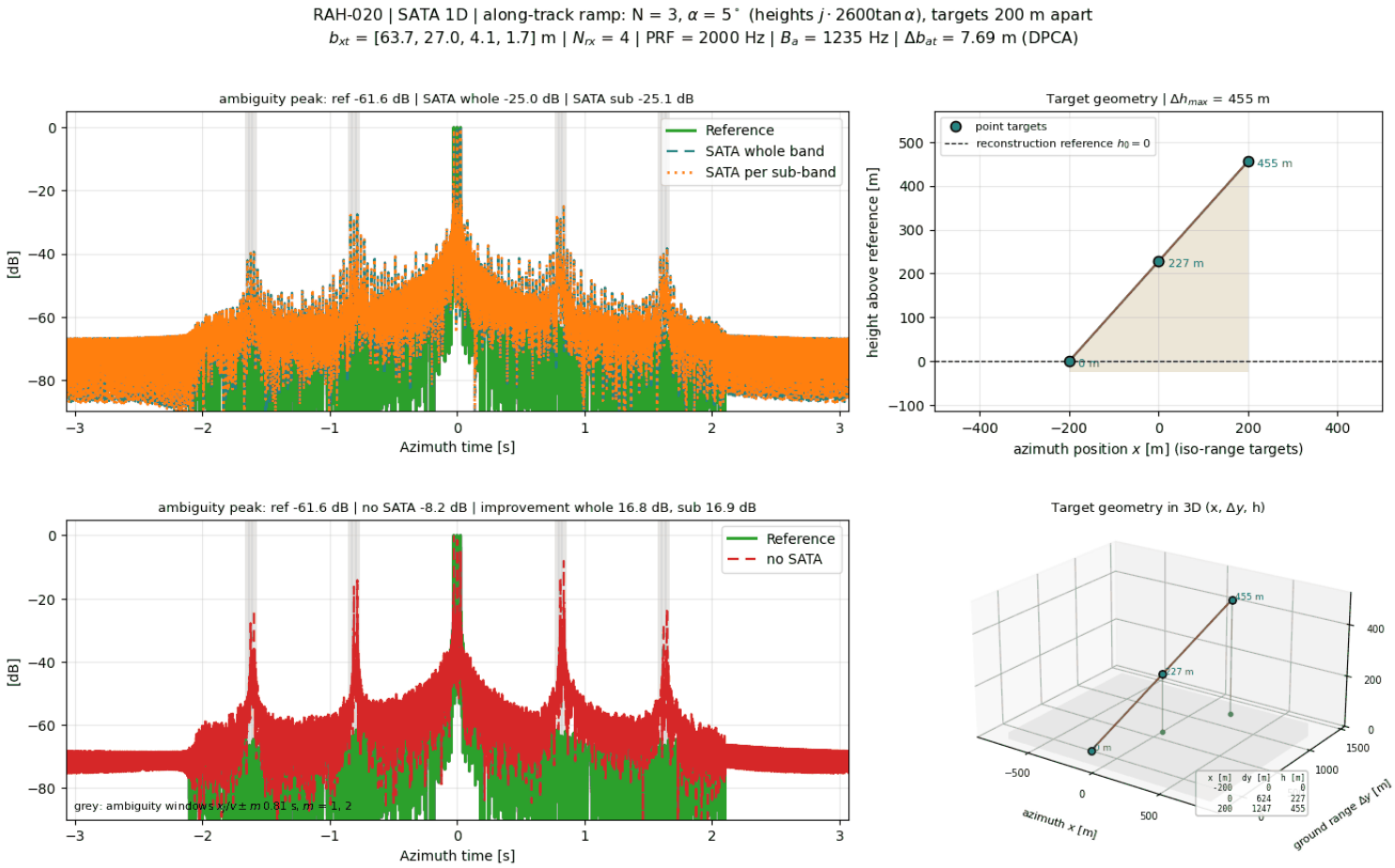


Figure 147: Along-track ramp, 200 m apart, large heights: RAH-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.2, SATA whole band −25.0, per sub-band −25.1 dB (improvement 16.8 dB).

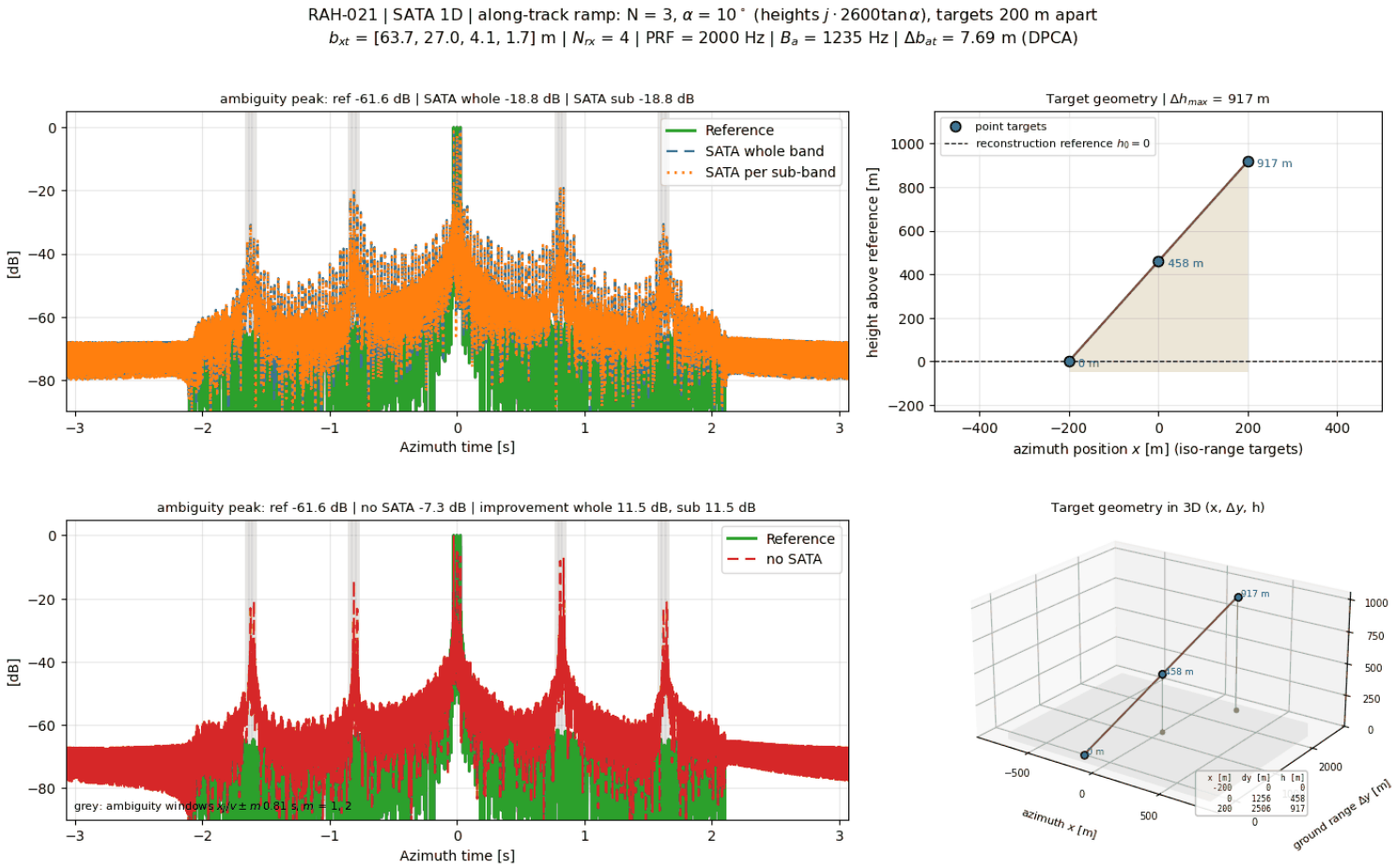


Figure 148: Along-track ramp, 200 m apart, large heights: RAH-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.3, SATA whole band −18.8, per sub-band −18.8 dB (improvement 11.5 dB).

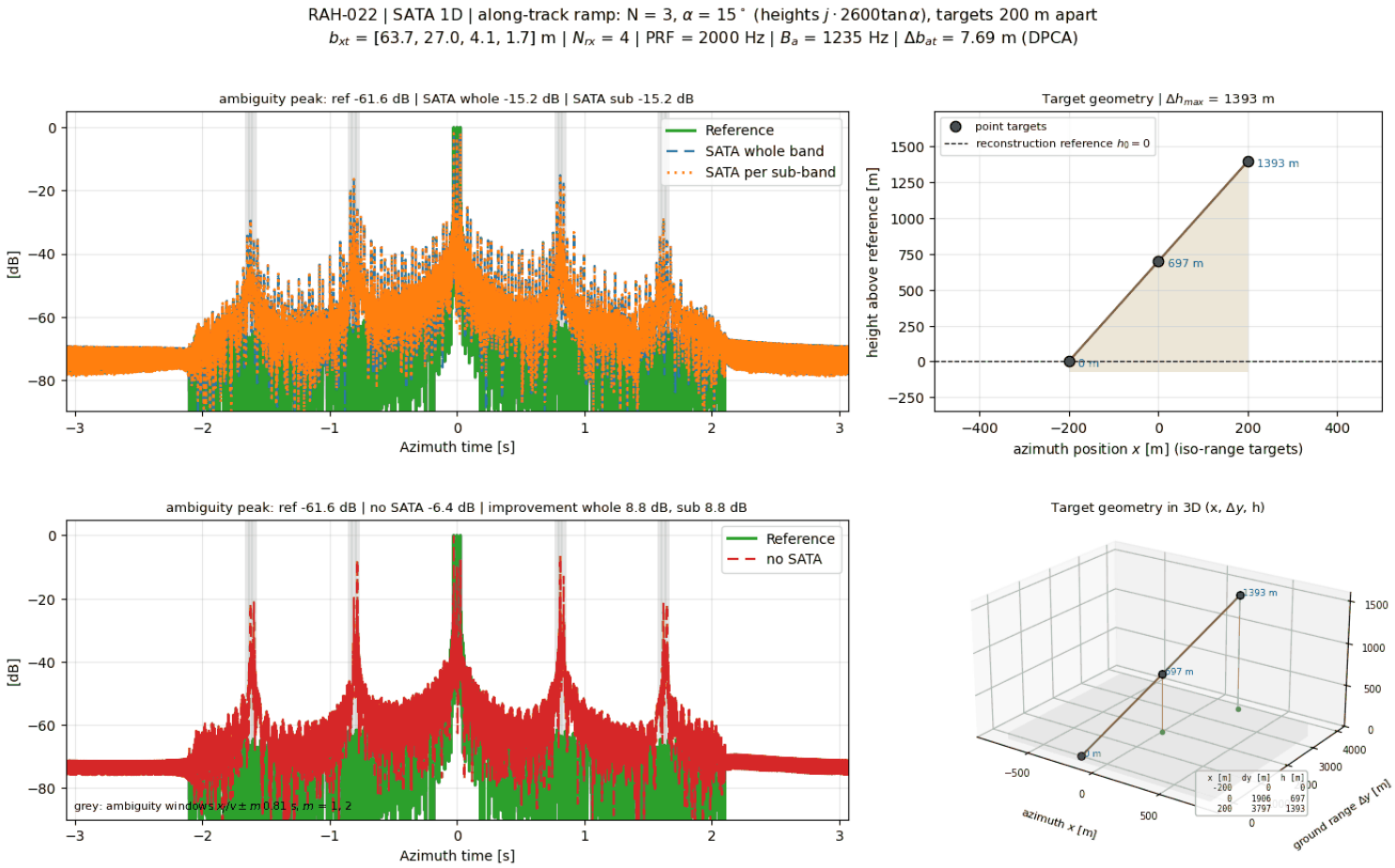


Figure 149: Along-track ramp, 200 m apart, large heights: RAH-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −15.2, per sub-band −15.2 dB (improvement 8.8 dB).

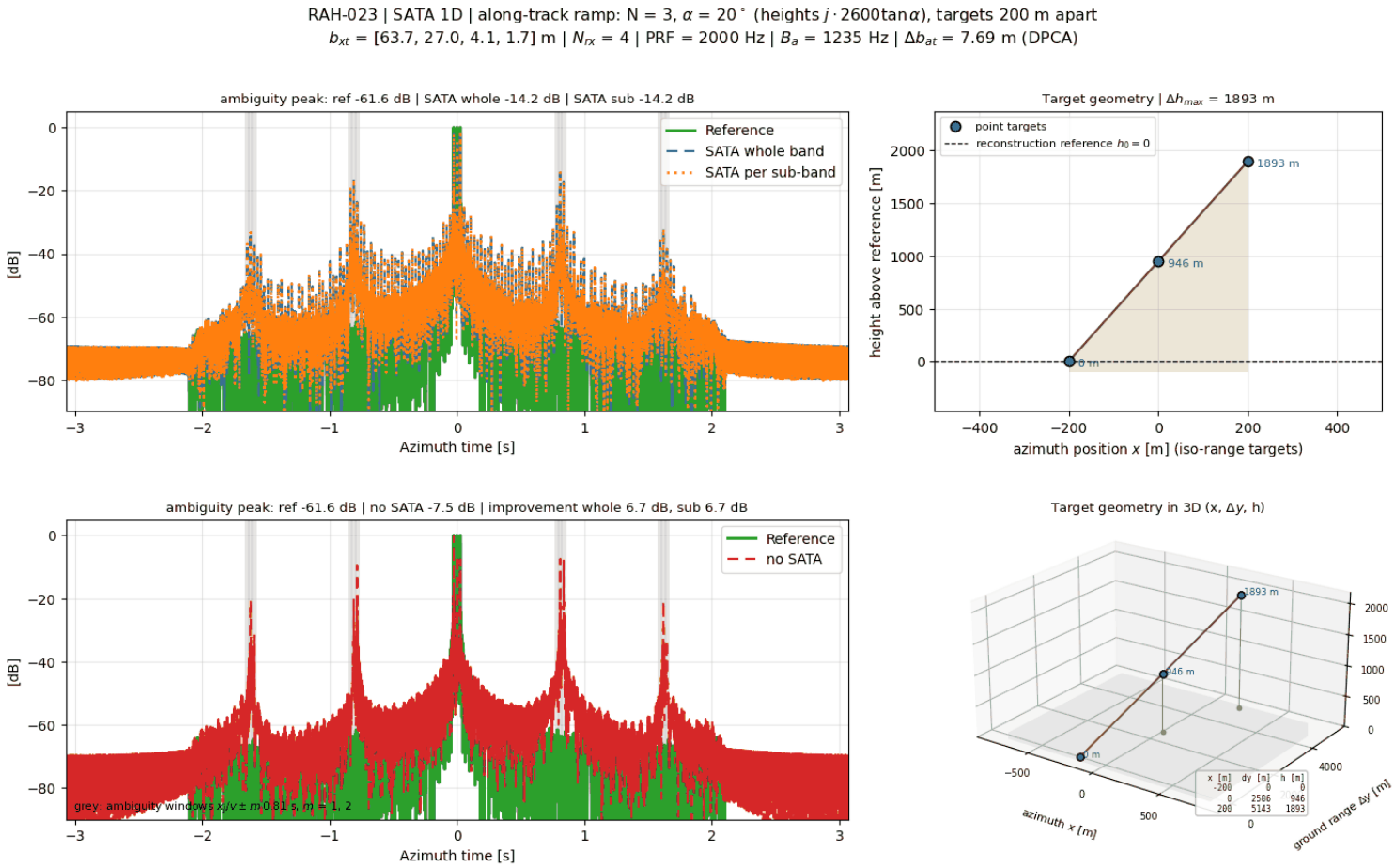


Figure 150: Along-track ramp, 200 m apart, large heights: RAH-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.5, SATA whole band −14.2, per sub-band −14.2 dB (improvement 6.7 dB).

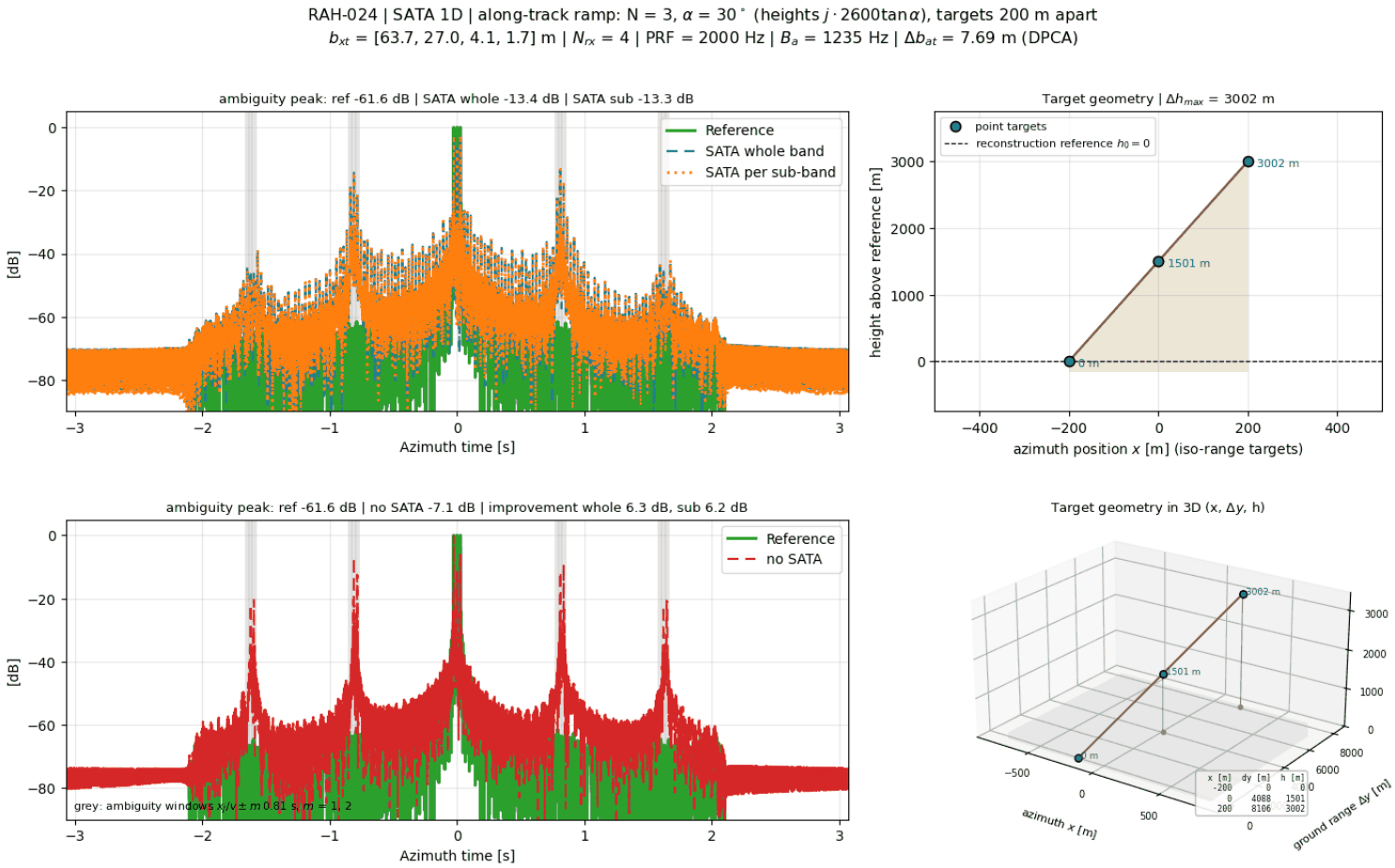


Figure 151: Along-track ramp, 200 m apart, large heights: RAH-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.1, SATA whole band −13.4, per sub-band −13.3 dB (improvement 6.3 dB).

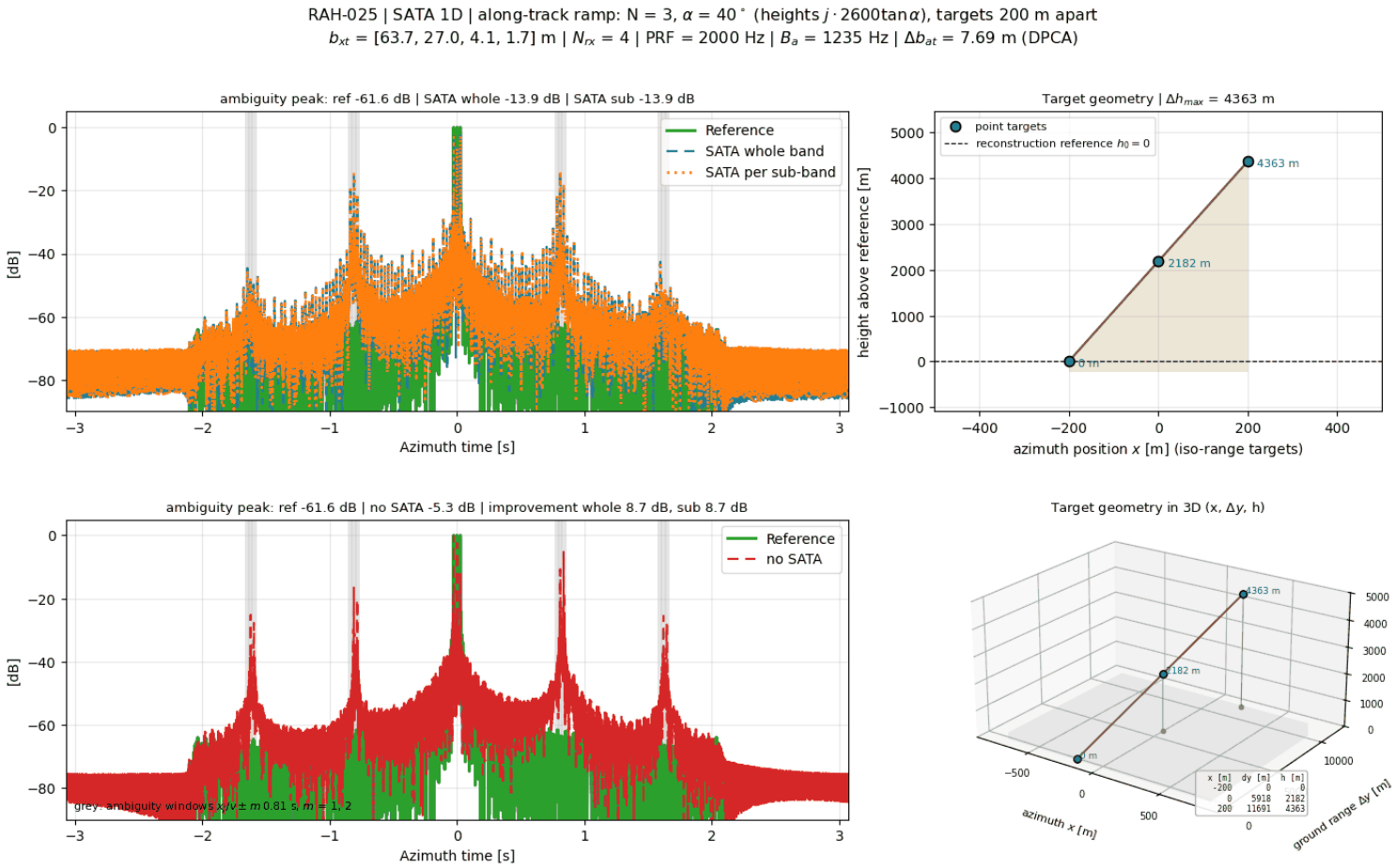


Figure 152: Along-track ramp, 200 m apart, large heights: RAH-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.3, SATA whole band −13.9, per sub-band −13.9 dB (improvement 8.7 dB).

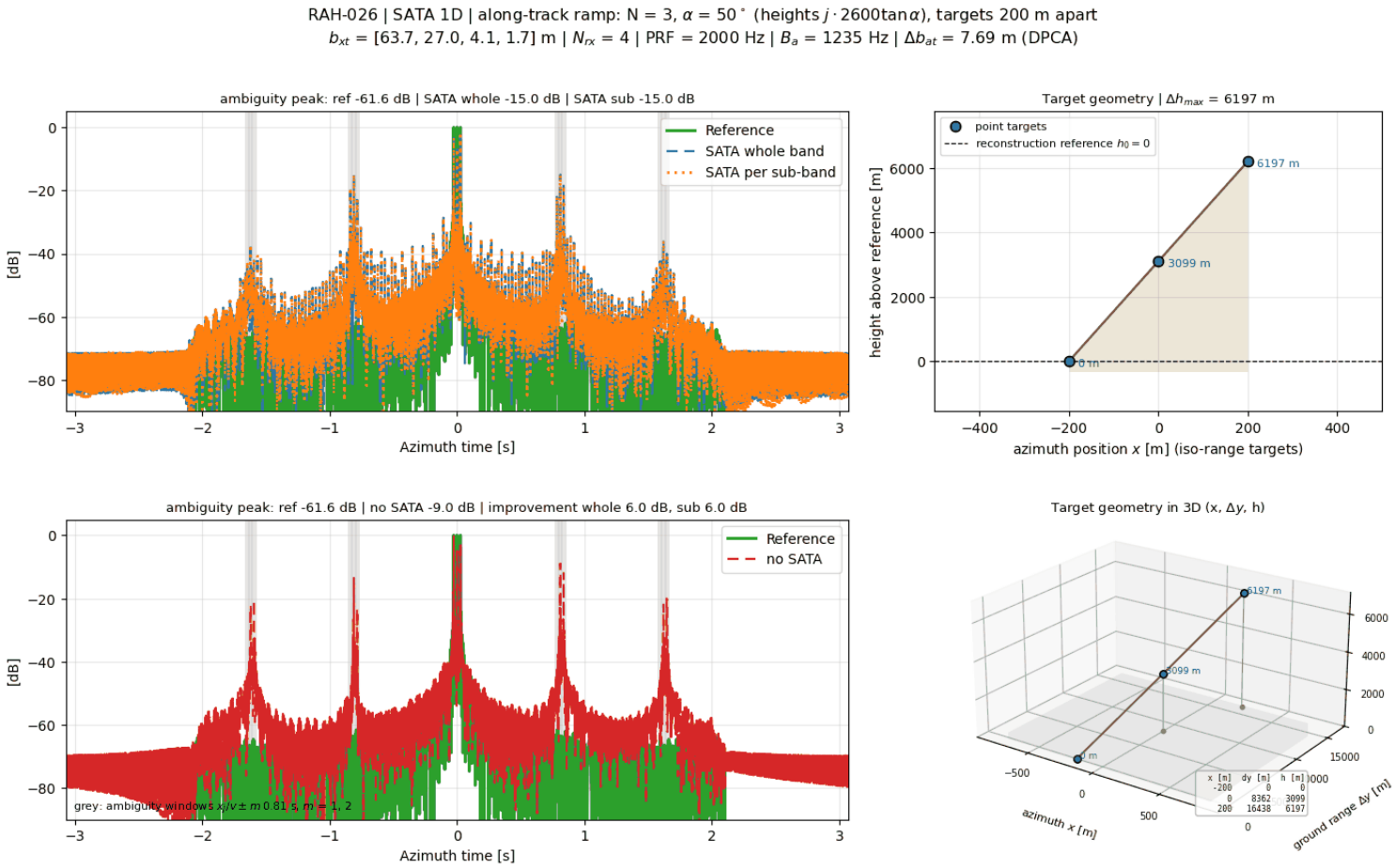


Figure 153: Along-track ramp, 200 m apart, large heights: RAH-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.0, SATA whole band −15.0, per sub-band −15.0 dB (improvement 6.0 dB).

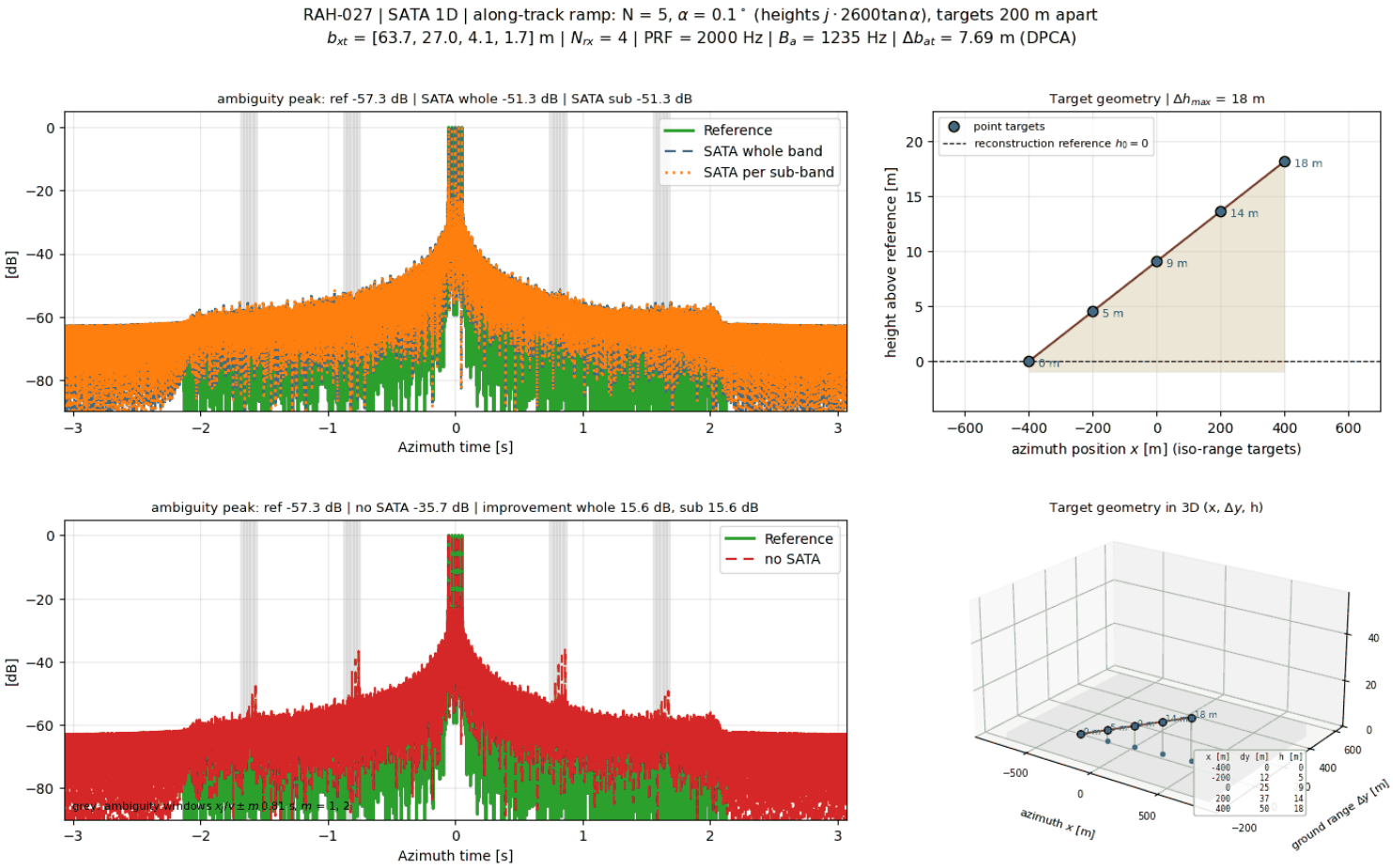


Figure 154: Along-track ramp, 200 m apart, large heights: RAH-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −35.7, SATA whole band −51.3, per sub-band −51.3 dB (improvement 15.6 dB).

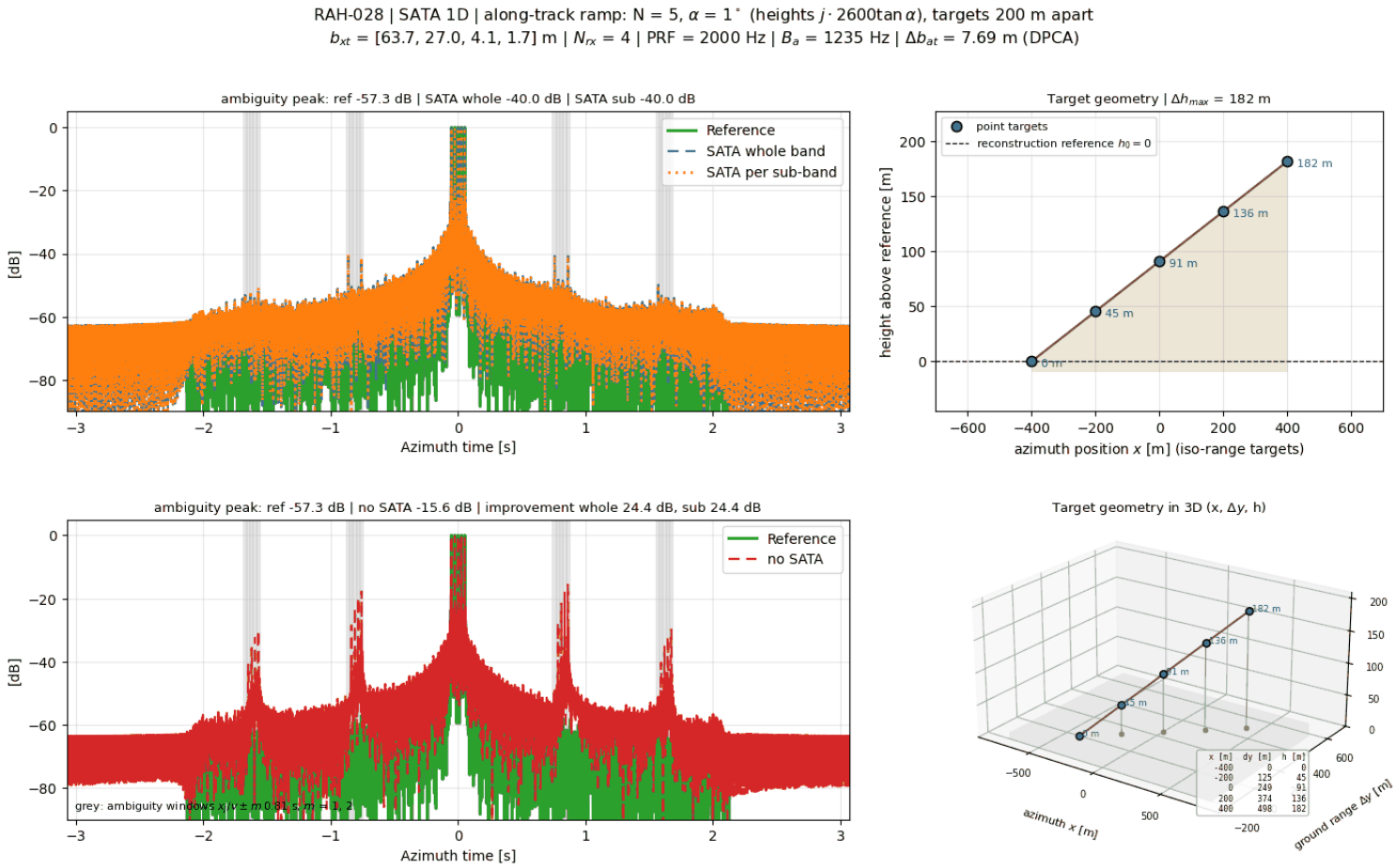


Figure 155: Along-track ramp, 200 m apart, large heights: RAH-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −15.6, SATA whole band −40.0, per sub-band −40.0 dB (improvement 24.4 dB).

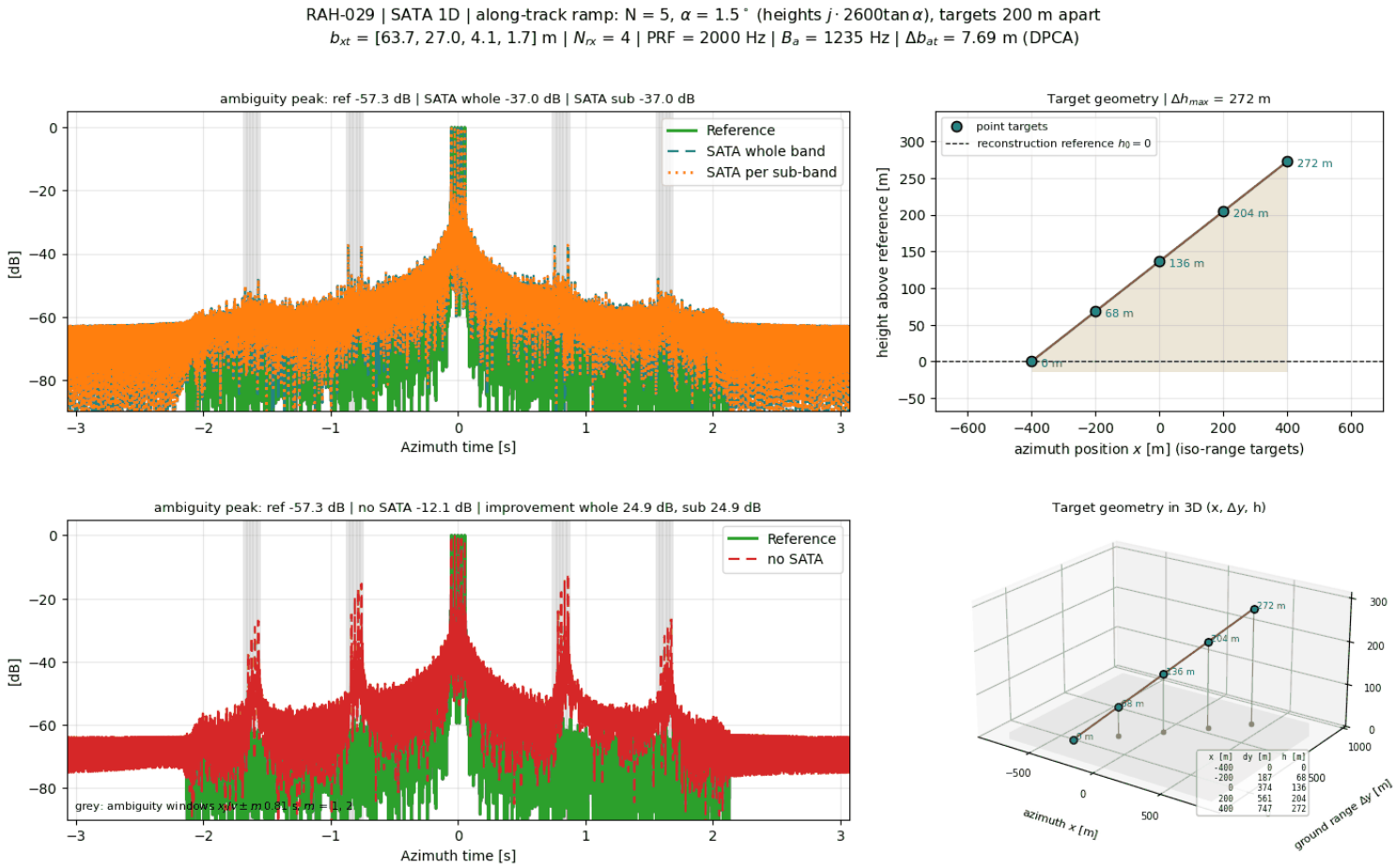


Figure 156: Along-track ramp, 200 m apart, large heights: RAH-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -12.1, SATA whole band -37.0, per sub-band -37.0 dB (improvement 24.9 dB).

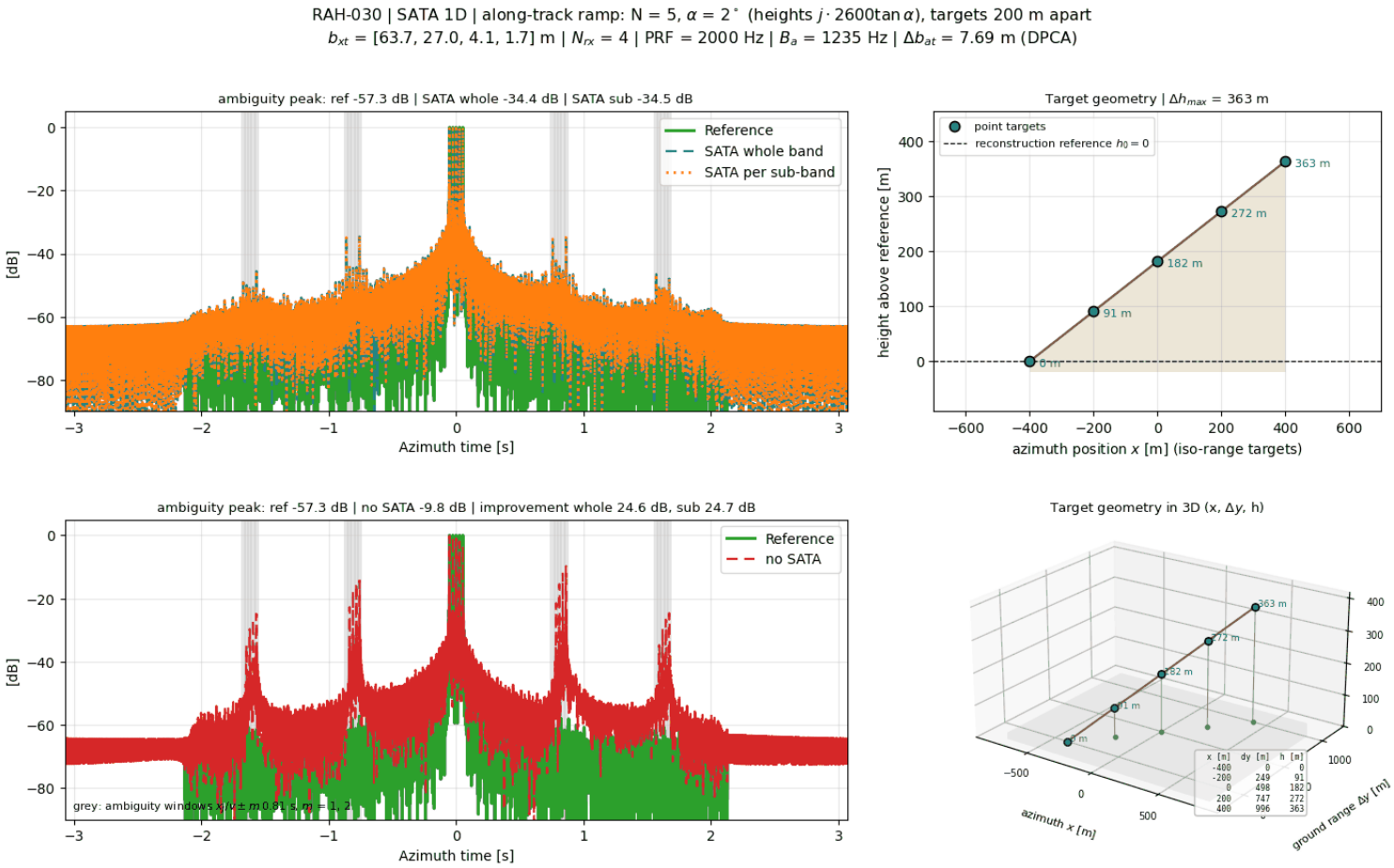


Figure 157: Along-track ramp, 200 m apart, large heights: RAH-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.8, SATA whole band −34.4, per sub-band −34.5 dB (improvement 24.6 dB).

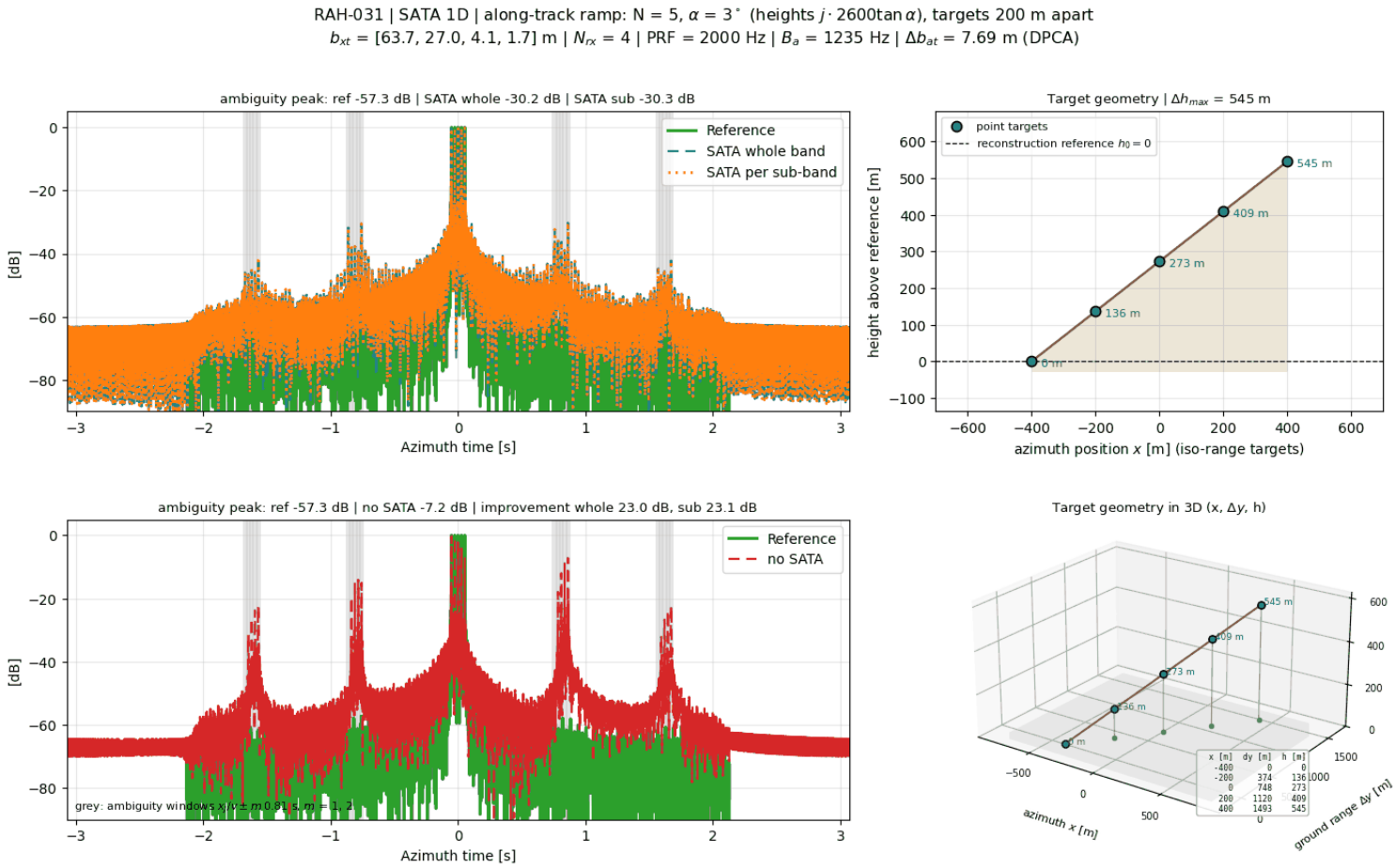


Figure 158: Along-track ramp, 200 m apart, large heights: RAH-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.2, SATA whole band −30.2, per sub-band −30.3 dB (improvement 23.0 dB).

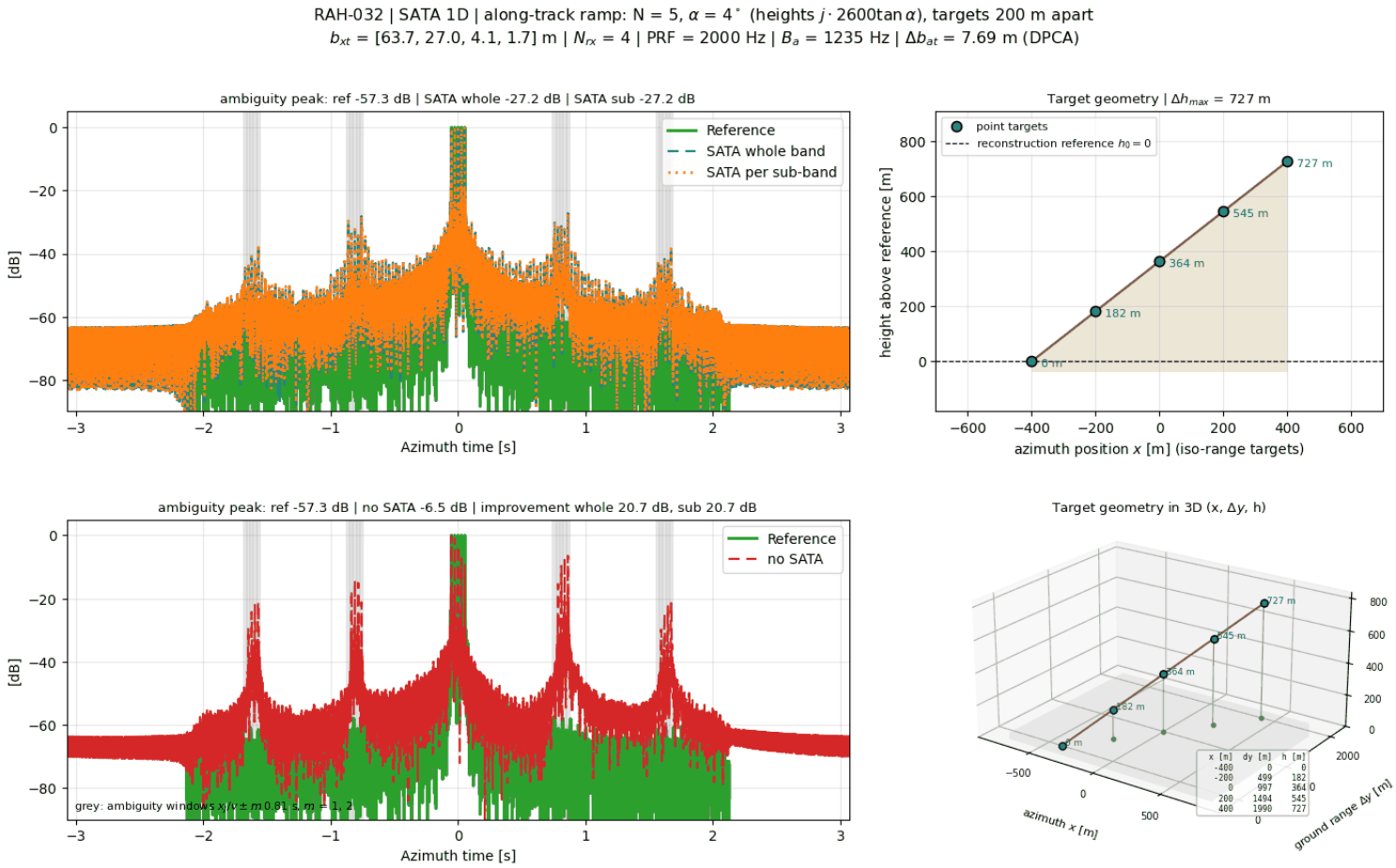


Figure 159: Along-track ramp, 200 m apart, large heights: RAH-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −27.2, per sub-band −27.2 dB (improvement 20.7 dB).

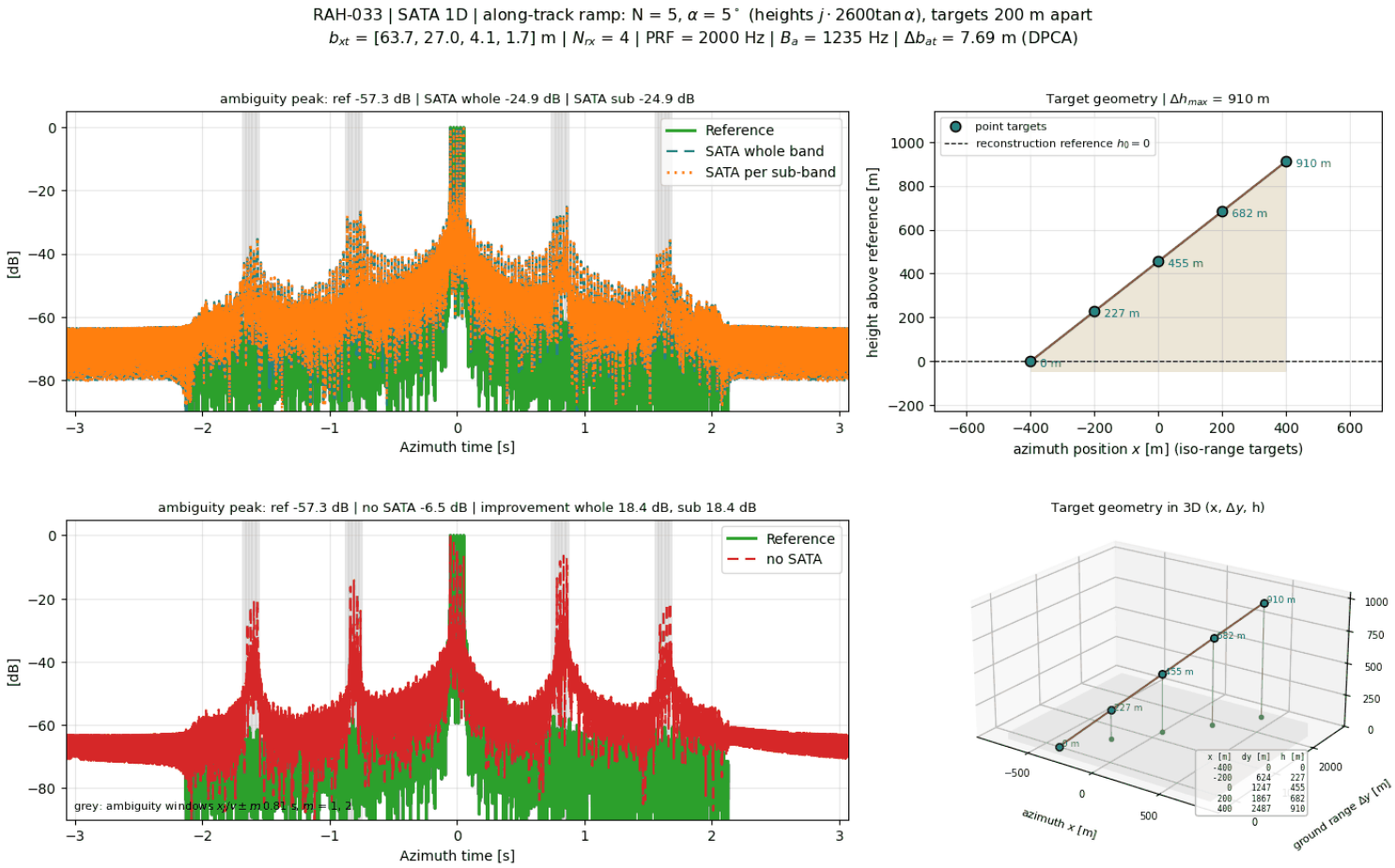


Figure 160: Along-track ramp, 200 m apart, large heights: RAH-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −24.9, per sub-band −24.9 dB (improvement 18.4 dB).

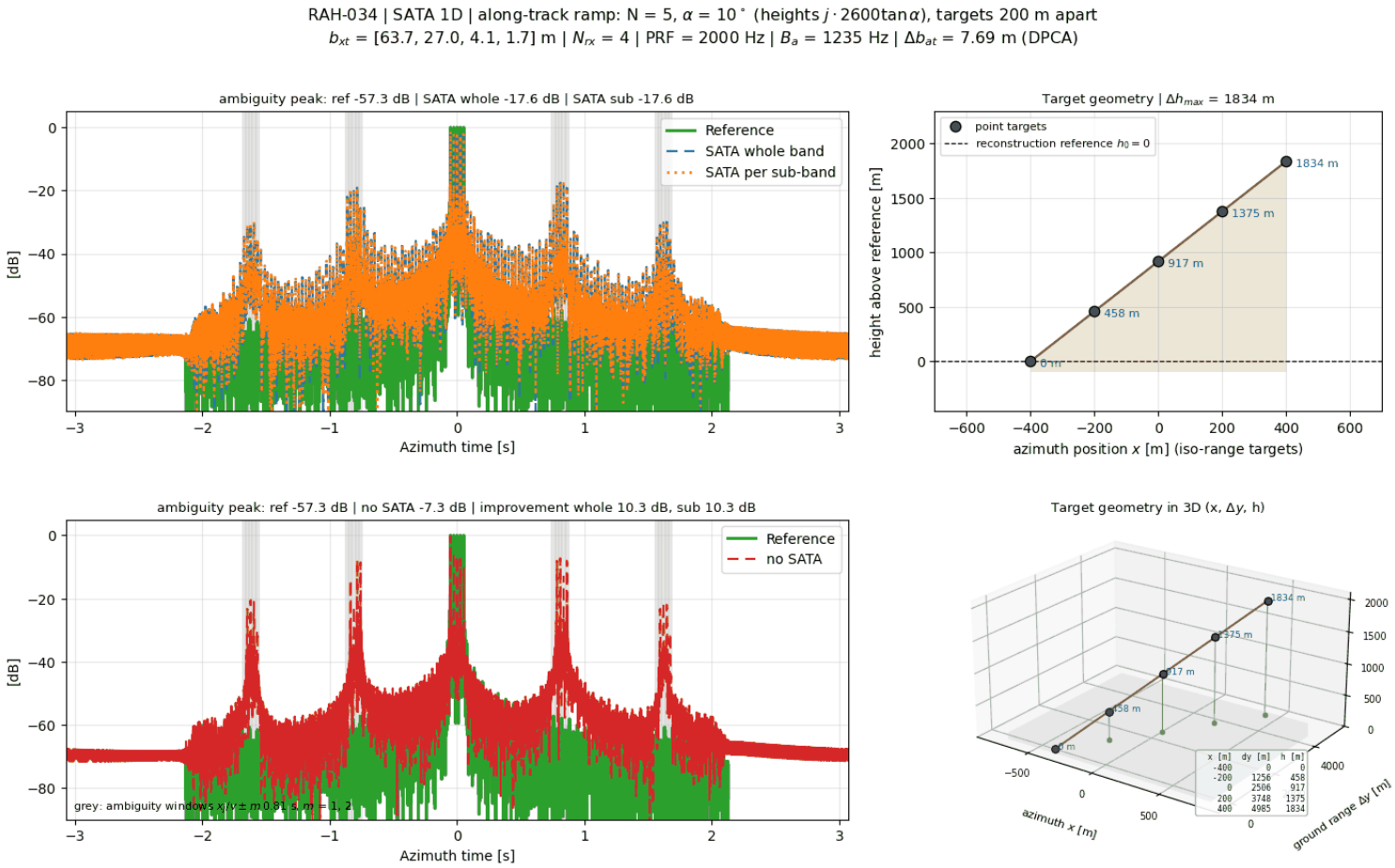


Figure 161: Along-track ramp, 200 m apart, large heights: RAH-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.3, SATA whole band −17.6, per sub-band −17.6 dB (improvement 10.3 dB).

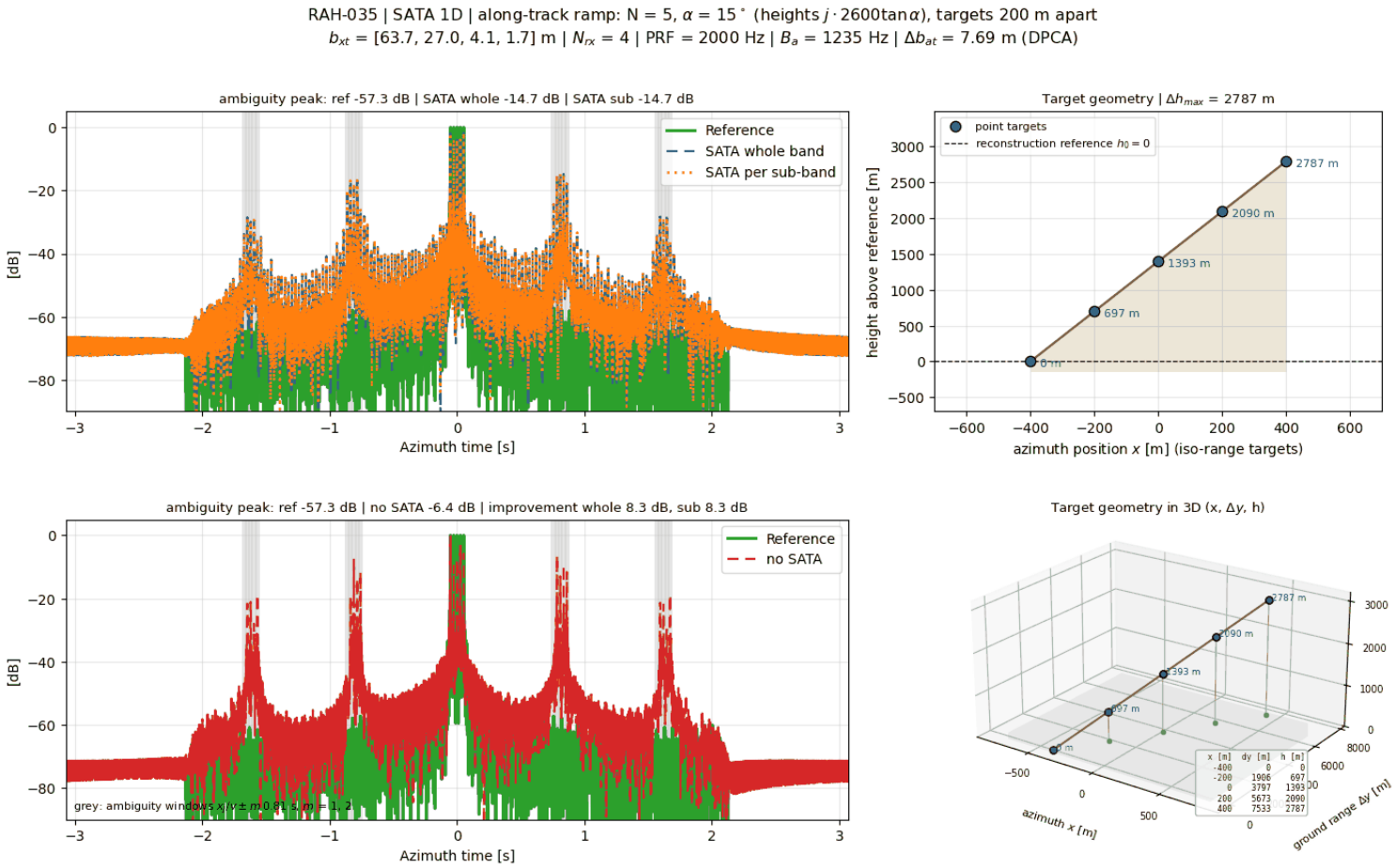


Figure 162: Along-track ramp, 200 m apart, large heights: RAH-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −14.7, per sub-band −14.7 dB (improvement 8.3 dB).

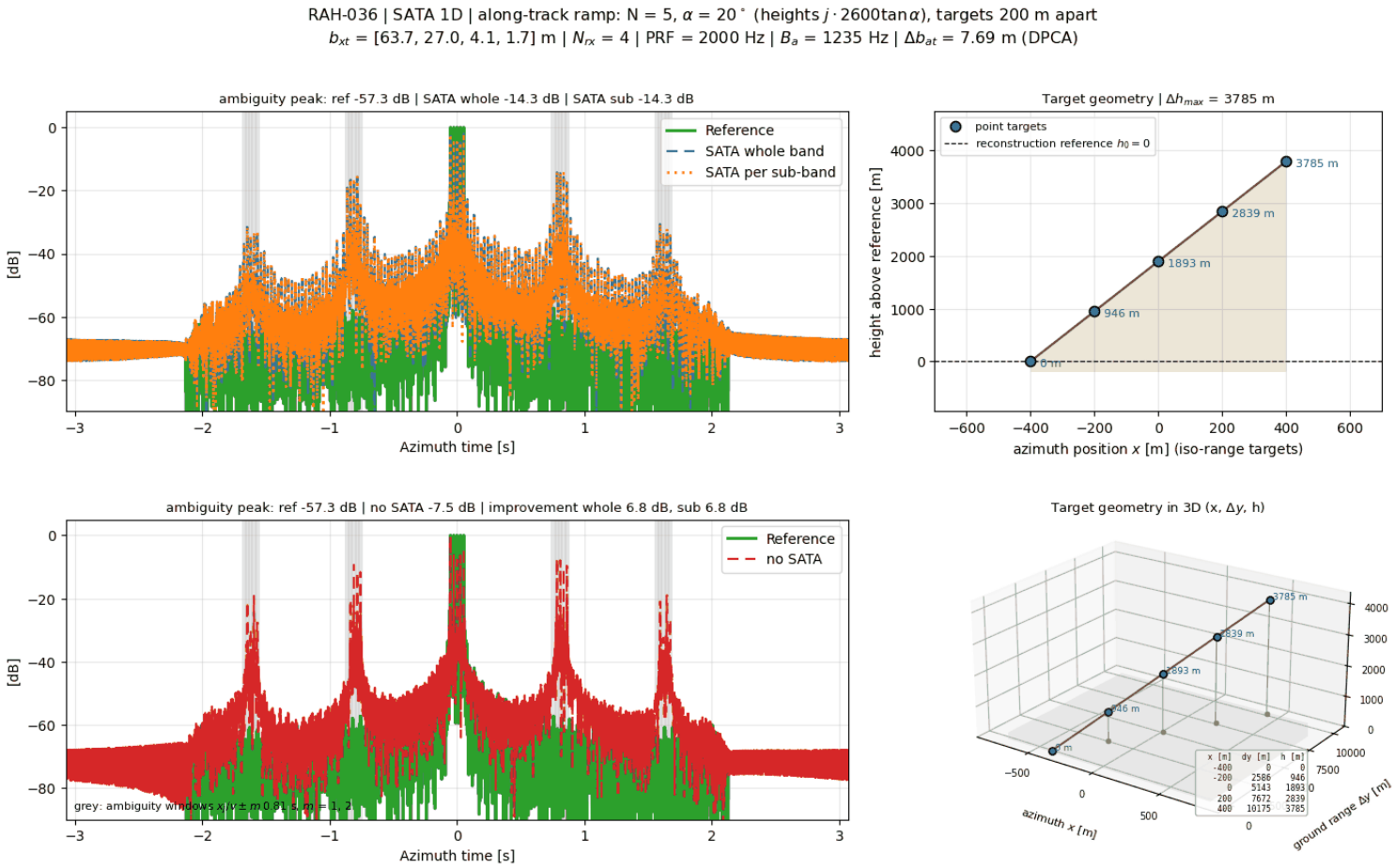


Figure 163: Along-track ramp, 200 m apart, large heights: RAH-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.5, SATA whole band −14.3, per sub-band −14.3 dB (improvement 6.8 dB).

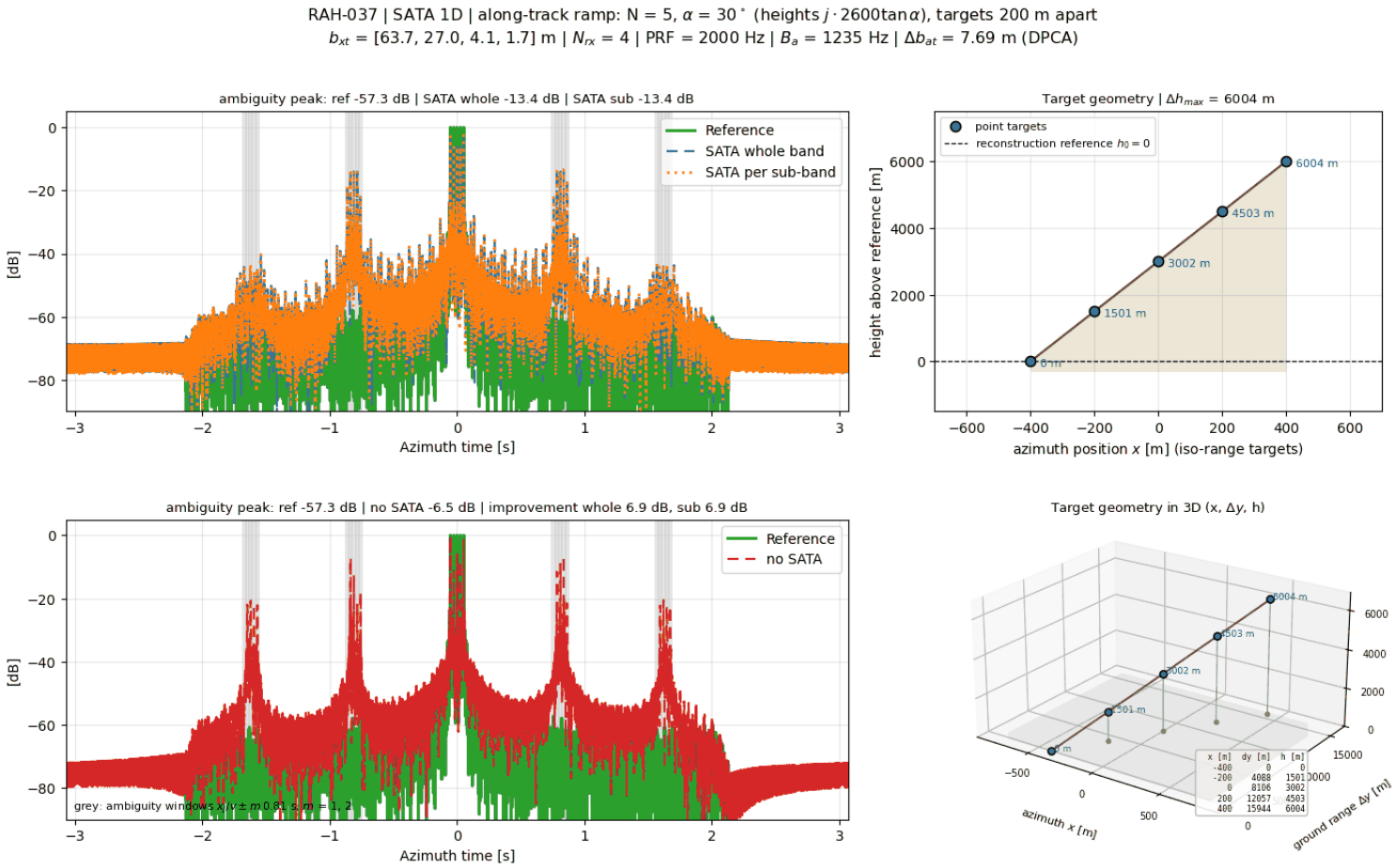


Figure 164: Along-track ramp, 200 m apart, large heights: RAH-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −13.4, per sub-band −13.4 dB (improvement 6.9 dB).

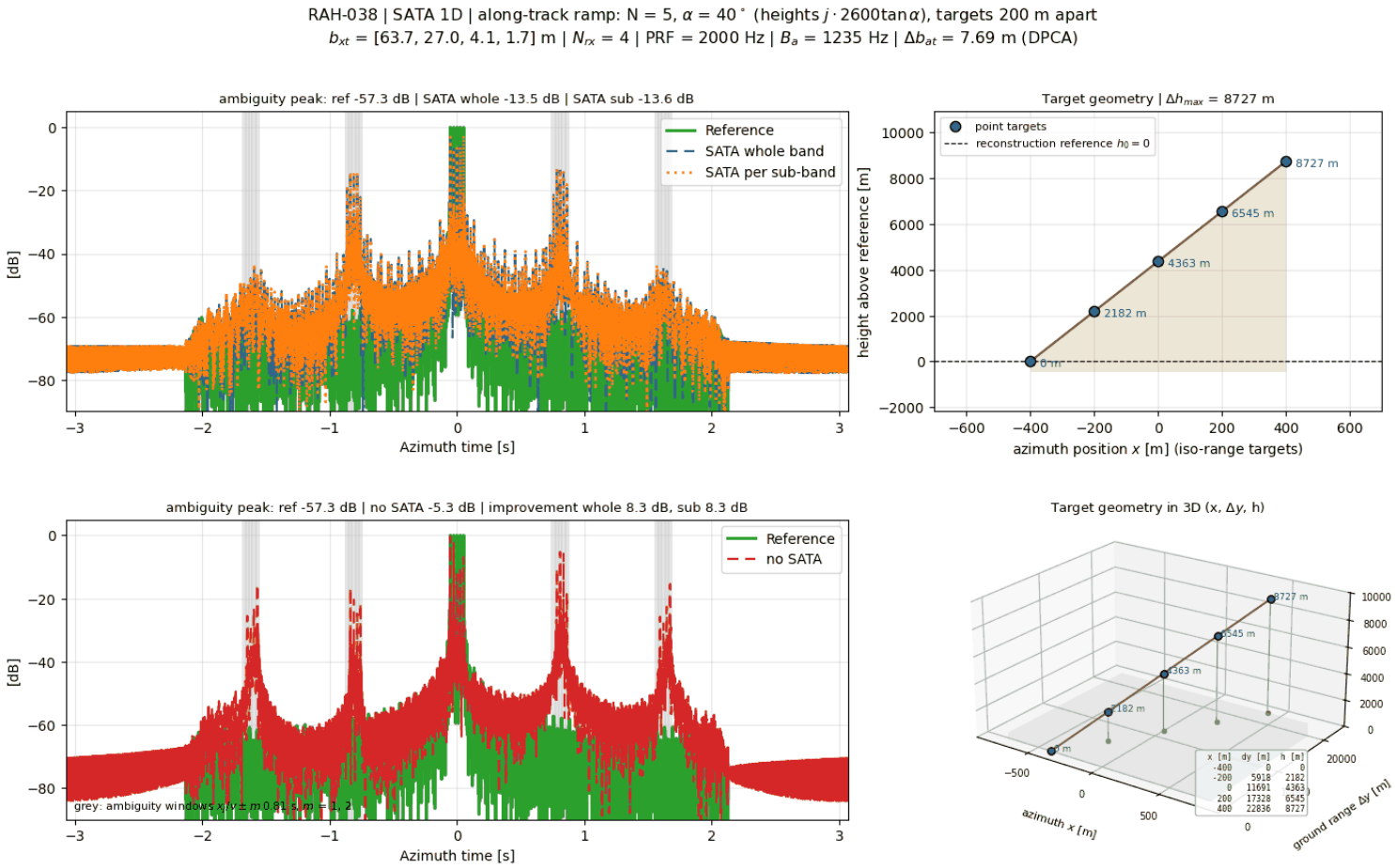


Figure 165: Along-track ramp, 200 m apart, large heights: RAH-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.3, SATA whole band −13.5, per sub-band −13.6 dB (improvement 8.3 dB).

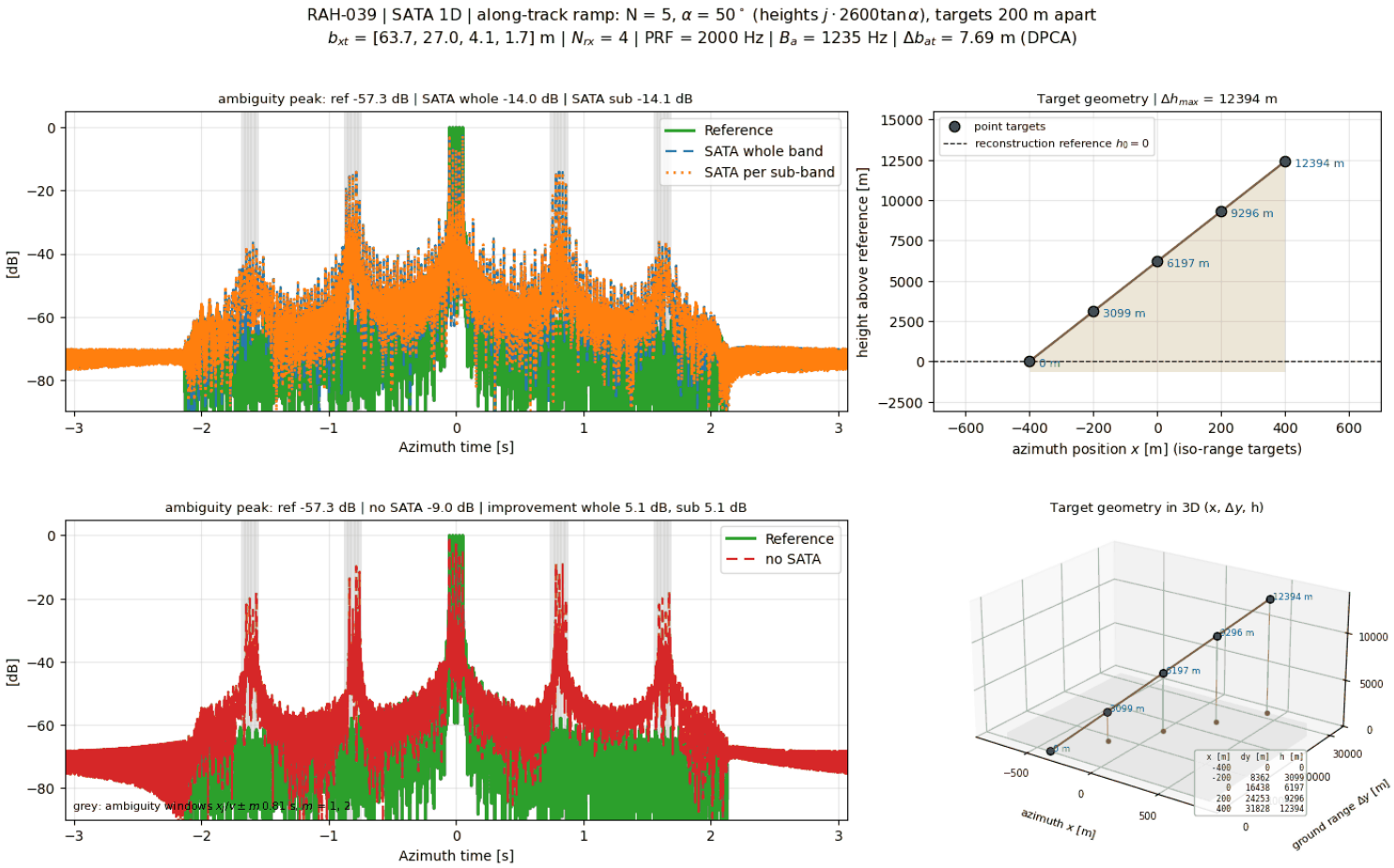


Figure 166: Along-track ramp, 200 m apart, large heights: RAH-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.0, SATA whole band −14.0, per sub-band −14.1 dB (improvement 5.1 dB).

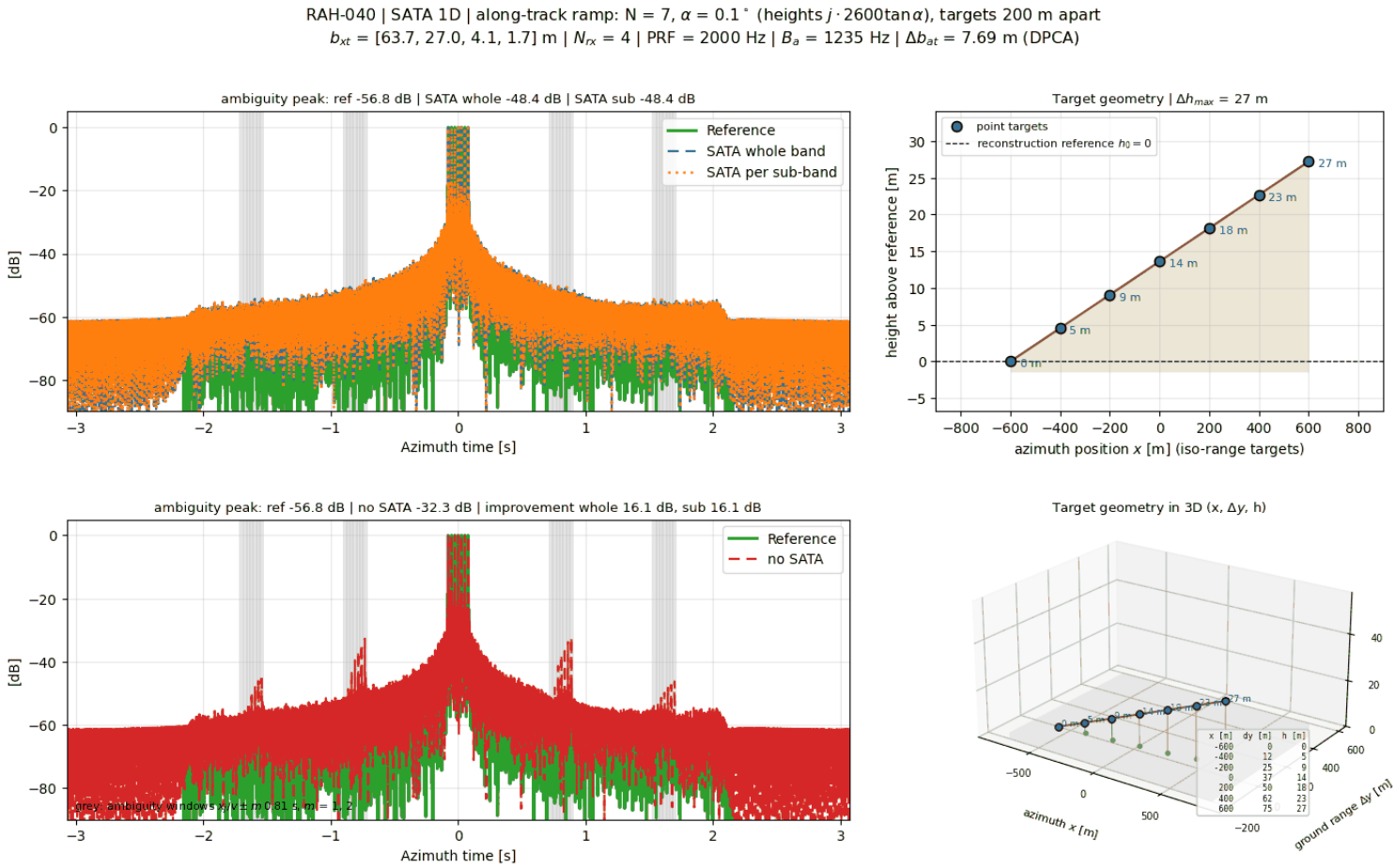


Figure 167: Along-track ramp, 200 m apart, large heights: RAH-040, $N = 7$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −32.3, SATA whole band −48.4, per sub-band −48.4 dB (improvement 16.1 dB).

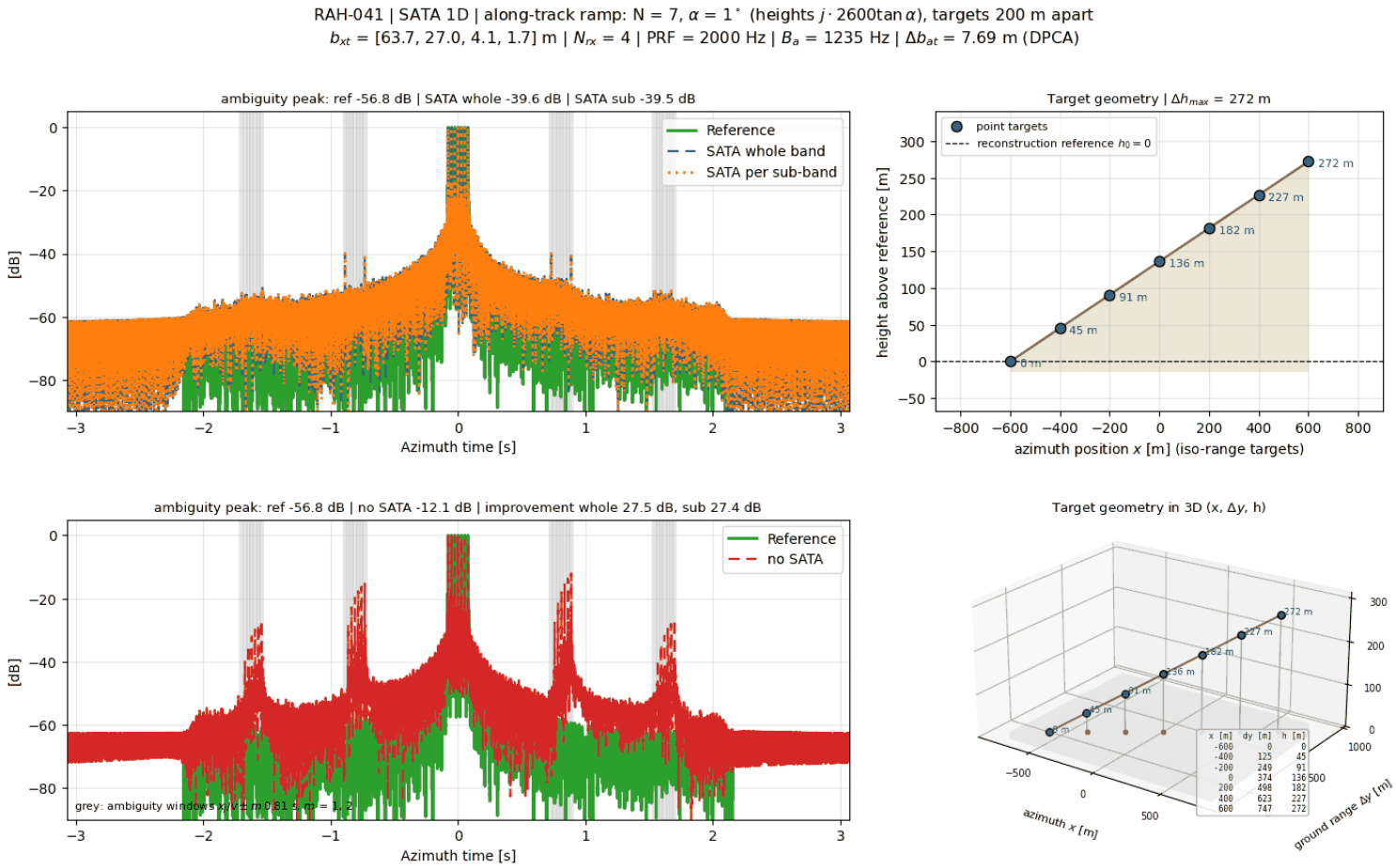


Figure 168: Along-track ramp, 200 m apart, large heights: RAH-041, $N = 7$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −12.1, SATA whole band −39.6, per sub-band −39.5 dB (improvement 27.5 dB).

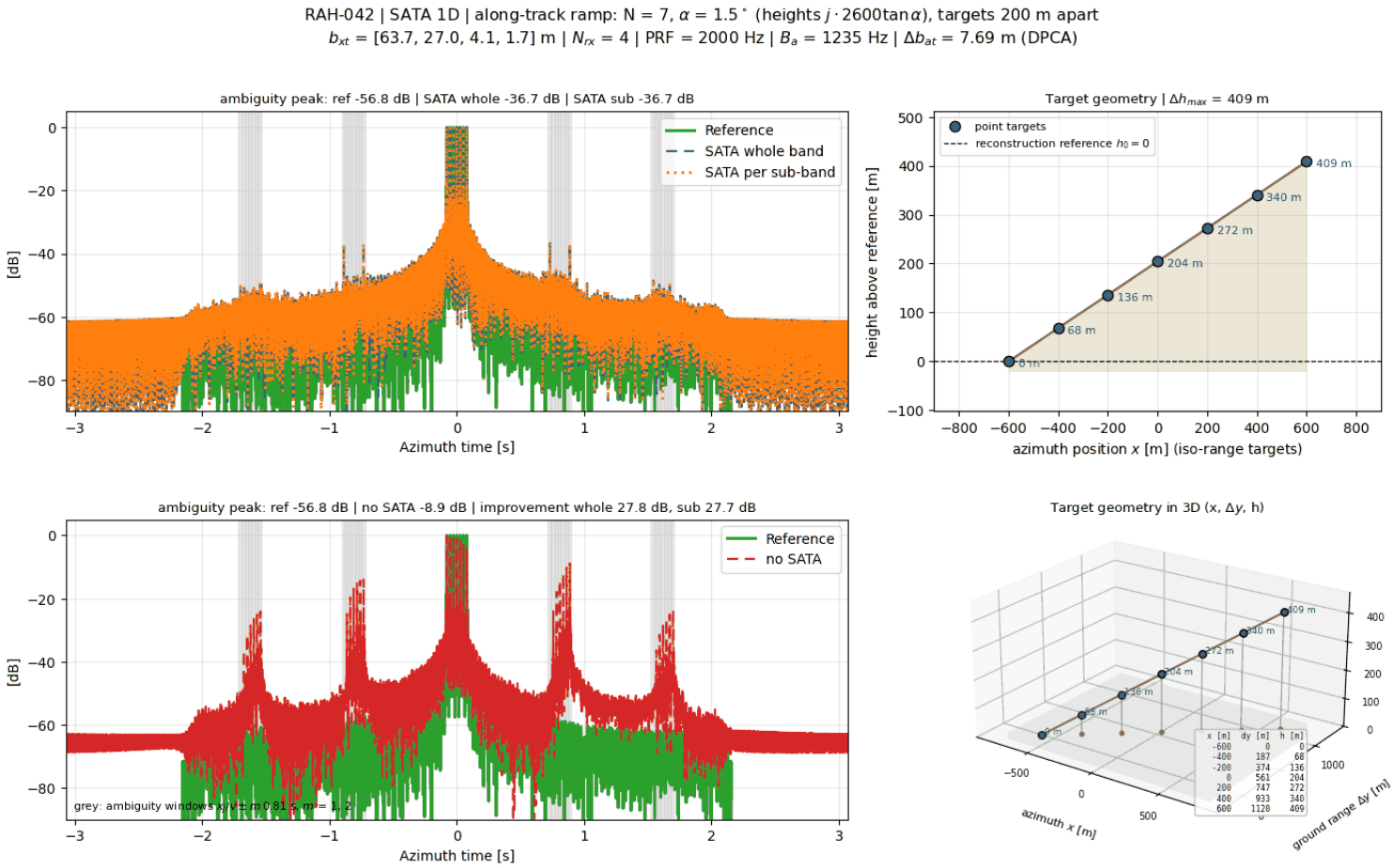


Figure 169: Along-track ramp, 200 m apart, large heights: RAH-042, $N = 7$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.9, SATA whole band −36.7, per sub-band −36.7 dB (improvement 27.8 dB).

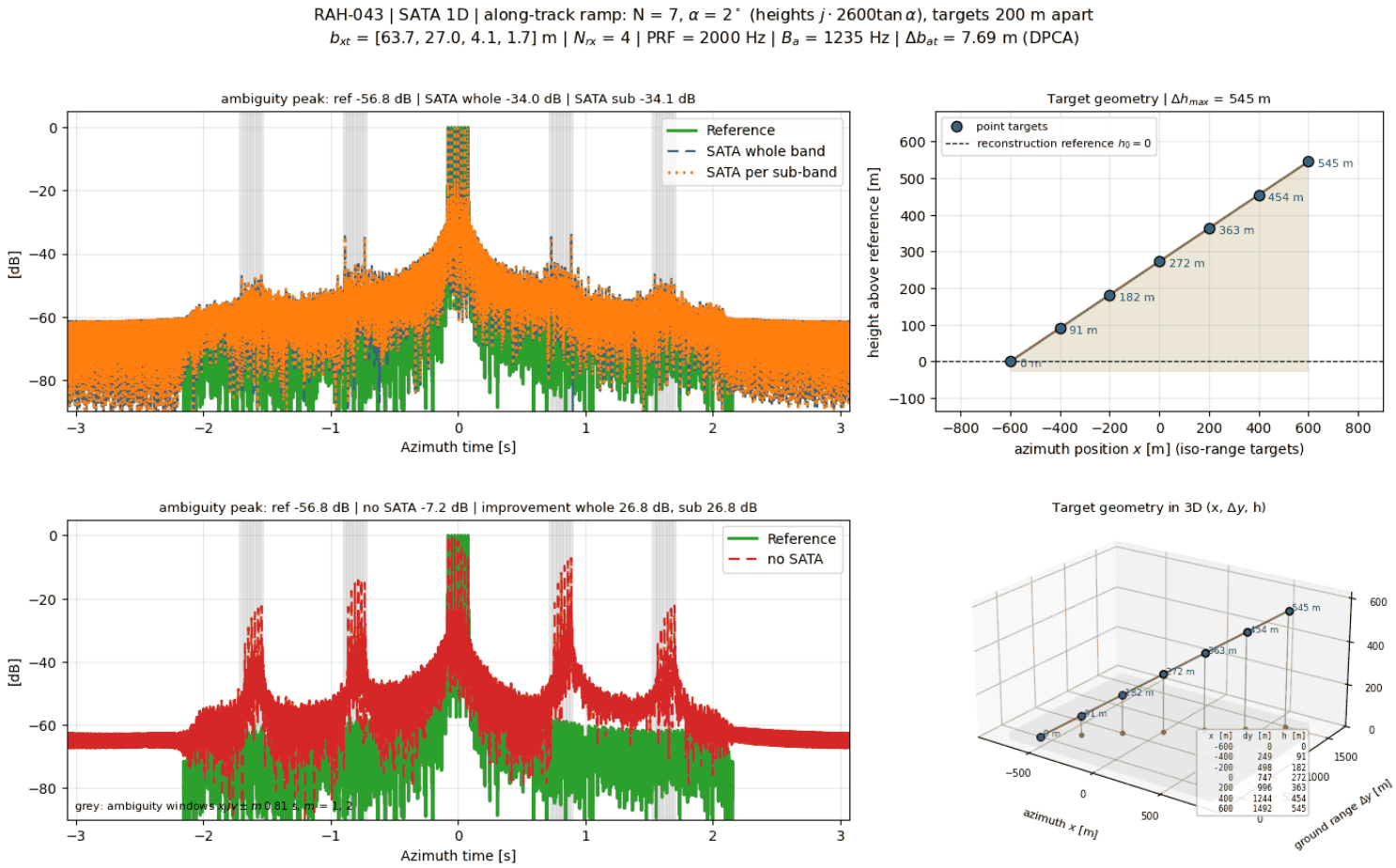


Figure 170: Along-track ramp, 200 m apart, large heights: RAH-043, $N = 7$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -7.2, SATA whole band -34.0, per sub-band -34.1 dB (improvement 26.8 dB).

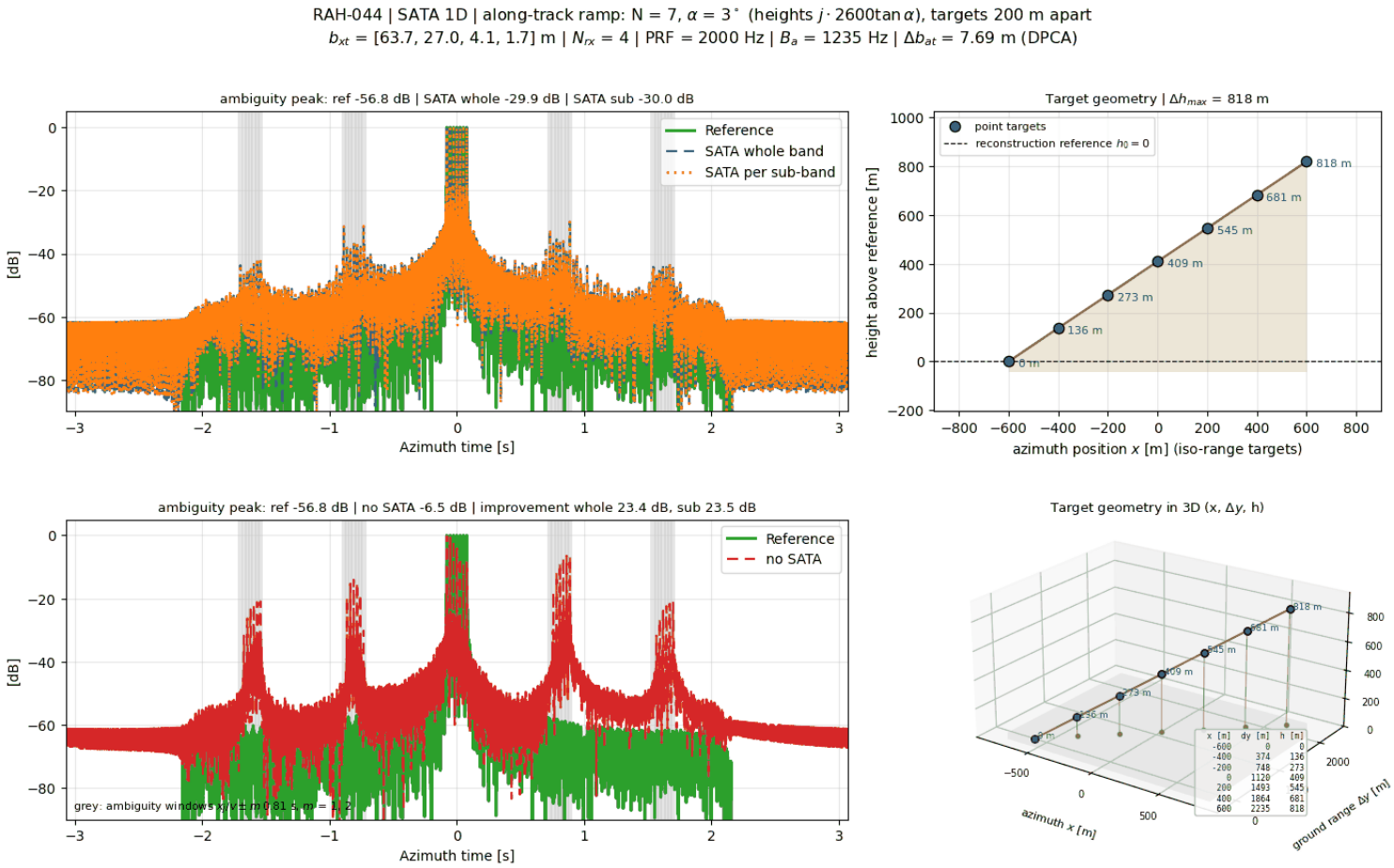


Figure 171: Along-track ramp, 200 m apart, large heights: RAH-044, $N = 7$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −29.9, per sub-band −30.0 dB (improvement 23.4 dB).

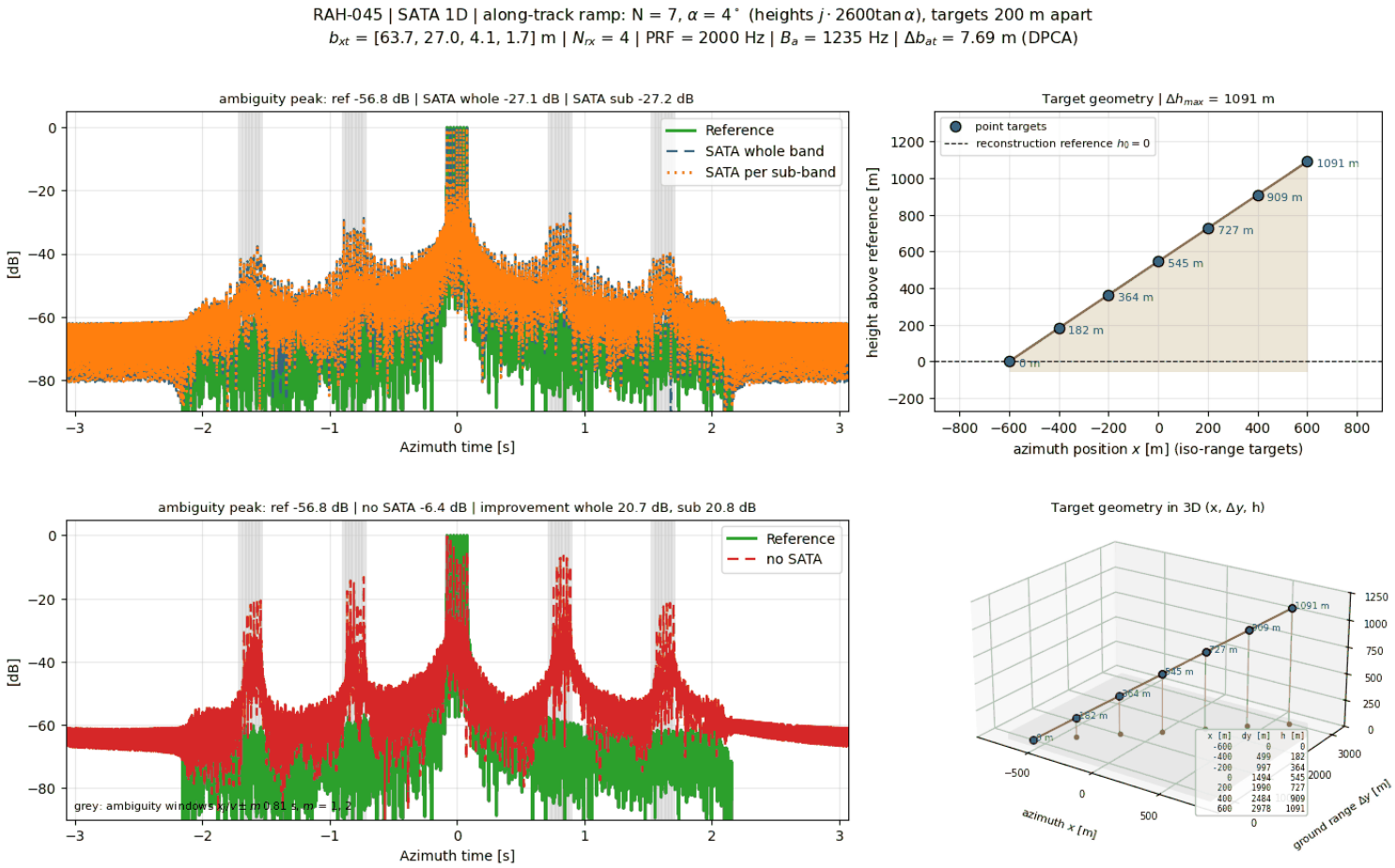


Figure 172: Along-track ramp, 200 m apart, large heights: RAH-045, $N = 7$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −27.1, per sub-band −27.2 dB (improvement 20.7 dB).

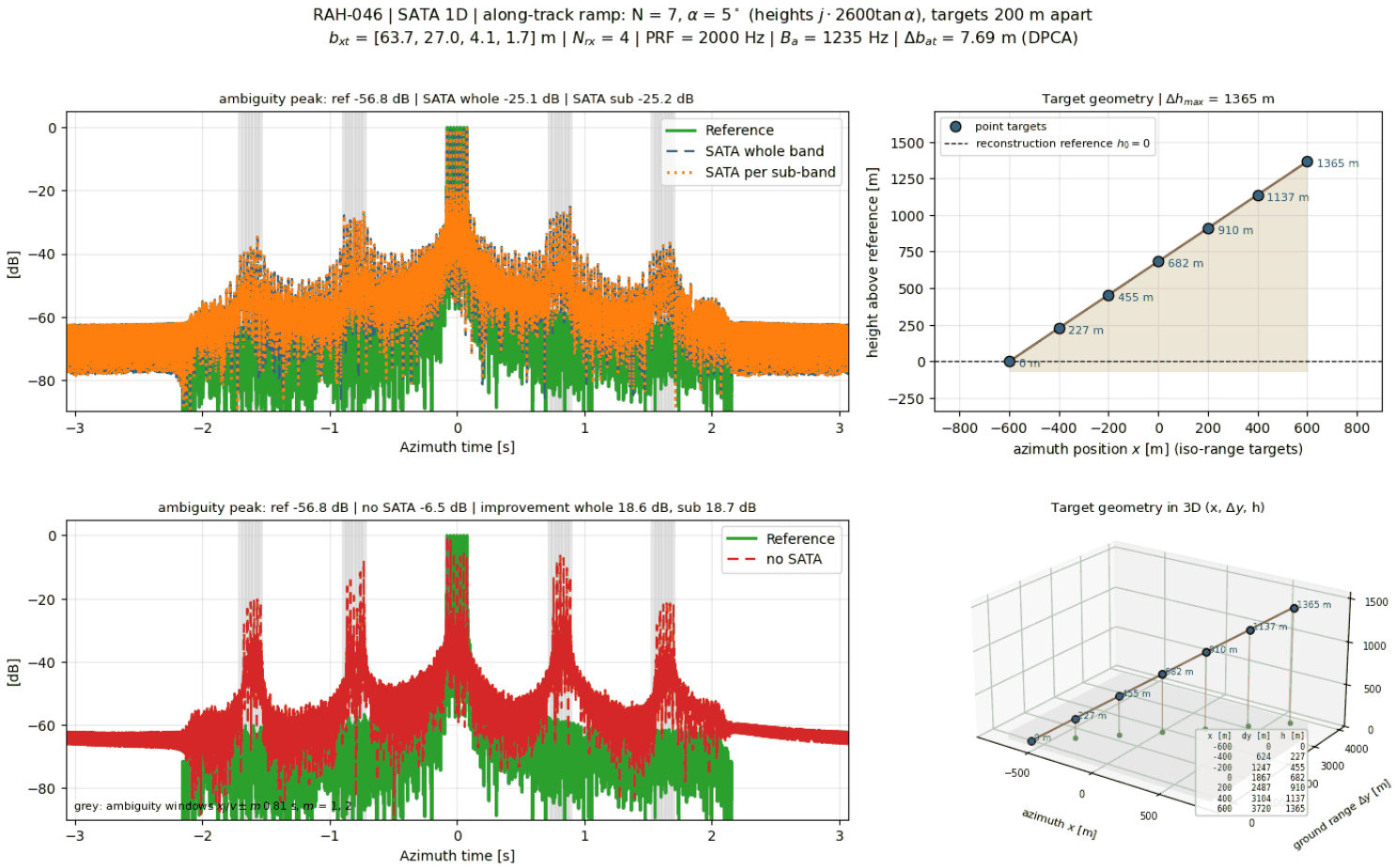


Figure 173: Along-track ramp, 200 m apart, large heights: RAH-046, $N = 7$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -6.5, SATA whole band -25.1, per sub-band -25.2 dB (improvement 18.6 dB).

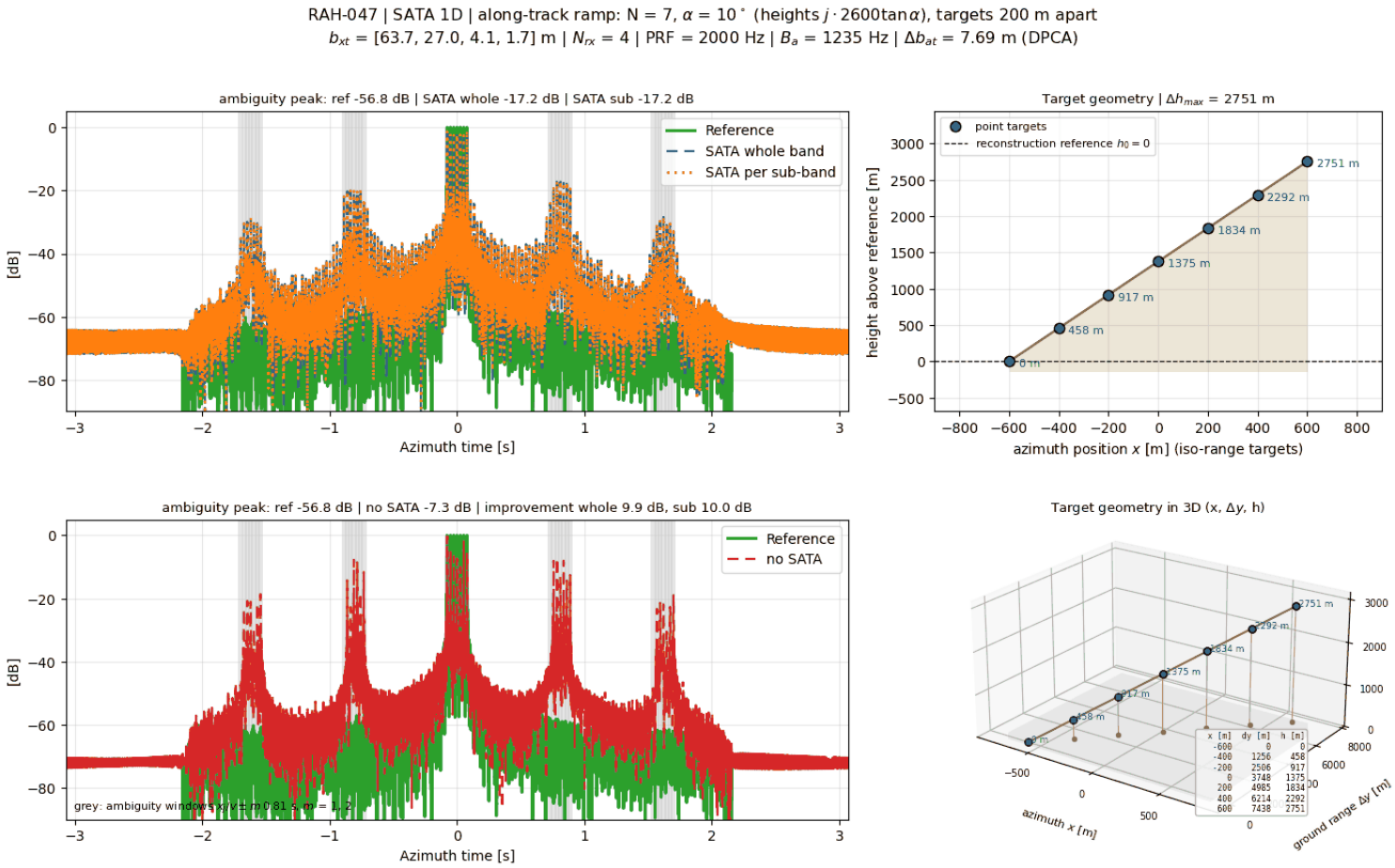


Figure 174: Along-track ramp, 200 m apart, large heights: RAH-047, $N = 7$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.3, SATA whole band −17.2, per sub-band −17.2 dB (improvement 9.9 dB).

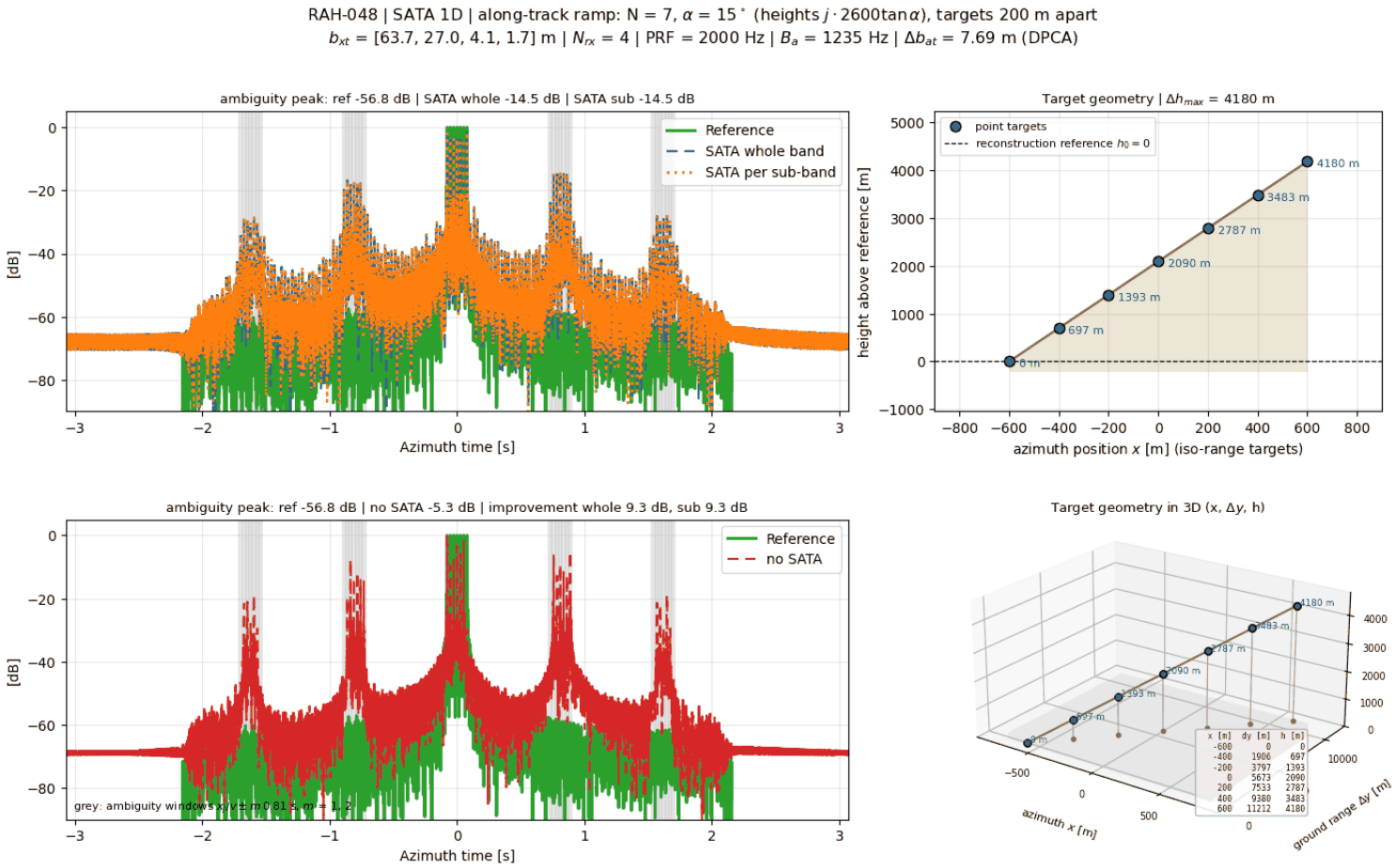


Figure 175: Along-track ramp, 200 m apart, large heights: RAH-048, $N = 7$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.3, SATA whole band −14.5, per sub-band −14.5 dB (improvement 9.3 dB).

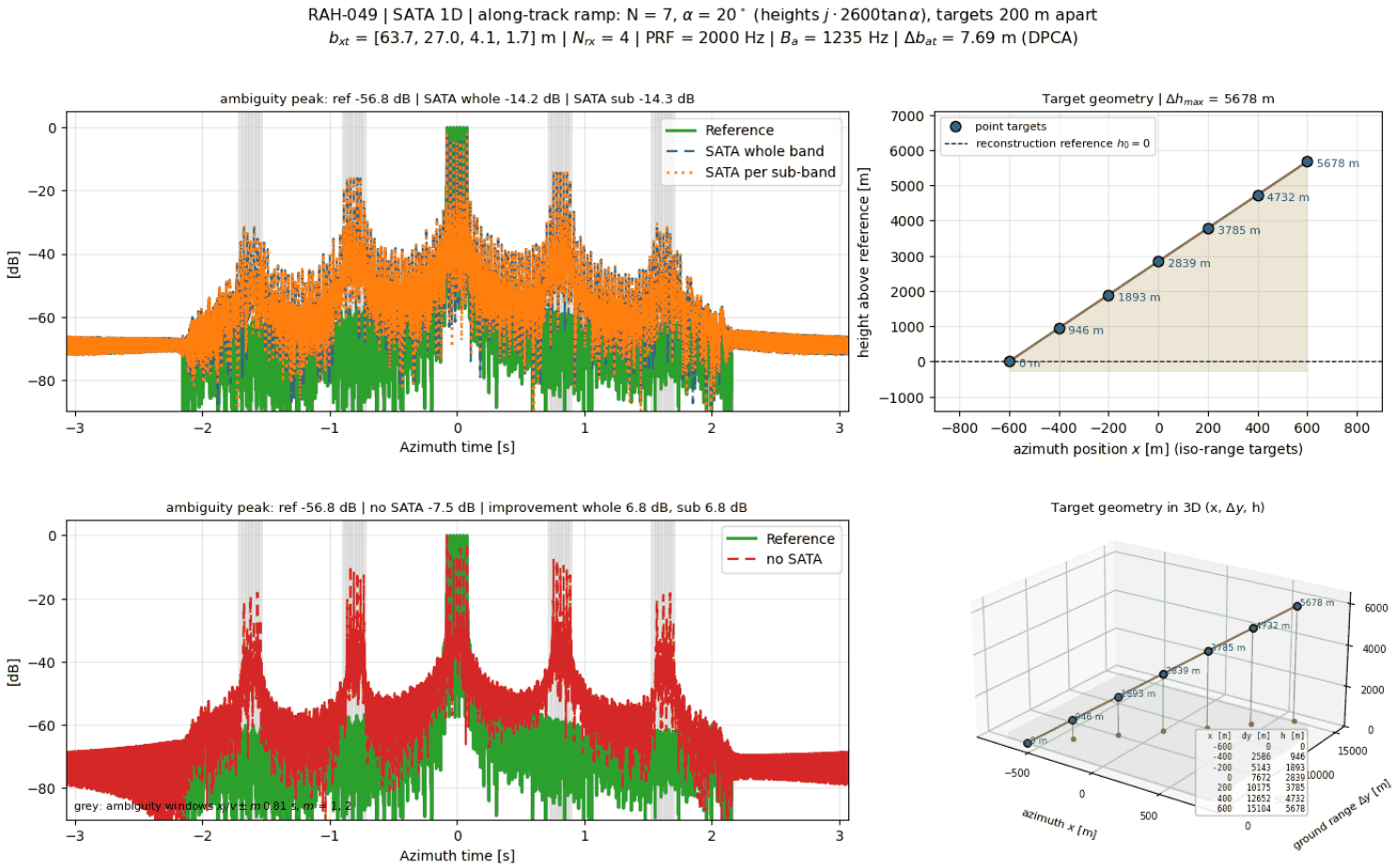


Figure 176: Along-track ramp, 200 m apart, large heights: RAH-049, $N = 7$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.5, SATA whole band −14.2, per sub-band −14.3 dB (improvement 6.8 dB).

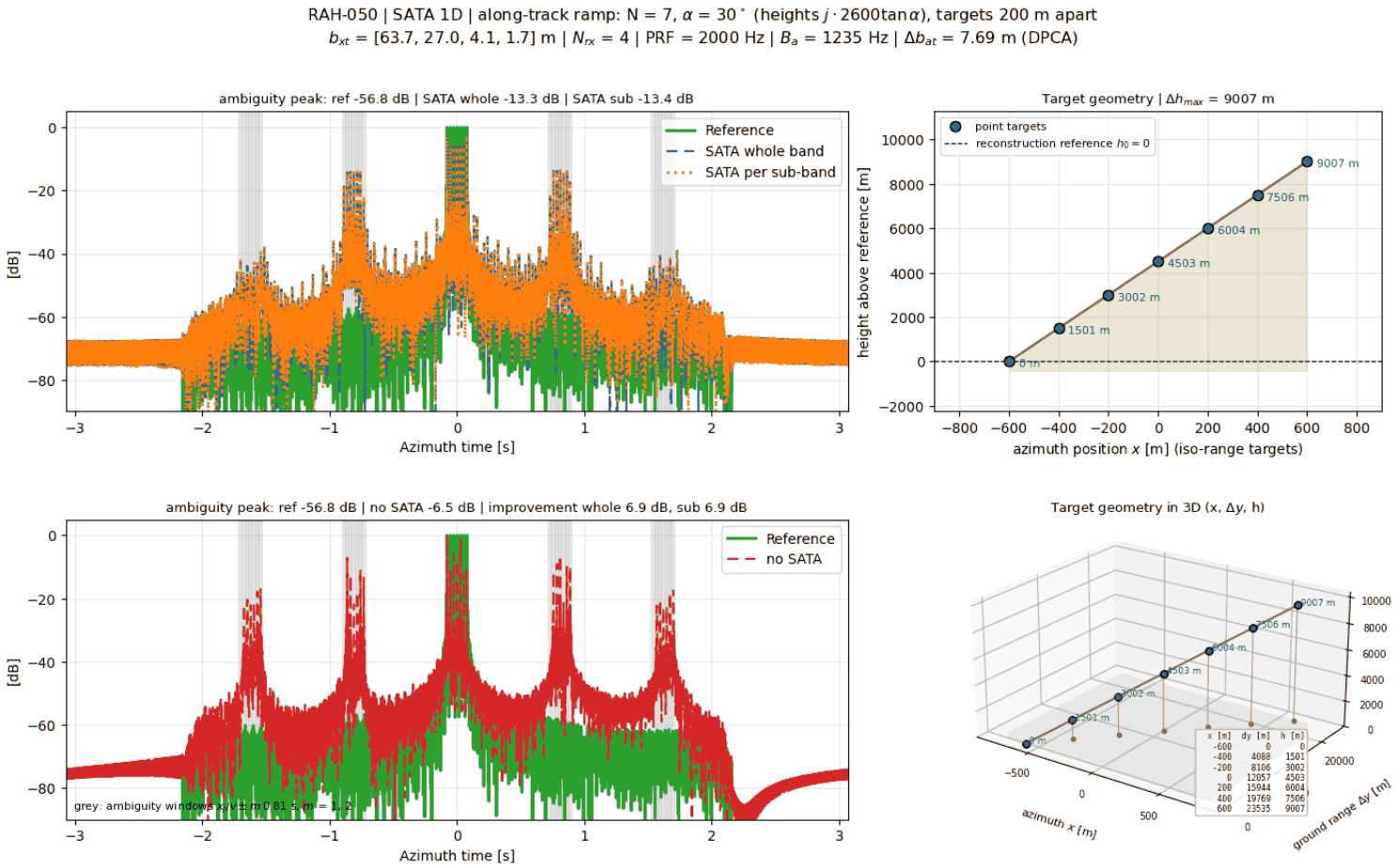


Figure 177: Along-track ramp, 200 m apart, large heights: RAH-050, $N = 7$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −13.3, per sub-band −13.4 dB (improvement 6.9 dB).

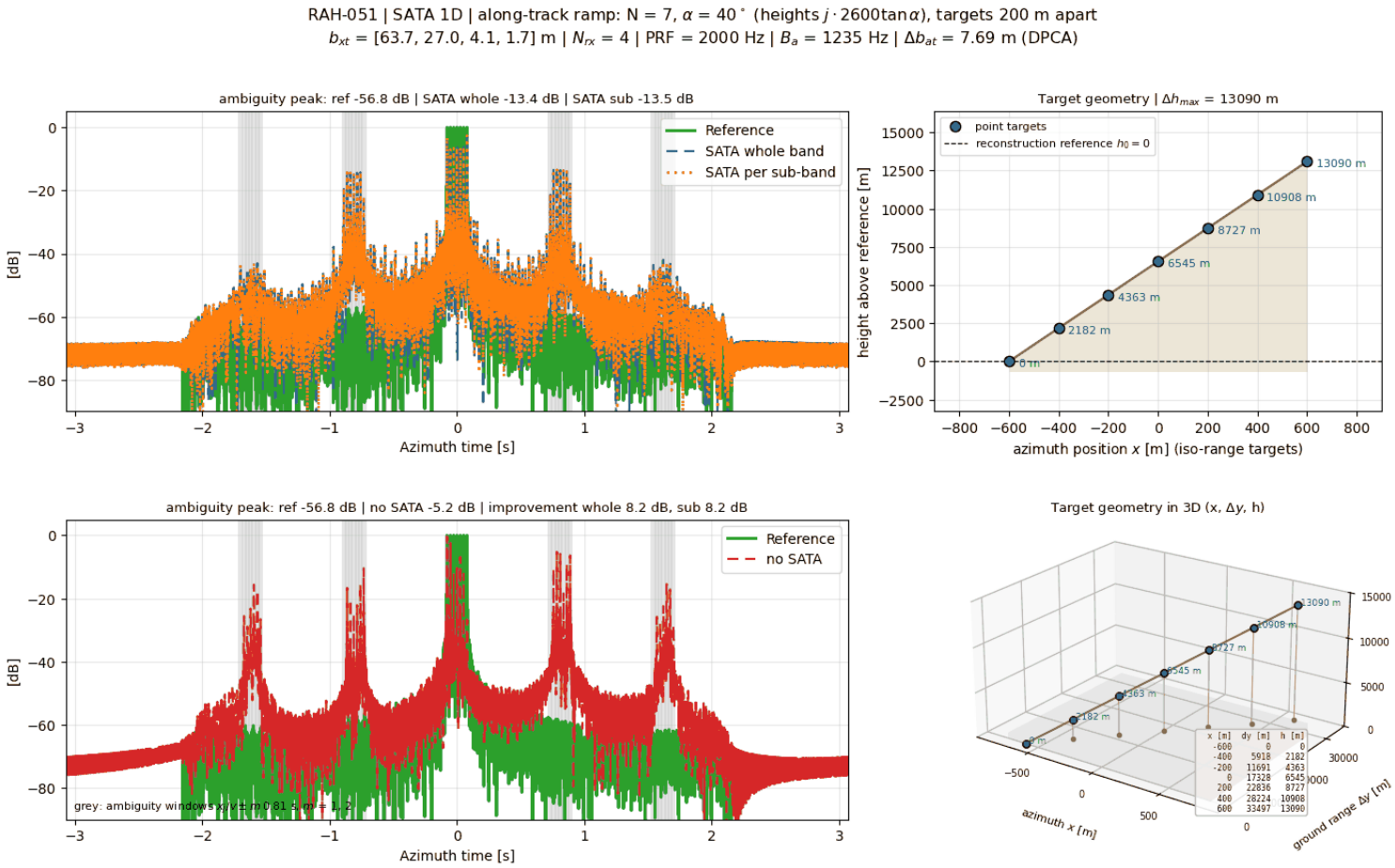


Figure 178: Along-track ramp, 200 m apart, large heights: RAH-051, $N = 7$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.2, SATA whole band −13.4, per sub-band −13.5 dB (improvement 8.2 dB).

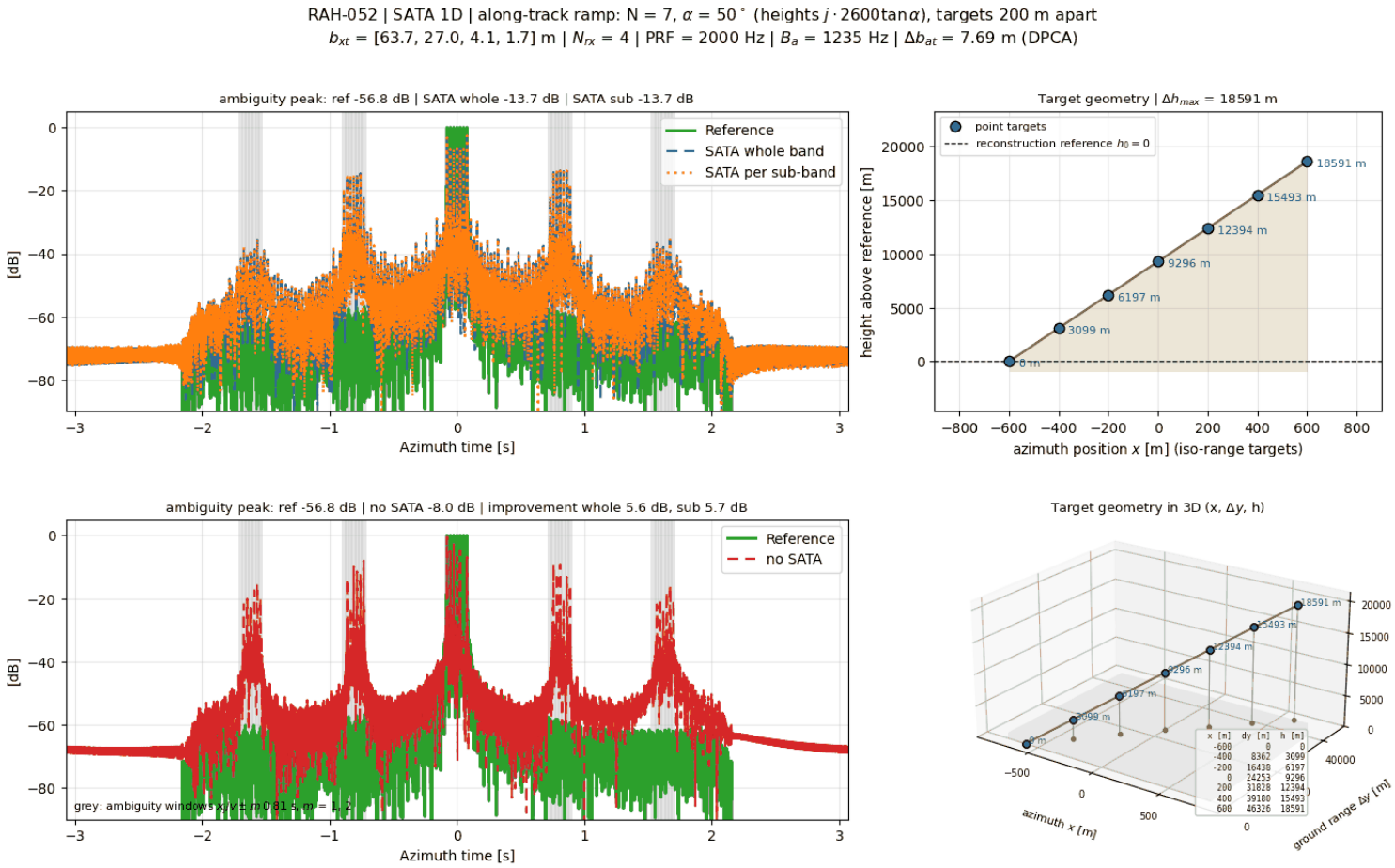


Figure 179: Along-track ramp, 200 m apart, large heights: RAH-052, $N = 7$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.0, SATA whole band −13.7, per sub-band −13.7 dB (improvement 5.6 dB).

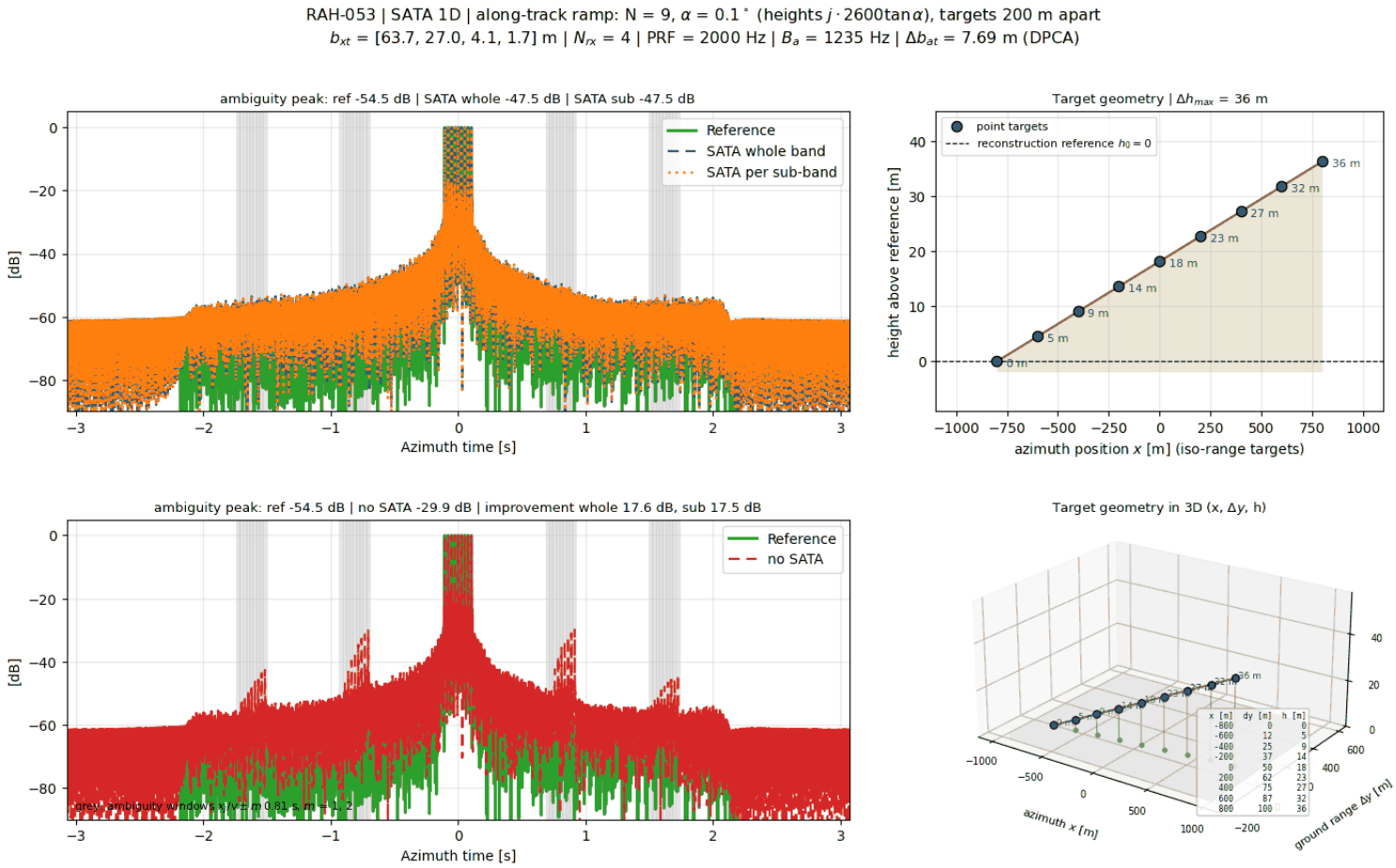


Figure 180: Along-track ramp, 200 m apart, large heights: RAH-053, $N = 9$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −29.9, SATA whole band −47.5, per sub-band −47.5 dB (improvement 17.6 dB).

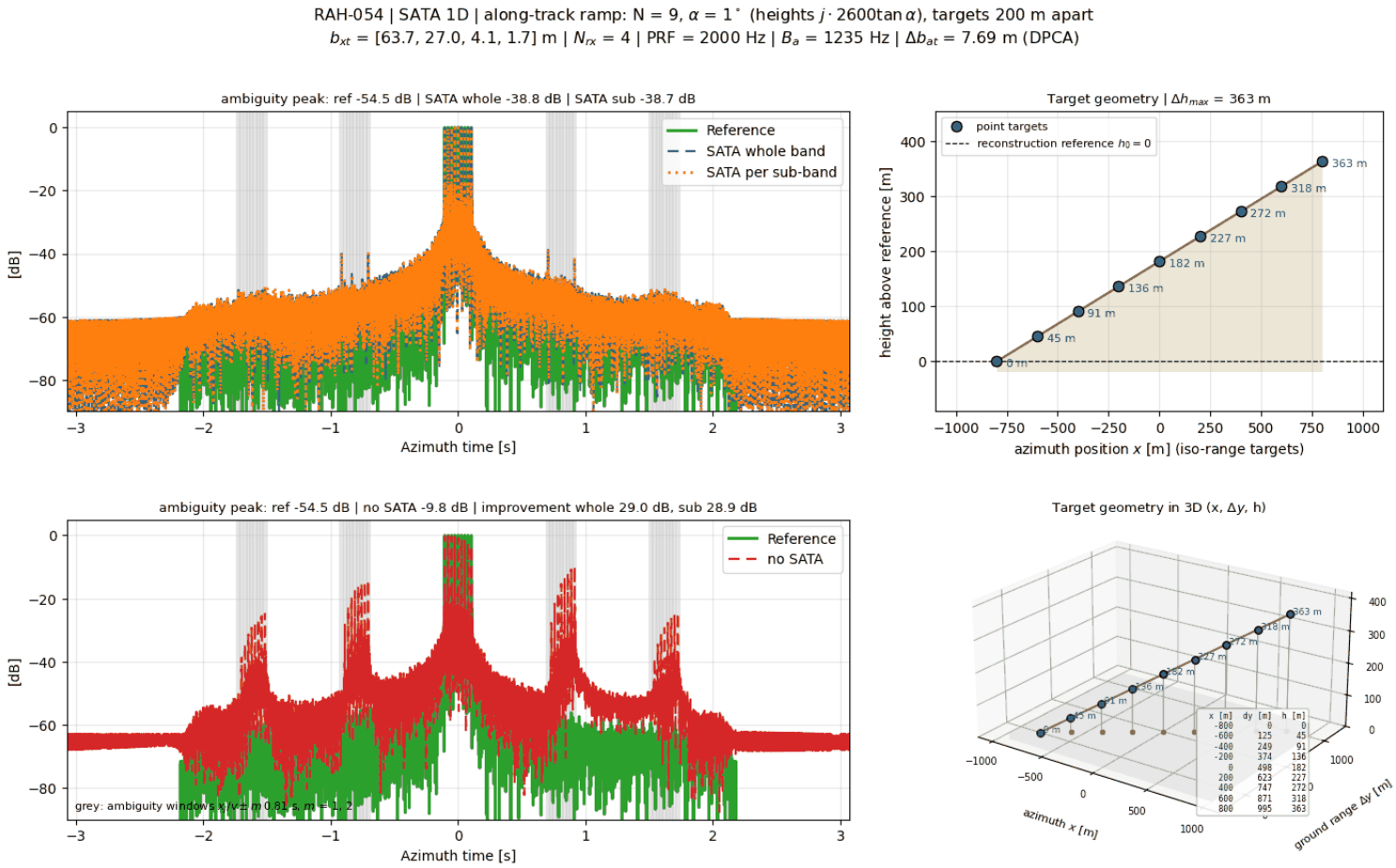


Figure 181: Along-track ramp, 200 m apart, large heights: RAH-054, $N = 9$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.8, SATA whole band −38.8, per sub-band −38.7 dB (improvement 29.0 dB).

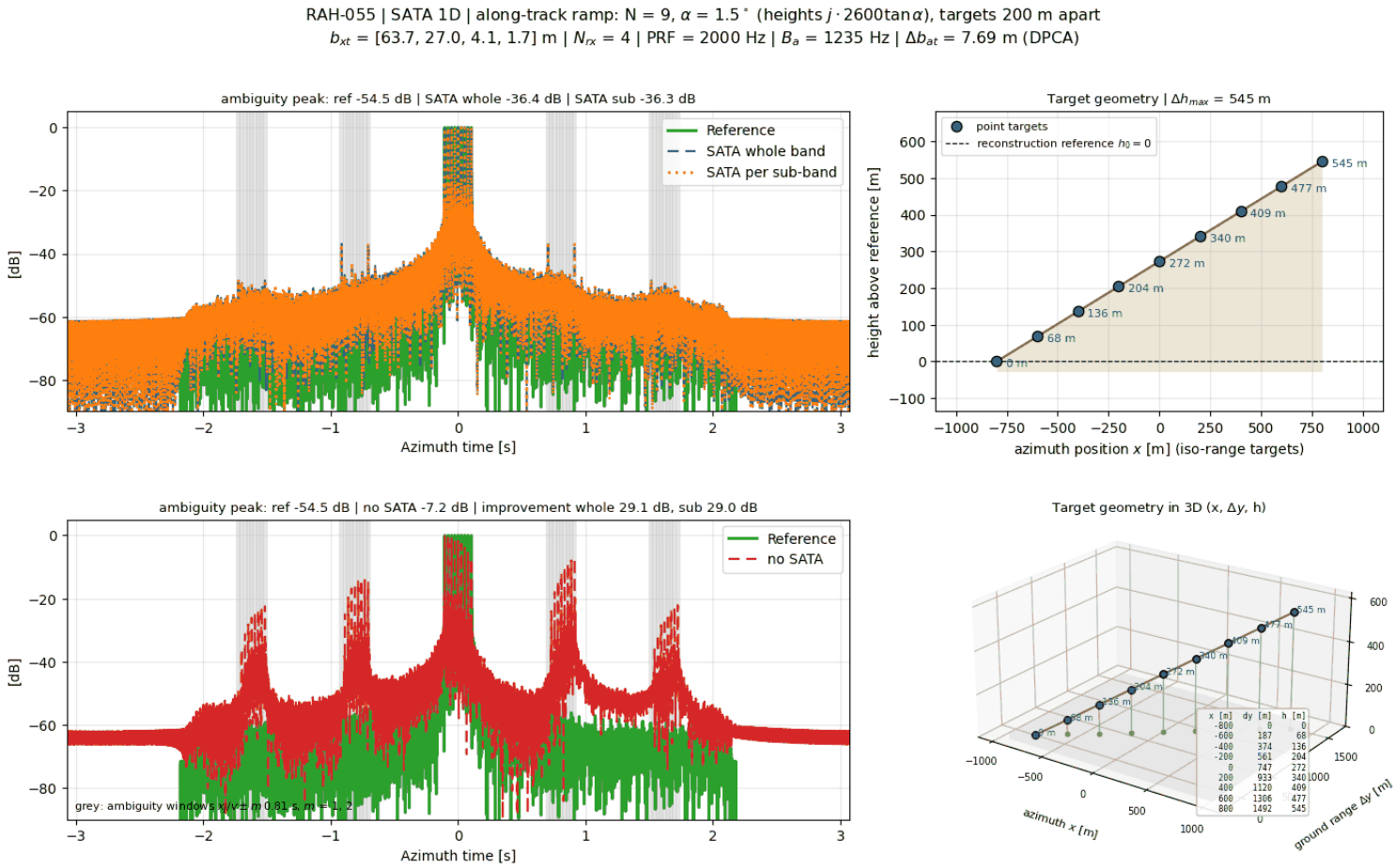


Figure 182: Along-track ramp, 200 m apart, large heights: RAH-055, $N = 9$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.2, SATA whole band −36.4, per sub-band −36.3 dB (improvement 29.1 dB).

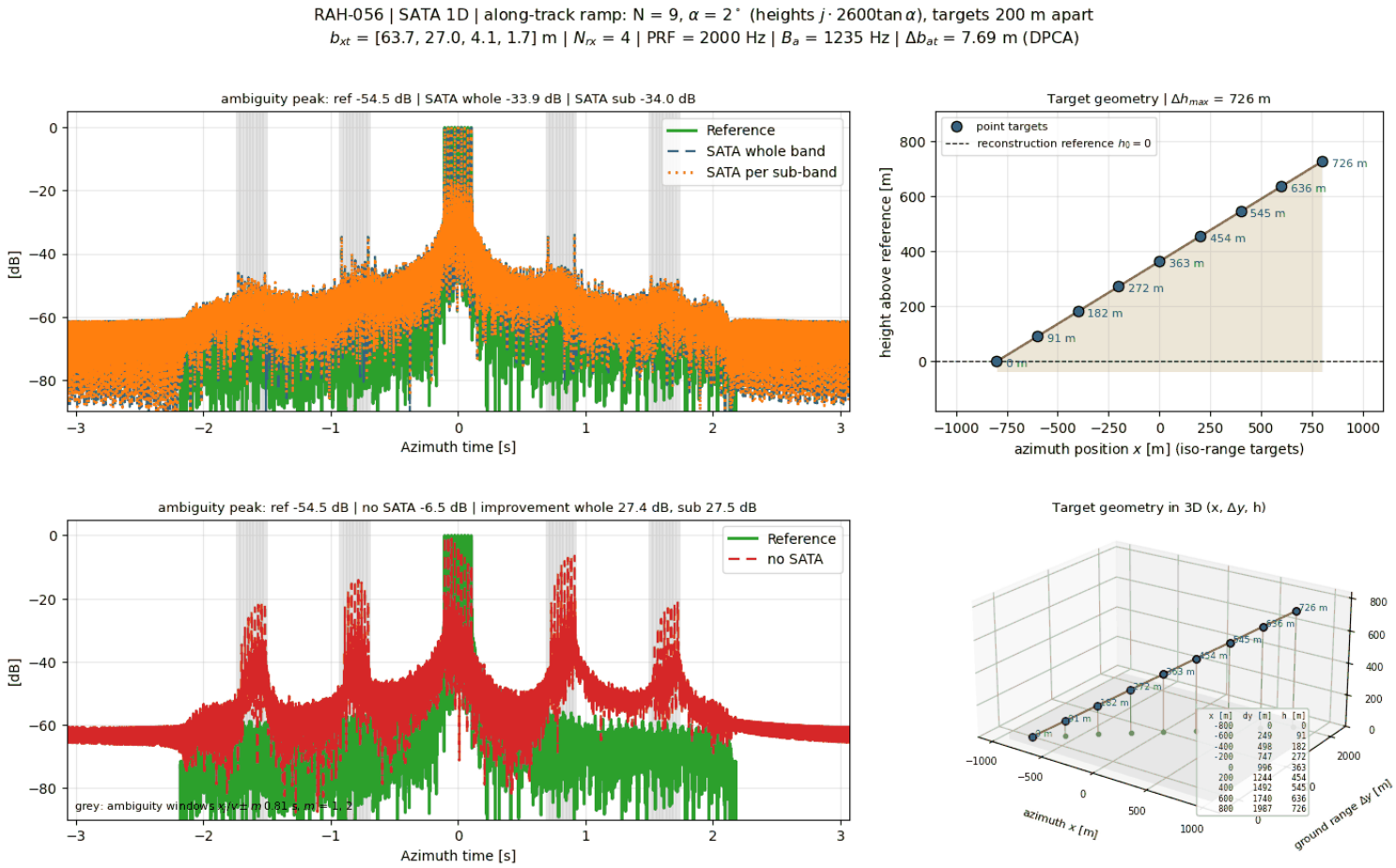


Figure 183: Along-track ramp, 200 m apart, large heights: RAH-056, $N = 9$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −33.9, per sub-band −34.0 dB (improvement 27.4 dB).

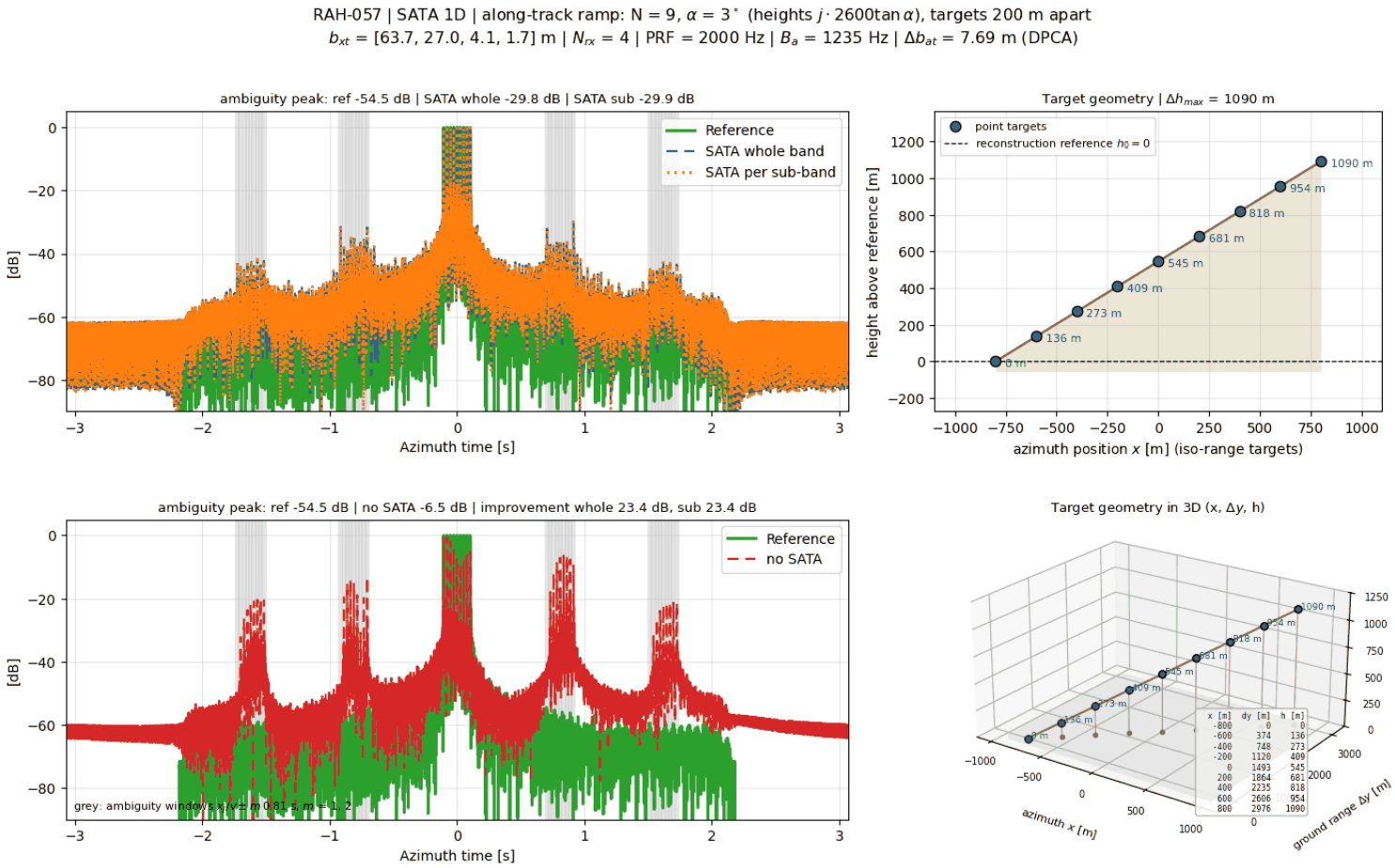


Figure 184: Along-track ramp, 200 m apart, large heights: RAH-057, $N = 9$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -6.5, SATA whole band -29.8, per sub-band -29.9 dB (improvement 23.4 dB).

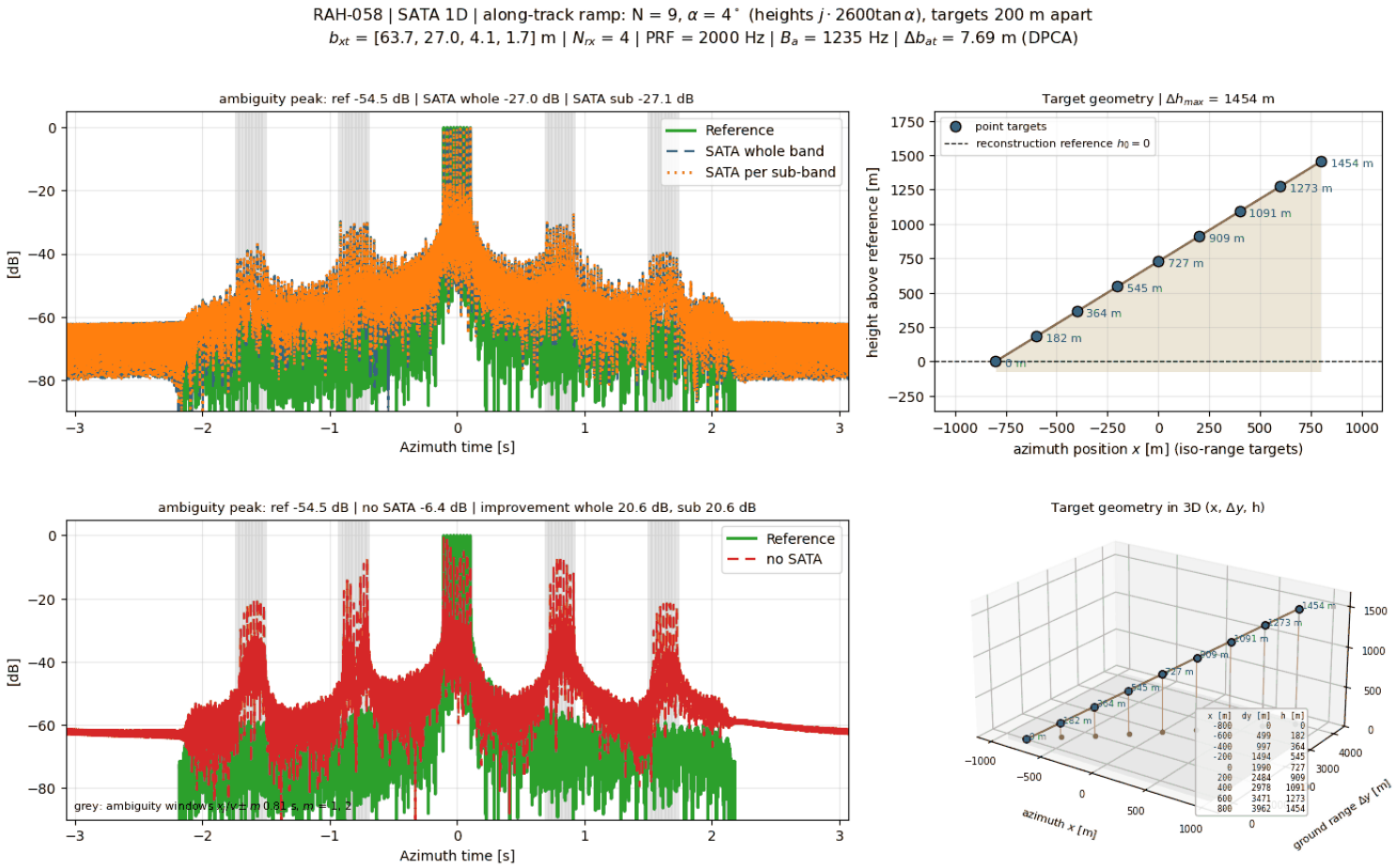


Figure 185: Along-track ramp, 200 m apart, large heights: RAH-058, $N = 9$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −27.0, per sub-band −27.1 dB (improvement 20.6 dB).

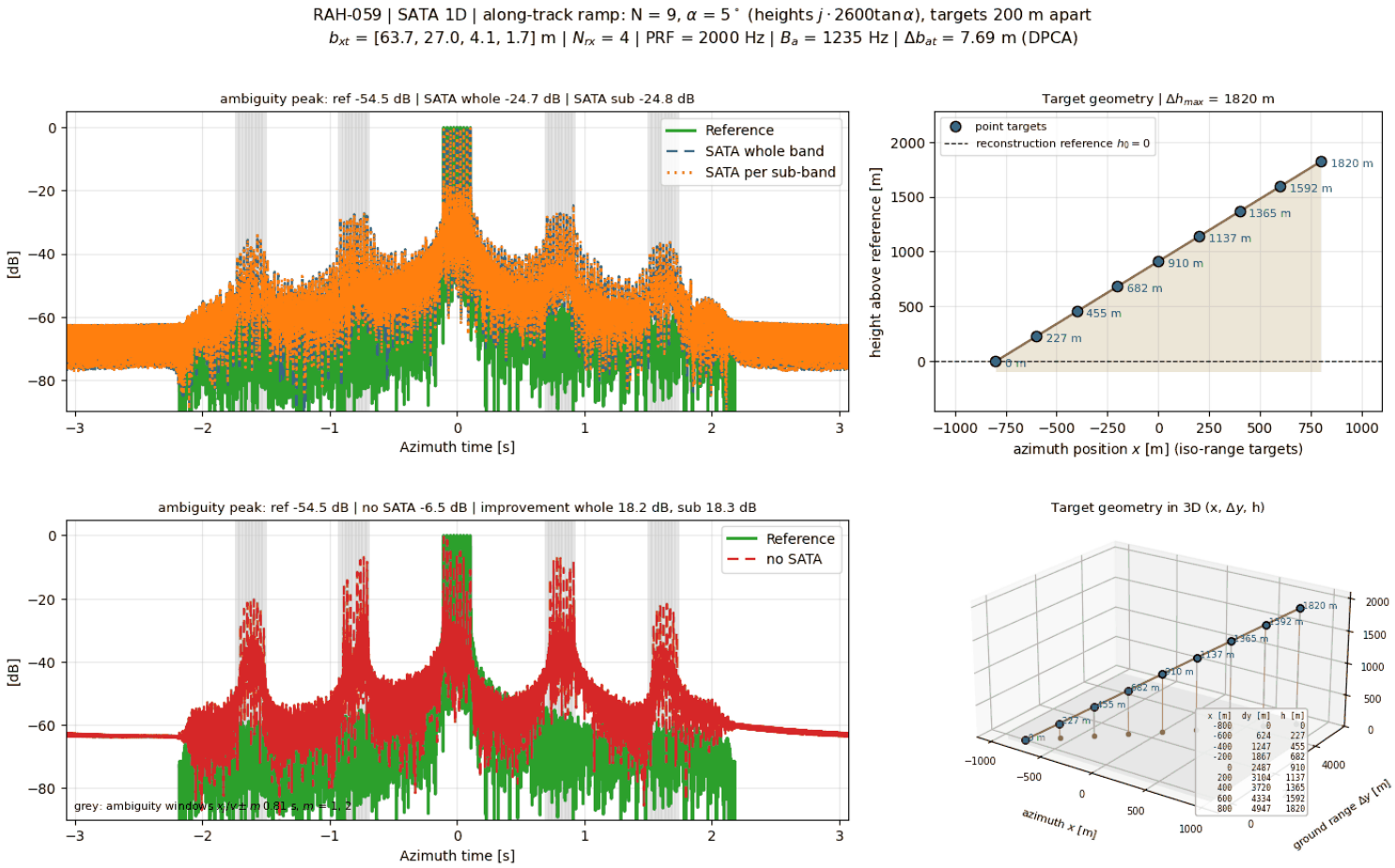


Figure 186: Along-track ramp, 200 m apart, large heights: RAH-059, $N = 9$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −24.7, per sub-band −24.8 dB (improvement 18.2 dB).

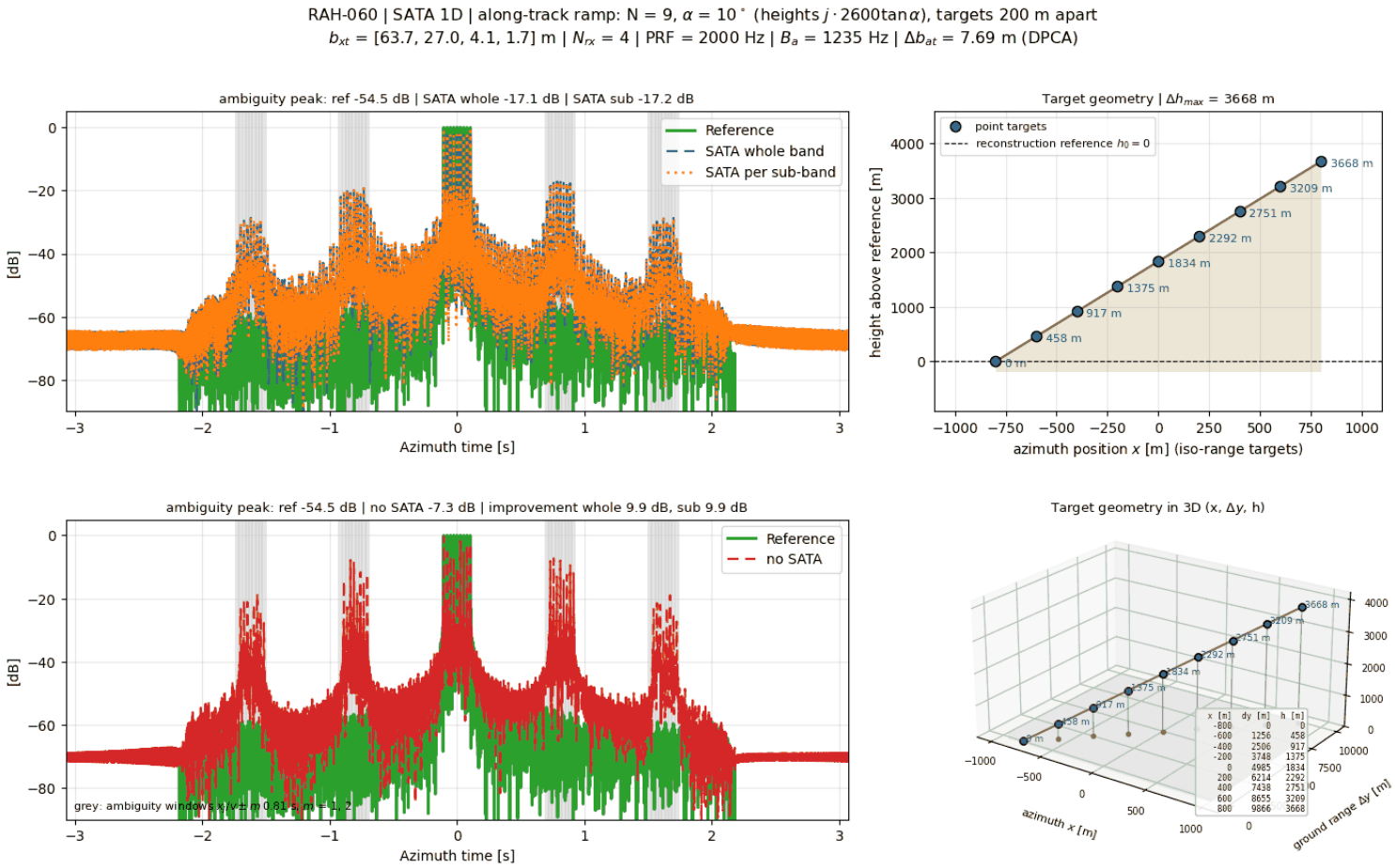


Figure 187: Along-track ramp, 200 m apart, large heights: RAH-060, $N = 9$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.3, SATA whole band −17.1, per sub-band −17.2 dB (improvement 9.9 dB).

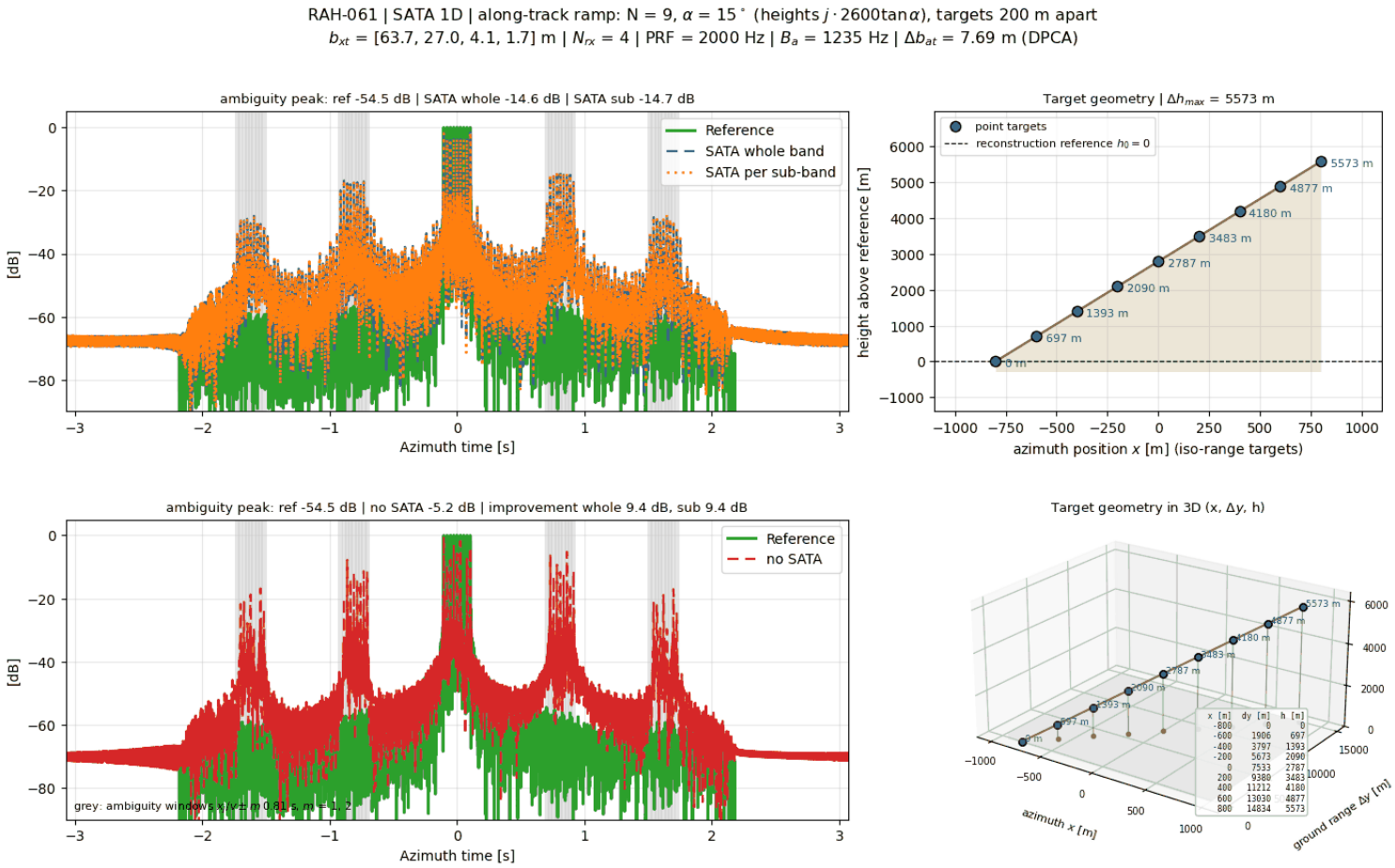


Figure 188: Along-track ramp, 200 m apart, large heights: RAH-061, $N = 9$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.2, SATA whole band −14.6, per sub-band −14.7 dB (improvement 9.4 dB).

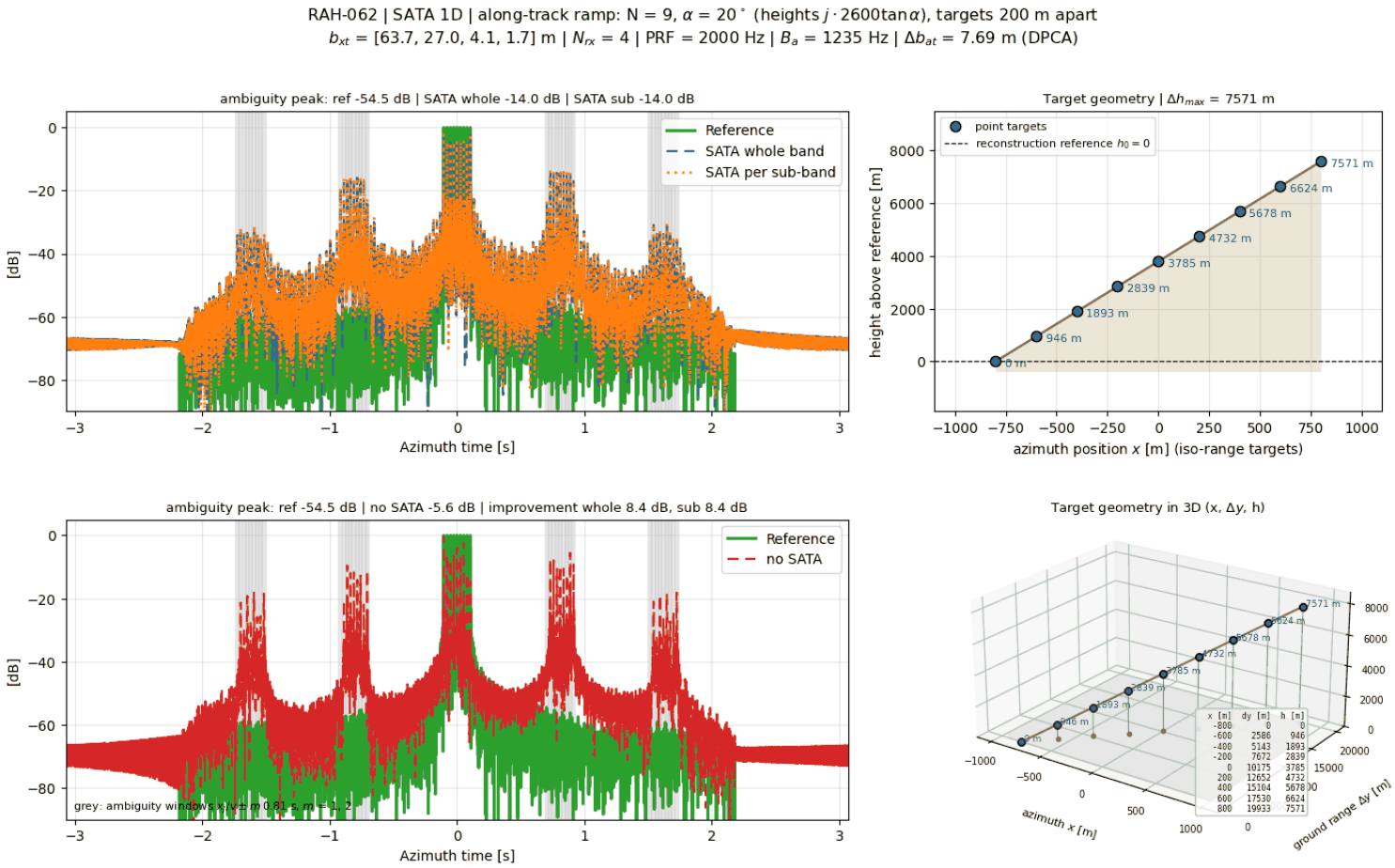


Figure 189: Along-track ramp, 200 m apart, large heights: RAH-062, $N = 9$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.6, SATA whole band −14.0, per sub-band −14.0 dB (improvement 8.4 dB).

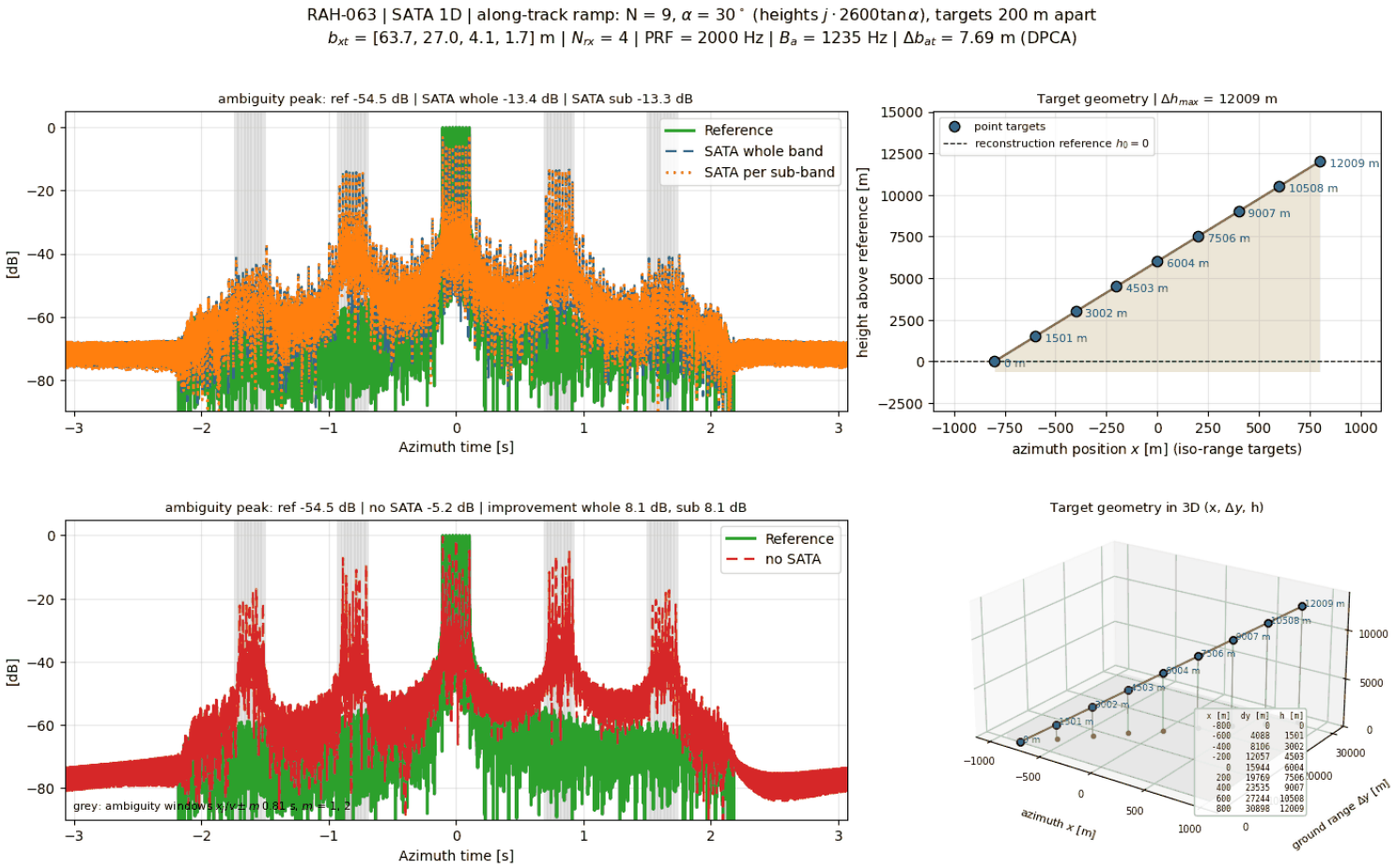


Figure 190: Along-track ramp, 200 m apart, large heights: RAH-063, $N = 9$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.2, SATA whole band −13.4, per sub-band −13.3 dB (improvement 8.1 dB).

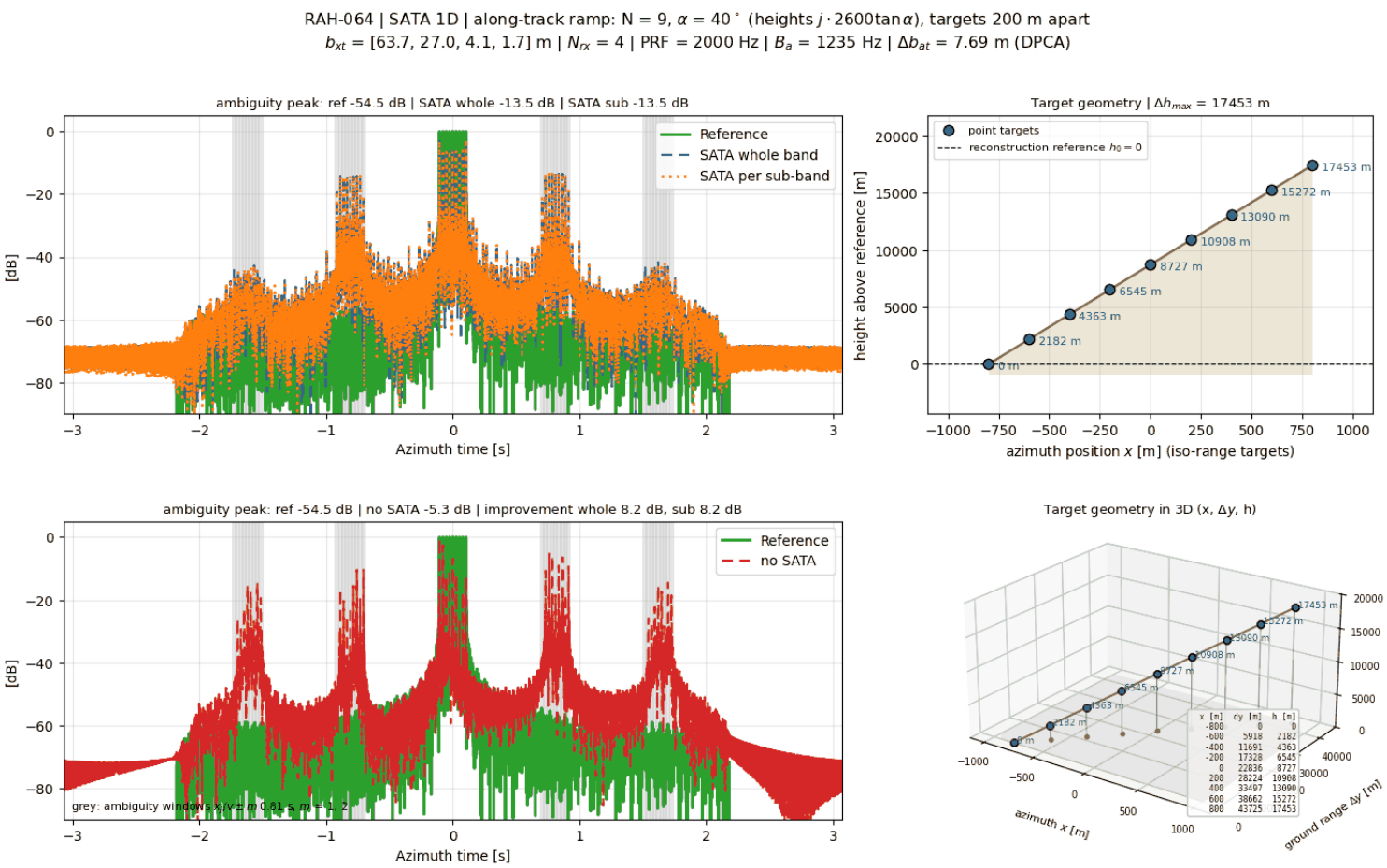


Figure 191: Along-track ramp, 200 m apart, large heights: RAH-064, $N = 9$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.3, SATA whole band −13.5, per sub-band −13.5 dB (improvement 8.2 dB).

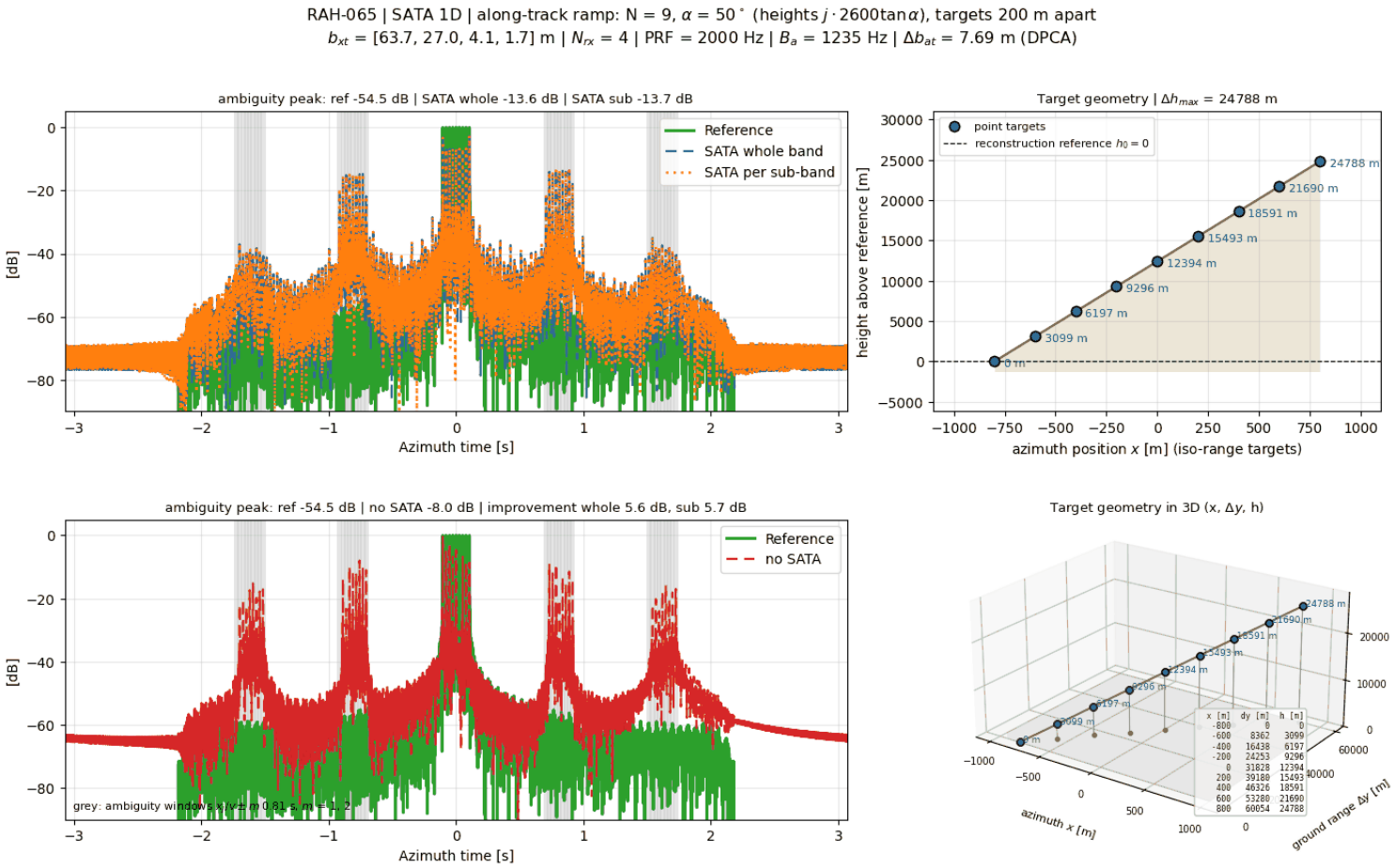


Figure 192: Along-track ramp, 200 m apart, large heights: RAH-065, $N = 9$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.0, SATA whole band −13.6, per sub-band −13.7 dB (improvement 5.6 dB).

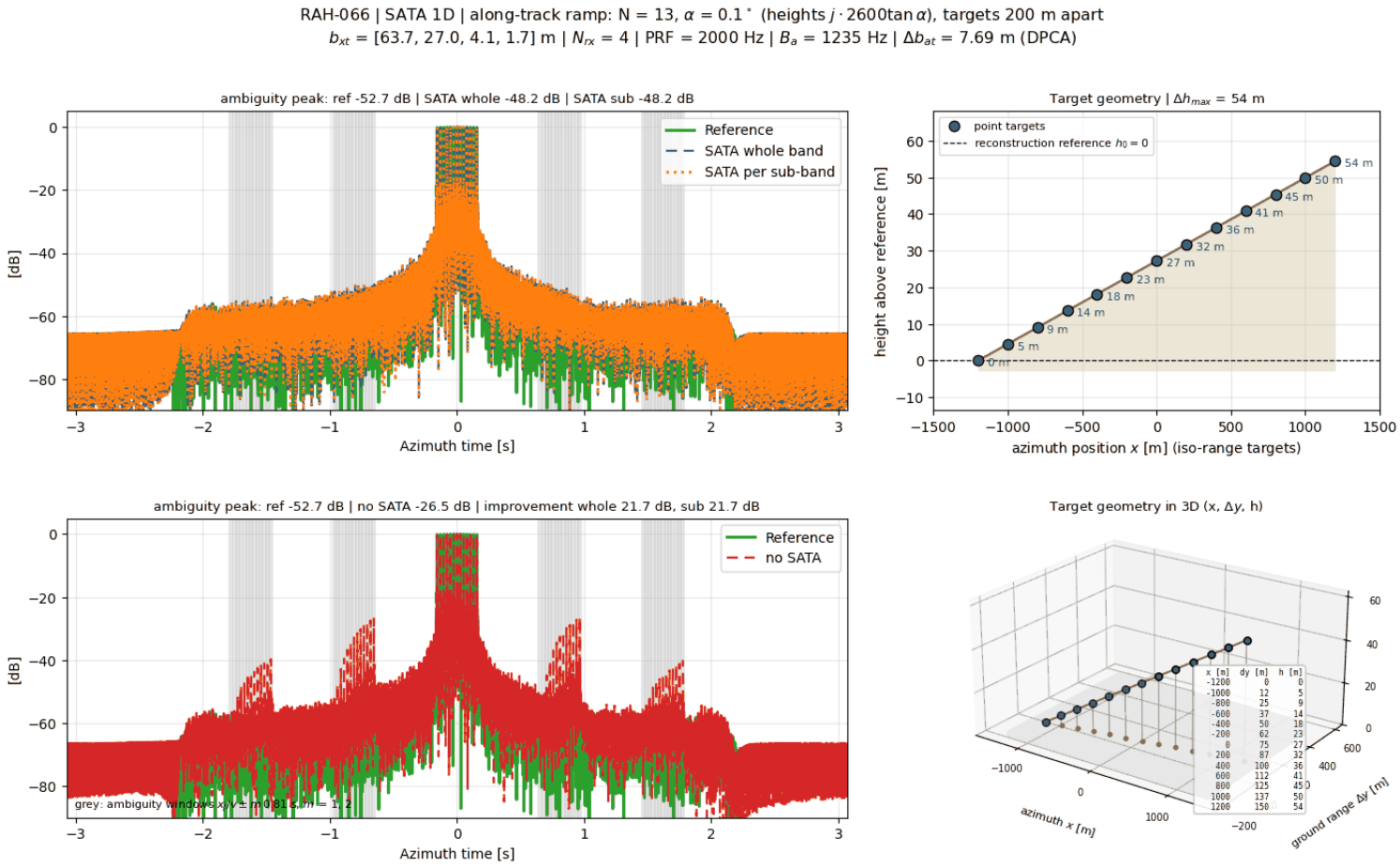


Figure 193: Along-track ramp, 200 m apart, large heights: RAH-066, $N = 13$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −26.5, SATA whole band −48.2, per sub-band −48.2 dB (improvement 21.7 dB).

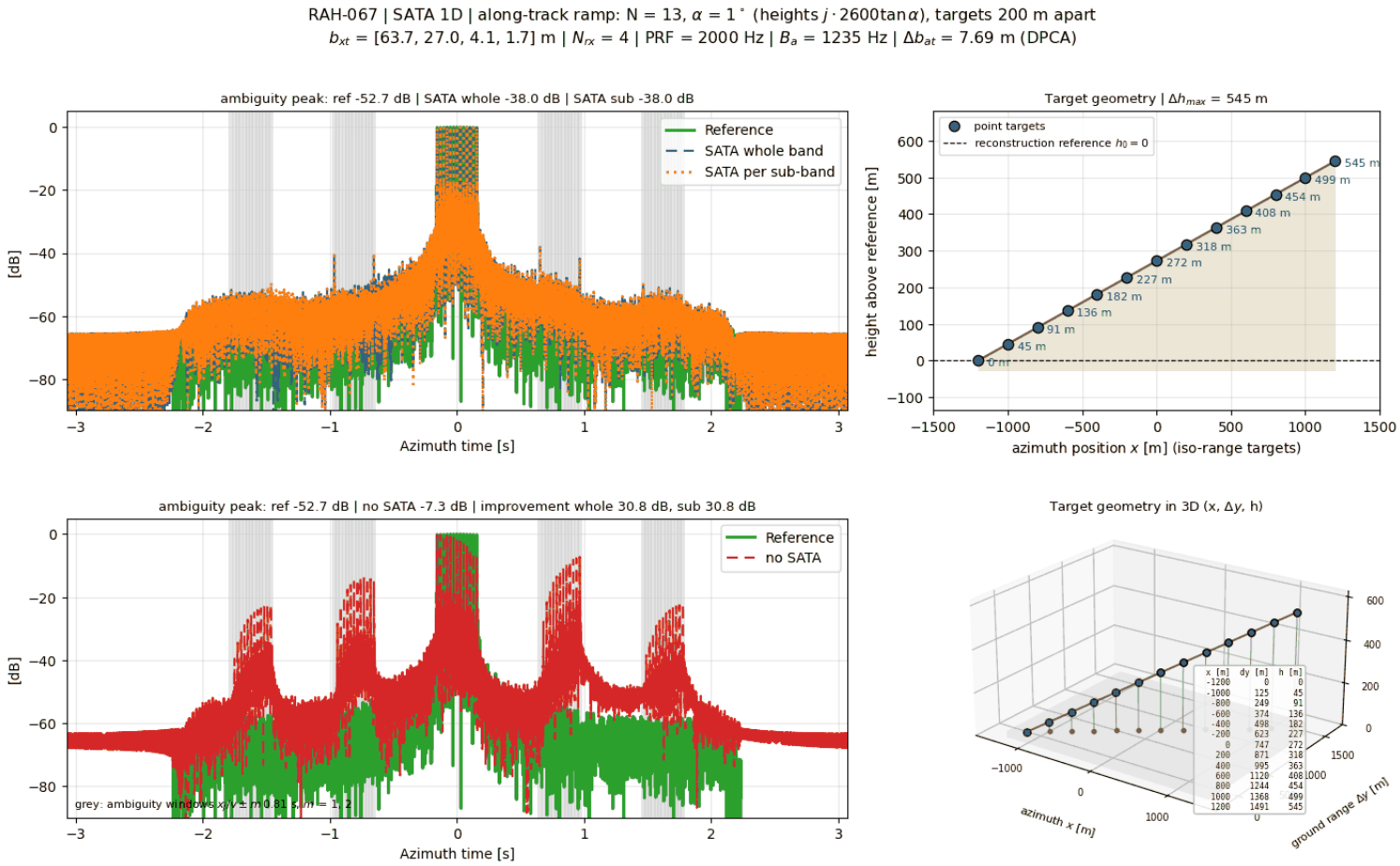


Figure 194: Along-track ramp, 200 m apart, large heights: RAH-067, $N = 13$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.3, SATA whole band −38.0, per sub-band −38.0 dB (improvement 30.8 dB).

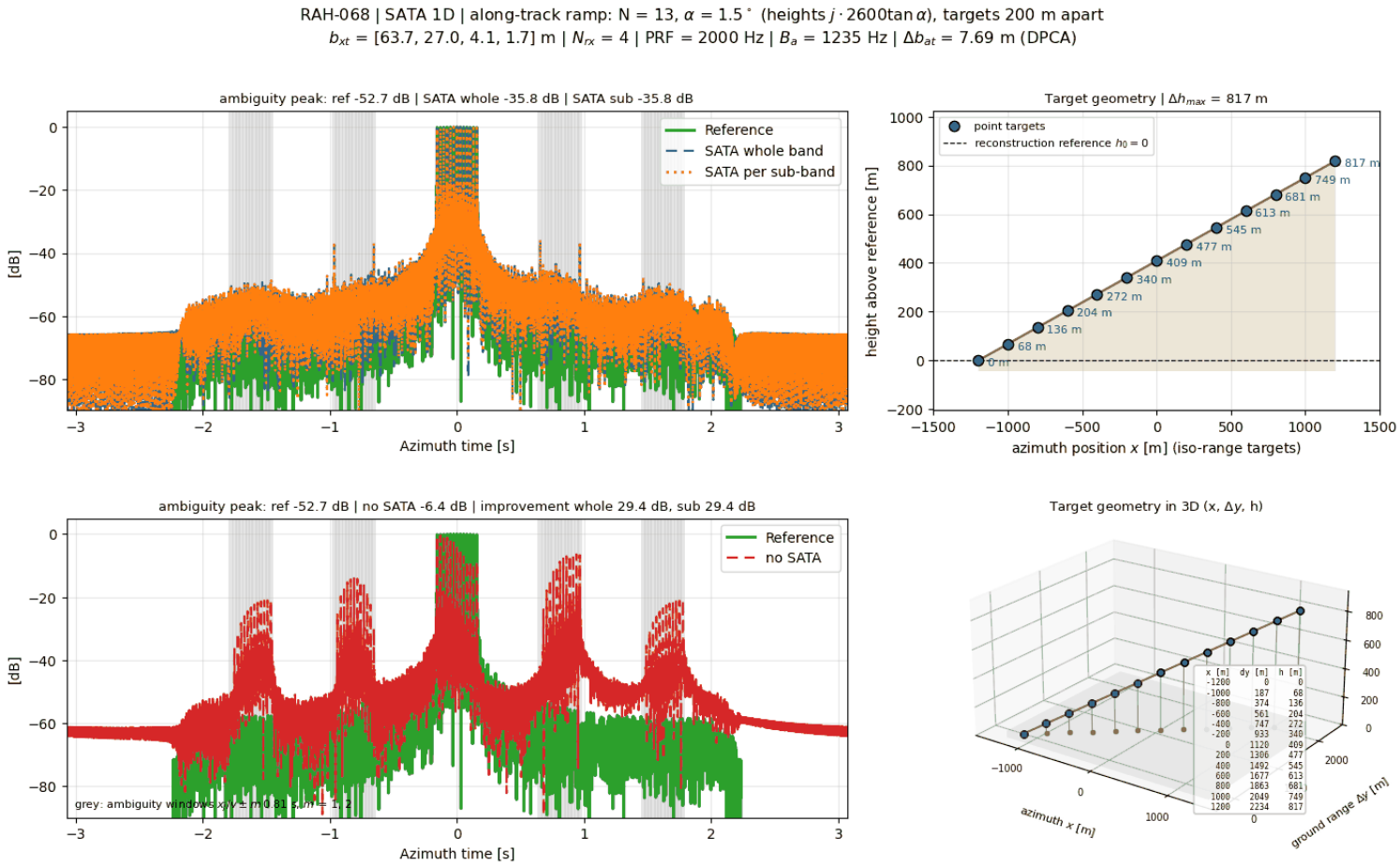


Figure 195: Along-track ramp, 200 m apart, large heights: RAH-068, $N = 13$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −35.8, per sub-band −35.8 dB (improvement 29.4 dB).

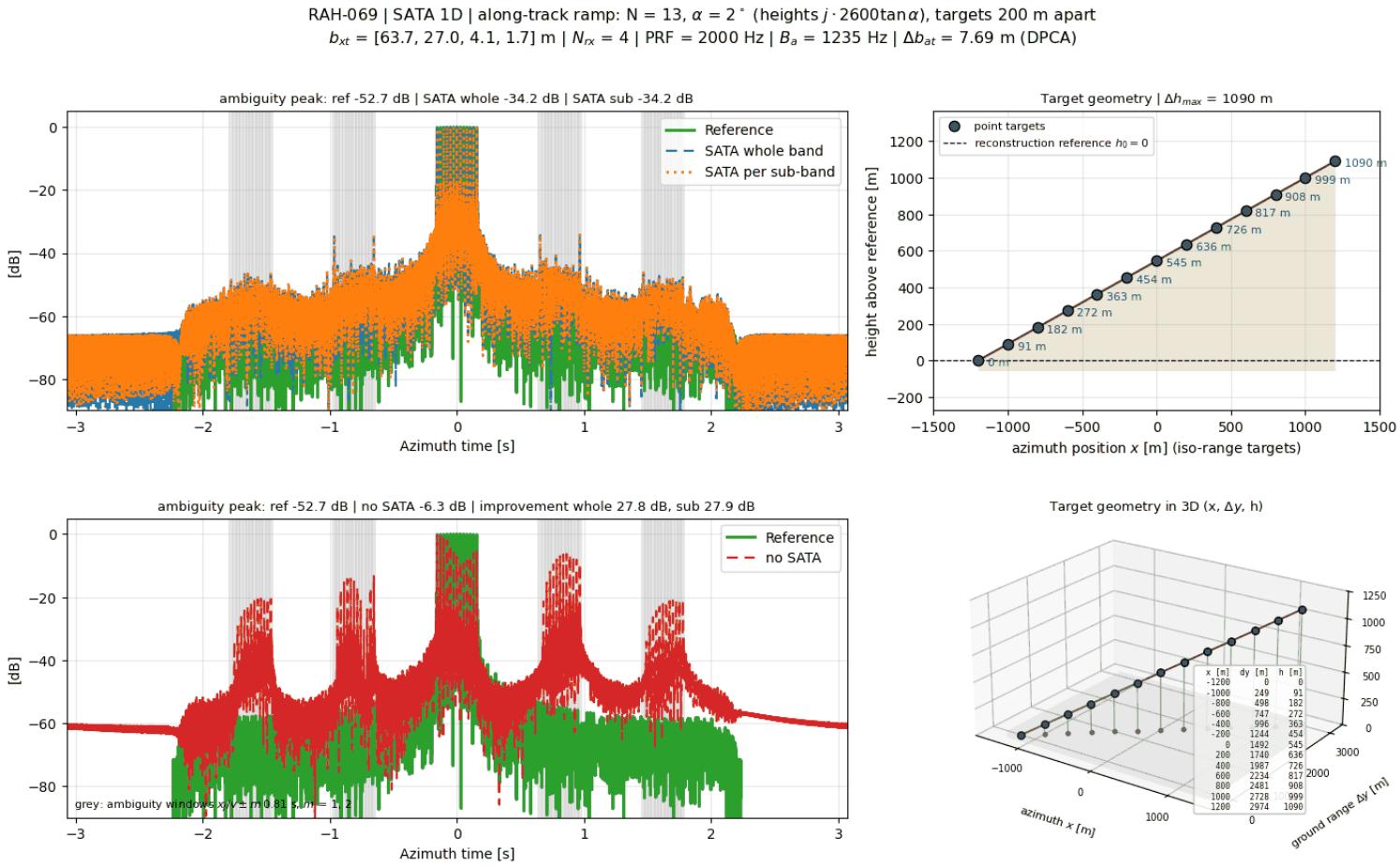


Figure 196: Along-track ramp, 200 m apart, large heights: RAH-069, $N = 13$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.3, SATA whole band −34.2, per sub-band −34.2 dB (improvement 27.8 dB).

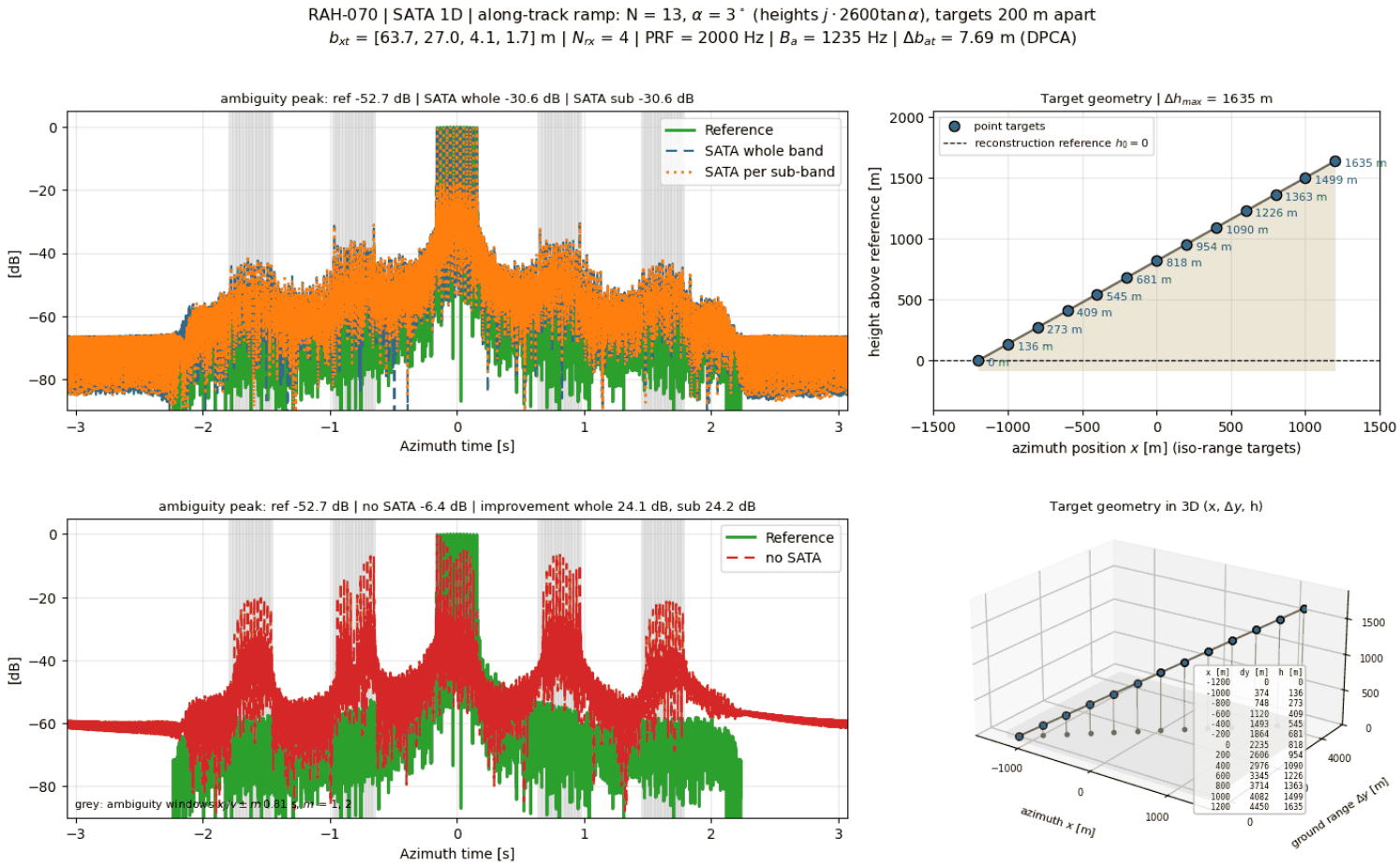


Figure 197: Along-track ramp, 200 m apart, large heights: RAH-070, $N = 13$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −30.6, per sub-band −30.6 dB (improvement 24.1 dB).

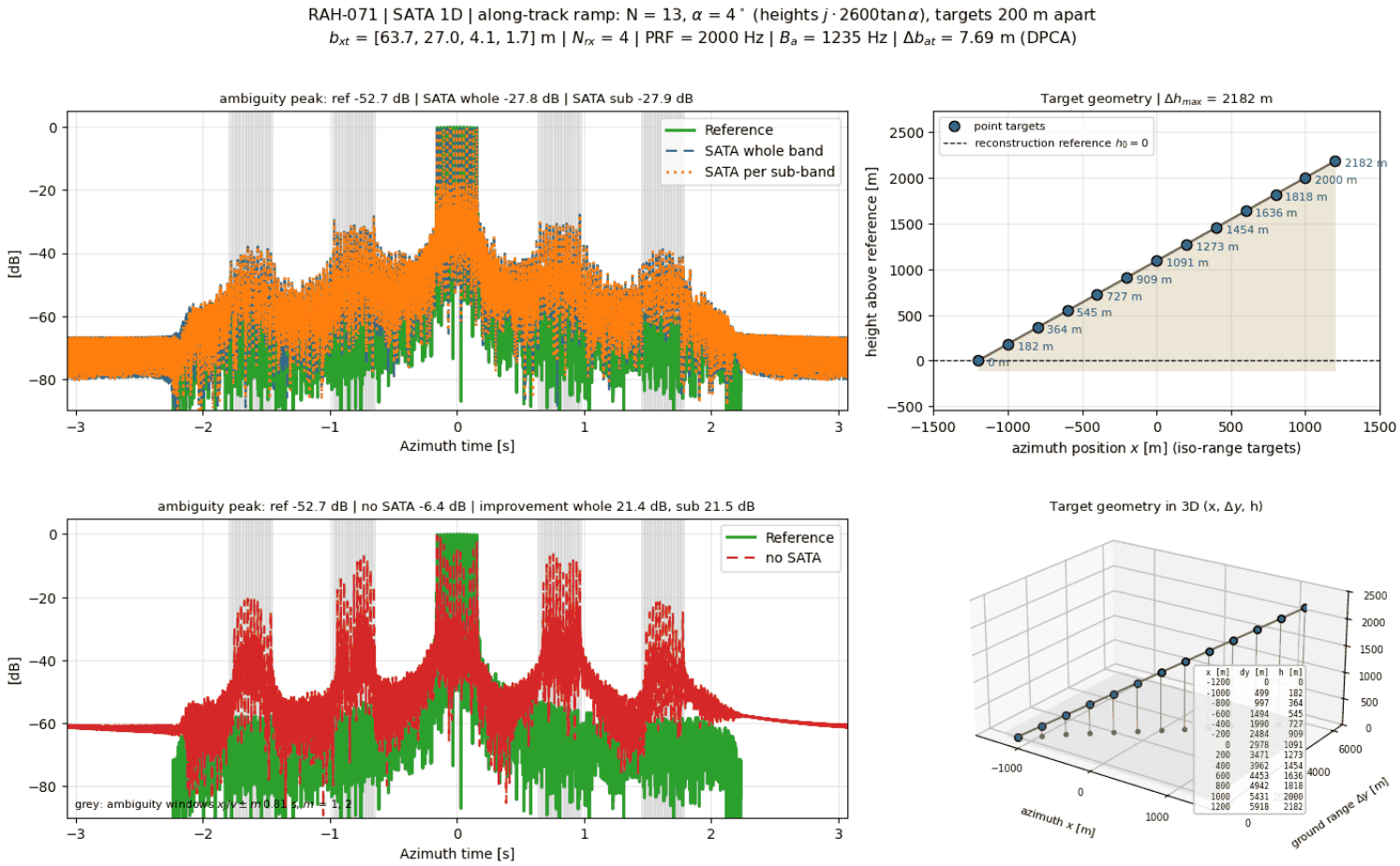


Figure 198: Along-track ramp, 200 m apart, large heights: RAH-071, $N = 13$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −27.8, per sub-band −27.9 dB (improvement 21.4 dB).

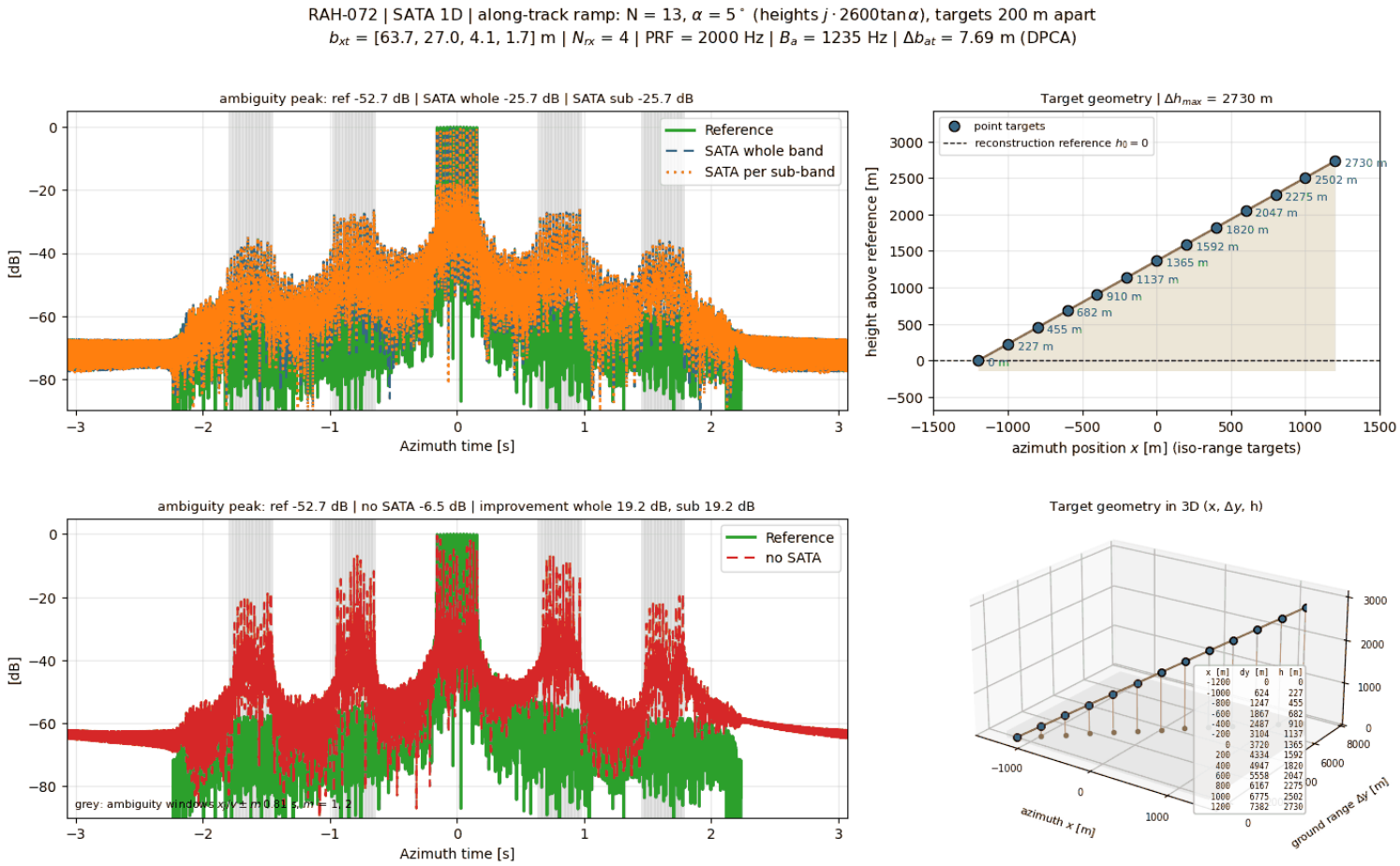


Figure 199: Along-track ramp, 200 m apart, large heights: RAH-072, $N = 13$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.5, SATA whole band −25.7, per sub-band −25.7 dB (improvement 19.2 dB).

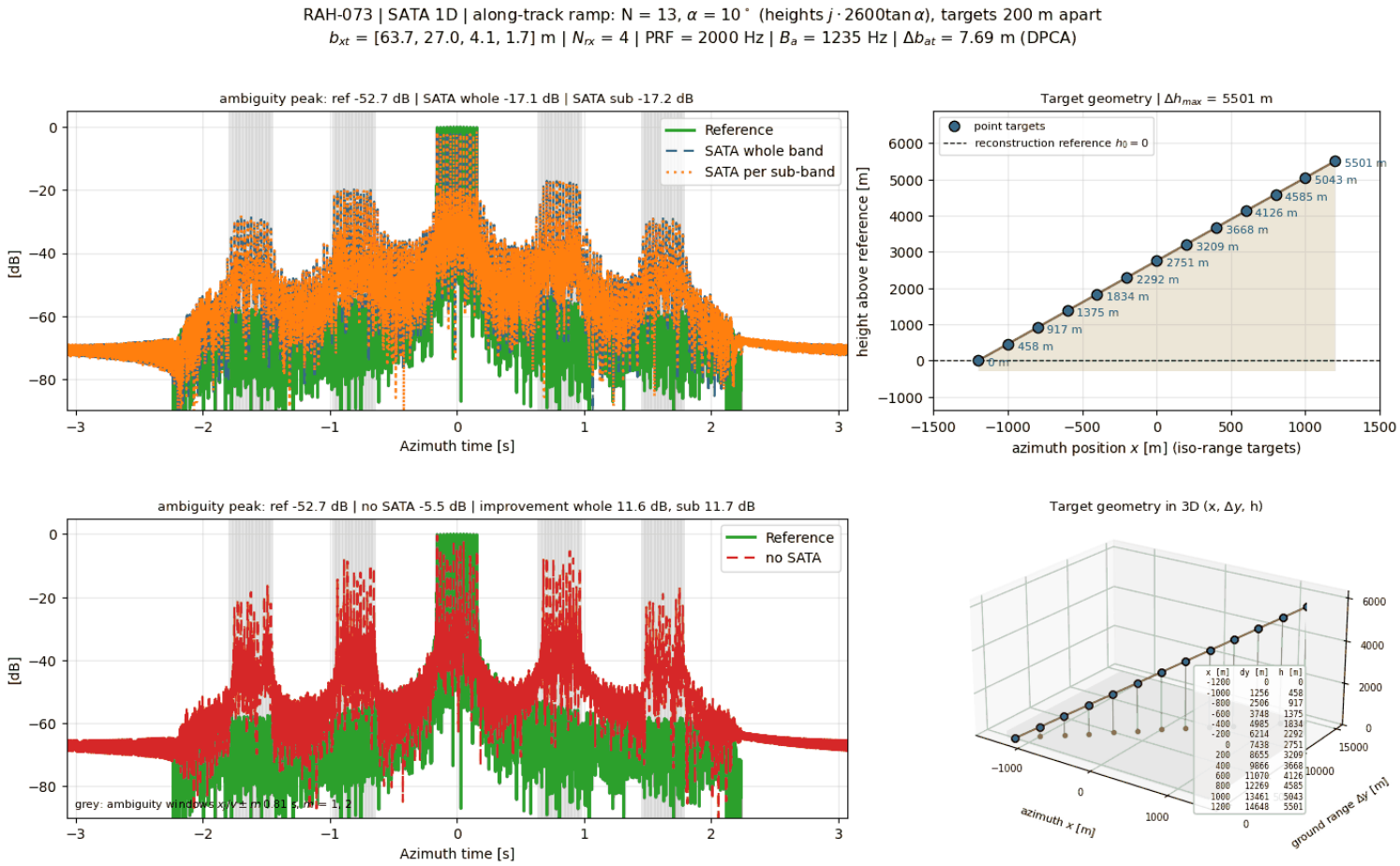


Figure 200: Along-track ramp, 200 m apart, large heights: RAH-073, $N = 13$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.5, SATA whole band −17.1, per sub-band −17.2 dB (improvement 11.6 dB).

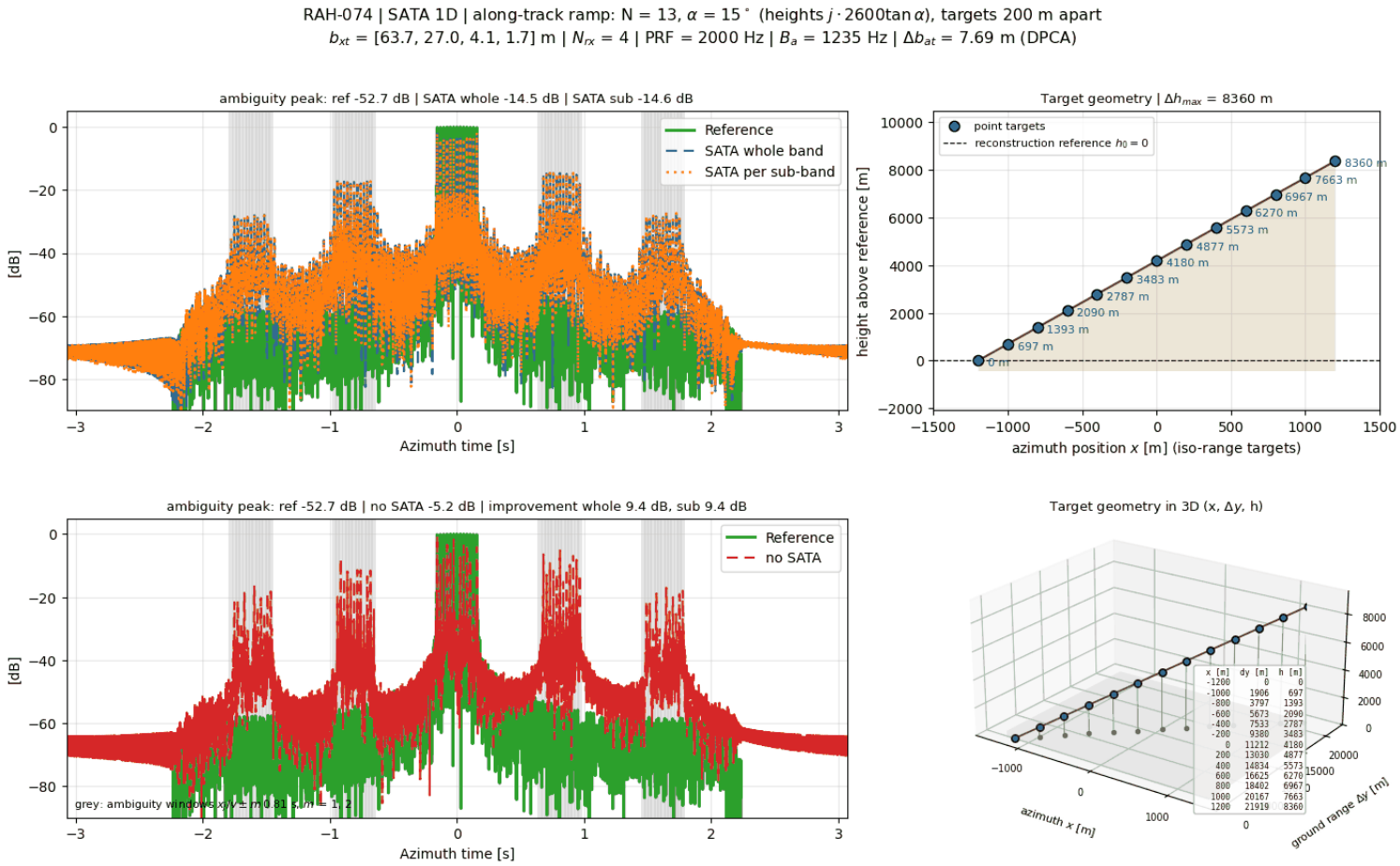


Figure 201: Along-track ramp, 200 m apart, large heights: RAH-074, $N = 13$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.2, SATA whole band −14.5, per sub-band −14.6 dB (improvement 9.4 dB).

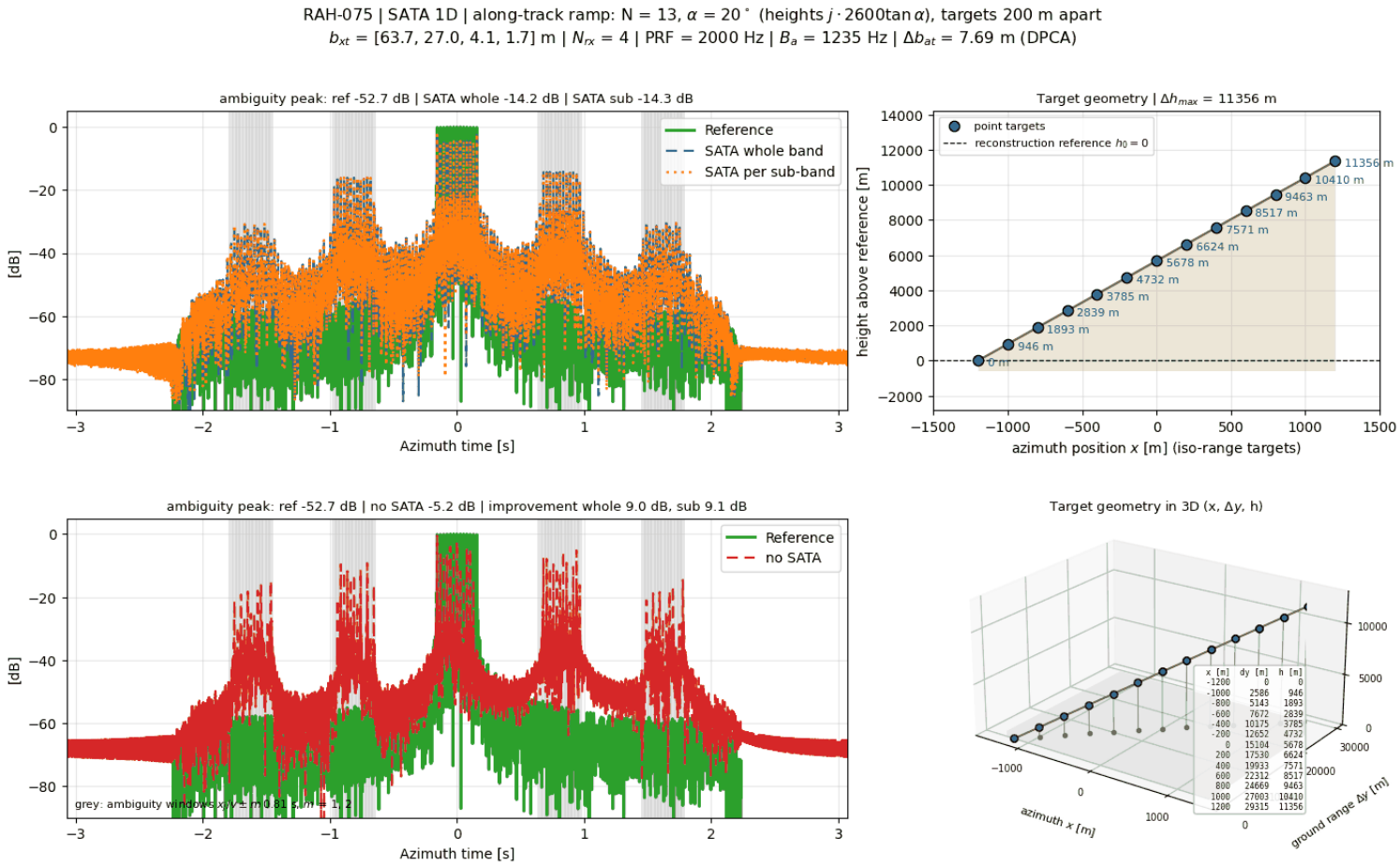


Figure 202: Along-track ramp, 200 m apart, large heights: RAH-075, $N = 13$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.2, SATA whole band −14.2, per sub-band −14.3 dB (improvement 9.0 dB).

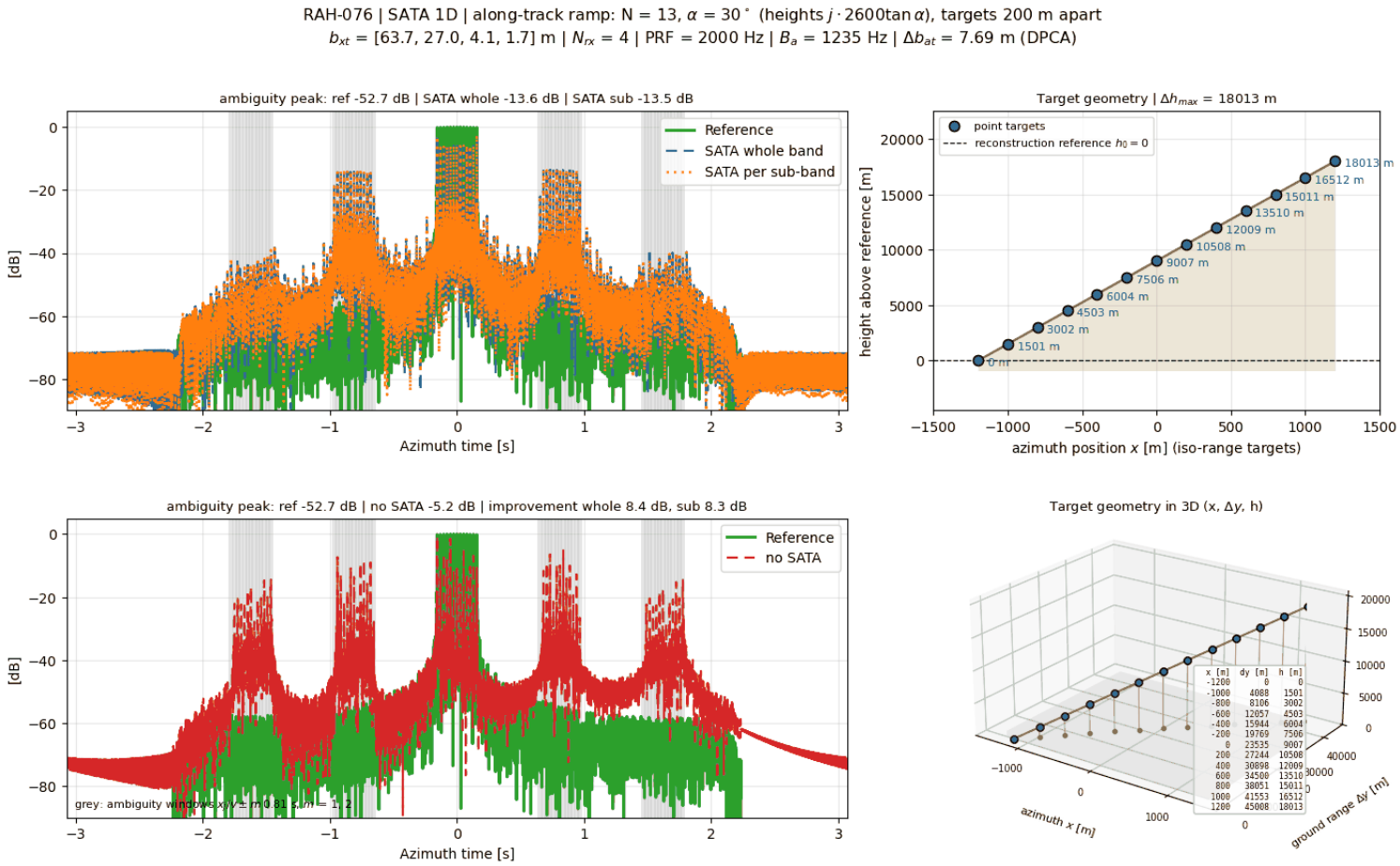


Figure 203: Along-track ramp, 200 m apart, large heights: RAH-076, $N = 13$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.2, SATA whole band −13.6, per sub-band −13.5 dB (improvement 8.4 dB).

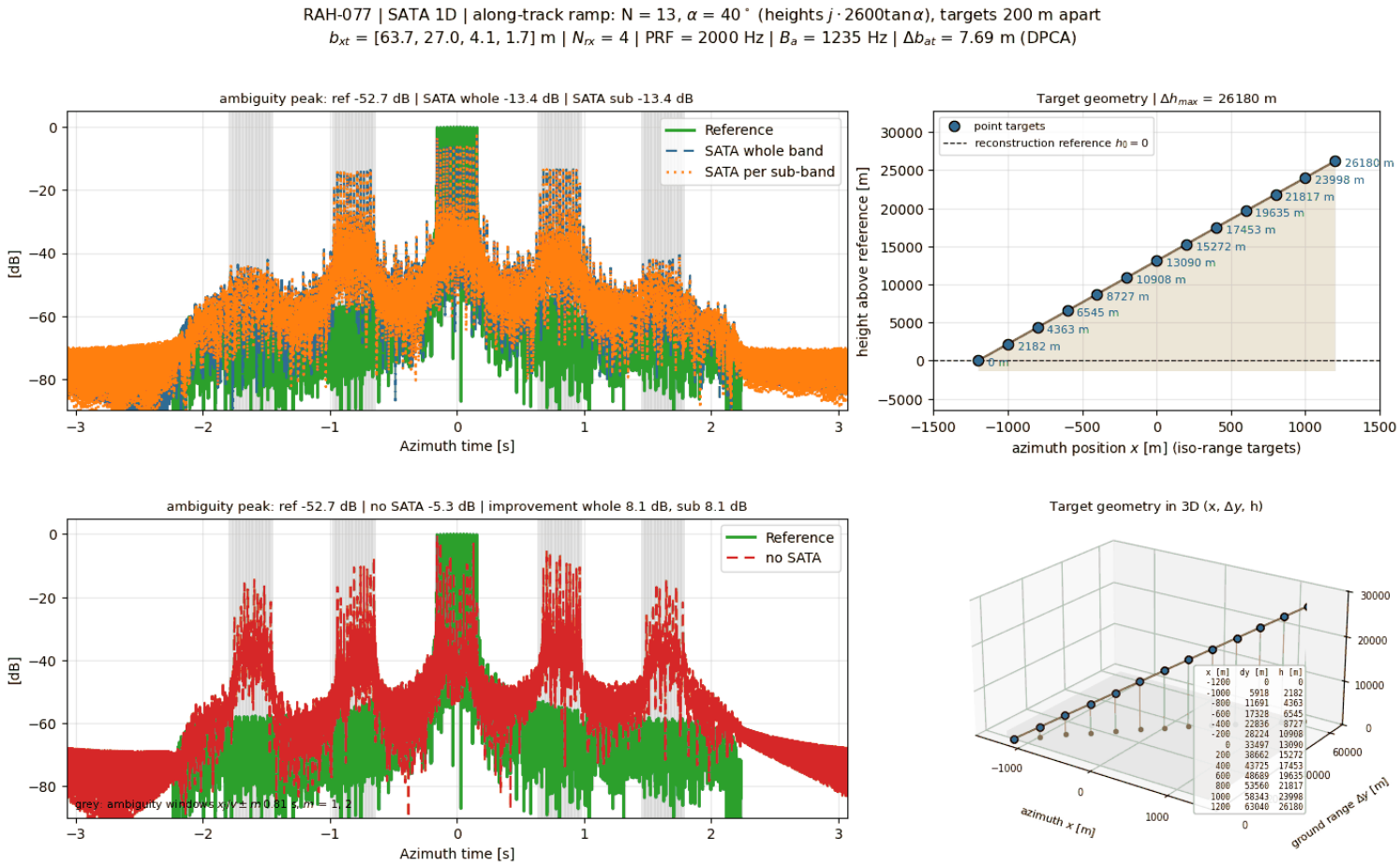


Figure 204: Along-track ramp, 200 m apart, large heights: RAH-077, $N = 13$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.3, SATA whole band −13.4, per sub-band −13.4 dB (improvement 8.1 dB).

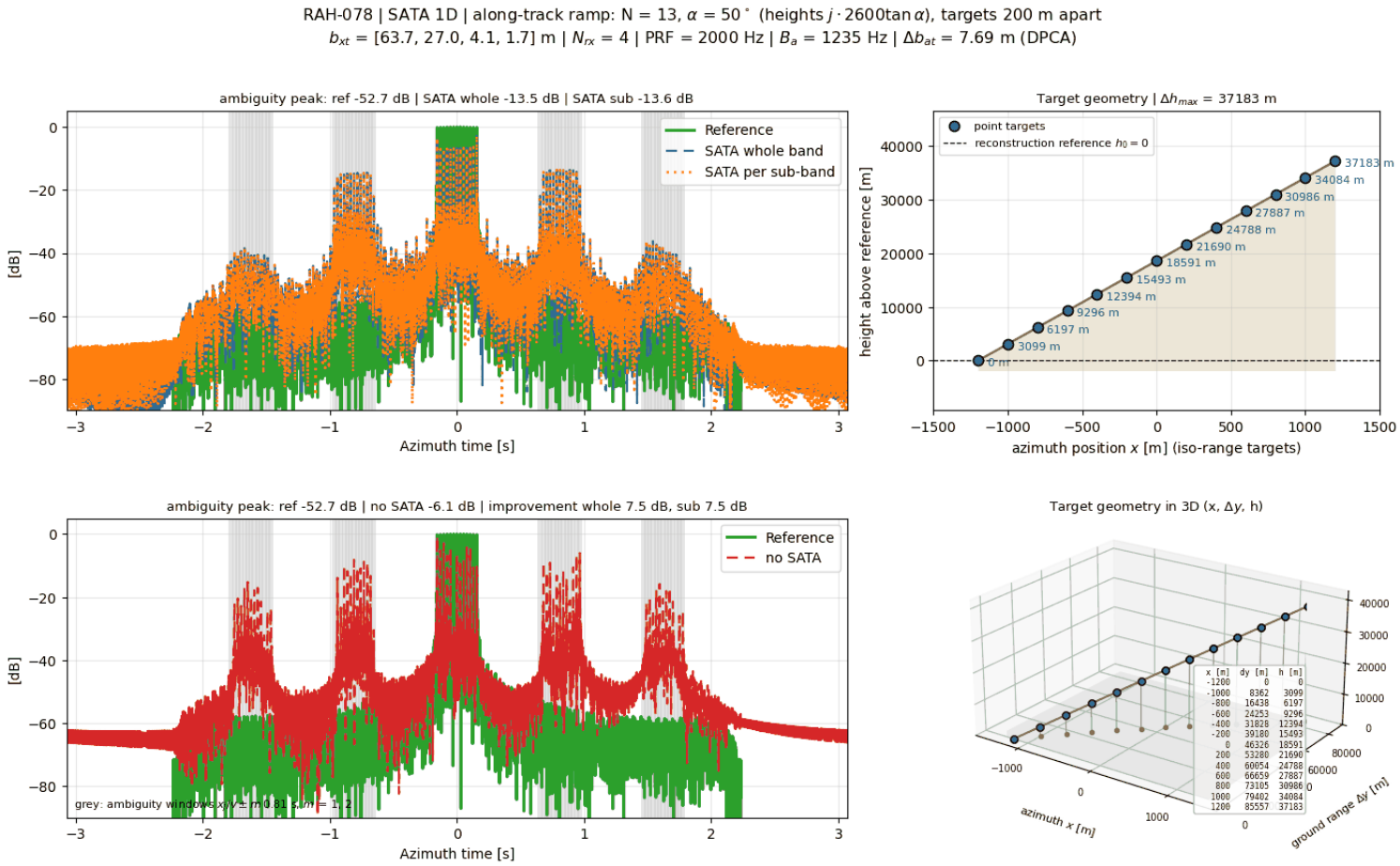


Figure 205: Along-track ramp, 200 m apart, large heights: RAH-078, $N = 13$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.1, SATA whole band −13.5, per sub-band −13.6 dB (improvement 7.5 dB).

7 IRFs, D: targets 2.6 km apart, large heights

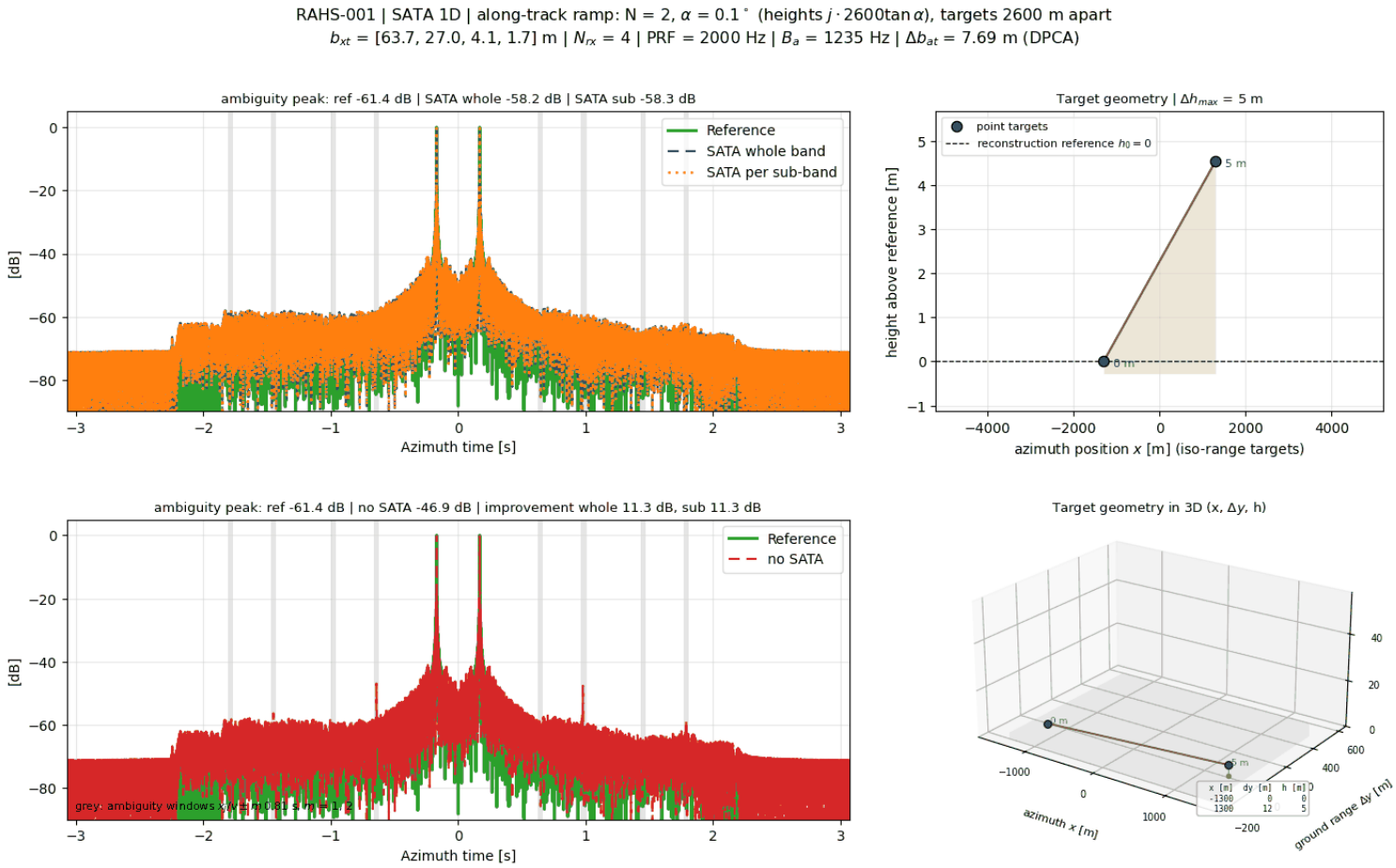


Figure 206: Along-track ramp, 2.6 km apart, large heights: RAHS-001, $N = 2$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −46.9, SATA whole band −58.2, per sub-band −58.3 dB (improvement 11.3 dB).

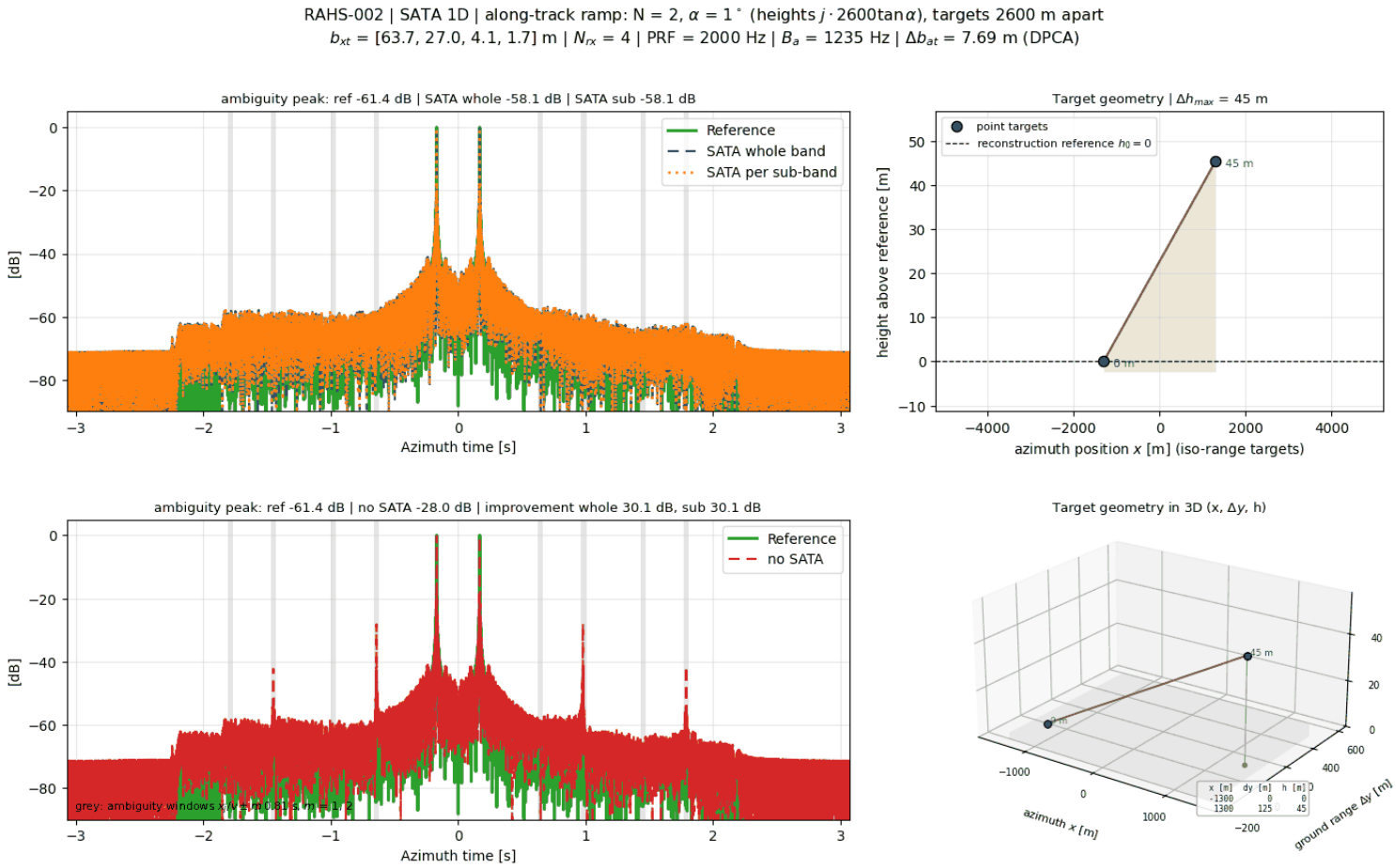


Figure 207: Along-track ramp, 2.6 km apart, large heights: RAHS-002, $N = 2$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −28.0, SATA whole band −58.1, per sub-band −58.1 dB (improvement 30.1 dB).

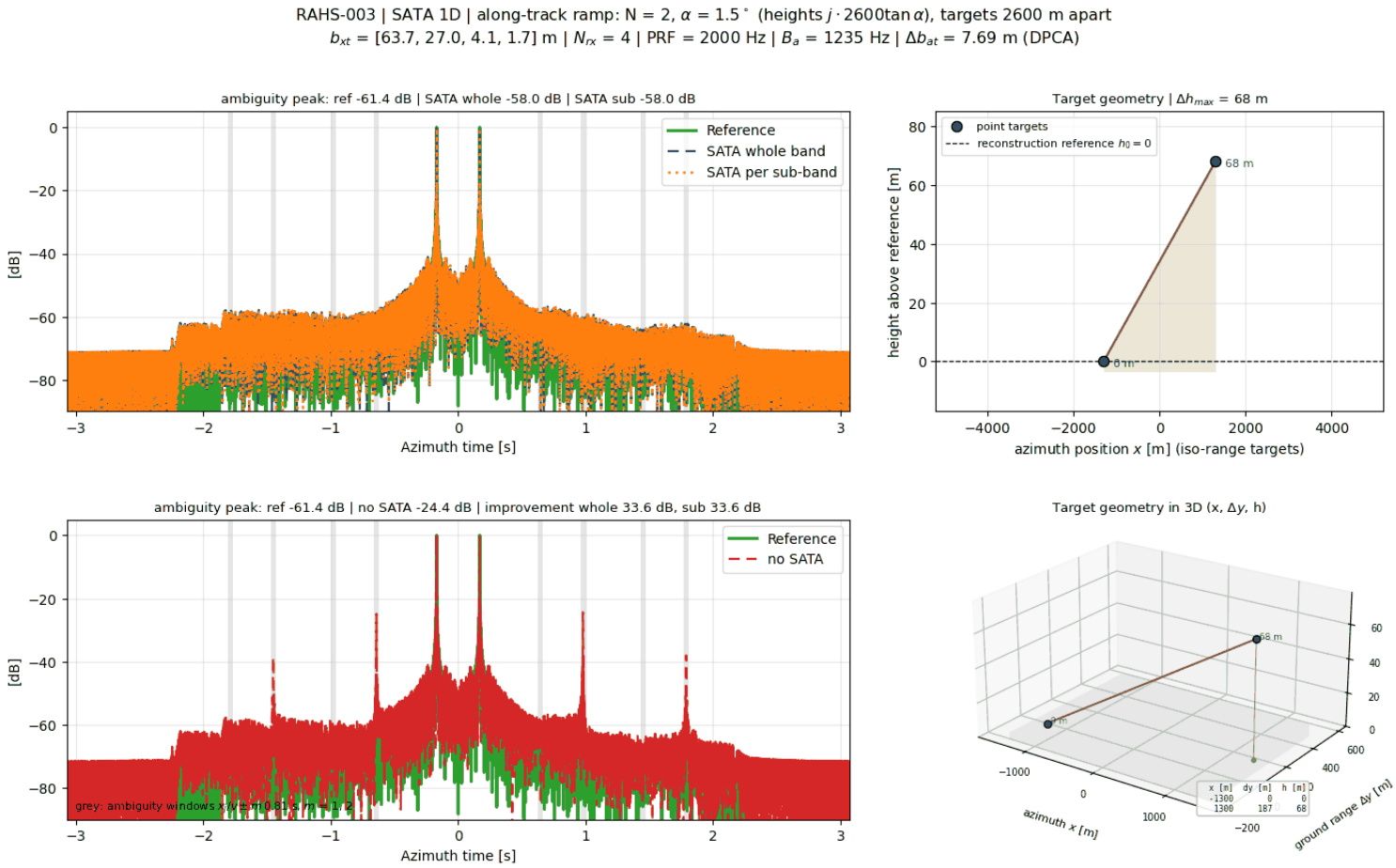


Figure 208: Along-track ramp, 2.6 km apart, large heights: RAHS-003, $N = 2$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -24.4, SATA whole band -58.0, per sub-band -58.0 dB (improvement 33.6 dB).

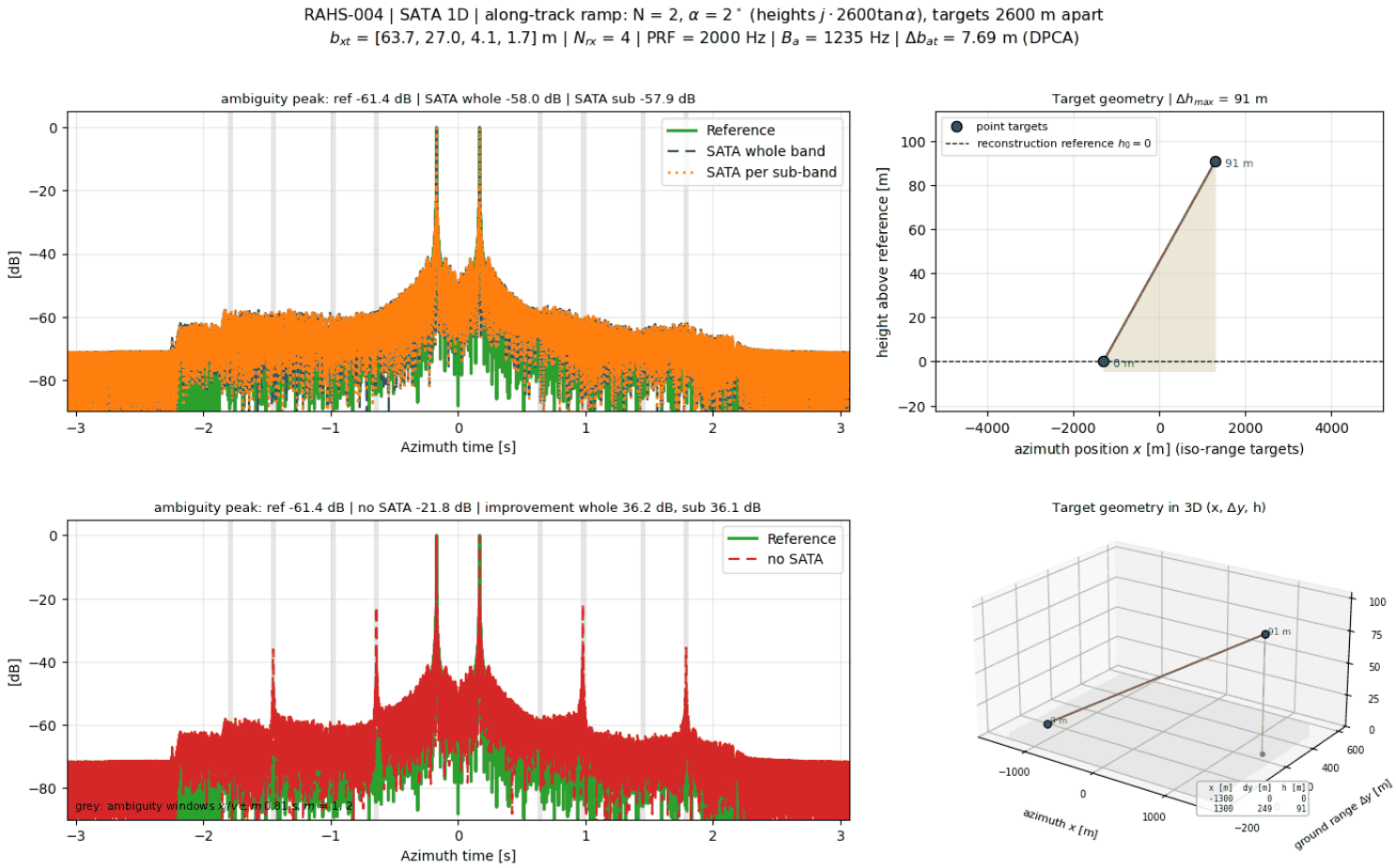


Figure 209: Along-track ramp, 2.6 km apart, large heights: RAHS-004, $N = 2$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan \alpha$). Ambiguity peak: no SATA −21.8, SATA whole band −58.0, per sub-band −57.9 dB (improvement 36.2 dB).

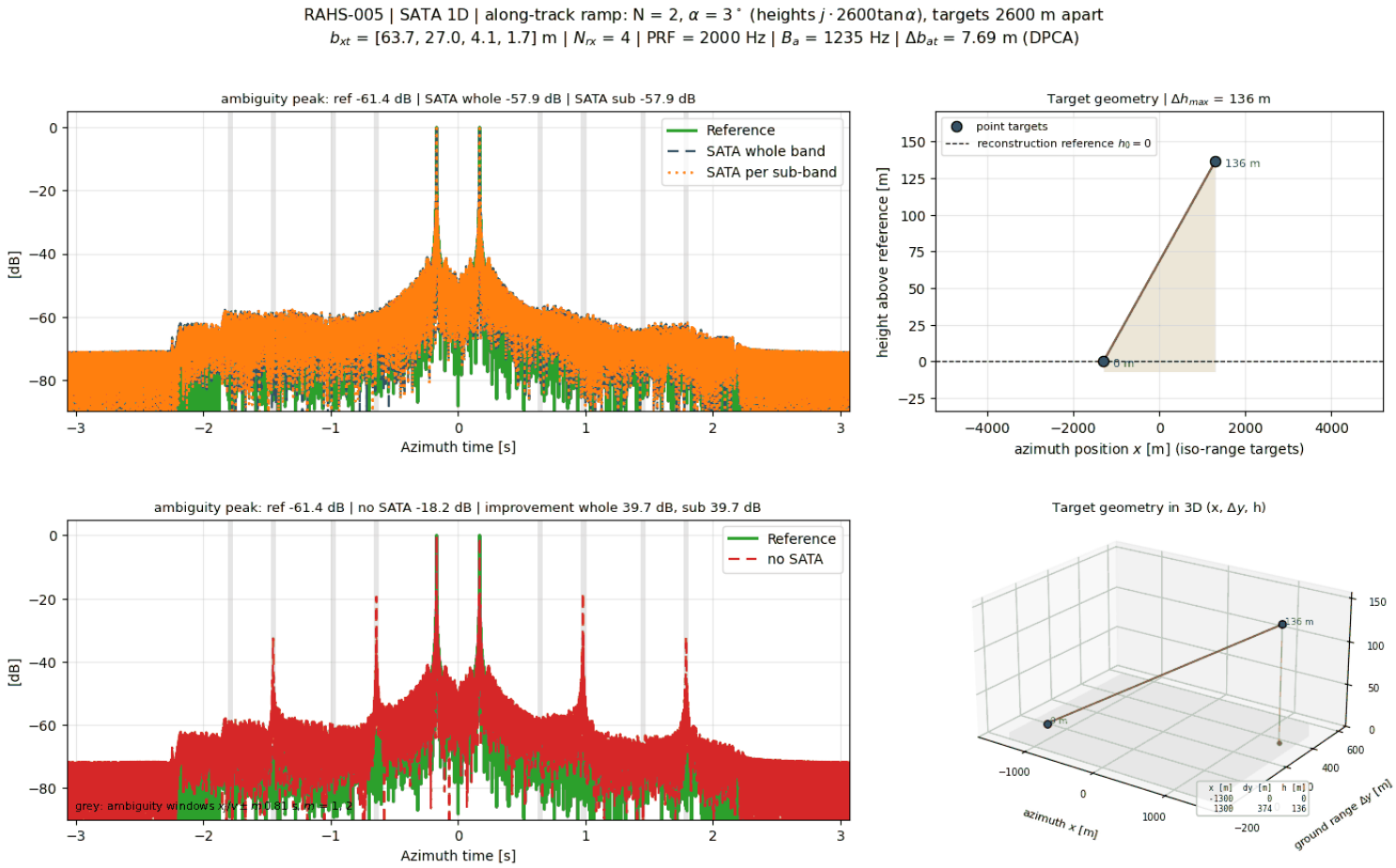


Figure 210: Along-track ramp, 2.6 km apart, large heights: RAHS-005, $N = 2$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan \alpha$). Ambiguity peak: no SATA −18.2, SATA whole band −57.9, per sub-band −57.9 dB (improvement 39.7 dB).

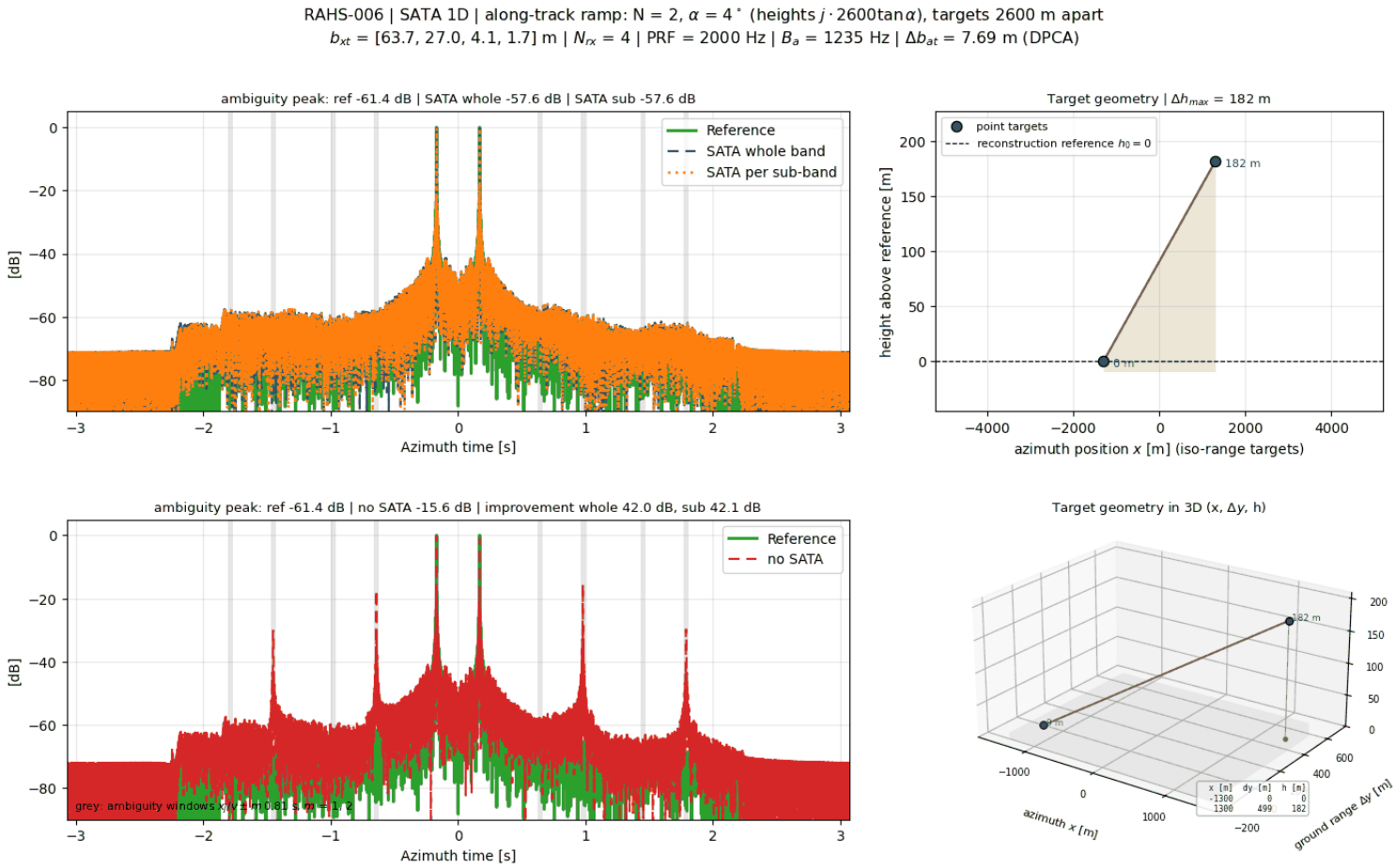


Figure 211: Along-track ramp, 2.6 km apart, large heights: RAHS-006, $N = 2$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan \alpha$). Ambiguity peak: no SATA −15.6, SATA whole band −57.6, per sub-band −57.6 dB (improvement 42.0 dB).

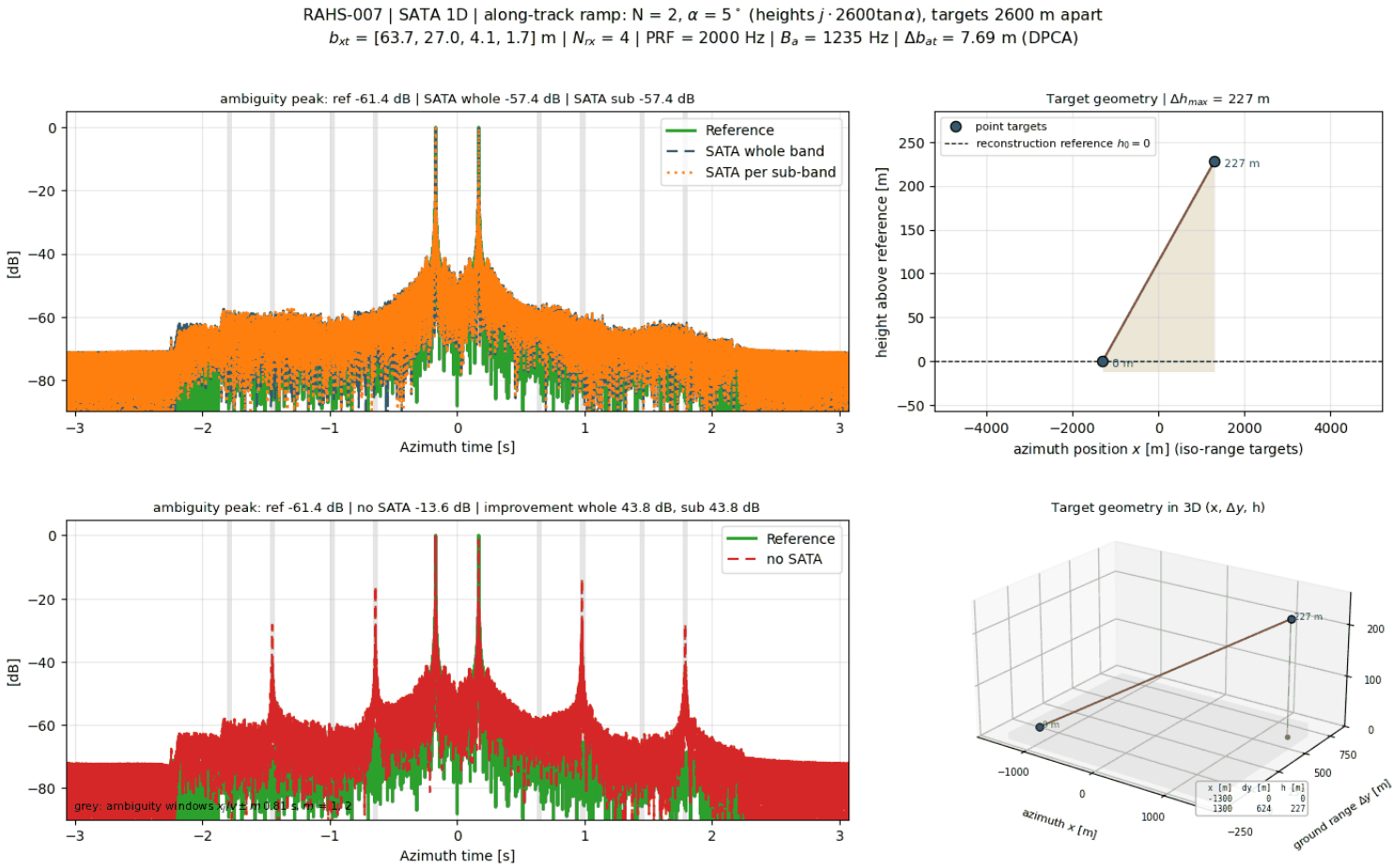


Figure 212: Along-track ramp, 2.6 km apart, large heights: RAHS-007, $N = 2$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -13.6, SATA whole band -57.4, per sub-band -57.4 dB (improvement 43.8 dB).

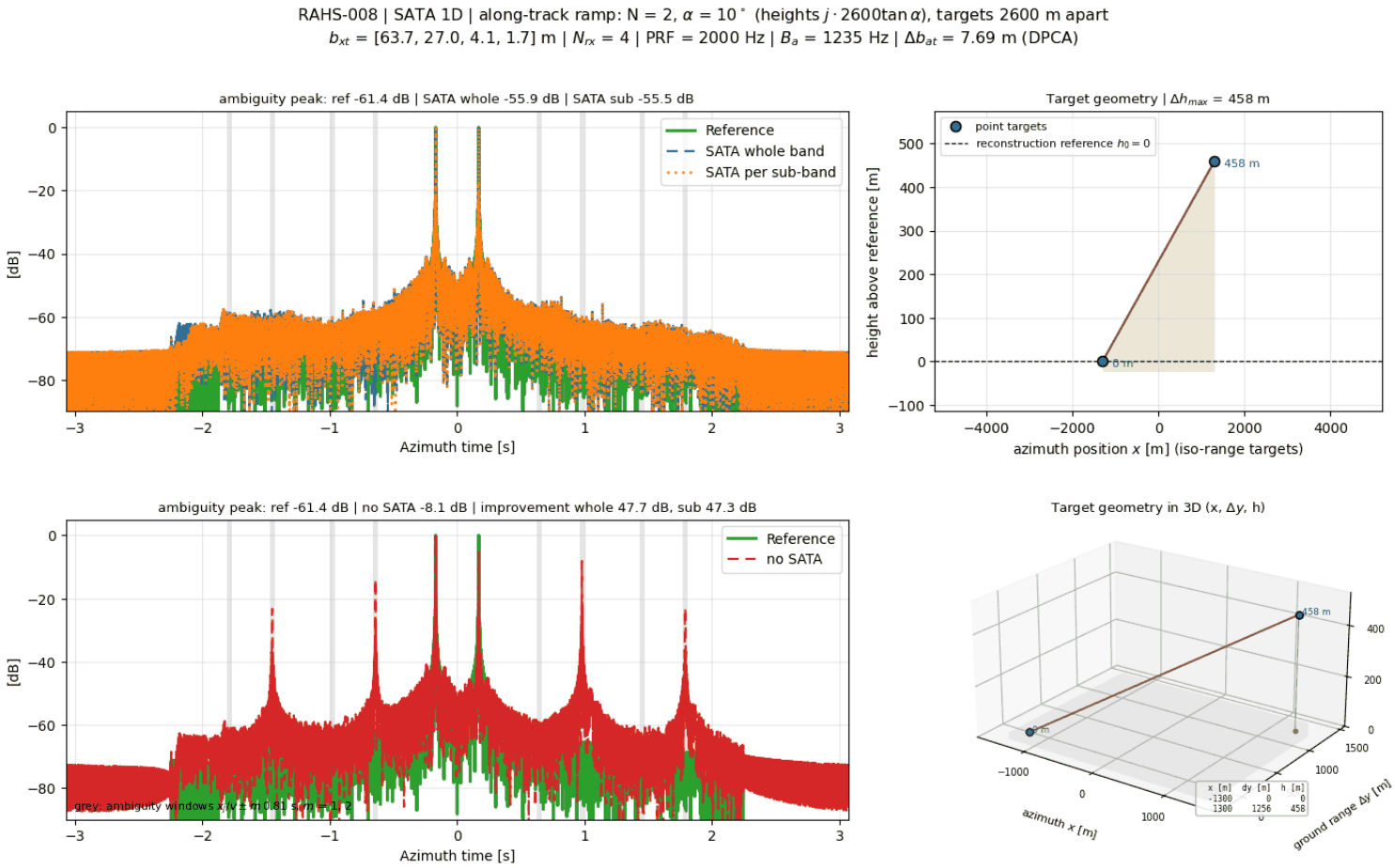


Figure 213: Along-track ramp, 2.6 km apart, large heights: RAHS-008, $N = 2$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.1, SATA whole band −55.9, per sub-band −55.5 dB (improvement 47.7 dB).

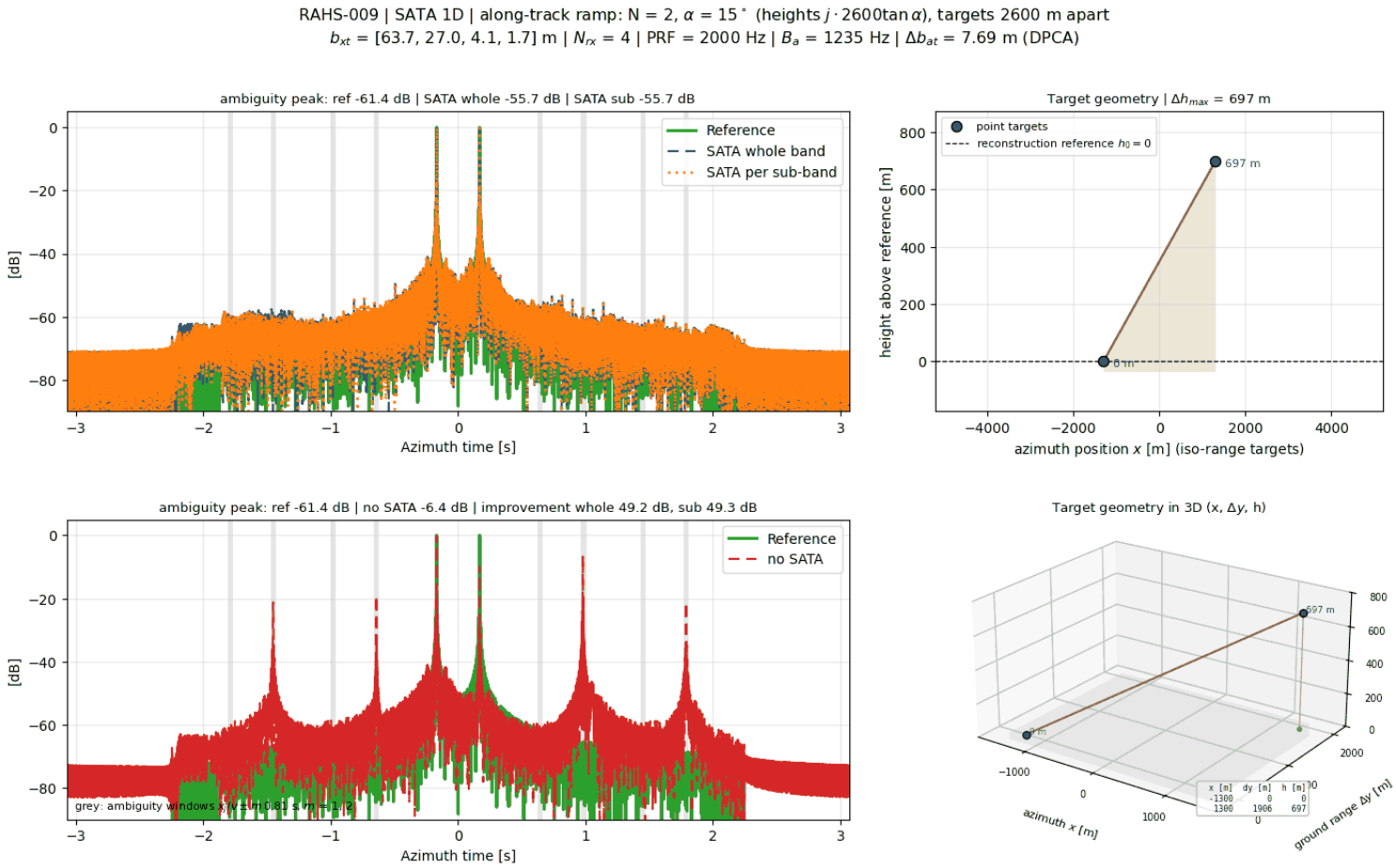


Figure 214: Along-track ramp, 2.6 km apart, large heights: RAHS-009, $N = 2$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −55.7, per sub-band −55.7 dB (improvement 49.2 dB).

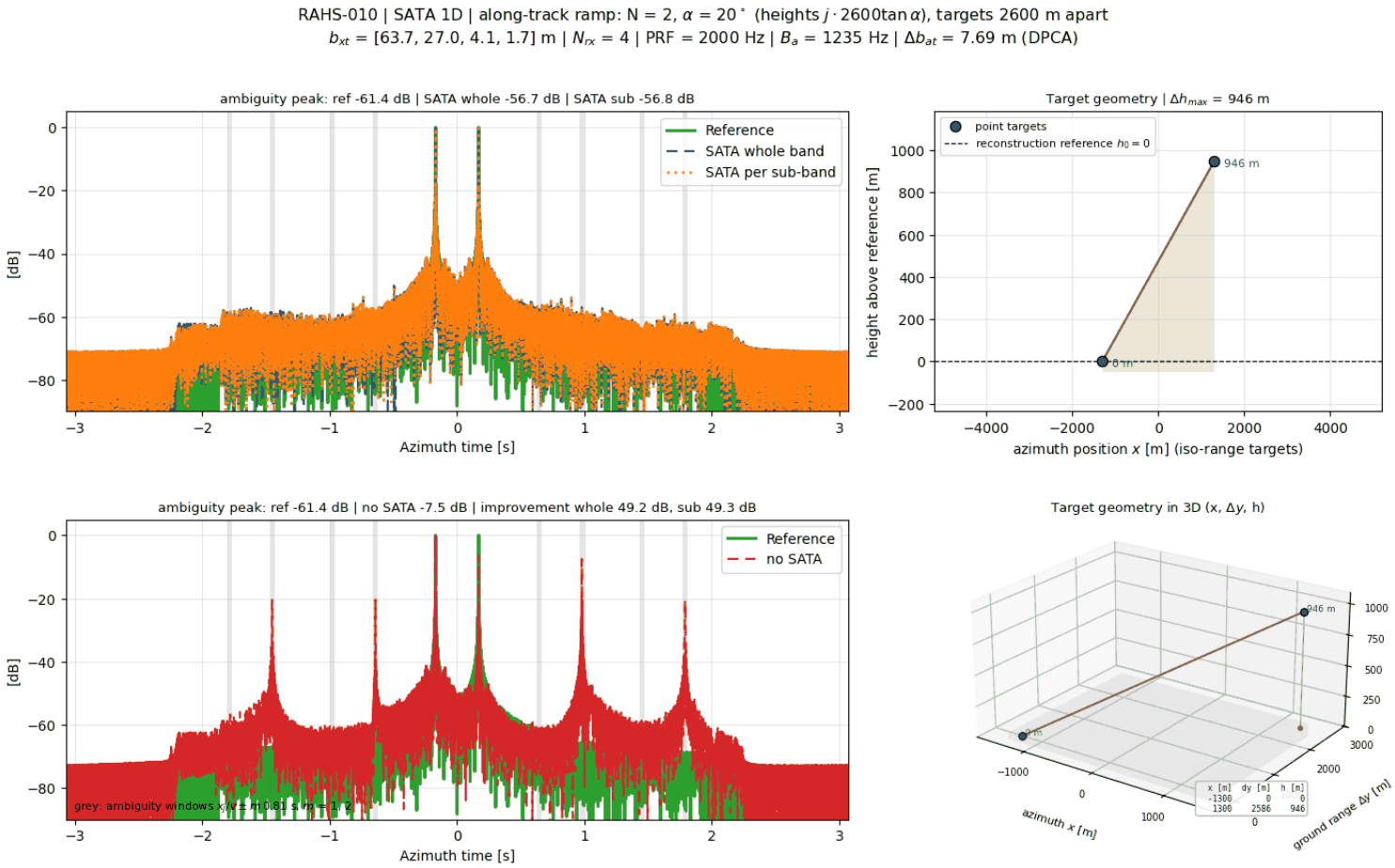


Figure 215: Along-track ramp, 2.6 km apart, large heights: RAHS-010, $N = 2$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.5, SATA whole band −56.7, per sub-band −56.8 dB (improvement 49.2 dB).

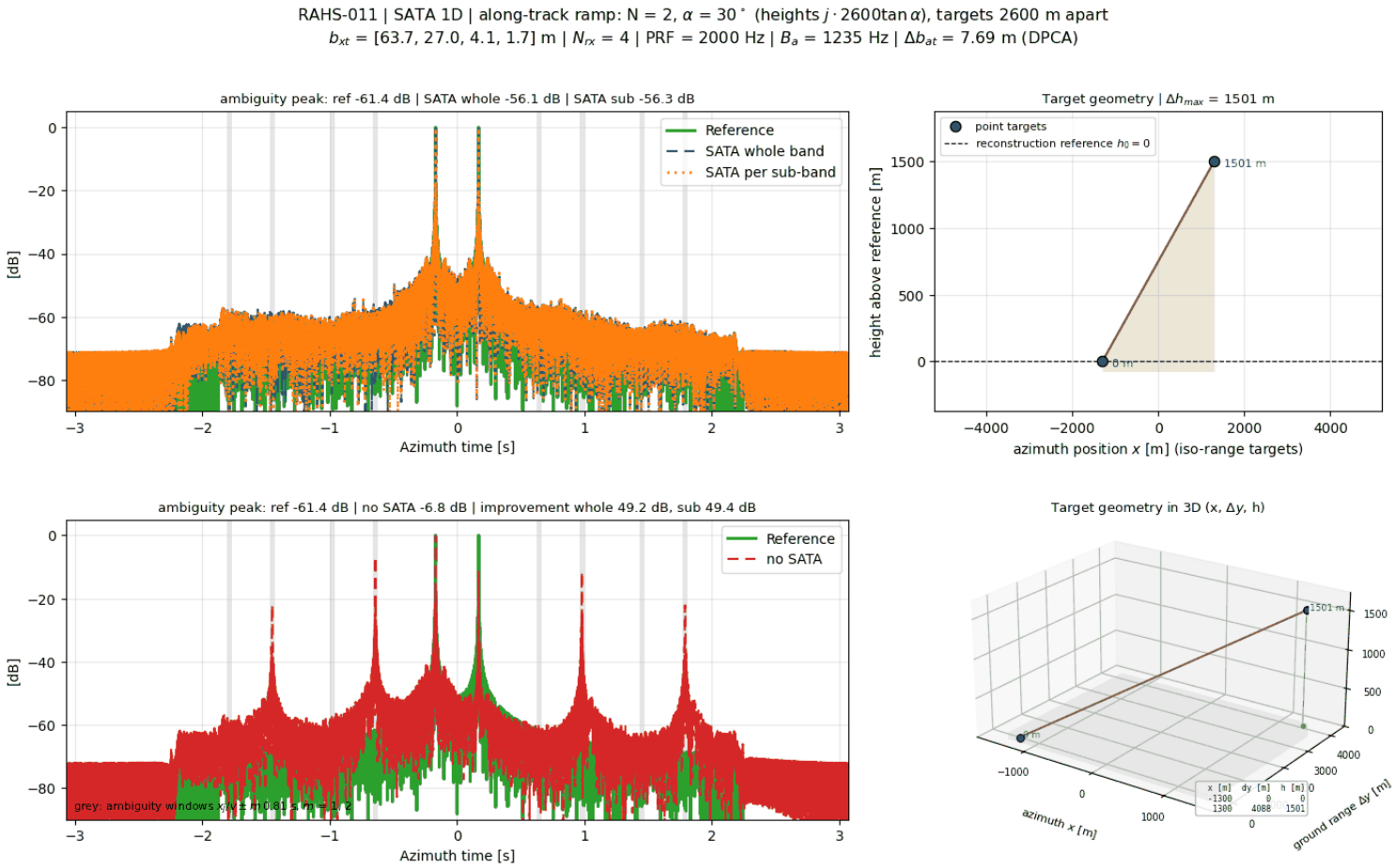


Figure 216: Along-track ramp, 2.6 km apart, large heights: RAHS-011, $N = 2$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.8, SATA whole band −56.1, per sub-band −56.3 dB (improvement 49.2 dB).

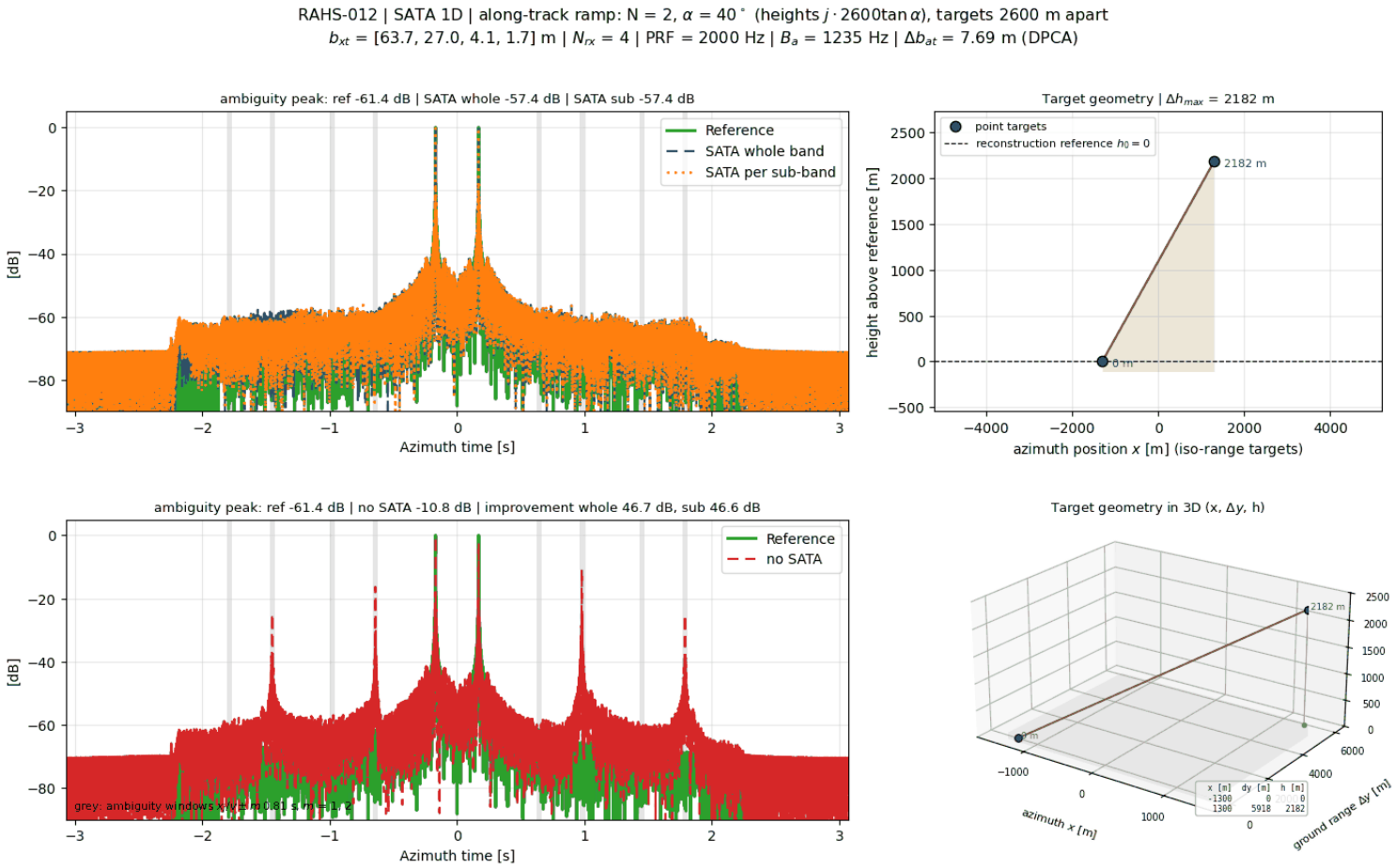


Figure 217: Along-track ramp, 2.6 km apart, large heights: RAHS-012, $N = 2$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -10.8, SATA whole band -57.4, per sub-band -57.4 dB (improvement 46.7 dB).

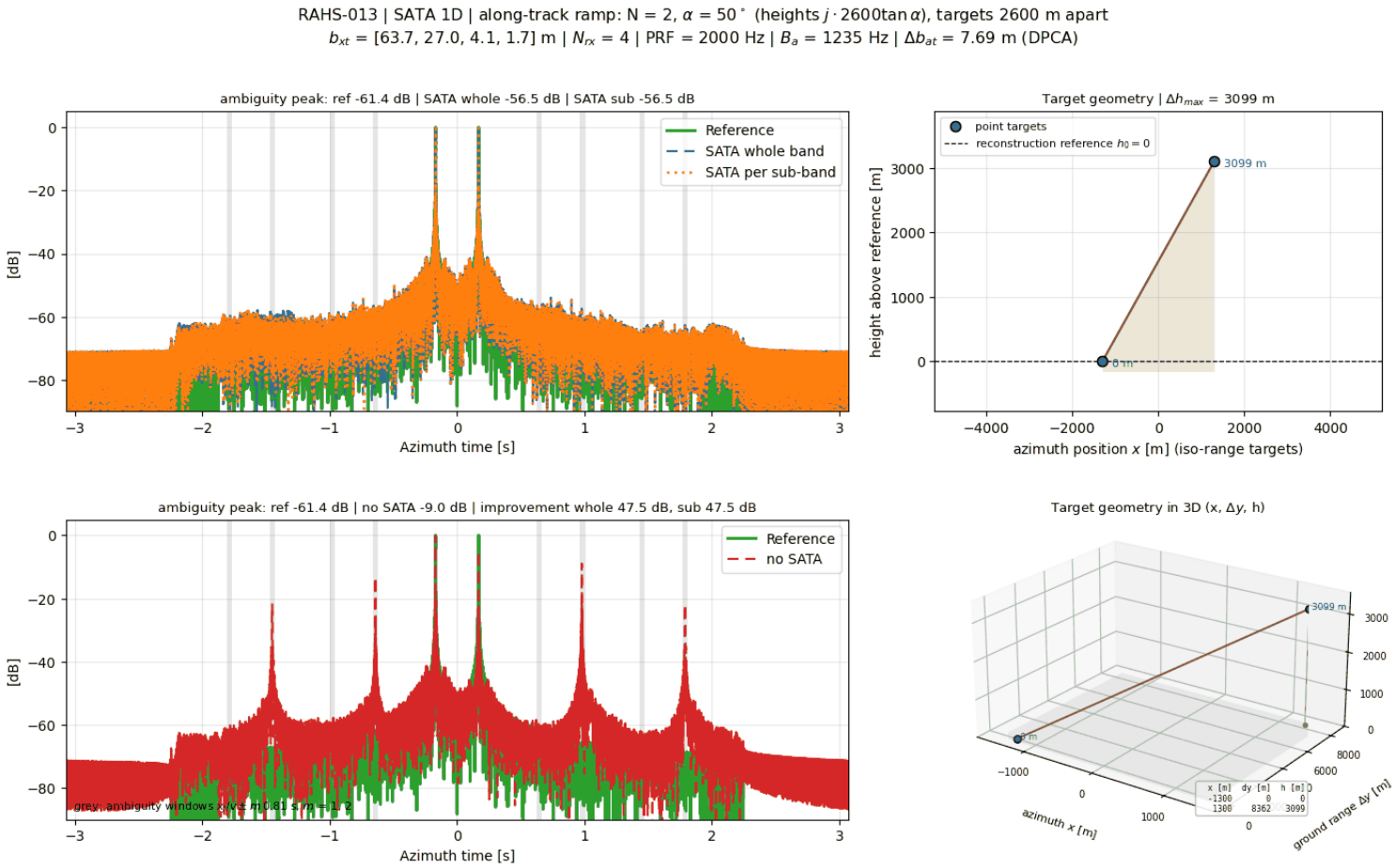


Figure 218: Along-track ramp, 2.6 km apart, large heights: RAHS-013, $N = 2$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.0, SATA whole band −56.5, per sub-band −56.5 dB (improvement 47.5 dB).

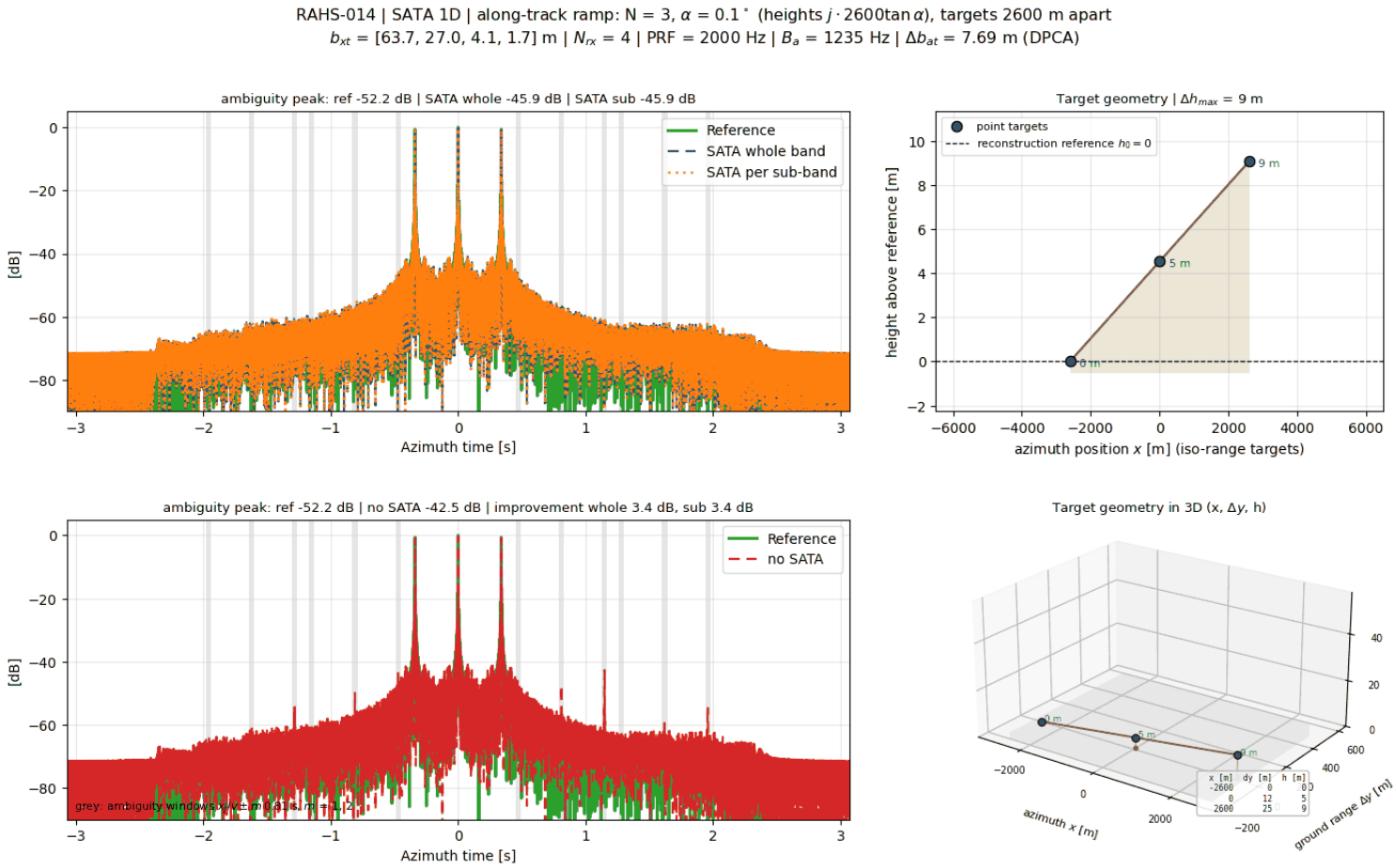


Figure 219: Along-track ramp, 2.6 km apart, large heights: RAHS-014, $N = 3$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -42.5, SATA whole band -45.9, per sub-band -45.9 dB (improvement 3.4 dB).

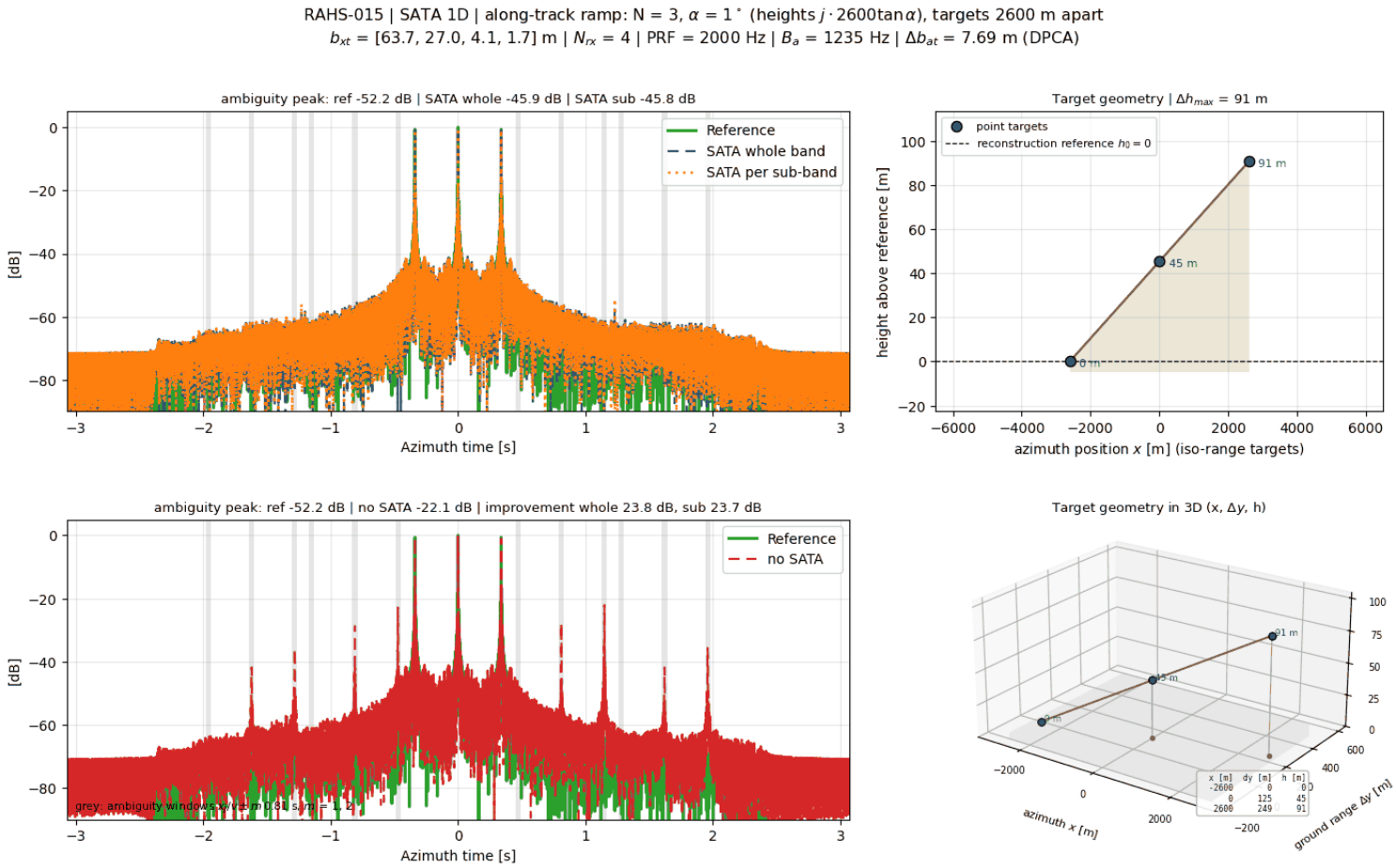


Figure 220: Along-track ramp, 2.6 km apart, large heights: RAHS-015, $N = 3$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan \alpha$). Ambiguity peak: no SATA -22.1, SATA whole band -45.9, per sub-band -45.8 dB (improvement 23.8 dB).

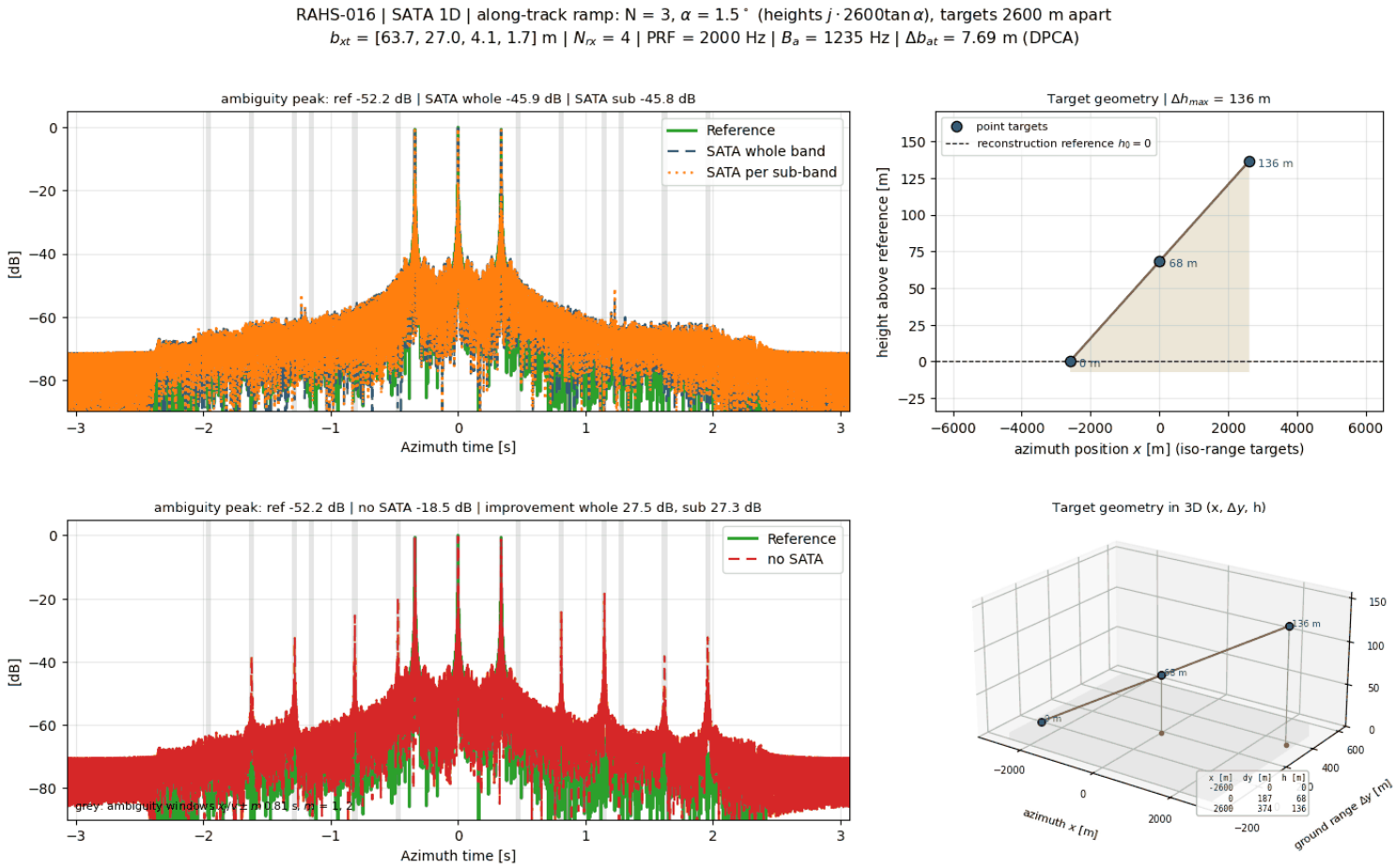


Figure 221: Along-track ramp, 2.6 km apart, large heights: RAHS-016, $N = 3$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −18.5, SATA whole band −45.9, per sub-band −45.8 dB (improvement 27.5 dB).

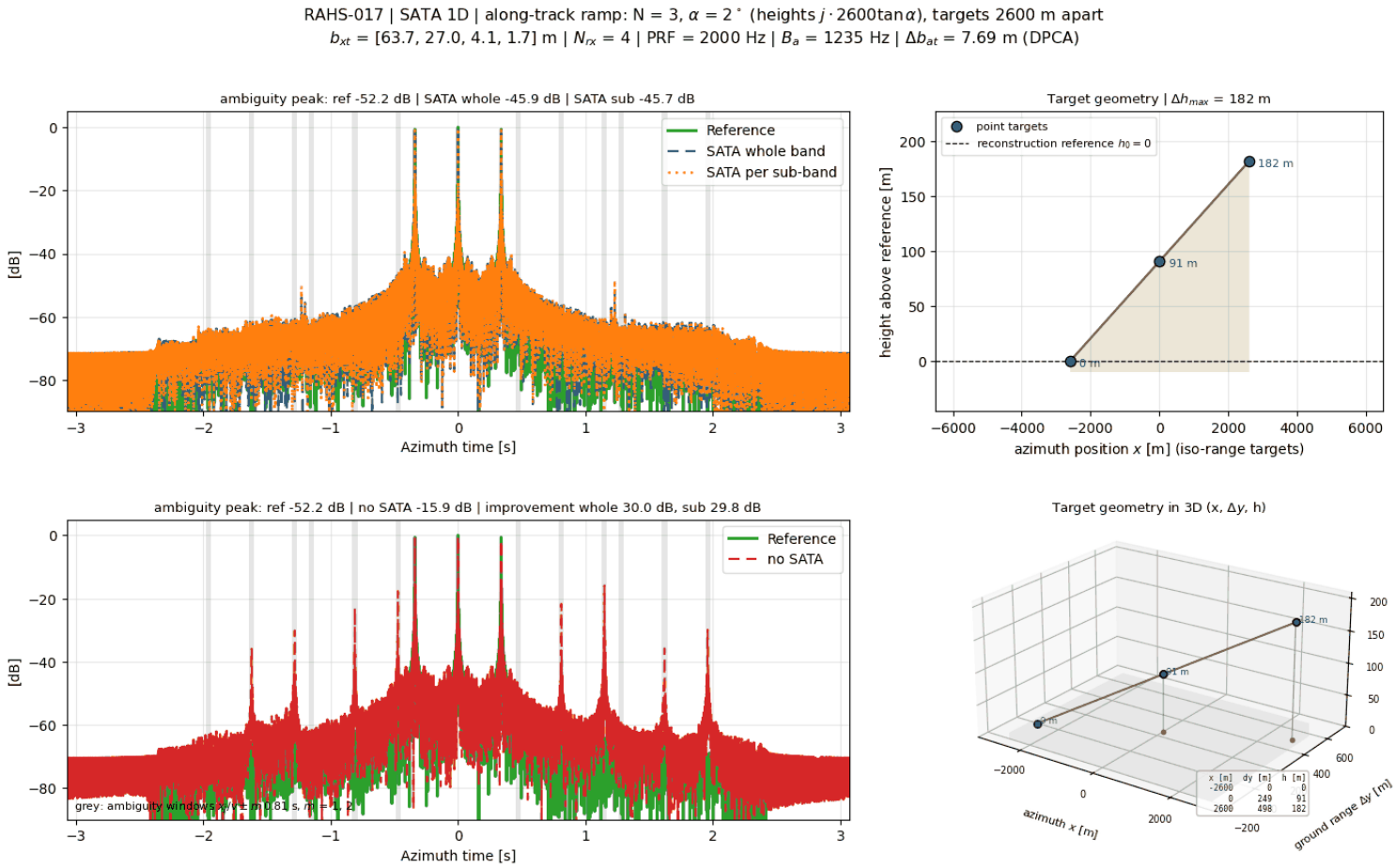


Figure 222: Along-track ramp, 2.6 km apart, large heights: RAHS-017, $N = 3$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −15.9, SATA whole band −45.9, per sub-band −45.7 dB (improvement 30.0 dB).

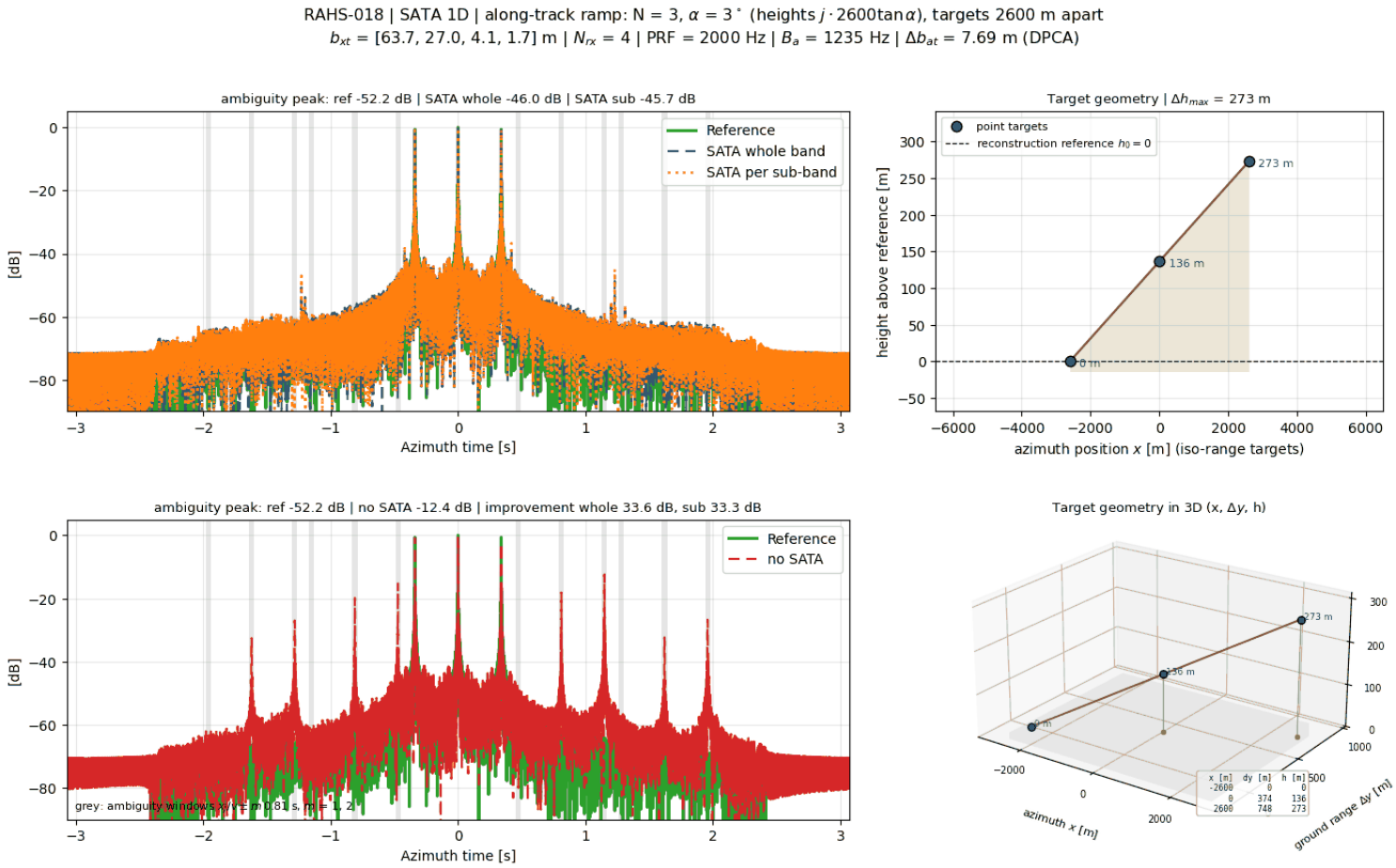


Figure 223: Along-track ramp, 2.6 km apart, large heights: RAHS-018, $N = 3$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -12.4, SATA whole band -46.0, per sub-band -45.7 dB (improvement 33.6 dB).

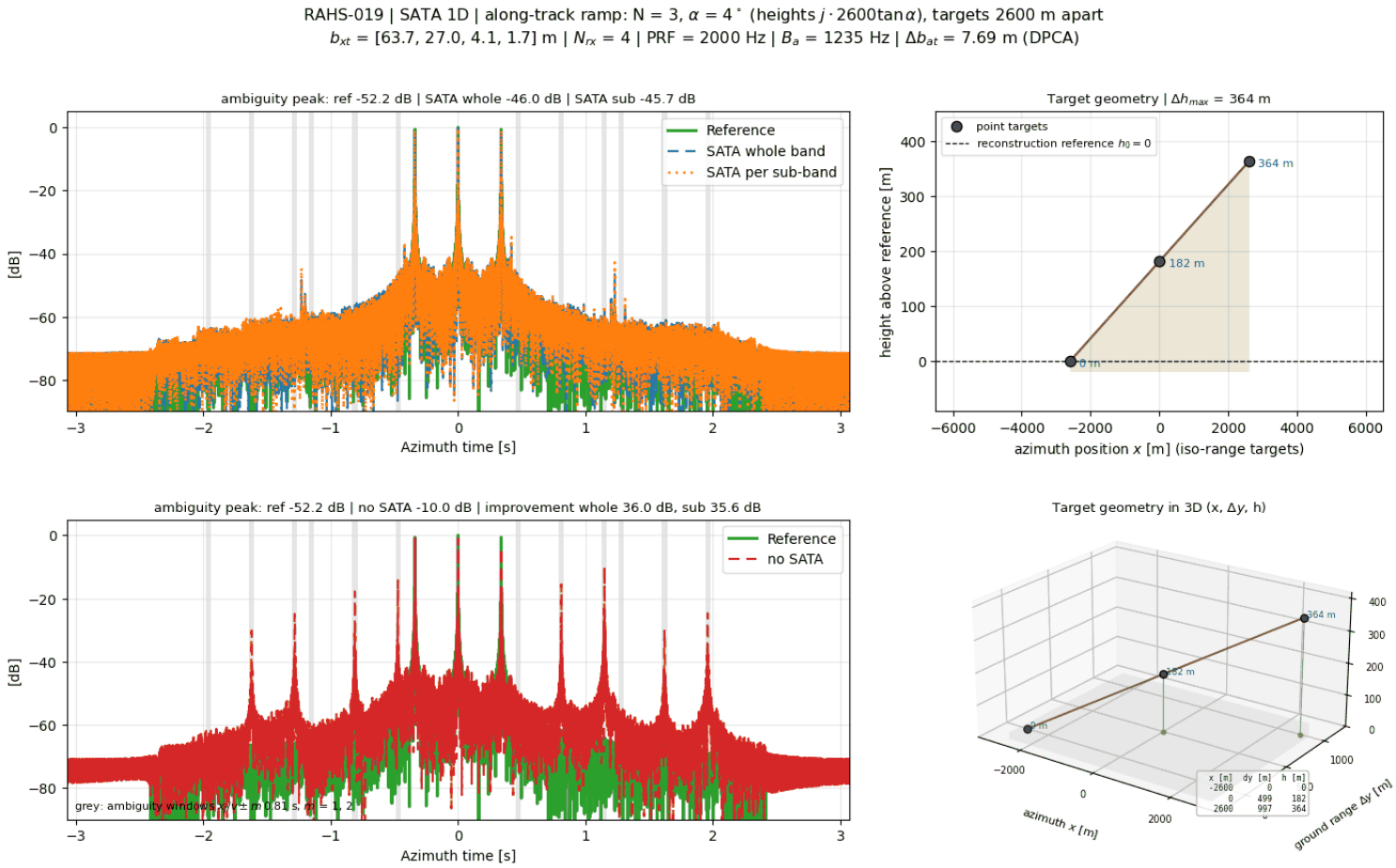


Figure 224: Along-track ramp, 2.6 km apart, large heights: RAHS-019, $N = 3$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -10.0, SATA whole band -46.0, per sub-band -45.7 dB (improvement 36.0 dB).

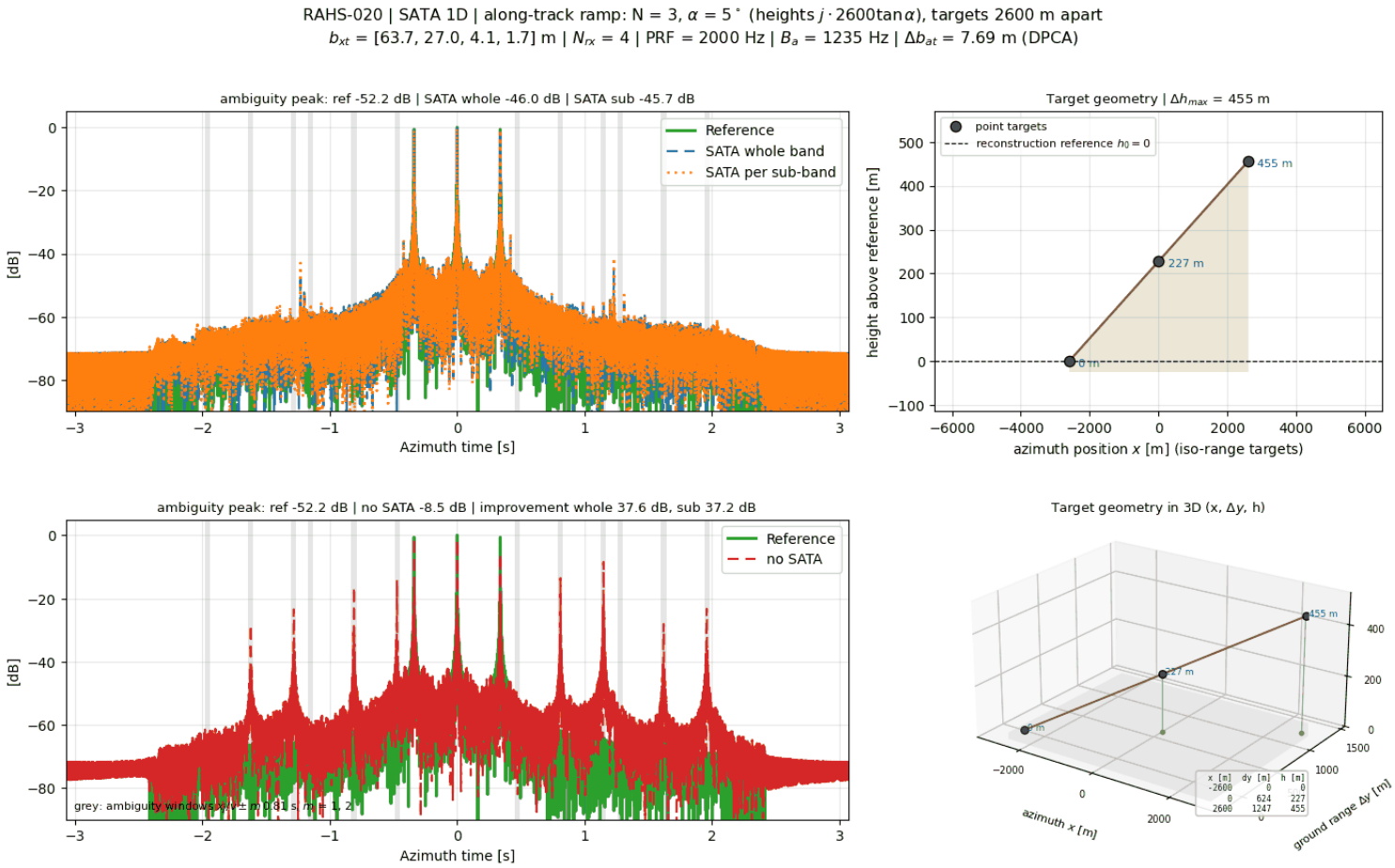


Figure 225: Along-track ramp, 2.6 km apart, large heights: RAHS-020, $N = 3$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.5, SATA whole band −46.0, per sub-band −45.7 dB (improvement 37.6 dB).

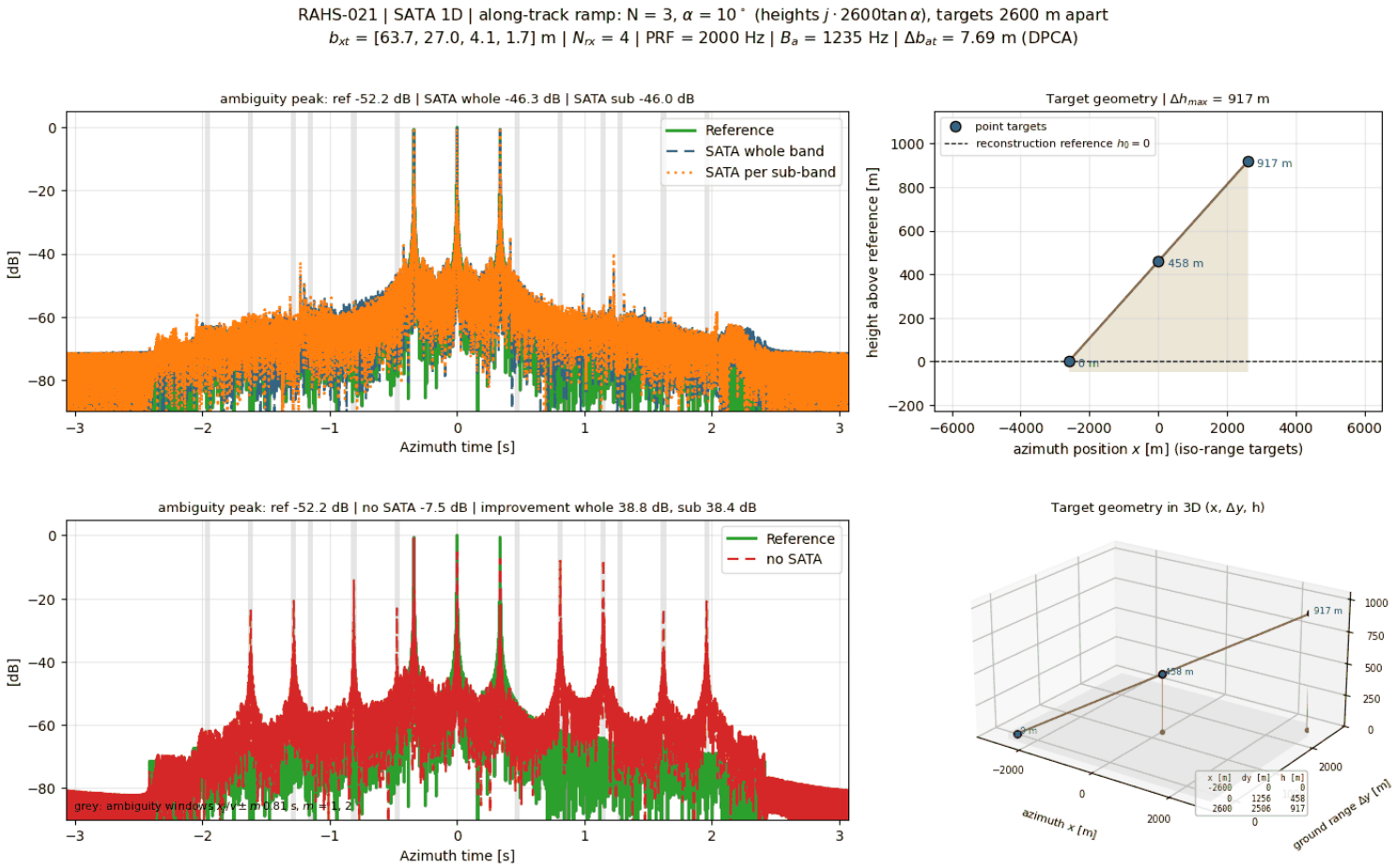


Figure 226: Along-track ramp, 2.6 km apart, large heights: RAHS-021, $N = 3$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -7.5, SATA whole band -46.3, per sub-band -46.0 dB (improvement 38.8 dB).

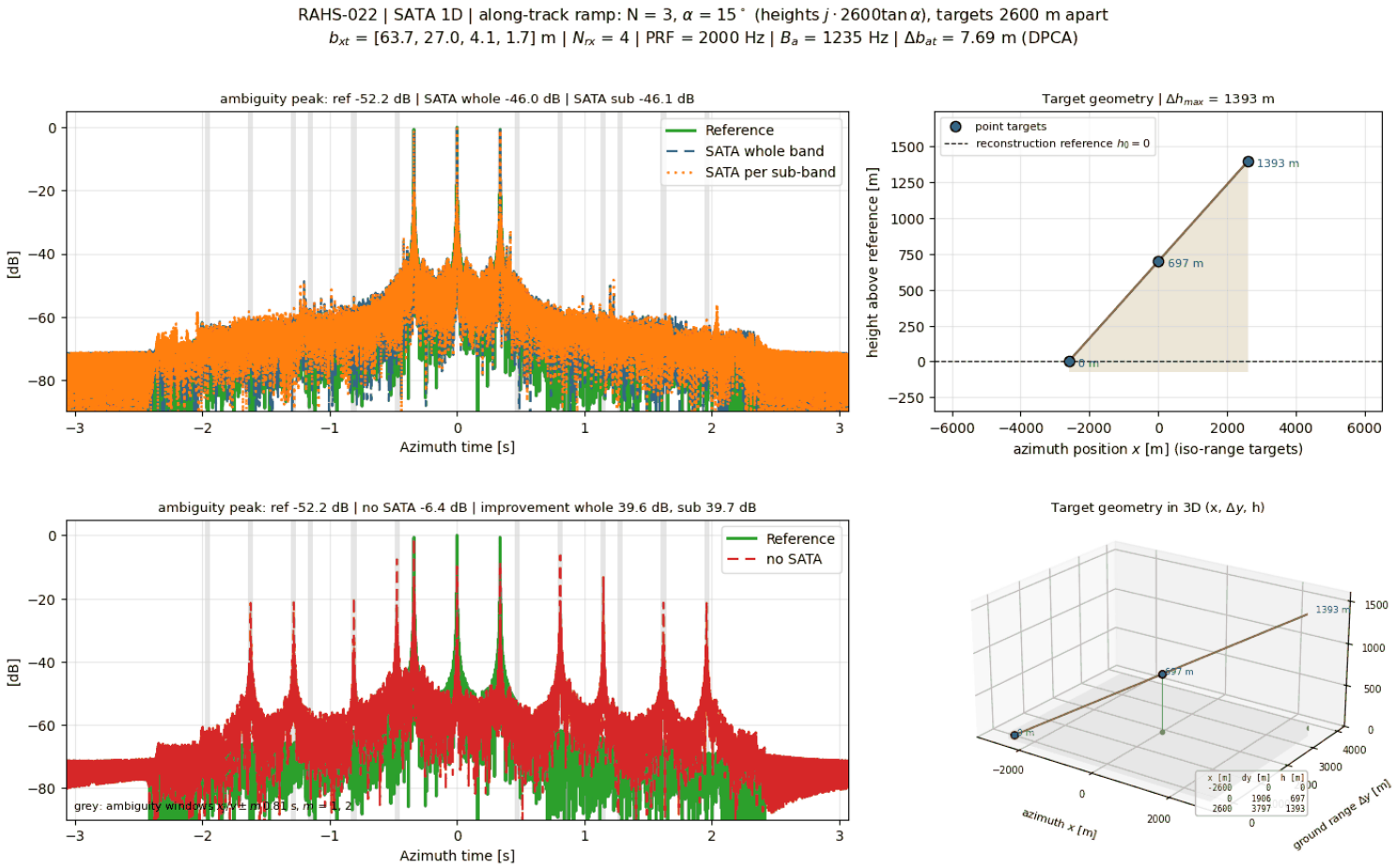


Figure 227: Along-track ramp, 2.6 km apart, large heights: RAHS-022, $N = 3$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.4, SATA whole band −46.0, per sub-band −46.1 dB (improvement 39.6 dB).

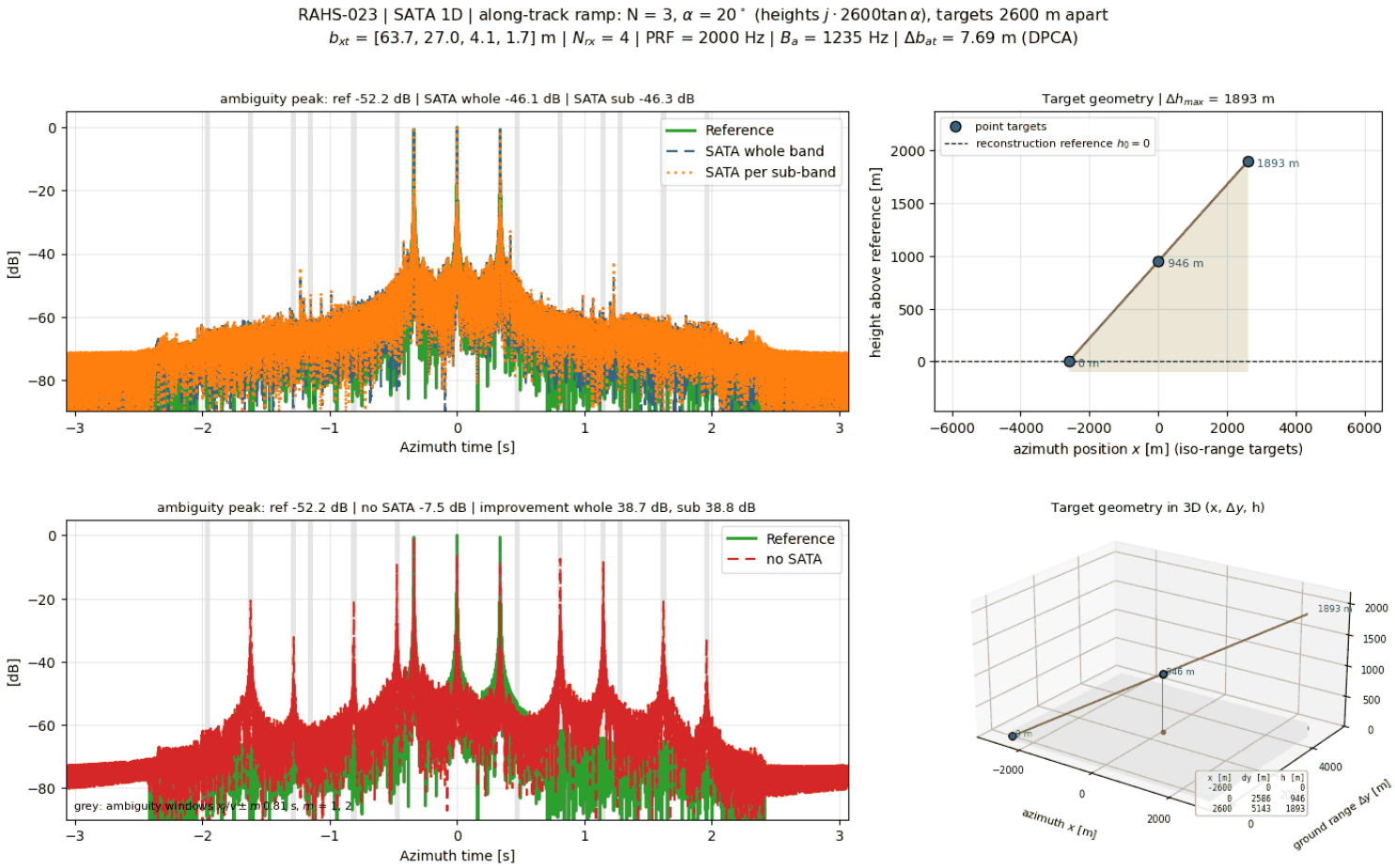


Figure 228: Along-track ramp, 2.6 km apart, large heights: RAHS-023, $N = 3$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.5, SATA whole band −46.1, per sub-band −46.3 dB (improvement 38.7 dB).

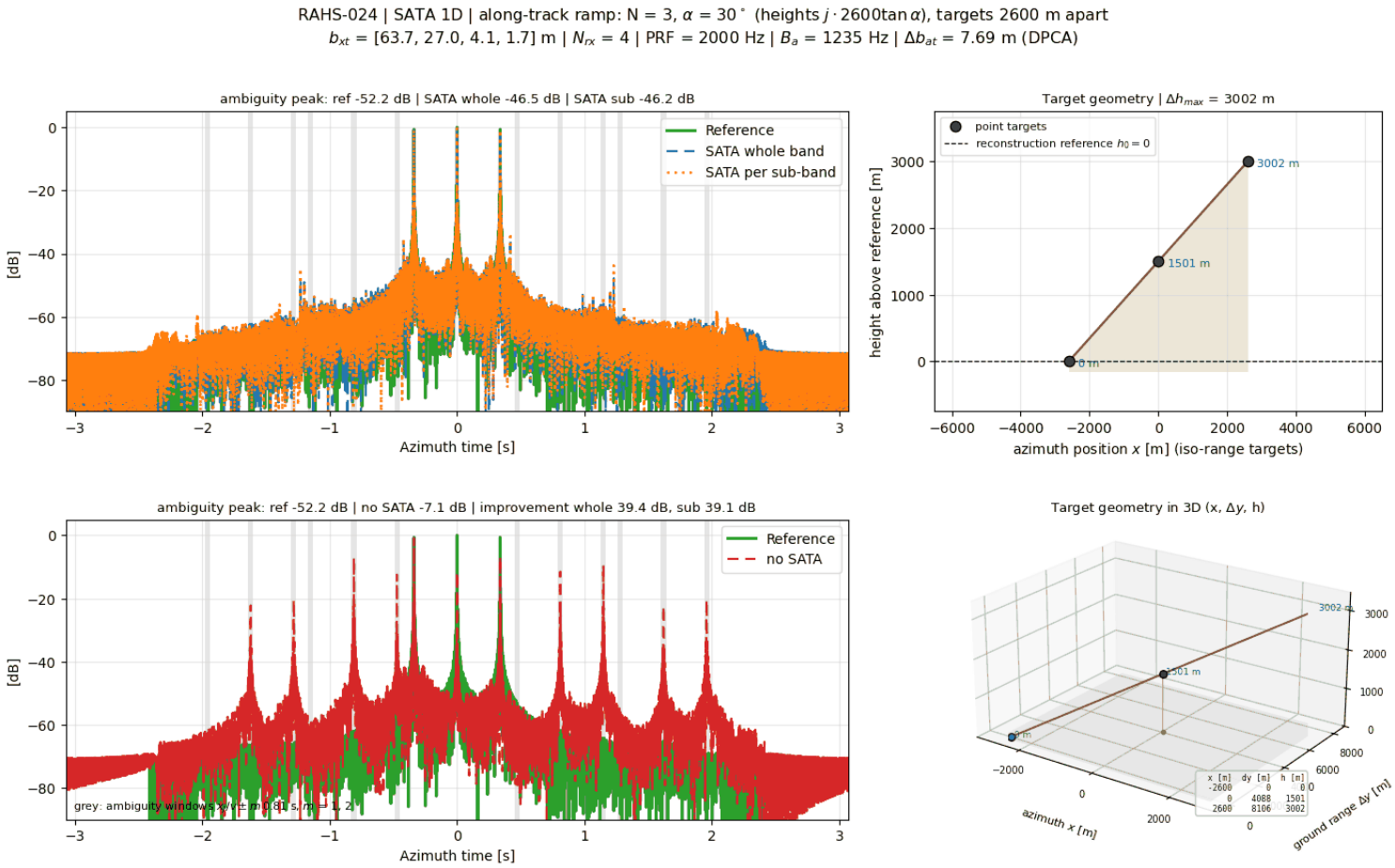


Figure 229: Along-track ramp, 2.6 km apart, large heights: RAHS-024, $N = 3$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -7.1, SATA whole band -46.5, per sub-band -46.2 dB (improvement 39.4 dB).

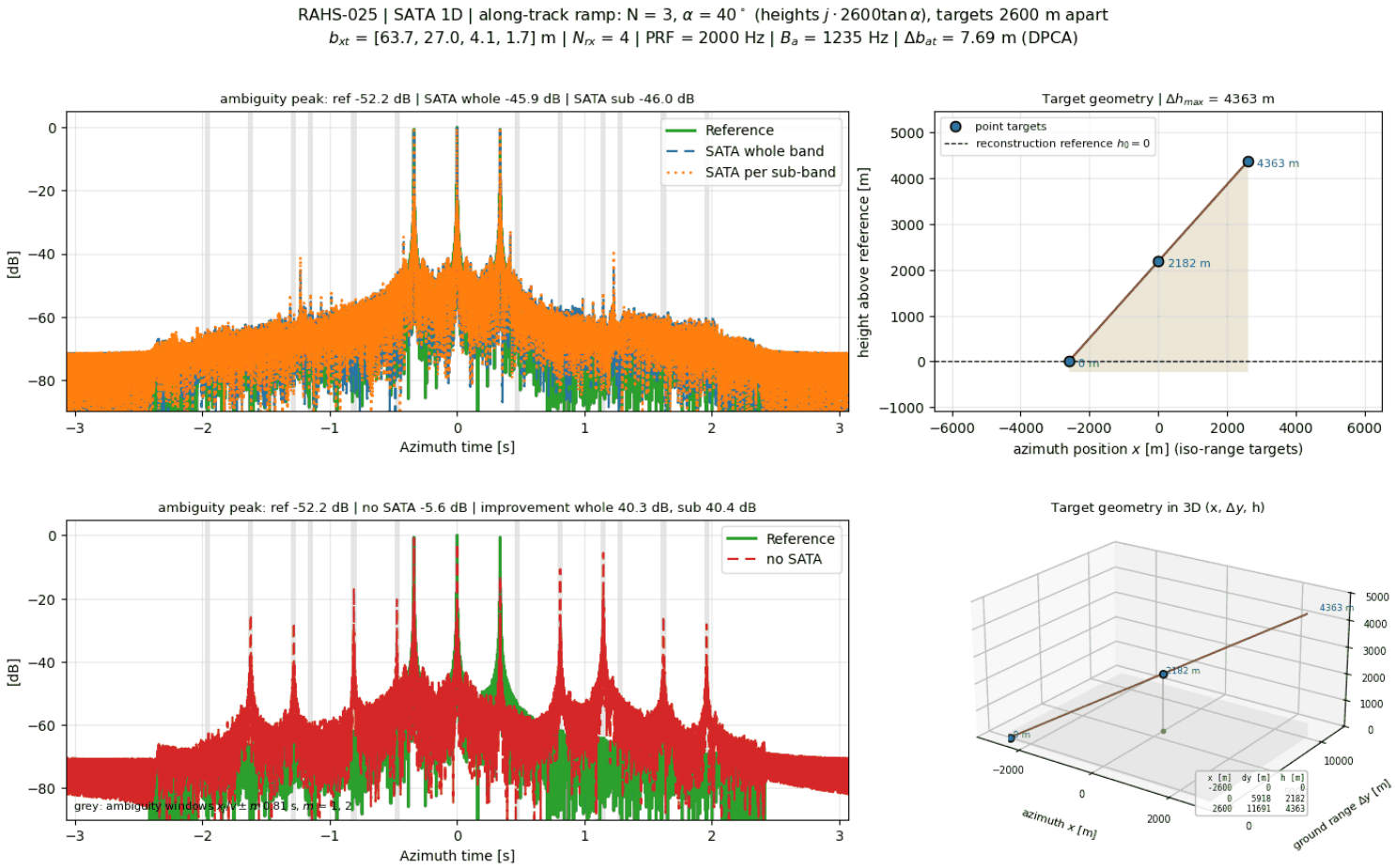


Figure 230: Along-track ramp, 2.6 km apart, large heights: RAHS-025, $N = 3$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.6, SATA whole band −45.9, per sub-band −46.0 dB (improvement 40.3 dB).

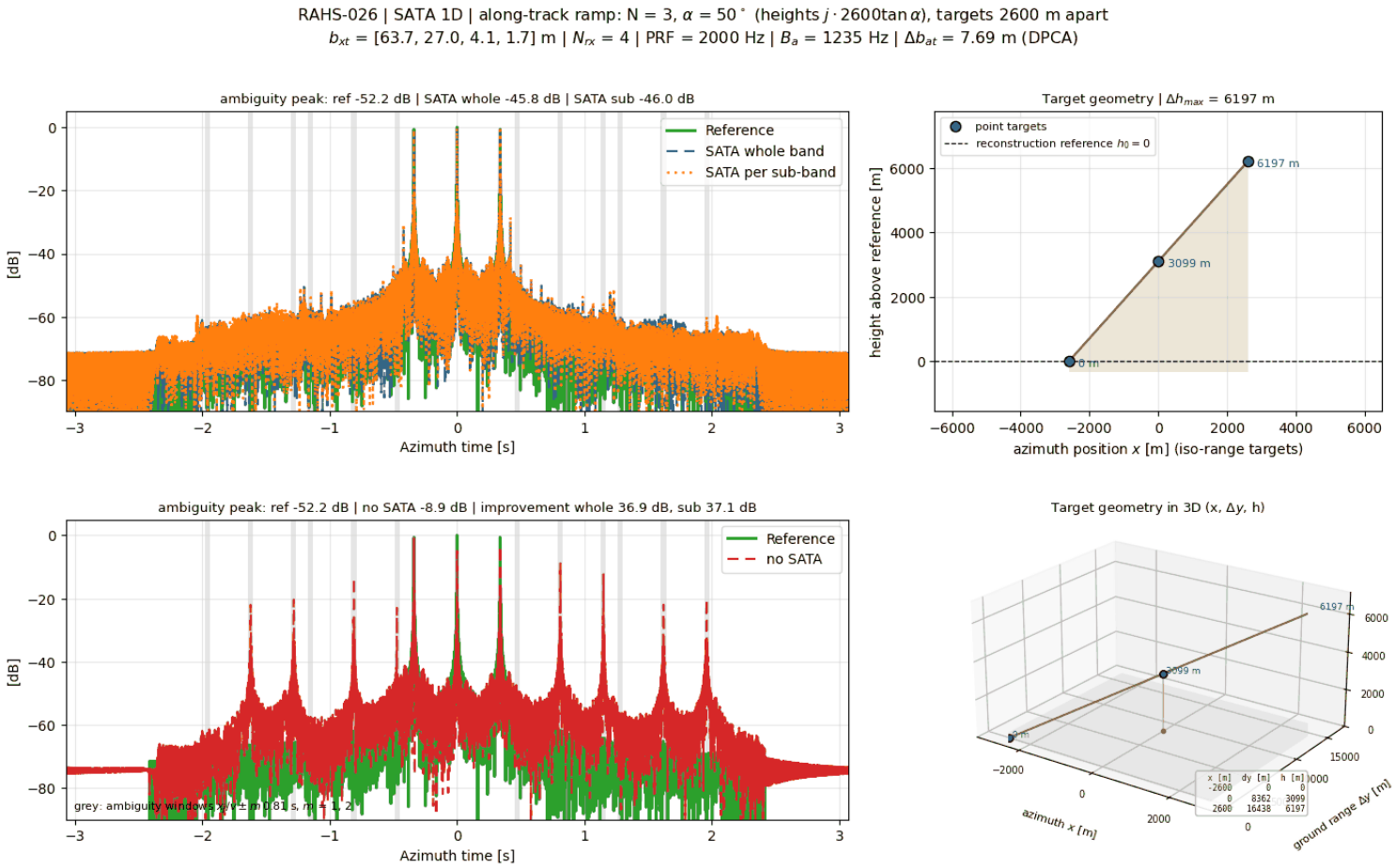


Figure 231: Along-track ramp, 2.6 km apart, large heights: RAHS-026, $N = 3$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.9, SATA whole band −45.8, per sub-band −46.0 dB (improvement 36.9 dB).

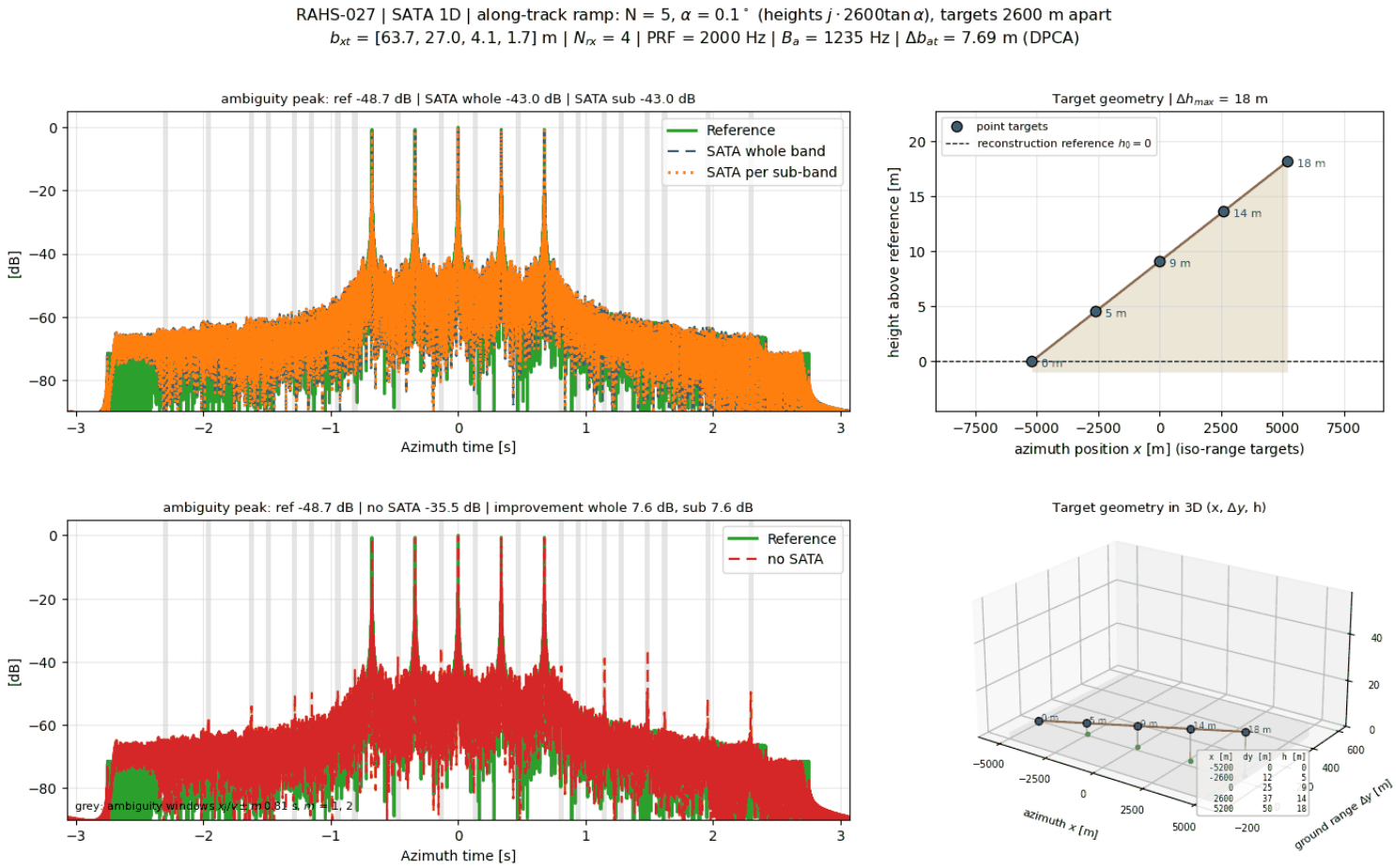


Figure 232: Along-track ramp, 2.6 km apart, large heights: RAHS-027, $N = 5$, $\alpha = 0.1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -35.5, SATA whole band -43.0, per sub-band -43.0 dB (improvement 7.6 dB).

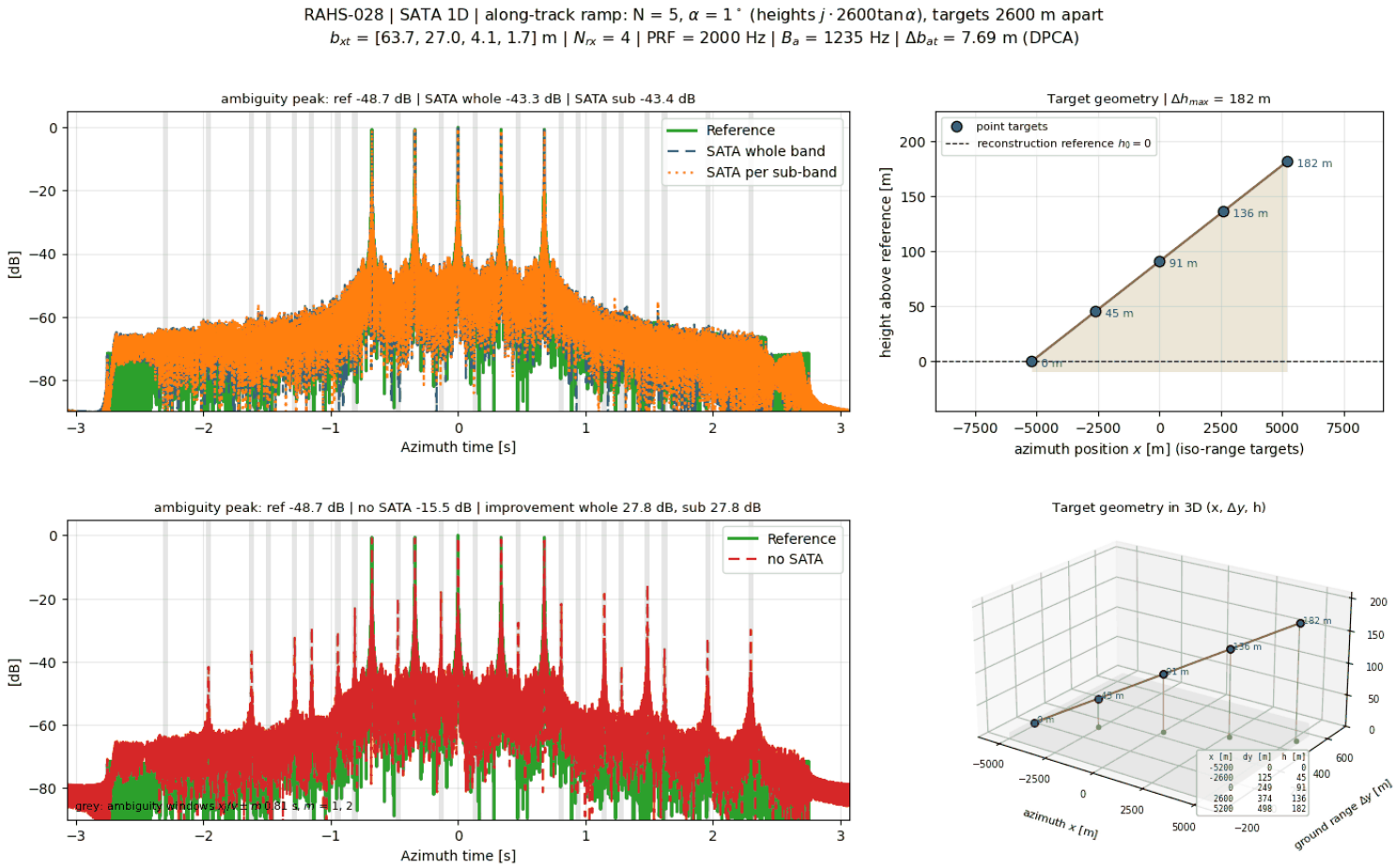


Figure 233: Along-track ramp, 2.6 km apart, large heights: RAHS-028, $N = 5$, $\alpha = 1°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -15.5, SATA whole band -43.3, per sub-band -43.4 dB (improvement 27.8 dB).

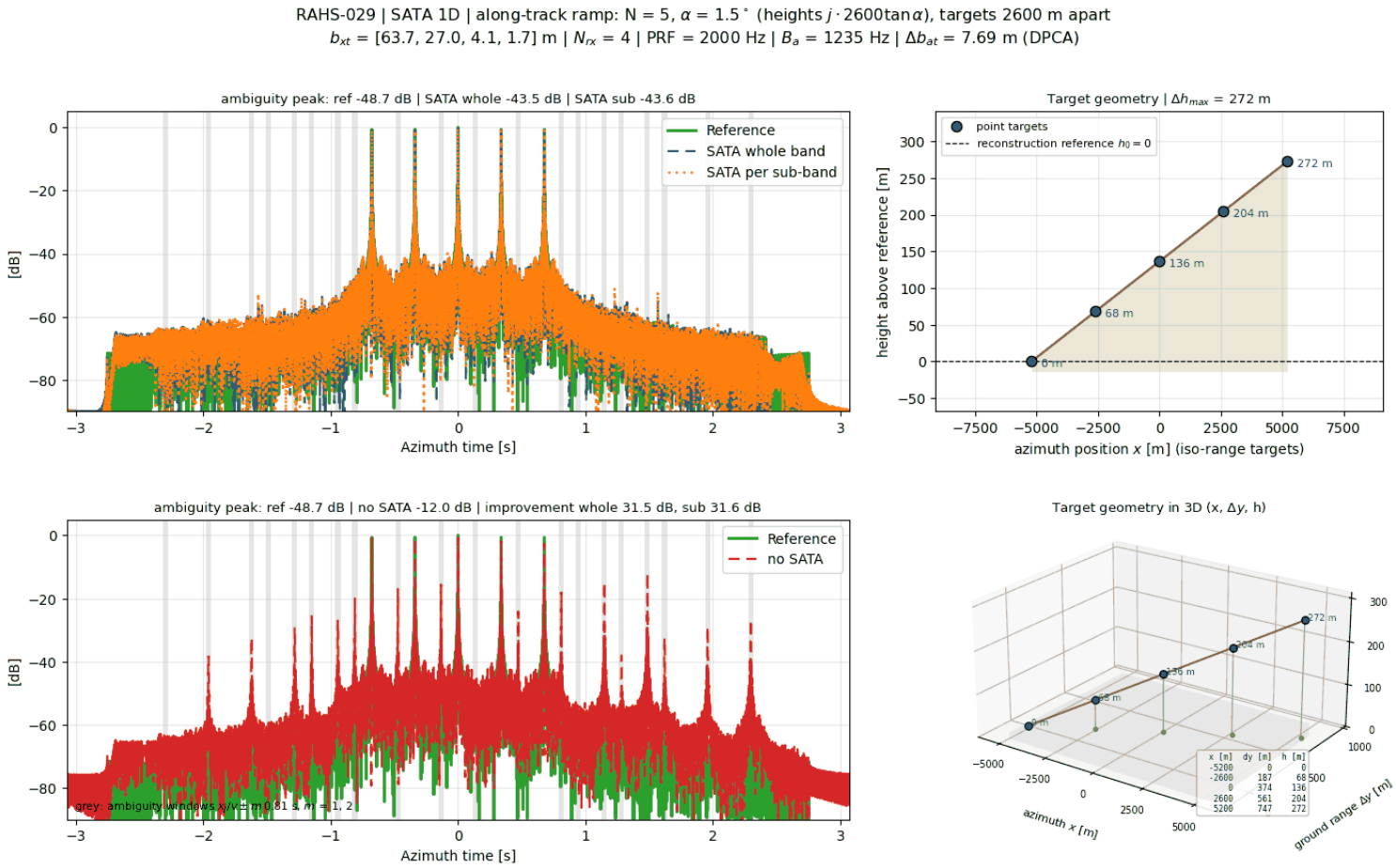


Figure 234: Along-track ramp, 2.6 km apart, large heights: RAHS-029, $N = 5$, $\alpha = 1.5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −12.0, SATA whole band −43.5, per sub-band −43.6 dB (improvement 31.5 dB).

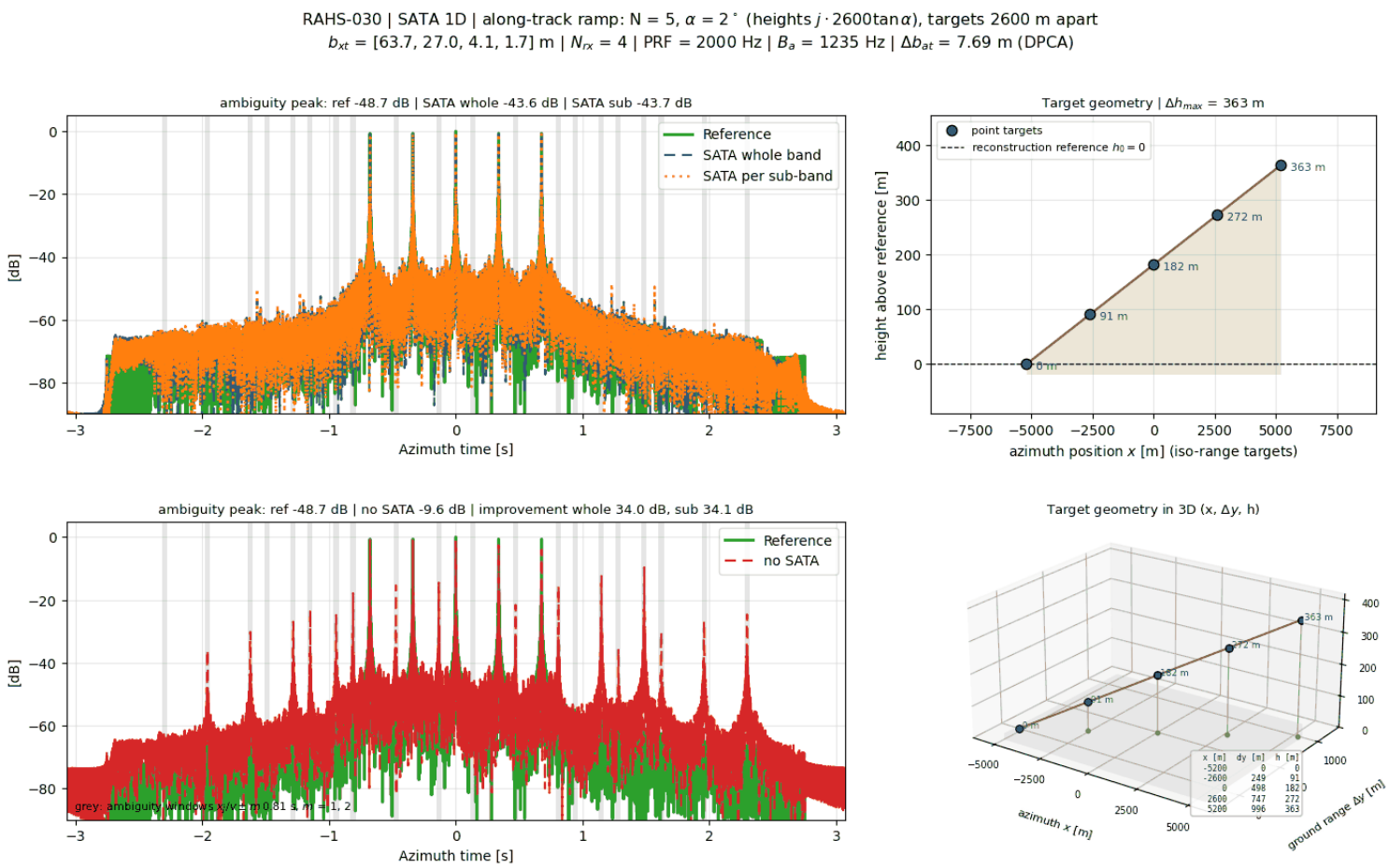


Figure 235: Along-track ramp, 2.6 km apart, large heights: RAHS-030, $N = 5$, $\alpha = 2°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −9.6, SATA whole band −43.6, per sub-band −43.7 dB (improvement 34.0 dB).

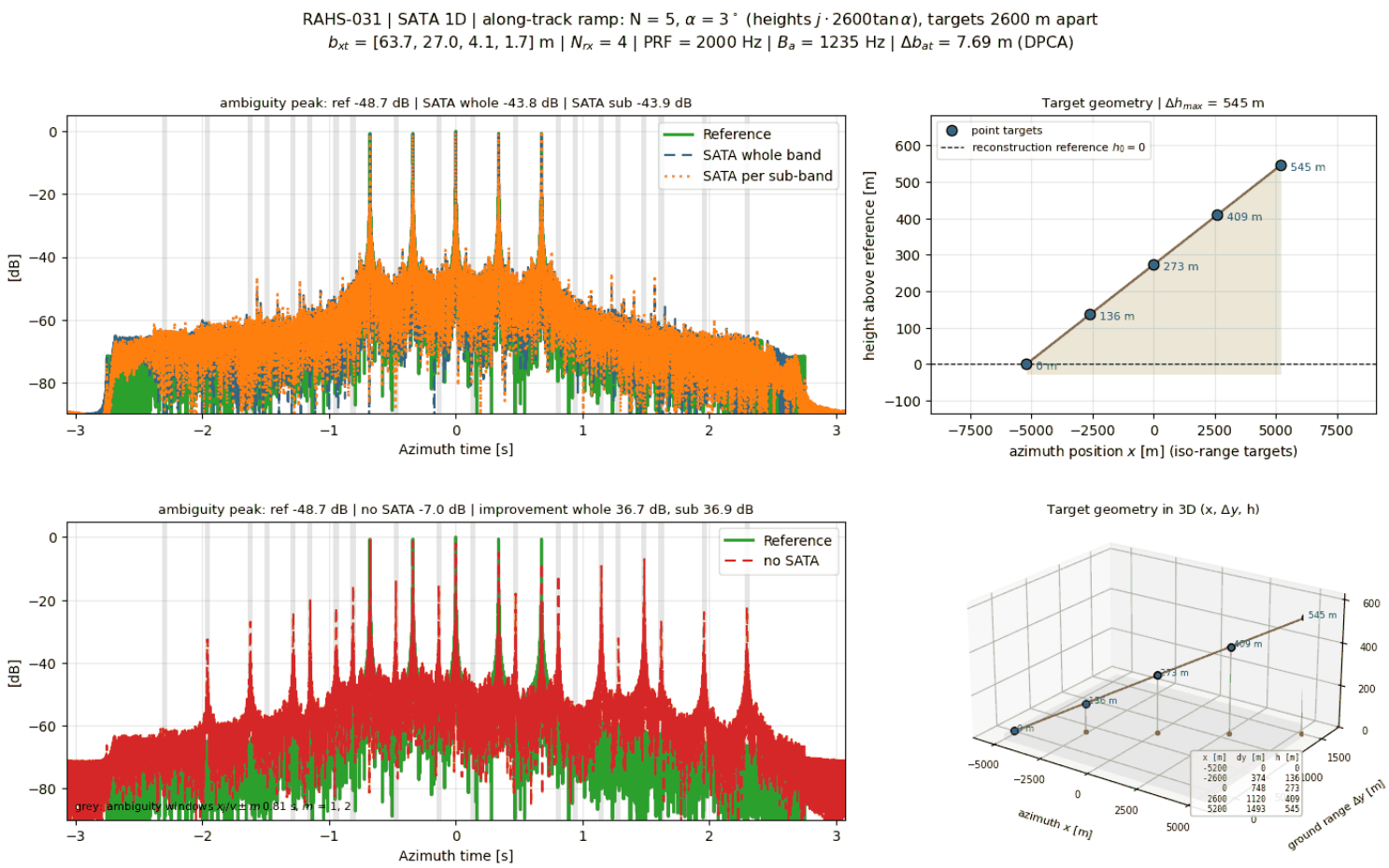


Figure 236: Along-track ramp, 2.6 km apart, large heights: RAHS-031, $N = 5$, $\alpha = 3°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA -7.0, SATA whole band -43.8, per sub-band -43.9 dB (improvement 36.7 dB).

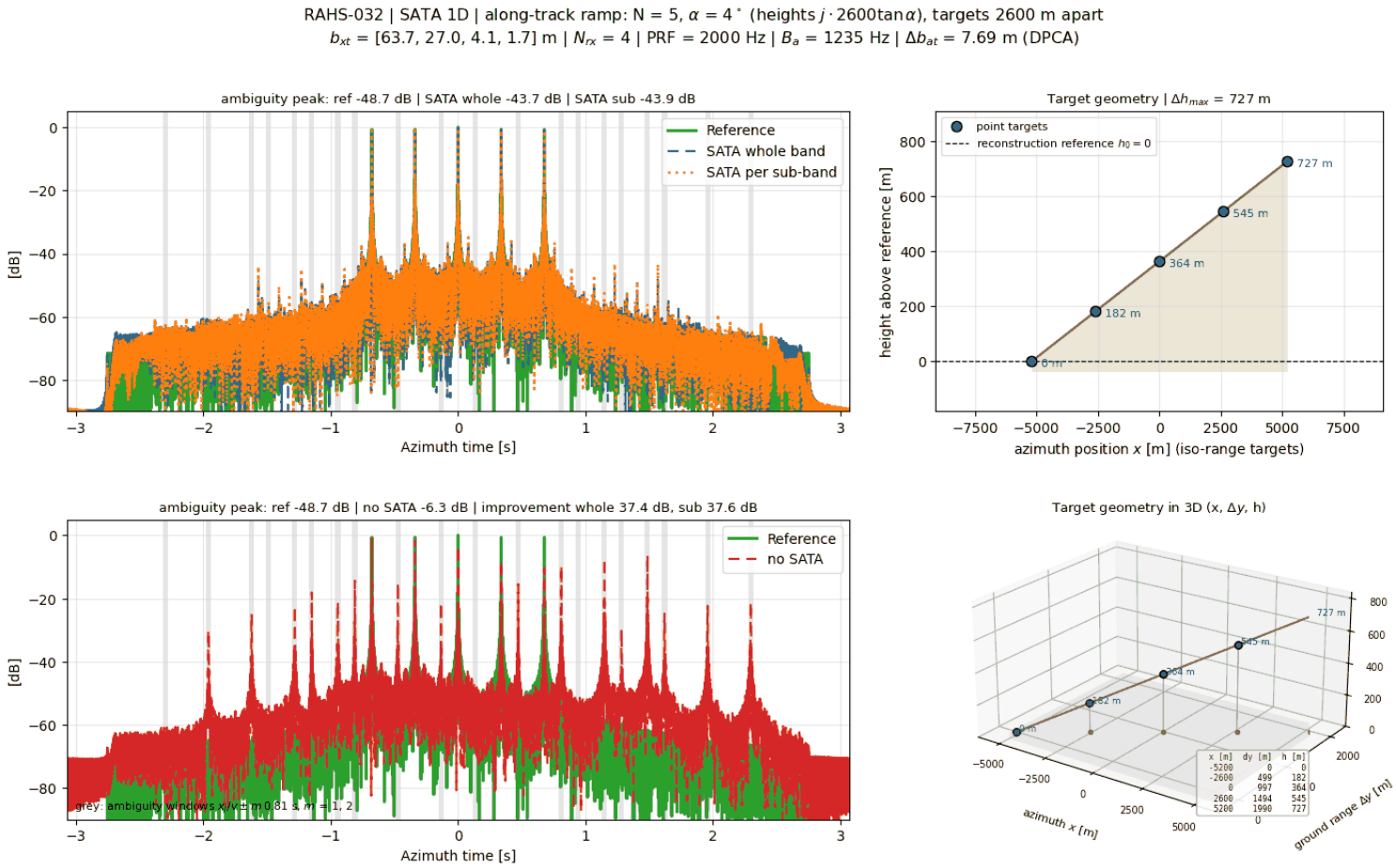


Figure 237: Along-track ramp, 2.6 km apart, large heights: RAHS-032, $N = 5$, $\alpha = 4°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.3, SATA whole band −43.7, per sub-band −43.9 dB (improvement 37.4 dB).

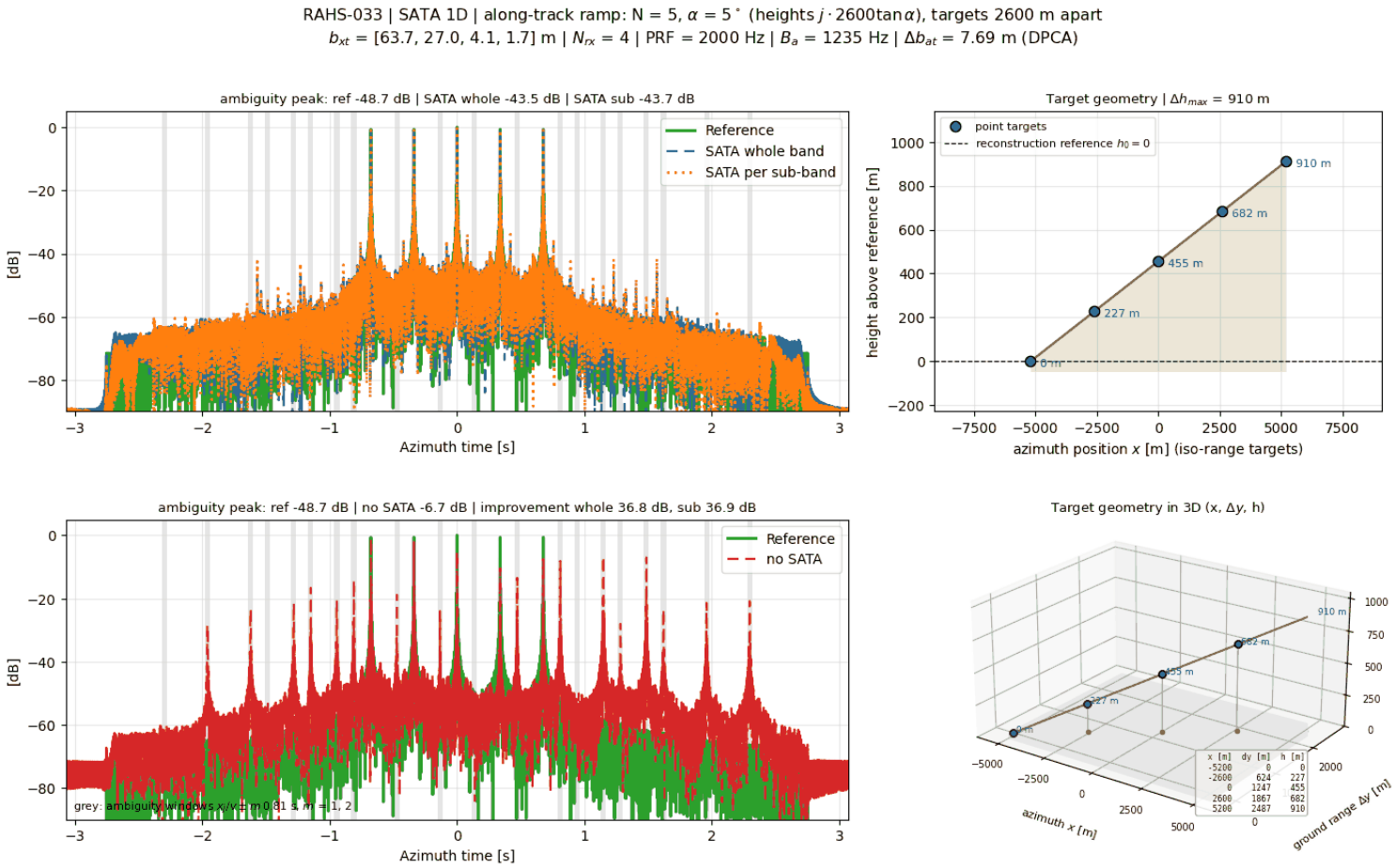


Figure 238: Along-track ramp, 2.6 km apart, large heights: RAHS-033, $N = 5$, $\alpha = 5°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.7, SATA whole band −43.5, per sub-band −43.7 dB (improvement 36.8 dB).

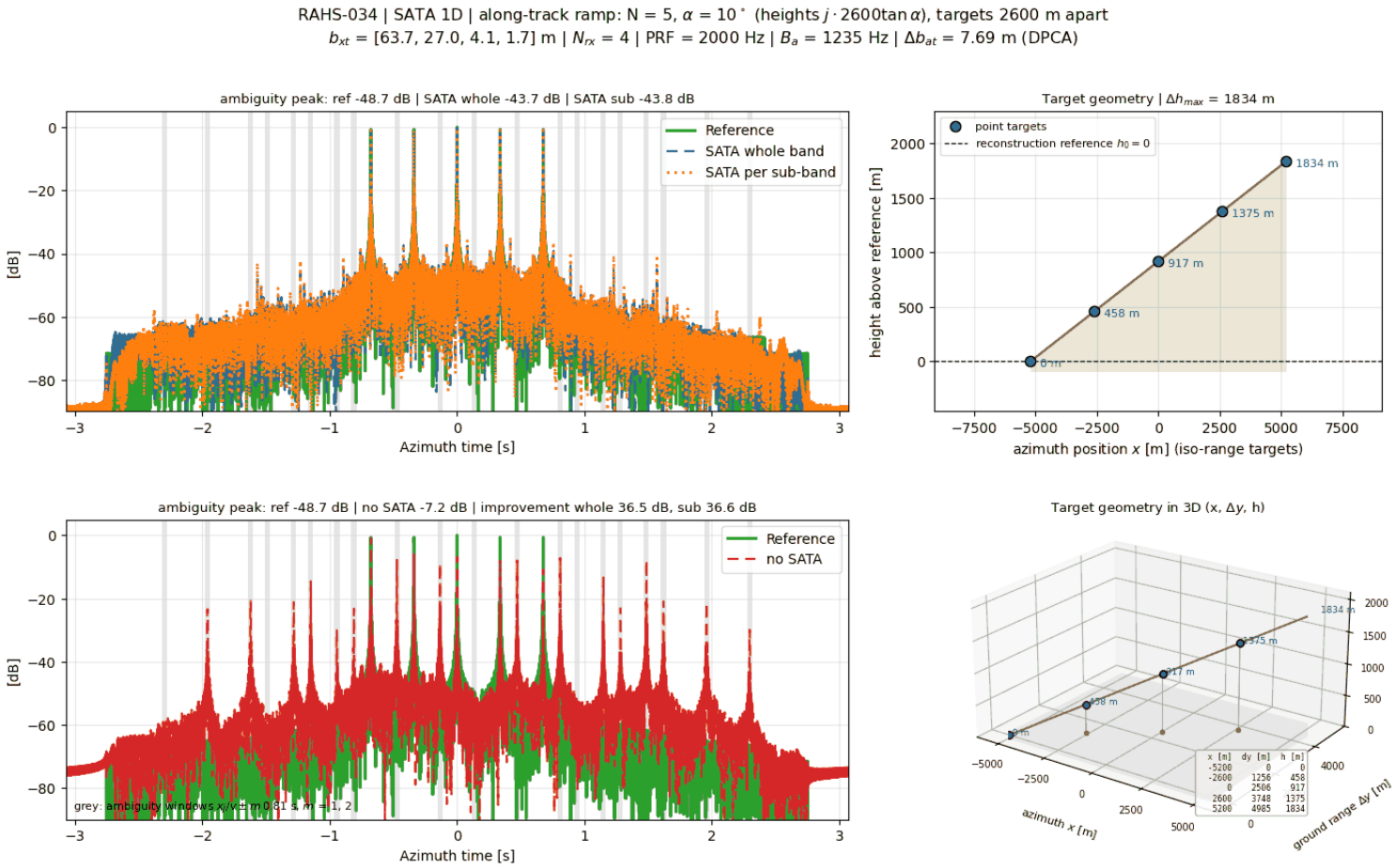


Figure 239: Along-track ramp, 2.6 km apart, large heights: RAHS-034, $N = 5$, $\alpha = 10°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.2, SATA whole band −43.7, per sub-band −43.8 dB (improvement 36.5 dB).

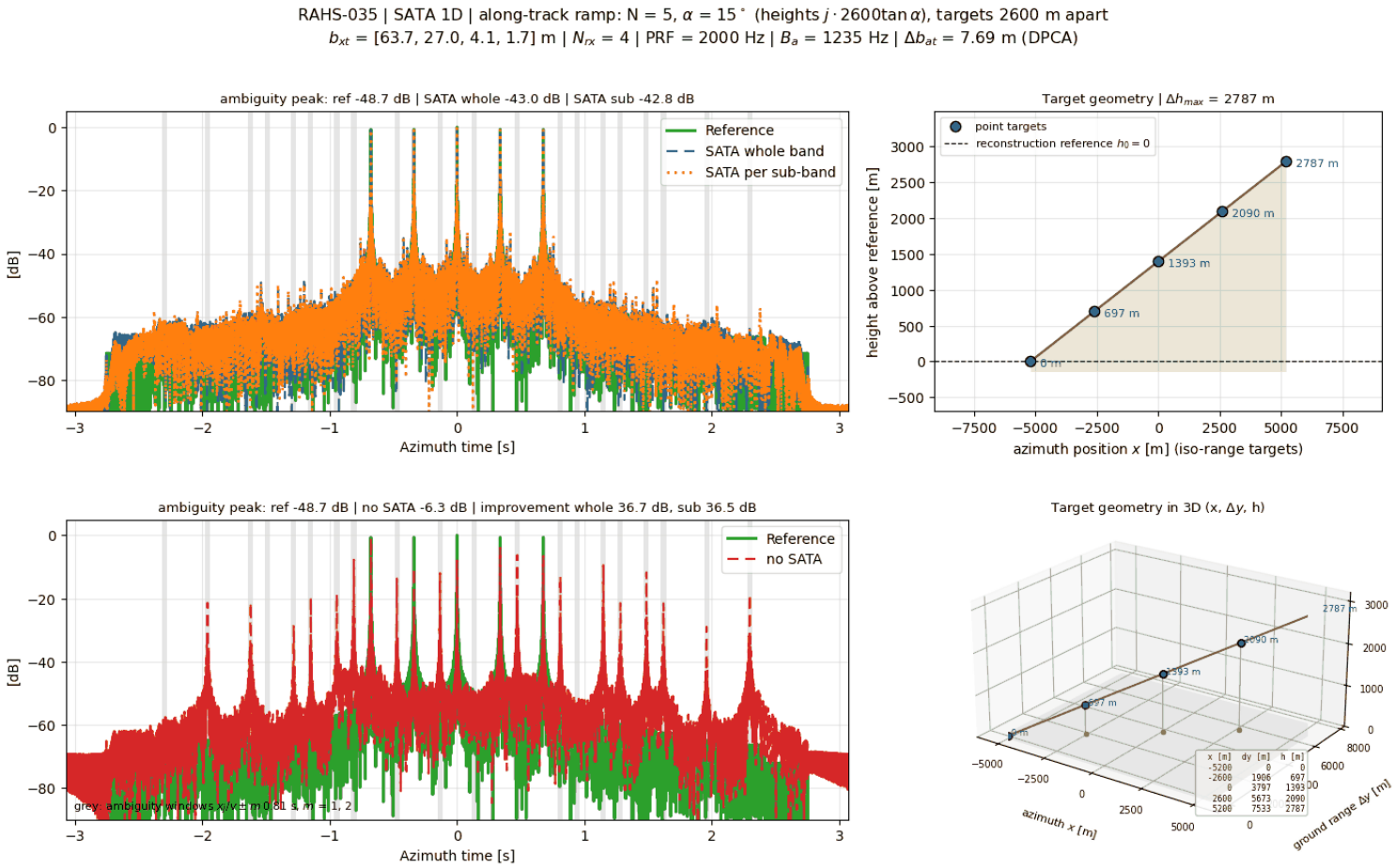


Figure 240: Along-track ramp, 2.6 km apart, large heights: RAHS-035, $N = 5$, $\alpha = 15°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.3, SATA whole band −43.0, per sub-band −42.8 dB (improvement 36.7 dB).

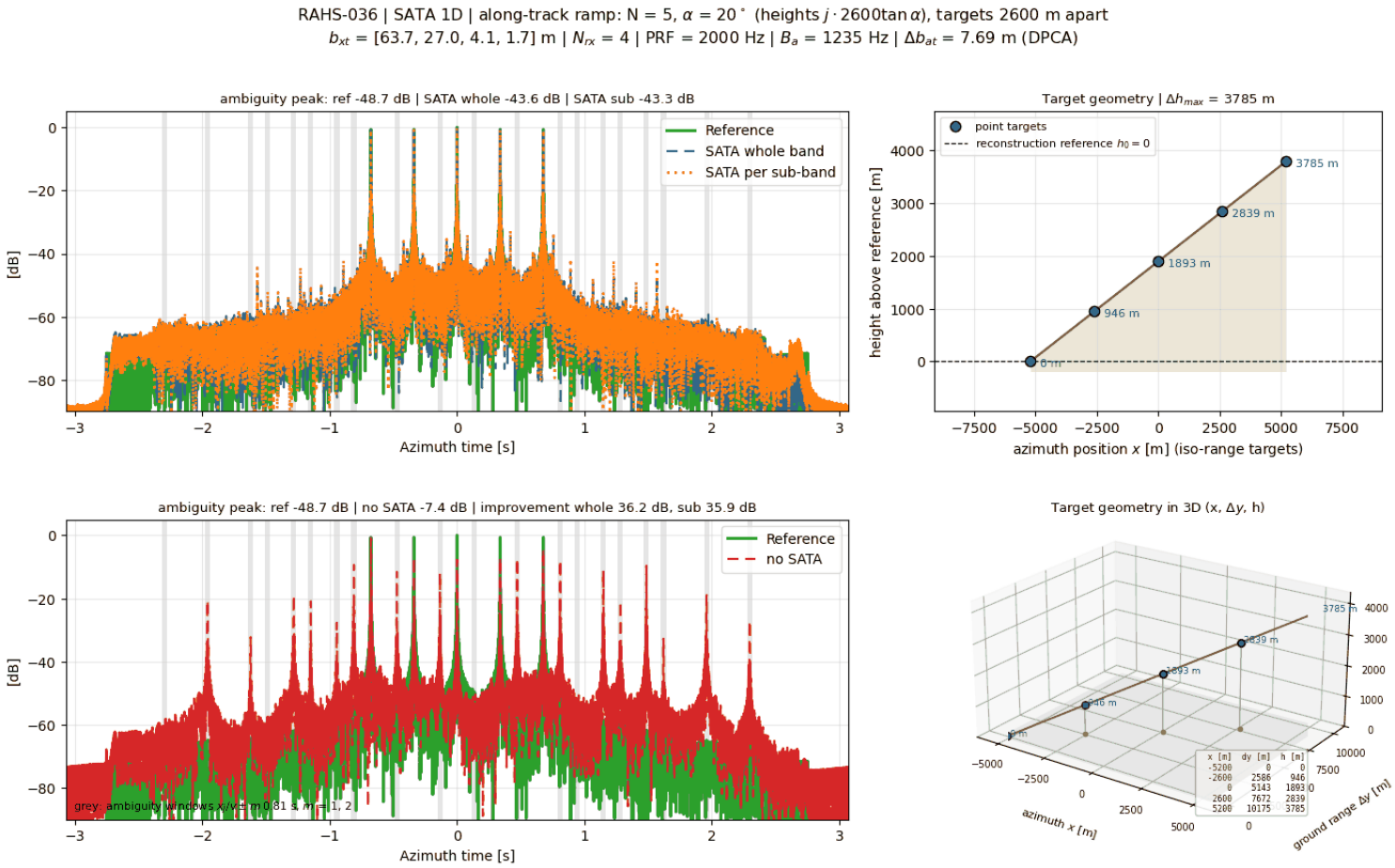


Figure 241: Along-track ramp, 2.6 km apart, large heights: RAHS-036, $N = 5$, $\alpha = 20°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −7.4, SATA whole band −43.6, per sub-band −43.3 dB (improvement 36.2 dB).

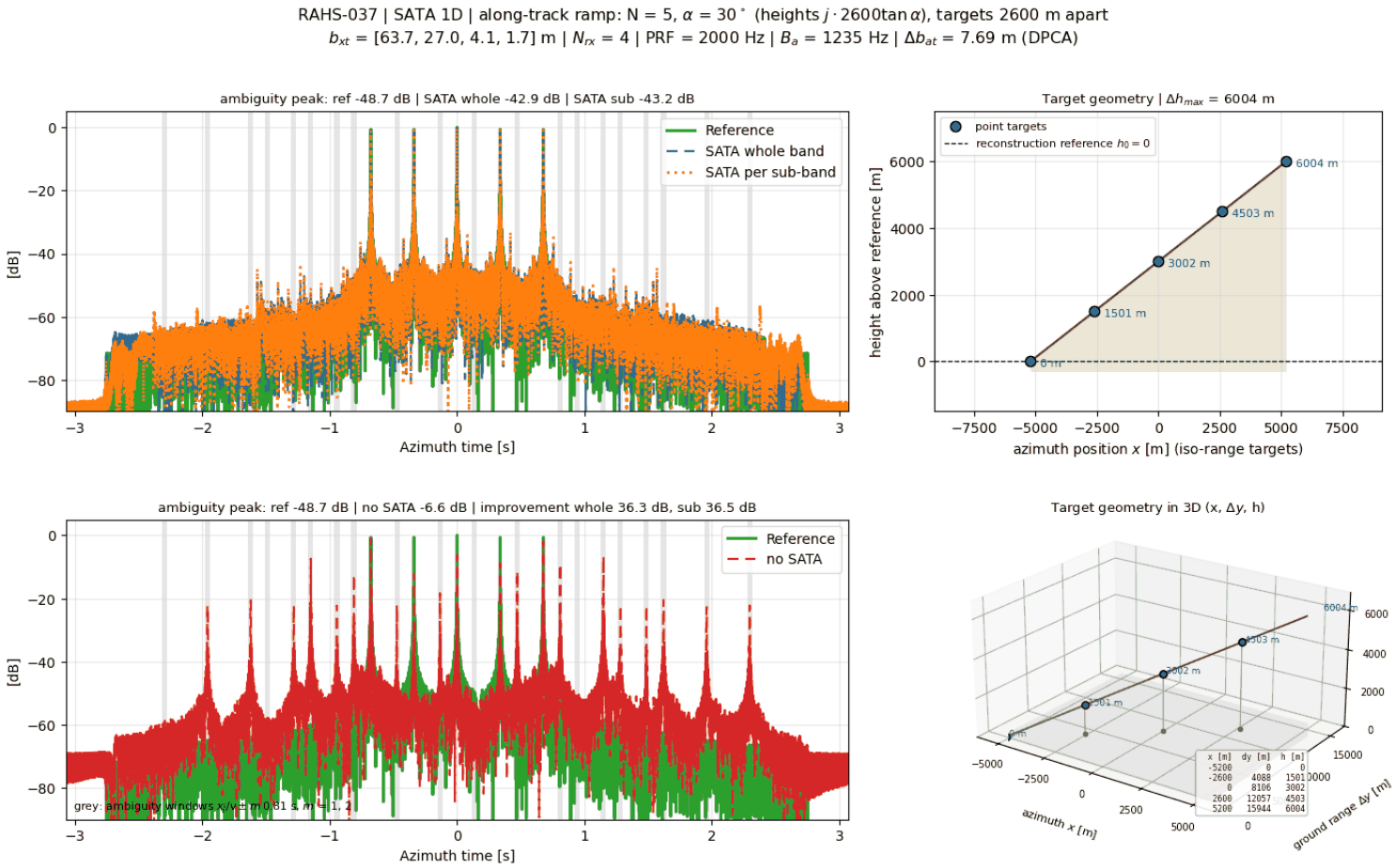


Figure 242: Along-track ramp, 2.6 km apart, large heights: RAHS-037, $N = 5$, $\alpha = 30°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −6.6, SATA whole band −42.9, per sub-band −43.2 dB (improvement 36.3 dB).

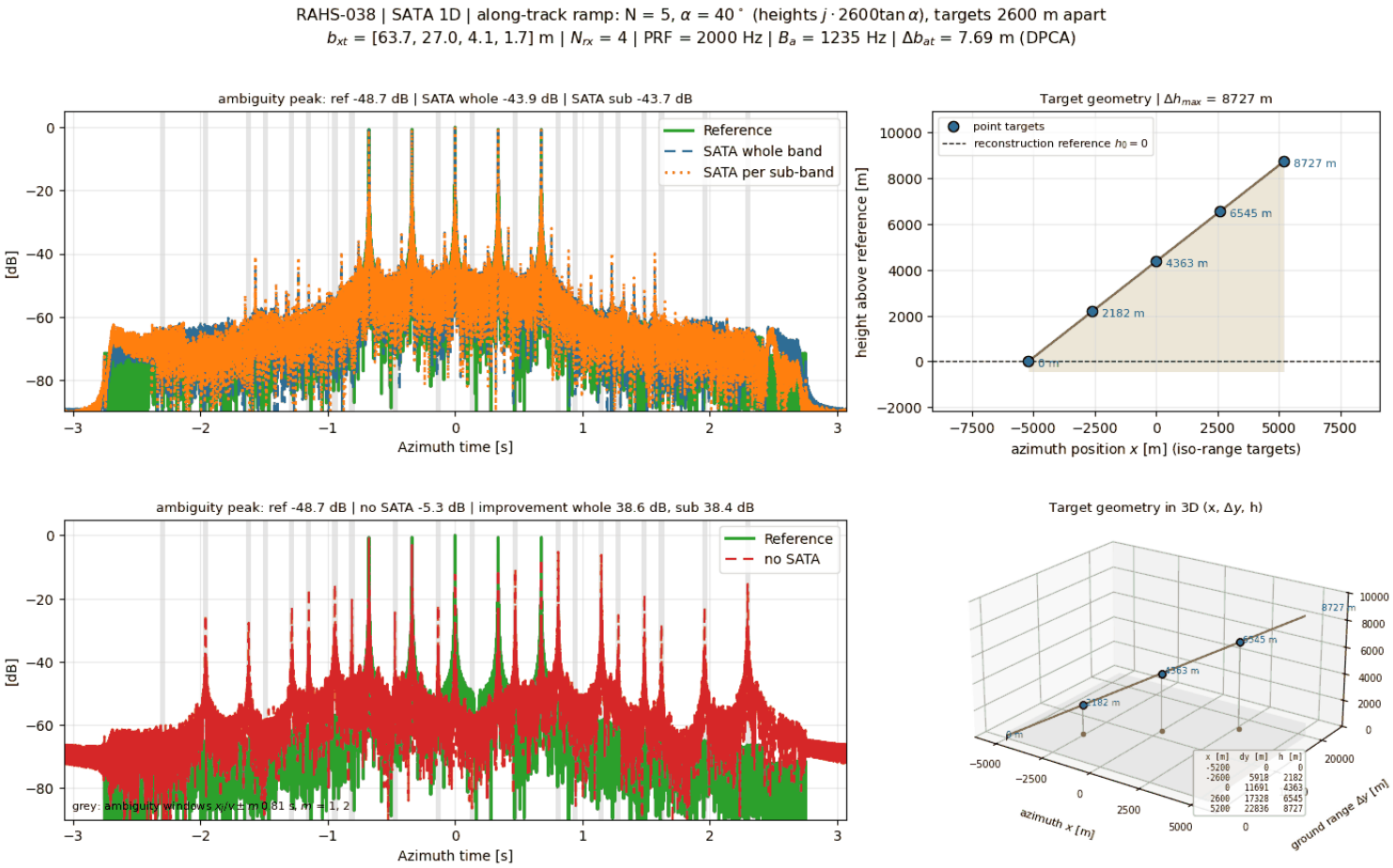


Figure 243: Along-track ramp, 2.6 km apart, large heights: RAHS-038, $N = 5$, $\alpha = 40°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −5.3, SATA whole band −43.9, per sub-band −43.7 dB (improvement 38.6 dB).

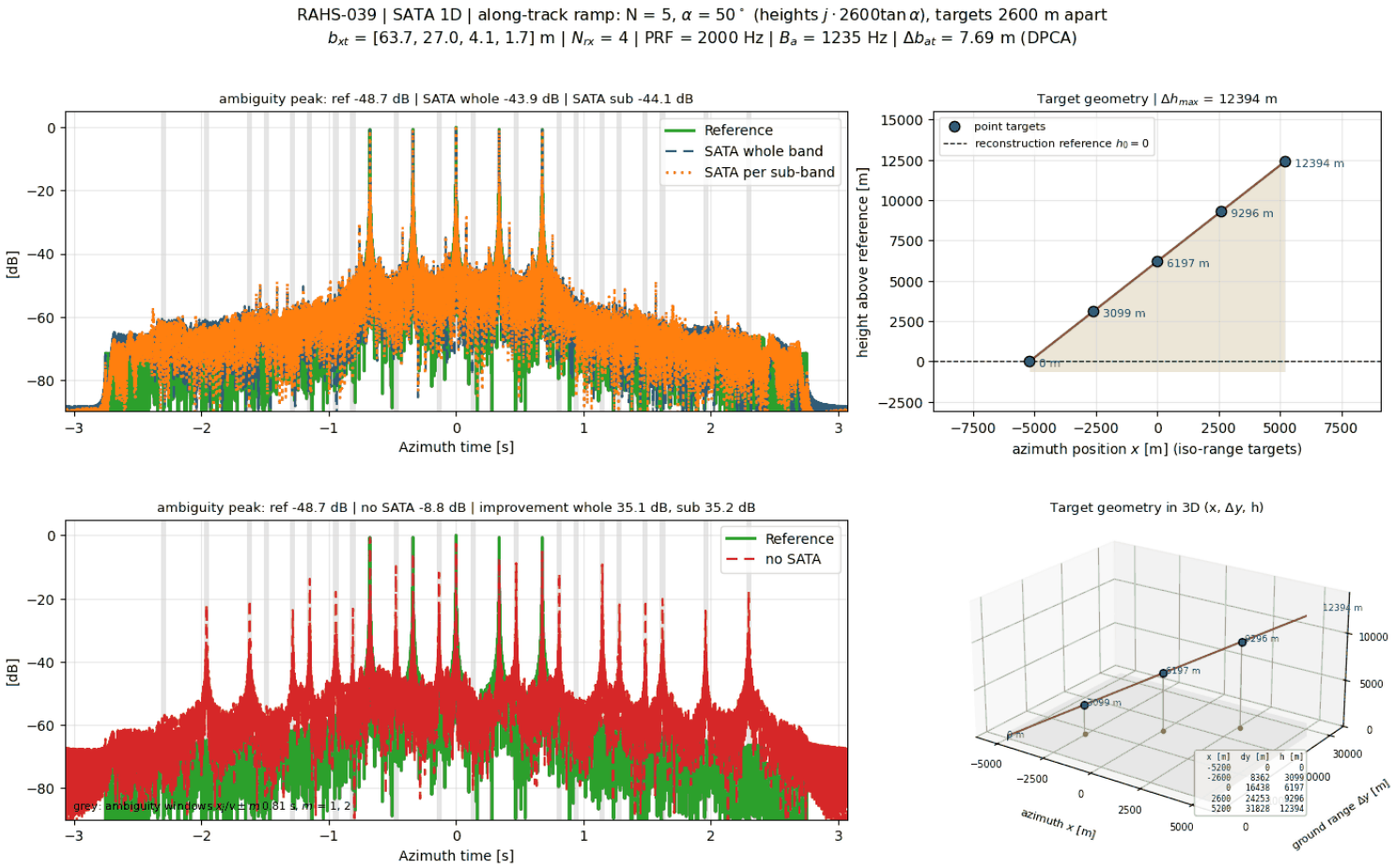


Figure 244: Along-track ramp, 2.6 km apart, large heights: RAHS-039, $N = 5$, $\alpha = 50°$ ($h_j = j \cdot 2600 \tan\alpha$). Ambiguity peak: no SATA −8.8, SATA whole band −43.9, per sub-band −44.1 dB (improvement 35.1 dB).